NANGATE

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Build Date: Thursday Feb 17 15:07 2011

Library

NangateOpenCellLibrary_typical

Corner

typical

AND2_X1

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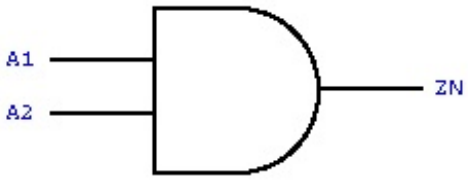
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	L
H	H	H
-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.02	0.10	0.07	0.16
	rise	0.02	0.17	0.05	0.20
A2 to ZN	**fall**	0.02	0.10	0.08	0.17
	rise	0.02	0.17	0.05	0.20

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A1	0.9181
A2	0.9746

Leakage [nW]
25.07

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	2.71	3.42	4.53	5.05
	rise	1.82	2.08	3.69	3.41
A2 to ZN	**fall**	3.04	3.80	4.90	5.42
	rise	1.83	1.91	3.04	3.11

AND2_X2

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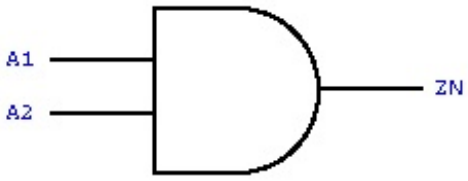
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	L
H	H	H
-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	**fall**	0.02	0.10	0.07	0.16
	rise	0.02	0.17	0.05	0.20
A2 to ZN	**fall**	0.02	0.10	0.08	0.17
	rise	0.02	0.17	0.04	0.19

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A1	1.6565
A2	1.7265

Leakage [nW]
50.35

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	**fall**	5.21	6.52	8.94	9.87
	rise	3.55	4.12	7.44	7.26
A2 to ZN	**fall**	5.87	7.27	9.70	10.61
	rise	3.61	3.95	6.09	6.02

AND2_X4

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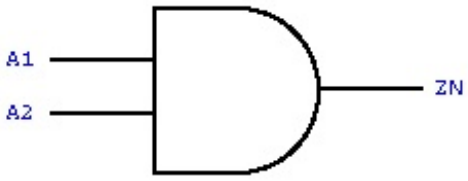
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	L
H	H	H
-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	**fall**	0.02	0.10	0.07	0.16
	rise	0.02	0.17	0.04	0.20
A2 to ZN	**fall**	0.02	0.10	0.08	0.17
	rise	0.02	0.17	0.04	0.19

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A1	3.1954
A2	3.5365

Leakage [nW]
100.71

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	**fall**	10.10	12.59	17.76	19.39
	rise	6.99	8.11	14.78	14.51
A2 to ZN	**fall**	11.38	14.11	19.18	20.83
	rise	7.00	8.06	11.99	12.18

AND3_X1

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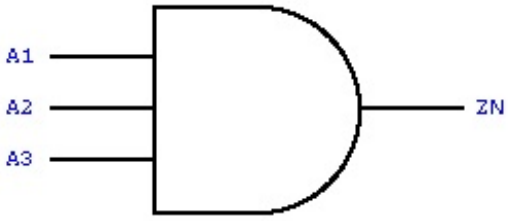
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	L
H	H	H	H
-	L	-	L
-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
A1 to ZN	**fall**	0.02	0.10	0.08	0.17
	rise	0.03	0.18	0.07	0.22
A2 to ZN	**fall**	0.03	0.11	0.08	0.17
	rise	0.03	0.19	0.07	0.22
A3 to ZN	**fall**	0.03	0.11	0.09	0.18
	rise	0.04	0.19	0.06	0.21

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A2 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A3 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A1	0.8797
A2	0.9275
A3	0.9648

Leakage [nW]
26.48

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
A1 to ZN	**fall**	2.85	3.63	4.51	5.05
	rise	2.03	2.15	3.83	3.54
A2 to ZN	**fall**	3.20	4.03	4.88	5.42
	rise	2.03	2.39	3.20	3.11
A3 to ZN	**fall**	3.48	4.37	5.26	5.82
	rise	2.03	2.19	2.99	2.96

AND3_X2

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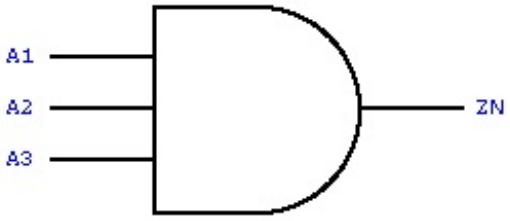
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.596 um^2
Equation	ZN = "((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	L
H	H	H	H
-	L	-	L
-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.544	0.3656	120.544
A1 to ZN	**fall**	0.02	0.10	0.07	0.16
	rise	0.03	0.18	0.06	0.22
A2 to ZN	**fall**	0.02	0.10	0.08	0.17
	rise	0.03	0.18	0.06	0.22
A3 to ZN	**fall**	0.03	0.11	0.09	0.18
	rise	0.03	0.18	0.05	0.21

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.544	0.3656	120.544
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A3 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A1	1.5994
A2	1.6489
A3	1.7001

Leakage [nW]
53.19

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.544	0.3656	120.544
A1 to ZN	**fall**	5.47	6.89	8.92	9.83
	rise	3.99	4.49	7.70	6.98
A2 to ZN	**fall**	6.14	7.70	9.65	10.55
	rise	4.02	4.62	6.50	6.09
A3 to ZN	**fall**	6.73	8.38	10.40	11.35
	rise	4.04	4.66	6.07	6.24

AND3_X4

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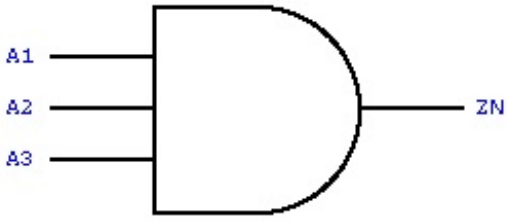
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Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	ZN = "((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	L
H	H	H	H
-	L	-	L
-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	241.089	0.3656	241.089
A1 to ZN	**fall**	0.02	0.10	0.07	0.16
	rise	0.03	0.18	0.06	0.22
A2 to ZN	**fall**	0.02	0.10	0.08	0.17
	rise	0.03	0.18	0.06	0.22
A3 to ZN	**fall**	0.03	0.11	0.09	0.17
	rise	0.03	0.18	0.05	0.21

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	241.089	0.3656	241.089
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A3 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A1	3.0851
A2	3.3007
A3	3.5818

Leakage [nW]
106.38

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	241.089	0.3656	241.089
A1 to ZN	**fall**	10.62	13.37	17.61	19.33
	rise	7.73	8.82	15.28	14.12
A2 to ZN	**fall**	12.03	15.00	19.05	20.75
	rise	7.67	8.81	12.68	12.32
A3 to ZN	**fall**	13.09	16.36	20.54	22.37
	rise	7.73	9.18	11.95	12.54

AND4_X1

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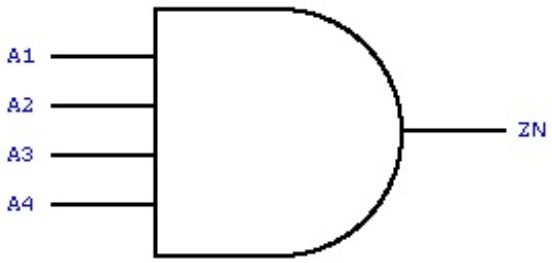
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Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	L
H	H	H	H	H
-	L	-	-	L
-	-	L	-	L
-	-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A1 to ZN	**fall**	0.03	0.11	0.08	0.17
	rise	0.04	0.20	0.09	0.24
A2 to ZN	**fall**	0.03	0.11	0.08	0.18
	rise	0.05	0.20	0.09	0.24
A3 to ZN	**fall**	0.03	0.11	0.09	0.18
	rise	0.05	0.20	0.08	0.24
A4 to ZN	**fall**	0.03	0.11	0.10	0.19
	rise	0.05	0.20	0.07	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A1 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A2 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A3 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A4 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A1	0.8565
A2	0.9023
A3	0.9241
A4	0.9445

Leakage [nW]
27.02

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.2722	0.3656	60.2722
A1 to ZN	fall	3.01	3.86	4.55	5.12
	rise	2.26	2.43	3.98	3.54
A2 to ZN	fall	3.34	4.26	4.91	5.49
	rise	2.27	2.50	3.42	3.01
A3 to ZN	fall	3.67	4.63	5.25	5.87
	rise	2.25	2.49	3.21	2.99
A4 to ZN	fall	3.90	4.95	5.61	6.24
	rise	2.25	2.60	3.07	3.04

AND4_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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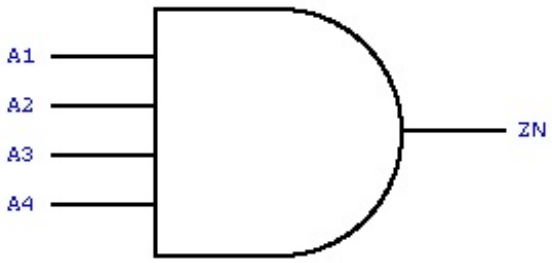
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	L
H	H	H	H	H
-	L	-	-	L
-	-	L	-	L
-	-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**120.392**	**0.3656**	**120.392**
A1 to ZN	**fall**	0.02	0.10	0.07	0.16
	rise	0.04	0.19	0.08	0.24
A2 to ZN	**fall**	0.03	0.11	0.08	0.17
	rise	0.04	0.20	0.08	0.24
A3 to ZN	**fall**	0.03	0.11	0.09	0.18
	rise	0.04	0.20	0.07	0.23
A4 to ZN	**fall**	0.03	0.11	0.09	0.18
	rise	0.05	0.20	0.06	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**120.392**	**0.3656**	**120.392**
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A3 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A4 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A1	1.5509
A2	1.6050
A3	1.6499
A4	1.6846

Leakage [nW]
54.27

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.392	0.3656	120.392
A1 to ZN	fall	5.66	7.17	8.88	9.80
	rise	4.39	4.82	7.87	7.07
A2 to ZN	fall	6.37	7.98	9.55	10.51
	rise	4.40	4.88	6.74	6.16
A3 to ZN	fall	6.98	8.72	10.25	11.28
	rise	4.40	5.03	6.36	6.26
A4 to ZN	fall	7.46	9.36	10.94	12.03
	rise	4.40	4.88	6.04	6.00

AND4_X4

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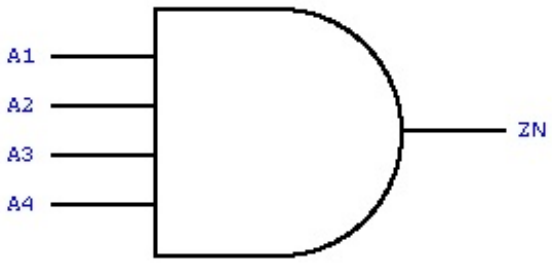
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	L
H	H	H	H	H
-	L	-	-	L
-	-	L	-	L
-	-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.02	0.10	0.07	0.16
	rise	0.04	0.19	0.08	0.24
A2 to ZN	**fall**	0.02	0.11	0.08	0.17
	rise	0.04	0.20	0.08	0.24
A3 to ZN	**fall**	0.03	0.11	0.08	0.18
	rise	0.04	0.20	0.07	0.23
A4 to ZN	**fall**	0.03	0.11	0.09	0.18
	rise	0.04	0.20	0.06	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A3 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A4 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A1	3.0146
A2	3.2606
A3	3.4860
A4	3.7662

Leakage [nW]
108.55

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	241.089	0.3656	241.089
A1 to ZN	fall	10.80	13.68	17.43	19.03
	rise	8.45	9.37	15.74	13.14
A2 to ZN	fall	12.15	15.32	18.77	20.43
	rise	8.48	9.61	13.31	12.19
A3 to ZN	fall	13.50	16.80	20.22	21.96
	rise	8.43	8.77	12.56	11.46
A4 to ZN	fall	14.50	18.08	21.52	23.47
	rise	8.38	9.82	11.79	12.07

ANTENNA_X1

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.266 um^2
Type	Physical
Input	A
PG Pins	VDD (primary_power), VSS (primary_ground)



Capacitance [fF]	
A	0.0234

Leakage [nW]
0.00

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1985**
A	**fall**	0.00	0.00
	rise	0.00	0.00

AOI211_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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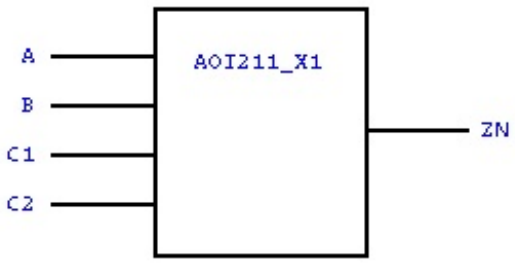
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((C1 & C2) | B) | A)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	L	L	-	H
L	L	-	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**14.4959**	**0.3656**	**14.4959**
A to ZN	**fall**	0.01	0.03	0.01	0.07
	rise	0.05	0.17	0.10	0.22
B to ZN	**fall**	0.01	0.03	0.01	0.06
	rise	0.05	0.16	0.11	0.24
C1 to ZN	**fall**	0.01	0.04	0.00	0.08
	rise	0.02	0.14	0.09	0.23
C2 to ZN	**fall**	0.01	0.04	0.01	0.08
	rise	0.03	0.14	0.10	0.23

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**14.4959**	**0.3656**	**14.4959**
A to ZN	**fall**	0.01	0.02	0.04	0.06
	rise	0.03	0.13	0.04	0.13
B to ZN	**fall**	0.01	0.02	0.04	0.06
	rise	0.03	0.13	0.05	0.13
C1 to ZN	**fall**	0.01	0.03	0.04	0.07
	rise	0.02	0.12	0.05	0.13
C2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.03	0.13	0.05	0.14

Capacitance [fF]	
A	1.6203
B	1.6584
C1	1.6552
C2	1.6795

Leakage [nW]
34.57

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	14.4959	0.3656	14.4959
A to ZN	fall	1.01	1.40	3.23	2.73
	rise	5.14	5.15	5.61	5.44
B to ZN	fall	0.90	1.08	2.82	2.40
	rise	4.40	4.44	5.20	4.89
C1 to ZN	fall	0.31	0.29	2.12	1.49
	rise	2.90	2.96	4.86	4.23
C2 to ZN	fall	0.32	0.30	1.49	1.11
	rise	3.63	3.70	5.58	5.13

AOI211_X2

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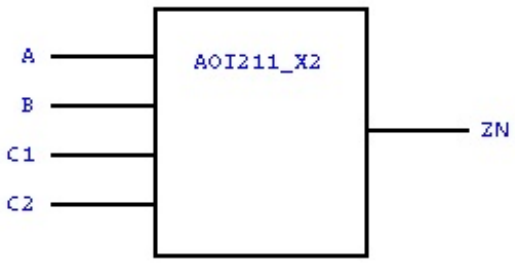
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((C1 & C2) | B) | A)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	L	L	-	H
L	L	-	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**28.9917**	**0.3656**	**28.9917**
A to ZN	**fall**	0.01	0.03	0.01	0.07
	rise	0.05	0.17	0.10	0.22
B to ZN	**fall**	0.01	0.03	0.01	0.06
	rise	0.05	0.16	0.11	0.24
C1 to ZN	**fall**	0.01	0.04	0.00	0.08
	rise	0.02	0.14	0.09	0.23
C2 to ZN	**fall**	0.01	0.04	0.01	0.08
	rise	0.03	0.14	0.10	0.23

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**28.9917**	**0.3656**	**28.9917**
A to ZN	**fall**	0.01	0.02	0.04	0.06
	rise	0.03	0.13	0.04	0.13
B to ZN	**fall**	0.00	0.02	0.04	0.06
	rise	0.03	0.13	0.05	0.13
C1 to ZN	**fall**	0.00	0.03	0.04	0.07
	rise	0.02	0.12	0.05	0.13
C2 to ZN	**fall**	0.00	0.03	0.03	0.05
	rise	0.03	0.13	0.05	0.14

Capacitance [fF]	
A	3.1142
B	3.4255
C1	3.1653
C2	3.4544

Leakage [nW]
69.13

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	28.9917	0.3656	28.9917
A to ZN	fall	1.94	2.70	6.37	5.35
	rise	10.38	10.52	11.35	10.99
B to ZN	fall	1.71	2.07	5.54	4.68
	rise	8.92	9.01	10.54	9.89
C1 to ZN	fall	0.64	0.59	4.24	2.96
	rise	5.80	5.86	9.74	8.49
C2 to ZN	fall	0.64	0.60	2.98	2.21
	rise	7.27	7.28	11.17	10.24

AOI211_X4

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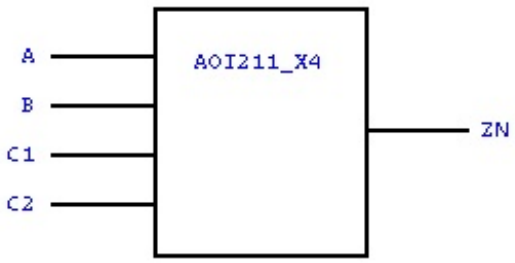
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	ZN = "!(!(!(((C1 & C2) | B) | A)))"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	L	L	-	H
L	L	-	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.03	0.11	0.06	0.14
	rise	0.10	0.25	0.16	0.31
B to ZN	**fall**	0.03	0.11	0.06	0.13
	rise	0.10	0.25	0.17	0.32
C1 to ZN	**fall**	0.04	0.12	0.06	0.14
	rise	0.08	0.22	0.16	0.30
C2 to ZN	**fall**	0.04	0.12	0.06	0.14
	rise	0.08	0.23	0.17	0.31

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
B to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
C1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
C2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A	1.6689
B	1.6963
C1	1.6327
C2	1.7500

Leakage [nW]
126.00

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	fall	10.69	13.31	13.65	16.15
	rise	15.65	18.03	16.47	19.70
B to ZN	fall	10.38	12.86	13.23	15.64
	rise	14.98	17.10	16.27	19.02
C1 to ZN	fall	9.56	11.93	12.57	15.07
	rise	13.50	16.12	16.36	18.85
C2 to ZN	fall	9.59	11.94	11.50	14.02
	rise	14.40	16.39	17.23	19.63

AOI21_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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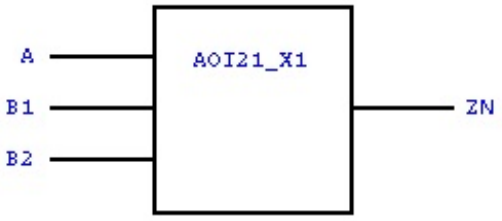
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!(A | (B1 & B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	L	-	H
L	-	L	H
H	-	-	L
-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to ZN	**fall**	0.01	0.04	0.01	0.09
	rise	0.03	0.16	0.08	0.24
B1 to ZN	**fall**	0.01	0.06	0.00	0.12
	rise	0.02	0.14	0.07	0.24
B2 to ZN	**fall**	0.01	0.06	0.01	0.11
	rise	0.02	0.15	0.08	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to ZN	**fall**	0.00	0.03	0.04	0.07
	rise	0.02	0.13	0.03	0.14
B1 to ZN	**fall**	0.00	0.05	0.04	0.08
	rise	0.01	0.13	0.04	0.14
B2 to ZN	**fall**	0.00	0.05	0.03	0.07
	rise	0.02	0.13	0.04	0.14

Capacitance [fF]	
A	1.6264
B1	1.6470
B2	1.6769

Leakage [nW]
27.86

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to ZN	**fall**	0.83	1.07	3.33	2.30
	rise	4.06	4.10	5.31	4.64
B1 to ZN	**fall**	0.26	0.29	2.54	1.28
	rise	2.55	2.66	4.94	3.94
B2 to ZN	**fall**	0.27	0.30	1.79	0.92
	rise	3.27	3.30	5.62	4.75

AOI21_X2

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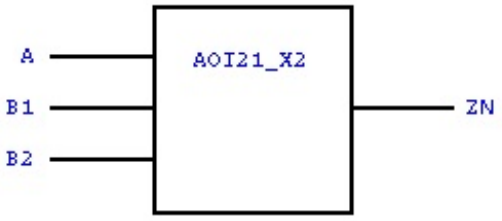
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!(A | (B1 & B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	L	-	H
L	-	L	H
H	-	-	L
-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**50.6592**	**0.3656**	**50.6592**
A to ZN	**fall**	0.01	0.04	0.01	0.09
	rise	0.03	0.16	0.08	0.24
B1 to ZN	**fall**	0.01	0.06	0.00	0.12
	rise	0.02	0.14	0.07	0.24
B2 to ZN	**fall**	0.01	0.06	0.01	0.11
	rise	0.02	0.15	0.08	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**50.6592**	**0.3656**	**50.6592**
A to ZN	**fall**	0.00	0.03	0.04	0.07
	rise	0.01	0.13	0.03	0.14
B1 to ZN	**fall**	0.00	0.05	0.03	0.08
	rise	0.01	0.13	0.04	0.14
B2 to ZN	**fall**	0.00	0.05	0.03	0.07
	rise	0.01	0.13	0.04	0.14

Capacitance [fF]	
A	3.1364
B1	3.1298
B2	3.4825

Leakage [nW]
55.72

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**50.6592**	**0.3656**	**50.6592**
A to ZN	**fall**	1.58	2.03	6.61	4.50
	rise	7.90	7.95	10.56	9.16
B1 to ZN	**fall**	0.53	0.60	5.13	2.55
	rise	5.04	5.26	9.91	7.83
B2 to ZN	**fall**	0.53	0.60	3.56	1.84
	rise	6.50	6.58	11.21	9.42

AOI21_X4

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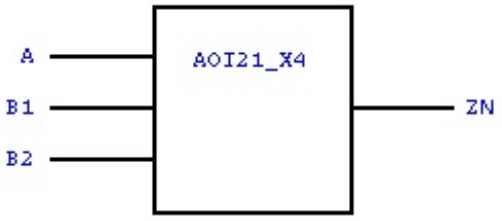
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(A | (B1 & B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	L	-	H
L	-	L	H
H	-	-	L
-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**101.013**	**0.3656**	**101.013**
A to ZN	**fall**	0.01	0.04	0.01	0.09
	rise	0.03	0.16	0.08	0.24
B1 to ZN	**fall**	0.01	0.06	0.00	0.12
	rise	0.02	0.14	0.07	0.24
B2 to ZN	**fall**	0.01	0.06	0.01	0.11
	rise	0.02	0.15	0.08	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**101.013**	**0.3656**	**101.013**
A to ZN	**fall**	0.00	0.03	0.04	0.07
	rise	0.01	0.13	0.03	0.14
B1 to ZN	**fall**	0.00	0.05	0.03	0.08
	rise	0.01	0.13	0.04	0.14
B2 to ZN	**fall**	0.00	0.05	0.03	0.07
	rise	0.01	0.13	0.04	0.14

Capacitance [fF]	
A	6.1393
B1	6.4019
B2	6.7132

Leakage [nW]
111.43

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**101.013**	**0.3656**	**101.013**
A to ZN	**fall**	3.15	4.05	13.20	8.96
	rise	16.16	16.25	21.38	18.73
B1 to ZN	**fall**	1.04	1.19	10.24	5.10
	rise	10.17	10.43	19.89	15.55
B2 to ZN	**fall**	1.08	1.19	7.17	3.67
	rise	13.01	13.07	22.47	18.72

AOI221_X1

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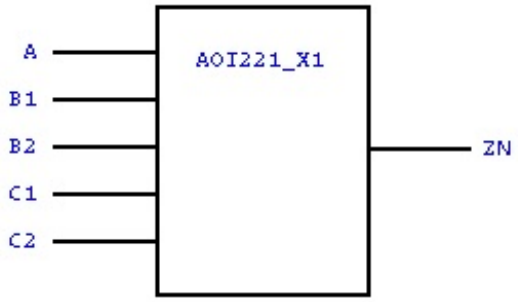
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "!(((C1 & C2) | A) | (B1 & B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	L	-	H
L	L	-	-	L	H
L	-	L	L	-	H
L	-	L	-	L	H
H	-	-	-	-	L
-	H	H	-	-	L
-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	13.8092	0.3656	13.8092
A to ZN	**fall**	0.01	0.02	0.01	0.06
	rise	0.06	0.16	0.12	0.24
B1 to ZN	**fall**	0.01	0.05	0.03	0.10
	rise	0.06	0.16	0.10	0.22
B2 to ZN	**fall**	0.02	0.05	0.03	0.09
	rise	0.06	0.17	0.11	0.23
C1 to ZN	**fall**	0.01	0.04	0.01	0.08
	rise	0.03	0.14	0.10	0.23
C2 to ZN	**fall**	0.01	0.04	0.01	0.08
	rise	0.04	0.15	0.12	0.24

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	13.8092	0.3656	13.8092
A to ZN	**fall**	0.01	0.02	0.04	0.06
	rise	0.03	0.13	0.05	0.13
B1 to ZN	**fall**	0.01	0.04	0.05	0.07
	rise	0.03	0.12	0.04	0.13
B2 to ZN	**fall**	0.01	0.04	0.04	0.06
	rise	0.03	0.13	0.05	0.13
C1 to ZN	**fall**	0.01	0.03	0.04	0.07
	rise	0.03	0.12	0.05	0.13
C2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.03	0.13	0.05	0.14

Capacitance [fF]	
A	1.6775
B1	1.5816
B2	1.6283
C1	1.6323
C2	1.7067

Leakage [nW]
41.74

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	13.8092	0.3656	13.8092
A to ZN	fall	0.83	0.96	2.83	2.36
	rise	4.66	4.71	5.54	5.24
B1 to ZN	fall	1.29	1.63	3.18	2.63
	rise	5.48	5.51	5.72	5.64
B2 to ZN	fall	1.28	1.63	2.51	2.22
	rise	6.19	6.26	6.45	6.42
C1 to ZN	fall	0.31	0.30	2.20	1.53
	rise	3.15	3.23	5.12	4.60
C2 to ZN	fall	0.31	0.30	1.54	1.13
	rise	3.88	3.93	5.86	5.43

AOI221_X2

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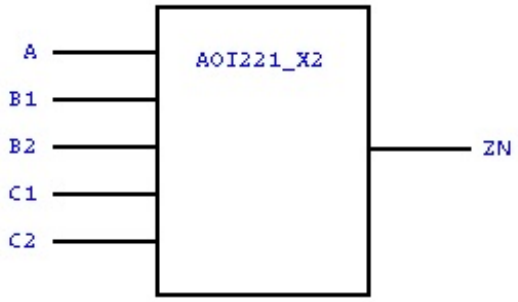
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.926 um^2
Equation	ZN = "!(((C1 & C2) | A) | (B1 & B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	L	-	H
L	L	-	-	L	H
L	-	L	L	-	H
L	-	L	-	L	H
H	-	-	-	-	L
-	H	H	-	-	L
-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	27.6184	0.3656	27.6184
A to ZN	**fall**	0.01	0.02	0.01	0.06
	rise	0.06	0.16	0.12	0.24
B1 to ZN	**fall**	0.01	0.05	0.03	0.10
	rise	0.05	0.16	0.10	0.22
B2 to ZN	**fall**	0.02	0.05	0.03	0.09
	rise	0.06	0.17	0.11	0.23
C1 to ZN	**fall**	0.01	0.04	0.00	0.08
	rise	0.03	0.14	0.10	0.23
C2 to ZN	**fall**	0.01	0.04	0.01	0.08
	rise	0.04	0.15	0.11	0.24

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	27.6184	0.3656	27.6184
A to ZN	**fall**	0.01	0.02	0.04	0.06
	rise	0.03	0.13	0.05	0.13
B1 to ZN	**fall**	0.01	0.04	0.05	0.07
	rise	0.03	0.12	0.04	0.13
B2 to ZN	**fall**	0.01	0.04	0.04	0.06
	rise	0.03	0.13	0.04	0.13
C1 to ZN	**fall**	0.01	0.03	0.04	0.07
	rise	0.03	0.12	0.05	0.13
C2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.03	0.13	0.05	0.14

Capacitance [fF]	
A	3.5047
B1	3.2261
B2	3.1353
C1	3.1450
C2	3.5015

Leakage [nW]
83.48

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	27.6184	0.3656	27.6184
A to ZN	fall	1.72	2.04	5.75	4.80
	rise	9.34	9.42	11.09	10.50
B1 to ZN	fall	2.59	3.27	6.39	5.28
	rise	10.93	10.97	11.41	11.22
B2 to ZN	fall	2.61	3.27	5.05	4.47
	rise	12.35	12.44	12.88	12.78
C1 to ZN	fall	0.62	0.59	4.42	3.05
	rise	6.35	6.52	10.32	9.19
C2 to ZN	fall	0.62	0.60	3.09	2.25
	rise	7.80	7.94	11.76	10.88

AOI221_X4

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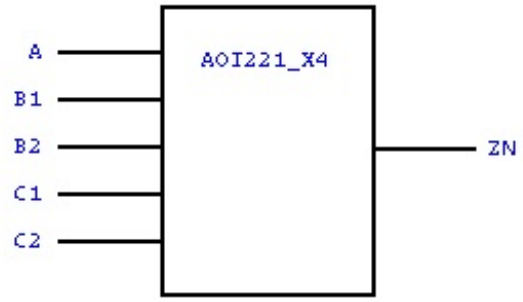
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(!(!(((C1 & C2) | A) | (B1 & B2))))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	L	-	H
L	L	-	-	L	H
L	-	L	L	-	H
L	-	L	-	L	H
H	-	-	-	-	L
-	H	H	-	-	L
-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.03	0.11	0.06	0.14
	rise	0.11	0.26	0.18	0.33
B1 to ZN	**fall**	0.05	0.13	0.09	0.17
	rise	0.11	0.26	0.16	0.31
B2 to ZN	**fall**	0.05	0.13	0.08	0.16
	rise	0.12	0.26	0.17	0.32
C1 to ZN	**fall**	0.04	0.12	0.07	0.15
	rise	0.08	0.23	0.17	0.31
C2 to ZN	**fall**	0.04	0.12	0.06	0.14
	rise	0.09	0.24	0.18	0.33

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
B1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
B2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
C1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
C2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A	1.6491
B1	1.5572
B2	1.6497
C1	1.6008
C2	1.6811

Leakage [nW]
131.58

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	fall	10.40	12.86	13.35	15.77
	rise	15.25	17.83	16.60	19.41
B1 to ZN	fall	11.03	13.74	13.89	16.39
	rise	16.08	18.55	16.56	19.13
B2 to ZN	fall	11.04	13.74	12.75	15.31
	rise	16.93	19.73	17.56	20.28
C1 to ZN	fall	9.60	11.96	12.72	15.20
	rise	13.91	16.40	16.73	19.45
C2 to ZN	fall	9.61	11.97	11.59	14.12
	rise	14.73	17.14	17.62	20.37

AOI222_X1

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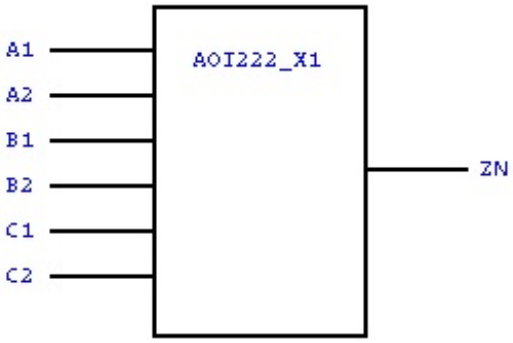
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.128 um^2
Equation	ZN = "!(((A1 & A2) | (B1 & B2)) | (C1 & C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	L	-	L	-	H
L	-	L	-	-	L	H
L	-	-	L	L	-	H
L	-	-	L	-	L	H
H	H	-	-	-	-	L
-	L	L	-	L	-	H
-	L	L	-	-	L	H
-	L	-	L	L	-	H
-	L	-	L	-	L	H
-	-	H	H	-	-	L
-	-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**13.0081**	**0.3656**	**13.0081**
A1 to ZN	**fall**	0.01	0.04	0.01	0.08
	rise	0.04	0.14	0.11	0.23
A2 to ZN	**fall**	0.01	0.04	0.01	0.07
	rise	0.05	0.15	0.13	0.24
B1 to ZN	**fall**	0.01	0.04	0.03	0.09
	rise	0.06	0.16	0.12	0.24
B2 to ZN	**fall**	0.02	0.04	0.03	0.09
	rise	0.07	0.17	0.13	0.25
C1 to ZN	**fall**	0.02	0.05	0.04	0.10
	rise	0.07	0.17	0.12	0.23
C2 to ZN	**fall**	0.02	0.05	0.04	0.09
	rise	0.08	0.18	0.13	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**13.0081**	**0.3656**	**13.0081**
A1 to ZN	**fall**	0.01	0.03	0.05	0.07
	rise	0.03	0.12	0.05	0.13
A2 to ZN	**fall**	0.01	0.03	0.04	0.06
	rise	0.04	0.13	0.06	0.14
B1 to ZN	**fall**	0.01	0.04	0.05	0.07
	rise	0.03	0.12	0.05	0.13
B2 to ZN	**fall**	0.01	0.04	0.04	0.06
	rise	0.04	0.13	0.05	0.13
C1 to ZN	**fall**	0.02	0.04	0.05	0.08
	rise	0.03	0.12	0.05	0.13
C2 to ZN	**fall**	0.02	0.04	0.04	0.06
	rise	0.04	0.13	0.05	0.13

Capacitance [fF]	
A1	1.6367
A2	1.6953
B1	1.5791
B2	1.6219
C1	1.5472
C2	1.5867

Leakage [nW]
47.40

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	13.0081	0.3656	13.0081
A1 to ZN	fall	0.32	0.29	2.27	1.59
	rise	3.38	3.47	5.42	4.91
A2 to ZN	fall	0.32	0.29	1.59	1.16
	rise	4.10	4.18	6.16	5.76
B1 to ZN	fall	1.13	1.27	2.93	2.39
	rise	5.01	5.07	5.57	5.35
B2 to ZN	fall	1.12	1.27	2.28	1.98
	rise	5.72	5.77	6.32	6.18
C1 to ZN	fall	1.63	2.03	3.49	3.04
	rise	6.35	6.43	6.40	6.39
C2 to ZN	fall	1.62	2.02	2.83	2.63
	rise	7.06	7.13	7.16	7.20

AOI222_X2

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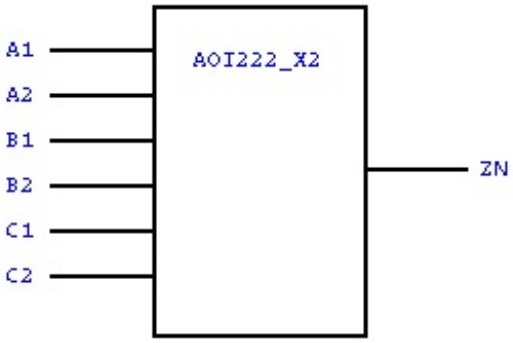
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	3.724 um^2
Equation	ZN = "!(((A1 & A2) | (B1 & B2)) | (C1 & C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	L	-	L	-	H
L	-	L	-	-	L	H
L	-	-	L	L	-	H
L	-	-	L	-	L	H
H	H	-	-	-	-	L
-	L	L	-	L	-	H
-	L	L	-	-	L	H
-	L	-	L	L	-	H
-	L	-	L	-	L	H
-	-	H	H	-	-	L
-	-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**25.6348**	**0.3656**	**25.6348**
A1 to ZN	**fall**	0.01	0.04	0.01	0.08
	rise	0.04	0.14	0.11	0.23
A2 to ZN	**fall**	0.01	0.04	0.01	0.07
	rise	0.05	0.15	0.13	0.24
B1 to ZN	**fall**	0.01	0.04	0.03	0.09
	rise	0.06	0.16	0.12	0.24
B2 to ZN	**fall**	0.02	0.04	0.03	0.09
	rise	0.07	0.17	0.13	0.25
C1 to ZN	**fall**	0.02	0.05	0.04	0.10
	rise	0.07	0.17	0.12	0.23
C2 to ZN	**fall**	0.02	0.05	0.04	0.09
	rise	0.08	0.18	0.13	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**25.6348**	**0.3656**	**25.6348**
A1 to ZN	**fall**	0.01	0.03	0.05	0.07
	rise	0.03	0.12	0.05	0.13
A2 to ZN	**fall**	0.01	0.03	0.04	0.06
	rise	0.04	0.13	0.06	0.14
B1 to ZN	**fall**	0.01	0.04	0.05	0.07
	rise	0.03	0.12	0.05	0.13
B2 to ZN	**fall**	0.01	0.04	0.04	0.06
	rise	0.04	0.13	0.05	0.13
C1 to ZN	**fall**	0.02	0.04	0.05	0.08
	rise	0.03	0.12	0.05	0.13
C2 to ZN	**fall**	0.02	0.04	0.04	0.06
	rise	0.04	0.13	0.05	0.13

Capacitance [fF]	
A1	3.1642
A2	3.2161
B1	3.0263
B2	3.4003
C1	2.9645
C2	3.3154

Leakage [nW]
94.80

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	25.6348	0.3656	25.6348
A1 to ZN	fall	0.63	0.58	4.56	3.18
	rise	6.97	7.16	11.06	10.08
A2 to ZN	fall	0.63	0.58	3.14	2.30
	rise	8.58	8.76	12.69	11.85
B1 to ZN	fall	2.25	2.55	5.88	4.78
	rise	10.24	10.35	11.38	11.00
B2 to ZN	fall	2.25	2.55	4.57	3.96
	rise	11.66	11.76	12.88	12.59
C1 to ZN	fall	3.24	4.05	6.99	6.08
	rise	13.00	13.18	13.09	13.12
C2 to ZN	fall	3.25	4.05	5.67	5.26
	rise	14.42	14.55	14.60	14.75

AOI222_X4

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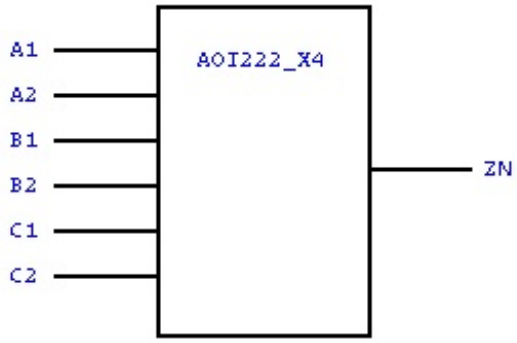
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.724 um^2
Equation	ZN = "!(!(!(((A1 & A2) | (B1 & B2)) | (C1 & C2))))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	L	-	L	-	H
L	-	L	-	-	L	H
L	-	-	L	L	-	H
L	-	-	L	-	L	H
H	H	-	-	-	-	L
-	L	L	-	L	-	H
-	L	L	-	-	L	H
-	L	-	L	L	-	H
-	L	-	L	-	L	H
-	-	H	H	-	-	L
-	-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.04	0.12	0.07	0.15
	rise	0.09	0.24	0.18	0.32
A2 to ZN	**fall**	0.04	0.12	0.07	0.15
	rise	0.10	0.25	0.19	0.34
B1 to ZN	**fall**	0.05	0.13	0.08	0.16
	rise	0.11	0.26	0.18	0.33
B2 to ZN	**fall**	0.05	0.13	0.08	0.16
	rise	0.12	0.27	0.19	0.34
C1 to ZN	**fall**	0.05	0.13	0.09	0.17
	rise	0.12	0.27	0.18	0.33
C2 to ZN	**fall**	0.05	0.13	0.09	0.17
	rise	0.13	0.28	0.19	0.34

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
B1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
B2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
C1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
C2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A1	1.5965
A2	1.7018
B1	1.5966
B2	1.6776
C1	1.5780
C2	1.6506

Leakage [nW]
134.85

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	fall	9.60	11.95	12.85	15.30
	rise	14.07	16.54	17.01	19.74
A2 to ZN	fall	9.59	11.96	11.65	14.18
	rise	14.85	17.62	17.86	20.77
B1 to ZN	fall	10.63	13.15	13.54	15.95
	rise	15.48	18.17	16.59	19.27
B2 to ZN	fall	10.64	13.15	12.40	14.86
	rise	16.25	19.02	17.45	20.56
C1 to ZN	fall	11.31	14.16	14.25	16.80
	rise	16.76	19.44	17.08	20.21
C2 to ZN	fall	11.30	14.16	13.04	15.69
	rise	17.54	20.39	18.02	21.09

AOI22_X1

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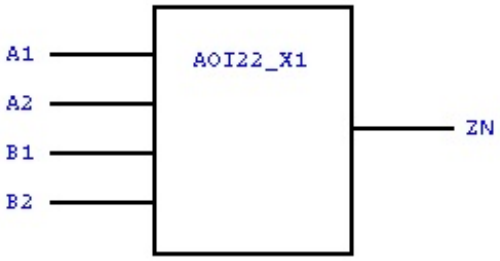
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!((A1 & A2) | (B1 & B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**24.6048**	**0.3656**	**24.6048**
A1 to ZN	**fall**	0.01	0.06	0.01	0.12
	rise	0.02	0.14	0.08	0.24
A2 to ZN	**fall**	0.01	0.06	0.01	0.11
	rise	0.03	0.15	0.09	0.24
B1 to ZN	**fall**	0.01	0.06	0.03	0.13
	rise	0.03	0.15	0.08	0.23
B2 to ZN	**fall**	0.01	0.07	0.03	0.12
	rise	0.03	0.16	0.09	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**24.6048**	**0.3656**	**24.6048**
A1 to ZN	**fall**	0.01	0.05	0.04	0.09
	rise	0.02	0.13	0.04	0.14
A2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.13	0.04	0.14
B1 to ZN	**fall**	0.01	0.05	0.04	0.09
	rise	0.02	0.13	0.04	0.13
B2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.13	0.04	0.14

Capacitance [fF]	
A1	1.6875
A2	1.6897
B1	1.5840
B2	1.6230

Leakage [nW]
32.61

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	24.6048	0.3656	24.6048
A1 to ZN	fall	0.25	0.29	2.67	1.32
	rise	2.80	2.86	5.27	4.21
A2 to ZN	fall	0.26	0.29	1.87	0.94
	rise	3.51	3.57	5.95	5.02
B1 to ZN	fall	1.07	1.29	3.29	2.20
	rise	4.21	4.26	5.15	4.63
B2 to ZN	fall	1.08	1.28	2.51	1.82
	rise	4.91	4.94	5.82	5.41

AOI22_X2

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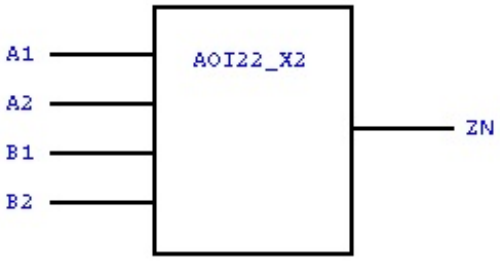
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!((A1 & A2) | (B1 & B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**49.1333**	**0.3656**	**49.1333**
A1 to ZN	**fall**	0.01	0.06	0.01	0.12
	rise	0.02	0.14	0.08	0.24
A2 to ZN	**fall**	0.01	0.06	0.01	0.11
	rise	0.03	0.15	0.09	0.24
B1 to ZN	**fall**	0.01	0.07	0.03	0.13
	rise	0.03	0.15	0.08	0.23
B2 to ZN	**fall**	0.01	0.07	0.03	0.12
	rise	0.03	0.16	0.09	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**49.1333**	**0.3656**	**49.1333**
A1 to ZN	**fall**	0.01	0.05	0.04	0.09
	rise	0.01	0.13	0.04	0.14
A2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.13	0.04	0.14
B1 to ZN	**fall**	0.01	0.05	0.04	0.09
	rise	0.01	0.13	0.04	0.14
B2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.13	0.04	0.14

Capacitance [fF]	
A1	3.1474
A2	3.4775
B1	2.9876
B2	3.4373

Leakage [nW]
65.22

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	49.1333	0.3656	49.1333
A1 to ZN	fall	0.51	0.58	5.34	2.64
	rise	5.62	5.85	10.62	8.47
A2 to ZN	fall	0.52	0.58	3.76	1.88
	rise	7.06	7.16	11.96	10.23
B1 to ZN	fall	2.13	2.57	6.56	4.37
	rise	8.72	8.80	10.59	9.48
B2 to ZN	fall	2.13	2.57	5.00	3.63
	rise	10.11	10.25	11.89	11.11

AOI22_X4

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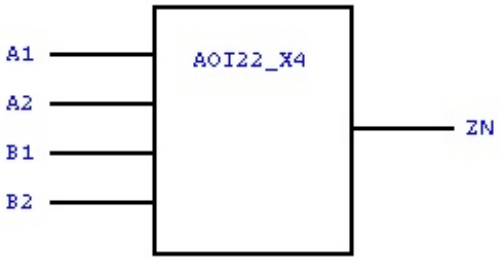
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Equation	ZN = "!((A1 & A2) | (B1 & B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**97.9614**	**0.3656**	**97.9614**
A1 to ZN	**fall**	0.01	0.06	0.00	0.12
	rise	0.02	0.15	0.08	0.24
A2 to ZN	**fall**	0.01	0.06	0.01	0.11
	rise	0.02	0.15	0.09	0.24
B1 to ZN	**fall**	0.01	0.06	0.03	0.13
	rise	0.03	0.15	0.08	0.23
B2 to ZN	**fall**	0.01	0.07	0.03	0.12
	rise	0.03	0.16	0.09	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**97.9614**	**0.3656**	**97.9614**
A1 to ZN	**fall**	0.01	0.05	0.04	0.09
	rise	0.01	0.13	0.04	0.14
A2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.13	0.04	0.14
B1 to ZN	**fall**	0.01	0.05	0.04	0.09
	rise	0.01	0.13	0.04	0.14
B2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.13	0.04	0.14

Capacitance [fF]	
A1	6.4186
A2	6.7809
B1	6.0901
B2	6.6051

Leakage [nW]
130.45

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	97.9614	0.3656	97.9614
A1 to ZN	fall	1.02	1.16	10.70	5.26
	rise	11.26	11.57	21.27	17.20
A2 to ZN	fall	1.05	1.16	7.53	3.75
	rise	14.11	14.50	23.93	20.49
B1 to ZN	fall	4.22	5.14	13.16	8.75
	rise	17.20	17.56	20.99	19.00
B2 to ZN	fall	4.25	5.14	10.03	7.26
	rise	19.98	20.32	23.60	22.25

BUF_X16

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	6.650 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**965.576**	**0.3656**	**965.576**
A to Z	**fall**	0.02	0.10	0.07	0.16
	rise	0.01	0.16	0.02	0.18

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**965.576**	**0.3656**	**965.576**
A to Z	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14

Capacitance [fF]	
A	12.4108

Leakage [nW]
344.49

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**965.576**	**0.3656**	**965.576**
A to Z	**fall**	37.46	46.64	71.85	78.78
	rise	24.79	27.23	57.22	59.88

BUF_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.6537**	**0.3656**	**60.6537**
A to Z	**fall**	0.02	0.10	0.08	0.16
	rise	0.02	0.16	0.03	0.18

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.6537**	**0.3656**	**60.6537**
A to Z	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14

Capacitance [fF]	
A	0.9747

Leakage [nW]
21.44

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.6537**	**0.3656**	**60.6537**
A to Z	**fall**	2.56	3.22	4.58	5.11
	rise	1.72	1.79	3.63	3.73

BUF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.064 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A to Z	**fall**	0.02	0.10	0.07	0.16
	rise	0.01	0.16	0.03	0.18

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A to Z	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14

Capacitance [fF]	
A	1.7792

Leakage [nW]
43.06

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A to Z	**fall**	4.99	6.15	9.10	10.07
	rise	3.42	3.63	7.24	7.14

BUF_X32

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	32
Cell Area	13.034 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**1904.30**	**0.3656**	**1904.30**
A to Z	**fall**	0.02	0.10	0.07	0.16
	rise	0.01	0.17	0.02	0.18

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**1904.30**	**0.3656**	**1904.30**
A to Z	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.15	0.01	0.15

Capacitance [fF]	
A	26.7039

Leakage [nW]
688.98

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**1904.30**	**0.3656**	**1904.30**
A to Z	**fall**	73.84	92.04	142.93	155.72
	rise	49.07	54.82	113.37	117.83

BUF_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	1.862 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to Z	**fall**	0.02	0.10	0.07	0.16
	rise	0.01	0.16	0.02	0.17

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to Z	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14

Capacitance [fF]	
A	3.4019

Leakage [nW]
86.12

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to Z	**fall**	9.48	11.71	18.00	19.67
	rise	6.40	7.45	14.30	14.47

BUF_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	3.458 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	484.009	0.3656	484.009
A to Z	**fall**	0.02	0.10	0.07	0.16
	rise	0.01	0.16	0.02	0.18

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	484.009	0.3656	484.009
A to Z	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14

Capacitance [fF]	
A	6.5852

Leakage [nW]
172.24

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	484.009	0.3656	484.009
A to Z	**fall**	18.66	23.18	35.94	39.24
	rise	12.52	14.70	29.01	30.76

CLKBUF_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
A to Z	**fall**	0.02	0.18	0.06	0.23
	rise	0.02	0.17	0.05	0.21

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
A to Z	**fall**	0.00	0.14	0.01	0.14
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A	0.7798

Leakage [nW]
11.21

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
A to Z	**fall**	2.23	2.61	3.39	3.85
	rise	1.37	1.59	2.72	2.55

CLKBUF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.064 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
A to Z	**fall**	0.02	0.18	0.06	0.23
	rise	0.02	0.17	0.05	0.20

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
A to Z	**fall**	0.00	0.14	0.01	0.14
	rise	0.00	0.14	0.01	0.14

Capacitance [fF]	
A	1.4059

Leakage [nW]
22.92

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
A to Z	**fall**	4.26	4.97	6.77	7.67
	rise	2.73	2.95	5.47	5.41

CLKBUF_X3

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	3
Cell Area	1.330 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	181.885	0.3656	181.885
A to Z	**fall**	0.02	0.18	0.07	0.24
	rise	0.02	0.17	0.06	0.21

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	181.885	0.3656	181.885
A to Z	**fall**	0.00	0.14	0.01	0.14
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A	1.4212

Leakage [nW]
30.56

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	181.885	0.3656	181.885
A to Z	**fall**	5.21	6.34	8.55	8.82
	rise	3.76	4.04	7.35	6.82

NANGATE CLKGATETST_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	3.990 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to GCK	**fall**	0.02	0.18	0.08	0.24
	rise	0.02	0.17	0.04	0.19

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to GCK	**fall**	0.00	0.14	0.01	0.14
	rise	0.00	0.14	0.01	0.14

Constraints Time [ns]		
Setup CK to E	**fall**	0.13
	rise	0.11
Hold CK to E	**fall**	0.12
	rise	0.08
Setup CK to SE	**fall**	0.14
	rise	0.11
Hold CK to SE	**fall**	0.12
	rise	0.08

Capacitance [fF]	
CK	1.8122
E	0.8780
SE	0.7768

Leakage [nW]
59.14

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	6309.76	8477.44
	rise	1992.42	4078.51
E	fall	2073.54	2504.59
	rise	987.20	1548.64
SE	fall	1700.72	2397.29
	rise	934.69	1662.77

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to GCK	fall	5.20	6.74	8.25	10.54
	rise	3.26	3.89	7.09	7.50

NANGATE CLKGATETST_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	4.256 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
CK to GCK	**fall**	0.02	0.18	0.07	0.24
	rise	0.02	0.17	0.04	0.19

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
CK to GCK	**fall**	0.00	0.14	0.01	0.14
	rise	0.00	0.14	0.01	0.14

Constraints Time [ns]		
Setup CK to E	**fall**	0.13
	rise	0.12
Hold CK to E	**fall**	0.11
	rise	0.08
Setup CK to SE	**fall**	0.14
	rise	0.11
Hold CK to SE	**fall**	0.12
	rise	0.08

Capacitance [fF]	
CK	2.8186
E	0.8722
SE	0.8109

Leakage [nW]
78.03

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	7264.48	9442.10
	rise	1191.23	3253.75
E	fall	2065.92	2508.38
	rise	935.93	1504.70
SE	fall	1692.84	2400.73
	rise	882.13	1627.15

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
CK to GCK	fall	7.46	9.72	11.94	15.27
	rise	4.25	5.36	9.72	10.68

NANGATE CLKGATETST_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	5.320 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.92**	**0.3656**	**242.92**
CK to GCK	**fall**	0.02	0.19	0.08	0.25
	rise	0.02	0.17	0.03	0.19

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.92**	**0.3656**	**242.92**
CK to GCK	**fall**	0.00	0.14	0.01	0.14
	rise	0.00	0.14	0.01	0.14

Constraints Time [ns]		
Setup CK to E	**fall**	0.15
	rise	0.16
Hold CK to E	**fall**	0.07
	rise	0.07
Setup CK to SE	**fall**	0.15
	rise	0.16
Hold CK to SE	**fall**	0.08
	rise	0.06

Capacitance [fF]	
CK	4.4389
E	0.9305
SE	0.8147

Leakage [nW]
126.65

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	10352.17	12351.35
	rise	121.65	2162.94
E	fall	2647.66	3064.37
	rise	1346.44	1889.89
SE	fall	2275.26	2955.71
	rise	1293.10	2001.61

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	242.92	0.3656	242.92
CK to GCK	fall	12.85	16.16	21.29	24.96
	rise	7.77	9.54	13.74	16.34

NANGATE CLKGATETST_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	7.714 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	484.619	0.3656	484.619
CK to GCK	**fall**	0.02	0.19	0.08	0.25
	rise	0.02	0.17	0.03	0.19

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	484.619	0.3656	484.619
CK to GCK	**fall**	0.00	0.14	0.01	0.14
	rise	0.00	0.14	0.01	0.14

Constraints Time [ns]		
Setup CK to E	**fall**	0.16
	rise	0.23
Hold CK to E	**fall**	0.00
	rise	0.06
Setup CK to SE	**fall**	0.17
	rise	0.23
Hold CK to SE	**fall**	0.01
	rise	0.05

Capacitance [fF]	
CK	7.9592
E	0.9015
SE	0.8013

Leakage [nW]
220.27

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	17648.55	19480.28
	rise	-1946.72	62.35
E	fall	3445.20	3869.00
	rise	2331.09	2877.56
SE	fall	3072.64	3758.47
	rise	2277.24	2989.91

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	484.619	0.3656	484.619
CK to GCK	fall	22.97	28.87	38.54	44.50
	rise	14.11	17.07	24.43	28.14

NANGATE CLKGATE_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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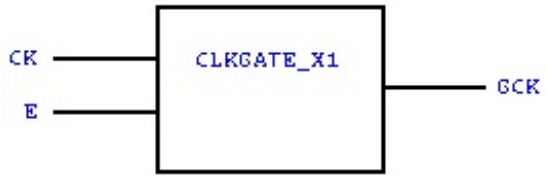
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	3.458 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	0.02	0.18	0.08	0.24
	rise	0.02	0.17	0.04	0.19

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	0.00	0.14	0.01	0.14
	rise	0.00	0.14	0.01	0.14

Constraints Time [ns]		
Setup CK to E	**fall**	0.10
	rise	0.10
Hold CK to E	**fall**	0.13
	rise	0.11

Capacitance [fF]	
CK	1.8379
E	0.9152

Leakage [nW]
48.65

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1985**
CK	**fall**	4942.58	7243.83
	rise	1307.26	3467.12
E	**fall**	1575.33	1868.05
	rise	672.73	920.23

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	5.07	7.03	8.20	10.90
	rise	3.11	3.54	6.66	7.18

NANGATE CLKGATE_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	3.724 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**121.307**	**0.3656**	**121.307**
CK to GCK	**fall**	0.02	0.18	0.07	0.24
	rise	0.02	0.17	0.04	0.19

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**121.307**	**0.3656**	**121.307**
CK to GCK	**fall**	0.00	0.14	0.01	0.14
	rise	0.00	0.14	0.01	0.14

Constraints Time [ns]		
Setup CK to E	**fall**	0.10
	rise	0.11
Hold CK to E	**fall**	0.12
	rise	0.11

Capacitance [fF]	
CK	2.5621
E	0.8932

Leakage [nW]
66.64

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1985**
CK	**fall**	6178.18	8462.60
	rise	1076.04	3215.24
E	**fall**	1764.11	2040.28
	rise	1036.47	1273.90

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**121.307**	**0.3656**	**121.307**
CK to GCK	**fall**	7.42	10.15	12.03	15.74
	rise	4.64	5.02	9.93	10.89

NANGATE CLKGATE_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
CK to GCK	**fall**	0.02	0.18	0.07	0.24
	rise	0.02	0.17	0.04	0.19

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
CK to GCK	**fall**	0.00	0.14	0.01	0.14
	rise	0.00	0.14	0.01	0.14

Constraints Time [ns]		
Setup CK to E	**fall**	0.11
	rise	0.11
Hold CK to E	**fall**	0.12
	rise	0.09

Capacitance [fF]	
CK	4.2539
E	0.8818

Leakage [nW]
109.65

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1985**
CK	**fall**	8248.96	10603.50
	rise	254.12	2377.85
E	**fall**	2512.20	2841.35
	rise	1609.84	1887.06

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
CK to GCK	**fall**	11.65	16.32	19.57	25.44
	rise	7.47	8.69	16.08	17.23

NANGATE CLKGATE_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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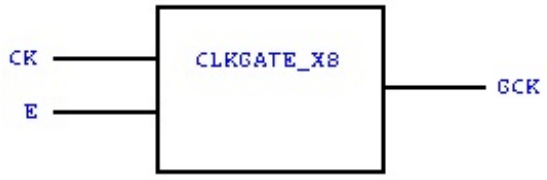
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	6.916 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	0.02	0.18	0.07	0.24
	rise	0.02	0.17	0.04	0.19

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	0.00	0.14	0.01	0.14
	rise	0.00	0.14	0.01	0.14

Constraints Time [ns]		
Setup CK to E	**fall**	0.12
	rise	0.12
Hold CK to E	**fall**	0.11
	rise	0.08

Capacitance [fF]	
CK	7.6543
E	1.1626

Leakage [nW]
197.71

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1985**
CK	**fall**	12817.11	15199.49
	rise	-1056.60	966.10
E	**fall**	4306.69	4877.54
	rise	2548.61	3012.51

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	20.54	29.70	35.00	45.70
	rise	13.23	15.20	28.42	29.97

DFFRS_X1

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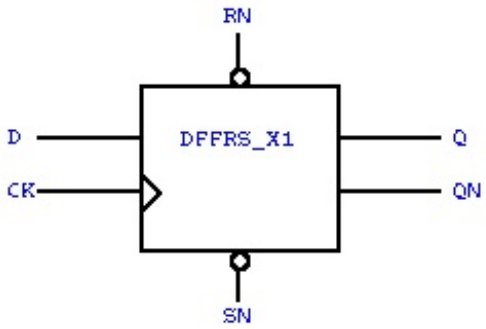
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, and drive strength X1
Strength	1
Cell Area	6.384 um^2
Equation	Q = "(!(!RN)*((D)+(!SN)))" QN = "(!(!SN)*(!(D)+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	RN	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	H	-	-	L	H
R	H	H	H	-	-	H	L
F	-	H	H	L	H	L	H
F	-	H	H	H	L	H	L
-	-	L	L	-	-	L	L
-	-	L	H	-	-	L	H
-	-	H	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.08	0.17	0.10	0.19
	rise	0.07	0.23	0.09	0.24
CK to QN	fall	0.10	*0.17*	0.11	*0.19*
	rise	0.11	*0.26*	0.13	*0.28*
RN to Q	fall	0.03	0.11	0.09	0.18
	rise	0.14	0.29	0.18	0.33
RN to QN	rise	0.06	*0.20*	0.13	*0.27*
SN to Q	rise	0.16	0.31	0.24	0.39
SN to QN	fall	0.02	*0.10*	0.08	*0.17*
	rise	0.03	*0.17*	0.05	*0.20*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
CK to QN	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*
RN to Q	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
RN to QN	rise	0.01	*0.14*	0.01	*0.14*
SN to Q	rise	0.01	0.14	0.01	0.14
SN to QN	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*

Constraints Time [ns]		
Setup CK to D	fall	0.10
	rise	0.06
Hold CK to D	fall	0.15
	rise	0.12
Recovery CK to RN	rise	0.08
Removal CK to RN	rise	0.23
Recovery CK to SN	rise	-0.04
Removal CK to SN	rise	0.45

Capacitance [fF]	
CK	0.9633
D	1.1480
RN	1.4071
SN	2.2119

Leakage [nW]
100.16

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	5658.63	7668.05
	rise	6937.10	8896.28
D	fall	2479.38	2962.99
	rise	1271.99	1729.32
RN	fall	686.42	692.22
	rise	-468.56	-478.96
SN	fall	1610.26	2023.29
	rise	-610.18	-220.11

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	5.85	6.04	6.85	6.93
	rise	5.65	6.03	6.65	7.01
CK to QN	fall	5.65	*6.11*	6.65	*7.08*
	rise	5.87	*5.91*	6.85	*7.01*
RN to Q	fall	3.54	5.49	5.03	6.62
	rise	4.15	4.55	5.15	5.29
RN to QN	rise	3.72	*5.41*	4.94	*6.20*
SN to Q	rise	5.53	5.85	7.44	7.38
SN to QN	fall	5.22	*5.99*	7.59	*8.26*
	rise	2.57	*3.10*	4.95	*5.62*

DFFRS_X2

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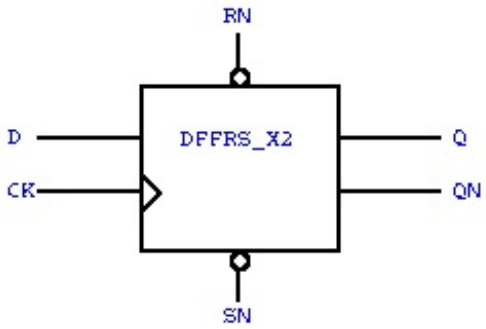
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, and drive strength X2
Strength	2
Cell Area	6.916 um^2
Equation	Q = "(!(!RN)*((D)+(!SN)))" QN = "(!(!SN)*(!(D)+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table

CK	D	RN	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	H	-	-	L	H
R	H	H	H	-	-	H	L
F	-	H	H	L	H	L	H
F	-	H	H	H	L	H	L
-	-	L	L	-	-	L	L
-	-	L	H	-	-	L	H
-	-	H	L	-	-	H	L

Propagation Delay [ns]

Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.08	0.16	0.10	0.18
	rise	0.08	0.23	0.10	0.25
CK to QN	fall	0.10	0.18	0.12	0.20
	rise	0.11	0.26	0.13	0.28
RN to Q	fall	0.03	0.11	0.09	0.19
	rise	0.15	0.30	0.18	0.34
RN to QN	rise	0.06	0.21	0.14	0.28
SN to Q	rise	0.17	0.32	0.25	0.41
SN to QN	fall	0.03	0.11	0.09	0.17
	rise	0.03	0.18	0.05	0.20

Output Transition [ns]

Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
CK to QN	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
RN to Q	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
RN to QN	rise	0.01	0.14	0.01	0.14
SN to Q	rise	0.01	0.14	0.01	0.14
SN to QN	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Constraints Time [ns]		
Setup CK to D	fall	0.10
	rise	0.06
Hold CK to D	fall	0.15
	rise	0.12
Recovery CK to RN	rise	0.02
Removal CK to RN	rise	0.23
Recovery CK to SN	rise	-0.03
Removal CK to SN	rise	0.46

Capacitance [fF]	
CK	0.9402
D	1.1622
RN	1.8443
SN	2.6209

Leakage [nW]
142.30

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	5572.76	7590.91
	rise	7609.45	9572.09
D	fall	2506.98	2989.06
	rise	1237.45	1694.13
RN	fall	1006.74	940.45
	rise	-869.32	-884.90
SN	fall	1895.30	2288.17
	rise	-979.42	-579.56

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	7.89	8.09	8.85	8.96
	rise	7.89	8.44	8.91	9.41
CK to QN	fall	7.89	8.44	8.91	9.41
	rise	7.89	8.09	8.85	8.96
RN to Q	fall	5.91	8.31	8.69	10.03
	rise	6.29	6.98	7.43	7.83
RN to QN	rise	6.15	8.21	8.38	9.39
SN to Q	rise	8.18	8.44	11.13	10.43
SN to QN	fall	7.43	8.41	11.15	11.34
	rise	4.20	5.35	7.56	8.23

DFFR_X1

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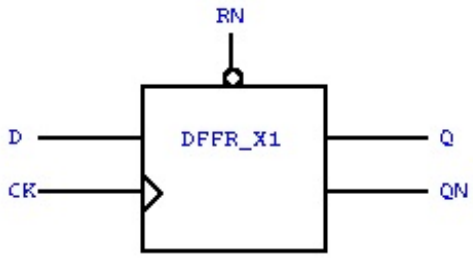
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, and drive strength X1
Strength	1
Cell Area	5.320 um^2
Equation	Q = "(!(!RN)*(D))" QN = "((!RN)+!(D))"
Clock	CK
Reset	RN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	RN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.08	0.16	0.10	0.18
	rise	0.10	0.24	0.12	0.26
CK to QN	fall	0.06	*0.15*	0.08	*0.17*
	rise	0.06	*0.21*	0.08	*0.23*
RN to Q	fall	0.03	0.11	0.09	0.18
RN to QN	rise	0.20	*0.35*	0.29	*0.43*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
CK to QN	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*
RN to Q	fall	0.01	0.07	0.01	0.07
RN to QN	rise	0.01	*0.14*	0.01	*0.14*

Constraints Time [ns]		
Setup CK to D	fall	0.10
	rise	0.06
Hold CK to D	fall	0.15
	rise	0.12
Recovery CK to RN	rise	-0.01
Removal CK to RN	rise	0.45

Capacitance [fF]	
CK	0.9766
D	1.1283
RN	1.7785

Leakage [nW]
86.21

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	5442.38	7448.13
	rise	6307.05	8264.08
D	fall	2561.20	3123.20
	rise	1320.11	1843.62
RN	fall	695.08	652.64
	rise	-583.13	-597.52

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	4.45	4.88	5.44	5.83
	rise	4.98	4.94	6.00	6.03
CK to QN	fall	4.98	*5.00*	5.99	*5.92*
	rise	4.46	*4.82*	5.44	*5.83*
RN to Q	fall	4.64	4.87	5.90	5.68
RN to QN	rise	4.63	*4.83*	5.90	*5.66*

DFFR_X2

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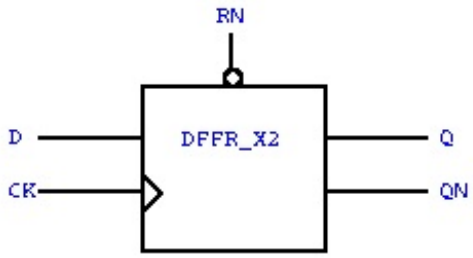
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, and drive strength X2
Strength	2
Cell Area	5.852 um^2
Equation	Q = "(!(!RN)*(D))" QN = "((!RN)+!(D))"
Clock	CK
Reset	RN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	RN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.09	0.17	0.11	0.19
	rise	0.13	0.26	0.15	0.28
CK to QN	fall	0.08	*0.19*	0.10	*0.21*
	rise	0.07	*0.22*	0.09	*0.24*
RN to Q	fall	0.02	0.10	0.08	0.17
RN to QN	rise	0.21	*0.37*	0.30	*0.45*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
CK to QN	fall	0.02	*0.07*	0.02	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*
RN to Q	fall	0.00	0.07	0.01	0.07
RN to QN	rise	0.01	*0.14*	0.01	*0.14*

Constraints Time [ns]		
Setup CK to D	fall	0.10
	rise	0.06
Hold CK to D	fall	0.15
	rise	0.12
Recovery CK to RN	rise	-0.05
Removal CK to RN	rise	0.43

Capacitance [fF]	
CK	0.9657
D	1.1284
RN	2.4751

Leakage [nW]
125.99

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	5387.37	7397.27
	rise	7359.82	9306.41
D	fall	2544.42	3108.77
	rise	1282.67	1809.65
RN	fall	1282.80	1189.66
	rise	-1087.28	-1102.92

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	6.60	7.02	7.59	8.15
	rise	8.49	7.62	9.56	8.62
CK to QN	fall	8.49	*7.64*	9.56	*8.53*
	rise	6.60	*7.08*	7.59	*8.21*
RN to Q	fall	7.58	7.48	10.13	9.29
RN to QN	rise	7.58	*7.54*	10.13	*9.18*

DFFS_X1

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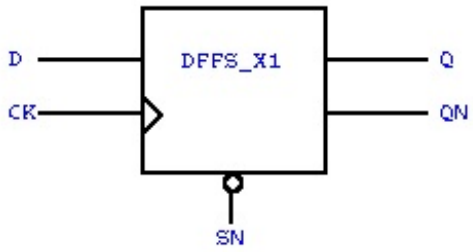
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, and drive strength X1
Strength	1
Cell Area	5.320 um^2
Equation	Q = "((!SN)+(D))" QN = "(!((!SN)+(D)))"
Clock	CK
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.08	0.16	0.10	0.18
	rise	0.08	0.23	0.11	0.25
CK to QN	fall	0.06	*0.15*	0.08	*0.18*
	rise	0.06	*0.21*	0.08	*0.23*
SN to Q	rise	0.21	0.34	0.29	0.43
SN to QN	fall	0.17	*0.30*	0.25	*0.38*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
CK to QN	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*
SN to Q	rise	0.01	0.14	0.01	0.14
SN to QN	fall	0.03	*0.09*	0.03	*0.09*

Constraints Time [ns]		
Setup CK to D	fall	0.10
	rise	0.06
Hold CK to D	fall	0.15
	rise	0.12
Recovery CK to SN	rise	-0.04
Removal CK to SN	rise	0.39

Capacitance [fF]	
CK	0.9652
D	1.1637
SN	1.3559

Leakage [nW]
84.40

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	6163.21	8134.10
	rise	5685.17	7549.44
D	fall	2410.57	2944.63
	rise	1012.82	1474.28
SN	fall	977.67	1398.45
	rise	-319.07	9.51

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	4.02	4.41	4.98	5.34
	rise	4.45	4.56	5.44	5.54
CK to QN	fall	4.45	*4.57*	5.44	*5.44*
	rise	4.02	*4.39*	4.97	*5.38*
SN to Q	rise	5.57	4.97	6.26	5.65
SN to QN	fall	5.56	*4.98*	6.26	*5.60*

DFFS_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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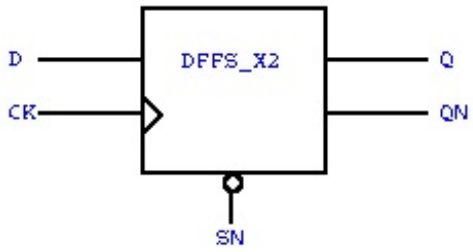
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, and drive strength X2
Strength	2
Cell Area	5.586 um^2
Equation	Q = "((!SN)+(D))" QN = "(!((!SN)+(D)))"
Clock	CK
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.09	0.17	0.12	0.19
	rise	0.11	0.25	0.14	0.27
CK to QN	fall	0.08	*0.19*	0.11	*0.22*
	rise	0.07	*0.22*	0.09	*0.25*
SN to Q	rise	0.32	0.43	0.40	0.52
SN to QN	fall	0.26	*0.42*	0.34	*0.50*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
CK to QN	fall	0.02	*0.07*	0.02	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*
SN to Q	rise	0.02	0.14	0.02	0.14
SN to QN	fall	0.06	*0.11*	0.06	*0.11*

Constraints Time [ns]		
Setup CK to D	fall	0.10
	rise	0.05
Hold CK to D	fall	0.15
	rise	0.12
Recovery CK to SN	rise	-0.03
Removal CK to SN	rise	0.36

Capacitance [fF]	
CK	0.9689
D	1.1630
SN	1.3336

Leakage [nW]
121.11

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	6300.74	8263.12
	rise	5729.34	7588.98
D	fall	2457.00	2987.96
	rise	1001.00	1460.08
SN	fall	968.50	1391.22
	rise	-318.92	11.67

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	6.18	6.64	7.13	7.45
	rise	8.04	7.29	8.99	8.16
CK to QN	fall	8.04	*7.07*	8.99	*7.95*
	rise	6.18	*6.76*	7.13	*7.57*
SN to Q	rise	11.72	8.78	12.36	9.55
SN to QN	fall	11.72	*8.90*	12.39	*9.51*

DFF_X1

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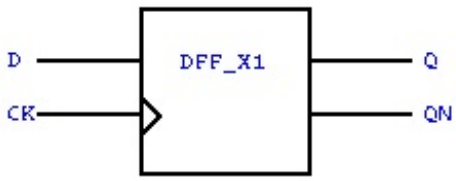
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with drive strength X1
Strength	1
Cell Area	4.522 um^2
Equation	Q = "(D)" QN = "(!(D))"
Clock	CK
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
CK	**D**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**	**QN**
R	L	-	-	L	H
R	H	-	-	H	L
F	-	L	H	L	H
F	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to Q	**fall**	0.08	0.15	0.10	0.18
	rise	0.08	0.22	0.11	0.25
CK to QN	**fall**	0.06	*0.15*	0.08	*0.17*
	rise	0.06	*0.21*	0.08	*0.23*

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
CK to QN	**fall**	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*

Constraints Time [ns]		
Setup CK to D	**fall**	0.10
	rise	0.06
Hold CK to D	**fall**	0.15
	rise	0.12

Capacitance [fF]	
CK	0.9497
D	1.1403

Leakage [nW]
79.11

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1985**
CK	**fall**	5970.75	7965.18
	rise	2534.50	4438.42
D	**fall**	2473.13	3050.30
	rise	1285.91	1800.09

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to Q	**fall**	3.98	4.28	4.94	5.26
	rise	4.36	4.47	5.34	5.29
CK to QN	**fall**	4.36	*4.39*	5.34	*5.39*
	rise	3.98	*4.38*	4.94	*5.35*

DFF_X2

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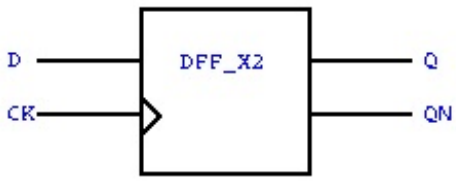
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with drive strength X2
Strength	2
Cell Area	5.054 um^2
Equation	Q = "(D)" QN = "(!(D))"
Clock	CK
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
CK	**D**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**	**QN**
R	L	-	-	L	H
R	H	-	-	H	L
F	-	L	H	L	H
F	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
CK to Q	**fall**	0.09	0.16	0.11	0.19
	rise	0.11	0.25	0.13	0.27
CK to QN	**fall**	0.08	*0.19*	0.10	*0.21*
	rise	0.07	*0.22*	0.09	*0.24*

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
CK to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
CK to QN	**fall**	0.02	*0.07*	0.02	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*

Constraints Time [ns]		
Setup CK to D	**fall**	0.10
	rise	0.05
Hold CK to D	**fall**	0.15
	rise	0.12

Capacitance [fF]	
CK	0.9305
D	1.1276

Leakage [nW]
115.10

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1985**
CK	**fall**	6009.28	7998.66
	rise	2551.45	4454.05
D	**fall**	2508.70	3083.40
	rise	1330.36	1839.82

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
CK to Q	**fall**	6.20	6.70	7.14	7.46
	rise	7.96	7.13	8.97	8.03
CK to QN	**fall**	7.95	*7.15*	8.97	*7.99*
	rise	6.19	*6.70*	7.15	*7.58*

DLH_X1

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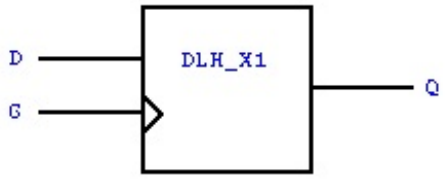
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	High enable Latch with drive strength X1
Strength	1
Cell Area	2.660 um^2
Equation	Q = "(D)"
Enable	G
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**G**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	H	-	-	L
H	H	-	-	H
-	L	L	H	L
-	L	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
D to Q	**fall**	0.05	0.14	0.11	0.21
	rise	0.03	0.18	0.05	0.21
G to Q	**fall**	0.05	0.14	0.07	0.16
	rise	0.05	0.20	0.06	0.21

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
D to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
G to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Constraints Time [ns]		
Setup G to D	**fall**	0.11
	rise	0.04
Hold G to D	**fall**	0.22
	rise	0.12

Capacitance [fF]	
D	0.9141
G	0.9855

Leakage [nW]
40.86

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1985**
D	**fall**	512.88	499.90
	rise	-395.33	-449.56
G	**fall**	3054.22	5152.49
	rise	2230.20	4328.69

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
D to Q	**fall**	3.73	4.62	4.70	5.36
	rise	2.37	3.00	3.40	3.58
G to Q	**fall**	5.01	5.92	7.08	8.04
	rise	4.65	5.27	6.61	7.36

DLH_X2

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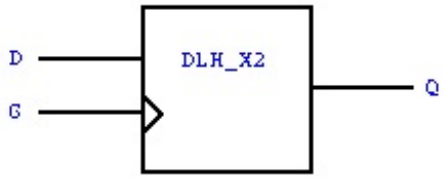
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	High enable Latch with drive strength X2
Strength	2
Cell Area	2.926 um^2
Equation	Q = "(D)"
Enable	G
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**G**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	H	-	-	L
H	H	-	-	H
-	L	L	H	L
-	L	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**120.697**	**0.3656**	**120.697**
D to Q	**fall**	0.06	0.15	0.12	0.22
	rise	0.03	0.18	0.06	0.21
G to Q	**fall**	0.06	0.15	0.07	0.17
	rise	0.05	0.20	0.07	0.21

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**120.697**	**0.3656**	**120.697**
D to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
G to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Constraints Time [ns]		
Setup G to D	**fall**	0.12
	rise	0.05
Hold G to D	**fall**	0.21
	rise	0.11

Capacitance [fF]	
D	1.1610
G	0.9870

Leakage [nW]
57.43

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1985**
D	**fall**	691.86	671.93
	rise	-532.28	-597.97
G	**fall**	2960.86	5068.82
	rise	2239.05	4330.36

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**120.697**	**0.3656**	**120.697**
D to Q	**fall**	6.34	7.09	8.01	8.06
	rise	3.99	4.83	5.87	5.45
G to Q	**fall**	7.26	8.05	9.43	10.17
	rise	6.34	7.02	8.29	9.48

DLL_X1

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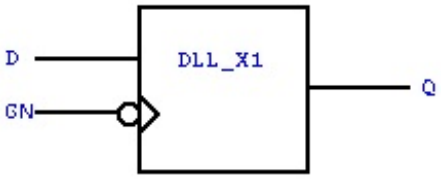
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Low enable Latch with drive strength X1
Strength	1
Cell Area	2.660 um^2
Equation	Q = "(D)"
Enable	GN
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**GN**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	L	-	-	L
H	L	-	-	H
-	H	L	H	L
-	H	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.2722	0.3656	60.2722
D to Q	**fall**	0.05	0.14	0.11	0.21
	rise	0.03	0.18	0.05	0.20
GN to Q	**fall**	0.07	0.15	0.12	0.21
	rise	0.04	0.19	0.10	0.25

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.2722	0.3656	60.2722
D to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
GN to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Constraints Time [ns]		
Setup GN to D	**fall**	0.11
	rise	0.08
Hold GN to D	**fall**	0.15
	rise	0.11

Capacitance [fF]	
D	0.8830
GN	0.9891

Leakage [nW]
40.86

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
D	**fall**	511.58	499.83
	rise	-364.51	-380.20
GN	**fall**	3283.07	5435.40
	rise	1793.48	3922.48

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.2722	0.3656	60.2722
D to Q	**fall**	3.79	4.59	4.73	5.33
	rise	2.53	3.11	3.54	3.91
GN to Q	**fall**	6.21	7.02	8.12	9.16
	rise	5.87	6.40	7.76	8.72

DLL_X2

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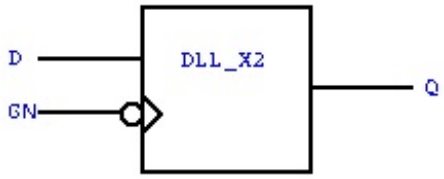
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Low enable Latch with drive strength X2
Strength	2
Cell Area	2.926 um^2
Equation	Q = "(D)"
Enable	GN
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**GN**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	L	-	-	L
H	L	-	-	H
-	H	L	H	L
-	H	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
D to Q	**fall**	0.05	0.15	0.12	0.22
	rise	0.03	0.18	0.06	0.21
GN to Q	**fall**	0.07	0.16	0.13	0.22
	rise	0.05	0.19	0.11	0.26

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
D to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
GN to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Constraints Time [ns]		
Setup GN to D	**fall**	0.11
	rise	0.08
Hold GN to D	**fall**	0.14
	rise	0.10

Capacitance [fF]	
D	1.1305
GN	0.9826

Leakage [nW]
57.43

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1985**
D	**fall**	692.11	672.06
	rise	-533.81	-552.47
GN	**fall**	3414.02	5570.16
	rise	1705.69	3823.06

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
D to Q	**fall**	6.14	6.82	7.79	7.81
	rise	3.93	5.00	5.88	5.76
GN to Q	**fall**	8.34	9.05	10.26	11.20
	rise	7.45	8.19	9.61	10.55

FA_X1

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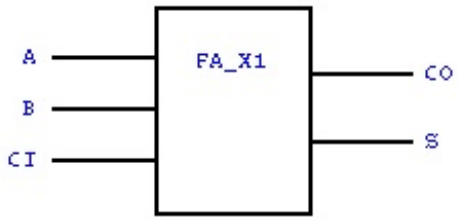
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	4.256 um^2
Equation	CO = "((A & B) | (CI & (A | B)))" S = "(CI ^ (A ^ B))"
Type	Combinational
Input	A, B, CI
Output	CO, S
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**CI**	**CO**	**S**
L	L	L	L	L
L	L	H	L	H
L	H	L	L	H
L	H	H	H	L
H	L	L	L	H
H	L	H	H	L
H	H	L	H	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A to CO	**fall**	0.07	*0.16*	0.14	*0.25*
	rise	0.04	*0.19*	0.08	*0.23*
A to S	**fall**	0.09	0.19	0.14	0.25
	rise	0.10	0.25	0.19	0.33
B to CO	**fall**	0.07	*0.16*	0.13	*0.24*
	rise	0.04	*0.19*	0.07	*0.22*
B to S	**fall**	0.08	0.19	0.16	0.26
	rise	0.11	0.25	0.18	0.32
CI to CO	**fall**	0.06	*0.16*	0.14	*0.24*
	rise	0.03	*0.18*	0.07	*0.22*
CI to S	**fall**	0.08	0.18	0.16	0.26
	rise	0.10	0.25	0.18	0.33

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A to CO	**fall**	0.01	*0.07*	0.02	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*
A to S	**fall**	0.01	0.07	0.02	0.07
	rise	0.01	0.14	0.01	0.14
B to CO	**fall**	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*
B to S	**fall**	0.01	0.07	0.02	0.07
	rise	0.01	0.14	0.01	0.14
CI to CO	**fall**	0.01	*0.07*	0.02	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*
CI to S	**fall**	0.01	0.07	0.02	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A	3.7457
B	3.4720
CI	2.7621

Leakage [nW]
75.76

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.2722	0.3656	60.2722
A to CO	fall	4.95	*5.00*	5.54	*5.42*
	rise	2.19	*2.61*	3.03	*3.02*
A to S	fall	4.01	4.61	4.52	4.93
	rise	3.10	3.34	3.91	3.68
B to CO	fall	5.16	*5.17*	5.56	*5.54*
	rise	2.32	*2.76*	2.95	*3.01*
B to S	fall	3.90	4.45	4.38	4.80
	rise	3.13	3.33	3.90	3.61
CI to CO	fall	4.61	*4.72*	5.50	*5.27*
	rise	2.60	*2.95*	3.58	*3.52*
CI to S	fall	3.61	4.21	4.46	4.77
	rise	3.15	3.40	4.21	3.97

FILLCELL_X16

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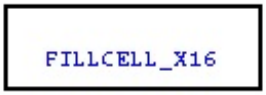
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	4.256 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.266 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	0.266 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X32

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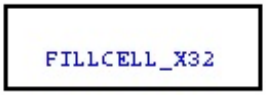
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	32
Cell Area	8.512 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	1.064 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	2.128 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



HA_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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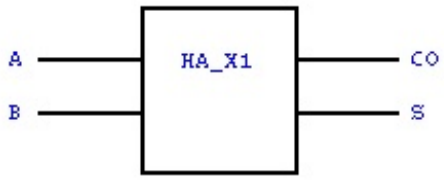
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.660 um^2
Equation	CO = "(A & B)" S = "(A ^ B)"
Type	Combinational
Input	A, B
Output	CO, S
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B**	**CO**	**S**
L	L	L	L
L	H	L	H
H	L	L	H
H	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A to CO	**fall**	0.02	0.10	0.07	0.16
	rise	0.02	0.17	0.05	0.20
A to S	**fall**	0.05	*0.13*	0.11	*0.25*
	rise	0.04	*0.34*	0.07	*0.43*
B to CO	**fall**	0.02	0.10	0.08	0.17
	rise	0.02	0.17	0.04	0.20
B to S	**fall**	0.05	*0.13*	0.11	*0.22*
	rise	0.03	*0.34*	0.08	*0.43*

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A to CO	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A to S	**fall**	0.01	*0.11*	0.04	*0.14*
	rise	0.02	*0.30*	0.04	*0.29*
B to CO	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
B to S	**fall**	0.01	*0.11*	0.03	*0.12*
	rise	0.02	*0.30*	0.04	*0.29*

Capacitance [fF]	
A	3.1859
B	3.4478

Leakage [nW]
61.23

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A to CO	fall	2.99	3.15	5.25	4.37
	rise	0.99	1.10	3.27	2.12
A to S	fall	3.22	*3.34*	4.75	*4.36*
	rise	3.23	*2.97*	4.91	*4.33*
B to CO	fall	3.69	3.78	5.90	5.12
	rise	0.87	0.98	2.41	1.68
B to S	fall	3.31	*3.54*	4.08	*4.15*
	rise	3.29	*3.29*	5.13	*4.34*

INV_X16

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	4.522 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	969.238	0.3656	969.238
A to ZN	**fall**	0.00	0.08	-0.01	0.15
	rise	0.01	0.15	0.04	0.26

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	969.238	0.3656	969.238
A to ZN	**fall**	0.00	0.07	0.03	0.10
	rise	0.00	0.14	0.03	0.15

Capacitance [fF]	
A	25.2281

Leakage [nW]
229.65

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	969.238	0.3656	969.238
A to ZN	**fall**	-0.00	-0.01	51.14	11.88
	rise	27.90	29.46	90.71	53.13

INV_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.532 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
A to ZN	**fall**	0.00	0.08	-0.01	0.15
	rise	0.01	0.15	0.04	0.26

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
A to ZN	**fall**	0.00	0.07	0.03	0.10
	rise	0.00	0.14	0.03	0.15

Capacitance [fF]	
A	1.7002

Leakage [nW]
14.35

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
A to ZN	**fall**	-0.00	-0.00	3.06	0.64
	rise	1.85	1.80	5.65	3.15

INV_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	0.798 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to ZN	**fall**	0.00	0.08	-0.01	0.15
	rise	0.01	0.15	0.04	0.26

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to ZN	**fall**	0.00	0.07	0.03	0.10
	rise	0.00	0.14	0.03	0.15

Capacitance [fF]	
A	3.2509

Leakage [nW]
28.71

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to ZN	**fall**	-0.00	-0.00	6.35	1.46
	rise	3.60	3.67	11.31	6.82

INV_X32

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	32
Cell Area	8.778 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**1923.83**	**0.3656**	**1923.83**
A to ZN	**fall**	0.00	0.08	-0.01	0.15
	rise	0.01	0.15	0.04	0.26

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**1923.83**	**0.3656**	**1923.83**
A to ZN	**fall**	0.00	0.07	0.03	0.10
	rise	0.00	0.14	0.03	0.15

Capacitance [fF]	
A	49.1915

Leakage [nW]
459.30

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**1923.83**	**0.3656**	**1923.83**
A to ZN	**fall**	-0.00	-0.02	101.59	24.79
	rise	54.77	57.59	179.80	105.38

INV_X4

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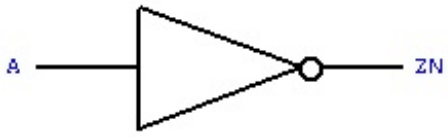
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	1.330 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.92**	**0.3656**	**242.92**
A to ZN	**fall**	0.00	0.08	-0.01	0.15
	rise	0.01	0.15	0.04	0.26

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.92**	**0.3656**	**242.92**
A to ZN	**fall**	0.00	0.07	0.03	0.10
	rise	0.00	0.14	0.03	0.15

Capacitance [fF]	
A	6.2584

Leakage [nW]
57.41

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.92**	**0.3656**	**242.92**
A to ZN	**fall**	-0.00	-0.00	12.73	2.93
	rise	7.03	6.89	22.59	13.46

INV_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	2.394 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**485.229**	**0.3656**	**485.229**
A to ZN	**fall**	0.00	0.08	-0.01	0.15
	rise	0.01	0.15	0.04	0.26

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**485.229**	**0.3656**	**485.229**
A to ZN	**fall**	0.00	0.07	0.03	0.10
	rise	0.00	0.14	0.03	0.15

Capacitance [fF]	
A	11.8107

Leakage [nW]
114.83

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**485.229**	**0.3656**	**485.229**
A to ZN	**fall**	-0.00	-0.01	25.66	5.89
	rise	13.72	15.35	45.49	26.34

LOGIC0_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.532 um^2
Equation	Z = "0"
Type	Constant
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)

LOGIC1_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.532 um^2
Equation	Z = "1"
Type	Constant
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)

NANGATE MUX2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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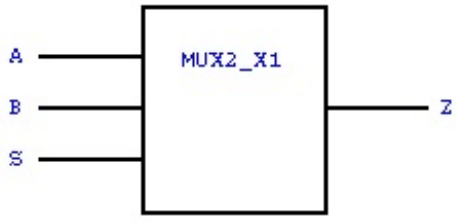
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.862 um^2
Equation	Z = "((S & B) | (A & !S))"
Type	Combinational
Input	A, B, S
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B**	**S**	**Z**
L	-	L	L
H	-	L	H
-	L	H	L
-	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5011	0.3656	60.5011
A to Z	**fall**	0.05	0.14	0.11	0.21
	rise	0.03	0.18	0.05	0.20
B to Z	**fall**	0.05	0.14	0.11	0.21
	rise	0.03	0.18	0.05	0.20
S to Z	**fall**	0.05	0.14	0.11	0.21
	rise	0.04	0.19	0.10	0.25

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5011	0.3656	60.5011
A to Z	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
B to Z	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
S to Z	**fall**	0.01	0.07	0.02	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A	0.9464
B	0.9448
S	1.9199

Leakage [nW]
35.93

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5011	0.3656	60.5011
A to Z	**fall**	3.38	4.05	4.27	4.73
	rise	2.33	2.54	3.29	3.26
B to Z	**fall**	3.63	4.40	4.49	5.07
	rise	2.07	2.21	3.05	2.97
S to Z	**fall**	3.80	4.54	6.57	7.20
	rise	3.48	3.75	6.17	6.33

MUX2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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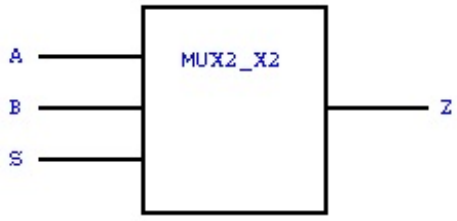
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	Z = "((S & B) | (A & !S))"
Type	Combinational
Input	A, B, S
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B**	**S**	**Z**
L	-	L	L
H	-	L	H
-	L	H	L
-	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A to Z	**fall**	0.05	0.14	0.10	0.20
	rise	0.03	0.18	0.06	0.21
B to Z	**fall**	0.05	0.14	0.12	0.22
	rise	0.03	0.17	0.05	0.20
S to Z	**fall**	0.06	0.15	0.10	0.19
	rise	0.05	0.20	0.11	0.26

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A to Z	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
B to Z	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
S to Z	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A	1.5923
B	1.7351
S	2.6240

Leakage [nW]
68.65

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A to Z	**fall**	6.82	8.56	8.21	9.63
	rise	4.51	5.29	7.74	7.26
B to Z	**fall**	6.67	8.36	10.23	10.94
	rise	3.47	3.93	5.72	5.41
S to Z	**fall**	7.77	9.44	11.30	12.55
	rise	6.52	7.13	9.68	10.04

NANGATE NAND2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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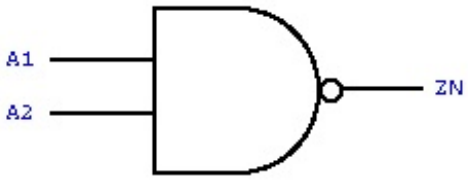
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	ZN = "!(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	59.3567	0.3656	59.3567
A1 to ZN	**fall**	0.01	0.13	0.01	0.21
	rise	0.01	0.15	0.04	0.25
A2 to ZN	**fall**	0.01	0.13	0.01	0.19
	rise	0.01	0.15	0.05	0.26

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	59.3567	0.3656	59.3567
A1 to ZN	**fall**	0.00	0.11	0.03	0.13
	rise	0.00	0.14	0.03	0.15
A2 to ZN	**fall**	0.00	0.11	0.03	0.12
	rise	0.01	0.14	0.03	0.15

Capacitance [fF]	
A1	1.5990
A2	1.6642

Leakage [nW]
17.39

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	59.3567	0.3656	59.3567
A1 to ZN	**fall**	0.21	0.26	3.27	0.92
	rise	2.19	2.18	5.34	3.47
A2 to ZN	**fall**	0.22	0.26	2.33	0.62
	rise	2.88	2.85	6.00	4.09

NANGATE NAND2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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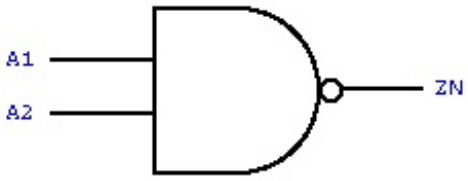
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "!(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	118.713	0.3656	118.713
A1 to ZN	**fall**	0.01	0.13	0.01	0.21
	rise	0.01	0.15	0.04	0.25
A2 to ZN	**fall**	0.01	0.13	0.01	0.19
	rise	0.01	0.15	0.05	0.26

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	118.713	0.3656	118.713
A1 to ZN	**fall**	0.00	0.11	0.03	0.13
	rise	0.00	0.14	0.03	0.15
A2 to ZN	**fall**	0.00	0.11	0.02	0.12
	rise	0.01	0.14	0.03	0.15

Capacitance [fF]	
A1	3.0531
A2	3.4510

Leakage [nW]
34.79

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	118.713	0.3656	118.713
A1 to ZN	**fall**	0.41	0.53	6.59	1.85
	rise	4.42	4.07	10.83	6.97
A2 to ZN	**fall**	0.44	0.53	4.71	1.24
	rise	5.86	5.81	12.10	8.40

NANGATE NAND2_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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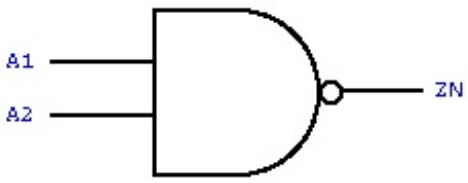
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "!(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	237.427	0.3656	237.427
A1 to ZN	**fall**	0.01	0.13	0.00	0.21
	rise	0.01	0.15	0.04	0.25
A2 to ZN	**fall**	0.01	0.13	0.01	0.19
	rise	0.01	0.15	0.05	0.26

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	237.427	0.3656	237.427
A1 to ZN	**fall**	0.00	0.11	0.03	0.13
	rise	0.00	0.14	0.03	0.15
A2 to ZN	**fall**	0.00	0.11	0.02	0.12
	rise	0.01	0.14	0.03	0.15

Capacitance [fF]	
A1	5.9550
A2	6.2019

Leakage [nW]
69.57

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	237.427	0.3656	237.427
A1 to ZN	**fall**	0.80	1.05	13.24	3.67
	rise	8.17	7.61	21.07	13.40
A2 to ZN	**fall**	0.86	1.05	9.44	2.44
	rise	11.32	10.78	23.81	16.91

NANGATE NAND3_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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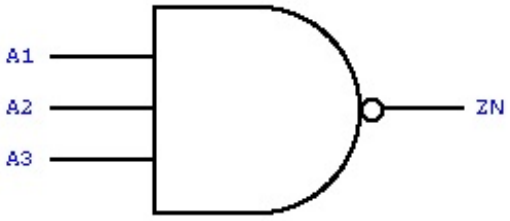
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	H
H	H	H	L
-	L	-	H
-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**58.3649**	**0.3656**	**58.3649**
A1 to ZN	**fall**	0.01	0.18	0.02	0.26
	rise	0.01	0.15	0.04	0.25
A2 to ZN	**fall**	0.01	0.18	0.03	0.25
	rise	0.01	0.15	0.05	0.25
A3 to ZN	**fall**	0.01	0.19	0.02	0.23
	rise	0.01	0.15	0.06	0.26

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**58.3649**	**0.3656**	**58.3649**
A1 to ZN	**fall**	0.01	0.15	0.04	0.16
	rise	0.01	0.14	0.04	0.15
A2 to ZN	**fall**	0.01	0.15	0.03	0.16
	rise	0.01	0.14	0.03	0.15
A3 to ZN	**fall**	0.01	0.15	0.03	0.16
	rise	0.01	0.14	0.03	0.15

Capacitance [fF]	
A1	1.5903
A2	1.6212
A3	1.6504

Leakage [nW]
18.10

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**58.3649**	**0.3656**	**58.3649**
A1 to ZN	**fall**	0.52	0.56	3.28	1.11
	rise	2.47	2.46	5.19	3.45
A2 to ZN	**fall**	0.53	0.56	2.42	0.82
	rise	3.20	3.01	5.85	4.37
A3 to ZN	**fall**	0.53	0.56	2.25	0.80
	rise	3.77	3.83	6.61	4.88

NANGATE NAND3_X2

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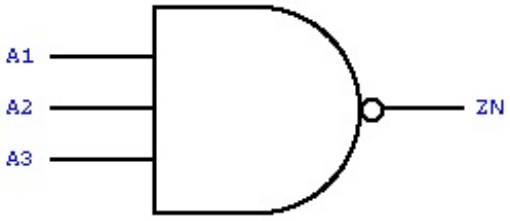
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	H
H	H	H	L
-	L	-	H
-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**116.272**	**0.3656**	**116.272**
A1 to ZN	**fall**	0.01	0.18	0.02	0.26
	rise	0.01	0.15	0.04	0.25
A2 to ZN	**fall**	0.01	0.18	0.03	0.25
	rise	0.01	0.15	0.05	0.25
A3 to ZN	**fall**	0.01	0.19	0.02	0.23
	rise	0.01	0.15	0.06	0.26

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**116.272**	**0.3656**	**116.272**
A1 to ZN	**fall**	0.01	0.15	0.04	0.16
	rise	0.01	0.14	0.04	0.15
A2 to ZN	**fall**	0.01	0.15	0.03	0.16
	rise	0.01	0.14	0.03	0.15
A3 to ZN	**fall**	0.01	0.15	0.03	0.16
	rise	0.01	0.14	0.03	0.15

Capacitance [fF]	
A1	2.9778
A2	3.2864
A3	3.5589

Leakage [nW]
36.21

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**116.272**	**0.3656**	**116.272**
A1 to ZN	**fall**	1.24	1.31	6.74	2.40
	rise	5.11	5.07	10.57	7.05
A2 to ZN	**fall**	1.24	1.31	5.03	1.83
	rise	6.57	6.85	11.85	8.89
A3 to ZN	**fall**	1.24	1.31	4.68	1.78
	rise	7.72	8.20	13.41	10.20

NANGATE NAND3_X4

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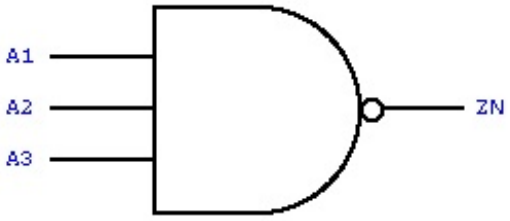
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	H
H	H	H	L
-	L	-	H
-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	233.154	0.3656	233.154
A1 to ZN	**fall**	0.01	0.18	0.02	0.26
	rise	0.01	0.15	0.04	0.25
A2 to ZN	**fall**	0.01	0.18	0.02	0.25
	rise	0.01	0.15	0.05	0.25
A3 to ZN	**fall**	0.01	0.19	0.02	0.23
	rise	0.01	0.15	0.06	0.26

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	233.154	0.3656	233.154
A1 to ZN	**fall**	0.01	0.15	0.04	0.16
	rise	0.01	0.14	0.04	0.15
A2 to ZN	**fall**	0.01	0.15	0.03	0.16
	rise	0.01	0.14	0.03	0.15
A3 to ZN	**fall**	0.01	0.15	0.03	0.16
	rise	0.01	0.14	0.03	0.15

Capacitance [fF]	
A1	6.2510
A2	6.9140
A3	7.1559

Leakage [nW]
72.42

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	233.154	0.3656	233.154
A1 to ZN	**fall**	2.42	2.62	13.49	4.81
	rise	9.98	9.79	21.01	13.77
A2 to ZN	**fall**	2.50	2.63	10.08	3.67
	rise	12.96	13.31	23.55	17.41
A3 to ZN	**fall**	2.50	2.63	9.46	3.59
	rise	15.17	15.24	26.57	19.47

NANGATE NAND4_X1

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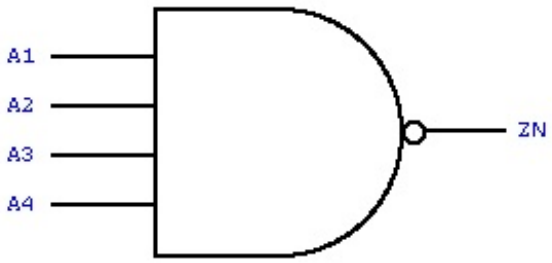
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	H
H	H	H	H	L
-	L	-	-	H
-	-	L	-	H
-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**55.9998**	**0.3656**	**55.9998**
A1 to ZN	**fall**	0.02	0.23	0.04	0.30
	rise	0.01	0.14	0.04	0.25
A2 to ZN	**fall**	0.02	0.23	0.04	0.30
	rise	0.01	0.15	0.05	0.25
A3 to ZN	**fall**	0.02	0.24	0.04	0.28
	rise	0.01	0.15	0.06	0.25
A4 to ZN	**fall**	0.02	0.24	0.03	0.27
	rise	0.02	0.15	0.07	0.25

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**55.9998**	**0.3656**	**55.9998**
A1 to ZN	**fall**	0.01	0.20	0.04	0.20
	rise	0.01	0.13	0.04	0.14
A2 to ZN	**fall**	0.01	0.20	0.04	0.20
	rise	0.01	0.14	0.04	0.15
A3 to ZN	**fall**	0.01	0.20	0.03	0.20
	rise	0.01	0.14	0.04	0.15
A4 to ZN	**fall**	0.01	0.20	0.03	0.20
	rise	0.01	0.14	0.04	0.15

Capacitance [fF]	
A1	1.5221
A2	1.5952
A3	1.6381
A4	1.6599

Leakage [nW]
18.13

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	55.9998	0.3656	55.9998
A1 to ZN	fall	1.02	1.04	3.44	1.52
	rise	2.75	2.72	5.15	3.64
A2 to ZN	fall	1.02	1.05	2.63	1.25
	rise	3.50	3.59	5.78	4.53
A3 to ZN	fall	1.03	1.04	2.45	1.21
	rise	4.10	4.10	6.51	5.05
A4 to ZN	fall	1.03	1.05	2.36	1.20
	rise	4.61	4.90	7.23	5.84

NANGATE NAND4_X2

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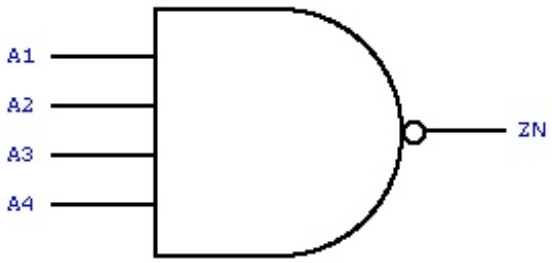
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	H
H	H	H	H	L
-	L	-	-	H
-	-	L	-	H
-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**111.542**	**0.3656**	**111.542**
A1 to ZN	**fall**	0.02	0.23	0.04	0.30
	rise	0.01	0.14	0.04	0.25
A2 to ZN	**fall**	0.02	0.23	0.04	0.30
	rise	0.01	0.15	0.05	0.25
A3 to ZN	**fall**	0.02	0.24	0.04	0.28
	rise	0.01	0.15	0.06	0.25
A4 to ZN	**fall**	0.02	0.24	0.03	0.27
	rise	0.02	0.15	0.07	0.25

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**111.542**	**0.3656**	**111.542**
A1 to ZN	**fall**	0.01	0.20	0.04	0.20
	rise	0.01	0.13	0.04	0.14
A2 to ZN	**fall**	0.01	0.20	0.04	0.20
	rise	0.01	0.14	0.04	0.15
A3 to ZN	**fall**	0.01	0.20	0.03	0.20
	rise	0.01	0.14	0.04	0.15
A4 to ZN	**fall**	0.01	0.20	0.03	0.20
	rise	0.01	0.14	0.04	0.15

Capacitance [fF]	
A1	2.9222
A2	3.2748
A3	3.4763
A4	3.8214

Leakage [nW]
36.25

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	111.542	0.3656	111.542
A1 to ZN	fall	2.02	2.09	6.88	3.04
	rise	5.95	5.78	10.74	7.62
A2 to ZN	fall	2.04	2.09	5.24	2.50
	rise	7.48	7.58	12.01	9.46
A3 to ZN	fall	2.05	2.08	4.91	2.42
	rise	8.74	8.73	13.52	10.69
A4 to ZN	fall	2.05	2.09	4.71	2.40
	rise	9.70	10.25	14.96	12.14

NANGATE NAND4_X4

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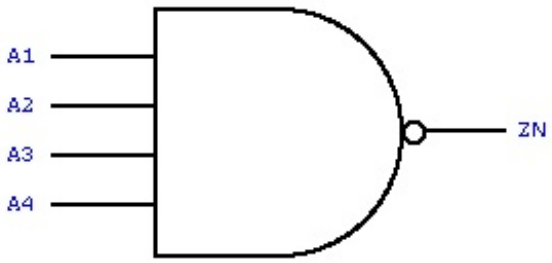
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.788 um^2
Equation	ZN = "!(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	H
H	H	H	H	L
-	L	-	-	H
-	-	L	-	H
-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**222.778**	**0.3656**	**222.778**
A1 to ZN	**fall**	0.01	0.23	0.04	0.30
	rise	0.01	0.14	0.04	0.25
A2 to ZN	**fall**	0.02	0.23	0.04	0.30
	rise	0.01	0.15	0.05	0.25
A3 to ZN	**fall**	0.02	0.24	0.03	0.29
	rise	0.01	0.15	0.06	0.25
A4 to ZN	**fall**	0.02	0.24	0.03	0.27
	rise	0.02	0.15	0.07	0.25

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**222.778**	**0.3656**	**222.778**
A1 to ZN	**fall**	0.01	0.20	0.04	0.20
	rise	0.01	0.13	0.04	0.14
A2 to ZN	**fall**	0.01	0.20	0.04	0.20
	rise	0.01	0.14	0.04	0.15
A3 to ZN	**fall**	0.01	0.20	0.03	0.20
	rise	0.01	0.14	0.04	0.15
A4 to ZN	**fall**	0.01	0.20	0.03	0.20
	rise	0.01	0.14	0.04	0.15

Capacitance [fF]	
A1	5.6520
A2	5.7950
A3	5.9052
A4	6.0955

Leakage [nW]
72.51

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	222.778	0.3656	222.778
A1 to ZN	fall	4.07	4.18	13.97	6.11
	rise	10.42	10.60	20.09	14.33
A2 to ZN	fall	4.09	4.18	10.52	4.96
	rise	13.88	13.51	22.99	17.48
A3 to ZN	fall	4.09	4.19	9.74	4.76
	rise	16.54	16.88	26.11	20.40
A4 to ZN	fall	4.05	4.18	9.23	4.72
	rise	18.70	20.35	29.26	24.14

NOR2_X1

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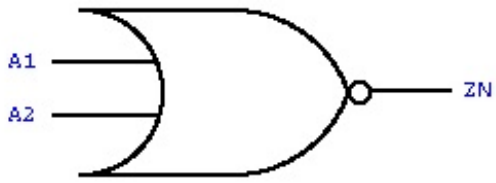
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	ZN = "!(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**26.7029**	**0.3656**	**26.7029**
A1 to ZN	**fall**	0.00	0.04	-0.01	0.09
	rise	0.01	0.15	0.07	0.24
A2 to ZN	**fall**	0.00	0.04	-0.00	0.09
	rise	0.02	0.15	0.06	0.23

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**26.7029**	**0.3656**	**26.7029**
A1 to ZN	**fall**	0.00	0.03	0.03	0.07
	rise	0.01	0.13	0.03	0.14
A2 to ZN	**fall**	0.00	0.03	0.03	0.07
	rise	0.01	0.13	0.03	0.14

Capacitance [fF]	
A1	1.7145
A2	1.6513

Leakage [nW]
21.20

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**26.7029**	**0.3656**	**26.7029**
A1 to ZN	**fall**	-0.00	-0.00	2.19	1.08
	rise	2.28	2.32	5.03	3.87
A2 to ZN	**fall**	0.09	0.27	2.62	1.50
	rise	3.12	3.14	4.72	3.76

NANGATE NOR2_X2

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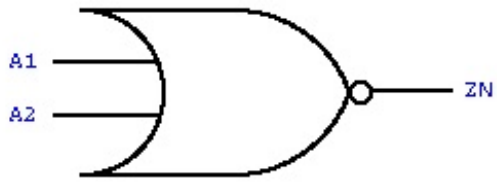
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "!(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	53.4058	0.3656	53.4058
A1 to ZN	**fall**	0.00	0.04	-0.02	0.09
	rise	0.01	0.15	0.06	0.24
A2 to ZN	**fall**	0.00	0.04	-0.00	0.09
	rise	0.02	0.15	0.06	0.23

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	53.4058	0.3656	53.4058
A1 to ZN	**fall**	0.00	0.03	0.03	0.07
	rise	0.01	0.13	0.03	0.14
A2 to ZN	**fall**	0.00	0.03	0.03	0.07
	rise	0.01	0.13	0.03	0.14

Capacitance [fF]	
A1	3.2933
A2	3.3469

Leakage [nW]
42.40

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	53.4058	0.3656	53.4058
A1 to ZN	**fall**	-0.00	-0.00	4.56	2.34
	rise	4.37	4.44	9.96	7.74
A2 to ZN	**fall**	0.34	0.72	5.41	3.17
	rise	6.02	6.09	9.29	7.36

NANGATE NOR2_X4

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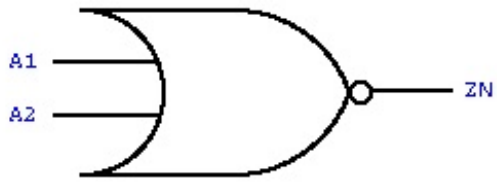
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "!(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	106.811	0.3656	106.811
A1 to ZN	**fall**	0.00	0.04	-0.02	0.09
	rise	0.01	0.15	0.06	0.24
A2 to ZN	**fall**	0.00	0.04	-0.01	0.09
	rise	0.02	0.15	0.06	0.23

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	106.811	0.3656	106.811
A1 to ZN	**fall**	0.00	0.03	0.03	0.07
	rise	0.01	0.13	0.03	0.14
A2 to ZN	**fall**	0.00	0.03	0.03	0.07
	rise	0.01	0.13	0.03	0.14

Capacitance [fF]	
A1	6.7731
A2	6.6834

Leakage [nW]
84.80

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	106.811	0.3656	106.811
A1 to ZN	**fall**	-0.00	-0.00	9.04	4.69
	rise	8.49	8.75	19.81	15.36
A2 to ZN	**fall**	0.68	1.43	10.84	6.35
	rise	11.93	12.03	18.48	14.70

NANGATE NOR3_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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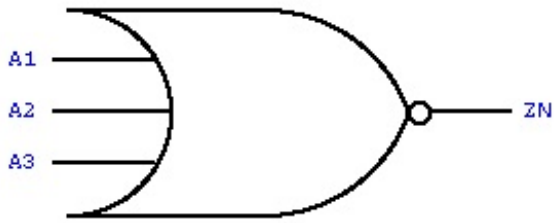
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	H
H	-	-	L
-	H	-	L
-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	16.0217	0.3656	16.0217
A1 to ZN	**fall**	0.00	0.02	-0.02	0.06
	rise	0.02	0.14	0.08	0.23
A2 to ZN	**fall**	0.01	0.02	-0.01	0.06
	rise	0.03	0.16	0.09	0.23
A3 to ZN	**fall**	0.00	0.03	0.00	0.06
	rise	0.04	0.16	0.08	0.22

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	16.0217	0.3656	16.0217
A1 to ZN	**fall**	0.00	0.02	0.03	0.06
	rise	0.02	0.13	0.04	0.14
A2 to ZN	**fall**	0.00	0.02	0.03	0.06
	rise	0.02	0.13	0.04	0.13
A3 to ZN	**fall**	0.00	0.02	0.03	0.06
	rise	0.02	0.13	0.03	0.13

Capacitance [fF]	
A1	1.7636
A2	1.6638
A3	1.6163

Leakage [nW]
26.83

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	16.0217	0.3656	16.0217
A1 to ZN	**fall**	-0.00	-0.00	1.70	1.23
	rise	2.56	2.66	4.74	4.09
A2 to ZN	**fall**	0.16	0.30	2.10	1.62
	rise	3.38	3.43	4.48	4.02
A3 to ZN	**fall**	0.22	0.64	2.51	1.95
	rise	4.11	4.09	4.89	4.50

NANGATE NOR3_X2

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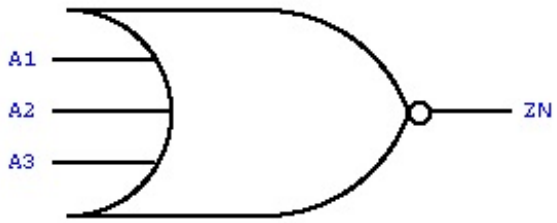
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	H
H	-	-	L
-	H	-	L
-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	31.7383	0.3656	31.7383
A1 to ZN	**fall**	0.00	0.02	-0.02	0.06
	rise	0.02	0.14	0.08	0.23
A2 to ZN	**fall**	0.01	0.03	-0.01	0.06
	rise	0.03	0.15	0.09	0.23
A3 to ZN	**fall**	0.00	0.03	0.00	0.06
	rise	0.03	0.16	0.08	0.22

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	31.7383	0.3656	31.7383
A1 to ZN	**fall**	0.00	0.02	0.03	0.06
	rise	0.02	0.13	0.04	0.14
A2 to ZN	**fall**	0.00	0.02	0.03	0.06
	rise	0.02	0.13	0.04	0.13
A3 to ZN	**fall**	0.00	0.02	0.03	0.06
	rise	0.02	0.13	0.03	0.13

Capacitance [fF]	
A1	3.3652
A2	3.4305
A3	3.4428

Leakage [nW]
53.66

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	31.7383	0.3656	31.7383
A1 to ZN	**fall**	-0.00	-0.00	3.59	2.63
	rise	5.22	5.43	9.62	8.29
A2 to ZN	**fall**	0.50	0.78	4.37	3.40
	rise	6.87	7.00	9.08	8.25
A3 to ZN	**fall**	0.60	1.47	5.22	4.06
	rise	8.32	8.46	9.89	9.10

NOR3_X4

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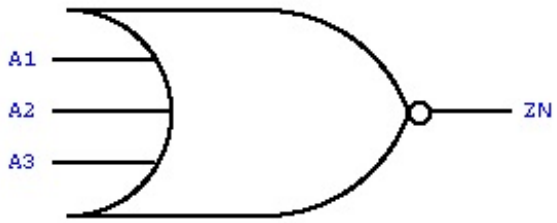
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.724 um^2
Equation	ZN = "!((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	H
H	-	-	L
-	H	-	L
-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	63.324	0.3656	63.324
A1 to ZN	**fall**	0.00	0.02	-0.02	0.06
	rise	0.02	0.14	0.08	0.24
A2 to ZN	**fall**	0.01	0.03	-0.01	0.06
	rise	0.03	0.16	0.09	0.24
A3 to ZN	**fall**	0.00	0.03	0.00	0.06
	rise	0.04	0.16	0.08	0.22

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	63.324	0.3656	63.324
A1 to ZN	**fall**	0.00	0.02	0.03	0.06
	rise	0.02	0.13	0.04	0.14
A2 to ZN	**fall**	0.00	0.02	0.03	0.06
	rise	0.02	0.13	0.04	0.13
A3 to ZN	**fall**	0.00	0.02	0.03	0.06
	rise	0.02	0.13	0.03	0.13

Capacitance [fF]	
A1	6.5149
A2	6.1714
A3	6.1054

Leakage [nW]
107.33

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	63.324	0.3656	63.324
A1 to ZN	**fall**	0.00	-0.00	7.11	5.25
	rise	10.33	10.64	19.16	16.78
A2 to ZN	**fall**	0.96	1.57	8.67	6.74
	rise	14.26	14.44	18.51	16.91
A3 to ZN	**fall**	1.17	2.91	10.29	8.01
	rise	17.56	17.64	20.34	19.09

NOR4_X1

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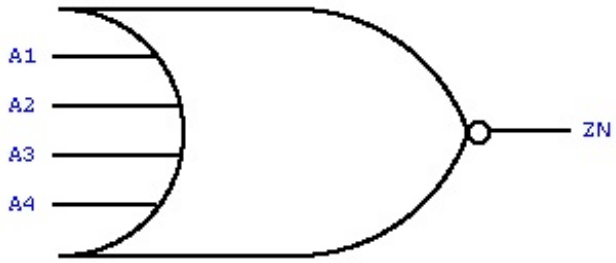
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	-	L
-	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**10.4713**	**0.3656**	**10.4713**
A1 to ZN	**fall**	0.00	0.02	-0.01	0.04
	rise	0.03	0.14	0.10	0.23
A2 to ZN	**fall**	0.01	0.02	-0.00	0.04
	rise	0.05	0.16	0.11	0.23
A3 to ZN	**fall**	0.01	0.02	0.00	0.05
	rise	0.06	0.17	0.11	0.23
A4 to ZN	**fall**	0.01	0.02	0.00	0.05
	rise	0.06	0.17	0.10	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**10.4713**	**0.3656**	**10.4713**
A1 to ZN	**fall**	0.00	0.01	0.03	0.05
	rise	0.03	0.13	0.05	0.13
A2 to ZN	**fall**	0.00	0.01	0.03	0.05
	rise	0.03	0.13	0.05	0.13
A3 to ZN	**fall**	0.00	0.02	0.04	0.05
	rise	0.03	0.13	0.05	0.13
A4 to ZN	**fall**	0.00	0.02	0.04	0.05
	rise	0.03	0.13	0.04	0.13

Capacitance [fF]	
A1	1.7368
A2	1.6741
A3	1.6360
A4	1.6059

Leakage [nW]
32.60

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	10.4713	0.3656	10.4713
A1 to ZN	fall	-0.00	-0.00	1.41	1.22
	rise	3.08	3.16	4.93	4.56
A2 to ZN	fall	0.24	0.29	1.79	1.60
	rise	3.90	3.90	4.74	4.51
A3 to ZN	fall	0.32	0.65	2.15	1.93
	rise	4.63	4.64	5.05	4.90
A4 to ZN	fall	0.27	0.80	2.40	2.15
	rise	5.35	5.39	5.54	5.45

NANGATE NOR4_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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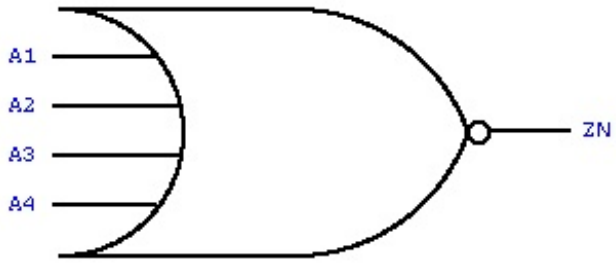
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	-	L
-	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**20.9045**	**0.3656**	**20.9045**
A1 to ZN	**fall**	0.00	0.02	-0.01	0.04
	rise	0.03	0.14	0.10	0.23
A2 to ZN	**fall**	0.01	0.02	-0.01	0.04
	rise	0.05	0.16	0.11	0.23
A3 to ZN	**fall**	0.01	0.02	0.00	0.05
	rise	0.06	0.17	0.10	0.23
A4 to ZN	**fall**	0.01	0.02	0.00	0.05
	rise	0.06	0.17	0.10	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**20.9045**	**0.3656**	**20.9045**
A1 to ZN	**fall**	0.00	0.01	0.03	0.05
	rise	0.03	0.13	0.05	0.13
A2 to ZN	**fall**	0.00	0.01	0.03	0.05
	rise	0.03	0.13	0.05	0.13
A3 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.03	0.13	0.04	0.13
A4 to ZN	**fall**	0.00	0.02	0.04	0.05
	rise	0.03	0.13	0.04	0.13

Capacitance [fF]	
A1	3.3907
A2	3.4099
A3	3.5192
A4	3.6150

Leakage [nW]
65.20

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	20.9045	0.3656	20.9045
A1 to ZN	fall	0.10	-0.00	3.00	2.62
	rise	6.01	6.16	9.72	9.00
A2 to ZN	fall	0.65	0.76	3.75	3.37
	rise	7.64	7.65	9.35	8.84
A3 to ZN	fall	0.80	1.47	4.47	4.03
	rise	9.11	9.15	9.97	9.65
A4 to ZN	fall	0.68	1.77	4.99	4.48
	rise	10.56	10.62	10.94	10.75

NOR4_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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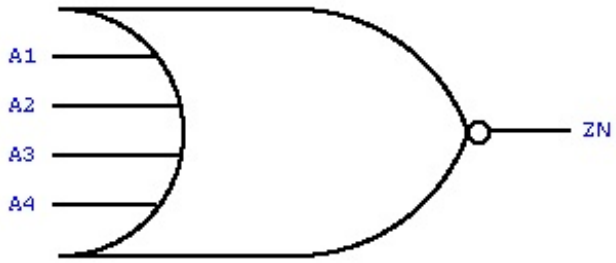
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.788 um^2
Equation	ZN = "!(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	-	L
-	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**41.5039**	**0.3656**	**41.5039**
A1 to ZN	**fall**	0.00	0.02	-0.02	0.04
	rise	0.03	0.14	0.10	0.23
A2 to ZN	**fall**	0.01	0.02	-0.00	0.04
	rise	0.05	0.16	0.11	0.23
A3 to ZN	**fall**	0.01	0.02	0.00	0.05
	rise	0.06	0.17	0.10	0.23
A4 to ZN	**fall**	0.01	0.02	0.00	0.05
	rise	0.06	0.17	0.10	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**41.5039**	**0.3656**	**41.5039**
A1 to ZN	**fall**	0.00	0.01	0.03	0.05
	rise	0.03	0.13	0.05	0.13
A2 to ZN	**fall**	0.00	0.01	0.03	0.05
	rise	0.03	0.13	0.05	0.13
A3 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.03	0.13	0.04	0.13
A4 to ZN	**fall**	0.00	0.02	0.04	0.05
	rise	0.03	0.13	0.04	0.13

Capacitance [fF]	
A1	6.5694
A2	6.1988
A3	6.0813
A4	6.0266

Leakage [nW]
130.41

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	41.5039	0.3656	41.5039
A1 to ZN	fall	0.29	-0.00	5.88	5.20
	rise	11.91	12.37	19.44	17.95
A2 to ZN	fall	1.35	1.51	7.41	6.70
	rise	15.74	15.94	18.98	18.07
A3 to ZN	fall	1.55	2.90	8.84	7.92
	rise	19.31	19.37	20.68	20.23
A4 to ZN	fall	1.33	3.48	9.78	8.78
	rise	22.55	22.85	23.06	22.80

OAI211_X1

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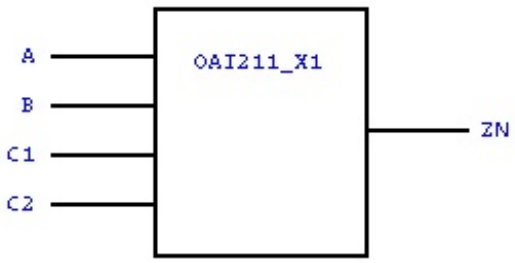
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((C1 | C2) & A) & B)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	-	-	-	H
H	H	H	-	L
H	H	-	H	L
-	L	-	-	H
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**25.5585**	**0.3656**	**25.5585**
A to ZN	**fall**	0.02	0.09	0.04	0.16
	rise	0.01	0.08	0.06	0.17
B to ZN	**fall**	0.02	0.10	0.03	0.13
	rise	0.02	0.08	0.07	0.17
C1 to ZN	**fall**	0.01	0.09	0.02	0.15
	rise	0.02	0.15	0.07	0.24
C2 to ZN	**fall**	0.01	0.09	0.03	0.16
	rise	0.02	0.15	0.07	0.23

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**25.5585**	**0.3656**	**25.5585**
A to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.01	0.07	0.05	0.10
B to ZN	**fall**	0.01	0.07	0.03	0.08
	rise	0.01	0.07	0.05	0.10
C1 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.01	0.13	0.04	0.14
C2 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.01	0.13	0.04	0.14

Capacitance [fF]	
A	1.6142
B	1.6573
C1	1.5952
C2	1.5557

Leakage [nW]
22.04

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	25.5585	0.3656	25.5585
A to ZN	fall	0.70	0.75	2.36	1.17
	rise	4.36	4.45	6.93	6.02
B to ZN	fall	0.70	0.76	2.20	1.15
	rise	4.92	4.98	7.71	6.71
C1 to ZN	fall	0.57	0.61	2.72	1.39
	rise	2.81	2.88	4.94	3.97
C2 to ZN	fall	1.01	1.09	3.07	1.83
	rise	3.62	3.66	4.72	4.09

OAI211_X2

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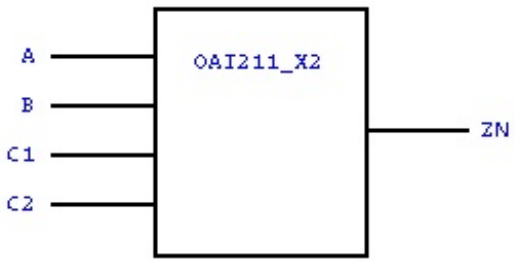
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((C1 | C2) & A) & B)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	-	-	-	H
H	H	H	-	L
H	H	-	H	L
-	L	-	-	H
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**50.8118**	**0.3656**	**50.8118**
A to ZN	**fall**	0.02	0.09	0.04	0.16
	rise	0.01	0.08	0.06	0.17
B to ZN	**fall**	0.02	0.10	0.03	0.13
	rise	0.02	0.08	0.07	0.17
C1 to ZN	**fall**	0.01	0.09	0.02	0.15
	rise	0.02	0.15	0.07	0.24
C2 to ZN	**fall**	0.01	0.09	0.03	0.16
	rise	0.02	0.15	0.07	0.23

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**50.8118**	**0.3656**	**50.8118**
A to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.01	0.07	0.05	0.10
B to ZN	**fall**	0.01	0.07	0.03	0.08
	rise	0.01	0.07	0.04	0.10
C1 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.01	0.13	0.04	0.14
C2 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.01	0.13	0.04	0.14

Capacitance [fF]	
A	3.3312
B	3.2079
C1	2.9955
C2	3.2210

Leakage [nW]
44.08

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	50.8118	0.3656	50.8118
A to ZN	fall	1.58	1.69	4.92	2.53
	rise	8.75	8.88	13.91	12.13
B to ZN	fall	1.58	1.69	4.61	2.49
	rise	9.85	10.25	15.47	13.43
C1 to ZN	fall	1.33	1.41	5.60	2.95
	rise	5.76	5.92	10.04	8.12
C2 to ZN	fall	2.19	2.35	6.31	3.84
	rise	7.37	7.40	9.59	8.28

OAI211_X4

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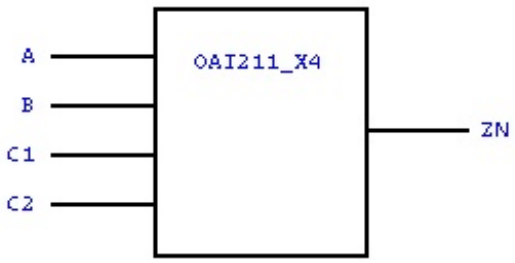
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Equation	ZN = "!(((C1 | C2) & A) & B)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	-	-	-	H
H	H	H	-	L
H	H	-	H	L
-	L	-	-	H
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**101.624**	**0.3656**	**101.624**
A to ZN	**fall**	0.02	0.10	0.04	0.16
	rise	0.01	0.08	0.06	0.17
B to ZN	**fall**	0.02	0.10	0.03	0.13
	rise	0.02	0.08	0.07	0.17
C1 to ZN	**fall**	0.01	0.09	0.02	0.16
	rise	0.02	0.15	0.07	0.24
C2 to ZN	**fall**	0.01	0.09	0.03	0.16
	rise	0.02	0.15	0.07	0.23

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**101.624**	**0.3656**	**101.624**
A to ZN	**fall**	0.01	0.08	0.03	0.09
	rise	0.01	0.07	0.05	0.10
B to ZN	**fall**	0.01	0.08	0.03	0.08
	rise	0.01	0.07	0.04	0.10
C1 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.01	0.13	0.04	0.14
C2 to ZN	**fall**	0.01	0.08	0.04	0.10
	rise	0.01	0.13	0.04	0.14

Capacitance [fF]	
A	6.3640
B	6.5507
C1	6.2062
C2	6.4234

Leakage [nW]
88.16

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	101.624	0.3656	101.624
A to ZN	fall	3.15	3.36	9.86	5.03
	rise	17.71	17.92	28.04	24.25
B to ZN	fall	3.17	3.36	9.23	4.94
	rise	19.97	20.60	31.16	27.24
C1 to ZN	fall	2.68	2.81	11.25	5.86
	rise	11.64	11.45	20.24	16.07
C2 to ZN	fall	4.39	4.71	12.63	7.66
	rise	14.88	14.73	19.33	16.57

NANGATE OAI21_X1

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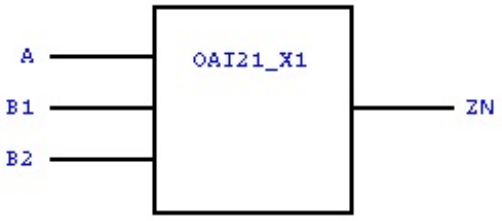
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!(A & (B1 | B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	-	-	H
H	H	-	L
H	-	H	L
-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**26.0544**	**0.3656**	**26.0544**
A to ZN	**fall**	0.01	0.07	0.02	0.12
	rise	0.01	0.08	0.06	0.17
B1 to ZN	**fall**	0.01	0.06	0.00	0.12
	rise	0.02	0.15	0.07	0.24
B2 to ZN	**fall**	0.01	0.06	0.02	0.13
	rise	0.02	0.15	0.07	0.23

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**26.0544**	**0.3656**	**26.0544**
A to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.07	0.04	0.10
B1 to ZN	**fall**	0.00	0.05	0.04	0.09
	rise	0.01	0.13	0.04	0.14
B2 to ZN	**fall**	0.01	0.05	0.04	0.09
	rise	0.01	0.13	0.03	0.14

Capacitance [fF]	
A	1.6707
B1	1.6621
B2	1.5719

Leakage [nW]
22.62

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**26.0544**	**0.3656**	**26.0544**
A to ZN	**fall**	0.38	0.45	2.31	1.08
	rise	4.06	4.21	7.12	6.11
B1 to ZN	**fall**	0.28	0.31	2.56	1.26
	rise	2.56	2.59	4.96	3.85
B2 to ZN	**fall**	0.67	0.77	2.95	1.69
	rise	3.38	3.40	4.69	3.92

OAI21_X2

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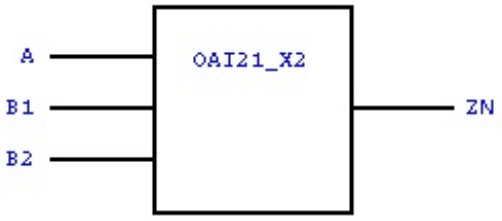
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!(A & (B1 | B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	-	-	H
H	H	-	L
H	-	H	L
-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	52.1088	0.3656	52.1088
A to ZN	**fall**	0.01	0.07	0.02	0.12
	rise	0.01	0.08	0.06	0.17
B1 to ZN	**fall**	0.01	0.06	0.00	0.12
	rise	0.02	0.15	0.07	0.24
B2 to ZN	**fall**	0.01	0.06	0.02	0.13
	rise	0.02	0.15	0.07	0.23

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	52.1088	0.3656	52.1088
A to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.07	0.04	0.10
B1 to ZN	**fall**	0.00	0.05	0.03	0.09
	rise	0.01	0.13	0.04	0.14
B2 to ZN	**fall**	0.01	0.05	0.03	0.09
	rise	0.01	0.13	0.03	0.14

Capacitance [fF]	
A	3.1807
B1	3.1008
B2	3.3289

Leakage [nW]
45.24

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	52.1088	0.3656	52.1088
A to ZN	**fall**	0.75	0.90	4.68	2.16
	rise	8.02	8.23	14.15	12.10
B1 to ZN	**fall**	0.54	0.63	5.11	2.51
	rise	5.14	5.13	9.96	7.68
B2 to ZN	**fall**	1.33	1.55	5.93	3.38
	rise	6.76	6.77	9.39	8.00

OAI21_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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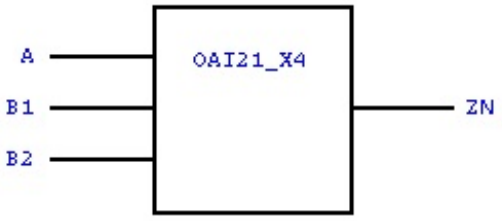
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(A & (B1 | B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	-	-	H
H	H	-	L
H	-	H	L
-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**104.065**	**0.3656**	**104.065**
A to ZN	**fall**	0.01	0.07	0.02	0.12
	rise	0.01	0.08	0.06	0.17
B1 to ZN	**fall**	0.01	0.06	0.00	0.12
	rise	0.02	0.15	0.07	0.24
B2 to ZN	**fall**	0.01	0.06	0.01	0.13
	rise	0.02	0.15	0.06	0.23

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**104.065**	**0.3656**	**104.065**
A to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.07	0.04	0.10
B1 to ZN	**fall**	0.00	0.05	0.03	0.09
	rise	0.01	0.13	0.04	0.14
B2 to ZN	**fall**	0.01	0.05	0.03	0.09
	rise	0.01	0.13	0.03	0.14

Capacitance [fF]	
A	6.1947
B1	6.3515
B2	6.5004

Leakage [nW]
90.48

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**104.065**	**0.3656**	**104.065**
A to ZN	**fall**	1.49	1.78	9.39	4.30
	rise	16.08	16.36	28.34	23.99
B1 to ZN	**fall**	1.08	1.25	10.29	4.99
	rise	10.26	10.25	19.99	15.52
B2 to ZN	**fall**	2.67	3.09	11.87	6.72
	rise	13.52	13.53	18.84	15.56

OAI221_X1

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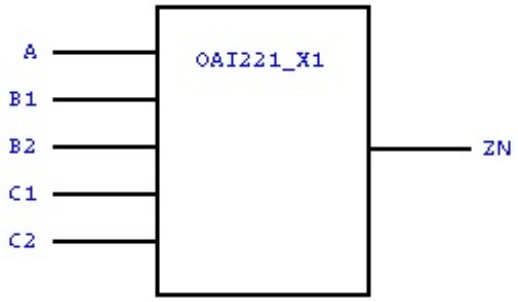
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "!(((C1 | C2) & A) & (B1 | B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	-	-	-	H
H	H	-	H	-	L
H	H	-	-	H	L
H	-	H	H	-	L
H	-	H	-	H	L
-	L	L	-	-	H
-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	22.1634	0.3656	22.1634
A to ZN	**fall**	0.02	0.09	0.05	0.15
	rise	0.01	0.07	0.06	0.16
B1 to ZN	**fall**	0.02	0.09	0.03	0.12
	rise	0.04	0.15	0.11	0.24
B2 to ZN	**fall**	0.02	0.09	0.04	0.13
	rise	0.04	0.15	0.10	0.23
C1 to ZN	**fall**	0.02	0.08	0.03	0.15
	rise	0.02	0.13	0.07	0.22
C2 to ZN	**fall**	0.02	0.08	0.04	0.15
	rise	0.02	0.14	0.07	0.21

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	22.1634	0.3656	22.1634
A to ZN	**fall**	0.01	0.07	0.03	0.08
	rise	0.01	0.07	0.06	0.10
B1 to ZN	**fall**	0.01	0.07	0.03	0.08
	rise	0.03	0.13	0.06	0.14
B2 to ZN	**fall**	0.01	0.07	0.03	0.08
	rise	0.03	0.13	0.05	0.14
C1 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.02	0.12	0.06	0.14
C2 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.02	0.12	0.05	0.13

Capacitance [fF]	
A	1.6302
B1	1.6554
B2	1.6054
C1	1.5697
C2	1.5833

Leakage [nW]
33.94

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	22.1634	0.3656	22.1634
A to ZN	fall	0.87	0.94	2.53	1.45
	rise	4.48	4.52	7.17	6.27
B1 to ZN	fall	0.75	0.80	1.80	1.17
	rise	5.22	5.38	7.29	6.62
B2 to ZN	fall	1.18	1.27	2.14	1.61
	rise	6.00	6.16	7.03	6.75
C1 to ZN	fall	0.74	0.80	2.85	1.66
	rise	3.03	3.02	5.21	4.31
C2 to ZN	fall	1.17	1.27	3.21	2.10
	rise	3.84	3.84	4.99	4.36

OAI221_X2

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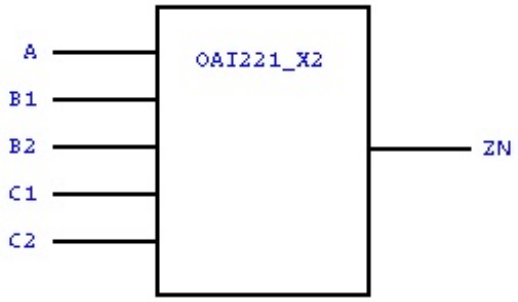
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.926 um^2
Equation	ZN = "!(((C1 | C2) & A) & (B1 | B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	-	-	-	H
H	H	-	H	-	L
H	H	-	-	H	L
H	-	H	H	-	L
H	-	H	-	H	L
-	L	L	-	-	H
-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	43.869	0.3656	43.869
A to ZN	**fall**	0.02	0.09	0.05	0.15
	rise	0.01	0.07	0.06	0.16
B1 to ZN	**fall**	0.02	0.09	0.03	0.12
	rise	0.04	0.15	0.11	0.24
B2 to ZN	**fall**	0.02	0.09	0.04	0.13
	rise	0.04	0.15	0.10	0.23
C1 to ZN	**fall**	0.02	0.08	0.03	0.15
	rise	0.02	0.13	0.07	0.22
C2 to ZN	**fall**	0.02	0.08	0.04	0.15
	rise	0.02	0.14	0.07	0.21

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	43.869	0.3656	43.869
A to ZN	**fall**	0.01	0.07	0.03	0.08
	rise	0.01	0.07	0.06	0.10
B1 to ZN	**fall**	0.01	0.07	0.03	0.08
	rise	0.03	0.13	0.06	0.14
B2 to ZN	**fall**	0.01	0.07	0.03	0.08
	rise	0.03	0.13	0.05	0.14
C1 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.02	0.12	0.06	0.14
C2 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.02	0.12	0.05	0.13

Capacitance [fF]	
A	3.5364
B1	3.4774
B2	3.1247
C1	2.9964
C2	3.2452

Leakage [nW]
67.88

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	43.869	0.3656	43.869
A to ZN	fall	1.92	2.06	5.18	3.05
	rise	9.35	9.51	14.70	13.01
B1 to ZN	fall	1.69	1.79	3.76	2.52
	rise	10.74	11.11	14.87	13.52
B2 to ZN	fall	2.54	2.72	4.44	3.38
	rise	12.31	12.61	14.37	13.74
C1 to ZN	fall	1.69	1.79	5.87	3.49
	rise	6.30	6.34	10.68	8.78
C2 to ZN	fall	2.52	2.72	6.60	4.38
	rise	7.91	7.95	10.21	8.94

NANGATE OAI221_X4

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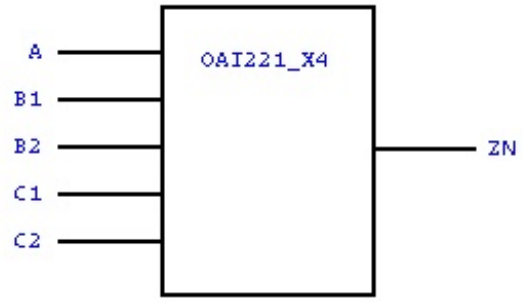
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(!(!(((C1 | C2) & A) & (B1 | B2))))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	-	-	-	H
H	H	-	H	-	L
H	H	-	-	H	L
H	-	H	H	-	L
H	-	H	-	H	L
-	L	L	-	-	H
-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.06	0.14	0.10	0.18
	rise	0.04	0.19	0.11	0.26
B1 to ZN	**fall**	0.06	0.14	0.08	0.16
	rise	0.08	0.22	0.16	0.31
B2 to ZN	**fall**	0.06	0.14	0.09	0.17
	rise	0.08	0.23	0.15	0.30
C1 to ZN	**fall**	0.05	0.13	0.09	0.17
	rise	0.06	0.21	0.13	0.28
C2 to ZN	**fall**	0.06	0.14	0.10	0.18
	rise	0.07	0.21	0.13	0.27

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14
B1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
B2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
C1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
C2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A	1.6434
B1	1.6728
B2	1.6530
C1	1.5482
C2	1.5845

Leakage [nW]
116.35

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	fall	10.45	12.93	12.61	15.11
	rise	15.08	16.74	18.08	20.58
B1 to ZN	fall	10.32	12.78	11.77	14.36
	rise	15.82	18.03	18.77	21.37
B2 to ZN	fall	10.81	13.35	12.17	14.68
	rise	16.62	19.00	18.05	20.80
C1 to ZN	fall	10.27	12.77	13.46	15.90
	rise	13.50	15.23	16.45	19.05
C2 to ZN	fall	10.77	13.34	13.85	16.25
	rise	14.30	16.18	15.73	18.22

OAI222_X1

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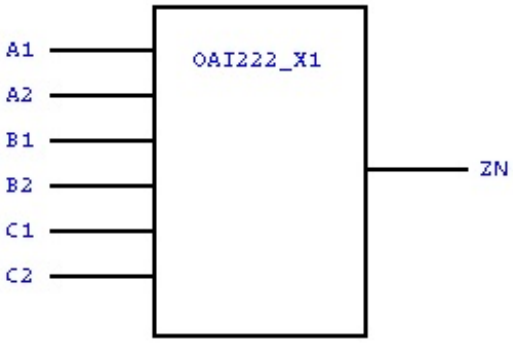
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.128 um^2
Equation	ZN = "!(((A1 | A2) & (B1 | B2)) & (C1 | C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	-	-	-	H
H	-	H	-	H	-	L
H	-	H	-	-	H	L
H	-	-	H	H	-	L
H	-	-	H	-	H	L
-	H	H	-	H	-	L
-	H	H	-	-	H	L
-	H	-	H	H	-	L
-	H	-	H	-	H	L
-	-	L	L	-	-	H
-	-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**20.0653**	**0.3656**	**20.0653**
A1 to ZN	**fall**	0.02	0.08	0.04	0.14
	rise	0.02	0.12	0.07	0.21
A2 to ZN	**fall**	0.02	0.08	0.05	0.15
	rise	0.03	0.13	0.07	0.20
B1 to ZN	**fall**	0.02	0.08	0.05	0.14
	rise	0.03	0.13	0.10	0.23
B2 to ZN	**fall**	0.03	0.09	0.06	0.15
	rise	0.04	0.14	0.09	0.22
C1 to ZN	**fall**	0.03	0.09	0.04	0.12
	rise	0.04	0.14	0.11	0.24
C2 to ZN	**fall**	0.03	0.09	0.05	0.13
	rise	0.05	0.15	0.11	0.23

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**20.0653**	**0.3656**	**20.0653**
A1 to ZN	**fall**	0.01	0.06	0.04	0.09
	rise	0.03	0.12	0.07	0.14
A2 to ZN	**fall**	0.02	0.07	0.04	0.09
	rise	0.03	0.12	0.06	0.13
B1 to ZN	**fall**	0.01	0.06	0.04	0.08
	rise	0.03	0.13	0.07	0.14
B2 to ZN	**fall**	0.02	0.07	0.04	0.08
	rise	0.03	0.13	0.06	0.14
C1 to ZN	**fall**	0.01	0.06	0.03	0.07
	rise	0.04	0.13	0.07	0.14
C2 to ZN	**fall**	0.02	0.07	0.03	0.08
	rise	0.04	0.13	0.06	0.14

Capacitance [fF]	
A1	1.5766
A2	1.5946
B1	1.6175
B2	1.6202
C1	1.5964
C2	1.5711

Leakage [nW]
43.18

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	20.0653	0.3656	20.0653
A1 to ZN	fall	0.92	0.99	2.99	1.93
	rise	3.24	3.25	5.49	4.57
A2 to ZN	fall	1.34	1.45	3.36	2.37
	rise	4.05	4.04	5.24	4.60
B1 to ZN	fall	0.92	0.99	2.13	1.42
	rise	4.80	4.89	6.83	6.17
B2 to ZN	fall	1.35	1.45	2.47	1.86
	rise	5.59	5.65	6.59	6.27
C1 to ZN	fall	0.92	0.99	1.84	1.34
	rise	6.05	6.25	8.08	7.56
C2 to ZN	fall	1.35	1.45	2.19	1.77
	rise	6.83	6.99	7.84	7.64

OAI222_X2

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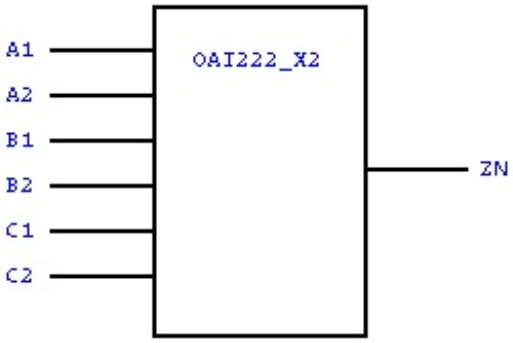
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	3.724 um^2
Equation	ZN = "!(((A1 | A2) & (B1 | B2)) & (C1 | C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	-	-	-	H
H	-	H	-	H	-	L
H	-	H	-	-	H	L
H	-	-	H	H	-	L
H	-	-	H	-	H	L
-	H	H	-	H	-	L
-	H	H	-	-	H	L
-	H	-	H	H	-	L
-	H	-	H	-	H	L
-	-	L	L	-	-	H
-	-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**39.5966**	**0.3656**	**39.5966**
A1 to ZN	**fall**	0.02	0.08	0.04	0.15
	rise	0.02	0.12	0.08	0.21
A2 to ZN	**fall**	0.02	0.08	0.06	0.15
	rise	0.03	0.13	0.07	0.20
B1 to ZN	**fall**	0.02	0.08	0.05	0.14
	rise	0.03	0.13	0.10	0.23
B2 to ZN	**fall**	0.03	0.09	0.06	0.15
	rise	0.04	0.14	0.09	0.22
C1 to ZN	**fall**	0.03	0.09	0.04	0.12
	rise	0.04	0.14	0.12	0.24
C2 to ZN	**fall**	0.03	0.09	0.05	0.13
	rise	0.05	0.15	0.11	0.23

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**39.5966**	**0.3656**	**39.5966**
A1 to ZN	**fall**	0.01	0.06	0.04	0.09
	rise	0.03	0.12	0.07	0.14
A2 to ZN	**fall**	0.02	0.07	0.04	0.09
	rise	0.03	0.12	0.06	0.13
B1 to ZN	**fall**	0.01	0.06	0.04	0.08
	rise	0.03	0.13	0.07	0.14
B2 to ZN	**fall**	0.02	0.07	0.04	0.08
	rise	0.03	0.13	0.06	0.14
C1 to ZN	**fall**	0.01	0.06	0.03	0.07
	rise	0.04	0.13	0.07	0.14
C2 to ZN	**fall**	0.02	0.07	0.03	0.08
	rise	0.04	0.13	0.06	0.14

Capacitance [fF]	
A1	3.2282
A2	3.0009
B1	3.3038
B2	2.9994
C1	3.3635
C2	3.0741

Leakage [nW]
86.36

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	39.5966	0.3656	39.5966
A1 to ZN	fall	2.01	2.16	6.15	4.02
	rise	6.83	6.89	11.30	9.45
A2 to ZN	fall	2.86	3.08	6.89	4.89
	rise	8.44	8.41	10.82	9.60
B1 to ZN	fall	2.03	2.16	4.44	3.03
	rise	9.92	10.04	13.96	12.68
B2 to ZN	fall	2.88	3.08	5.11	3.90
	rise	11.50	11.60	13.47	12.79
C1 to ZN	fall	2.03	2.15	3.87	2.85
	rise	12.44	12.81	16.50	15.43
C2 to ZN	fall	2.88	3.08	4.55	3.71
	rise	14.01	14.35	16.01	15.61

OAI222_X4

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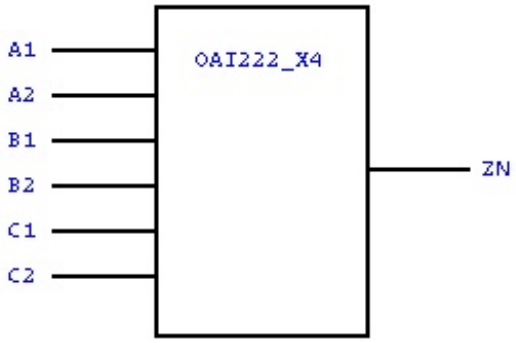
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.724 um^2
Equation	ZN = "!(!(!(((A1 | A2) & (B1 | B2)) & (C1 | C2))))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	-	-	-	H
H	-	H	-	H	-	L
H	-	H	-	-	H	L
H	-	-	H	H	-	L
H	-	-	H	-	H	L
-	H	H	-	H	-	L
-	H	H	-	-	H	L
-	H	-	H	H	-	L
-	H	-	H	-	H	L
-	-	L	L	-	-	H
-	-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.06	0.14	0.10	0.18
	rise	0.06	0.21	0.14	0.28
A2 to ZN	**fall**	0.06	0.14	0.11	0.19
	rise	0.07	0.22	0.13	0.28
B1 to ZN	**fall**	0.06	0.14	0.11	0.18
	rise	0.08	0.22	0.15	0.30
B2 to ZN	**fall**	0.07	0.15	0.11	0.19
	rise	0.08	0.23	0.15	0.29
C1 to ZN	**fall**	0.07	0.14	0.09	0.17
	rise	0.08	0.23	0.17	0.31
C2 to ZN	**fall**	0.07	0.15	0.10	0.18
	rise	0.09	0.23	0.16	0.31

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
B1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
B2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
C1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
C2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A1	1.5789
A2	1.5742
B1	1.6236
B2	1.6127
C1	1.6576
C2	1.6416

Leakage [nW]
127.98

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	fall	10.48	12.97	13.59	16.05
	rise	13.71	15.74	16.72	19.49
A2 to ZN	fall	10.98	13.54	13.98	16.42
	rise	14.48	16.62	15.98	18.89
B1 to ZN	fall	10.49	12.97	12.33	14.82
	rise	15.37	17.94	18.34	21.12
B2 to ZN	fall	10.98	13.55	12.71	15.21
	rise	16.17	18.47	17.56	20.49
C1 to ZN	fall	10.49	12.98	11.83	14.39
	rise	16.73	19.39	19.67	22.72
C2 to ZN	fall	10.98	13.54	12.20	14.74
	rise	17.50	20.24	18.94	22.08

OAI22_X1

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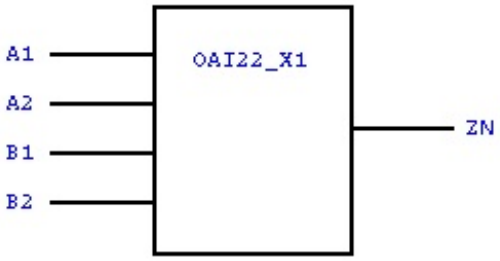
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!((A1 | A2) & (B1 | B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	L	-	-	H
H	-	H	-	L
H	-	-	H	L
-	H	H	-	L
-	H	-	H	L
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**23.2315**	**0.3656**	**23.2315**
A1 to ZN	**fall**	0.01	0.06	0.02	0.12
	rise	0.02	0.13	0.07	0.23
A2 to ZN	**fall**	0.01	0.06	0.03	0.12
	rise	0.02	0.14	0.07	0.22
B1 to ZN	**fall**	0.01	0.06	0.02	0.11
	rise	0.03	0.14	0.09	0.24
B2 to ZN	**fall**	0.01	0.06	0.03	0.12
	rise	0.03	0.15	0.09	0.23

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**23.2315**	**0.3656**	**23.2315**
A1 to ZN	**fall**	0.01	0.05	0.04	0.08
	rise	0.02	0.13	0.05	0.14
A2 to ZN	**fall**	0.01	0.05	0.04	0.08
	rise	0.02	0.13	0.05	0.13
B1 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.13	0.05	0.14
B2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.13	0.04	0.14

Capacitance [fF]	
A1	1.6710
A2	1.5842
B1	1.6654
B2	1.6156

Leakage [nW]
34.03

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	23.2315	0.3656	23.2315
A1 to ZN	fall	0.43	0.50	2.69	1.55
	rise	2.73	2.76	5.19	4.22
A2 to ZN	fall	0.82	0.95	3.09	1.98
	rise	3.55	3.53	4.90	4.19
B1 to ZN	fall	0.43	0.50	1.81	1.07
	rise	4.21	4.30	6.46	5.70
B2 to ZN	fall	0.83	0.95	2.16	1.49
	rise	4.99	5.06	6.17	5.71

OAI22_X2

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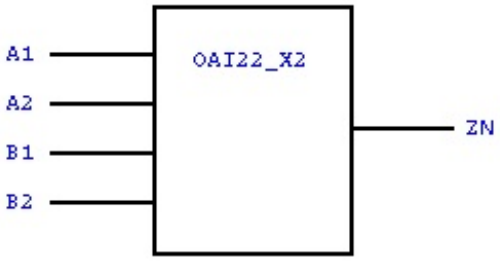
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!((A1 | A2) & (B1 | B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	L	-	-	H
H	-	H	-	L
H	-	-	H	L
-	H	H	-	L
-	H	-	H	L
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**46.3104**	**0.3656**	**46.3104**
A1 to ZN	**fall**	0.01	0.06	0.01	0.12
	rise	0.02	0.13	0.07	0.23
A2 to ZN	**fall**	0.01	0.06	0.03	0.12
	rise	0.02	0.14	0.07	0.22
B1 to ZN	**fall**	0.01	0.06	0.02	0.11
	rise	0.03	0.14	0.09	0.24
B2 to ZN	**fall**	0.01	0.06	0.03	0.12
	rise	0.03	0.15	0.09	0.23

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**46.3104**	**0.3656**	**46.3104**
A1 to ZN	**fall**	0.01	0.05	0.04	0.08
	rise	0.02	0.13	0.05	0.14
A2 to ZN	**fall**	0.01	0.05	0.04	0.08
	rise	0.02	0.13	0.04	0.13
B1 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.13	0.05	0.14
B2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.13	0.04	0.14

Capacitance [fF]	
A1	3.1330
A2	3.3971
B1	3.1607
B2	3.3331

Leakage [nW]
68.05

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	46.3104	0.3656	46.3104
A1 to ZN	fall	0.85	1.00	5.39	3.09
	rise	5.60	5.63	10.56	8.56
A2 to ZN	fall	1.64	1.90	6.20	3.96
	rise	7.24	7.16	9.96	8.61
B1 to ZN	fall	0.86	0.99	3.62	2.13
	rise	8.59	8.76	13.12	11.67
B2 to ZN	fall	1.65	1.90	4.33	2.97
	rise	10.17	10.39	12.49	11.73

OAI22_X4

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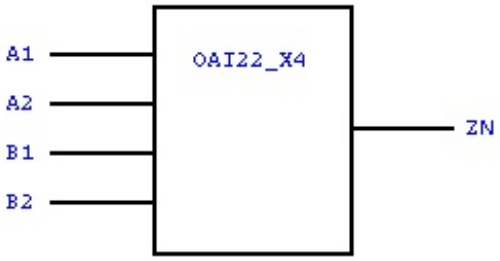
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Equation	ZN = "!((A1 | A2) & (B1 | B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	L	-	-	H
H	-	H	-	L
H	-	-	H	L
-	H	H	-	L
-	H	-	H	L
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**92.4683**	**0.3656**	**92.4683**
A1 to ZN	**fall**	0.01	0.06	0.01	0.12
	rise	0.02	0.13	0.07	0.23
A2 to ZN	**fall**	0.01	0.06	0.03	0.12
	rise	0.02	0.14	0.07	0.22
B1 to ZN	**fall**	0.01	0.06	0.02	0.11
	rise	0.03	0.15	0.09	0.24
B2 to ZN	**fall**	0.01	0.06	0.03	0.12
	rise	0.03	0.15	0.09	0.23

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**92.4683**	**0.3656**	**92.4683**
A1 to ZN	**fall**	0.01	0.05	0.04	0.08
	rise	0.02	0.13	0.05	0.14
A2 to ZN	**fall**	0.01	0.05	0.04	0.08
	rise	0.02	0.13	0.04	0.13
B1 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.13	0.05	0.14
B2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.13	0.04	0.14

Capacitance [fF]	
A1	6.4481
A2	6.5231
B1	6.5212
B2	6.4817

Leakage [nW]
136.10

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	92.4683	0.3656	92.4683
A1 to ZN	fall	1.69	1.99	10.75	6.15
	rise	11.33	11.57	21.24	17.14
A2 to ZN	fall	3.25	3.80	12.42	7.87
	rise	14.59	14.58	20.12	17.26
B1 to ZN	fall	1.71	1.98	7.27	4.23
	rise	17.30	17.70	26.32	23.50
B2 to ZN	fall	3.29	3.81	8.64	5.91
	rise	20.46	20.84	25.08	23.70

OAI33_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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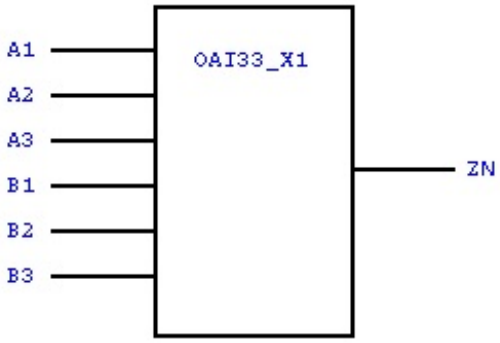
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.862 um^2
Equation	ZN = "!(((A1 | A2) | A3) & ((B1 | B2) | B3))"
Type	Combinational
Input	A1, A2, A3, B1, B2, B3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**A3**	**B1**	**B2**	**B3**	**ZN**
L	L	L	-	-	-	H
H	-	-	H	-	-	L
H	-	-	-	H	-	L
H	-	-	-	-	H	L
-	H	-	H	-	-	L
-	H	-	-	H	-	L
-	H	-	-	-	H	L
-	-	H	H	-	-	L
-	-	H	-	H	-	L
-	-	H	-	-	H	L
-	-	-	L	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**11.4822**	**0.3656**	**11.4822**
A1 to ZN	**fall**	0.01	0.04	0.02	0.08
	rise	0.03	0.12	0.09	0.20
A2 to ZN	**fall**	0.01	0.04	0.03	0.09
	rise	0.04	0.13	0.10	0.20
A3 to ZN	**fall**	0.02	0.04	0.04	0.09
	rise	0.04	0.13	0.09	0.19
B1 to ZN	**fall**	0.01	0.04	0.03	0.08
	rise	0.05	0.14	0.13	0.23
B2 to ZN	**fall**	0.02	0.04	0.04	0.09
	rise	0.06	0.15	0.13	0.23
B3 to ZN	**fall**	0.02	0.05	0.04	0.09
	rise	0.07	0.16	0.12	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**11.4822**	**0.3656**	**11.4822**
A1 to ZN	**fall**	0.01	0.03	0.04	0.06
	rise	0.03	0.12	0.07	0.14
A2 to ZN	**fall**	0.01	0.03	0.04	0.07
	rise	0.03	0.12	0.06	0.13
A3 to ZN	**fall**	0.01	0.03	0.04	0.07
	rise	0.03	0.12	0.06	0.13
B1 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.05	0.13	0.07	0.14
B2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.05	0.13	0.07	0.14
B3 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.05	0.13	0.06	0.14

Capacitance [fF]	
A1	1.6799
A2	1.6175
A3	1.5734
B1	1.6513
B2	1.6120
B3	1.5815

Leakage [nW]
48.31

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	11.4822	0.3656	11.4822
A1 to ZN	fall	0.55	0.60	2.37	1.94
	rise	3.25	3.31	5.27	4.75
A2 to ZN	fall	0.97	1.08	2.76	2.33
	rise	4.07	4.09	5.05	4.68
A3 to ZN	fall	1.19	1.45	3.14	2.67
	rise	4.80	4.85	5.44	5.15
B1 to ZN	fall	0.54	0.60	1.59	1.33
	rise	5.41	5.45	7.23	6.95
B2 to ZN	fall	0.98	1.09	1.95	1.73
	rise	6.19	6.24	7.04	6.90
B3 to ZN	fall	1.23	1.45	2.29	2.06
	rise	6.92	6.98	7.42	7.38

NANGATE OR2_X1

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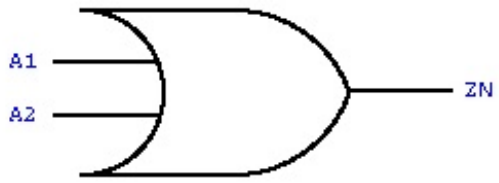
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	L
H	-	H
-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.04	0.12	0.11	0.20
	rise	0.02	0.16	0.02	0.18
A2 to ZN	**fall**	0.04	0.13	0.10	0.20
	rise	0.02	0.16	0.03	0.18

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14
A2 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14

Capacitance [fF]	
A1	0.9468
A2	0.9419

Leakage [nW]
22.69

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	2.71	3.42	4.50	4.74
	rise	1.77	1.80	3.32	3.19
A2 to ZN	**fall**	3.10	3.81	4.04	4.59
	rise	1.98	2.28	3.55	3.48

OR2_X2

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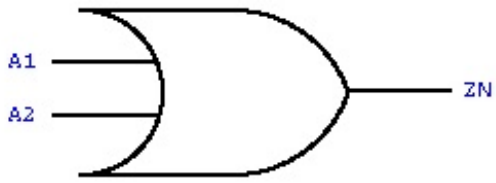
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	L
H	-	H
-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	0.04	0.12	0.10	0.20
	rise	0.01	0.16	0.02	0.17
A2 to ZN	**fall**	0.04	0.13	0.10	0.19
	rise	0.02	0.16	0.03	0.18

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14
A2 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14

Capacitance [fF]	
A1	1.7459
A2	1.6943

Leakage [nW]
45.66

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	5.24	6.53	8.89	9.26
	rise	3.49	3.57	6.64	6.63
A2 to ZN	**fall**	6.04	7.33	8.00	8.97
	rise	3.90	4.52	7.09	6.94

NANGATE OR2_X4

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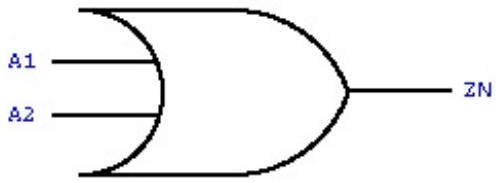
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	L
H	-	H
-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.03	0.12	0.10	0.19
	rise	0.01	0.16	0.02	0.17
A2 to ZN	**fall**	0.04	0.12	0.10	0.19
	rise	0.02	0.16	0.03	0.18

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14
A2 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14

Capacitance [fF]	
A1	3.3850
A2	3.4547

Leakage [nW]
91.31

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	10.23	12.77	17.69	18.29
	rise	6.47	7.18	12.91	13.15
A2 to ZN	**fall**	11.78	14.38	15.85	17.70
	rise	7.23	8.64	13.82	12.94

NANGATE OR3_X1

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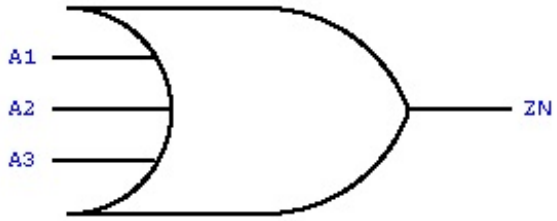
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	L
H	-	-	H
-	H	-	H
-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.06	0.16	0.14	0.24
	rise	0.02	0.16	0.03	0.18
A2 to ZN	**fall**	0.07	0.17	0.14	0.24
	rise	0.02	0.16	0.03	0.18
A3 to ZN	**fall**	0.07	0.17	0.12	0.23
	rise	0.02	0.17	0.04	0.19

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.01	0.07	0.02	0.07
	rise	0.00	0.14	0.01	0.14
A2 to ZN	**fall**	0.01	0.07	0.02	0.07
	rise	0.00	0.14	0.01	0.14
A3 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A1	0.9591
A2	0.9401
A3	0.9216

Leakage [nW]
24.41

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	3.16	3.79	4.72	4.84
	rise	1.77	1.79	3.12	3.13
A2 to ZN	**fall**	3.55	4.18	4.35	4.72
	rise	2.00	2.21	3.31	3.16
A3 to ZN	**fall**	3.90	4.55	4.41	4.91
	rise	2.12	2.38	3.54	3.40

NANGATE OR3_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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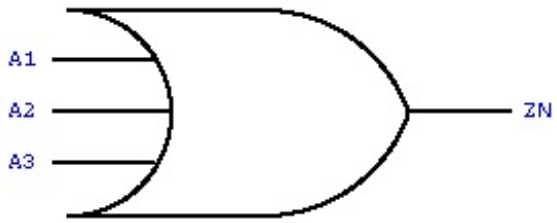
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.596 um^2
Equation	ZN = "((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	L
H	-	-	H
-	H	-	H
-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
A1 to ZN	**fall**	0.05	0.15	0.13	0.23
	rise	0.02	0.16	0.02	0.17
A2 to ZN	**fall**	0.06	0.16	0.13	0.23
	rise	0.02	0.16	0.03	0.18
A3 to ZN	**fall**	0.07	0.16	0.12	0.22
	rise	0.02	0.17	0.03	0.18

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
A1 to ZN	**fall**	0.01	0.07	0.02	0.07
	rise	0.00	0.14	0.01	0.14
A2 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14
A3 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14

Capacitance [fF]	
A1	1.7480
A2	1.6930
A3	1.6730

Leakage [nW]
49.16

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
A1 to ZN	**fall**	5.97	7.17	9.22	9.32
	rise	3.45	3.51	6.22	5.96
A2 to ZN	**fall**	6.81	7.97	8.52	9.08
	rise	3.88	4.40	6.56	6.01
A3 to ZN	**fall**	7.51	8.71	8.63	9.50
	rise	4.14	4.73	7.08	6.43

NANGATE OR3_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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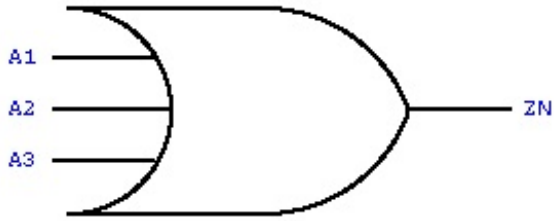
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	ZN = "((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	L
H	-	-	H
-	H	-	H
-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.05	0.15	0.13	0.23
	rise	0.01	0.16	0.02	0.17
A2 to ZN	**fall**	0.06	0.16	0.13	0.23
	rise	0.02	0.16	0.03	0.18
A3 to ZN	**fall**	0.06	0.16	0.12	0.22
	rise	0.02	0.16	0.03	0.18

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.01	0.07	0.02	0.07
	rise	0.00	0.14	0.01	0.14
A2 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14
A3 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.14	0.01	0.14

Capacitance [fF]	
A1	3.3749
A2	3.3772
A3	3.3947

Leakage [nW]
98.33

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	11.54	13.73	18.15	18.11
	rise	6.50	6.85	12.08	12.06
A2 to ZN	**fall**	13.16	15.33	16.64	17.62
	rise	7.40	8.48	12.94	12.41
A3 to ZN	**fall**	14.62	16.79	16.77	18.43
	rise	7.85	9.15	13.76	12.90

OR4_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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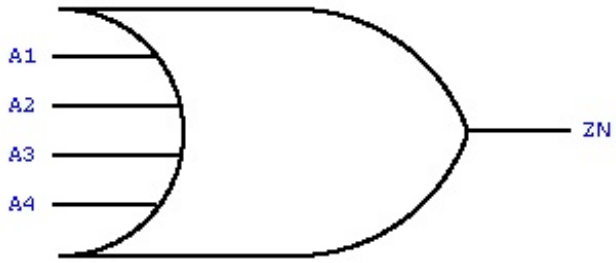
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	L
H	-	-	-	H
-	H	-	-	H
-	-	H	-	H
-	-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.08	0.19	0.16	0.27
	rise	0.02	0.16	0.03	0.18
A2 to ZN	**fall**	0.09	0.20	0.17	0.28
	rise	0.02	0.17	0.03	0.18
A3 to ZN	**fall**	0.10	0.21	0.16	0.27
	rise	0.02	0.17	0.04	0.19
A4 to ZN	**fall**	0.11	0.22	0.15	0.26
	rise	0.02	0.17	0.04	0.19

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.02	0.08	0.02	0.08
	rise	0.00	0.14	0.01	0.14
A2 to ZN	**fall**	0.02	0.08	0.02	0.08
	rise	0.00	0.14	0.01	0.14
A3 to ZN	**fall**	0.02	0.08	0.02	0.08
	rise	0.01	0.14	0.01	0.14
A4 to ZN	**fall**	0.02	0.08	0.02	0.08
	rise	0.01	0.14	0.01	0.14

Capacitance [fF]	
A1	0.9412
A2	0.9383
A3	0.9238
A4	0.9142

Leakage [nW]
26.73

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	fall	3.56	3.99	4.90	4.89
	rise	1.80	2.01	3.01	2.92
A2 to ZN	fall	3.97	4.39	4.68	4.80
	rise	2.00	2.17	3.18	2.97
A3 to ZN	fall	4.31	4.75	4.71	4.96
	rise	2.14	2.48	3.38	3.21
A4 to ZN	fall	4.67	5.11	4.87	5.22
	rise	2.15	2.51	3.53	3.52

NANGATE OR4_X2

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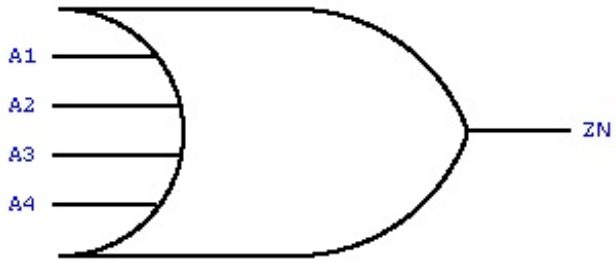
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	L
H	-	-	-	H
-	H	-	-	H
-	-	H	-	H
-	-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**120.85**	**0.3656**	**120.85**
A1 to ZN	**fall**	0.07	0.18	0.15	0.26
	rise	0.02	0.16	0.02	0.17
A2 to ZN	**fall**	0.09	0.19	0.16	0.27
	rise	0.02	0.16	0.03	0.18
A3 to ZN	**fall**	0.10	0.20	0.15	0.26
	rise	0.02	0.17	0.03	0.18
A4 to ZN	**fall**	0.10	0.21	0.15	0.25
	rise	0.02	0.17	0.04	0.19

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**120.85**	**0.3656**	**120.85**
A1 to ZN	**fall**	0.01	0.08	0.02	0.08
	rise	0.00	0.14	0.01	0.14
A2 to ZN	**fall**	0.01	0.08	0.02	0.08
	rise	0.00	0.14	0.01	0.14
A3 to ZN	**fall**	0.01	0.08	0.02	0.08
	rise	0.00	0.14	0.01	0.14
A4 to ZN	**fall**	0.01	0.08	0.02	0.08
	rise	0.00	0.14	0.01	0.14

Capacitance [fF]	
A1	1.7231
A2	1.6751
A3	1.6484
A4	1.6369

Leakage [nW]
53.87

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	fall	6.74	7.53	9.60	9.37
	rise	3.51	3.46	5.94	5.70
A2 to ZN	fall	7.55	8.33	9.03	9.19
	rise	3.93	4.44	6.38	5.89
A3 to ZN	fall	8.26	9.06	9.17	9.52
	rise	4.21	4.83	6.78	6.19
A4 to ZN	fall	8.98	9.78	9.46	10.05
	rise	4.24	5.22	7.08	6.82

OR4_X4

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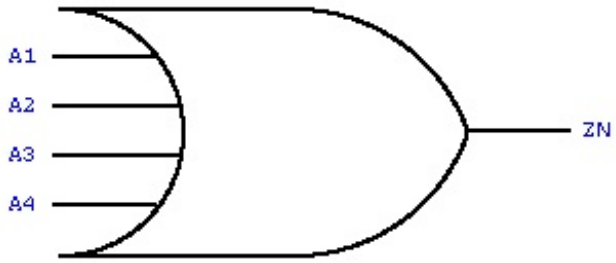
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	L
H	-	-	-	H
-	H	-	-	H
-	-	H	-	H
-	-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**241.699**	**0.3656**	**241.699**
A1 to ZN	**fall**	0.07	0.17	0.15	0.26
	rise	0.02	0.16	0.02	0.17
A2 to ZN	**fall**	0.09	0.19	0.16	0.26
	rise	0.02	0.16	0.03	0.18
A3 to ZN	**fall**	0.10	0.20	0.15	0.26
	rise	0.02	0.17	0.03	0.18
A4 to ZN	**fall**	0.10	0.20	0.14	0.25
	rise	0.02	0.17	0.04	0.19

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1985**	
Load Capacitance [fF]		**0.3656**	**241.699**	**0.3656**	**241.699**
A1 to ZN	**fall**	0.01	0.08	0.02	0.08
	rise	0.00	0.14	0.01	0.14
A2 to ZN	**fall**	0.01	0.08	0.02	0.08
	rise	0.00	0.14	0.01	0.14
A3 to ZN	**fall**	0.01	0.08	0.02	0.08
	rise	0.00	0.14	0.01	0.14
A4 to ZN	**fall**	0.01	0.08	0.02	0.08
	rise	0.00	0.14	0.01	0.14

Capacitance [fF]	
A1	3.3483
A2	3.3756
A3	3.5047
A4	3.6125

Leakage [nW]
107.74

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	fall	13.16	14.76	18.99	18.50
	rise	6.67	6.53	11.73	11.40
A2 to ZN	fall	14.79	16.35	17.87	18.12
	rise	7.53	8.52	12.48	11.38
A3 to ZN	fall	16.30	17.83	18.06	18.77
	rise	8.04	9.84	13.30	12.27
A4 to ZN	fall	17.68	19.26	18.78	19.81
	rise	8.09	9.91	13.89	13.46

SDFFRS_X1

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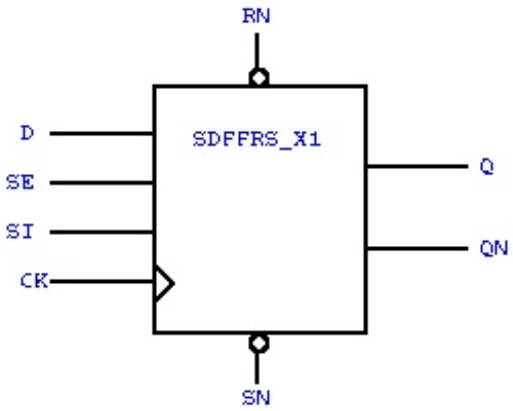
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, scan, and drive strength X1
Strength	1
Cell Area	7.714 um^2
Equation	Q = "(!(!RN)*((((SE * SI) + (D * !SE)))+(!SN)))" QN = "(!(!SN)*(!((((SE * SI) + (D * !SE)))+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table									
CK	D	RN	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	H	-	-	L	H
R	H	H	L	-	H	-	-	H	L
R	-	H	H	L	H	-	-	L	H
R	-	H	H	H	H	-	-	H	L
F	-	H	-	-	H	L	H	L	H
F	-	H	-	-	H	H	L	H	L
-	-	L	-	-	L	-	-	L	L
-	-	L	-	-	H	-	-	L	H
-	-	H	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.10	*0.17*	0.12	*0.19*
	rise	0.11	*0.25*	0.13	*0.27*
CK to QN	fall	0.08	0.16	0.10	0.18
	rise	0.08	0.23	0.09	0.25
RN to Q	fall	0.02	*0.10*	0.08	*0.17*
	rise	0.03	*0.17*	0.05	*0.20*
RN to QN	rise	0.16	0.31	0.24	0.39
SN to Q	rise	0.06	*0.20*	0.14	*0.28*
SN to QN	fall	0.03	0.11	0.10	0.19
	rise	0.14	0.29	0.18	0.33

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*
CK to QN	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
RN to Q	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*
RN to QN	rise	0.01	0.14	0.01	0.14
SN to Q	rise	0.01	*0.14*	0.01	*0.14*
SN to QN	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Constraints Time [ns]		
Setup CK to D	fall	0.14
	rise	0.08
Hold CK to D	fall	0.14
	rise	0.07
Recovery CK to RN	rise	-0.03
Removal CK to RN	rise	0.45
Setup CK to SE	fall	0.13
	rise	0.10
Hold CK to SE	fall	0.12
	rise	0.09
Setup CK to SI	fall	0.15
	rise	0.09
Hold CK to SI	fall	0.13
	rise	0.06
Recovery CK to SN	rise	0.03
Removal CK to SN	rise	0.23

Capacitance [fF]	
CK	0.9546
D	1.1437
RN	2.2485
SE	2.1438
SI	0.8603
SN	1.5234

Leakage [nW]
122.72

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	5660.07	7636.74
	rise	6880.49	8821.76
D	fall	3050.03	3630.60
	rise	1730.19	2238.88
RN	fall	1595.07	2010.47
	rise	-707.66	-302.62
SE	fall	4043.77	6803.95
	rise	1833.91	4435.95
SI	fall	2725.97	3064.62
	rise	1382.61	1671.89
SN	fall	695.32	652.53
	rise	-582.00	-597.24

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	5.72	*6.02*	6.70	*7.09*
	rise	5.78	*5.87*	6.75	*6.91*
CK to QN	fall	5.79	5.94	6.75	6.85
	rise	5.72	6.14	6.70	7.05
RN to Q	fall	5.24	*5.98*	7.62	*8.25*
	rise	2.59	*3.19*	4.98	*5.64*
RN to QN	rise	5.57	5.81	7.50	7.42
SN to Q	rise	3.99	*5.69*	5.19	*6.61*
SN to QN	fall	3.85	5.89	5.36	7.10
	rise	4.17	4.74	4.83	5.26

SDFFRS_X2

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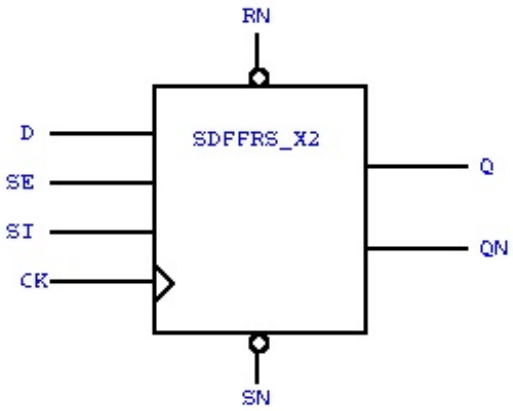
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, scan, and drive strength X2
Strength	2
Cell Area	8.246 um^2
Equation	Q = "(!(!RN)*((((SE * SI) + (D * !SE)))+(!SN)))" QN = "(!(!SN)*(!((((SE * SI) + (D * !SE)))+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table									
CK	D	RN	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	H	-	-	L	H
R	H	H	L	-	H	-	-	H	L
R	-	H	H	L	H	-	-	L	H
R	-	H	H	H	H	-	-	H	L
F	-	H	-	-	H	L	H	L	H
F	-	H	-	-	H	H	L	H	L
-	-	L	-	-	L	-	-	L	L
-	-	L	-	-	H	-	-	L	H
-	-	H	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.11	0.19	0.13	0.21
	rise	0.11	0.26	0.14	0.28
CK to QN	fall	0.08	0.17	0.11	0.19
	rise	0.08	0.23	0.10	0.25
RN to Q	fall	0.02	0.10	0.08	0.17
	rise	0.02	0.17	0.05	0.21
RN to QN	rise	0.18	0.33	0.26	0.41
SN to Q	rise	0.06	0.20	0.13	0.27
SN to QN	fall	0.03	0.11	0.09	0.18
	rise	0.15	0.30	0.18	0.34

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
CK to QN	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
RN to Q	fall	0.00	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
RN to QN	rise	0.01	0.14	0.01	0.14
SN to Q	rise	0.01	0.14	0.01	0.14
SN to QN	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Constraints Time [ns]		
Setup CK to D	fall	0.14
	rise	0.08
Hold CK to D	fall	0.14
	rise	0.07
Recovery CK to RN	rise	-0.00
Removal CK to RN	rise	0.45
Setup CK to SE	fall	0.13
	rise	0.10
Hold CK to SE	fall	0.12
	rise	0.08
Setup CK to SI	fall	0.15
	rise	0.09
Hold CK to SI	fall	0.13
	rise	0.06
Recovery CK to SN	rise	0.07
Removal CK to SN	rise	0.23

Capacitance [fF]	
CK	0.9382
D	1.1240
RN	2.6328
SE	2.0164
SI	0.8634
SN	1.8350

Leakage [nW]
162.30

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	5790.60	7760.09
	rise	7557.10	9491.10
D	fall	3106.32	3680.83
	rise	1686.66	2191.00
RN	fall	1893.91	2355.87
	rise	-828.64	-417.47
SE	fall	4019.33	6787.10
	rise	1833.06	4434.47
SI	fall	2771.83	3107.53
	rise	1339.33	1625.33
SN	fall	995.14	1001.04
	rise	-671.19	-685.98

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	8.10	8.79	9.08	9.75
	rise	8.37	8.67	9.34	9.60
CK to QN	fall	8.37	8.67	9.34	9.60
	rise	8.10	8.79	9.08	9.75
RN to Q	fall	7.23	8.25	10.86	11.13
	rise	3.88	5.20	8.69	8.67
RN to QN	rise	7.94	8.19	10.86	10.14
SN to Q	rise	6.17	8.00	8.23	9.30
SN to QN	fall	5.98	8.36	8.60	10.09
	rise	6.03	6.66	7.77	7.56

SDFFR_X1

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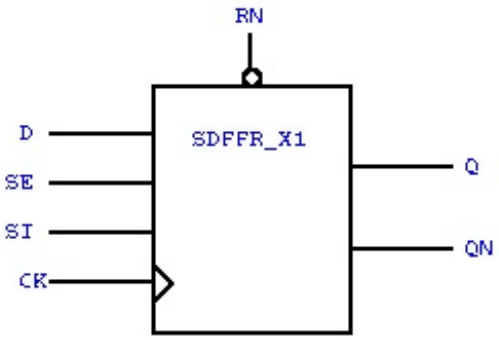
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, scan, and drive strength X1
Strength	1
Cell Area	6.650 um^2
Equation	Q = "(!(!RN)*(((SE * SI) + (D * !SE))))" QN = "((!RN)+!(((SE * SI) + (D * !SE))))"
Clock	CK
Reset	RN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	RN	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	-	-	L	H
R	H	H	L	-	-	-	H	L
R	-	H	H	L	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	H	-	-	L	H	L	H
F	-	H	-	-	H	L	H	L
-	-	L	-	-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
CK to Q	fall	0.06	0.16	0.08	0.18
	rise	0.07	0.22	0.09	0.24
CK to QN	fall	0.09	0.17	0.11	0.19
	rise	0.09	0.24	0.12	0.26
RN to Q	fall	0.15	0.25	0.24	0.33
RN to QN	rise	0.18	0.32	0.27	0.41

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
CK to Q	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
CK to QN	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
RN to Q	fall	0.01	0.07	0.01	0.07
RN to QN	rise	0.01	0.14	0.01	0.14

Constraints Time [ns]		
Setup CK to D	fall	0.14
	rise	0.08
Hold CK to D	fall	0.14
	rise	0.07
Recovery CK to RN	rise	-0.02
Removal CK to RN	rise	0.37
Setup CK to SE	fall	0.13
	rise	0.10
Hold CK to SE	fall	0.12
	rise	0.08
Setup CK to SI	fall	0.15
	rise	0.09
Hold CK to SI	fall	0.13
	rise	0.06

Capacitance [fF]	
CK	1.0263
D	1.1535
RN	1.4910
SE	2.0008
SI	0.8756

Leakage [nW]
105.26

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	5721.73	7694.68
	rise	5063.04	7006.09
D	fall	2701.33	3204.61
	rise	1242.67	1674.90
RN	fall	1076.69	1168.90
	rise	-492.74	-438.71
SE	fall	3853.87	6477.47
	rise	1561.41	4006.60
SI	fall	2595.00	2949.83
	rise	1075.91	1376.53

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
CK to Q	fall	5.13	5.17	6.12	6.12
	rise	4.74	5.16	5.70	6.18
CK to QN	fall	4.74	5.16	5.70	6.18
	rise	5.13	5.17	6.12	6.12
RN to Q	fall	5.19	5.18	5.57	5.49
RN to QN	rise	5.19	5.18	5.57	5.49

SDFFR_X2

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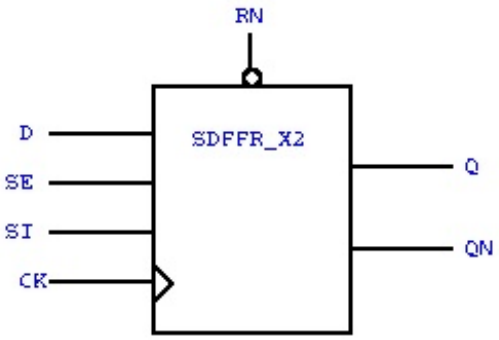
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, scan, and drive strength X2
Strength	2
Cell Area	6.916 um^2
Equation	Q = "(!(!RN)*(((SE * SI) + (D * !SE))))" QN = "((!RN)+!(((SE * SI) + (D * !SE))))"
Clock	CK
Reset	RN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	RN	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	-	-	L	H
R	H	H	L	-	-	-	H	L
R	-	H	H	L	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	H	-	-	L	H	L	H
F	-	H	-	-	H	L	H	L
-	-	L	-	-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.09	*0.20*	0.11	*0.22*
	rise	0.08	*0.23*	0.10	*0.25*
CK to QN	fall	0.10	0.18	0.12	0.20
	rise	0.12	0.26	0.14	0.28
RN to Q	fall	0.17	*0.28*	0.25	*0.36*
RN to QN	rise	0.20	0.34	0.29	0.42

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.02	*0.07*	0.02	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*
CK to QN	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
RN to Q	fall	0.02	*0.08*	0.02	*0.08*
RN to QN	rise	0.01	0.14	0.01	0.14

Constraints Time [ns]		
Setup CK to D	fall	0.14
	rise	0.08
Hold CK to D	fall	0.14
	rise	0.07
Recovery CK to RN	rise	-0.02
Removal CK to RN	rise	0.06
Setup CK to SE	fall	0.13
	rise	0.10
Hold CK to SE	fall	0.12
	rise	0.08
Setup CK to SI	fall	0.15
	rise	0.09
Hold CK to SI	fall	0.13
	rise	0.06

Capacitance [fF]	
CK	0.9629
D	1.1373
RN	1.5279
SE	1.9225
SI	0.8786

Leakage [nW]
141.96

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	5624.33	7598.60
	rise	4866.47	6815.34
D	fall	2665.43	3169.08
	rise	1215.56	1649.20
RN	fall	1039.06	1132.92
	rise	-491.21	-435.24
SE	fall	3830.23	6454.77
	rise	1530.98	3973.94
SI	fall	2558.88	2915.10
	rise	1048.92	1350.32

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	8.79	*7.86*	9.78	*8.88*
	rise	6.92	*7.59*	7.90	*8.39*
CK to QN	fall	6.92	7.58	7.90	8.56
	rise	8.79	7.98	9.77	8.94
RN to Q	fall	8.78	*7.80*	9.36	*8.24*
RN to QN	rise	8.79	7.81	9.36	8.22

SDFFS_X1

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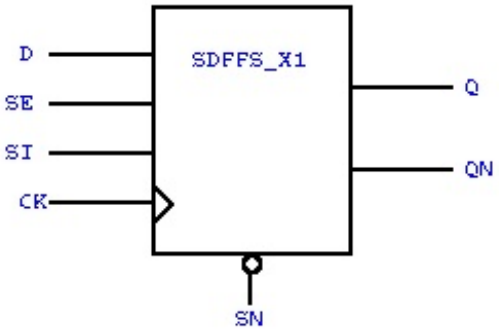
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, scan, and drive strength X1
Strength	1
Cell Area	6.650 um^2
Equation	Q = "((!SN)+(((SE * SI) + (D * !SE))))" QN = "(!((!SN)+(((SE * SI) + (D * !SE)))))"
Clock	CK
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	H	-	-	L	H
R	H	L	-	H	-	-	H	L
R	-	H	L	H	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	-	-	H	L	H	L	H
F	-	-	-	H	H	L	H	L
-	-	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.06	*0.16*	0.08	*0.18*
	rise	0.06	*0.21*	0.08	*0.23*
CK to QN	fall	0.08	0.16	0.11	0.18
	rise	0.10	0.24	0.12	0.26
SN to Q	rise	0.20	*0.35*	0.28	*0.43*
SN to QN	fall	0.03	0.11	0.09	0.18

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*
CK to QN	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
SN to Q	rise	0.01	*0.14*	0.01	*0.14*
SN to QN	fall	0.01	0.07	0.01	0.07

Constraints Time [ns]		
Setup CK to D	fall	0.14
	rise	0.08
Hold CK to D	fall	0.14
	rise	0.07
Setup CK to SE	fall	0.13
	rise	0.10
Hold CK to SE	fall	0.11
	rise	0.08
Setup CK to SI	fall	0.15
	rise	0.09
Hold CK to SI	fall	0.13
	rise	0.06
Recovery CK to SN	rise	-0.01
Removal CK to SN	rise	0.44

Capacitance [fF]	
CK	0.9791
D	1.1499
SE	2.0027
SI	0.8992
SN	1.3384

Leakage [nW]
107.72

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	5643.70	7620.19
	rise	6418.16	8349.60
D	fall	2941.97	3445.21
	rise	1395.11	1825.22
SE	fall	4230.27	6786.92
	rise	1861.63	4241.37
SI	fall	2836.92	3192.27
	rise	1226.33	1525.35
SN	fall	697.09	652.90
	rise	-583.98	-597.92

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	5.09	*5.04*	6.09	*6.11*
	rise	4.60	*5.00*	5.57	*6.00*
CK to QN	fall	4.60	5.05	5.57	5.97
	rise	5.09	5.12	6.09	6.02
SN to Q	rise	4.73	*4.90*	5.98	*5.80*
SN to QN	fall	4.72	4.97	5.98	5.71

SDFFS_X2

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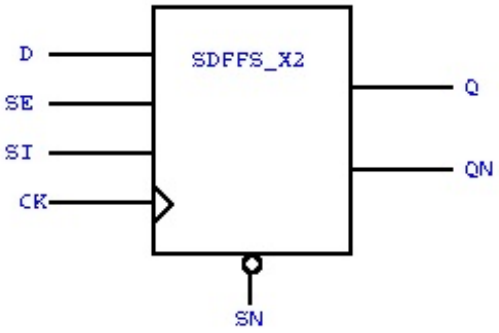
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, scan, and drive strength X2
Strength	2
Cell Area	7.182 um^2
Equation	Q = "((!SN)+(((SE * SI) + (D * !SE))))" QN = "(!((!SN)+(((SE * SI) + (D * !SE)))))"
Clock	CK
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	H	-	-	L	H
R	H	L	-	H	-	-	H	L
R	-	H	L	H	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	-	-	H	L	H	L	H
F	-	-	-	H	H	L	H	L
-	-	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.08	*0.19*	0.10	*0.21*
	rise	0.07	*0.23*	0.09	*0.25*
CK to QN	fall	0.10	0.18	0.12	0.20
	rise	0.13	0.27	0.15	0.28
SN to Q	rise	0.23	*0.38*	0.31	*0.46*
SN to QN	fall	0.02	0.10	0.08	0.17

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.02	*0.07*	0.02	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*
CK to QN	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
SN to Q	rise	0.01	*0.14*	0.01	*0.14*
SN to QN	fall	0.00	0.07	0.01	0.07

Constraints Time [ns]		
Setup CK to D	fall	0.15
	rise	0.09
Hold CK to D	fall	0.13
	rise	0.07
Setup CK to SE	fall	0.14
	rise	0.11
Hold CK to SE	fall	0.11
	rise	0.07
Setup CK to SI	fall	0.16
	rise	0.10
Hold CK to SI	fall	0.12
	rise	0.06
Recovery CK to SN	rise	-0.03
Removal CK to SN	rise	0.44

Capacitance [fF]	
CK	0.9616
D	1.1194
SE	1.9122
SI	0.9129
SN	2.2245

Leakage [nW]
140.59

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	5753.61	7742.01
	rise	7723.91	9631.35
D	fall	3111.44	3614.77
	rise	1443.41	1872.28
SE	fall	4299.81	6865.27
	rise	1970.12	4356.19
SI	fall	3006.95	3361.45
	rise	1274.45	1572.66
SN	fall	1263.67	1269.63
	rise	-843.06	-856.30

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	8.52	*7.74*	9.53	*8.51*
	rise	7.31	*8.01*	8.29	*8.85*
CK to QN	fall	7.31	8.00	8.29	9.02
	rise	8.52	7.74	9.53	8.45
SN to Q	rise	7.69	*7.68*	10.29	*9.20*
SN to QN	fall	7.69	7.63	10.29	9.31

SDFF_X1

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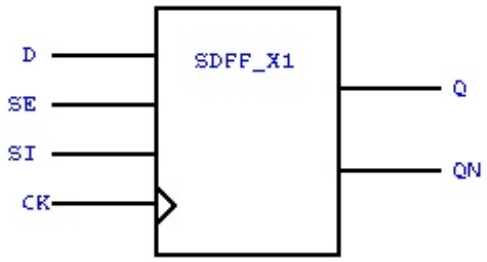
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with scan, and drive strength X1
Strength	1
Cell Area	6.118 um^2
Equation	Q = "(((SE * SI) + (D * !SE)))" QN = "(!(((SE * SI) + (D * !SE))))"
Clock	CK
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	-	-	L	H
R	H	L	-	-	-	H	L
R	-	H	L	-	-	L	H
R	-	H	H	-	-	H	L
F	-	-	-	L	H	L	H
F	-	-	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.06	0.15	0.08	0.17
	rise	0.06	0.21	0.07	0.22
CK to QN	fall	0.08	*0.15*	0.10	*0.17*
	rise	0.09	*0.23*	0.10	*0.25*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14
CK to QN	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*

Constraints Time [ns]		
Setup CK to D	fall	0.14
	rise	0.09
Hold CK to D	fall	0.13
	rise	0.07
Setup CK to SE	fall	0.14
	rise	0.10
Hold CK to SE	fall	0.11
	rise	0.08
Setup CK to SI	fall	0.16
	rise	0.10
Hold CK to SI	fall	0.12
	rise	0.06

Capacitance [fF]	
CK	0.9589
D	1.1197
SE	1.8899
SI	0.9188

Leakage [nW]
100.68

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	5298.10	7334.49
	rise	2685.58	4665.91
D	fall	2913.11	3416.59
	rise	1419.36	1850.53
SE	fall	3996.28	6635.22
	rise	1759.10	4224.11
SI	fall	2807.29	3164.14
	rise	1250.85	1551.48

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	4.59	4.62	5.64	5.62
	rise	4.23	4.65	5.24	5.66
CK to QN	fall	4.23	*4.54*	5.24	*5.63*
	rise	4.59	*4.63*	5.63	*5.64*

SDFF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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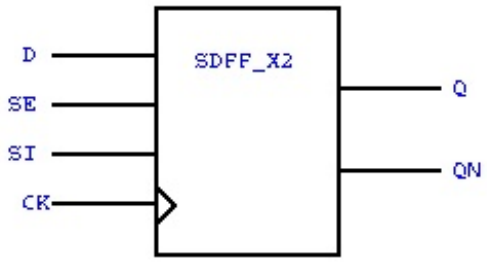
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with scan, and drive strength X2
Strength	2
Cell Area	6.384 um^2
Equation	Q = "(((SE * SI) + (D * !SE)))" QN = "(!(((SE * SI) + (D * !SE))))"
Clock	CK
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	-	-	L	H
R	H	L	-	-	-	H	L
R	-	H	L	-	-	L	H
R	-	H	H	-	-	H	L
F	-	-	-	L	H	L	H
F	-	-	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.08	*0.19*	0.10	*0.21*
	rise	0.07	*0.22*	0.09	*0.24*
CK to QN	fall	0.09	0.16	0.11	0.18
	rise	0.11	0.25	0.13	0.27

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.02	*0.07*	0.02	*0.07*
	rise	0.01	*0.14*	0.01	*0.14*
CK to QN	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.14	0.01	0.14

Constraints Time [ns]		
Setup CK to D	fall	0.14
	rise	0.09
Hold CK to D	fall	0.13
	rise	0.07
Setup CK to SE	fall	0.14
	rise	0.10
Hold CK to SE	fall	0.11
	rise	0.08
Setup CK to SI	fall	0.15
	rise	0.10
Hold CK to SI	fall	0.12
	rise	0.06

Capacitance [fF]	
CK	0.9839
D	1.1266
SE	1.8532
SI	0.8982

Leakage [nW]
136.68

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
CK	fall	5364.33	7384.13
	rise	2761.35	4727.14
D	fall	2919.02	3421.86
	rise	1422.61	1851.48
SE	fall	3986.46	6626.70
	rise	1763.78	4229.85
SI	fall	2814.44	3168.70
	rise	1253.68	1552.88

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	8.23	*7.42*	9.24	*8.40*
	rise	6.34	*6.88*	7.35	*7.95*
CK to QN	fall	6.34	6.95	7.35	7.83
	rise	8.23	7.37	9.24	8.40

TBUF_X16

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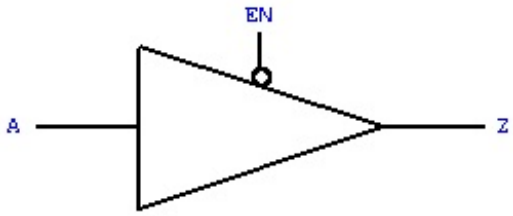
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	6.916 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		13.3976	833.3787	13.3976	833.3787
A to Z	**fall**	*0.04*	0.13	*0.10*	0.21
	rise	0.03	*0.18*	0.05	*0.22*
EN to Z	**fall**	*0.04*	0.13	*0.09*	0.20
	rise	0.04	*0.20*	0.10	*0.26*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		13.3976	833.3787	13.3976	833.3787
A to Z	**fall**	*0.01*	0.07	*0.02*	0.07
	rise	0.01	*0.14*	0.02	*0.14*
EN to Z	**fall**	*0.01*	0.07	*0.01*	0.07
	rise	0.01	*0.14*	0.01	*0.14*

Capacitance [fF]	
A	6.5205
EN	4.9283
Z	13.0667

Leakage [nW]
197.27

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		13.3976	833.3787	13.3976	833.3787
A to Z	**fall**	*28.10*	37.15	*45.95*	47.94
	rise	19.39	*23.18*	28.45	*33.35*
EN to Z	**fall**	*23.09*	30.63	*27.60*	37.62
	rise	26.76	*30.52*	29.09	*33.59*

TBUF_X1

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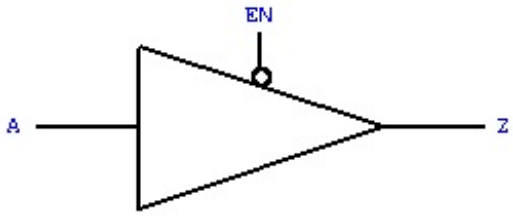
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.128 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		1.4185	52.6287	1.4185	52.6287
A to Z	**fall**	*0.03*	0.11	*0.08*	0.17
	rise	0.02	*0.16*	0.03	*0.19*
EN to Z	**fall**	*0.03*	0.11	*0.08*	0.17
	rise	0.03	*0.18*	0.08	*0.23*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		1.4185	52.6287	1.4185	52.6287
A to Z	**fall**	*0.00*	0.07	*0.01*	0.07
	rise	0.01	*0.14*	0.01	*0.14*
EN to Z	**fall**	*0.01*	0.07	*0.01*	0.07
	rise	0.01	*0.14*	0.01	*0.14*

Capacitance [fF]	
A	1.8794
EN	1.7266
Z	1.0540

Leakage [nW]
32.75

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		1.4185	52.6287	1.4185	52.6287
A to Z	**fall**	*4.29*	5.13	*5.89*	7.76
	rise	1.95	*2.21*	4.44	*5.16*
EN to Z	**fall**	*3.98*	4.68	*6.61*	8.23
	rise	4.20	*4.23*	5.64	*6.34*

TBUF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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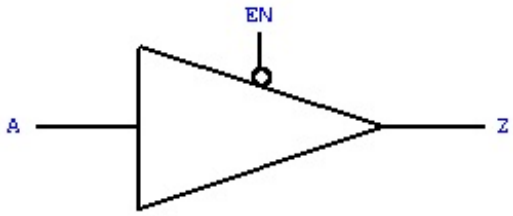
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		2.00	105.2434	2.00	105.2434
A to Z	**fall**	*0.03*	0.11	*0.07*	0.17
	rise	0.02	*0.16*	0.03	*0.19*
EN to Z	**fall**	*0.02*	0.10	*0.07*	0.17
	rise	0.03	*0.18*	0.09	*0.24*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		2.00	105.2434	2.00	105.2434
A to Z	**fall**	*0.00*	0.07	*0.01*	0.07
	rise	0.00	*0.14*	0.01	*0.14*
EN to Z	**fall**	*0.00*	0.07	*0.01*	0.07
	rise	0.00	*0.14*	0.01	*0.14*

Capacitance [fF]	
A	3.3251
EN	2.7379
Z	1.6364

Leakage [nW]
58.90

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		2.00	105.2434	2.00	105.2434
A to Z	**fall**	*7.56*	9.22	*10.95*	14.72
	rise	3.61	*4.26*	8.53	*9.93*
EN to Z	**fall**	*5.57*	7.02	*9.01*	12.02
	rise	6.43	*6.83*	7.53	*8.86*

TBUF_X4

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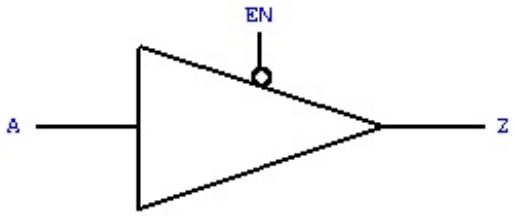
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		3.5816	210.1313	3.5816	210.1313
A to Z	**fall**	*0.03*	0.11	*0.08*	0.18
	rise	0.02	*0.17*	0.03	*0.19*
EN to Z	**fall**	*0.03*	0.11	*0.08*	0.18
	rise	0.04	*0.19*	0.10	*0.25*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		3.5816	210.1313	3.5816	210.1313
A to Z	**fall**	*0.01*	0.07	*0.01*	0.07
	rise	0.00	*0.14*	0.01	*0.14*
EN to Z	**fall**	*0.01*	0.07	*0.01*	0.07
	rise	0.01	*0.14*	0.01	*0.14*

Capacitance [fF]	
A	3.3801
EN	2.4363
Z	3.2223

Leakage [nW]
72.12

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		3.5816	210.1313	3.5816	210.1313
A to Z	**fall**	*8.52*	11.45	*13.83*	17.51
	rise	5.54	*6.94*	10.01	*12.89*
EN to Z	**fall**	*8.17*	10.30	*10.28*	14.02
	rise	9.27	*9.58*	10.26	*12.07*

TBUF_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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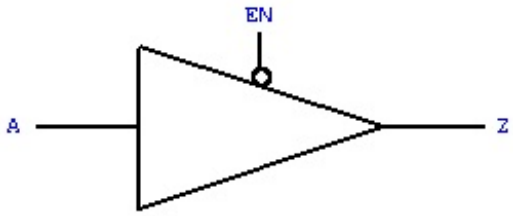
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	4.788 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		7.0967	419.3423	7.0967	419.3423
A to Z	**fall**	*0.03*	0.11	*0.08*	0.18
	rise	0.02	*0.17*	0.04	*0.20*
EN to Z	**fall**	*0.03*	0.11	*0.08*	0.18
	rise	0.04	*0.19*	0.10	*0.25*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		7.0967	419.3423	7.0967	419.3423
A to Z	**fall**	*0.01*	0.07	*0.01*	0.07
	rise	0.01	*0.14*	0.01	*0.14*
EN to Z	**fall**	*0.01*	0.07	*0.01*	0.07
	rise	0.01	*0.14*	0.01	*0.14*

Capacitance [fF]	
A	6.7197
EN	4.9857
Z	6.7443

Leakage [nW]
144.43

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		7.0967	419.3423	7.0967	419.3423
A to Z	**fall**	*17.47*	23.37	*28.23*	35.46
	rise	11.11	*13.54*	20.12	*25.48*
EN to Z	**fall**	*16.61*	20.96	*20.76*	28.42
	rise	19.03	*20.69*	21.04	*24.03*

TINV_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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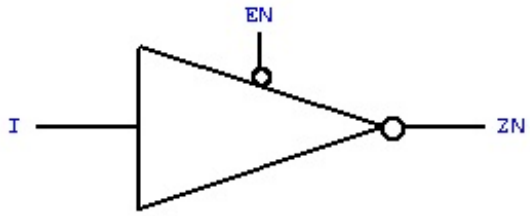
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!I", when !(EN)
Type	Tristate
Input	EN, I
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
I	**EN**	**ZN**
L	L	H
H	L	L
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		1.1641	23.4202	1.1641	23.4202
EN to ZN	**fall**	0.01	*0.07*	0.06	*0.12*
	rise	*0.01*	0.14	*-0.01*	0.23
I to ZN	**fall**	0.01	*0.07*	0.01	*0.12*
	rise	*0.02*	0.15	*0.07*	0.23

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		1.1641	23.4202	1.1641	23.4202
EN to ZN	**fall**	0.01	*0.05*	0.01	*0.05*
	rise	*0.01*	0.13	*0.05*	0.15
I to ZN	**fall**	0.00	*0.05*	0.02	*0.07*
	rise	*0.01*	0.13	*0.03*	0.14

Capacitance [fF]	
EN	1.7520
I	1.4446
ZN	0.7990

Leakage [nW]
17.89

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		1.1641	23.4202	1.1641	23.4202
EN to ZN	**fall**	2.50	*2.77*	3.54	*4.77*
	rise	*3.04*	3.06	*2.89*	5.21
I to ZN	**fall**	0.45	*0.64*	1.81	*1.12*
	rise	*2.52*	2.56	*3.61*	3.08

TLAT_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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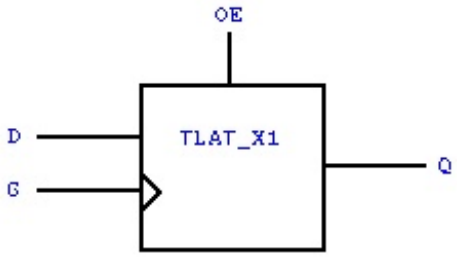
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	High enable Latch with drive strength X1
Strength	1
Cell Area	3.458 um^2
Equation	Q = "(D)", when !(!OE)
Enable	G
Type	Sequential
Input	D, OE
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
OE	**D**	**G**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	-	-	-	-	Z
H	L	H	-	-	L
H	H	H	-	-	H
H	-	L	L	H	L
H	-	L	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		1.1559	23.3749	1.1559	23.3749
D to Q	**fall**	*0.05*	0.11	*0.11*	0.17
	rise	0.04	*0.17*	0.06	*0.19*
G to Q	**fall**	*0.05*	0.11	*0.07*	0.13
	rise	0.06	*0.19*	0.07	*0.21*
OE to Q	**fall**	*0.00*	0.06	*-0.05*	0.11
	rise	0.02	*0.15*	0.02	*0.16*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		1.1559	23.3749	1.1559	23.3749
D to Q	**fall**	*0.01*	0.05	*0.01*	0.05
	rise	0.01	*0.13*	0.01	*0.13*
G to Q	**fall**	*0.01*	0.05	*0.01*	0.05
	rise	0.01	*0.13*	0.01	*0.13*
OE to Q	**fall**	*0.00*	0.05	*0.04*	0.09
	rise	0.01	*0.13*	0.02	*0.13*

Constraints Time [ns]		
Setup G to D	**fall**	0.10
	rise	0.03
Hold G to D	**fall**	0.23
	rise	0.13

Capacitance [fF]	
D	1.1398
G	1.0160
OE	1.4972
Q	0.7919

Leakage [nW]
48.00

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1985
D	fall	1478.92	1712.71
	rise	280.25	511.91
G	fall	2851.60	4985.85
	rise	2827.51	4953.59
OE	fall	1685.39	3769.90
	rise	883.38	2976.23

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		1.1559	23.3749	1.1559	23.3749
D to Q	fall	*4.19*	4.96	*4.84*	5.94
	rise	4.20	*4.45*	5.13	*5.35*
G to Q	fall	*5.09*	5.87	*7.13*	8.04
	rise	6.41	*6.68*	8.54	*8.86*
OE to Q	fall	*0.51*	0.86	*1.23*	2.96
	rise	1.47	*1.52*	2.89	*3.58*

NANGATE XNOR2_X1

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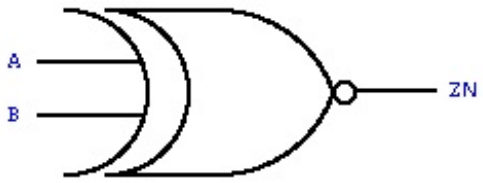
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "!(A ^ B)"
Type	Combinational
Input	A, B
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**ZN**
L	L	H
L	H	L
H	L	L
H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	26.0162	0.3656	26.0162
A to ZN	**fall**	0.03	0.08	0.09	0.15
	rise	0.03	0.15	0.07	0.24
B to ZN	**fall**	0.03	0.09	0.09	0.15
	rise	0.03	0.15	0.07	0.23

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	26.0162	0.3656	26.0162
A to ZN	**fall**	0.01	0.05	0.04	0.09
	rise	0.01	0.13	0.04	0.14
B to ZN	**fall**	0.01	0.05	0.04	0.09
	rise	0.01	0.13	0.03	0.14

Capacitance [fF]	
A	2.2328
B	2.5736

Leakage [nW]
36.44

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	26.0162	0.3656	26.0162
A to ZN	**fall**	2.39	2.33	3.95	3.65
	rise	3.39	3.21	5.31	4.71
B to ZN	**fall**	2.58	2.82	4.26	4.04
	rise	3.49	3.35	4.57	4.19

NANGATE XNOR2_X2

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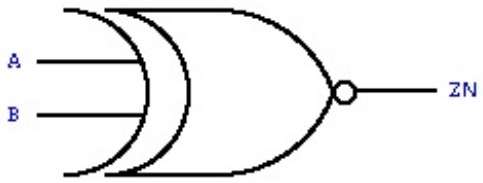
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.660 um^2
Equation	ZN = "!(A ^ B)"
Type	Combinational
Input	A, B
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**ZN**
L	L	H
L	H	L
H	L	L
H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	52.0325	0.3656	52.0325
A to ZN	**fall**	0.03	0.08	0.08	0.14
	rise	0.03	0.15	0.07	0.24
B to ZN	**fall**	0.03	0.08	0.09	0.15
	rise	0.03	0.15	0.07	0.23

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	52.0325	0.3656	52.0325
A to ZN	**fall**	0.01	0.05	0.04	0.09
	rise	0.01	0.13	0.04	0.14
B to ZN	**fall**	0.01	0.05	0.03	0.09
	rise	0.01	0.13	0.03	0.14

Capacitance [fF]	
A	4.0038
B	4.8369

Leakage [nW]
73.10

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	52.0325	0.3656	52.0325
A to ZN	**fall**	4.47	4.32	7.65	7.02
	rise	6.79	6.70	10.69	9.52
B to ZN	**fall**	4.90	5.32	8.30	7.80
	rise	7.09	6.98	9.26	8.48

XOR2_X1

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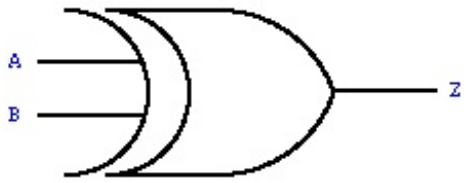
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	Z = "(A ^ B)"
Type	Combinational
Input	A, B
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**Z**
L	L	L
L	H	H
H	L	H
H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	25.3296	0.3656	25.3296
A to Z	**fall**	0.04	0.08	0.11	0.16
	rise	0.04	0.16	0.07	0.24
B to Z	**fall**	0.05	0.09	0.11	0.15
	rise	0.03	0.16	0.08	0.24

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	25.3296	0.3656	25.3296
A to Z	**fall**	0.01	0.05	0.04	0.08
	rise	0.02	0.13	0.04	0.14
B to Z	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.13	0.04	0.14

Capacitance [fF]	
A	2.2321
B	2.4115

Leakage [nW]
36.16

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	25.3296	0.3656	25.3296
A to Z	**fall**	2.43	2.52	4.07	3.67
	rise	3.40	3.29	5.06	4.59
B to Z	**fall**	2.53	2.63	3.37	3.21
	rise	3.64	3.66	5.41	4.90

XOR2_X2

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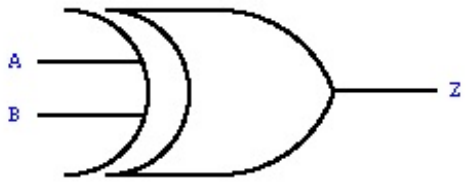
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_typical_typical**: Vdd= **1.10**V, Tj= **25.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	Z = "(A ^ B)"
Type	Combinational
Input	A, B
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**Z**
L	L	L
L	H	H
H	L	H
H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	50.5066	0.3656	50.5066
A to Z	**fall**	0.05	0.08	0.11	0.16
	rise	0.04	0.16	0.07	0.24
B to Z	**fall**	0.05	0.09	0.11	0.15
	rise	0.03	0.16	0.08	0.24

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	50.5066	0.3656	50.5066
A to Z	**fall**	0.01	0.05	0.04	0.08
	rise	0.01	0.13	0.04	0.14
B to Z	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.13	0.04	0.14

Capacitance [fF]	
A	4.3305
B	4.5009

Leakage [nW]
72.59

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1985	
Load Capacitance [fF]		0.3656	50.5066	0.3656	50.5066
A to Z	**fall**	5.03	5.24	8.30	7.54
	rise	6.92	6.75	10.26	9.37
B to Z	**fall**	5.23	5.46	6.87	6.61
	rise	7.38	7.42	10.95	9.97

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Library

NangateOpenCellLibrary_slow

Corner

slow

AND2_X1

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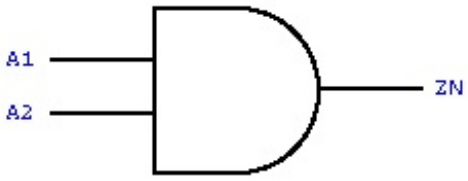
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	L
H	H	H
-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.08	0.24	0.28	0.48
	rise	0.06	0.58	0.17	0.70
A2 to ZN	**fall**	0.08	0.25	0.30	0.50
	rise	0.07	0.58	0.18	0.71

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.01	0.13	0.03	0.14
	rise	0.02	0.50	0.03	0.50
A2 to ZN	**fall**	0.01	0.13	0.03	0.14
	rise	0.02	0.50	0.03	0.50

Capacitance [fF]	
A1	0.8948
A2	0.9392

Leakage [nW]
15.15

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	1.97	2.55	2.27	2.91
	rise	1.39	1.58	1.77	1.85
A2 to ZN	**fall**	2.20	2.84	2.53	3.19
	rise	1.39	1.62	1.57	1.65

AND2_X2

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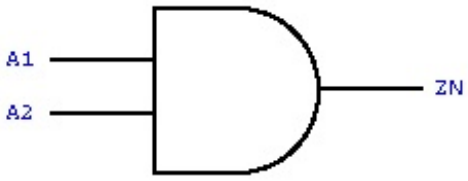
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	L
H	H	H
-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	**fall**	0.07	0.23	0.27	0.46
	rise	0.06	0.57	0.16	0.69
A2 to ZN	**fall**	0.08	0.24	0.29	0.48
	rise	0.06	0.58	0.17	0.69

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	**fall**	0.01	0.13	0.03	0.14
	rise	0.02	0.50	0.03	0.50
A2 to ZN	**fall**	0.01	0.13	0.03	0.14
	rise	0.02	0.50	0.03	0.50

Capacitance [fF]	
A1	1.6150
A2	1.6677

Leakage [nW]
30.47

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	**fall**	3.81	4.86	4.45	5.61
	rise	2.77	3.14	3.54	3.66
A2 to ZN	**fall**	4.28	5.44	4.97	6.17
	rise	2.76	3.20	3.12	3.33

AND2_X4

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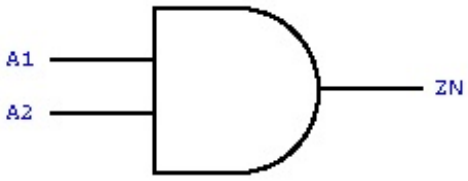
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	L
H	H	H
-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	**fall**	0.07	0.23	0.26	0.46
	rise	0.06	0.57	0.16	0.68
A2 to ZN	**fall**	0.08	0.24	0.28	0.48
	rise	0.06	0.57	0.17	0.69

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	**fall**	0.01	0.13	0.03	0.14
	rise	0.01	0.50	0.03	0.50
A2 to ZN	**fall**	0.01	0.13	0.03	0.14
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A1	3.1244
A2	3.4067

Leakage [nW]
60.93

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	**fall**	7.34	9.40	8.73	10.92
	rise	5.41	6.24	6.95	7.22
A2 to ZN	**fall**	8.27	10.55	9.69	12.04
	rise	5.39	6.10	6.13	6.51

AND3_X1

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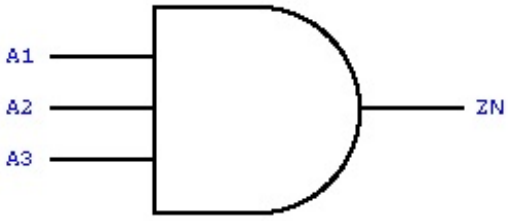
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	L
H	H	H	H
-	L	-	L
-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
A1 to ZN	**fall**	0.08	0.25	0.29	0.50
	rise	0.09	0.61	0.22	0.75
A2 to ZN	**fall**	0.09	0.27	0.31	0.52
	rise	0.10	0.62	0.24	0.77
A3 to ZN	**fall**	0.10	0.27	0.33	0.53
	rise	0.10	0.62	0.22	0.75

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
A1 to ZN	**fall**	0.02	0.13	0.03	0.14
	rise	0.02	0.50	0.04	0.50
A2 to ZN	**fall**	0.02	0.13	0.03	0.14
	rise	0.02	0.50	0.03	0.50
A3 to ZN	**fall**	0.02	0.13	0.03	0.14
	rise	0.02	0.50	0.03	0.50

Capacitance [fF]	
A1	0.8695
A2	0.9092
A3	0.9285

Leakage [nW]
14.32

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
A1 to ZN	**fall**	2.06	2.70	2.34	3.02
	rise	1.52	1.80	1.90	1.95
A2 to ZN	**fall**	2.30	3.01	2.60	3.31
	rise	1.52	1.78	1.70	1.82
A3 to ZN	**fall**	2.51	3.25	2.83	3.56
	rise	1.53	1.75	1.63	1.77

AND3_X2

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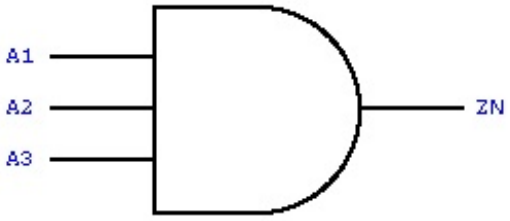
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.596 um^2
Equation	ZN = "((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	L
H	H	H	H
-	L	-	L
-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
A1 to ZN	**fall**	0.08	0.25	0.28	0.48
	rise	0.09	0.61	0.21	0.74
A2 to ZN	**fall**	0.09	0.26	0.30	0.50
	rise	0.09	0.61	0.23	0.75
A3 to ZN	**fall**	0.09	0.26	0.32	0.52
	rise	0.09	0.61	0.21	0.73

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
A1 to ZN	**fall**	0.01	0.13	0.03	0.14
	rise	0.02	0.50	0.04	0.50
A2 to ZN	**fall**	0.01	0.13	0.03	0.14
	rise	0.02	0.50	0.03	0.50
A3 to ZN	**fall**	0.02	0.13	0.03	0.14
	rise	0.02	0.50	0.03	0.50

Capacitance [fF]	
A1	1.5701
A2	1.6240
A3	1.6413

Leakage [nW]
28.79

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
A1 to ZN	**fall**	3.97	5.14	4.57	5.81
	rise	3.03	3.51	3.81	3.90
A2 to ZN	**fall**	4.46	5.74	5.08	6.37
	rise	3.05	3.48	3.41	3.60
A3 to ZN	**fall**	4.85	6.24	5.54	6.87
	rise	3.04	3.55	3.26	3.61

AND3_X4

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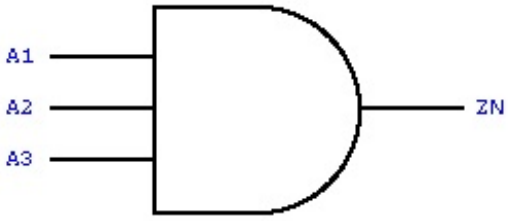
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	ZN = "((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	L
H	H	H	H
-	L	-	L
-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.07	0.24	0.28	0.48
	rise	0.08	0.60	0.20	0.73
A2 to ZN	**fall**	0.08	0.25	0.30	0.50
	rise	0.09	0.61	0.22	0.75
A3 to ZN	**fall**	0.09	0.26	0.31	0.51
	rise	0.09	0.61	0.20	0.73

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.01	0.13	0.03	0.14
	rise	0.02	0.50	0.04	0.50
A2 to ZN	**fall**	0.01	0.13	0.03	0.14
	rise	0.02	0.50	0.03	0.50
A3 to ZN	**fall**	0.01	0.13	0.03	0.14
	rise	0.02	0.50	0.03	0.50

Capacitance [fF]	
A1	3.0235
A2	3.2350
A3	3.4388

Leakage [nW]
57.56

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	7.69	9.97	8.93	11.34
	rise	5.82	6.93	7.38	7.67
A2 to ZN	**fall**	8.64	11.19	9.98	12.46
	rise	5.82	6.87	6.55	7.04
A3 to ZN	**fall**	9.50	12.18	10.86	13.46
	rise	5.80	6.74	6.27	6.99

AND4_X1

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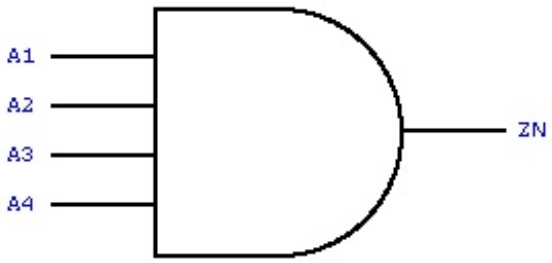
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	L
H	H	H	H	H
-	L	-	-	L
-	-	L	-	L
-	-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A1 to ZN	**fall**	0.09	0.26	0.31	0.51
	rise	0.12	0.65	0.27	0.81
A2 to ZN	**fall**	0.10	0.28	0.32	0.53
	rise	0.13	0.66	0.29	0.82
A3 to ZN	**fall**	0.11	0.28	0.34	0.55
	rise	0.14	0.67	0.27	0.81
A4 to ZN	**fall**	0.11	0.29	0.35	0.56
	rise	0.14	0.67	0.25	0.79

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A1 to ZN	**fall**	0.02	0.13	0.04	0.14
	rise	0.03	0.50	0.04	0.50
A2 to ZN	**fall**	0.02	0.13	0.03	0.14
	rise	0.03	0.50	0.04	0.50
A3 to ZN	**fall**	0.02	0.14	0.03	0.14
	rise	0.03	0.50	0.03	0.50
A4 to ZN	**fall**	0.02	0.14	0.03	0.14
	rise	0.03	0.50	0.03	0.50

Capacitance [fF]	
A1	0.8475
A2	0.9011
A3	0.9062
A4	0.9080

Leakage [nW]
13.63

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.2722	0.3656	60.2722
A1 to ZN	fall	2.16	2.87	2.43	3.16
	rise	1.64	1.90	2.02	2.11
A2 to ZN	fall	2.43	3.17	2.68	3.45
	rise	1.64	1.90	1.83	1.93
A3 to ZN	fall	2.64	3.43	2.91	3.70
	rise	1.65	1.96	1.76	1.88
A4 to ZN	fall	2.82	3.66	3.12	3.93
	rise	1.64	1.93	1.71	1.87

AND4_X2

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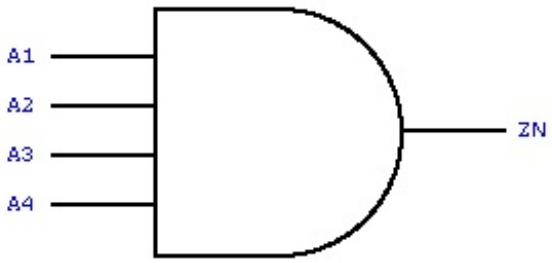
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	L
H	H	H	H	H
-	L	-	-	L
-	-	L	-	L
-	-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**120.392**	**0.3656**	**120.392**
A1 to ZN	**fall**	0.08	0.25	0.29	0.50
	rise	0.11	0.64	0.25	0.79
A2 to ZN	**fall**	0.09	0.26	0.31	0.51
	rise	0.12	0.65	0.27	0.81
A3 to ZN	**fall**	0.10	0.27	0.33	0.53
	rise	0.13	0.66	0.26	0.79
A4 to ZN	**fall**	0.10	0.28	0.34	0.54
	rise	0.13	0.66	0.24	0.77

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**120.392**	**0.3656**	**120.392**
A1 to ZN	**fall**	0.01	0.13	0.03	0.14
	rise	0.02	0.50	0.04	0.50
A2 to ZN	**fall**	0.02	0.13	0.03	0.14
	rise	0.02	0.50	0.04	0.50
A3 to ZN	**fall**	0.02	0.13	0.03	0.14
	rise	0.02	0.50	0.03	0.50
A4 to ZN	**fall**	0.02	0.14	0.03	0.14
	rise	0.02	0.50	0.03	0.50

Capacitance [fF]	
A1	1.5276
A2	1.6002
A3	1.6125
A4	1.6270

Leakage [nW]
27.40

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.392	0.3656	120.392
A1 to ZN	fall	4.08	5.34	4.67	5.96
	rise	3.19	3.76	3.97	4.14
A2 to ZN	fall	4.58	5.95	5.17	6.53
	rise	3.19	3.81	3.57	3.88
A3 to ZN	fall	5.02	6.48	5.63	7.03
	rise	3.20	3.73	3.43	3.73
A4 to ZN	fall	5.38	6.93	6.02	7.49
	rise	3.19	3.81	3.33	3.68

AND4_X4

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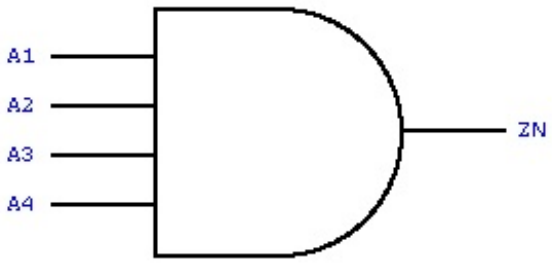
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	L
H	H	H	H	H
-	L	-	-	L
-	-	L	-	L
-	-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.08	0.25	0.28	0.49
	rise	0.11	0.64	0.24	0.78
A2 to ZN	**fall**	0.09	0.26	0.30	0.51
	rise	0.12	0.65	0.26	0.80
A3 to ZN	**fall**	0.09	0.27	0.32	0.52
	rise	0.12	0.65	0.25	0.79
A4 to ZN	**fall**	0.10	0.28	0.33	0.54
	rise	0.13	0.65	0.23	0.77

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.01	0.13	0.03	0.14
	rise	0.02	0.50	0.04	0.50
A2 to ZN	**fall**	0.01	0.13	0.03	0.14
	rise	0.02	0.50	0.03	0.50
A3 to ZN	**fall**	0.02	0.13	0.03	0.14
	rise	0.02	0.50	0.03	0.50
A4 to ZN	**fall**	0.02	0.14	0.03	0.14
	rise	0.02	0.50	0.03	0.50

Capacitance [fF]	
A1	2.9728
A2	3.2292
A3	3.4087
A4	3.6267

Leakage [nW]
54.79

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	241.089	0.3656	241.089
A1 to ZN	fall	7.80	10.21	9.02	11.46
	rise	6.17	7.24	7.76	8.22
A2 to ZN	fall	8.81	11.42	10.04	12.59
	rise	6.16	7.51	6.94	7.40
A3 to ZN	fall	9.68	12.49	10.95	13.59
	rise	6.17	7.47	6.63	7.33
A4 to ZN	fall	10.35	13.39	11.73	14.53
	rise	6.16	7.35	6.40	7.16

ANTENNA_X1

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.266 um^2
Type	Physical
Input	A
PG Pins	VDD (primary_power), VSS (primary_ground)



Capacitance [fF]	
A	0.0234

Leakage [nW]
0.00

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0023**	**0.50**
A	**fall**	0.00	0.00
	rise	0.00	0.00

AOI211_X1

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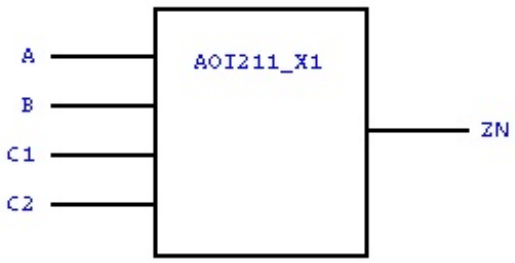
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((C1 & C2) | B) | A)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	L	L	-	H
L	L	-	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**14.4959**	**0.3656**	**14.4959**
A to ZN	**fall**	0.01	0.05	0.09	0.23
	rise	0.21	0.66	0.40	0.86
B to ZN	**fall**	0.01	0.05	0.08	0.22
	rise	0.20	0.64	0.40	0.87
C1 to ZN	**fall**	0.02	0.09	0.05	0.27
	rise	0.10	0.54	0.31	0.78
C2 to ZN	**fall**	0.02	0.09	0.08	0.27
	rise	0.13	0.57	0.36	0.82

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**14.4959**	**0.3656**	**14.4959**
A to ZN	**fall**	0.01	0.04	0.10	0.14
	rise	0.11	0.50	0.13	0.50
B to ZN	**fall**	0.01	0.04	0.10	0.14
	rise	0.11	0.50	0.14	0.50
C1 to ZN	**fall**	0.01	0.07	0.10	0.17
	rise	0.09	0.48	0.15	0.48
C2 to ZN	**fall**	0.01	0.07	0.07	0.13
	rise	0.11	0.50	0.16	0.50

Capacitance [fF]	
A	1.5583
B	1.5902
C1	1.6012
C2	1.6078

Leakage [nW]
17.41

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	14.4959	0.3656	14.4959
A to ZN	fall	0.58	0.95	1.45	1.30
	rise	3.81	3.84	3.71	3.74
B to ZN	fall	0.59	0.77	1.27	1.11
	rise	3.29	3.32	3.27	3.25
C1 to ZN	fall	0.29	0.23	0.82	0.56
	rise	2.20	2.22	2.67	2.45
C2 to ZN	fall	0.28	0.23	0.53	0.42
	rise	2.75	2.79	3.25	3.06

AOI211_X2

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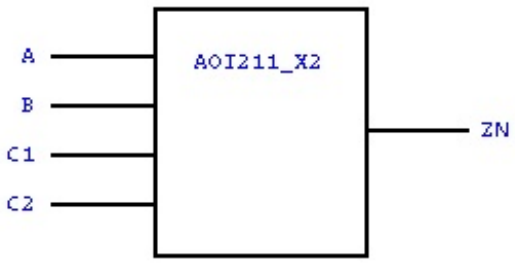
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((C1 & C2) | B) | A)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	L	L	-	H
L	L	-	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**28.9917**	**0.3656**	**28.9917**
A to ZN	**fall**	0.01	0.05	0.09	0.23
	rise	0.21	0.66	0.39	0.86
B to ZN	**fall**	0.01	0.05	0.07	0.22
	rise	0.19	0.64	0.40	0.87
C1 to ZN	**fall**	0.02	0.09	0.05	0.27
	rise	0.10	0.54	0.30	0.79
C2 to ZN	**fall**	0.02	0.09	0.08	0.27
	rise	0.13	0.57	0.35	0.82

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**28.9917**	**0.3656**	**28.9917**
A to ZN	**fall**	0.01	0.04	0.10	0.14
	rise	0.10	0.50	0.13	0.50
B to ZN	**fall**	0.01	0.04	0.10	0.14
	rise	0.10	0.50	0.14	0.50
C1 to ZN	**fall**	0.01	0.07	0.09	0.17
	rise	0.08	0.48	0.15	0.48
C2 to ZN	**fall**	0.01	0.07	0.07	0.13
	rise	0.10	0.50	0.15	0.50

Capacitance [fF]	
A	2.9924
B	3.2929
C1	3.0560
C2	3.3006

Leakage [nW]
34.81

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	28.9917	0.3656	28.9917
A to ZN	fall	1.12	1.83	2.84	2.53
	rise	7.72	7.79	7.50	7.59
B to ZN	fall	1.13	1.46	2.47	2.15
	rise	6.67	6.75	6.62	6.60
C1 to ZN	fall	0.59	0.46	1.65	1.13
	rise	4.39	4.44	5.34	4.90
C2 to ZN	fall	0.59	0.46	1.06	0.84
	rise	5.50	5.59	6.49	6.17

AOI211_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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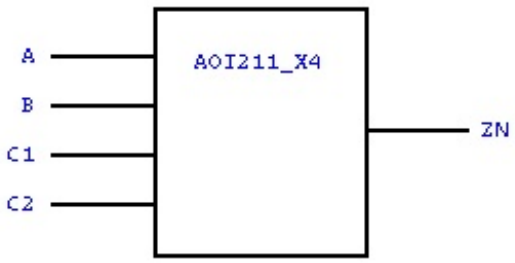
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	ZN = "!(!(!(((C1 & C2) | B) | A)))"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	L	L	-	H
L	L	-	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.09	0.26	0.24	0.40
	rise	0.42	0.93	0.62	1.13
B to ZN	**fall**	0.09	0.25	0.23	0.39
	rise	0.40	0.92	0.63	1.15
C1 to ZN	**fall**	0.11	0.27	0.24	0.40
	rise	0.30	0.82	0.55	1.07
C2 to ZN	**fall**	0.12	0.28	0.24	0.41
	rise	0.33	0.85	0.59	1.10

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.01	0.13	0.01	0.13
	rise	0.02	0.50	0.02	0.50
B to ZN	**fall**	0.01	0.13	0.01	0.13
	rise	0.02	0.50	0.02	0.50
C1 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
C2 to ZN	**fall**	0.01	0.13	0.01	0.13
	rise	0.02	0.50	0.02	0.50

Capacitance [fF]	
A	1.6075
B	1.6285
C1	1.5822
C2	1.6707

Leakage [nW]
77.92

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	fall	7.77	9.92	8.32	10.31
	rise	11.08	13.25	11.05	13.15
B to ZN	fall	7.58	9.60	8.11	10.04
	rise	10.57	12.70	10.64	12.74
C1 to ZN	fall	6.93	8.89	7.56	9.54
	rise	9.49	11.53	10.16	12.16
C2 to ZN	fall	6.99	8.89	7.27	9.18
	rise	10.07	12.08	10.72	12.93

AOI21_X1

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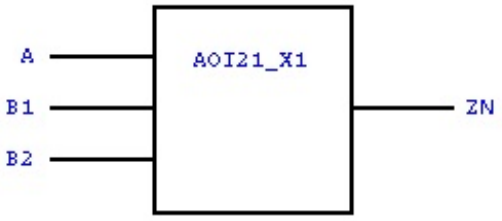
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!(A | (B1 & B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	L	-	H
L	-	L	H
H	-	-	L
-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to ZN	**fall**	0.01	0.07	0.07	0.28
	rise	0.11	0.60	0.30	0.84
B1 to ZN	**fall**	0.02	0.14	0.04	0.36
	rise	0.06	0.55	0.24	0.81
B2 to ZN	**fall**	0.02	0.14	0.07	0.34
	rise	0.08	0.57	0.28	0.83

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to ZN	**fall**	0.01	0.06	0.10	0.16
	rise	0.06	0.50	0.10	0.50
B1 to ZN	**fall**	0.01	0.11	0.09	0.20
	rise	0.05	0.49	0.12	0.49
B2 to ZN	**fall**	0.01	0.11	0.07	0.16
	rise	0.06	0.50	0.12	0.50

Capacitance [fF]	
A	1.5716
B1	1.6028
B2	1.6112

Leakage [nW]
16.33

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to ZN	**fall**	0.54	0.78	1.34	1.03
	rise	3.03	3.06	2.97	3.03
B1 to ZN	**fall**	0.24	0.23	0.91	0.46
	rise	1.92	1.96	2.45	2.15
B2 to ZN	**fall**	0.23	0.23	0.57	0.34
	rise	2.47	2.54	2.99	2.77

AOI21_X2

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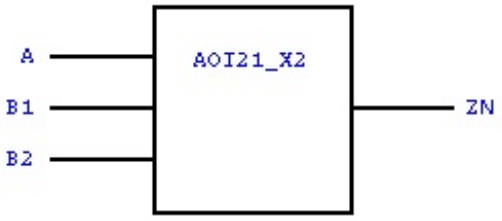
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!(A | (B1 & B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	L	-	H
L	-	L	H
H	-	-	L
-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**50.6592**	**0.3656**	**50.6592**
A to ZN	**fall**	0.01	0.07	0.07	0.28
	rise	0.10	0.59	0.29	0.84
B1 to ZN	**fall**	0.02	0.14	0.04	0.36
	rise	0.06	0.55	0.23	0.81
B2 to ZN	**fall**	0.02	0.14	0.07	0.34
	rise	0.08	0.57	0.27	0.83

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**50.6592**	**0.3656**	**50.6592**
A to ZN	**fall**	0.01	0.06	0.10	0.16
	rise	0.05	0.50	0.10	0.50
B1 to ZN	**fall**	0.01	0.11	0.09	0.20
	rise	0.04	0.49	0.12	0.49
B2 to ZN	**fall**	0.01	0.11	0.07	0.16
	rise	0.05	0.50	0.12	0.50

Capacitance [fF]	
A	3.0152
B1	3.0404
B2	3.3554

Leakage [nW]
32.65

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**50.6592**	**0.3656**	**50.6592**
A to ZN	**fall**	1.05	1.50	2.64	1.98
	rise	5.90	6.01	5.83	5.89
B1 to ZN	**fall**	0.47	0.47	1.84	0.93
	rise	3.82	3.91	4.89	4.29
B2 to ZN	**fall**	0.46	0.47	1.15	0.68
	rise	4.92	5.08	5.95	5.54

AOI21_X4

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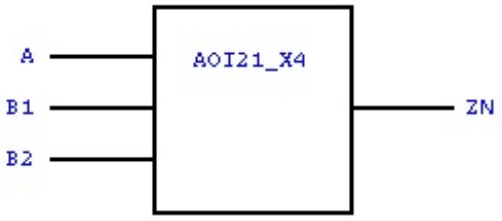
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(A | (B1 & B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	L	-	H
L	-	L	H
H	-	-	L
-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	101.013	0.3656	101.013
A to ZN	**fall**	0.01	0.07	0.07	0.28
	rise	0.10	0.59	0.29	0.84
B1 to ZN	**fall**	0.02	0.14	0.04	0.36
	rise	0.06	0.55	0.23	0.81
B2 to ZN	**fall**	0.02	0.14	0.06	0.34
	rise	0.08	0.57	0.27	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	101.013	0.3656	101.013
A to ZN	**fall**	0.01	0.06	0.10	0.16
	rise	0.05	0.50	0.10	0.50
B1 to ZN	**fall**	0.01	0.11	0.09	0.20
	rise	0.04	0.49	0.12	0.49
B2 to ZN	**fall**	0.01	0.11	0.07	0.16
	rise	0.05	0.50	0.12	0.50

Capacitance [fF]	
A	5.9318
B1	6.2392
B2	6.4606

Leakage [nW]
65.31

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	101.013	0.3656	101.013
A to ZN	**fall**	2.05	2.99	5.27	3.95
	rise	12.07	12.26	11.88	12.08
B1 to ZN	**fall**	0.94	0.93	3.68	1.85
	rise	7.68	7.86	9.84	8.63
B2 to ZN	**fall**	0.95	0.92	2.32	1.37
	rise	9.87	10.07	11.95	11.13

AOI221_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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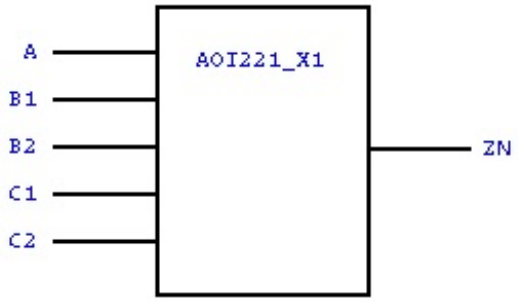
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "!(((C1 & C2) | A) | (B1 & B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	L	-	H
L	L	-	-	L	H
L	-	L	L	-	H
L	-	L	-	L	H
H	-	-	-	-	L
-	H	H	-	-	L
-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	13.8092	0.3656	13.8092
A to ZN	**fall**	0.01	0.05	0.08	0.21
	rise	0.23	0.65	0.44	0.88
B1 to ZN	**fall**	0.03	0.10	0.13	0.30
	rise	0.22	0.64	0.41	0.85
B2 to ZN	**fall**	0.03	0.10	0.15	0.29
	rise	0.25	0.67	0.45	0.88
C1 to ZN	**fall**	0.02	0.08	0.05	0.26
	rise	0.13	0.56	0.36	0.80
C2 to ZN	**fall**	0.02	0.09	0.08	0.26
	rise	0.16	0.59	0.40	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	13.8092	0.3656	13.8092
A to ZN	**fall**	0.01	0.04	0.11	0.14
	rise	0.12	0.50	0.16	0.50
B1 to ZN	**fall**	0.02	0.08	0.12	0.17
	rise	0.11	0.48	0.13	0.48
B2 to ZN	**fall**	0.02	0.08	0.09	0.13
	rise	0.12	0.50	0.14	0.50
C1 to ZN	**fall**	0.01	0.07	0.12	0.17
	rise	0.11	0.48	0.16	0.48
C2 to ZN	**fall**	0.01	0.07	0.09	0.13
	rise	0.12	0.50	0.17	0.50

Capacitance [fF]	
A	1.6066
B1	1.5487
B2	1.5885
C1	1.5790
C2	1.6357

Leakage [nW]
21.45

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	13.8092	0.3656	13.8092
A to ZN	fall	0.56	0.69	1.22	1.05
	rise	3.48	3.53	3.47	3.47
B1 to ZN	fall	0.81	1.13	1.48	1.35
	rise	4.04	4.08	3.90	3.94
B2 to ZN	fall	0.81	1.13	1.23	1.22
	rise	4.59	4.65	4.51	4.58
C1 to ZN	fall	0.29	0.23	0.83	0.57
	rise	2.38	2.43	2.86	2.67
C2 to ZN	fall	0.28	0.23	0.54	0.42
	rise	2.94	3.00	3.43	3.29

AOI221_X2

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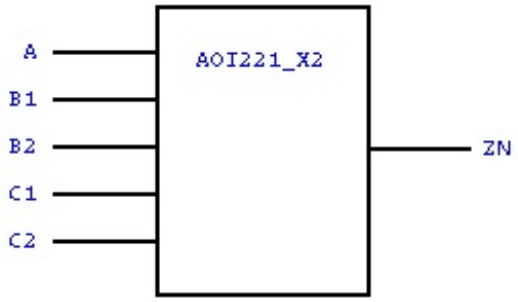
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.926 um^2
Equation	ZN = "!(((C1 & C2) | A) | (B1 & B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	L	-	H
L	L	-	-	L	H
L	-	L	L	-	H
L	-	L	-	L	H
H	-	-	-	-	L
-	H	H	-	-	L
-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	27.6184	0.3656	27.6184
A to ZN	**fall**	0.01	0.05	0.07	0.22
	rise	0.23	0.65	0.44	0.88
B1 to ZN	**fall**	0.03	0.10	0.13	0.30
	rise	0.21	0.64	0.40	0.85
B2 to ZN	**fall**	0.03	0.10	0.14	0.29
	rise	0.24	0.67	0.44	0.88
C1 to ZN	**fall**	0.02	0.08	0.05	0.26
	rise	0.13	0.56	0.35	0.80
C2 to ZN	**fall**	0.02	0.09	0.08	0.26
	rise	0.16	0.59	0.39	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	27.6184	0.3656	27.6184
A to ZN	**fall**	0.01	0.04	0.11	0.14
	rise	0.12	0.50	0.15	0.50
B1 to ZN	**fall**	0.02	0.08	0.12	0.17
	rise	0.10	0.48	0.13	0.48
B2 to ZN	**fall**	0.02	0.08	0.09	0.13
	rise	0.12	0.50	0.14	0.50
C1 to ZN	**fall**	0.01	0.07	0.12	0.17
	rise	0.10	0.48	0.16	0.48
C2 to ZN	**fall**	0.01	0.07	0.09	0.13
	rise	0.12	0.50	0.16	0.50

Capacitance [fF]	
A	3.3731
B1	3.1610
B2	3.0562
C1	3.0383
C2	3.3551

Leakage [nW]
42.91

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	27.6184	0.3656	27.6184
A to ZN	fall	1.17	1.45	2.51	2.16
	rise	6.99	7.09	6.95	6.98
B1 to ZN	fall	1.64	2.27	2.98	2.73
	rise	8.06	8.14	7.75	7.86
B2 to ZN	fall	1.64	2.28	2.48	2.45
	rise	9.15	9.26	8.99	9.12
C1 to ZN	fall	0.57	0.46	1.68	1.14
	rise	4.80	4.90	5.76	5.38
C2 to ZN	fall	0.57	0.46	1.08	0.83
	rise	5.91	6.03	6.90	6.62

AOI221_X4

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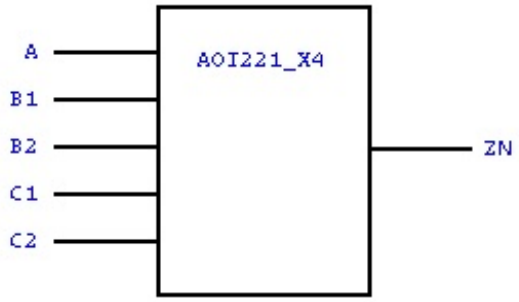
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(!(!(((C1 & C2) | A) | (B1 & B2))))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	L	-	H
L	L	-	-	L	H
L	-	L	L	-	H
L	-	L	-	L	H
H	-	-	-	-	L
-	H	H	-	-	L
-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.10	0.26	0.24	0.40
	rise	0.44	0.95	0.67	1.18
B1 to ZN	**fall**	0.13	0.29	0.30	0.46
	rise	0.43	0.94	0.63	1.15
B2 to ZN	**fall**	0.14	0.30	0.30	0.46
	rise	0.46	0.97	0.67	1.19
C1 to ZN	**fall**	0.12	0.28	0.25	0.41
	rise	0.34	0.85	0.59	1.10
C2 to ZN	**fall**	0.12	0.28	0.26	0.42
	rise	0.37	0.89	0.62	1.14

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.01	0.13	0.01	0.13
	rise	0.02	0.50	0.02	0.50
B1 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
B2 to ZN	**fall**	0.01	0.13	0.01	0.13
	rise	0.02	0.50	0.02	0.50
C1 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
C2 to ZN	**fall**	0.01	0.13	0.01	0.13
	rise	0.02	0.50	0.02	0.50

Capacitance [fF]	
A	1.5807
B1	1.5238
B2	1.6105
C1	1.5523
C2	1.6067

Leakage [nW]
82.68

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	fall	7.62	9.61	8.13	10.07
	rise	10.80	12.90	10.87	13.03
B1 to ZN	fall	7.83	10.07	8.43	10.50
	rise	11.39	13.52	11.29	13.51
B2 to ZN	fall	7.84	10.07	8.12	10.20
	rise	11.90	14.16	11.93	14.17
C1 to ZN	fall	6.98	8.90	7.61	9.57
	rise	9.81	11.78	10.45	12.47
C2 to ZN	fall	7.00	8.90	7.25	9.21
	rise	10.35	12.43	11.00	13.13

AOI222_X1

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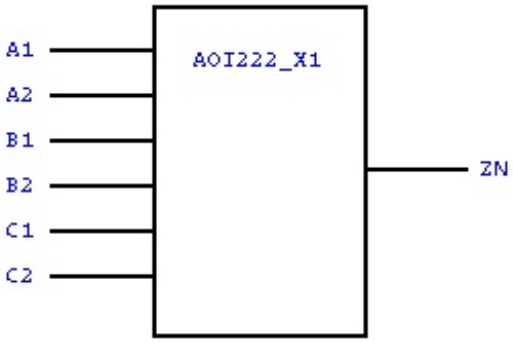
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.128 um^2
Equation	ZN = "!(((A1 & A2) | (B1 & B2)) | (C1 & C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	L	-	L	-	H
L	-	L	-	-	L	H
L	-	-	L	L	-	H
L	-	-	L	-	L	H
H	H	-	-	-	-	L
-	L	L	-	L	-	H
-	L	L	-	-	L	H
-	L	-	L	L	-	H
-	L	-	L	-	L	H
-	-	H	H	-	-	L
-	-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**13.0081**	**0.3656**	**13.0081**
A1 to ZN	**fall**	0.02	0.08	0.06	0.25
	rise	0.16	0.56	0.40	0.80
A2 to ZN	**fall**	0.02	0.08	0.09	0.26
	rise	0.19	0.58	0.44	0.83
B1 to ZN	**fall**	0.03	0.09	0.12	0.28
	rise	0.25	0.64	0.46	0.87
B2 to ZN	**fall**	0.03	0.10	0.14	0.28
	rise	0.27	0.67	0.50	0.91
C1 to ZN	**fall**	0.03	0.10	0.15	0.30
	rise	0.28	0.67	0.48	0.89
C2 to ZN	**fall**	0.03	0.10	0.16	0.30
	rise	0.31	0.70	0.52	0.92

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**13.0081**	**0.3656**	**13.0081**
A1 to ZN	**fall**	0.02	0.07	0.13	0.18
	rise	0.12	0.48	0.17	0.48
A2 to ZN	**fall**	0.02	0.07	0.10	0.13
	rise	0.14	0.50	0.18	0.50
B1 to ZN	**fall**	0.03	0.08	0.13	0.18
	rise	0.13	0.48	0.16	0.48
B2 to ZN	**fall**	0.03	0.08	0.09	0.13
	rise	0.14	0.50	0.17	0.50
C1 to ZN	**fall**	0.03	0.09	0.13	0.18
	rise	0.13	0.48	0.14	0.48
C2 to ZN	**fall**	0.03	0.09	0.10	0.14
	rise	0.14	0.50	0.16	0.50

Capacitance [fF]	
A1	1.5851
A2	1.6274
B1	1.5417
B2	1.5776
C1	1.5171
C2	1.5512

Leakage [nW]
25.24

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	13.0081	0.3656	13.0081
A1 to ZN	fall	0.29	0.23	0.84	0.58
	rise	2.55	2.62	3.04	2.88
A2 to ZN	fall	0.29	0.23	0.54	0.42
	rise	3.10	3.18	3.62	3.50
B1 to ZN	fall	0.77	0.92	1.35	1.19
	rise	3.72	3.77	3.65	3.67
B2 to ZN	fall	0.77	0.92	1.09	1.05
	rise	4.27	4.34	4.25	4.30
C1 to ZN	fall	0.99	1.37	1.71	1.63
	rise	4.68	4.74	4.53	4.59
C2 to ZN	fall	1.00	1.37	1.46	1.49
	rise	5.22	5.31	5.14	5.22

AOI222_X2

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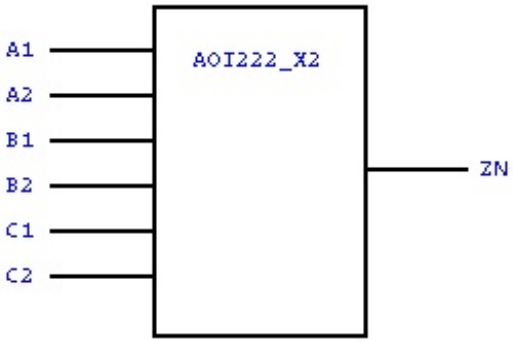
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	3.724 um^2
Equation	ZN = "!(((A1 & A2) | (B1 & B2)) | (C1 & C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	L	-	L	-	H
L	-	L	-	-	L	H
L	-	-	L	L	-	H
L	-	-	L	-	L	H
H	H	-	-	-	-	L
-	L	L	-	L	-	H
-	L	L	-	-	L	H
-	L	-	L	L	-	H
-	L	-	L	-	L	H
-	-	H	H	-	-	L
-	-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**25.6348**	**0.3656**	**25.6348**
A1 to ZN	**fall**	0.02	0.08	0.06	0.25
	rise	0.15	0.55	0.39	0.80
A2 to ZN	**fall**	0.02	0.08	0.09	0.26
	rise	0.19	0.59	0.44	0.83
B1 to ZN	**fall**	0.03	0.09	0.12	0.28
	rise	0.25	0.64	0.46	0.87
B2 to ZN	**fall**	0.03	0.10	0.14	0.28
	rise	0.27	0.67	0.50	0.91
C1 to ZN	**fall**	0.03	0.10	0.15	0.30
	rise	0.28	0.68	0.48	0.89
C2 to ZN	**fall**	0.03	0.10	0.16	0.29
	rise	0.31	0.70	0.52	0.93

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**25.6348**	**0.3656**	**25.6348**
A1 to ZN	**fall**	0.02	0.07	0.13	0.18
	rise	0.12	0.48	0.17	0.48
A2 to ZN	**fall**	0.02	0.07	0.10	0.13
	rise	0.14	0.50	0.18	0.50
B1 to ZN	**fall**	0.03	0.08	0.13	0.18
	rise	0.12	0.48	0.16	0.48
B2 to ZN	**fall**	0.03	0.08	0.10	0.14
	rise	0.14	0.50	0.17	0.50
C1 to ZN	**fall**	0.03	0.09	0.13	0.18
	rise	0.12	0.48	0.14	0.48
C2 to ZN	**fall**	0.03	0.09	0.10	0.14
	rise	0.14	0.50	0.16	0.50

Capacitance [fF]	
A1	3.0630
A2	3.0820
B1	2.9532
B2	3.3069
C1	2.9123
C2	3.2204

Leakage [nW]
50.48

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	25.6348	0.3656	25.6348
A1 to ZN	fall	0.59	0.46	1.69	1.16
	rise	5.25	5.39	6.25	5.92
A2 to ZN	fall	0.58	0.46	1.06	0.82
	rise	6.49	6.67	7.51	7.29
B1 to ZN	fall	1.55	1.83	2.71	2.39
	rise	7.59	7.70	7.45	7.49
B2 to ZN	fall	1.55	1.83	2.19	2.09
	rise	8.68	8.83	8.64	8.75
C1 to ZN	fall	1.98	2.74	3.42	3.26
	rise	9.56	9.68	9.27	9.39
C2 to ZN	fall	2.00	2.74	2.91	2.97
	rise	10.66	10.82	10.49	10.65

AOI222_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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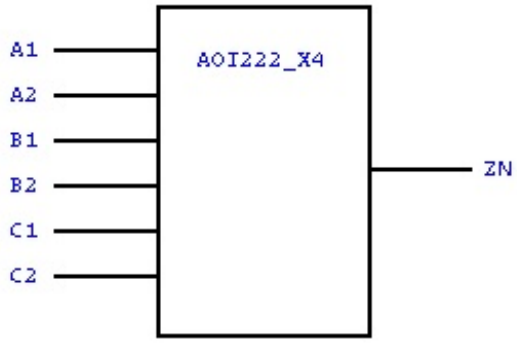
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.724 um^2
Equation	ZN = "!(!(!(((A1 & A2) | (B1 & B2)) | (C1 & C2))))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	L	-	L	-	H
L	-	L	-	-	L	H
L	-	-	L	L	-	H
L	-	-	L	-	L	H
H	H	-	-	-	-	L
-	L	L	-	L	-	H
-	L	L	-	-	L	H
-	L	-	L	L	-	H
-	L	-	L	-	L	H
-	-	H	H	-	-	L
-	-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.12	0.28	0.26	0.43
	rise	0.37	0.88	0.62	1.14
A2 to ZN	**fall**	0.13	0.29	0.27	0.43
	rise	0.40	0.92	0.66	1.17
B1 to ZN	**fall**	0.13	0.30	0.30	0.46
	rise	0.45	0.97	0.68	1.19
B2 to ZN	**fall**	0.14	0.30	0.29	0.46
	rise	0.48	1.00	0.71	1.23
C1 to ZN	**fall**	0.14	0.30	0.32	0.48
	rise	0.48	1.00	0.70	1.21
C2 to ZN	**fall**	0.14	0.31	0.31	0.47
	rise	0.51	1.03	0.73	1.25

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
A2 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
B1 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
B2 to ZN	**fall**	0.01	0.13	0.01	0.13
	rise	0.02	0.50	0.02	0.50
C1 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
C2 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50

Capacitance [fF]	
A1	1.5520
A2	1.6386
B1	1.5616
B2	1.6334
C1	1.5495
C2	1.6154

Leakage [nW]
87.51

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	fall	6.95	8.89	7.62	9.59
	rise	9.94	12.09	10.58	12.77
A2 to ZN	fall	6.97	8.89	7.26	9.21
	rise	10.48	12.76	11.15	13.45
B1 to ZN	fall	7.60	9.69	8.20	10.21
	rise	10.94	13.16	10.97	13.29
B2 to ZN	fall	7.62	9.69	7.88	9.88
	rise	11.48	13.82	11.56	13.94
C1 to ZN	fall	7.99	10.32	8.61	10.76
	rise	11.85	14.08	11.75	14.07
C2 to ZN	fall	8.00	10.32	8.29	10.46
	rise	12.38	14.74	12.36	14.80

AOI22_X1

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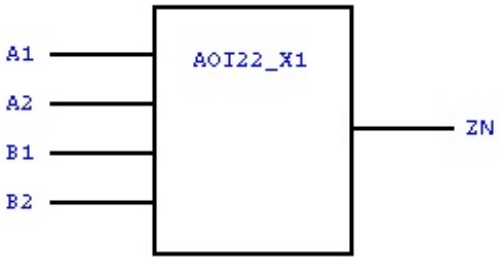
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!((A1 & A2) | (B1 & B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**24.6048**	**0.3656**	**24.6048**
A1 to ZN	**fall**	0.02	0.13	0.05	0.35
	rise	0.08	0.56	0.28	0.82
A2 to ZN	**fall**	0.02	0.14	0.07	0.34
	rise	0.10	0.57	0.32	0.84
B1 to ZN	**fall**	0.03	0.15	0.11	0.38
	rise	0.11	0.59	0.31	0.83
B2 to ZN	**fall**	0.03	0.15	0.13	0.36
	rise	0.13	0.61	0.34	0.85

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**24.6048**	**0.3656**	**24.6048**
A1 to ZN	**fall**	0.01	0.12	0.11	0.20
	rise	0.06	0.49	0.13	0.49
A2 to ZN	**fall**	0.01	0.12	0.08	0.16
	rise	0.07	0.50	0.13	0.50
B1 to ZN	**fall**	0.02	0.12	0.11	0.20
	rise	0.06	0.49	0.10	0.49
B2 to ZN	**fall**	0.02	0.12	0.08	0.17
	rise	0.07	0.50	0.11	0.50

Capacitance [fF]	
A1	1.6443
A2	1.6277
B1	1.5482
B2	1.5798

Leakage [nW]
19.17

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	24.6048	0.3656	24.6048
A1 to ZN	fall	0.23	0.23	0.93	0.46
	rise	2.11	2.17	2.65	2.38
A2 to ZN	fall	0.23	0.23	0.58	0.34
	rise	2.65	2.73	3.18	3.01
B1 to ZN	fall	0.74	0.95	1.41	1.10
	rise	3.12	3.17	3.01	3.09
B2 to ZN	fall	0.74	0.95	1.10	0.98
	rise	3.65	3.74	3.58	3.71

AOI22_X2

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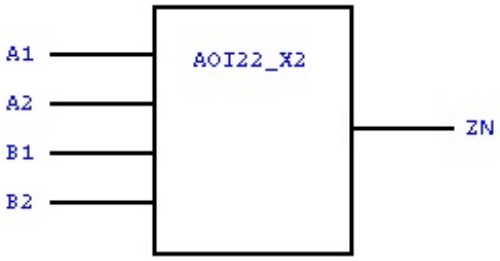
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!((A1 & A2) | (B1 & B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**49.1333**	**0.3656**	**49.1333**
A1 to ZN	**fall**	0.02	0.13	0.04	0.35
	rise	0.08	0.55	0.27	0.82
A2 to ZN	**fall**	0.02	0.14	0.07	0.34
	rise	0.10	0.57	0.31	0.84
B1 to ZN	**fall**	0.03	0.15	0.11	0.38
	rise	0.11	0.59	0.31	0.84
B2 to ZN	**fall**	0.03	0.15	0.13	0.36
	rise	0.13	0.61	0.34	0.86

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**49.1333**	**0.3656**	**49.1333**
A1 to ZN	**fall**	0.01	0.12	0.11	0.20
	rise	0.06	0.49	0.12	0.49
A2 to ZN	**fall**	0.01	0.12	0.08	0.16
	rise	0.07	0.50	0.13	0.50
B1 to ZN	**fall**	0.02	0.12	0.11	0.20
	rise	0.06	0.49	0.10	0.49
B2 to ZN	**fall**	0.02	0.12	0.08	0.17
	rise	0.07	0.50	0.10	0.50

Capacitance [fF]	
A1	3.0632
A2	3.3510
B1	2.9310
B2	3.3470

Leakage [nW]
38.33

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	49.1333	0.3656	49.1333
A1 to ZN	fall	0.47	0.46	1.87	0.93
	rise	4.25	4.40	5.34	4.81
A2 to ZN	fall	0.47	0.46	1.17	0.67
	rise	5.33	5.51	6.40	6.05
B1 to ZN	fall	1.46	1.88	2.80	2.19
	rise	6.47	6.59	6.23	6.42
B2 to ZN	fall	1.46	1.89	2.19	1.96
	rise	7.53	7.70	7.37	7.66

AOI22_X4

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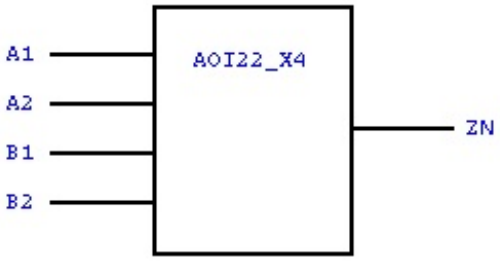
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Equation	ZN = "!((A1 & A2) | (B1 & B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**97.9614**	**0.3656**	**97.9614**
A1 to ZN	**fall**	0.02	0.13	0.04	0.35
	rise	0.08	0.55	0.27	0.82
A2 to ZN	**fall**	0.02	0.14	0.07	0.34
	rise	0.09	0.57	0.31	0.84
B1 to ZN	**fall**	0.03	0.15	0.10	0.38
	rise	0.11	0.59	0.30	0.83
B2 to ZN	**fall**	0.03	0.15	0.13	0.36
	rise	0.13	0.61	0.33	0.86

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**97.9614**	**0.3656**	**97.9614**
A1 to ZN	**fall**	0.01	0.12	0.11	0.20
	rise	0.05	0.49	0.12	0.49
A2 to ZN	**fall**	0.01	0.12	0.08	0.16
	rise	0.06	0.50	0.13	0.50
B1 to ZN	**fall**	0.02	0.12	0.11	0.20
	rise	0.05	0.49	0.10	0.49
B2 to ZN	**fall**	0.02	0.12	0.08	0.17
	rise	0.06	0.50	0.10	0.50

Capacitance [fF]	
A1	6.2502
A2	6.5399
B1	5.9695
B2	6.4141

Leakage [nW]
76.66

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	97.9614	0.3656	97.9614
A1 to ZN	fall	0.93	0.92	3.77	1.86
	rise	8.51	8.82	10.70	9.66
A2 to ZN	fall	0.93	0.92	2.36	1.34
	rise	10.69	11.04	12.82	12.14
B1 to ZN	fall	2.90	3.79	5.63	4.38
	rise	12.76	12.92	12.31	12.63
B2 to ZN	fall	2.92	3.79	4.40	3.92
	rise	14.89	15.22	14.58	15.11

BUF_X16

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	6.650 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	965.576	0.3656	965.576
A to Z	**fall**	0.06	0.22	0.25	0.44
	rise	0.03	0.55	0.11	0.64

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	965.576	0.3656	965.576
A to Z	**fall**	0.01	0.13	0.03	0.14
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A	12.2707

Leakage [nW]
251.31

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	965.576	0.3656	965.576
A to Z	**fall**	27.46	35.07	33.97	42.22
	rise	19.29	22.15	25.62	27.26

BUF_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.6537**	**0.3656**	**60.6537**
A to Z	**fall**	0.07	0.23	0.27	0.46
	rise	0.04	0.55	0.13	0.65

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.6537**	**0.3656**	**60.6537**
A to Z	**fall**	0.01	0.13	0.03	0.14
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A	0.9346

Leakage [nW]
15.64

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.6537**	**0.3656**	**60.6537**
A to Z	**fall**	1.88	2.41	2.22	2.82
	rise	1.32	1.48	1.71	1.78

BUF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.064 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A to Z	**fall**	0.07	0.23	0.26	0.45
	rise	0.04	0.55	0.12	0.64

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A to Z	**fall**	0.01	0.13	0.03	0.14
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A	1.7039

Leakage [nW]
31.42

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A to Z	**fall**	3.65	4.62	4.37	5.47
	rise	2.62	2.83	3.39	3.53

BUF_X32

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	32
Cell Area	13.034 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	1904.30	0.3656	1904.30
A to Z	**fall**	0.06	0.22	0.25	0.44
	rise	0.03	0.55	0.11	0.63

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	1904.30	0.3656	1904.30
A to Z	**fall**	0.01	0.13	0.03	0.14
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A	25.7111

Leakage [nW]
502.61

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	1904.30	0.3656	1904.30
A to Z	**fall**	54.65	69.50	67.53	83.77
	rise	38.39	42.46	51.22	54.74

BUF_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	1.862 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to Z	**fall**	0.06	0.22	0.25	0.44
	rise	0.03	0.55	0.11	0.63

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to Z	**fall**	0.01	0.13	0.03	0.14
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A	3.2519

Leakage [nW]
62.83

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to Z	**fall**	6.94	8.78	8.46	10.55
	rise	4.97	5.46	6.55	6.95

BUF_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	3.458 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	484.009	0.3656	484.009
A to Z	**fall**	0.06	0.22	0.25	0.44
	rise	0.03	0.55	0.11	0.63

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	484.009	0.3656	484.009
A to Z	**fall**	0.01	0.13	0.03	0.14
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A	6.3598

Leakage [nW]
125.65

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	484.009	0.3656	484.009
A to Z	**fall**	13.69	17.42	16.94	21.00
	rise	9.88	10.82	13.07	13.90

CLKBUF_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
A to Z	**fall**	0.07	0.39	0.25	0.59
	rise	0.05	0.57	0.20	0.72

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
A to Z	**fall**	0.01	0.28	0.03	0.28
	rise	0.01	0.50	0.04	0.50

Capacitance [fF]	
A	0.7577

Leakage [nW]
8.44

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
A to Z	**fall**	1.62	1.95	1.73	2.20
	rise	1.09	1.22	1.27	1.40

CLKBUF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.064 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to Z	**fall**	0.06	0.38	0.24	0.58
	rise	0.05	0.56	0.17	0.70

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to Z	**fall**	0.01	0.28	0.03	0.28
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A	1.3612

Leakage [nW]
17.24

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to Z	**fall**	3.12	3.73	3.39	4.27
	rise	2.15	2.37	2.53	2.78

CLKBUF_X3

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	3
Cell Area	1.330 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	181.885	0.3656	181.885
A to Z	**fall**	0.08	0.40	0.27	0.62
	rise	0.05	0.57	0.20	0.73

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	181.885	0.3656	181.885
A to Z	**fall**	0.01	0.28	0.04	0.28
	rise	0.01	0.50	0.04	0.50

Capacitance [fF]	
A	1.3799

Leakage [nW]
22.98

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	181.885	0.3656	181.885
A to Z	**fall**	3.82	4.75	4.29	5.23
	rise	2.94	3.32	3.38	3.69

NANGATE CLKGATETST_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	3.990 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to GCK	**fall**	0.08	0.40	0.28	0.62
	rise	0.05	0.57	0.14	0.67

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to GCK	**fall**	0.01	0.28	0.03	0.28
	rise	0.01	0.50	0.03	0.50

Constraints Time [ns]		
Setup CK to E	**fall**	0.42
	rise	0.38
Hold CK to E	**fall**	0.19
	rise	0.19
Setup CK to SE	**fall**	0.42
	rise	0.36
Hold CK to SE	**fall**	0.21
	rise	0.22

Capacitance [fF]	
CK	1.7438
E	0.8532
SE	0.7594

Leakage [nW]
39.66

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	4680.82	5167.36
	rise	1505.52	1892.18
E	fall	1534.33	1582.32
	rise	731.42	803.77
SE	fall	1270.24	1436.76
	rise	686.93	839.06

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to GCK	fall	3.85	5.03	4.33	5.88
	rise	2.49	3.11	3.37	3.81

NANGATE CLKGATETST_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	4.256 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
CK to GCK	**fall**	0.07	0.39	0.27	0.61
	rise	0.05	0.57	0.13	0.66

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
CK to GCK	**fall**	0.01	0.28	0.03	0.28
	rise	0.01	0.50	0.03	0.50

Constraints Time [ns]		
Setup CK to E	**fall**	0.40
	rise	0.40
Hold CK to E	**fall**	0.17
	rise	0.19
Setup CK to SE	**fall**	0.41
	rise	0.38
Hold CK to SE	**fall**	0.19
	rise	0.21

Capacitance [fF]	
CK	2.7292
E	0.8473
SE	0.7919

Leakage [nW]
51.99

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	5362.79	5840.11
	rise	911.24	1296.71
E	fall	1527.76	1577.30
	rise	687.03	763.54
SE	fall	1263.00	1433.76
	rise	641.79	800.81

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
CK to GCK	fall	5.43	7.26	6.12	8.52
	rise	3.41	4.08	4.44	5.11

NANGATE CLKGATETST_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	5.320 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.92**	**0.3656**	**242.92**
CK to GCK	**fall**	0.08	0.40	0.29	0.63
	rise	0.06	0.57	0.15	0.68

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.92**	**0.3656**	**242.92**
CK to GCK	**fall**	0.01	0.28	0.03	0.28
	rise	0.01	0.50	0.03	0.50

Constraints Time [ns]		
Setup CK to E	**fall**	0.45
	rise	0.60
Hold CK to E	**fall**	-0.00
	rise	0.19
Setup CK to SE	**fall**	0.46
	rise	0.59
Hold CK to SE	**fall**	0.02
	rise	0.23

Capacitance [fF]	
CK	4.2449
E	0.9026
SE	0.7911

Leakage [nW]
83.31

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	7521.03	7860.18
	rise	122.16	506.82
E	fall	1939.03	1984.00
	rise	975.69	1045.85
SE	fall	1675.49	1836.19
	rise	931.49	1079.60

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	242.92	0.3656	242.92
CK to GCK	fall	9.49	12.10	10.88	14.05
	rise	6.09	7.54	7.04	8.57

NANGATE CLKGATETST_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	7.714 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	0.08	0.41	0.29	0.64
	rise	0.06	0.58	0.15	0.68

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	0.01	0.28	0.03	0.28
	rise	0.01	0.50	0.03	0.50

Constraints Time [ns]		
Setup CK to E	**fall**	0.48
	rise	0.92
Hold CK to E	**fall**	-0.10
	rise	0.18
Setup CK to SE	**fall**	0.49
	rise	0.90
Hold CK to SE	**fall**	-0.09
	rise	0.21

Capacitance [fF]	
CK	7.6412
E	0.8718
SE	0.7786

Leakage [nW]
144.25

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	12228.00	12413.97
	rise	-1413.17	-1037.67
E	fall	2458.02	2504.70
	rise	1568.04	1640.30
SE	fall	2194.09	2356.53
	rise	1523.73	1673.49

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	484.619	0.3656	484.619
CK to GCK	fall	16.82	21.59	19.45	25.04
	rise	10.86	13.44	12.50	15.09

NANGATE CLKGATE_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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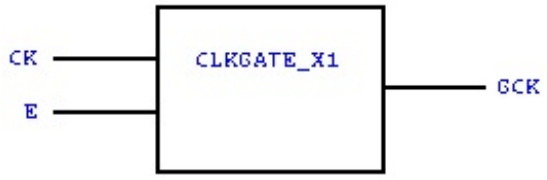
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	3.458 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	0.08	0.40	0.28	0.62
	rise	0.05	0.57	0.14	0.67

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	0.01	0.28	0.03	0.28
	rise	0.01	0.50	0.03	0.50

Constraints Time [ns]		
Setup CK to E	**fall**	0.34
	rise	0.30
Hold CK to E	**fall**	0.28
	rise	0.26

Capacitance [fF]	
CK	1.7756
E	0.8922

Leakage [nW]
32.73

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0023**	**0.50**
CK	**fall**	3694.05	4199.46
	rise	999.83	1414.26
E	**fall**	1171.48	1202.77
	rise	484.06	484.88

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	3.71	5.25	4.22	6.13
	rise	2.40	2.68	3.15	3.38

NANGATE CLKGATE_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	3.724 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**121.307**	**0.3656**	**121.307**
CK to GCK	**fall**	0.07	0.40	0.27	0.62
	rise	0.05	0.57	0.14	0.67

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**121.307**	**0.3656**	**121.307**
CK to GCK	**fall**	0.01	0.28	0.03	0.28
	rise	0.01	0.50	0.03	0.50

Constraints Time [ns]		
Setup CK to E	**fall**	0.35
	rise	0.33
Hold CK to E	**fall**	0.24
	rise	0.26

Capacitance [fF]	
CK	2.4774
E	0.8685

Leakage [nW]
44.47

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0023**	**0.50**
CK	**fall**	4610.12	5103.95
	rise	830.39	1228.72
E	**fall**	1316.88	1345.41
	rise	769.48	765.92

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**121.307**	**0.3656**	**121.307**
CK to GCK	**fall**	5.40	7.58	6.13	8.86
	rise	3.60	4.12	4.68	5.20

NANGATE CLKGATE_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
CK to GCK	**fall**	0.07	0.39	0.27	0.61
	rise	0.05	0.56	0.13	0.67

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
CK to GCK	**fall**	0.01	0.28	0.03	0.28
	rise	0.01	0.50	0.03	0.50

Constraints Time [ns]		
Setup CK to E	**fall**	0.40
	rise	0.35
Hold CK to E	**fall**	0.23
	rise	0.28

Capacitance [fF]	
CK	4.1151
E	0.8623

Leakage [nW]
73.08

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0023**	**0.50**
CK	**fall**	6133.81	6633.97
	rise	217.59	604.00
E	**fall**	1856.55	1895.51
	rise	1188.95	1189.28

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
CK to GCK	**fall**	8.50	12.18	9.75	14.27
	rise	5.87	6.81	7.53	8.44

NANGATE CLKGATE_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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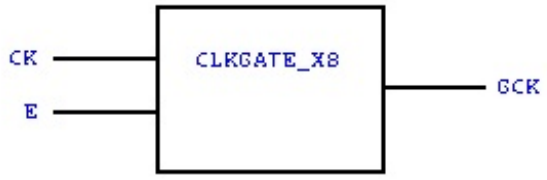
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	6.916 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	0.07	0.40	0.27	0.62
	rise	0.05	0.57	0.13	0.67

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	0.01	0.28	0.03	0.28
	rise	0.01	0.50	0.03	0.50

Constraints Time [ns]		
Setup CK to E	**fall**	0.43
	rise	0.36
Hold CK to E	**fall**	0.22
	rise	0.31

Capacitance [fF]	
CK	7.4645
E	1.1388

Leakage [nW]
131.52

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0023**	**0.50**
CK	**fall**	9483.43	9967.40
	rise	-753.82	-403.12
E	**fall**	3123.91	3207.70
	rise	1862.87	1881.73

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	14.99	22.11	17.35	25.77
	rise	10.49	12.25	13.22	15.44

DFFRS_X1

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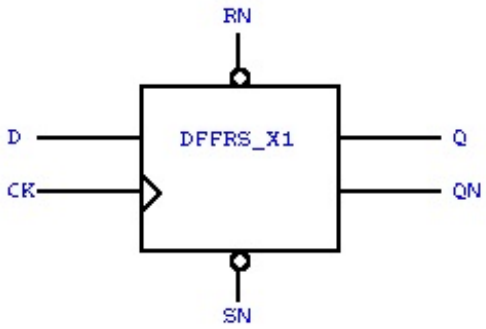
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, and drive strength X1
Strength	1
Cell Area	6.384 um^2
Equation	Q = "(!(!RN)*((D)+(!SN)))" QN = "(!(!SN)*(!(D)+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table

CK	D	RN	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	H	-	-	L	H
R	H	H	H	-	-	H	L
F	-	H	H	L	H	L	H
F	-	H	H	H	L	H	L
-	-	L	L	-	-	L	L
-	-	L	H	-	-	L	H
-	-	H	L	-	-	H	L

Propagation Delay [ns]

Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.26	0.45	0.37	0.56
	rise	0.24	0.76	0.35	0.87
CK to QN	fall	0.32	*0.48*	0.42	*0.58*
	rise	0.35	*0.86*	0.46	*0.97*
RN to Q	fall	0.10	0.28	0.32	0.54
	rise	0.45	0.97	0.62	1.14
RN to QN	rise	0.19	*0.69*	0.44	*0.94*
SN to Q	rise	0.61	1.14	0.86	1.39
SN to QN	fall	0.09	*0.25*	0.31	*0.50*
	rise	0.07	*0.59*	0.19	*0.71*

Output Transition [ns]

Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.02	0.14	0.02	0.14
	rise	0.02	0.50	0.02	0.50
CK to QN	fall	0.02	*0.13*	0.02	*0.13*
	rise	0.02	*0.50*	0.02	*0.50*
RN to Q	fall	0.02	0.14	0.04	0.14
	rise	0.02	0.50	0.04	0.50
RN to QN	rise	0.02	*0.50*	0.02	*0.50*
SN to Q	rise	0.03	0.50	0.03	0.50
SN to QN	fall	0.02	*0.13*	0.03	*0.14*
	rise	0.02	*0.50*	0.03	*0.50*

Constraints Time [ns]		
Setup CK to D	fall	0.37
	rise	0.21
Hold CK to D	fall	0.33
	rise	0.32
Recovery CK to RN	rise	0.18
Removal CK to RN	rise	0.85
Recovery CK to SN	rise	-0.11
Removal CK to SN	rise	1.31

Capacitance [fF]	
CK	0.9209
D	1.1147
RN	1.3874
SN	2.2011

Leakage [nW]
62.61

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	4245.25	4683.00
	rise	5055.23	5381.81
D	fall	1844.80	1893.53
	rise	953.21	972.60
RN	fall	510.03	511.33
	rise	-346.30	-355.44
SN	fall	1204.16	1275.86
	rise	-473.73	-427.81

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	4.29	4.48	4.48	4.68
	rise	4.17	4.53	4.33	4.70
CK to QN	fall	4.17	*4.55*	4.33	*4.68*
	rise	4.29	*4.50*	4.48	*4.68*
RN to Q	fall	2.56	3.70	2.81	3.94
	rise	3.12	3.54	3.28	3.62
RN to QN	rise	2.69	*3.93*	2.92	*4.08*
SN to Q	rise	4.10	4.44	4.54	4.77
SN to QN	fall	3.85	*4.42*	4.31	*4.94*
	rise	1.97	*2.52*	2.47	*2.85*

DFFRS_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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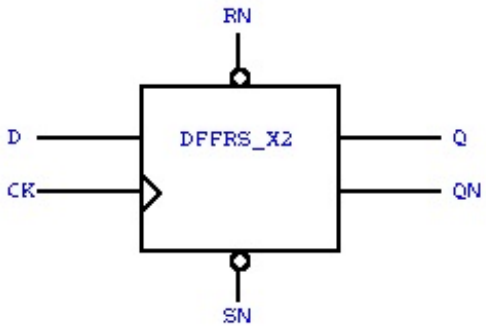
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, and drive strength X2
Strength	2
Cell Area	6.916 um^2
Equation	Q = "(!(!RN)*((D)+(!SN)))" QN = "(!(!SN)*(!(D)+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table

CK	D	RN	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	H	-	-	L	H
R	H	H	H	-	-	H	L
F	-	H	H	L	H	L	H
F	-	H	H	H	L	H	L
-	-	L	L	-	-	L	L
-	-	L	H	-	-	L	H
-	-	H	L	-	-	H	L

Propagation Delay [ns]

Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.26	0.44	0.37	0.55
	rise	0.26	0.79	0.37	0.89
CK to QN	fall	0.34	0.50	0.45	0.61
	rise	0.35	0.85	0.46	0.96
RN to Q	fall	0.10	0.29	0.34	0.56
	rise	0.47	0.99	0.64	1.16
RN to QN	rise	0.20	0.70	0.46	0.96
SN to Q	rise	0.66	1.19	0.91	1.44
SN to QN	fall	0.09	0.26	0.31	0.51
	rise	0.07	0.59	0.20	0.72

Output Transition [ns]

Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.02	0.14	0.02	0.14
	rise	0.02	0.50	0.02	0.50
CK to QN	fall	0.02	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
RN to Q	fall	0.02	0.14	0.04	0.14
	rise	0.02	0.50	0.03	0.50
RN to QN	rise	0.02	0.50	0.02	0.50
SN to Q	rise	0.04	0.50	0.03	0.50
SN to QN	fall	0.02	0.14	0.04	0.14
	rise	0.02	0.50	0.03	0.50

Constraints Time [ns]		
Setup CK to D	fall	0.37
	rise	0.21
Hold CK to D	fall	0.33
	rise	0.32
Recovery CK to RN	rise	0.06
Removal CK to RN	rise	0.85
Recovery CK to SN	rise	-0.09
Removal CK to SN	rise	1.33

Capacitance [fF]	
CK	0.8972
D	1.1354
RN	1.8236
SN	2.5875

Leakage [nW]
89.54

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	4189.86	4632.53
	rise	5497.33	5829.02
D	fall	1864.13	1912.57
	rise	926.05	947.14
RN	fall	754.14	701.51
	rise	-640.42	-654.62
SN	fall	1417.36	1470.73
	rise	-724.15	-639.50

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	5.71	6.03	5.88	6.25
	rise	5.67	6.27	5.84	6.44
CK to QN	fall	5.67	6.27	5.84	6.44
	rise	5.71	6.03	5.88	6.25
RN to Q	fall	4.23	5.72	4.77	6.08
	rise	4.65	5.42	4.80	5.47
RN to QN	rise	4.41	5.95	4.87	6.18
SN to Q	rise	5.88	6.37	6.58	6.80
SN to QN	fall	5.37	6.23	6.15	6.88
	rise	3.18	4.13	3.81	4.39

DFFR_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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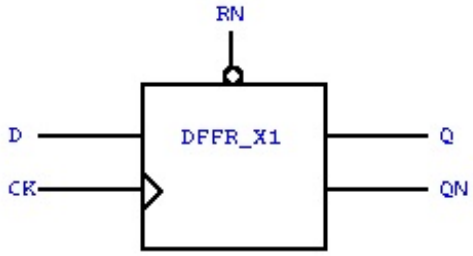
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, and drive strength X1
Strength	1
Cell Area	5.320 um^2
Equation	Q = "(!(!RN)*(D))" QN = "((!RN)+!(D))"
Clock	CK
Reset	RN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	RN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.25	0.42	0.37	0.53
	rise	0.33	0.83	0.44	0.93
CK to QN	fall	0.21	*0.44*	0.32	*0.55*
	rise	0.18	*0.70*	0.30	*0.82*
RN to Q	fall	0.09	0.27	0.31	0.52
RN to QN	rise	0.79	*1.32*	1.04	*1.57*

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.02	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
CK to QN	fall	0.03	*0.16*	0.04	*0.16*
	rise	0.02	*0.50*	0.02	*0.50*
RN to Q	fall	0.02	0.13	0.03	0.14
RN to QN	rise	0.03	*0.50*	0.03	*0.50*

Constraints Time [ns]		
Setup CK to D	fall	0.37
	rise	0.20
Hold CK to D	fall	0.34
	rise	0.31
Recovery CK to RN	rise	-0.05
Removal CK to RN	rise	1.44

Capacitance [fF]	
CK	0.9363
D	1.0944
RN	1.7843

Leakage [nW]
56.18

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	4068.54	4501.93
	rise	4373.78	4692.31
D	fall	1887.73	1953.30
	rise	979.82	1018.53
RN	fall	522.64	486.86
	rise	-433.00	-445.38

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	3.24	3.63	3.43	3.80
	rise	3.54	3.73	3.71	3.91
CK to QN	fall	3.54	*3.74*	3.71	*3.92*
	rise	3.25	*3.61*	3.44	*3.81*
RN to Q	fall	3.23	3.54	3.51	3.71
RN to QN	rise	3.23	*3.52*	3.51	*3.69*

DFFR_X2

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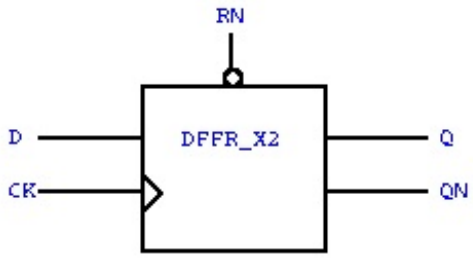
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, and drive strength X2
Strength	2
Cell Area	5.852 um^2
Equation	Q = "(!(!RN)*(D))" QN = "((!RN)+!(D))"
Clock	CK
Reset	RN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	RN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.28	0.44	0.39	0.55
	rise	0.45	0.92	0.56	1.03
CK to QN	fall	0.30	*0.59*	0.40	*0.70*
	rise	0.20	*0.73*	0.32	*0.85*
RN to Q	fall	0.08	0.25	0.29	0.49
RN to QN	rise	0.83	*1.36*	1.08	*1.61*

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.02	0.13	0.02	0.13
	rise	0.03	0.50	0.03	0.50
CK to QN	fall	0.05	*0.18*	0.05	*0.18*
	rise	0.02	*0.50*	0.02	*0.50*
RN to Q	fall	0.01	0.14	0.03	0.14
RN to QN	rise	0.04	*0.51*	0.04	*0.51*

Constraints Time [ns]		
Setup CK to D	fall	0.37
	rise	0.20
Hold CK to D	fall	0.34
	rise	0.30
Recovery CK to RN	rise	-0.19
Removal CK to RN	rise	1.38

Capacitance [fF]	
CK	0.9233
D	1.1030
RN	2.4837

Leakage [nW]
83.54

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	4023.72	4457.48
	rise	4854.52	5058.35
D	fall	1874.69	1942.90
	rise	952.30	992.89
RN	fall	962.52	887.30
	rise	-802.78	-815.95

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	4.63	5.32	4.83	5.49
	rise	5.52	5.56	5.68	5.69
CK to QN	fall	5.52	*5.53*	5.68	*5.74*
	rise	4.63	*5.30*	4.82	*5.47*
RN to Q	fall	5.09	5.55	5.70	5.92
RN to QN	rise	5.09	*5.58*	5.70	*5.93*

DFFS_X1

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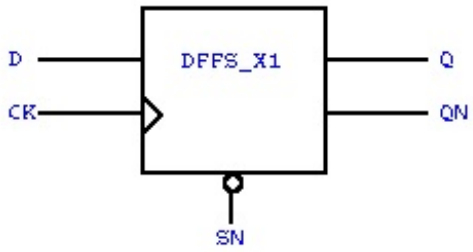
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, and drive strength X1
Strength	1
Cell Area	5.320 um^2
Equation	Q = "((!SN)+(D))" QN = "(!((!SN)+(D)))"
Clock	CK
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.24	0.40	0.37	0.53
	rise	0.30	0.79	0.42	0.91
CK to QN	fall	0.21	*0.44*	0.33	*0.56*
	rise	0.17	*0.69*	0.30	*0.82*
SN to Q	rise	0.84	1.28	1.09	1.54
SN to QN	fall	0.66	*1.06*	0.91	*1.31*

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.02	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
CK to QN	fall	0.04	*0.15*	0.04	*0.15*
	rise	0.02	*0.50*	0.02	*0.50*
SN to Q	rise	0.04	0.50	0.04	0.50
SN to QN	fall	0.12	*0.22*	0.12	*0.22*

Constraints Time [ns]		
Setup CK to D	fall	0.37
	rise	0.20
Hold CK to D	fall	0.36
	rise	0.32
Recovery CK to SN	rise	-0.09
Removal CK to SN	rise	1.09

Capacitance [fF]	
CK	0.9281
D	1.1439
SN	1.3653

Leakage [nW]
55.81

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	4613.91	5039.81
	rise	3507.23	3841.36
D	fall	1792.94	1859.40
	rise	750.86	765.46
SN	fall	735.17	818.77
	rise	-232.42	-187.94

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	2.93	3.29	3.11	3.47
	rise	3.12	3.39	3.29	3.53
CK to QN	fall	3.12	*3.37*	3.29	*3.52*
	rise	2.93	*3.29*	3.12	*3.46*
SN to Q	rise	3.57	3.53	3.72	3.68
SN to QN	fall	3.56	*3.55*	3.72	*3.68*

DFFS_X2

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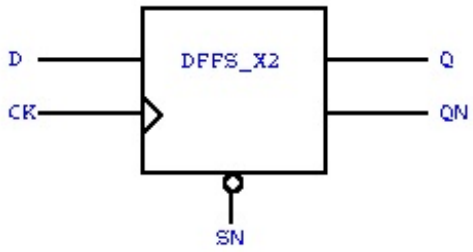
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, and drive strength X2
Strength	2
Cell Area	5.586 um^2
Equation	Q = "((!SN)+(D))" QN = "(!((!SN)+(D)))"
Clock	CK
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.27	0.42	0.40	0.55
	rise	0.42	0.89	0.54	1.01
CK to QN	fall	0.30	*0.59*	0.42	*0.72*
	rise	0.20	*0.73*	0.33	*0.86*
SN to Q	rise	1.27	1.65	1.52	1.90
SN to QN	fall	1.02	*1.54*	1.27	*1.79*

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.02	0.13	0.02	0.13
	rise	0.03	0.50	0.03	0.50
CK to QN	fall	0.06	*0.18*	0.06	*0.18*
	rise	0.02	*0.50*	0.02	*0.50*
SN to Q	rise	0.06	0.50	0.06	0.50
SN to QN	fall	0.21	*0.30*	0.21	*0.30*

Constraints Time [ns]		
Setup CK to D	fall	0.38
	rise	0.19
Hold CK to D	fall	0.36
	rise	0.32
Recovery CK to SN	rise	-0.07
Removal CK to SN	rise	0.28

Capacitance [fF]	
CK	0.9294
D	1.1422
SN	1.3346

Leakage [nW]
81.79

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	4698.99	5126.57
	rise	3366.74	3709.05
D	fall	1826.14	1891.73
	rise	742.10	756.62
SN	fall	729.09	813.12
	rise	-232.70	-187.95

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	4.34	4.96	4.50	5.11
	rise	5.12	5.19	5.27	5.37
CK to QN	fall	5.12	*5.20*	5.27	*5.39*
	rise	4.32	*4.98*	4.50	*5.13*
SN to Q	rise	6.63	5.89	7.09	6.04
SN to QN	fall	6.63	*5.88*	7.09	*6.05*

DFF_X1

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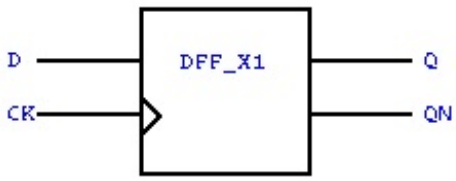
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with drive strength X1
Strength	1
Cell Area	4.522 um^2
Equation	Q = "(D)" QN = "(!(D))"
Clock	CK
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
CK	D	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	-	-	L	H
R	H	-	-	H	L
F	-	L	H	L	H
F	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to Q	fall	0.23	0.39	0.35	0.51
	rise	0.29	0.78	0.41	0.90
CK to QN	fall	0.21	*0.44*	0.32	*0.55*
	rise	0.17	*0.69*	0.29	*0.81*

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to Q	fall	0.02	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
CK to QN	fall	0.03	*0.15*	0.03	*0.15*
	rise	0.02	*0.50*	0.02	*0.50*

Constraints Time [ns]		
Setup CK to D	fall	0.37
	rise	0.20
Hold CK to D	fall	0.35
	rise	0.32

Capacitance [fF]	
CK	0.9096
D	1.1091

Leakage [nW]
54.44

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	4461.29	4889.04
	rise	1924.44	2260.55
D	fall	1820.56	1897.08
	rise	953.06	984.49

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to Q	fall	2.90	3.25	3.08	3.43
	rise	3.06	3.29	3.23	3.45
CK to QN	fall	3.06	*3.29*	3.23	*3.44*
	rise	2.90	*3.24*	3.08	*3.43*

DFF_X2

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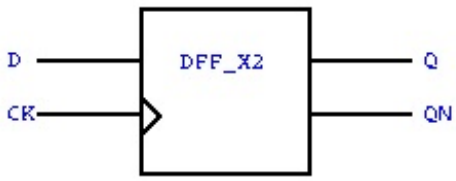
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with drive strength X2
Strength	2
Cell Area	5.054 um^2
Equation	Q = "(D)" QN = "(!(D))"
Clock	CK
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
CK	**D**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**	**QN**
R	L	-	-	L	H
R	H	-	-	H	L
F	-	L	H	L	H
F	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
CK to Q	**fall**	0.26	0.42	0.38	0.54
	rise	0.42	0.88	0.53	1.00
CK to QN	**fall**	0.30	*0.59*	0.41	*0.71*
	rise	0.19	*0.72*	0.32	*0.85*

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
CK to Q	**fall**	0.02	0.13	0.02	0.13
	rise	0.03	0.50	0.03	0.50
CK to QN	**fall**	0.06	*0.18*	0.06	*0.18*
	rise	0.02	*0.50*	0.02	*0.50*

Constraints Time [ns]		
Setup CK to D	**fall**	0.38
	rise	0.19
Hold CK to D	**fall**	0.35
	rise	0.31

Capacitance [fF]	
CK	0.8903
D	1.0921

Leakage [nW]
80.73

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0023**	**0.50**
CK	**fall**	4488.87	4914.42
	rise	1937.06	2273.68
D	**fall**	1847.59	1920.56
	rise	985.37	1016.34

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
CK to Q	**fall**	4.32	4.99	4.51	5.12
	rise	5.08	5.11	5.24	5.27
CK to QN	**fall**	5.08	*5.13*	5.24	*5.28*
	rise	4.32	*4.94*	4.51	*5.13*

DLH_X1

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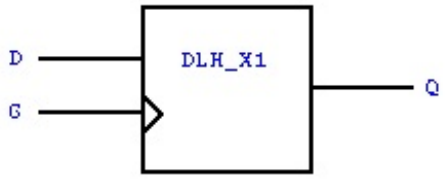
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	High enable Latch with drive strength X1
Strength	1
Cell Area	2.660 um^2
Equation	Q = "(D)"
Enable	G
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**G**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	H	-	-	L
H	H	-	-	H
-	L	L	H	L
-	L	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
D to Q	**fall**	0.20	0.42	0.42	0.65
	rise	0.08	0.60	0.21	0.73
G to Q	**fall**	0.18	0.40	0.28	0.49
	rise	0.14	0.66	0.24	0.76

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
D to Q	**fall**	0.03	0.15	0.04	0.15
	rise	0.02	0.50	0.03	0.50
G to Q	**fall**	0.03	0.15	0.03	0.15
	rise	0.02	0.50	0.02	0.50

Constraints Time [ns]		
Setup G to D	**fall**	0.39
	rise	0.14
Hold G to D	**fall**	0.58
	rise	0.41

Capacitance [fF]	
D	0.8897
G	0.9475

Leakage [nW]
28.42

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0023**	**0.50**
D	**fall**	386.07	372.42
	rise	-290.51	-343.88
G	**fall**	2293.64	2751.05
	rise	1688.71	2085.03

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
D to Q	**fall**	2.57	3.40	2.68	3.48
	rise	1.76	2.34	1.89	2.37
G to Q	**fall**	3.55	4.39	3.96	4.76
	rise	3.47	4.02	3.91	4.46

DLH_X2

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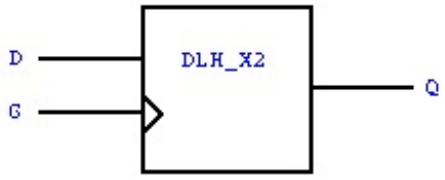
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	High enable Latch with drive strength X2
Strength	2
Cell Area	2.926 um^2
Equation	Q = "(D)"
Enable	G
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**G**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	H	-	-	L
H	H	-	-	H
-	L	L	H	L
-	L	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**120.697**	**0.3656**	**120.697**
D to Q	**fall**	0.22	0.45	0.44	0.68
	rise	0.08	0.60	0.22	0.75
G to Q	**fall**	0.20	0.43	0.30	0.53
	rise	0.14	0.66	0.24	0.76

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**120.697**	**0.3656**	**120.697**
D to Q	**fall**	0.04	0.16	0.05	0.16
	rise	0.02	0.50	0.03	0.50
G to Q	**fall**	0.04	0.16	0.04	0.16
	rise	0.02	0.50	0.02	0.50

Constraints Time [ns]		
Setup G to D	**fall**	0.42
	rise	0.15
Hold G to D	**fall**	0.56
	rise	0.42

Capacitance [fF]	
D	1.1319
G	0.9469

Leakage [nW]
40.26

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0023**	**0.50**
D	**fall**	520.21	500.37
	rise	-396.10	-462.08
G	**fall**	2218.08	2675.30
	rise	1700.43	2095.57

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**120.697**	**0.3656**	**120.697**
D to Q	**fall**	4.17	5.21	4.44	5.29
	rise	2.94	3.84	3.18	3.76
G to Q	**fall**	4.92	5.95	5.33	6.32
	rise	4.68	5.49	5.10	5.95

DLL_X1

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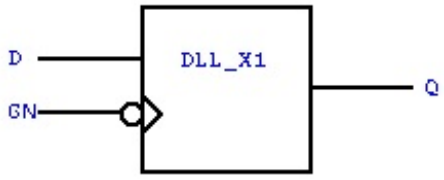
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Low enable Latch with drive strength X1
Strength	1
Cell Area	2.660 um^2
Equation	Q = "(D)"
Enable	GN
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**GN**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	L	-	-	L
H	L	-	-	H
-	H	L	H	L
-	H	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
D to Q	**fall**	0.20	0.41	0.41	0.64
	rise	0.08	0.59	0.21	0.73
GN to Q	**fall**	0.24	0.46	0.45	0.66
	rise	0.14	0.65	0.36	0.87

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
D to Q	**fall**	0.03	0.15	0.04	0.15
	rise	0.02	0.50	0.03	0.50
GN to Q	**fall**	0.03	0.15	0.03	0.15
	rise	0.02	0.50	0.02	0.50

Constraints Time [ns]		
Setup GN to D	**fall**	0.38
	rise	0.20
Hold GN to D	**fall**	0.33
	rise	0.35

Capacitance [fF]	
D	0.8546
GN	0.9457

Leakage [nW]
28.43

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0023**	**0.50**
D	**fall**	385.43	372.45
	rise	-276.20	-332.63
GN	**fall**	2469.75	2945.17
	rise	1358.54	1762.43

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
D to Q	**fall**	2.63	3.39	2.73	3.46
	rise	1.90	2.47	2.04	2.45
GN to Q	**fall**	4.46	5.21	4.85	5.70
	rise	4.37	4.90	4.69	5.35

DLL_X2

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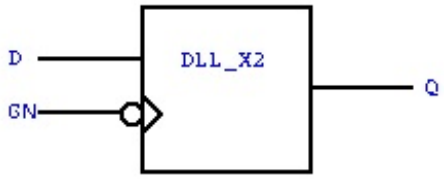
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Low enable Latch with drive strength X2
Strength	2
Cell Area	2.926 um^2
Equation	Q = "(D)"
Enable	GN
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**GN**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	L	-	-	L
H	L	-	-	H
-	H	L	H	L
-	H	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
D to Q	**fall**	0.21	0.44	0.44	0.67
	rise	0.08	0.60	0.22	0.74
GN to Q	**fall**	0.26	0.48	0.47	0.69
	rise	0.14	0.66	0.36	0.88

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
D to Q	**fall**	0.04	0.15	0.04	0.16
	rise	0.02	0.50	0.03	0.50
GN to Q	**fall**	0.04	0.15	0.04	0.15
	rise	0.02	0.50	0.02	0.50

Constraints Time [ns]		
Setup GN to D	**fall**	0.41
	rise	0.21
Hold GN to D	**fall**	0.31
	rise	0.37

Capacitance [fF]	
D	1.0999
GN	0.9415

Leakage [nW]
40.26

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0023**	**0.50**
D	**fall**	520.04	500.67
	rise	-382.36	-459.26
GN	**fall**	2571.58	3047.58
	rise	1297.05	1696.50

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
D to Q	**fall**	4.05	5.02	4.31	5.10
	rise	2.92	3.81	3.17	3.80
GN to Q	**fall**	5.72	6.70	6.15	7.18
	rise	5.51	6.42	5.87	6.78

FA_X1

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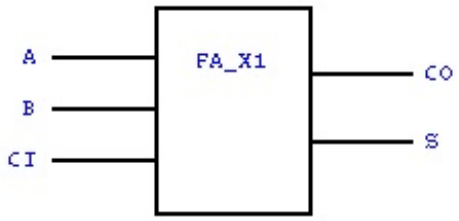
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	4.256 um^2
Equation	CO = "((A & B) | (CI & (A | B)))" S = "(CI ^ (A ^ B))"
Type	Combinational
Input	A, B, CI
Output	CO, S
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**CI**	**CO**	**S**
L	L	L	L	L
L	L	H	L	H
L	H	L	L	H
L	H	H	H	L
H	L	L	L	H
H	L	H	H	L
H	H	L	H	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A to CO	**fall**	0.26	*0.51*	0.51	*0.77*
	rise	0.10	*0.62*	0.26	*0.79*
A to S	**fall**	0.36	0.62	0.55	0.82
	rise	0.39	0.88	0.65	1.14
B to CO	**fall**	0.26	*0.51*	0.49	*0.75*
	rise	0.10	*0.62*	0.25	*0.78*
B to S	**fall**	0.35	0.61	0.57	0.84
	rise	0.40	0.90	0.64	1.13
CI to CO	**fall**	0.23	*0.48*	0.49	*0.75*
	rise	0.09	*0.61*	0.24	*0.77*
CI to S	**fall**	0.32	0.59	0.56	0.83
	rise	0.38	0.87	0.64	1.13

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A to CO	**fall**	0.04	*0.17*	0.05	*0.17*
	rise	0.02	*0.50*	0.04	*0.50*
A to S	**fall**	0.05	0.17	0.05	0.17
	rise	0.03	0.50	0.03	0.50
B to CO	**fall**	0.04	*0.16*	0.04	*0.16*
	rise	0.02	*0.50*	0.03	*0.50*
B to S	**fall**	0.05	0.17	0.05	0.17
	rise	0.03	0.50	0.03	0.50
CI to CO	**fall**	0.04	*0.17*	0.05	*0.17*
	rise	0.02	*0.50*	0.04	*0.50*
CI to S	**fall**	0.05	0.17	0.05	0.17
	rise	0.03	0.50	0.04	0.50

Capacitance [fF]	
A	3.7010
B	3.4732
CI	2.7393

Leakage [nW]
46.59

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.2722	0.3656	60.2722
A to CO	fall	3.42	*3.74*	3.54	*3.79*
	rise	1.47	*1.99*	1.64	*2.01*
A to S	fall	2.77	3.39	2.83	3.37
	rise	2.16	2.52	2.33	2.49
B to CO	fall	3.59	*3.87*	3.64	*3.89*
	rise	1.56	*2.09*	1.66	*2.05*
B to S	fall	2.70	3.33	2.73	3.28
	rise	2.19	2.51	2.34	2.47
CI to CO	fall	3.15	*3.49*	3.36	*3.61*
	rise	1.78	*2.26*	1.97	*2.29*
CI to S	fall	2.49	3.11	2.61	3.15
	rise	2.19	2.56	2.42	2.61

FILLCELL_X16

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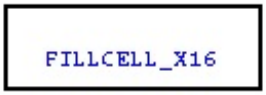
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	4.256 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X1

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.266 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X2

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	0.266 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X32

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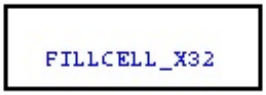
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	32
Cell Area	8.512 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X4

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	1.064 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X8

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	2.128 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



HA_X1

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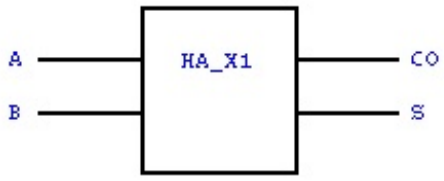
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.660 um^2
Equation	CO = "(A & B)" S = "(A ^ B)"
Type	Combinational
Input	A, B
Output	CO, S
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B**	**CO**	**S**
L	L	L	L
L	H	L	H
H	L	L	H
H	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A to CO	**fall**	0.07	0.24	0.28	0.47
	rise	0.06	0.58	0.17	0.69
A to S	**fall**	0.17	*0.37*	0.40	*0.68*
	rise	0.12	*1.27*	0.24	*1.48*
B to CO	**fall**	0.08	0.25	0.30	0.49
	rise	0.06	0.58	0.18	0.70
B to S	**fall**	0.18	*0.38*	0.39	*0.62*
	rise	0.11	*1.27*	0.28	*1.50*

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A to CO	**fall**	0.01	0.13	0.03	0.14
	rise	0.02	0.50	0.03	0.50
A to S	**fall**	0.03	*0.26*	0.09	*0.32*
	rise	0.06	*1.12*	0.12	*1.13*
B to CO	**fall**	0.01	0.13	0.03	0.14
	rise	0.02	0.50	0.03	0.50
B to S	**fall**	0.03	*0.26*	0.07	*0.27*
	rise	0.06	*1.12*	0.12	*1.11*

Capacitance [fF]	
A	3.1624
B	3.3960

Leakage [nW]
40.09

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A to CO	fall	2.24	2.11	2.66	2.54
	rise	0.77	0.88	1.35	1.08
A to S	fall	2.36	*2.55*	2.63	*2.87*
	rise	2.40	*2.31*	2.69	*2.61*
B to CO	fall	2.76	2.65	3.18	3.09
	rise	0.68	0.78	0.98	0.81
B to S	fall	2.43	*2.66*	2.39	*2.74*
	rise	2.47	*2.51*	2.86	*2.77*

INV_X16

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	4.522 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	969.238	0.3656	969.238
A to ZN	**fall**	0.01	0.15	-0.01	0.42
	rise	0.02	0.53	0.13	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	969.238	0.3656	969.238
A to ZN	**fall**	0.00	0.13	0.07	0.21
	rise	0.01	0.50	0.08	0.50

Capacitance [fF]	
A	24.6946

Leakage [nW]
167.51

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	969.238	0.3656	969.238
A to ZN	**fall**	-0.00	-0.02	13.95	0.85
	rise	21.31	22.48	34.29	25.71

INV_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.532 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
A to ZN	**fall**	0.01	0.15	-0.00	0.42
	rise	0.02	0.53	0.14	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
A to ZN	**fall**	0.00	0.13	0.07	0.21
	rise	0.01	0.50	0.08	0.50

Capacitance [fF]	
A	1.6437

Leakage [nW]
10.46

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
A to ZN	**fall**	-0.00	-0.00	0.78	-0.01
	rise	1.40	1.49	2.18	1.61

INV_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	0.798 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to ZN	**fall**	0.01	0.15	-0.00	0.42
	rise	0.02	0.53	0.14	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to ZN	**fall**	0.00	0.13	0.07	0.21
	rise	0.01	0.50	0.08	0.50

Capacitance [fF]	
A	3.1360

Leakage [nW]
20.94

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to ZN	**fall**	-0.00	-0.00	1.72	0.10
	rise	2.74	2.89	4.33	3.26

INV_X32

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	32
Cell Area	8.778 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	1923.83	0.3656	1923.83
A to ZN	**fall**	0.01	0.15	-0.01	0.42
	rise	0.02	0.53	0.13	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	1923.83	0.3656	1923.83
A to ZN	**fall**	0.00	0.13	0.07	0.21
	rise	0.01	0.50	0.08	0.50

Capacitance [fF]	
A	48.5578

Leakage [nW]
335.02

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	1923.83	0.3656	1923.83
A to ZN	**fall**	-0.00	-0.04	28.06	1.84
	rise	42.15	44.25	68.17	51.05

INV_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	1.330 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	242.92	0.3656	242.92
A to ZN	**fall**	0.01	0.15	-0.01	0.42
	rise	0.02	0.53	0.13	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	242.92	0.3656	242.92
A to ZN	**fall**	0.00	0.13	0.07	0.21
	rise	0.01	0.50	0.08	0.50

Capacitance [fF]	
A	6.1142

Leakage [nW]
41.88

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	242.92	0.3656	242.92
A to ZN	**fall**	-0.00	-0.00	3.47	0.21
	rise	5.34	5.65	8.55	6.44

INV_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	2.394 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	485.229	0.3656	485.229
A to ZN	**fall**	0.01	0.15	-0.01	0.42
	rise	0.02	0.53	0.13	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	485.229	0.3656	485.229
A to ZN	**fall**	0.00	0.13	0.07	0.21
	rise	0.01	0.50	0.08	0.50

Capacitance [fF]	
A	11.8074

Leakage [nW]
83.75

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	485.229	0.3656	485.229
A to ZN	**fall**	-0.00	-0.01	7.00	0.42
	rise	10.57	11.24	17.12	12.84

NANGATE

LOGIC0_X1

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.532 um^2
Equation	Z = "0"
Type	Constant
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)

LOGIC1_X1

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.532 um^2
Equation	Z = "1"
Type	Constant
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)

MUX2_X1

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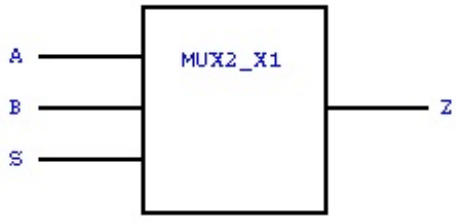
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.862 um^2
Equation	Z = "((S & B) | (A & !S))"
Type	Combinational
Input	A, B, S
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B**	**S**	**Z**
L	-	L	L
H	-	L	H
-	L	H	L
-	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.5011**	**0.3656**	**60.5011**
A to Z	**fall**	0.19	0.41	0.41	0.64
	rise	0.08	0.59	0.21	0.73
B to Z	**fall**	0.20	0.41	0.41	0.64
	rise	0.07	0.59	0.21	0.73
S to Z	**fall**	0.18	0.40	0.39	0.62
	rise	0.13	0.64	0.33	0.85

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.5011**	**0.3656**	**60.5011**
A to Z	**fall**	0.03	0.15	0.04	0.15
	rise	0.02	0.50	0.03	0.50
B to Z	**fall**	0.03	0.15	0.04	0.15
	rise	0.02	0.50	0.03	0.50
S to Z	**fall**	0.03	0.15	0.05	0.15
	rise	0.02	0.50	0.04	0.50

Capacitance [fF]	
A	0.9254
B	0.9198
S	1.8935

Leakage [nW]
24.27

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.5011**	**0.3656**	**60.5011**
A to Z	**fall**	2.36	3.00	2.44	3.06
	rise	1.77	2.05	1.90	2.07
B to Z	**fall**	2.53	3.25	2.61	3.31
	rise	1.56	1.84	1.69	1.82
S to Z	**fall**	2.68	3.36	3.31	3.93
	rise	2.61	2.85	3.18	3.45

MUX2_X2

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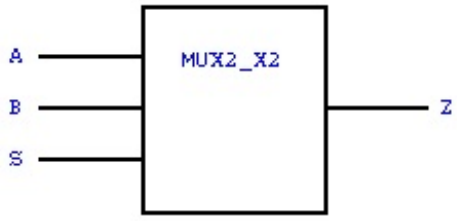
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	Z = "((S & B) | (A & !S))"
Type	Combinational
Input	A, B, S
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B**	**S**	**Z**
L	-	L	L
H	-	L	H
-	L	H	L
-	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A to Z	**fall**	0.18	0.39	0.40	0.62
	rise	0.08	0.60	0.22	0.74
B to Z	**fall**	0.19	0.41	0.44	0.67
	rise	0.07	0.59	0.19	0.72
S to Z	**fall**	0.23	0.44	0.35	0.58
	rise	0.15	0.67	0.38	0.90

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A to Z	**fall**	0.03	0.15	0.04	0.15
	rise	0.02	0.50	0.03	0.50
B to Z	**fall**	0.03	0.15	0.04	0.15
	rise	0.02	0.50	0.03	0.50
S to Z	**fall**	0.03	0.15	0.04	0.15
	rise	0.02	0.50	0.03	0.50

Capacitance [fF]	
A	1.5570
B	1.6710
S	2.5967

Leakage [nW]
45.53

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A to Z	**fall**	4.80	6.31	4.77	6.30
	rise	3.33	4.05	3.99	4.25
B to Z	**fall**	4.68	6.21	5.50	6.78
	rise	2.65	3.03	2.99	3.19
S to Z	**fall**	5.45	6.95	6.23	7.61
	rise	4.84	5.41	5.45	6.00

NANGATE NAND2_X1

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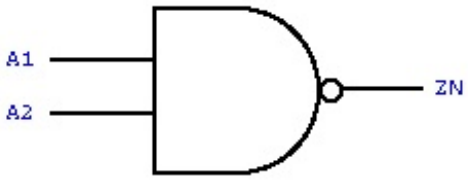
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	ZN = "!(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	59.3567	0.3656	59.3567
A1 to ZN	**fall**	0.02	0.30	0.04	0.57
	rise	0.03	0.52	0.16	0.83
A2 to ZN	**fall**	0.02	0.30	0.06	0.54
	rise	0.03	0.53	0.19	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	59.3567	0.3656	59.3567
A1 to ZN	**fall**	0.01	0.25	0.08	0.29
	rise	0.02	0.49	0.09	0.49
A2 to ZN	**fall**	0.01	0.25	0.07	0.27
	rise	0.02	0.50	0.09	0.50

Capacitance [fF]	
A1	1.5682
A2	1.6132

Leakage [nW]
11.38

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	59.3567	0.3656	59.3567
A1 to ZN	**fall**	0.20	0.21	1.05	0.32
	rise	1.64	1.68	2.30	1.84
A2 to ZN	**fall**	0.20	0.22	0.65	0.24
	rise	2.16	2.23	2.78	2.41

NANGATE NAND2_X2

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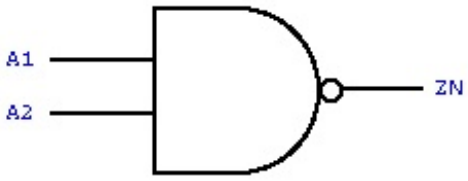
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "!(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	118.713	0.3656	118.713
A1 to ZN	**fall**	0.02	0.30	0.03	0.57
	rise	0.02	0.52	0.16	0.83
A2 to ZN	**fall**	0.02	0.30	0.06	0.54
	rise	0.03	0.53	0.19	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	118.713	0.3656	118.713
A1 to ZN	**fall**	0.01	0.25	0.08	0.29
	rise	0.02	0.49	0.09	0.50
A2 to ZN	**fall**	0.01	0.25	0.06	0.27
	rise	0.02	0.50	0.09	0.50

Capacitance [fF]	
A1	3.0027
A2	3.3561

Leakage [nW]
22.76

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	118.713	0.3656	118.713
A1 to ZN	**fall**	0.40	0.43	2.13	0.65
	rise	3.35	3.40	4.67	3.72
A2 to ZN	**fall**	0.40	0.43	1.31	0.47
	rise	4.38	4.48	5.63	4.87

NANGATE NAND2_X4

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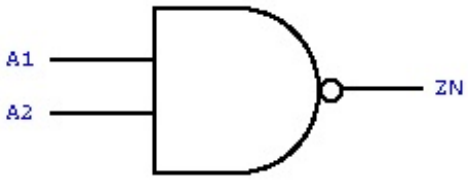
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "!(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	237.427	0.3656	237.427
A1 to ZN	**fall**	0.01	0.30	0.03	0.57
	rise	0.02	0.52	0.15	0.83
A2 to ZN	**fall**	0.02	0.30	0.05	0.54
	rise	0.03	0.53	0.18	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	237.427	0.3656	237.427
A1 to ZN	**fall**	0.01	0.25	0.08	0.29
	rise	0.01	0.49	0.09	0.49
A2 to ZN	**fall**	0.01	0.25	0.06	0.27
	rise	0.02	0.50	0.08	0.50

Capacitance [fF]	
A1	5.8842
A2	6.0406

Leakage [nW]
45.53

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	237.427	0.3656	237.427
A1 to ZN	**fall**	0.76	0.86	4.33	1.30
	rise	6.17	6.44	8.90	6.85
A2 to ZN	**fall**	0.81	0.86	2.59	0.94
	rise	8.49	8.96	10.95	9.75

NANGATE NAND3_X1

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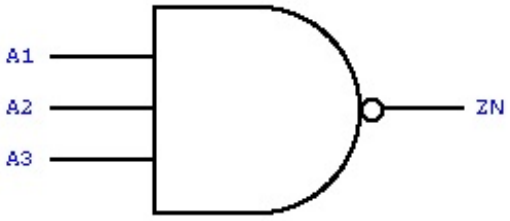
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	H
H	H	H	L
-	L	-	H
-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**58.3649**	**0.3656**	**58.3649**
A1 to ZN	**fall**	0.03	0.45	0.07	0.69
	rise	0.03	0.52	0.17	0.82
A2 to ZN	**fall**	0.03	0.46	0.10	0.69
	rise	0.04	0.53	0.20	0.83
A3 to ZN	**fall**	0.04	0.46	0.10	0.65
	rise	0.04	0.54	0.22	0.84

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**58.3649**	**0.3656**	**58.3649**
A1 to ZN	**fall**	0.02	0.38	0.09	0.39
	rise	0.02	0.49	0.10	0.49
A2 to ZN	**fall**	0.02	0.38	0.08	0.39
	rise	0.03	0.50	0.10	0.50
A3 to ZN	**fall**	0.02	0.38	0.07	0.39
	rise	0.03	0.50	0.10	0.50

Capacitance [fF]	
A1	1.5714
A2	1.5991
A3	1.5977

Leakage [nW]
10.62

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**58.3649**	**0.3656**	**58.3649**
A1 to ZN	**fall**	0.44	0.45	1.20	0.54
	rise	1.85	1.87	2.43	2.03
A2 to ZN	**fall**	0.44	0.45	0.82	0.45
	rise	2.40	2.43	2.93	2.60
A3 to ZN	**fall**	0.44	0.45	0.68	0.44
	rise	2.81	3.00	3.38	3.11

NANGATE NAND3_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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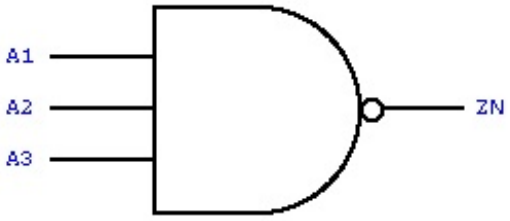
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	H
H	H	H	L
-	L	-	H
-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**116.272**	**0.3656**	**116.272**
A1 to ZN	**fall**	0.03	0.45	0.07	0.69
	rise	0.03	0.52	0.17	0.82
A2 to ZN	**fall**	0.03	0.46	0.10	0.69
	rise	0.04	0.53	0.20	0.83
A3 to ZN	**fall**	0.04	0.46	0.10	0.65
	rise	0.04	0.54	0.22	0.84

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**116.272**	**0.3656**	**116.272**
A1 to ZN	**fall**	0.02	0.38	0.09	0.39
	rise	0.02	0.49	0.10	0.49
A2 to ZN	**fall**	0.02	0.38	0.08	0.39
	rise	0.03	0.50	0.10	0.50
A3 to ZN	**fall**	0.02	0.38	0.07	0.39
	rise	0.03	0.50	0.10	0.50

Capacitance [fF]	
A1	2.9483
A2	3.2335
A3	3.4603

Leakage [nW]
21.26

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**116.272**	**0.3656**	**116.272**
A1 to ZN	**fall**	1.01	1.03	2.54	1.21
	rise	3.85	3.88	5.02	4.21
A2 to ZN	**fall**	1.03	1.03	1.78	1.05
	rise	4.95	5.02	6.00	5.36
A3 to ZN	**fall**	1.02	1.03	1.50	1.01
	rise	5.76	6.16	6.91	6.38

NANGATE NAND3_X4

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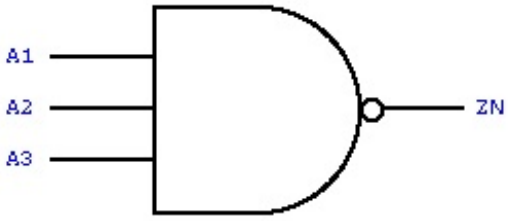
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	H
H	H	H	L
-	L	-	H
-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	233.154	0.3656	233.154
A1 to ZN	**fall**	0.03	0.45	0.07	0.69
	rise	0.03	0.52	0.17	0.82
A2 to ZN	**fall**	0.03	0.46	0.09	0.69
	rise	0.04	0.53	0.20	0.83
A3 to ZN	**fall**	0.03	0.46	0.09	0.65
	rise	0.04	0.54	0.22	0.84

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	233.154	0.3656	233.154
A1 to ZN	**fall**	0.02	0.38	0.09	0.39
	rise	0.02	0.49	0.10	0.49
A2 to ZN	**fall**	0.02	0.38	0.08	0.39
	rise	0.02	0.50	0.10	0.50
A3 to ZN	**fall**	0.02	0.38	0.07	0.39
	rise	0.03	0.50	0.10	0.50

Capacitance [fF]	
A1	6.1997
A2	6.8325
A3	6.9671

Leakage [nW]
42.51

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	233.154	0.3656	233.154
A1 to ZN	**fall**	2.02	2.07	5.13	2.43
	rise	7.50	7.48	9.87	8.15
A2 to ZN	**fall**	2.03	2.06	3.58	2.09
	rise	9.70	10.00	11.83	10.70
A3 to ZN	**fall**	2.05	2.07	3.03	2.03
	rise	11.32	12.02	13.62	12.74

NANGATE NAND4_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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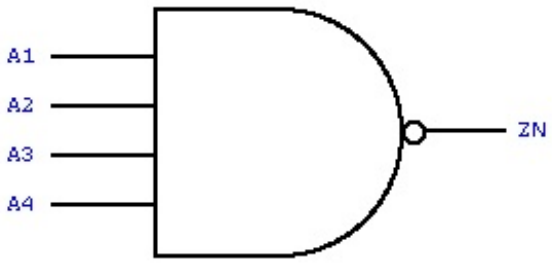
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	H
H	H	H	H	L
-	L	-	-	H
-	-	L	-	H
-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**55.9998**	**0.3656**	**55.9998**
A1 to ZN	**fall**	0.04	0.59	0.11	0.81
	rise	0.04	0.51	0.19	0.81
A2 to ZN	**fall**	0.05	0.60	0.14	0.83
	rise	0.04	0.51	0.22	0.82
A3 to ZN	**fall**	0.06	0.61	0.14	0.80
	rise	0.05	0.52	0.24	0.82
A4 to ZN	**fall**	0.06	0.61	0.13	0.76
	rise	0.05	0.53	0.25	0.83

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**55.9998**	**0.3656**	**55.9998**
A1 to ZN	**fall**	0.03	0.50	0.11	0.50
	rise	0.03	0.48	0.11	0.48
A2 to ZN	**fall**	0.03	0.50	0.09	0.50
	rise	0.03	0.48	0.11	0.48
A3 to ZN	**fall**	0.03	0.50	0.08	0.50
	rise	0.03	0.49	0.11	0.49
A4 to ZN	**fall**	0.03	0.50	0.07	0.50
	rise	0.04	0.49	0.11	0.49

Capacitance [fF]	
A1	1.5116
A2	1.5839
A3	1.6121
A4	1.6037

Leakage [nW]
9.71

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	55.9998	0.3656	55.9998
A1 to ZN	fall	0.81	0.82	1.48	0.90
	rise	2.06	2.06	2.59	2.22
A2 to ZN	fall	0.82	0.81	1.12	0.81
	rise	2.61	2.69	3.09	2.79
A3 to ZN	fall	0.82	0.82	1.00	0.79
	rise	3.06	3.21	3.55	3.37
A4 to ZN	fall	0.82	0.82	0.93	0.79
	rise	3.43	3.69	3.97	3.85

NANGATE NAND4_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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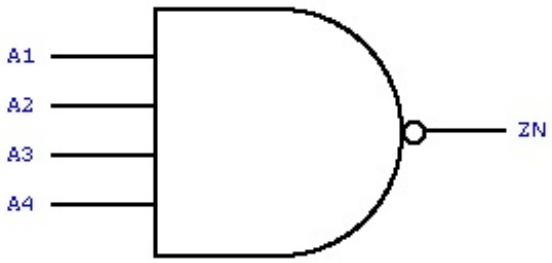
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	H
H	H	H	H	L
-	L	-	-	H
-	-	L	-	H
-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**111.542**	**0.3656**	**111.542**
A1 to ZN	**fall**	0.04	0.59	0.11	0.81
	rise	0.04	0.51	0.19	0.81
A2 to ZN	**fall**	0.05	0.60	0.14	0.83
	rise	0.04	0.51	0.22	0.82
A3 to ZN	**fall**	0.06	0.61	0.14	0.80
	rise	0.05	0.52	0.24	0.82
A4 to ZN	**fall**	0.06	0.61	0.13	0.76
	rise	0.06	0.53	0.26	0.83

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**111.542**	**0.3656**	**111.542**
A1 to ZN	**fall**	0.03	0.50	0.11	0.50
	rise	0.03	0.48	0.11	0.48
A2 to ZN	**fall**	0.03	0.50	0.09	0.50
	rise	0.03	0.48	0.11	0.48
A3 to ZN	**fall**	0.03	0.50	0.08	0.50
	rise	0.04	0.49	0.11	0.49
A4 to ZN	**fall**	0.03	0.50	0.07	0.50
	rise	0.04	0.49	0.11	0.49

Capacitance [fF]	
A1	2.9121
A2	3.2627
A3	3.4280
A4	3.7125

Leakage [nW]
19.42

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	111.542	0.3656	111.542
A1 to ZN	fall	1.63	1.63	2.95	1.79
	rise	4.48	4.44	5.52	4.77
A2 to ZN	fall	1.64	1.63	2.24	1.63
	rise	5.57	5.71	6.55	6.06
A3 to ZN	fall	1.64	1.63	2.00	1.59
	rise	6.49	6.77	7.46	7.09
A4 to ZN	fall	1.64	1.63	1.86	1.57
	rise	7.20	7.74	8.30	8.06

NANGATE NAND4_X4

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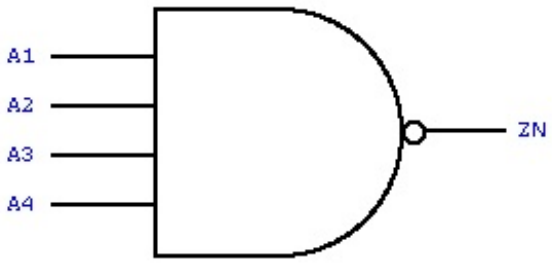
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.788 um^2
Equation	ZN = "!(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	H
H	H	H	H	L
-	L	-	-	H
-	-	L	-	H
-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**222.778**	**0.3656**	**222.778**
A1 to ZN	**fall**	0.04	0.59	0.10	0.81
	rise	0.03	0.50	0.18	0.80
A2 to ZN	**fall**	0.05	0.60	0.13	0.83
	rise	0.04	0.51	0.21	0.81
A3 to ZN	**fall**	0.06	0.61	0.13	0.80
	rise	0.05	0.52	0.23	0.82
A4 to ZN	**fall**	0.06	0.61	0.13	0.77
	rise	0.05	0.53	0.25	0.83

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**222.778**	**0.3656**	**222.778**
A1 to ZN	**fall**	0.03	0.50	0.10	0.50
	rise	0.02	0.47	0.11	0.47
A2 to ZN	**fall**	0.03	0.50	0.09	0.50
	rise	0.03	0.48	0.11	0.48
A3 to ZN	**fall**	0.03	0.50	0.08	0.50
	rise	0.03	0.48	0.11	0.49
A4 to ZN	**fall**	0.03	0.50	0.06	0.50
	rise	0.04	0.49	0.11	0.49

Capacitance [fF]	
A1	5.6415
A2	5.7942
A3	5.8606
A4	5.9480

Leakage [nW]
38.85

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	222.778	0.3656	222.778
A1 to ZN	fall	3.24	3.26	6.02	3.58
	rise	7.85	7.95	10.02	8.65
A2 to ZN	fall	3.28	3.27	4.46	3.23
	rise	10.38	10.74	12.30	11.45
A3 to ZN	fall	3.27	3.26	3.92	3.16
	rise	12.31	12.97	14.26	13.62
A4 to ZN	fall	3.28	3.26	3.65	3.13
	rise	13.89	15.32	16.10	15.70

NANGATE NOR2_X1

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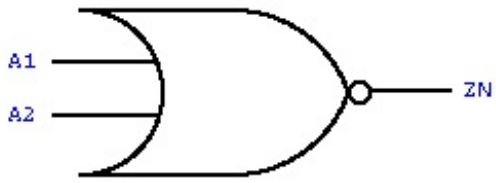
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	ZN = "!(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**26.7029**	**0.3656**	**26.7029**
A1 to ZN	**fall**	0.01	0.07	0.01	0.27
	rise	0.05	0.57	0.22	0.83
A2 to ZN	**fall**	0.01	0.07	0.04	0.28
	rise	0.07	0.58	0.23	0.82

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**26.7029**	**0.3656**	**26.7029**
A1 to ZN	**fall**	0.00	0.06	0.08	0.16
	rise	0.04	0.50	0.10	0.50
A2 to ZN	**fall**	0.01	0.06	0.08	0.16
	rise	0.04	0.50	0.09	0.50

Capacitance [fF]	
A1	1.6419
A2	1.5912

Leakage [nW]
12.82

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**26.7029**	**0.3656**	**26.7029**
A1 to ZN	**fall**	-0.00	-0.00	0.60	0.16
	rise	1.74	1.79	2.35	2.03
A2 to ZN	**fall**	0.01	0.20	0.83	0.43
	rise	2.33	2.36	2.37	2.35

NOR2_X2

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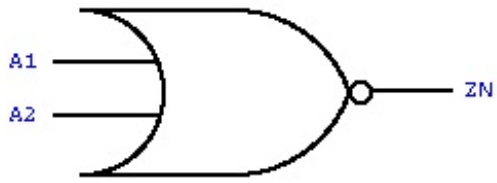
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "!(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**53.4058**	**0.3656**	**53.4058**
A1 to ZN	**fall**	0.01	0.07	0.00	0.27
	rise	0.05	0.57	0.21	0.83
A2 to ZN	**fall**	0.01	0.07	0.03	0.28
	rise	0.06	0.58	0.22	0.82

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**53.4058**	**0.3656**	**53.4058**
A1 to ZN	**fall**	0.00	0.06	0.08	0.16
	rise	0.03	0.50	0.10	0.50
A2 to ZN	**fall**	0.01	0.06	0.07	0.16
	rise	0.03	0.50	0.08	0.50

Capacitance [fF]	
A1	3.1489
A2	3.2340

Leakage [nW]
25.65

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**53.4058**	**0.3656**	**53.4058**
A1 to ZN	**fall**	-0.00	-0.00	1.35	0.46
	rise	3.34	3.44	4.56	3.97
A2 to ZN	**fall**	0.14	0.54	1.80	1.01
	rise	4.52	4.59	4.60	4.55

NOR2_X4

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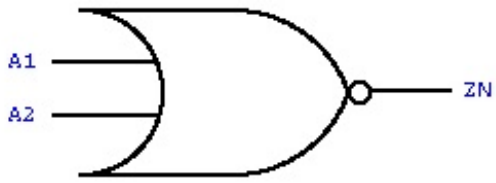
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "!(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	106.811	0.3656	106.811
A1 to ZN	**fall**	0.01	0.07	-0.00	0.27
	rise	0.05	0.57	0.20	0.83
A2 to ZN	**fall**	0.01	0.07	0.03	0.28
	rise	0.06	0.58	0.22	0.82

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	106.811	0.3656	106.811
A1 to ZN	**fall**	0.00	0.06	0.08	0.16
	rise	0.03	0.50	0.10	0.50
A2 to ZN	**fall**	0.00	0.06	0.07	0.16
	rise	0.03	0.50	0.08	0.50

Capacitance [fF]	
A1	6.4984
A2	6.4365

Leakage [nW]
51.29

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	106.811	0.3656	106.811
A1 to ZN	**fall**	-0.00	-0.00	2.74	0.92
	rise	6.54	6.84	9.03	7.72
A2 to ZN	**fall**	0.30	1.09	3.63	2.01
	rise	8.91	9.08	9.09	9.01

NANGATE NOR3_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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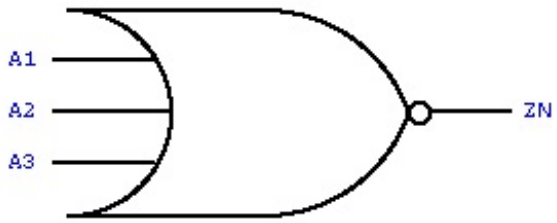
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	H
H	-	-	L
-	H	-	L
-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	16.0217	0.3656	16.0217
A1 to ZN	**fall**	0.01	0.04	0.01	0.21
	rise	0.09	0.58	0.28	0.82
A2 to ZN	**fall**	0.01	0.05	0.04	0.22
	rise	0.13	0.62	0.30	0.85
A3 to ZN	**fall**	0.01	0.05	0.05	0.22
	rise	0.14	0.63	0.30	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	16.0217	0.3656	16.0217
A1 to ZN	**fall**	0.00	0.04	0.08	0.14
	rise	0.07	0.50	0.13	0.50
A2 to ZN	**fall**	0.01	0.04	0.08	0.14
	rise	0.07	0.50	0.12	0.50
A3 to ZN	**fall**	0.01	0.04	0.08	0.14
	rise	0.07	0.50	0.10	0.50

Capacitance [fF]	
A1	1.6810
A2	1.5984
A3	1.5555

Leakage [nW]
13.48

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	16.0217	0.3656	16.0217
A1 to ZN	**fall**	-0.00	-0.00	0.49	0.24
	rise	1.95	1.98	2.47	2.25
A2 to ZN	**fall**	0.08	0.20	0.74	0.53
	rise	2.53	2.56	2.55	2.53
A3 to ZN	**fall**	0.02	0.40	0.92	0.73
	rise	3.05	3.07	2.92	2.99

NOR3_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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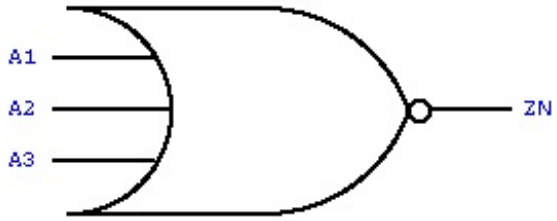
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	H
H	-	-	L
-	H	-	L
-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**31.7383**	**0.3656**	**31.7383**
A1 to ZN	**fall**	0.01	0.05	0.01	0.21
	rise	0.08	0.58	0.28	0.82
A2 to ZN	**fall**	0.01	0.05	0.04	0.22
	rise	0.13	0.62	0.30	0.85
A3 to ZN	**fall**	0.01	0.05	0.05	0.22
	rise	0.14	0.63	0.30	0.83

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**31.7383**	**0.3656**	**31.7383**
A1 to ZN	**fall**	0.00	0.04	0.08	0.14
	rise	0.07	0.50	0.13	0.50
A2 to ZN	**fall**	0.01	0.04	0.08	0.14
	rise	0.07	0.50	0.12	0.50
A3 to ZN	**fall**	0.01	0.04	0.08	0.14
	rise	0.07	0.50	0.10	0.50

Capacitance [fF]	
A1	3.2041
A2	3.2848
A3	3.3270

Leakage [nW]
26.97

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**31.7383**	**0.3656**	**31.7383**
A1 to ZN	**fall**	0.07	-0.00	1.13	0.62
	rise	3.98	4.05	5.03	4.59
A2 to ZN	**fall**	0.28	0.54	1.62	1.19
	rise	5.14	5.17	5.19	5.14
A3 to ZN	**fall**	0.18	0.94	1.98	1.59
	rise	6.19	6.24	5.93	6.08

NOR3_X4

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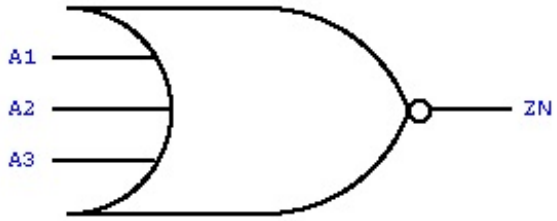
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.724 um^2
Equation	ZN = "!((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	H
H	-	-	L
-	H	-	L
-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	63.324	0.3656	63.324
A1 to ZN	**fall**	0.01	0.05	0.01	0.21
	rise	0.08	0.57	0.27	0.82
A2 to ZN	**fall**	0.01	0.05	0.04	0.22
	rise	0.13	0.62	0.30	0.85
A3 to ZN	**fall**	0.01	0.05	0.05	0.22
	rise	0.14	0.63	0.31	0.84

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	63.324	0.3656	63.324
A1 to ZN	**fall**	0.00	0.04	0.08	0.14
	rise	0.06	0.50	0.13	0.50
A2 to ZN	**fall**	0.01	0.04	0.08	0.14
	rise	0.06	0.50	0.11	0.50
A3 to ZN	**fall**	0.01	0.04	0.08	0.14
	rise	0.06	0.50	0.10	0.50

Capacitance [fF]	
A1	6.2290
A2	5.9343
A3	5.8938

Leakage [nW]
53.94

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	63.324	0.3656	63.324
A1 to ZN	**fall**	0.20	-0.00	2.25	1.25
	rise	7.88	8.04	10.00	9.11
A2 to ZN	**fall**	0.55	1.07	3.21	2.36
	rise	10.71	10.84	10.70	10.65
A3 to ZN	**fall**	0.33	1.86	3.89	3.12
	rise	13.06	13.13	12.48	12.84

NOR4_X1

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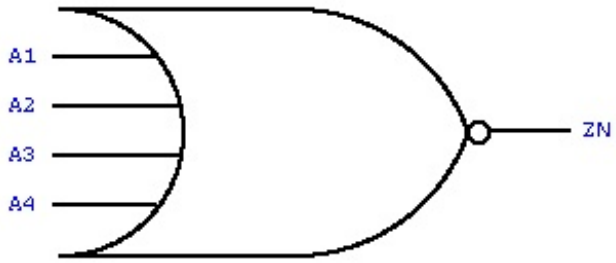
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	-	L
-	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**10.4713**	**0.3656**	**10.4713**
A1 to ZN	**fall**	0.01	0.03	0.02	0.16
	rise	0.13	0.57	0.35	0.80
A2 to ZN	**fall**	0.01	0.04	0.05	0.18
	rise	0.20	0.64	0.39	0.85
A3 to ZN	**fall**	0.01	0.04	0.06	0.18
	rise	0.24	0.68	0.41	0.87
A4 to ZN	**fall**	0.01	0.04	0.07	0.19
	rise	0.25	0.69	0.42	0.87

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**10.4713**	**0.3656**	**10.4713**
A1 to ZN	**fall**	0.01	0.03	0.09	0.12
	rise	0.11	0.50	0.16	0.50
A2 to ZN	**fall**	0.01	0.03	0.08	0.12
	rise	0.12	0.50	0.16	0.50
A3 to ZN	**fall**	0.01	0.03	0.09	0.12
	rise	0.12	0.50	0.15	0.50
A4 to ZN	**fall**	0.01	0.03	0.09	0.13
	rise	0.12	0.50	0.14	0.50

Capacitance [fF]	
A1	1.6459
A2	1.6003
A3	1.5720
A4	1.5431

Leakage [nW]
14.19

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	10.4713	0.3656	10.4713
A1 to ZN	fall	-0.00	-0.00	0.42	0.29
	rise	2.35	2.38	2.86	2.66
A2 to ZN	fall	0.16	0.20	0.68	0.58
	rise	2.93	2.94	2.98	2.89
A3 to ZN	fall	0.09	0.37	0.87	0.78
	rise	3.44	3.46	3.38	3.35
A4 to ZN	fall	0.01	0.40	0.97	0.90
	rise	3.95	3.97	3.86	3.84

NOR4_X2

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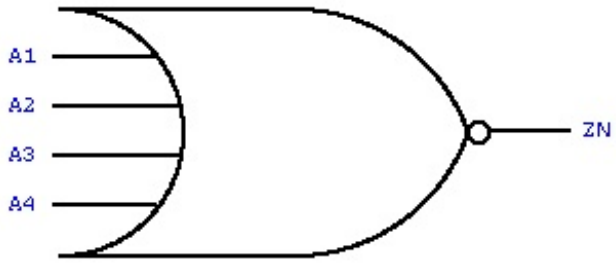
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	-	L
-	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**20.9045**	**0.3656**	**20.9045**
A1 to ZN	**fall**	0.01	0.03	0.02	0.17
	rise	0.12	0.57	0.34	0.80
A2 to ZN	**fall**	0.01	0.04	0.04	0.18
	rise	0.19	0.64	0.38	0.85
A3 to ZN	**fall**	0.01	0.04	0.06	0.18
	rise	0.23	0.68	0.40	0.87
A4 to ZN	**fall**	0.01	0.04	0.07	0.19
	rise	0.24	0.69	0.41	0.87

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**20.9045**	**0.3656**	**20.9045**
A1 to ZN	**fall**	0.01	0.03	0.09	0.12
	rise	0.11	0.50	0.16	0.50
A2 to ZN	**fall**	0.01	0.03	0.08	0.12
	rise	0.11	0.50	0.15	0.50
A3 to ZN	**fall**	0.01	0.03	0.09	0.12
	rise	0.11	0.50	0.14	0.50
A4 to ZN	**fall**	0.01	0.03	0.09	0.13
	rise	0.11	0.50	0.13	0.50

Capacitance [fF]	
A1	3.2168
A2	3.2580
A3	3.3802
A4	3.4882

Leakage [nW]
28.39

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	20.9045	0.3656	20.9045
A1 to ZN	fall	0.14	-0.00	0.98	0.71
	rise	4.58	4.64	5.60	5.21
A2 to ZN	fall	0.46	0.53	1.51	1.29
	rise	5.74	5.76	5.84	5.67
A3 to ZN	fall	0.31	0.88	1.89	1.70
	rise	6.77	6.81	6.65	6.60
A4 to ZN	fall	0.14	0.94	2.09	1.93
	rise	7.79	7.83	7.61	7.57

NOR4_X4

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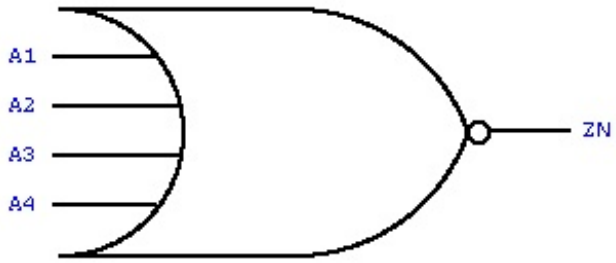
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.788 um^2
Equation	ZN = "!(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	-	L
-	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**41.5039**	**0.3656**	**41.5039**
A1 to ZN	**fall**	0.01	0.03	0.02	0.16
	rise	0.11	0.56	0.33	0.79
A2 to ZN	**fall**	0.01	0.04	0.04	0.18
	rise	0.19	0.64	0.38	0.85
A3 to ZN	**fall**	0.01	0.04	0.06	0.18
	rise	0.24	0.68	0.41	0.88
A4 to ZN	**fall**	0.01	0.04	0.07	0.19
	rise	0.25	0.70	0.42	0.88

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**41.5039**	**0.3656**	**41.5039**
A1 to ZN	**fall**	0.01	0.03	0.09	0.12
	rise	0.10	0.50	0.16	0.50
A2 to ZN	**fall**	0.01	0.03	0.08	0.12
	rise	0.11	0.50	0.15	0.50
A3 to ZN	**fall**	0.01	0.03	0.09	0.12
	rise	0.11	0.50	0.14	0.50
A4 to ZN	**fall**	0.01	0.03	0.09	0.13
	rise	0.11	0.50	0.13	0.50

Capacitance [fF]	
A1	6.2549
A2	5.9374
A3	5.8395
A4	5.8320

Leakage [nW]
56.77

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	41.5039	0.3656	41.5039
A1 to ZN	fall	0.34	-0.00	1.95	1.42
	rise	9.09	9.22	11.17	10.36
A2 to ZN	fall	0.92	1.06	2.99	2.56
	rise	11.83	11.90	11.98	11.68
A3 to ZN	fall	0.59	1.69	3.70	3.33
	rise	14.33	14.43	14.03	13.94
A4 to ZN	fall	0.28	1.83	4.07	3.79
	rise	16.67	16.73	16.28	16.23

OAI211_X1

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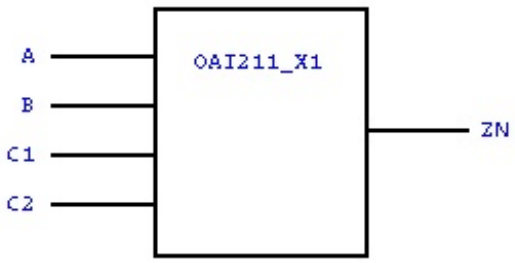
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((C1 | C2) & A) & B)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	-	-	-	H
H	H	H	-	L
H	H	-	H	L
-	L	-	-	H
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**25.5585**	**0.3656**	**25.5585**
A to ZN	**fall**	0.05	0.23	0.15	0.46
	rise	0.05	0.26	0.23	0.57
B to ZN	**fall**	0.05	0.24	0.14	0.41
	rise	0.06	0.27	0.25	0.58
C1 to ZN	**fall**	0.03	0.21	0.08	0.44
	rise	0.08	0.57	0.26	0.83
C2 to ZN	**fall**	0.04	0.22	0.11	0.45
	rise	0.09	0.58	0.27	0.82

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**25.5585**	**0.3656**	**25.5585**
A to ZN	**fall**	0.03	0.18	0.08	0.22
	rise	0.03	0.25	0.12	0.28
B to ZN	**fall**	0.03	0.18	0.07	0.21
	rise	0.04	0.25	0.12	0.29
C1 to ZN	**fall**	0.02	0.18	0.10	0.24
	rise	0.06	0.50	0.13	0.50
C2 to ZN	**fall**	0.03	0.18	0.10	0.25
	rise	0.06	0.50	0.11	0.50

Capacitance [fF]	
A	1.5837
B	1.6008
C1	1.5644
C2	1.5054

Leakage [nW]
12.78

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	25.5585	0.3656	25.5585
A to ZN	fall	0.55	0.59	0.83	0.59
	rise	3.25	3.36	3.77	3.58
B to ZN	fall	0.56	0.59	0.73	0.56
	rise	3.66	3.83	4.23	4.12
C1 to ZN	fall	0.46	0.47	1.08	0.65
	rise	2.12	2.15	2.61	2.35
C2 to ZN	fall	0.77	0.84	1.33	0.95
	rise	2.70	2.71	2.68	2.68

NANGATE OAI211_X2

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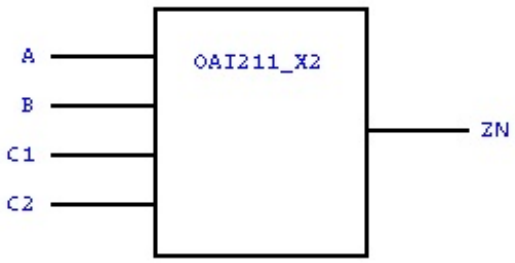
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((C1 | C2) & A) & B)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	-	-	-	H
H	H	H	-	L
H	H	-	H	L
-	L	-	-	H
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**50.8118**	**0.3656**	**50.8118**
A to ZN	**fall**	0.05	0.23	0.15	0.46
	rise	0.05	0.26	0.23	0.57
B to ZN	**fall**	0.05	0.24	0.14	0.41
	rise	0.05	0.27	0.25	0.57
C1 to ZN	**fall**	0.03	0.21	0.08	0.44
	rise	0.07	0.57	0.26	0.83
C2 to ZN	**fall**	0.04	0.22	0.11	0.45
	rise	0.09	0.58	0.26	0.82

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**50.8118**	**0.3656**	**50.8118**
A to ZN	**fall**	0.03	0.18	0.08	0.22
	rise	0.03	0.25	0.12	0.28
B to ZN	**fall**	0.03	0.18	0.07	0.21
	rise	0.04	0.25	0.12	0.29
C1 to ZN	**fall**	0.02	0.18	0.10	0.24
	rise	0.05	0.50	0.13	0.50
C2 to ZN	**fall**	0.03	0.18	0.10	0.25
	rise	0.05	0.50	0.11	0.50

Capacitance [fF]	
A	3.2715
B	3.0965
C1	2.9409
C2	3.1193

Leakage [nW]
25.56

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	50.8118	0.3656	50.8118
A to ZN	fall	1.25	1.31	1.82	1.32
	rise	6.52	6.74	7.56	7.21
B to ZN	fall	1.25	1.31	1.61	1.27
	rise	7.34	7.68	8.48	8.28
C1 to ZN	fall	1.06	1.08	2.32	1.43
	rise	4.33	4.36	5.30	4.74
C2 to ZN	fall	1.67	1.83	2.81	2.05
	rise	5.48	5.50	5.46	5.45

OAI211_X4

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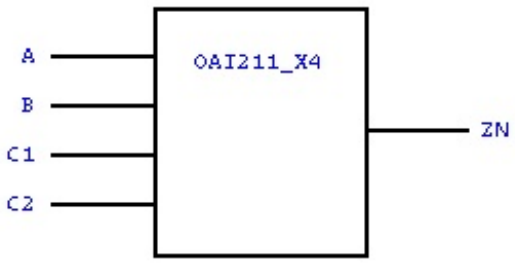
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Equation	ZN = "!(((C1 | C2) & A) & B)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	-	-	-	H
H	H	H	-	L
H	H	-	H	L
-	L	-	-	H
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**101.624**	**0.3656**	**101.624**
A to ZN	**fall**	0.05	0.24	0.15	0.46
	rise	0.05	0.26	0.23	0.57
B to ZN	**fall**	0.05	0.24	0.14	0.41
	rise	0.05	0.27	0.25	0.58
C1 to ZN	**fall**	0.03	0.21	0.08	0.44
	rise	0.07	0.57	0.25	0.83
C2 to ZN	**fall**	0.04	0.22	0.11	0.45
	rise	0.09	0.58	0.26	0.82

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**101.624**	**0.3656**	**101.624**
A to ZN	**fall**	0.02	0.18	0.08	0.22
	rise	0.03	0.25	0.12	0.29
B to ZN	**fall**	0.02	0.18	0.07	0.21
	rise	0.04	0.25	0.12	0.29
C1 to ZN	**fall**	0.02	0.18	0.10	0.24
	rise	0.05	0.50	0.13	0.50
C2 to ZN	**fall**	0.03	0.18	0.10	0.25
	rise	0.05	0.50	0.11	0.50

Capacitance [fF]	
A	6.2610
B	6.3264
C1	6.0906
C2	6.2291

Leakage [nW]
51.12

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	101.624	0.3656	101.624
A to ZN	fall	2.50	2.63	3.63	2.64
	rise	13.22	13.52	15.31	14.71
B to ZN	fall	2.49	2.62	3.21	2.54
	rise	14.85	15.40	17.13	16.78
C1 to ZN	fall	2.11	2.16	4.64	2.87
	rise	8.79	8.90	10.74	9.66
C2 to ZN	fall	3.33	3.66	5.62	4.10
	rise	11.09	11.06	11.04	11.07

OAI21_X1

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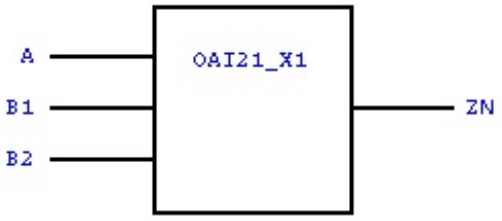
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!(A & (B1 | B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	-	-	H
H	H	-	L
H	-	H	L
-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**26.0544**	**0.3656**	**26.0544**
A to ZN	**fall**	0.03	0.15	0.11	0.36
	rise	0.04	0.26	0.23	0.57
B1 to ZN	**fall**	0.02	0.14	0.04	0.36
	rise	0.07	0.57	0.24	0.83
B2 to ZN	**fall**	0.02	0.15	0.08	0.37
	rise	0.08	0.58	0.25	0.82

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**26.0544**	**0.3656**	**26.0544**
A to ZN	**fall**	0.01	0.12	0.07	0.16
	rise	0.03	0.25	0.12	0.28
B1 to ZN	**fall**	0.01	0.12	0.09	0.20
	rise	0.05	0.50	0.12	0.50
B2 to ZN	**fall**	0.01	0.12	0.09	0.20
	rise	0.05	0.50	0.10	0.50

Capacitance [fF]	
A	1.6127
B1	1.6170
B2	1.5345

Leakage [nW]
14.00

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**26.0544**	**0.3656**	**26.0544**
A to ZN	**fall**	0.31	0.35	0.65	0.40
	rise	3.04	3.16	3.64	3.45
B1 to ZN	**fall**	0.23	0.24	0.91	0.46
	rise	1.94	1.97	2.47	2.16
B2 to ZN	**fall**	0.49	0.60	1.15	0.75
	rise	2.53	2.54	2.52	2.50

OAI21_X2

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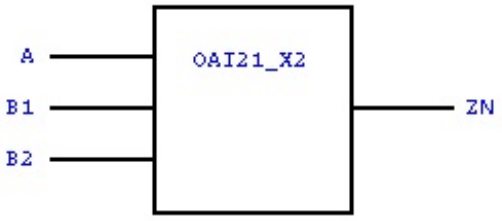
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!(A & (B1 | B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	-	-	H
H	H	-	L
H	-	H	L
-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**52.1088**	**0.3656**	**52.1088**
A to ZN	**fall**	0.03	0.15	0.11	0.36
	rise	0.04	0.26	0.22	0.57
B1 to ZN	**fall**	0.02	0.14	0.04	0.36
	rise	0.06	0.57	0.23	0.83
B2 to ZN	**fall**	0.02	0.15	0.07	0.37
	rise	0.08	0.58	0.24	0.82

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**52.1088**	**0.3656**	**52.1088**
A to ZN	**fall**	0.01	0.12	0.07	0.16
	rise	0.03	0.25	0.11	0.28
B1 to ZN	**fall**	0.01	0.12	0.09	0.20
	rise	0.04	0.50	0.12	0.50
B2 to ZN	**fall**	0.01	0.12	0.09	0.20
	rise	0.04	0.50	0.09	0.50

Capacitance [fF]	
A	3.0685
B1	3.0102
B2	3.2609

Leakage [nW]
27.99

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**52.1088**	**0.3656**	**52.1088**
A to ZN	**fall**	0.61	0.70	1.34	0.80
	rise	5.98	6.24	7.21	6.83
B1 to ZN	**fall**	0.45	0.47	1.84	0.91
	rise	3.88	3.93	4.94	4.33
B2 to ZN	**fall**	0.98	1.20	2.31	1.49
	rise	5.05	5.08	5.04	5.01

OAI21_X4

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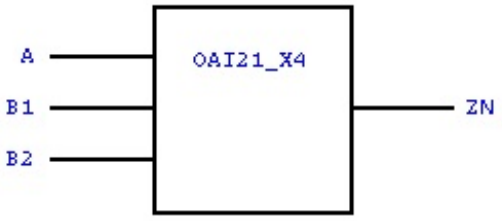
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(A & (B1 | B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	-	-	H
H	H	-	L
H	-	H	L
-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**104.065**	**0.3656**	**104.065**
A to ZN	**fall**	0.03	0.15	0.10	0.36
	rise	0.04	0.26	0.22	0.57
B1 to ZN	**fall**	0.02	0.14	0.04	0.37
	rise	0.06	0.57	0.23	0.83
B2 to ZN	**fall**	0.02	0.15	0.07	0.37
	rise	0.08	0.58	0.24	0.82

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**104.065**	**0.3656**	**104.065**
A to ZN	**fall**	0.01	0.12	0.07	0.16
	rise	0.03	0.25	0.11	0.28
B1 to ZN	**fall**	0.01	0.12	0.09	0.20
	rise	0.04	0.50	0.12	0.50
B2 to ZN	**fall**	0.01	0.12	0.09	0.20
	rise	0.04	0.50	0.09	0.50

Capacitance [fF]	
A	6.0144
B1	6.1732
B2	6.3560

Leakage [nW]
55.97

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**104.065**	**0.3656**	**104.065**
A to ZN	**fall**	1.21	1.40	2.67	1.60
	rise	12.01	12.46	14.47	13.85
B1 to ZN	**fall**	0.89	0.95	3.70	1.82
	rise	7.77	7.89	9.91	8.68
B2 to ZN	**fall**	1.96	2.39	4.62	2.98
	rise	10.11	10.18	10.09	10.04

NANGATE OAI221_X1

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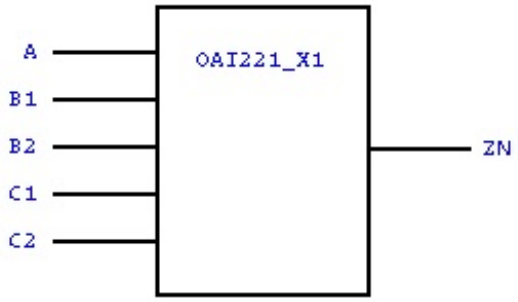
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "!(((C1 | C2) & A) & (B1 | B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	-	-	-	H
H	H	-	H	-	L
H	H	-	-	H	L
H	-	H	H	-	L
H	-	H	-	H	L
-	L	L	-	-	H
-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	22.1634	0.3656	22.1634
A to ZN	**fall**	0.06	0.22	0.18	0.44
	rise	0.05	0.23	0.24	0.53
B1 to ZN	**fall**	0.05	0.21	0.15	0.39
	rise	0.14	0.57	0.37	0.83
B2 to ZN	**fall**	0.06	0.22	0.17	0.40
	rise	0.15	0.58	0.36	0.82
C1 to ZN	**fall**	0.04	0.20	0.11	0.42
	rise	0.08	0.50	0.26	0.77
C2 to ZN	**fall**	0.04	0.21	0.14	0.43
	rise	0.09	0.52	0.27	0.76

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	22.1634	0.3656	22.1634
A to ZN	**fall**	0.03	0.17	0.08	0.20
	rise	0.04	0.23	0.15	0.28
B1 to ZN	**fall**	0.03	0.16	0.07	0.19
	rise	0.10	0.50	0.17	0.50
B2 to ZN	**fall**	0.03	0.17	0.07	0.19
	rise	0.10	0.50	0.14	0.50
C1 to ZN	**fall**	0.03	0.16	0.10	0.23
	rise	0.07	0.47	0.17	0.47
C2 to ZN	**fall**	0.03	0.17	0.10	0.23
	rise	0.07	0.47	0.13	0.47

Capacitance [fF]	
A	1.5968
B1	1.5935
B2	1.5627
C1	1.5349
C2	1.5367

Leakage [nW]
18.29

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	22.1634	0.3656	22.1634
A to ZN	fall	0.67	0.72	0.96	0.74
	rise	3.34	3.39	3.89	3.72
B1 to ZN	fall	0.58	0.61	0.71	0.62
	rise	3.89	4.06	4.37	4.28
B2 to ZN	fall	0.88	0.97	1.01	0.93
	rise	4.45	4.63	4.49	4.63
C1 to ZN	fall	0.57	0.61	1.19	0.80
	rise	2.28	2.31	2.79	2.51
C2 to ZN	fall	0.88	0.97	1.44	1.10
	rise	2.85	2.88	2.86	2.85

NANGATE OAI221_X2

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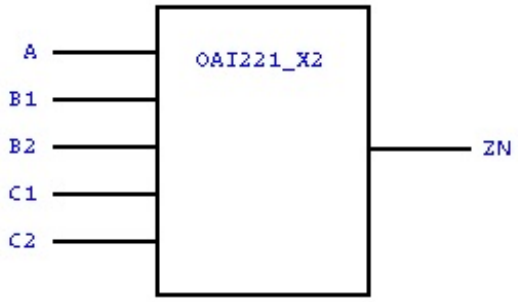
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.926 um^2
Equation	ZN = "!(((C1 | C2) & A) & (B1 | B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	-	-	-	H
H	H	-	H	-	L
H	H	-	-	H	L
H	-	H	H	-	L
H	-	H	-	H	L
-	L	L	-	-	H
-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	43.869	0.3656	43.869
A to ZN	**fall**	0.06	0.22	0.18	0.44
	rise	0.05	0.24	0.24	0.54
B1 to ZN	**fall**	0.05	0.21	0.15	0.39
	rise	0.14	0.57	0.37	0.83
B2 to ZN	**fall**	0.06	0.22	0.17	0.40
	rise	0.15	0.58	0.36	0.82
C1 to ZN	**fall**	0.04	0.20	0.12	0.42
	rise	0.08	0.50	0.26	0.77
C2 to ZN	**fall**	0.04	0.21	0.14	0.43
	rise	0.09	0.52	0.27	0.76

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	43.869	0.3656	43.869
A to ZN	**fall**	0.03	0.17	0.08	0.20
	rise	0.04	0.23	0.15	0.28
B1 to ZN	**fall**	0.03	0.16	0.07	0.19
	rise	0.10	0.50	0.17	0.50
B2 to ZN	**fall**	0.03	0.17	0.07	0.19
	rise	0.10	0.50	0.14	0.50
C1 to ZN	**fall**	0.03	0.16	0.10	0.23
	rise	0.07	0.47	0.17	0.47
C2 to ZN	**fall**	0.03	0.17	0.10	0.23
	rise	0.07	0.47	0.13	0.47

Capacitance [fF]	
A	3.4752
B1	3.3409
B2	3.0372
C1	2.9361
C2	3.1441

Leakage [nW]
36.59

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	43.869	0.3656	43.869
A to ZN	fall	1.48	1.58	2.03	1.61
	rise	6.97	7.17	8.07	7.78
B1 to ZN	fall	1.29	1.35	1.55	1.39
	rise	8.00	8.34	8.94	8.77
B2 to ZN	fall	1.91	2.09	2.15	2.01
	rise	9.12	9.47	9.20	9.48
C1 to ZN	fall	1.28	1.35	2.52	1.75
	rise	4.74	4.80	5.76	5.20
C2 to ZN	fall	1.90	2.09	3.02	2.35
	rise	5.89	5.90	5.90	5.85

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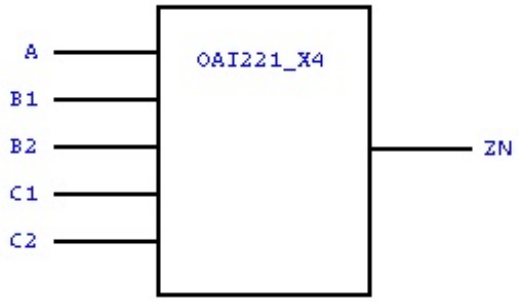
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(!(!(((C1 | C2) & A) & (B1 | B2))))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	-	-	-	H
H	H	-	H	-	L
H	H	-	-	H	L
H	-	H	H	-	L
H	-	H	-	H	L
-	L	L	-	-	H
-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.17	0.34	0.35	0.51
	rise	0.15	0.66	0.39	0.91
B1 to ZN	**fall**	0.17	0.33	0.30	0.47
	rise	0.29	0.81	0.56	1.07
B2 to ZN	**fall**	0.18	0.34	0.32	0.48
	rise	0.31	0.82	0.54	1.06
C1 to ZN	**fall**	0.15	0.31	0.31	0.47
	rise	0.23	0.75	0.48	0.99
C2 to ZN	**fall**	0.16	0.33	0.33	0.49
	rise	0.25	0.76	0.47	0.98

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
B1 to ZN	**fall**	0.01	0.13	0.01	0.13
	rise	0.02	0.50	0.02	0.50
B2 to ZN	**fall**	0.01	0.13	0.01	0.13
	rise	0.02	0.50	0.02	0.50
C1 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
C2 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50

Capacitance [fF]	
A	1.6104
B1	1.6028
B2	1.5960
C1	1.5175
C2	1.5402

Leakage [nW]
82.77

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	fall	7.54	9.55	7.80	9.80
	rise	11.06	12.48	11.38	13.31
B1 to ZN	fall	7.41	9.43	7.52	9.52
	rise	11.28	13.42	11.91	14.01
B2 to ZN	fall	7.76	9.83	7.83	9.88
	rise	11.83	13.87	11.93	14.08
C1 to ZN	fall	7.40	9.42	8.05	10.07
	rise	9.66	11.40	10.26	12.15
C2 to ZN	fall	7.74	9.83	8.34	10.38
	rise	10.24	11.88	10.25	12.25

OAI222_X1

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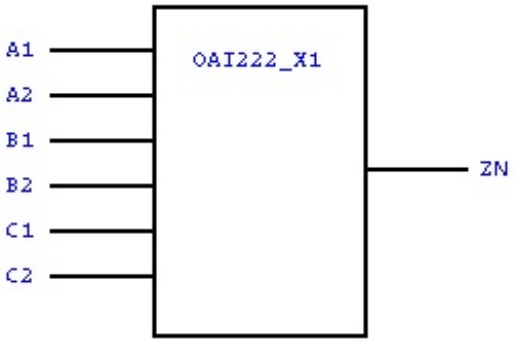
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.128 um^2
Equation	ZN = "!(((A1 | A2) & (B1 | B2)) & (C1 | C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	-	-	-	H
H	-	H	-	H	-	L
H	-	H	-	-	H	L
H	-	-	H	H	-	L
H	-	-	H	-	H	L
-	H	H	-	H	-	L
-	H	H	-	-	H	L
-	H	-	H	H	-	L
-	H	-	H	-	H	L
-	-	L	L	-	-	H
-	-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**20.0653**	**0.3656**	**20.0653**
A1 to ZN	**fall**	0.04	0.19	0.15	0.42
	rise	0.08	0.47	0.27	0.73
A2 to ZN	**fall**	0.05	0.20	0.17	0.43
	rise	0.09	0.48	0.28	0.72
B1 to ZN	**fall**	0.06	0.21	0.18	0.42
	rise	0.12	0.51	0.34	0.77
B2 to ZN	**fall**	0.07	0.22	0.21	0.44
	rise	0.14	0.52	0.34	0.77
C1 to ZN	**fall**	0.06	0.21	0.18	0.39
	rise	0.16	0.55	0.40	0.81
C2 to ZN	**fall**	0.07	0.22	0.20	0.40
	rise	0.17	0.56	0.39	0.80

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**20.0653**	**0.3656**	**20.0653**
A1 to ZN	**fall**	0.03	0.16	0.10	0.22
	rise	0.09	0.46	0.21	0.47
A2 to ZN	**fall**	0.04	0.16	0.11	0.22
	rise	0.09	0.46	0.17	0.46
B1 to ZN	**fall**	0.03	0.16	0.09	0.19
	rise	0.12	0.48	0.21	0.49
B2 to ZN	**fall**	0.04	0.16	0.09	0.20
	rise	0.12	0.48	0.18	0.48
C1 to ZN	**fall**	0.03	0.16	0.07	0.18
	rise	0.14	0.50	0.21	0.50
C2 to ZN	**fall**	0.04	0.16	0.07	0.19
	rise	0.14	0.50	0.18	0.50

Capacitance [fF]	
A1	1.5425
A2	1.5500
B1	1.5794
B2	1.5915
C1	1.5371
C2	1.5257

Leakage [nW]
23.15

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	20.0653	0.3656	20.0653
A1 to ZN	fall	0.68	0.74	1.30	0.96
	rise	2.43	2.46	2.97	2.70
A2 to ZN	fall	1.00	1.10	1.55	1.26
	rise	3.00	3.02	3.04	3.00
B1 to ZN	fall	0.69	0.74	0.91	0.79
	rise	3.59	3.68	4.06	3.92
B2 to ZN	fall	1.00	1.11	1.20	1.10
	rise	4.15	4.25	4.17	4.24
C1 to ZN	fall	0.69	0.74	0.79	0.75
	rise	4.51	4.71	4.98	4.94
C2 to ZN	fall	1.00	1.11	1.10	1.07
	rise	5.07	5.28	5.12	5.28

OAI222_X2

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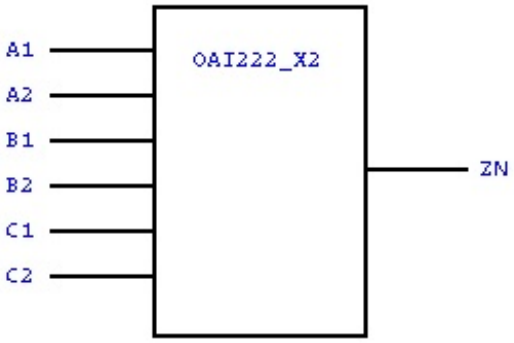
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	3.724 um^2
Equation	ZN = "!(((A1 | A2) & (B1 | B2)) & (C1 | C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	-	-	-	H
H	-	H	-	H	-	L
H	-	H	-	-	H	L
H	-	-	H	H	-	L
H	-	-	H	-	H	L
-	H	H	-	H	-	L
-	H	H	-	-	H	L
-	H	-	H	H	-	L
-	H	-	H	-	H	L
-	-	L	L	-	-	H
-	-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**39.5966**	**0.3656**	**39.5966**
A1 to ZN	**fall**	0.04	0.19	0.15	0.42
	rise	0.08	0.47	0.27	0.73
A2 to ZN	**fall**	0.05	0.20	0.17	0.43
	rise	0.10	0.48	0.28	0.72
B1 to ZN	**fall**	0.06	0.21	0.18	0.42
	rise	0.12	0.51	0.34	0.77
B2 to ZN	**fall**	0.07	0.22	0.21	0.44
	rise	0.14	0.52	0.34	0.76
C1 to ZN	**fall**	0.06	0.21	0.18	0.39
	rise	0.16	0.55	0.40	0.81
C2 to ZN	**fall**	0.07	0.22	0.20	0.40
	rise	0.17	0.56	0.39	0.80

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**39.5966**	**0.3656**	**39.5966**
A1 to ZN	**fall**	0.03	0.16	0.10	0.22
	rise	0.10	0.46	0.21	0.47
A2 to ZN	**fall**	0.04	0.16	0.11	0.22
	rise	0.10	0.46	0.17	0.46
B1 to ZN	**fall**	0.03	0.16	0.09	0.19
	rise	0.12	0.48	0.21	0.48
B2 to ZN	**fall**	0.04	0.16	0.09	0.20
	rise	0.12	0.48	0.18	0.48
C1 to ZN	**fall**	0.03	0.16	0.07	0.18
	rise	0.14	0.50	0.21	0.50
C2 to ZN	**fall**	0.04	0.16	0.07	0.19
	rise	0.14	0.50	0.18	0.50

Capacitance [fF]	
A1	3.1543
A2	2.9119
B1	3.2264
B2	2.9582
C1	3.2431
C2	2.9952

Leakage [nW]
46.31

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	39.5966	0.3656	39.5966
A1 to ZN	fall	1.51	1.62	2.73	2.06
	rise	5.12	5.16	6.20	5.63
A2 to ZN	fall	2.13	2.35	3.24	2.65
	rise	6.27	6.30	6.34	6.27
B1 to ZN	fall	1.52	1.62	1.97	1.72
	rise	7.40	7.59	8.35	8.04
B2 to ZN	fall	2.14	2.36	2.55	2.34
	rise	8.53	8.72	8.58	8.71
C1 to ZN	fall	1.52	1.62	1.72	1.65
	rise	9.27	9.67	10.22	10.10
C2 to ZN	fall	2.15	2.35	2.34	2.27
	rise	10.38	10.79	10.50	10.79

OAI222_X4

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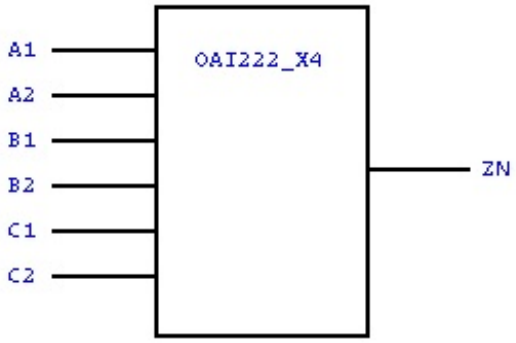
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.724 um^2
Equation	ZN = "!(!(!(((A1 | A2) & (B1 | B2)) & (C1 | C2))))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	-	-	-	H
H	-	H	-	H	-	L
H	-	H	-	-	H	L
H	-	-	H	H	-	L
H	-	-	H	-	H	L
-	H	H	-	H	-	L
-	H	H	-	-	H	L
-	H	-	H	H	-	L
-	H	-	H	-	H	L
-	-	L	L	-	-	H
-	-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.17	0.33	0.33	0.50
	rise	0.24	0.76	0.49	1.01
A2 to ZN	**fall**	0.18	0.34	0.35	0.52
	rise	0.26	0.77	0.48	0.99
B1 to ZN	**fall**	0.18	0.34	0.35	0.52
	rise	0.29	0.80	0.54	1.06
B2 to ZN	**fall**	0.19	0.35	0.37	0.54
	rise	0.30	0.81	0.53	1.05
C1 to ZN	**fall**	0.18	0.35	0.33	0.50
	rise	0.32	0.84	0.59	1.10
C2 to ZN	**fall**	0.20	0.36	0.35	0.51
	rise	0.33	0.85	0.57	1.09

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
A2 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
B1 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
B2 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
C1 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
C2 to ZN	**fall**	0.01	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50

Capacitance [fF]	
A1	1.5392
A2	1.5367
B1	1.5847
B2	1.5848
C1	1.5907
C2	1.5880

Leakage [nW]
86.59

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	fall	7.53	9.56	8.17	10.20
	rise	9.79	11.69	10.39	12.53
A2 to ZN	fall	7.89	9.97	8.45	10.51
	rise	10.35	12.28	10.39	12.45
B1 to ZN	fall	7.54	9.57	7.74	9.77
	rise	10.93	13.09	11.54	13.78
B2 to ZN	fall	7.88	9.97	8.06	10.12
	rise	11.50	13.69	11.58	13.88
C1 to ZN	fall	7.54	9.57	7.59	9.62
	rise	11.84	14.24	12.46	14.91
C2 to ZN	fall	7.88	9.97	7.93	9.99
	rise	12.41	14.77	12.55	15.00

OAI22_X1

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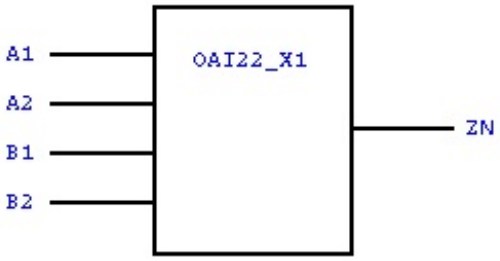
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!((A1 | A2) & (B1 | B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	L	-	-	H
H	-	H	-	L
H	-	-	H	L
-	H	H	-	L
-	H	-	H	L
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**23.2315**	**0.3656**	**23.2315**
A1 to ZN	**fall**	0.02	0.13	0.08	0.35
	rise	0.07	0.51	0.25	0.78
A2 to ZN	**fall**	0.03	0.14	0.10	0.36
	rise	0.08	0.53	0.25	0.77
B1 to ZN	**fall**	0.03	0.14	0.11	0.35
	rise	0.11	0.56	0.32	0.82
B2 to ZN	**fall**	0.03	0.15	0.13	0.35
	rise	0.12	0.57	0.32	0.81

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**23.2315**	**0.3656**	**23.2315**
A1 to ZN	**fall**	0.01	0.11	0.09	0.19
	rise	0.06	0.48	0.16	0.48
A2 to ZN	**fall**	0.02	0.11	0.09	0.20
	rise	0.06	0.48	0.12	0.48
B1 to ZN	**fall**	0.01	0.11	0.07	0.15
	rise	0.08	0.50	0.16	0.50
B2 to ZN	**fall**	0.02	0.11	0.07	0.16
	rise	0.08	0.50	0.13	0.50

Capacitance [fF]	
A1	1.6172
A2	1.5449
B1	1.6002
B2	1.5722

Leakage [nW]
19.22

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	23.2315	0.3656	23.2315
A1 to ZN	fall	0.33	0.37	1.02	0.62
	rise	2.06	2.10	2.61	2.32
A2 to ZN	fall	0.60	0.72	1.25	0.91
	rise	2.64	2.66	2.66	2.63
B1 to ZN	fall	0.33	0.37	0.60	0.46
	rise	3.16	3.27	3.66	3.52
B2 to ZN	fall	0.60	0.72	0.87	0.75
	rise	3.72	3.84	3.75	3.84

OAI22_X2

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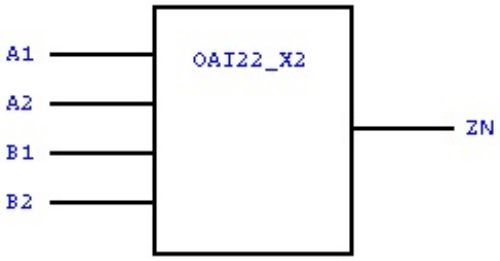
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!((A1 | A2) & (B1 | B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	L	-	-	H
H	-	H	-	L
H	-	-	H	L
-	H	H	-	L
-	H	-	H	L
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**46.3104**	**0.3656**	**46.3104**
A1 to ZN	**fall**	0.02	0.13	0.07	0.35
	rise	0.07	0.51	0.24	0.78
A2 to ZN	**fall**	0.03	0.14	0.10	0.36
	rise	0.08	0.53	0.25	0.77
B1 to ZN	**fall**	0.03	0.14	0.11	0.35
	rise	0.11	0.56	0.32	0.82
B2 to ZN	**fall**	0.03	0.15	0.13	0.35
	rise	0.12	0.57	0.32	0.81

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**46.3104**	**0.3656**	**46.3104**
A1 to ZN	**fall**	0.01	0.11	0.09	0.19
	rise	0.06	0.48	0.16	0.48
A2 to ZN	**fall**	0.02	0.11	0.09	0.20
	rise	0.06	0.48	0.12	0.48
B1 to ZN	**fall**	0.01	0.11	0.07	0.15
	rise	0.08	0.50	0.16	0.50
B2 to ZN	**fall**	0.02	0.11	0.07	0.16
	rise	0.08	0.50	0.13	0.50

Capacitance [fF]	
A1	3.0296
A2	3.3186
B1	3.0306
B2	3.2365

Leakage [nW]
38.43

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	46.3104	0.3656	46.3104
A1 to ZN	fall	0.65	0.74	2.04	1.24
	rise	4.23	4.32	5.35	4.75
A2 to ZN	fall	1.18	1.45	2.51	1.81
	rise	5.39	5.42	5.44	5.37
B1 to ZN	fall	0.65	0.74	1.21	0.91
	rise	6.44	6.67	7.44	7.16
B2 to ZN	fall	1.20	1.45	1.75	1.51
	rise	7.57	7.79	7.61	7.81

OAI22_X4

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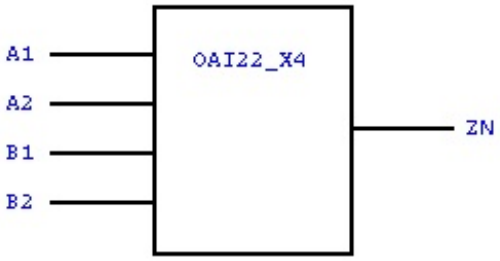
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Equation	ZN = "!((A1 | A2) & (B1 | B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	L	-	-	H
H	-	H	-	L
H	-	-	H	L
-	H	H	-	L
-	H	-	H	L
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**92.4683**	**0.3656**	**92.4683**
A1 to ZN	**fall**	0.02	0.13	0.07	0.35
	rise	0.06	0.51	0.24	0.78
A2 to ZN	**fall**	0.03	0.14	0.10	0.36
	rise	0.08	0.53	0.25	0.77
B1 to ZN	**fall**	0.03	0.14	0.11	0.35
	rise	0.10	0.56	0.32	0.82
B2 to ZN	**fall**	0.03	0.15	0.13	0.35
	rise	0.12	0.57	0.32	0.81

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**92.4683**	**0.3656**	**92.4683**
A1 to ZN	**fall**	0.01	0.11	0.09	0.19
	rise	0.05	0.48	0.16	0.48
A2 to ZN	**fall**	0.02	0.11	0.09	0.20
	rise	0.05	0.48	0.12	0.48
B1 to ZN	**fall**	0.01	0.11	0.07	0.15
	rise	0.07	0.50	0.15	0.50
B2 to ZN	**fall**	0.02	0.11	0.07	0.16
	rise	0.07	0.50	0.13	0.50

Capacitance [fF]	
A1	6.2546
A2	6.3838
B1	6.2633
B2	6.3107

Leakage [nW]
76.86

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	92.4683	0.3656	92.4683
A1 to ZN	fall	1.31	1.48	4.09	2.48
	rise	8.53	8.71	10.79	9.58
A2 to ZN	fall	2.36	2.90	5.03	3.62
	rise	10.86	10.92	10.96	10.81
B1 to ZN	fall	1.30	1.48	2.43	1.83
	rise	12.97	13.44	14.97	14.41
B2 to ZN	fall	2.39	2.90	3.50	3.01
	rise	15.22	15.64	15.30	15.71

OAI33_X1

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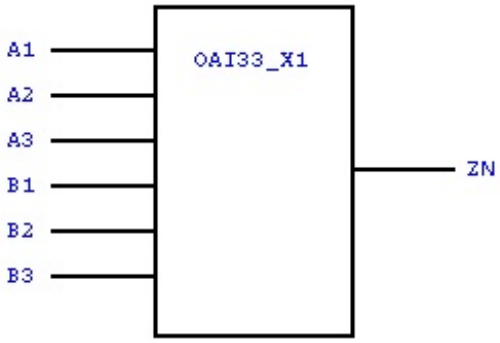
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.862 um^2
Equation	ZN = "!(((A1 | A2) | A3) & ((B1 | B2) | B3))"
Type	Combinational
Input	A1, A2, A3, B1, B2, B3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**A3**	**B1**	**B2**	**B3**	**ZN**
L	L	L	-	-	-	H
H	-	-	H	-	-	L
H	-	-	-	H	-	L
H	-	-	-	-	H	L
-	H	-	H	-	-	L
-	H	-	-	H	-	L
-	H	-	-	-	H	L
-	-	H	H	-	-	L
-	-	H	-	H	-	L
-	-	H	-	-	H	L
-	-	-	L	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**11.4822**	**0.3656**	**11.4822**
A1 to ZN	**fall**	0.02	0.08	0.10	0.26
	rise	0.11	0.46	0.32	0.71
A2 to ZN	**fall**	0.03	0.09	0.13	0.27
	rise	0.15	0.50	0.34	0.73
A3 to ZN	**fall**	0.03	0.09	0.14	0.28
	rise	0.17	0.52	0.34	0.72
B1 to ZN	**fall**	0.03	0.09	0.14	0.27
	rise	0.21	0.56	0.45	0.80
B2 to ZN	**fall**	0.04	0.10	0.16	0.28
	rise	0.25	0.60	0.47	0.83
B3 to ZN	**fall**	0.04	0.10	0.17	0.29
	rise	0.27	0.62	0.46	0.82

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**11.4822**	**0.3656**	**11.4822**
A1 to ZN	**fall**	0.02	0.07	0.10	0.16
	rise	0.12	0.45	0.21	0.46
A2 to ZN	**fall**	0.02	0.07	0.09	0.16
	rise	0.12	0.45	0.18	0.46
A3 to ZN	**fall**	0.03	0.07	0.10	0.16
	rise	0.12	0.45	0.17	0.46
B1 to ZN	**fall**	0.02	0.07	0.07	0.12
	rise	0.17	0.50	0.22	0.50
B2 to ZN	**fall**	0.02	0.07	0.07	0.12
	rise	0.17	0.50	0.21	0.50
B3 to ZN	**fall**	0.03	0.07	0.07	0.12
	rise	0.17	0.50	0.20	0.50

Capacitance [fF]	
A1	1.6119
A2	1.5632
A3	1.5310
B1	1.5802
B2	1.5558
B3	1.5350

Leakage [nW]
23.23

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	11.4822	0.3656	11.4822
A1 to ZN	fall	0.43	0.44	1.00	0.82
	rise	2.45	2.48	2.97	2.75
A2 to ZN	fall	0.67	0.78	1.26	1.11
	rise	3.02	3.04	3.07	3.00
A3 to ZN	fall	0.77	1.02	1.44	1.31
	rise	3.54	3.56	3.46	3.46
B1 to ZN	fall	0.41	0.44	0.62	0.59
	rise	4.05	4.15	4.51	4.43
B2 to ZN	fall	0.69	0.79	0.92	0.90
	rise	4.61	4.72	4.67	4.70
B3 to ZN	fall	0.80	1.03	1.12	1.12
	rise	5.13	5.24	5.09	5.18

OR2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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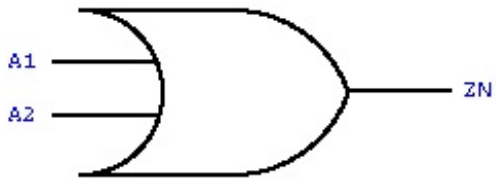
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	L
H	-	H
-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.15	0.35	0.38	0.60
	rise	0.04	0.55	0.13	0.65
A2 to ZN	**fall**	0.16	0.36	0.37	0.59
	rise	0.04	0.56	0.15	0.67

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.03	0.15	0.04	0.15
	rise	0.01	0.50	0.03	0.50
A2 to ZN	**fall**	0.03	0.15	0.04	0.15
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A1	0.9046
A2	0.9101

Leakage [nW]
17.73

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	1.89	2.54	2.30	2.84
	rise	1.34	1.50	1.66	1.69
A2 to ZN	**fall**	2.17	2.82	2.23	2.88
	rise	1.49	1.73	1.79	1.89

NANGATE OR2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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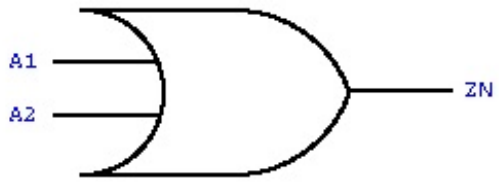
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	L
H	-	H
-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	0.14	0.34	0.36	0.58
	rise	0.04	0.55	0.12	0.64
A2 to ZN	**fall**	0.15	0.35	0.36	0.57
	rise	0.04	0.55	0.14	0.66

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	0.02	0.14	0.04	0.15
	rise	0.01	0.50	0.03	0.50
A2 to ZN	**fall**	0.02	0.14	0.04	0.15
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A1	1.6527
A2	1.6232

Leakage [nW]
35.64

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	3.67	4.86	4.54	5.49
	rise	2.65	2.96	3.30	3.44
A2 to ZN	**fall**	4.24	5.44	4.38	5.58
	rise	2.94	3.43	3.56	3.67

NANGATE OR2_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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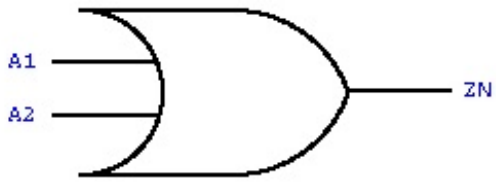
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	L
H	-	H
-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.13	0.33	0.35	0.57
	rise	0.04	0.55	0.12	0.64
A2 to ZN	**fall**	0.15	0.34	0.35	0.56
	rise	0.04	0.55	0.14	0.66

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.02	0.14	0.04	0.15
	rise	0.01	0.50	0.03	0.50
A2 to ZN	**fall**	0.02	0.14	0.03	0.15
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A1	3.2101
A2	3.2996

Leakage [nW]
71.26

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	7.11	9.53	8.98	10.80
	rise	4.97	5.65	6.26	6.56
A2 to ZN	**fall**	8.31	10.67	8.60	10.97
	rise	5.51	6.32	6.75	7.26

NANGATE OR3_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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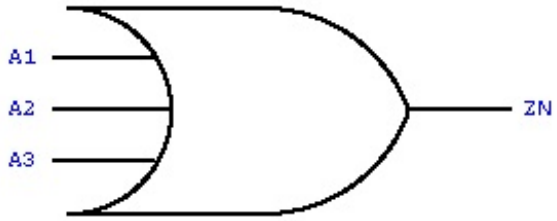
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	L
H	-	-	H
-	H	-	H
-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.24	0.49	0.48	0.74
	rise	0.04	0.55	0.14	0.66
A2 to ZN	**fall**	0.28	0.53	0.50	0.76
	rise	0.05	0.56	0.16	0.68
A3 to ZN	**fall**	0.29	0.54	0.49	0.74
	rise	0.05	0.56	0.17	0.69

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.04	0.17	0.05	0.17
	rise	0.01	0.50	0.03	0.50
A2 to ZN	**fall**	0.04	0.17	0.05	0.17
	rise	0.01	0.50	0.03	0.50
A3 to ZN	**fall**	0.04	0.17	0.05	0.17
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A1	0.9110
A2	0.9074
A3	0.8921

Leakage [nW]
18.53

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	2.11	2.77	2.50	3.00
	rise	1.34	1.46	1.63	1.64
A2 to ZN	**fall**	2.39	3.05	2.49	3.07
	rise	1.50	1.70	1.77	1.86
A3 to ZN	**fall**	2.64	3.30	2.65	3.28
	rise	1.57	1.93	1.87	2.00

NANGATE OR3_X2

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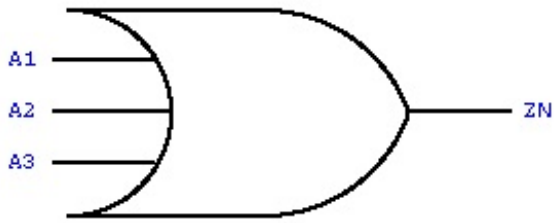
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.596 um^2
Equation	ZN = "((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	L
H	-	-	H
-	H	-	H
-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
A1 to ZN	**fall**	0.22	0.46	0.46	0.71
	rise	0.04	0.55	0.13	0.65
A2 to ZN	**fall**	0.26	0.50	0.47	0.73
	rise	0.04	0.56	0.15	0.67
A3 to ZN	**fall**	0.27	0.51	0.46	0.71
	rise	0.04	0.56	0.16	0.68

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
A1 to ZN	**fall**	0.04	0.16	0.05	0.17
	rise	0.01	0.50	0.03	0.50
A2 to ZN	**fall**	0.04	0.16	0.05	0.17
	rise	0.01	0.50	0.03	0.50
A3 to ZN	**fall**	0.04	0.16	0.04	0.17
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A1	1.6634
A2	1.6228
A3	1.6060

Leakage [nW]
37.23

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
A1 to ZN	**fall**	4.05	5.26	4.87	5.76
	rise	2.65	2.89	3.20	3.37
A2 to ZN	**fall**	4.61	5.83	4.82	5.88
	rise	2.95	3.37	3.48	3.63
A3 to ZN	**fall**	5.13	6.36	5.12	6.30
	rise	3.07	3.69	3.69	3.89

NANGATE OR3_X4

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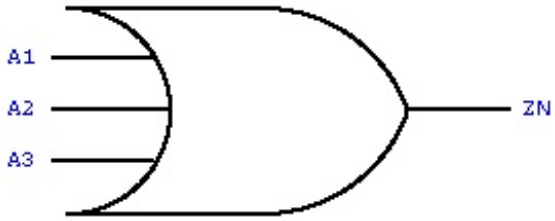
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	ZN = "((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	L
H	-	-	H
-	H	-	H
-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.21	0.45	0.44	0.69
	rise	0.04	0.55	0.12	0.65
A2 to ZN	**fall**	0.25	0.49	0.46	0.71
	rise	0.04	0.56	0.14	0.66
A3 to ZN	**fall**	0.26	0.50	0.45	0.70
	rise	0.04	0.56	0.15	0.67

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.04	0.16	0.05	0.16
	rise	0.01	0.50	0.03	0.50
A2 to ZN	**fall**	0.04	0.16	0.05	0.16
	rise	0.01	0.50	0.03	0.50
A3 to ZN	**fall**	0.04	0.16	0.04	0.16
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A1	3.1883
A2	3.2343
A3	3.2732

Leakage [nW]
74.45

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	7.76	10.10	9.48	11.12
	rise	5.01	5.57	6.11	6.49
A2 to ZN	**fall**	8.87	11.24	9.31	11.36
	rise	5.56	6.53	6.63	7.06
A3 to ZN	**fall**	9.93	12.28	9.91	12.19
	rise	5.80	7.17	7.08	7.57

NANGATE OR4_X1

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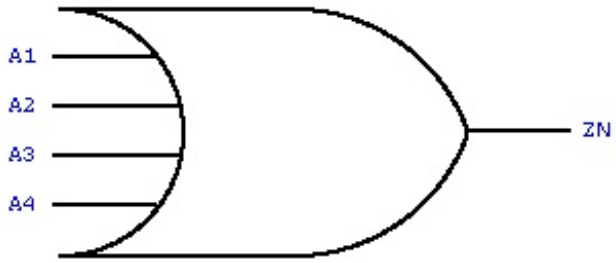
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	L
H	-	-	-	H
-	H	-	-	H
-	-	H	-	H
-	-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.32	0.62	0.56	0.86
	rise	0.04	0.56	0.15	0.67
A2 to ZN	**fall**	0.39	0.69	0.60	0.91
	rise	0.05	0.56	0.16	0.68
A3 to ZN	**fall**	0.43	0.73	0.62	0.92
	rise	0.05	0.56	0.17	0.69
A4 to ZN	**fall**	0.44	0.74	0.63	0.93
	rise	0.05	0.56	0.18	0.70

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.06	0.19	0.06	0.19
	rise	0.01	0.50	0.03	0.50
A2 to ZN	**fall**	0.06	0.19	0.06	0.19
	rise	0.01	0.50	0.03	0.50
A3 to ZN	**fall**	0.06	0.19	0.06	0.19
	rise	0.02	0.50	0.03	0.50
A4 to ZN	**fall**	0.06	0.19	0.06	0.19
	rise	0.02	0.50	0.03	0.50

Capacitance [fF]	
A1	0.8908
A2	0.9015
A3	0.8924
A4	0.8839

Leakage [nW]
19.11

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	fall	2.31	2.85	2.64	3.05
	rise	1.37	1.51	1.61	1.68
A2 to ZN	fall	2.59	3.13	2.67	3.13
	rise	1.50	1.68	1.74	1.84
A3 to ZN	fall	2.85	3.39	2.87	3.34
	rise	1.57	1.91	1.85	1.98
A4 to ZN	fall	3.09	3.64	3.08	3.58
	rise	1.57	2.05	1.92	2.08

OR4_X2

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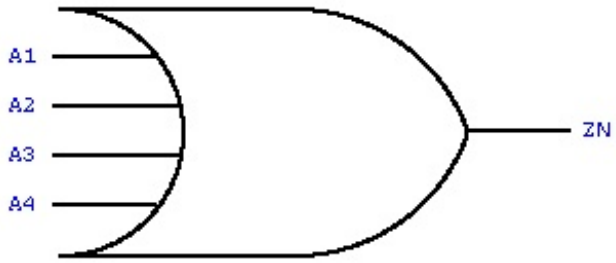
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	L
H	-	-	-	H
-	H	-	-	H
-	-	H	-	H
-	-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**120.85**	**0.3656**	**120.85**
A1 to ZN	**fall**	0.29	0.58	0.53	0.82
	rise	0.04	0.55	0.14	0.66
A2 to ZN	**fall**	0.36	0.65	0.57	0.86
	rise	0.04	0.56	0.15	0.67
A3 to ZN	**fall**	0.40	0.69	0.59	0.88
	rise	0.04	0.56	0.16	0.68
A4 to ZN	**fall**	0.41	0.70	0.59	0.88
	rise	0.04	0.56	0.17	0.69

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**120.85**	**0.3656**	**120.85**
A1 to ZN	**fall**	0.05	0.19	0.06	0.19
	rise	0.01	0.50	0.03	0.50
A2 to ZN	**fall**	0.05	0.19	0.06	0.19
	rise	0.01	0.50	0.03	0.50
A3 to ZN	**fall**	0.05	0.19	0.06	0.19
	rise	0.01	0.50	0.03	0.50
A4 to ZN	**fall**	0.05	0.19	0.06	0.19
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A1	1.6318
A2	1.6020
A3	1.5784
A4	1.5715

Leakage [nW]
38.41

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	fall	4.39	5.40	5.08	5.83
	rise	2.69	2.95	3.18	3.28
A2 to ZN	fall	4.96	5.98	5.16	5.98
	rise	2.97	3.43	3.45	3.56
A3 to ZN	fall	5.48	6.50	5.53	6.40
	rise	3.11	3.76	3.67	3.97
A4 to ZN	fall	5.99	7.00	5.97	6.88
	rise	3.08	4.17	3.78	4.16

OR4_X4

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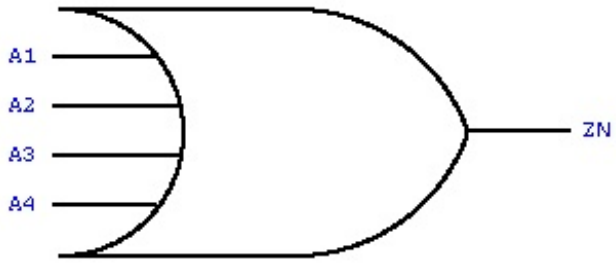
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	L
H	-	-	-	H
-	H	-	-	H
-	-	H	-	H
-	-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**241.699**	**0.3656**	**241.699**
A1 to ZN	**fall**	0.28	0.57	0.52	0.81
	rise	0.04	0.55	0.13	0.65
A2 to ZN	**fall**	0.35	0.64	0.56	0.85
	rise	0.04	0.56	0.15	0.67
A3 to ZN	**fall**	0.39	0.68	0.58	0.87
	rise	0.04	0.56	0.16	0.68
A4 to ZN	**fall**	0.40	0.69	0.58	0.87
	rise	0.04	0.56	0.16	0.69

Output Transition [ns]					
Input Transition [ns]		**0.0023**		**0.50**	
Load Capacitance [fF]		**0.3656**	**241.699**	**0.3656**	**241.699**
A1 to ZN	**fall**	0.05	0.18	0.06	0.19
	rise	0.01	0.50	0.03	0.50
A2 to ZN	**fall**	0.05	0.18	0.06	0.19
	rise	0.01	0.50	0.03	0.50
A3 to ZN	**fall**	0.05	0.18	0.06	0.19
	rise	0.01	0.50	0.03	0.50
A4 to ZN	**fall**	0.05	0.18	0.06	0.18
	rise	0.01	0.50	0.03	0.50

Capacitance [fF]	
A1	3.1597
A2	3.2145
A3	3.3478
A4	3.4854

Leakage [nW]
76.80

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	fall	8.60	10.61	10.04	11.49
	rise	5.14	5.68	6.08	6.26
A2 to ZN	fall	9.74	11.74	10.16	11.77
	rise	5.69	6.66	6.65	7.08
A3 to ZN	fall	10.77	12.79	10.89	12.61
	rise	5.93	7.30	7.07	7.69
A4 to ZN	fall	11.82	13.80	11.76	13.59
	rise	5.89	7.86	7.33	8.26

SDFFRS_X1

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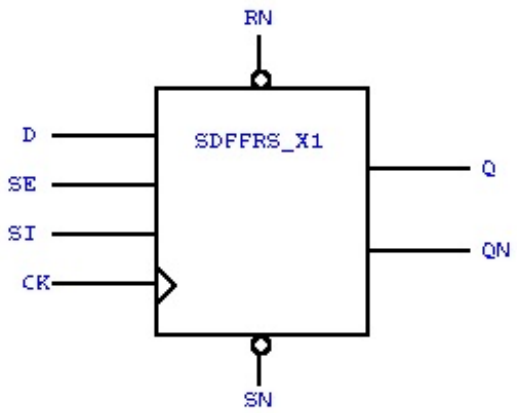
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, scan, and drive strength X1
Strength	1
Cell Area	7.714 um^2
Equation	Q = "(!(!RN)*((((SE * SI) + (D * !SE)))+(!SN)))" QN = "(!(!SN)*(!((((SE * SI) + (D * !SE)))+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table									
CK	D	RN	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	H	-	-	L	H
R	H	H	L	-	H	-	-	H	L
R	-	H	H	L	H	-	-	L	H
R	-	H	H	H	H	-	-	H	L
F	-	H	-	-	H	L	H	L	H
F	-	H	-	-	H	H	L	H	L
-	-	L	-	-	L	-	-	L	L
-	-	L	-	-	H	-	-	L	H
-	-	H	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.32	*0.48*	0.43	*0.59*
	rise	0.35	*0.85*	0.46	*0.97*
CK to QN	fall	0.25	0.44	0.37	0.56
	rise	0.25	0.77	0.35	0.88
RN to Q	fall	0.09	*0.25*	0.31	*0.50*
	rise	0.07	*0.59*	0.19	*0.71*
RN to QN	rise	0.60	1.13	0.86	1.38
SN to Q	rise	0.20	*0.71*	0.47	*0.96*
SN to QN	fall	0.11	0.30	0.35	0.57
	rise	0.45	0.97	0.62	1.14

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.02	*0.13*	0.02	*0.13*
	rise	0.02	*0.50*	0.02	*0.50*
CK to QN	fall	0.02	0.14	0.02	0.14
	rise	0.02	0.50	0.02	0.50
RN to Q	fall	0.02	*0.13*	0.03	*0.14*
	rise	0.02	*0.50*	0.03	*0.50*
RN to QN	rise	0.03	0.50	0.03	0.50
SN to Q	rise	0.02	*0.50*	0.02	*0.50*
SN to QN	fall	0.02	0.14	0.04	0.14
	rise	0.02	0.50	0.03	0.50

Constraints Time [ns]		
Setup CK to D	fall	0.44
	rise	0.34
Hold CK to D	fall	0.28
	rise	0.14
Recovery CK to RN	rise	-0.09
Removal CK to RN	rise	1.31
Setup CK to SE	fall	0.47
	rise	0.33
Hold CK to SE	fall	0.21
	rise	0.13
Setup CK to SI	fall	0.49
	rise	0.36
Hold CK to SI	fall	0.26
	rise	0.16
Recovery CK to SN	rise	0.10
Removal CK to SN	rise	0.85

Capacitance [fF]	
CK	0.9118
D	1.1105
RN	2.2312
SE	2.0653
SI	0.8437
SN	1.5218

Leakage [nW]
76.84

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	4242.26	4670.69
	rise	4993.67	5319.48
D	fall	2274.80	2338.74
	rise	1290.23	1318.61
RN	fall	1193.06	1265.30
	rise	-528.86	-446.41
SE	fall	3045.95	3734.70
	rise	1372.20	1924.52
SI	fall	2033.27	2064.61
	rise	1031.75	1039.62
SN	fall	521.33	486.25
	rise	-430.36	-444.10

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	4.17	*4.56*	4.34	*4.71*
	rise	4.23	*4.44*	4.41	*4.61*
CK to QN	fall	4.23	4.44	4.41	4.62
	rise	4.17	4.55	4.34	4.71
RN to Q	fall	3.85	*4.41*	4.31	*4.92*
	rise	2.00	*2.53*	2.50	*2.85*
RN to QN	rise	4.11	4.41	4.56	4.75
SN to Q	rise	2.88	*4.17*	3.11	*4.30*
SN to QN	fall	2.78	3.97	3.04	4.20
	rise	3.10	3.59	3.19	3.62

SDFFRS_X2

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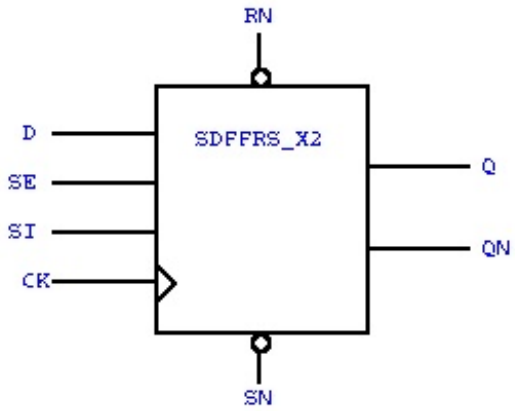
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, scan, and drive strength X2
Strength	2
Cell Area	8.246 um^2
Equation	Q = "(!(!RN)*((((SE * SI) + (D * !SE)))+(!SN)))" QN = "(!(!SN)*(!((((SE * SI) + (D * !SE)))+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table									
CK	D	RN	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	H	-	-	L	H
R	H	H	L	-	H	-	-	H	L
R	-	H	H	L	H	-	-	L	H
R	-	H	H	H	H	-	-	H	L
F	-	H	-	-	H	L	H	L	H
F	-	H	-	-	H	H	L	H	L
-	-	L	-	-	L	-	-	L	L
-	-	L	-	-	H	-	-	L	H
-	-	H	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.36	0.52	0.47	0.63
	rise	0.36	0.87	0.48	0.98
CK to QN	fall	0.27	0.46	0.38	0.57
	rise	0.27	0.79	0.38	0.90
RN to Q	fall	0.08	0.25	0.29	0.49
	rise	0.07	0.58	0.19	0.71
RN to QN	rise	0.69	1.22	0.94	1.47
SN to Q	rise	0.20	0.70	0.46	0.95
SN to QN	fall	0.10	0.29	0.33	0.55
	rise	0.47	0.99	0.63	1.15

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.02	0.14	0.02	0.14
	rise	0.02	0.50	0.02	0.50
CK to QN	fall	0.02	0.14	0.02	0.14
	rise	0.02	0.50	0.02	0.50
RN to Q	fall	0.02	0.13	0.04	0.14
	rise	0.02	0.50	0.04	0.50
RN to QN	rise	0.03	0.50	0.03	0.50
SN to Q	rise	0.02	0.50	0.02	0.50
SN to QN	fall	0.02	0.14	0.04	0.14
	rise	0.02	0.50	0.04	0.50

Constraints Time [ns]		
Setup CK to D	fall	0.45
	rise	0.34
Hold CK to D	fall	0.28
	rise	0.15
Recovery CK to RN	rise	-0.04
Removal CK to RN	rise	1.31
Setup CK to SE	fall	0.46
	rise	0.33
Hold CK to SE	fall	0.21
	rise	0.13
Setup CK to SI	fall	0.49
	rise	0.36
Hold CK to SI	fall	0.26
	rise	0.16
Recovery CK to SN	rise	0.14
Removal CK to SN	rise	0.84

Capacitance [fF]	
CK	0.8975
D	1.0992
RN	2.6348
SE	1.9638
SI	0.8438
SN	1.8111

Leakage [nW]
102.57

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	4329.92	4756.82
	rise	5407.70	5736.65
D	fall	2316.11	2378.36
	rise	1258.80	1287.17
RN	fall	1407.33	1513.98
	rise	-599.43	-511.72
SE	fall	3029.14	3720.39
	rise	1371.87	1925.31
SI	fall	2067.67	2096.85
	rise	1000.18	1007.00
SN	fall	736.68	738.14
	rise	-489.49	-502.20

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	5.83	6.53	5.99	6.69
	rise	6.05	6.45	6.23	6.63
CK to QN	fall	6.05	6.45	6.23	6.63
	rise	5.83	6.53	5.99	6.69
RN to Q	fall	5.22	6.12	5.97	6.76
	rise	3.01	3.86	4.02	4.53
RN to QN	rise	5.71	6.24	6.41	6.67
SN to Q	rise	4.45	5.96	4.84	6.15
SN to QN	fall	4.27	5.81	4.77	6.15
	rise	4.48	5.21	4.76	5.38

SDFFR_X1

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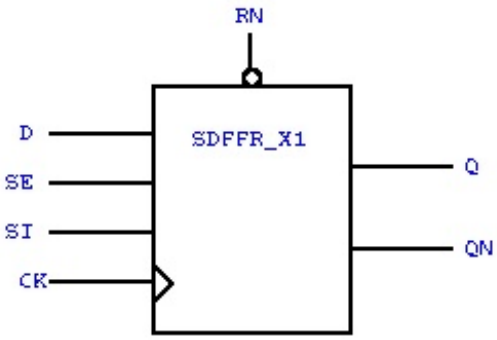
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, scan, and drive strength X1
Strength	1
Cell Area	6.650 um^2
Equation	Q = "(!(!RN)*(((SE * SI) + (D * !SE))))" QN = "((!RN)+!(((SE * SI) + (D * !SE))))"
Clock	CK
Reset	RN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	RN	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	-	-	L	H
R	H	H	L	-	-	-	H	L
R	-	H	H	L	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	H	-	-	L	H	L	H
F	-	H	-	-	H	L	H	L
-	-	L	-	-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
CK to Q	fall	0.22	0.47	0.33	0.58
	rise	0.19	0.71	0.31	0.83
CK to QN	fall	0.28	0.45	0.40	0.57
	rise	0.34	0.82	0.45	0.93
RN to Q	fall	0.53	0.78	0.81	1.06
RN to QN	rise	0.64	1.13	0.92	1.41

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
CK to Q	fall	0.04	0.16	0.04	0.16
	rise	0.02	0.50	0.02	0.50
CK to QN	fall	0.02	0.14	0.02	0.14
	rise	0.02	0.50	0.02	0.50
RN to Q	fall	0.04	0.16	0.04	0.16
RN to QN	rise	0.02	0.50	0.02	0.50

Constraints Time [ns]		
Setup CK to D	fall	0.46
	rise	0.34
Hold CK to D	fall	0.29
	rise	0.16
Recovery CK to RN	rise	-0.02
Removal CK to RN	rise	1.02
Setup CK to SE	fall	0.47
	rise	0.33
Hold CK to SE	fall	0.20
	rise	0.16
Setup CK to SI	fall	0.50
	rise	0.36
Hold CK to SI	fall	0.26
	rise	0.18

Capacitance [fF]	
CK	0.9880
D	1.1291
RN	1.4904
SE	1.9539
SI	0.8582

Leakage [nW]
68.51

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	4278.38	4705.27
	rise	3321.26	3673.66
D	fall	2010.40	2066.12
	rise	929.39	946.41
RN	fall	802.98	812.51
	rise	-363.72	-362.72
SE	fall	2900.61	3557.42
	rise	1164.83	1674.58
SI	fall	1929.09	1965.19
	rise	808.54	813.61

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
CK to Q	fall	3.60	3.83	3.76	3.99
	rise	3.44	3.88	3.62	4.05
CK to QN	fall	3.44	3.88	3.62	4.05
	rise	3.60	3.83	3.76	3.99
RN to Q	fall	3.64	3.89	3.70	3.91
RN to QN	rise	3.64	3.89	3.70	3.91

SDFFR_X2

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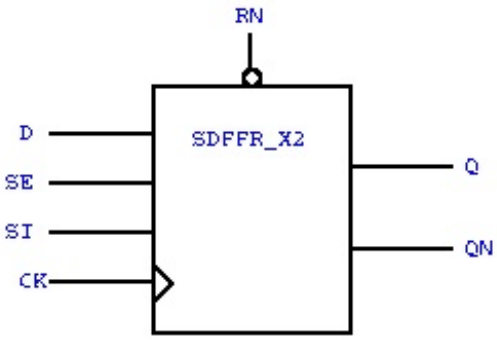
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, scan, and drive strength X2
Strength	2
Cell Area	6.916 um^2
Equation	Q = "(!(!RN)*(((SE * SI) + (D * !SE))))" QN = "((!RN)+!(((SE * SI) + (D * !SE))))"
Clock	CK
Reset	RN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	RN	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	-	-	L	H
R	H	H	L	-	-	-	H	L
R	-	H	H	L	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	H	-	-	L	H	L	H
F	-	H	-	-	H	L	H	L
-	-	L	-	-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.31	*0.62*	0.42	*0.72*
	rise	0.22	*0.75*	0.34	*0.87*
CK to QN	fall	0.30	0.46	0.41	0.57
	rise	0.44	0.91	0.55	1.02
RN to Q	fall	0.60	*0.91*	0.88	*1.19*
RN to QN	rise	0.73	1.20	1.01	1.48

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.06	*0.18*	0.06	*0.18*
	rise	0.02	*0.50*	0.02	*0.50*
CK to QN	fall	0.02	0.13	0.02	0.13
	rise	0.03	0.50	0.03	0.50
RN to Q	fall	0.06	*0.19*	0.06	*0.18*
RN to QN	rise	0.03	0.50	0.03	0.50

Constraints Time [ns]		
Setup CK to D	fall	0.45
	rise	0.34
Hold CK to D	fall	0.29
	rise	0.16
Recovery CK to RN	rise	-0.01
Removal CK to RN	rise	0.23
Setup CK to SE	fall	0.47
	rise	0.33
Hold CK to SE	fall	0.21
	rise	0.16
Setup CK to SI	fall	0.49
	rise	0.36
Hold CK to SI	fall	0.27
	rise	0.18

Capacitance [fF]	
CK	0.9213
D	1.1182
RN	1.5347
SE	1.8834
SI	0.8608

Leakage [nW]
94.47

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	4189.86	4618.87
	rise	3199.63	3545.60
D	fall	1984.17	2039.72
	rise	909.12	926.73
RN	fall	774.36	784.34
	rise	-364.24	-362.01
SE	fall	2882.82	3539.72
	rise	1141.54	1650.56
SI	fall	1902.78	1939.49
	rise	788.16	793.85

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	5.64	*5.75*	5.79	*5.89*
	rise	4.85	*5.54*	5.03	*5.75*
CK to QN	fall	4.85	5.58	5.03	5.73
	rise	5.64	5.76	5.79	5.91
RN to Q	fall	5.65	*5.71*	5.73	*5.74*
RN to QN	rise	5.65	5.73	5.73	5.77

SDFFS_X1

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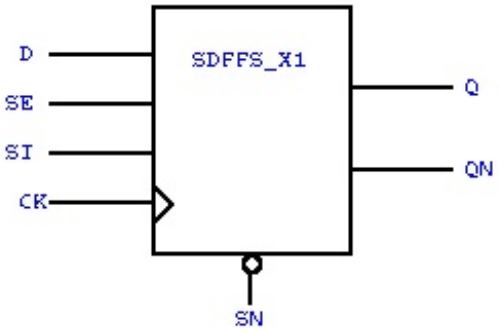
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, scan, and drive strength X1
Strength	1
Cell Area	6.650 um^2
Equation	Q = "((!SN)+(((SE * SI) + (D * !SE))))" QN = "(!((!SN)+(((SE * SI) + (D * !SE)))))"
Clock	CK
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	H	-	-	L	H
R	H	L	-	H	-	-	H	L
R	-	H	L	H	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	-	-	H	L	H	L	H
F	-	-	-	H	H	L	H	L
-	-	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.22	*0.45*	0.32	*0.56*
	rise	0.19	*0.71*	0.31	*0.82*
CK to QN	fall	0.26	0.43	0.38	0.55
	rise	0.34	0.83	0.45	0.94
SN to Q	rise	0.78	*1.30*	1.03	*1.55*
SN to QN	fall	0.09	0.27	0.32	0.52

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.04	*0.16*	0.04	*0.16*
	rise	0.02	*0.50*	0.02	*0.50*
CK to QN	fall	0.02	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
SN to Q	rise	0.03	*0.51*	0.03	*0.51*
SN to QN	fall	0.02	0.14	0.03	0.14

Constraints Time [ns]		
Setup CK to D	fall	0.46
	rise	0.35
Hold CK to D	fall	0.28
	rise	0.15
Setup CK to SE	fall	0.49
	rise	0.35
Hold CK to SE	fall	0.18
	rise	0.14
Setup CK to SI	fall	0.50
	rise	0.37
Hold CK to SI	fall	0.25
	rise	0.16
Recovery CK to SN	rise	-0.06
Removal CK to SN	rise	1.43

Capacitance [fF]	
CK	0.9401
D	1.1310
SE	1.9577
SI	0.8746
SN	1.3357

Leakage [nW]
69.90

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	4209.53	4633.82
	rise	4440.20	4758.40
D	fall	2186.86	2239.53
	rise	1029.38	1045.90
SE	fall	3171.41	3806.41
	rise	1381.20	1868.24
SI	fall	2106.98	2143.92
	rise	906.52	913.34
SN	fall	523.00	487.30
	rise	-432.12	-444.47

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	3.61	*3.83*	3.78	*4.00*
	rise	3.34	*3.75*	3.53	*3.93*
CK to QN	fall	3.35	3.76	3.53	3.93
	rise	3.61	3.83	3.78	4.00
SN to Q	rise	3.30	*3.60*	3.57	*3.76*
SN to QN	fall	3.30	3.59	3.57	3.75

SDFFS_X2

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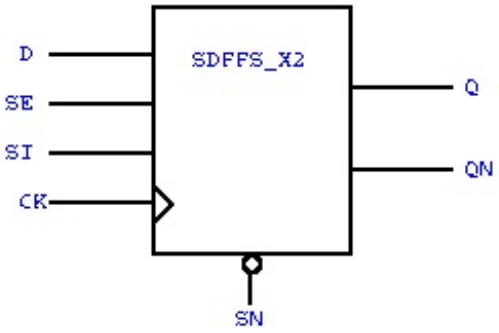
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, scan, and drive strength X2
Strength	2
Cell Area	7.182 um^2
Equation	Q = "((!SN)+(((SE * SI) + (D * !SE))))" QN = "(!((!SN)+(((SE * SI) + (D * !SE)))))"
Clock	CK
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	H	-	-	L	H
R	H	L	-	H	-	-	H	L
R	-	H	L	H	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	-	-	H	L	H	L	H
F	-	-	-	H	H	L	H	L
-	-	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.31	*0.60*	0.41	*0.71*
	rise	0.21	*0.74*	0.33	*0.86*
CK to QN	fall	0.30	0.47	0.42	0.58
	rise	0.47	0.94	0.58	1.05
SN to Q	rise	0.87	*1.40*	1.12	*1.65*
SN to QN	fall	0.08	0.25	0.28	0.49

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.06	*0.18*	0.06	*0.18*
	rise	0.02	*0.50*	0.02	*0.50*
CK to QN	fall	0.02	0.13	0.02	0.13
	rise	0.02	0.50	0.02	0.50
SN to Q	rise	0.04	*0.51*	0.04	*0.51*
SN to QN	fall	0.01	0.13	0.03	0.14

Constraints Time [ns]		
Setup CK to D	fall	0.46
	rise	0.37
Hold CK to D	fall	0.25
	rise	0.15
Setup CK to SE	fall	0.51
	rise	0.37
Hold CK to SE	fall	0.18
	rise	0.13
Setup CK to SI	fall	0.50
	rise	0.40
Hold CK to SI	fall	0.23
	rise	0.16
Recovery CK to SN	rise	-0.19
Removal CK to SN	rise	1.40

Capacitance [fF]	
CK	0.9208
D	1.0856
SE	1.8602
SI	0.8873
SN	2.2140

Leakage [nW]
94.03

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	4287.42	4715.70
	rise	5080.97	5303.11
D	fall	2313.43	2365.56
	rise	1066.12	1082.02
SE	fall	3223.74	3861.21
	rise	1462.69	1951.47
SI	fall	2233.13	2270.41
	rise	943.05	949.91
SN	fall	937.71	936.73
	rise	-615.48	-625.51

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	5.56	*5.68*	5.73	*5.84*
	rise	5.13	*5.92*	5.31	*6.09*
CK to QN	fall	5.13	5.94	5.31	6.12
	rise	5.56	5.68	5.73	5.81
SN to Q	rise	5.11	*5.61*	5.73	*5.98*
SN to QN	fall	5.11	5.58	5.73	5.94

SDFF_X1

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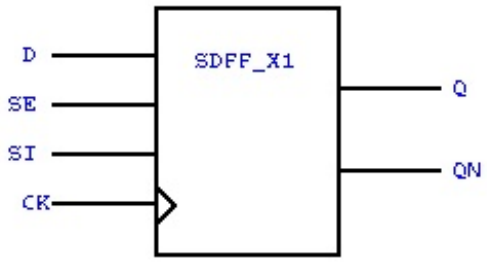
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with scan, and drive strength X1
Strength	1
Cell Area	6.118 um^2
Equation	Q = "(((SE * SI) + (D * !SE)))" QN = "(!(((SE * SI) + (D * !SE))))"
Clock	CK
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	-	-	L	H
R	H	L	-	-	-	H	L
R	-	H	L	-	-	L	H
R	-	H	H	-	-	H	L
F	-	-	-	L	H	L	H
F	-	-	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.20	0.44	0.31	0.55
	rise	0.16	0.68	0.27	0.79
CK to QN	fall	0.24	*0.40*	0.35	*0.51*
	rise	0.31	*0.79*	0.41	*0.90*

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.03	0.15	0.03	0.15
	rise	0.02	0.50	0.02	0.50
CK to QN	fall	0.02	*0.13*	0.02	*0.13*
	rise	0.02	*0.50*	0.02	*0.50*

Constraints Time [ns]		
Setup CK to D	fall	0.46
	rise	0.37
Hold CK to D	fall	0.25
	rise	0.14
Setup CK to SE	fall	0.49
	rise	0.36
Hold CK to SE	fall	0.20
	rise	0.13
Setup CK to SI	fall	0.50
	rise	0.39
Hold CK to SI	fall	0.23
	rise	0.16

Capacitance [fF]	
CK	0.9172
D	1.0971
SE	1.8395
SI	0.8957

Leakage [nW]
68.24

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	3935.28	4377.70
	rise	2035.47	2389.33
D	fall	2165.38	2217.58
	rise	1044.38	1062.56
SE	fall	2997.38	3657.09
	rise	1302.36	1820.93
SI	fall	2084.25	2120.30
	rise	920.36	927.81

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	3.23	3.43	3.40	3.60
	rise	3.07	3.45	3.27	3.64
CK to QN	fall	3.07	*3.45*	3.27	*3.64*
	rise	3.23	*3.43*	3.40	*3.60*

SDFF_X2

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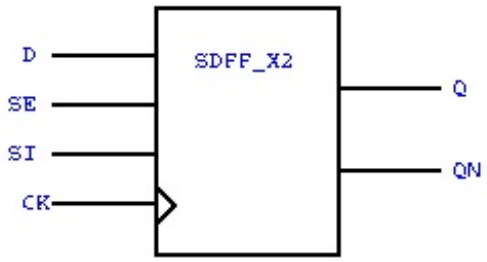
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with scan, and drive strength X2
Strength	2
Cell Area	6.384 um^2
Equation	Q = "(((SE * SI) + (D * !SE)))" QN = "(!(((SE * SI) + (D * !SE))))"
Clock	CK
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	-	-	L	H
R	H	L	-	-	-	H	L
R	-	H	L	-	-	L	H
R	-	H	H	-	-	H	L
F	-	-	-	L	H	L	H
F	-	-	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.30	*0.59*	0.40	*0.70*
	rise	0.19	*0.72*	0.30	*0.84*
CK to QN	fall	0.26	0.42	0.37	0.53
	rise	0.42	0.89	0.53	0.99

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.06	*0.18*	0.06	*0.18*
	rise	0.02	*0.50*	0.02	*0.50*
CK to QN	fall	0.02	0.13	0.02	0.13
	rise	0.03	0.50	0.03	0.50

Constraints Time [ns]		
Setup CK to D	fall	0.45
	rise	0.37
Hold CK to D	fall	0.24
	rise	0.15
Setup CK to SE	fall	0.50
	rise	0.37
Hold CK to SE	fall	0.19
	rise	0.13
Setup CK to SI	fall	0.50
	rise	0.40
Hold CK to SI	fall	0.22
	rise	0.16

Capacitance [fF]	
CK	0.9420
D	1.1069
SE	1.8150
SI	0.8748

Leakage [nW]
94.53

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
CK	fall	3983.98	4420.97
	rise	2092.82	2442.54
D	fall	2170.34	2221.56
	rise	1045.64	1063.43
SE	fall	2989.46	3649.50
	rise	1304.90	1824.66
SI	fall	2089.16	2123.26
	rise	921.99	929.50

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	5.25	*5.34*	5.43	*5.51*
	rise	4.44	*5.09*	4.62	*5.25*
CK to QN	fall	4.44	5.09	4.63	5.24
	rise	5.25	5.32	5.43	5.49

TBUF_X16

Databook Build Date: Thursday Feb 17 15:07 2011

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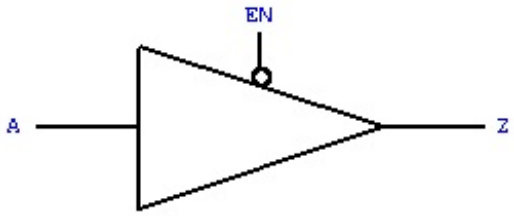
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	6.916 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		13.7499	833.7086	13.7499	833.7086
A to Z	**fall**	0.14	*0.36*	0.37	*0.62*
	rise	*0.07*	0.60	*0.20*	0.74
EN to Z	**fall**	0.15	*0.38*	0.35	*0.60*
	rise	*0.14*	0.67	*0.36*	0.88

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		13.7499	833.7086	13.7499	833.7086
A to Z	**fall**	0.03	*0.15*	0.05	*0.16*
	rise	*0.02*	0.50	*0.04*	0.50
EN to Z	**fall**	0.03	*0.15*	0.04	*0.16*
	rise	*0.02*	0.50	*0.02*	0.50

Capacitance [fF]	
A	6.2511
EN	4.7610
Z	13.3966

Leakage [nW]
149.18

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		13.7499	833.7086	13.7499	833.7086
A to Z	**fall**	19.97	*27.60*	24.93	*29.92*
	rise	*16.18*	19.42	*17.24*	20.97
EN to Z	**fall**	17.56	*22.73*	17.72	*23.75*
	rise	*21.30*	23.58	*21.15*	23.97

TBUF_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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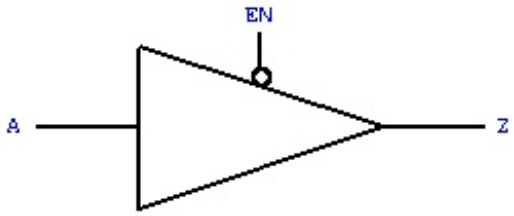
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.128 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		1.4425	52.6534	1.4425	52.6534
A to Z	**fall**	0.12	*0.30*	0.30	*0.50*
	rise	*0.05*	0.56	*0.14*	0.66
EN to Z	**fall**	0.10	*0.28*	0.29	*0.50*
	rise	*0.11*	0.61	*0.30*	0.80

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		1.4425	52.6534	1.4425	52.6534
A to Z	**fall**	0.02	*0.14*	0.03	*0.14*
	rise	*0.02*	0.50	*0.03*	0.50
EN to Z	**fall**	0.02	*0.14*	0.04	*0.14*
	rise	*0.02*	0.50	*0.02*	0.50

Capacitance [fF]	
A	1.8162
EN	1.6589
Z	1.0787

Leakage [nW]
22.76

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		1.4425	52.6534	1.4425	52.6534
A to Z	**fall**	3.21	*3.82*	3.27	*4.32*
	rise	*1.56*	1.74	*2.18*	2.27
EN to Z	**fall**	3.00	*3.52*	3.47	*4.36*
	rise	*3.19*	3.26	*3.37*	3.69

TBUF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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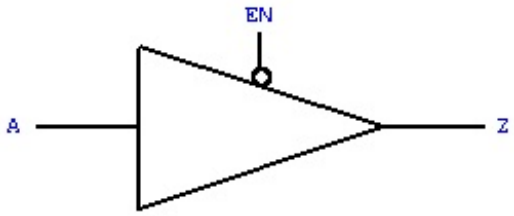
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		2.0402	105.2852	2.0402	105.2852
A to Z	**fall**	0.10	*0.28*	0.28	*0.48*
	rise	*0.05*	0.56	*0.12*	0.65
EN to Z	**fall**	0.08	*0.26*	0.26	*0.48*
	rise	*0.12*	0.63	*0.33*	0.84

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		2.0402	105.2852	2.0402	105.2852
A to Z	**fall**	0.01	*0.13*	0.02	*0.14*
	rise	*0.01*	0.50	*0.03*	0.50
EN to Z	**fall**	0.01	*0.14*	0.03	*0.14*
	rise	*0.01*	0.50	*0.02*	0.50

Capacitance [fF]	
A	3.2056
EN	2.6372
Z	1.6782

Leakage [nW]
40.67

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		2.0402	105.2852	2.0402	105.2852
A to Z	**fall**	5.71	*6.88*	5.82	*7.94*
	rise	*2.91*	3.29	*4.10*	4.47
EN to Z	**fall**	4.24	*5.28*	4.71	*6.47*
	rise	*4.91*	5.10	*4.89*	5.49

TBUF_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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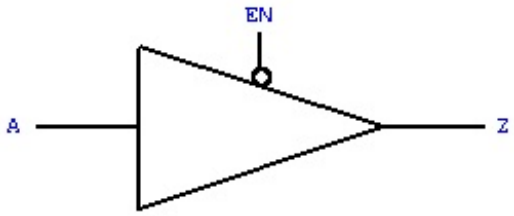
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		3.6634	210.2117	3.6634	210.2117
A to Z	**fall**	0.10	*0.30*	0.30	*0.53*
	rise	*0.05*	0.57	*0.14*	0.68
EN to Z	**fall**	0.11	*0.31*	0.30	*0.52*
	rise	*0.13*	0.64	*0.34*	0.86

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		3.6634	210.2117	3.6634	210.2117
A to Z	**fall**	0.02	*0.14*	0.04	*0.15*
	rise	*0.01*	0.50	*0.04*	0.50
EN to Z	**fall**	0.02	*0.14*	0.03	*0.15*
	rise	*0.02*	0.50	*0.02*	0.50

Capacitance [fF]	
A	3.2650
EN	2.3352
Z	3.3027

Leakage [nW]
51.95

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		3.6634	210.2117	3.6634	210.2117
A to Z	**fall**	6.48	*8.56*	7.52	*9.95*
	rise	*4.55*	5.23	*5.39*	6.32
EN to Z	**fall**	6.17	*7.68*	6.02	*8.28*
	rise	*7.19*	7.45	*7.09*	7.84

TBUF_X8

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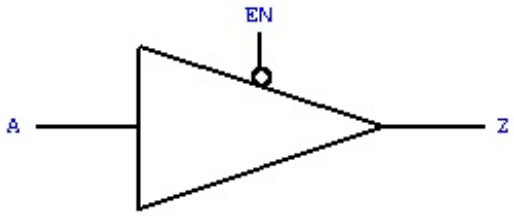
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	4.788 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		7.2637	419.5057	7.2637	419.5057
A to Z	**fall**	0.10	*0.30*	0.30	*0.53*
	rise	*0.06*	0.57	*0.15*	0.68
EN to Z	**fall**	0.11	*0.31*	0.30	*0.52*
	rise	*0.13*	0.64	*0.34*	0.86

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		7.2637	419.5057	7.2637	419.5057
A to Z	**fall**	0.02	*0.14*	0.04	*0.15*
	rise	*0.01*	0.50	*0.04*	0.50
EN to Z	**fall**	0.02	*0.14*	0.03	*0.15*
	rise	*0.02*	0.50	*0.02*	0.50

Capacitance [fF]	
A	6.4896
EN	4.8071
Z	6.9077

Leakage [nW]
104.04

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		7.2637	419.5057	7.2637	419.5057
A to Z	**fall**	13.25	*17.46*	15.37	*20.26*
	rise	*9.18*	10.56	*10.77*	12.52
EN to Z	**fall**	12.65	*15.64*	12.42	*16.84*
	rise	*14.74*	15.47	*14.44*	16.17

TINV_X1

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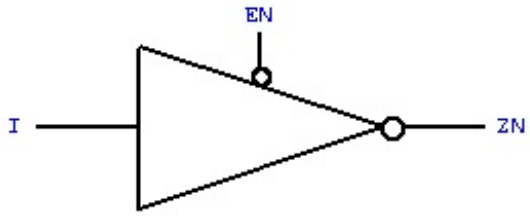
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!I", when !(EN)
Type	Tristate
Input	EN, I
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
I	**EN**	**ZN**
L	L	H
H	L	L
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		1.1805	23.439	1.1805	23.439
EN to ZN	**fall**	0.05	*0.18*	0.20	*0.37*
	rise	*0.04*	0.55	*0.08*	0.80
I to ZN	**fall**	0.02	*0.15*	0.08	*0.35*
	rise	*0.08*	0.59	*0.25*	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		1.1805	23.439	1.1805	23.439
EN to ZN	**fall**	0.02	*0.12*	0.04	*0.12*
	rise	*0.03*	0.50	*0.13*	0.50
I to ZN	**fall**	0.01	*0.12*	0.06	*0.16*
	rise	*0.04*	0.50	*0.08*	0.50

Capacitance [fF]	
EN	1.7213
I	1.4021
ZN	0.8178

Leakage [nW]
11.02

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		1.1805	23.439	1.1805	23.439
EN to ZN	**fall**	1.82	*2.07*	1.81	*2.49*
	rise	*2.29*	2.35	*2.13*	2.86
I to ZN	**fall**	0.28	*0.47*	0.65	*0.50*
	rise	*1.88*	1.92	*1.87*	1.95

TLAT_X1

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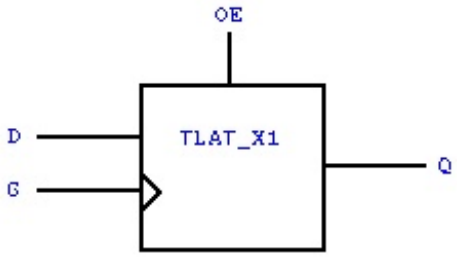
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	High enable Latch with drive strength X1
Strength	1
Cell Area	3.458 um^2
Equation	Q = "(D)", when !(!OE)
Enable	G
Type	Sequential
Input	D, OE
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
OE	**D**	**G**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	-	-	-	-	Z
H	L	H	-	-	L
H	H	H	-	-	H
H	-	L	L	H	L
H	-	L	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		1.1732	23.391	1.1732	23.391
D to Q	**fall**	0.19	*0.33*	0.40	*0.55*
	rise	*0.13*	0.64	*0.25*	0.76
G to Q	**fall**	0.17	*0.32*	0.26	*0.41*
	rise	*0.18*	0.69	*0.28*	0.79
OE to Q	**fall**	0.01	*0.13*	-0.06	*0.35*
	rise	*0.05*	0.56	*0.11*	0.63

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		1.1732	23.391	1.1732	23.391
D to Q	**fall**	0.02	*0.12*	0.02	*0.12*
	rise	*0.04*	0.50	*0.04*	0.50
G to Q	**fall**	0.02	*0.12*	0.02	*0.12*
	rise	*0.04*	0.50	*0.04*	0.50
OE to Q	**fall**	0.01	*0.12*	0.10	*0.21*
	rise	*0.04*	0.50	*0.05*	0.50

Constraints Time [ns]		
Setup G to D	**fall**	0.34
	rise	0.11
Hold G to D	**fall**	0.61
	rise	0.29

Capacitance [fF]	
D	1.1027
G	0.9725
OE	1.4591
Q	0.8080

Leakage [nW]
30.70

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0023	0.50
D	fall	1104.73	1116.45
	rise	201.02	201.01
G	fall	2138.94	2609.16
	rise	2142.70	2550.61
OE	fall	1271.54	1759.50
	rise	659.44	1104.75

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		1.1732	23.391	1.1732	23.391
D to Q	fall	3.03	*3.67*	2.96	*3.77*
	rise	*3.16*	3.34	*3.27*	3.42
G to Q	fall	3.74	*4.37*	4.14	*4.76*
	rise	*4.83*	5.03	*5.27*	5.44
OE to Q	fall	0.40	*0.65*	0.59	*1.10*
	rise	*1.12*	1.20	*1.46*	1.63

XNOR2_X1

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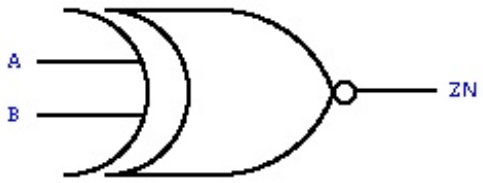
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "!(A ^ B)"
Type	Combinational
Input	A, B
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**ZN**
L	L	H
L	H	L
H	L	L
H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	26.0162	0.3656	26.0162
A to ZN	**fall**	0.10	0.23	0.32	0.46
	rise	0.09	0.57	0.24	0.83
B to ZN	**fall**	0.11	0.24	0.33	0.47
	rise	0.09	0.58	0.25	0.82

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	26.0162	0.3656	26.0162
A to ZN	**fall**	0.02	0.12	0.09	0.20
	rise	0.05	0.50	0.12	0.50
B to ZN	**fall**	0.02	0.12	0.09	0.20
	rise	0.05	0.50	0.10	0.50

Capacitance [fF]	
A	2.2236
B	2.5375

Leakage [nW]
23.35

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	26.0162	0.3656	26.0162
A to ZN	**fall**	1.72	1.73	2.06	2.08
	rise	2.54	2.49	2.90	2.72
B to ZN	**fall**	1.91	2.12	2.25	2.38
	rise	2.62	2.58	2.60	2.57

NANGATE XNOR2_X2

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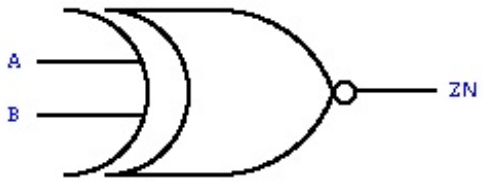
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.660 um^2
Equation	ZN = "!(A ^ B)"
Type	Combinational
Input	A, B
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**ZN**
L	L	H
L	H	L
H	L	L
H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	52.0325	0.3656	52.0325
A to ZN	**fall**	0.09	0.22	0.30	0.44
	rise	0.08	0.57	0.24	0.83
B to ZN	**fall**	0.10	0.23	0.32	0.46
	rise	0.09	0.58	0.25	0.82

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	52.0325	0.3656	52.0325
A to ZN	**fall**	0.02	0.12	0.09	0.20
	rise	0.04	0.50	0.12	0.50
B to ZN	**fall**	0.01	0.12	0.09	0.20
	rise	0.04	0.50	0.10	0.50

Capacitance [fF]	
A	4.0075
B	4.7822

Leakage [nW]
46.83

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	52.0325	0.3656	52.0325
A to ZN	**fall**	3.22	3.22	3.95	3.93
	rise	5.10	4.96	5.82	5.51
B to ZN	**fall**	3.62	4.00	4.31	4.53
	rise	5.30	5.29	5.25	5.24

XOR2_X1

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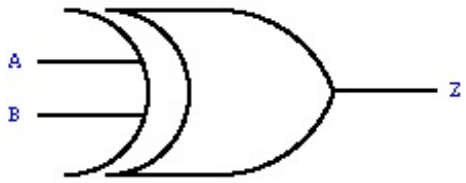
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	Z = "(A ^ B)"
Type	Combinational
Input	A, B
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**Z**
L	L	L
L	H	H
H	L	H
H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	25.3296	0.3656	25.3296
A to Z	**fall**	0.17	0.26	0.40	0.50
	rise	0.12	0.60	0.24	0.81
B to Z	**fall**	0.18	0.27	0.39	0.49
	rise	0.11	0.59	0.28	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	25.3296	0.3656	25.3296
A to Z	**fall**	0.03	0.11	0.09	0.20
	rise	0.06	0.50	0.12	0.50
B to Z	**fall**	0.03	0.11	0.07	0.16
	rise	0.06	0.50	0.12	0.50

Capacitance [fF]	
A	2.2397
B	2.3539

Leakage [nW]
24.93

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	25.3296	0.3656	25.3296
A to Z	**fall**	1.78	1.85	2.16	2.13
	rise	2.54	2.44	2.83	2.69
B to Z	**fall**	1.84	1.92	1.86	1.95
	rise	2.74	2.77	3.11	2.98

XOR2_X2

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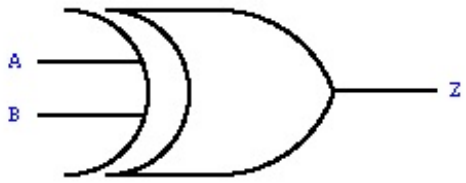
Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_slow_slow**: Vdd= **.95**V, Tj= **125.0** deg. C .

Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	Z = "(A ^ B)"
Type	Combinational
Input	A, B
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**Z**
L	L	L
L	H	H
H	L	H
H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	50.5066	0.3656	50.5066
A to Z	**fall**	0.17	0.26	0.40	0.51
	rise	0.12	0.60	0.24	0.81
B to Z	**fall**	0.18	0.28	0.39	0.50
	rise	0.11	0.59	0.28	0.83

Output Transition [ns]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	50.5066	0.3656	50.5066
A to Z	**fall**	0.03	0.11	0.09	0.20
	rise	0.06	0.50	0.12	0.50
B to Z	**fall**	0.03	0.11	0.07	0.16
	rise	0.06	0.50	0.12	0.50

Capacitance [fF]	
A	4.3460
B	4.3807

Leakage [nW]
50.03

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0023		0.50	
Load Capacitance [fF]		0.3656	50.5066	0.3656	50.5066
A to Z	**fall**	3.67	3.83	4.43	4.39
	rise	5.16	4.97	5.76	5.48
B to Z	**fall**	3.79	3.98	3.84	4.04
	rise	5.58	5.65	6.31	6.07

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Library

NangateOpenCellLibrary_fast

Corner

fast

AND2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

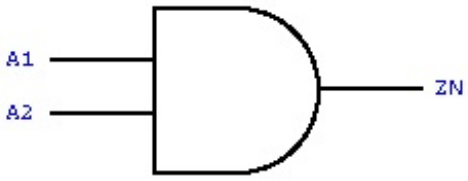
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	L
H	H	H
-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.01	0.07	0.04	0.11
	rise	0.02	0.10	0.03	0.12
A2 to ZN	**fall**	0.01	0.08	0.05	0.12
	rise	0.02	0.10	0.02	0.11

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A1	0.9204
A2	0.9814

Leakage [nW]
89.64

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	3.84	4.42	10.92	10.92
	rise	2.60	2.70	10.36	9.25
A2 to ZN	**fall**	4.27	4.92	11.98	11.49
	rise	2.62	2.54	8.61	8.46

AND2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

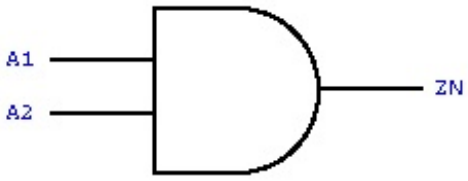
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	L
H	H	H
-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	**fall**	0.01	0.07	0.04	0.11
	rise	0.01	0.10	0.03	0.12
A2 to ZN	**fall**	0.01	0.07	0.05	0.11
	rise	0.02	0.10	0.02	0.11

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A1	1.6685
A2	1.7779

Leakage [nW]
180.07

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	**fall**	7.33	8.40	21.80	21.64
	rise	5.07	4.67	20.77	18.27
A2 to ZN	**fall**	8.24	9.41	23.78	22.75
	rise	5.07	5.28	17.30	16.32

AND2_X4

Databook Build Date: Thursday Feb 17 15:07 2011

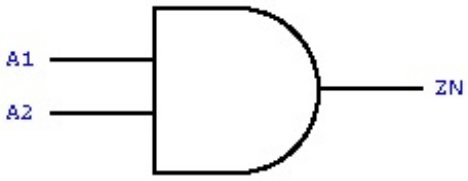
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	L
H	H	H
-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	**fall**	0.01	0.07	0.04	0.11
	rise	0.01	0.10	0.03	0.12
A2 to ZN	**fall**	0.01	0.07	0.05	0.11
	rise	0.02	0.10	0.02	0.11

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A1	3.2011
A2	3.6354

Leakage [nW]
360.14

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	**fall**	14.12	16.23	43.26	42.86
	rise	9.95	8.48	41.47	36.56
A2 to ZN	**fall**	16.02	18.28	47.40	45.05
	rise	9.96	10.30	34.33	32.46

AND3_X1

Databook Build Date: Thursday Feb 17 15:07 2011

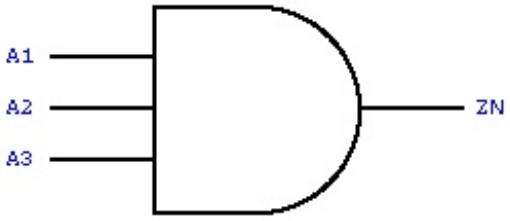
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	L
H	H	H	H
-	L	-	L
-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
A1 to ZN	**fall**	0.01	0.08	0.04	0.11
	rise	0.02	0.11	0.05	0.14
A2 to ZN	**fall**	0.01	0.08	0.04	0.11
	rise	0.02	0.11	0.04	0.13
A3 to ZN	**fall**	0.02	0.08	0.05	0.12
	rise	0.02	0.11	0.03	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A1	0.9082
A2	0.9296
A3	0.9832

Leakage [nW]
92.60

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
A1 to ZN	**fall**	4.02	4.70	10.55	10.33
	rise	3.22	2.93	10.40	8.16
A2 to ZN	**fall**	4.56	5.24	11.42	10.81
	rise	3.21	3.11	8.92	7.57
A3 to ZN	**fall**	4.99	5.71	12.35	11.77
	rise	3.21	2.99	8.65	8.07

AND3_X2

Databook Build Date: Thursday Feb 17 15:07 2011

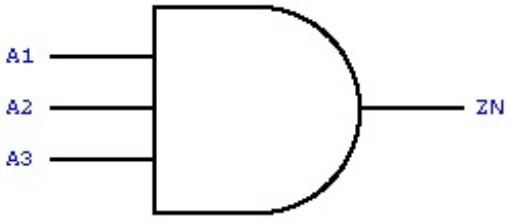
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.596 um^2
Equation	ZN = "((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	L
H	H	H	H
-	L	-	L
-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
A1 to ZN	**fall**	0.01	0.07	0.03	0.10
	rise	0.02	0.11	0.05	0.13
A2 to ZN	**fall**	0.01	0.08	0.04	0.11
	rise	0.02	0.11	0.04	0.13
A3 to ZN	**fall**	0.02	0.08	0.05	0.12
	rise	0.02	0.11	0.03	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A1	1.6559
A2	1.6574
A3	1.7570

Leakage [nW]
185.95

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
A1 to ZN	**fall**	7.72	8.90	20.78	20.31
	rise	6.33	5.87	20.84	16.99
A2 to ZN	**fall**	8.72	9.98	22.66	21.26
	rise	6.34	5.87	18.00	15.29
A3 to ZN	**fall**	9.58	10.93	24.66	23.22
	rise	6.34	5.80	17.46	15.90

AND3_X4

Databook Build Date: Thursday Feb 17 15:07 2011

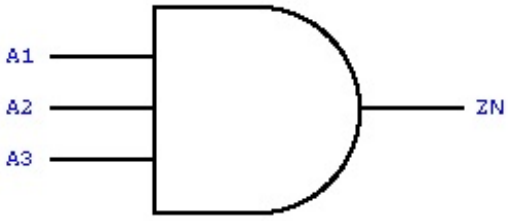
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	ZN = "((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	L
H	H	H	H
-	L	-	L
-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	241.089	0.3656	241.089
A1 to ZN	**fall**	0.01	0.07	0.03	0.10
	rise	0.02	0.11	0.05	0.13
A2 to ZN	**fall**	0.01	0.08	0.04	0.11
	rise	0.02	0.11	0.04	0.13
A3 to ZN	**fall**	0.01	0.08	0.05	0.12
	rise	0.02	0.11	0.03	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	241.089	0.3656	241.089
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A1	3.2032
A2	3.3418
A3	3.6402

Leakage [nW]
371.92

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	241.089	0.3656	241.089
A1 to ZN	**fall**	15.00	17.25	41.44	40.21
	rise	12.24	11.18	41.54	33.34
A2 to ZN	**fall**	17.02	19.44	44.86	42.05
	rise	12.26	11.54	35.85	30.83
A3 to ZN	**fall**	18.80	21.34	48.90	46.02
	rise	12.26	11.09	34.65	32.69

AND4_X1

Databook Build Date: Thursday Feb 17 15:07 2011

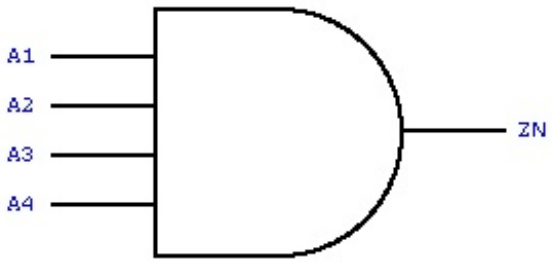
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	L
H	H	H	H	H
-	L	-	-	L
-	-	L	-	L
-	-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A1 to ZN	**fall**	0.01	0.08	0.03	0.10
	rise	0.03	0.12	0.06	0.15
A2 to ZN	**fall**	0.02	0.08	0.04	0.11
	rise	0.03	0.12	0.06	0.15
A3 to ZN	**fall**	0.02	0.08	0.05	0.11
	rise	0.03	0.12	0.05	0.14
A4 to ZN	**fall**	0.02	0.08	0.05	0.12
	rise	0.03	0.12	0.04	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.01	0.08	0.01	0.08
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.01	0.08	0.01	0.08
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.01	0.08	0.01	0.08
A4 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.01	0.08	0.01	0.08

Capacitance [fF]	
A1	0.8976
A2	0.9178
A3	0.9260
A4	0.9654

Leakage [nW]
93.06

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.2722	0.3656	60.2722
A1 to ZN	fall	4.28	4.99	10.30	9.98
	rise	3.91	3.31	10.42	7.66
A2 to ZN	fall	4.79	5.54	11.06	10.42
	rise	3.92	3.25	9.20	6.85
A3 to ZN	fall	5.26	6.05	11.95	11.24
	rise	3.94	3.28	8.93	7.41
A4 to ZN	fall	5.65	6.50	12.84	12.14
	rise	3.93	3.44	8.85	7.16

AND4_X2

Databook Build Date: Thursday Feb 17 15:07 2011

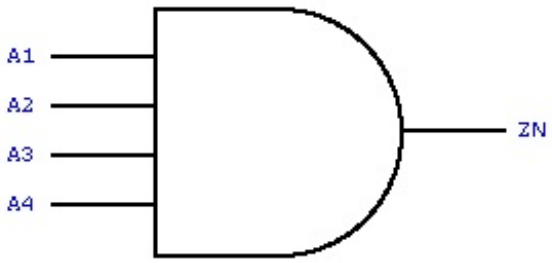
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	L
H	H	H	H	H
-	L	-	-	L
-	-	L	-	L
-	-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.392**	**0.3656**	**120.392**
A1 to ZN	**fall**	0.01	0.08	0.03	0.10
	rise	0.03	0.11	0.06	0.15
A2 to ZN	**fall**	0.01	0.08	0.04	0.11
	rise	0.03	0.12	0.06	0.15
A3 to ZN	**fall**	0.02	0.08	0.04	0.11
	rise	0.03	0.12	0.05	0.14
A4 to ZN	**fall**	0.02	0.08	0.05	0.12
	rise	0.03	0.12	0.04	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.392**	**0.3656**	**120.392**
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.01	0.08	0.01	0.08
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.01	0.08	0.01	0.08
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.01	0.08	0.01	0.08
A4 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.01	0.08	0.01	0.08

Capacitance [fF]	
A1	1.6449
A2	1.6545
A3	1.6679
A4	1.7417

Leakage [nW]
186.82

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.392	0.3656	120.392
A1 to ZN	fall	8.04	9.26	20.26	19.37
	rise	7.49	5.62	20.78	15.49
A2 to ZN	fall	9.03	10.36	21.73	20.22
	rise	7.55	6.73	18.33	14.44
A3 to ZN	fall	10.03	11.38	23.48	21.88
	rise	7.55	5.62	17.94	14.12
A4 to ZN	fall	10.77	12.28	25.49	23.75
	rise	7.59	6.69	17.66	14.98

AND4_X4

Databook Build Date: Thursday Feb 17 15:07 2011

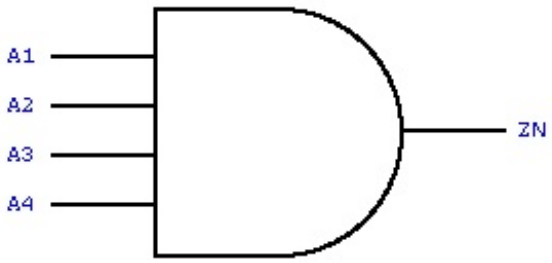
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	L
H	H	H	H	H
-	L	-	-	L
-	-	L	-	L
-	-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.01	0.07	0.03	0.10
	rise	0.02	0.11	0.06	0.15
A2 to ZN	**fall**	0.01	0.08	0.04	0.11
	rise	0.03	0.12	0.06	0.15
A3 to ZN	**fall**	0.02	0.08	0.04	0.11
	rise	0.03	0.12	0.05	0.14
A4 to ZN	**fall**	0.02	0.08	0.05	0.12
	rise	0.03	0.12	0.03	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.01	0.08	0.01	0.08
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.01	0.08	0.01	0.08
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.01	0.08	0.01	0.08
A4 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.01	0.08	0.01	0.08

Capacitance [fF]	
A1	3.2142
A2	3.3135
A3	3.4883
A4	3.8240

Leakage [nW]
373.64

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	241.089	0.3656	241.089
A1 to ZN	fall	15.31	17.66	40.04	38.00
	rise	14.75	11.81	41.36	31.37
A2 to ZN	fall	17.47	19.88	43.09	39.67
	rise	14.76	13.10	36.32	27.42
A3 to ZN	fall	19.38	21.94	46.42	43.00
	rise	14.63	12.71	35.74	28.45
A4 to ZN	fall	20.92	23.75	50.47	46.78
	rise	14.76	10.94	35.12	30.41

ANTENNA_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.266 um^2
Type	Physical
Input	A
PG Pins	VDD (primary_power), VSS (primary_ground)



Capacitance [fF]	
A	0.0236

Leakage [nW]
0.00

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1462**
A	**fall**	0.00	0.00
	rise	0.00	0.00

AOI211_X1

Databook Build Date: Thursday Feb 17 15:07 2011

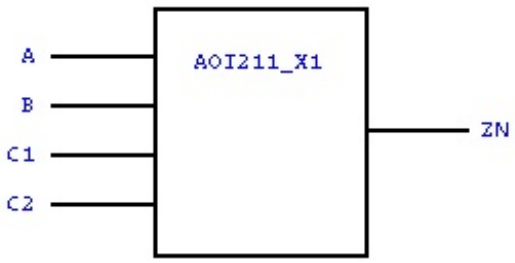
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((C1 & C2) | B) | A)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	L	L	-	H
L	L	-	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**14.4959**	**0.3656**	**14.4959**
A to ZN	**fall**	0.01	0.02	-0.00	0.03
	rise	0.03	0.08	0.06	0.12
B to ZN	**fall**	0.01	0.02	-0.01	0.03
	rise	0.02	0.08	0.07	0.13
C1 to ZN	**fall**	0.01	0.03	-0.01	0.05
	rise	0.01	0.07	0.06	0.13
C2 to ZN	**fall**	0.01	0.03	-0.01	0.04
	rise	0.02	0.07	0.07	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**14.4959**	**0.3656**	**14.4959**
A to ZN	**fall**	0.01	0.02	0.04	0.05
	rise	0.01	0.07	0.03	0.07
B to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.07	0.03	0.07
C1 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.06	0.03	0.08
C2 to ZN	**fall**	0.00	0.02	0.02	0.04
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A	1.6605
B	1.7095
C1	1.6991
C2	1.7401

Leakage [nW]
102.24

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	14.4959	0.3656	14.4959
A to ZN	fall	1.52	1.94	8.75	8.02
	rise	6.66	6.76	13.06	10.93
B to ZN	fall	1.29	1.48	7.45	7.10
	rise	5.68	5.71	11.72	9.99
C1 to ZN	fall	0.35	0.38	6.07	5.24
	rise	3.73	3.85	11.27	9.34
C2 to ZN	fall	0.35	0.38	4.74	4.43
	rise	4.69	4.76	12.26	10.55

AOI211_X2

Databook Build Date: Thursday Feb 17 15:07 2011

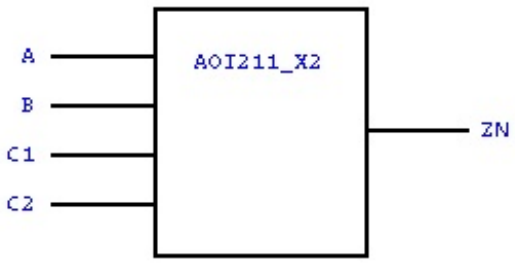
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((C1 & C2) | B) | A)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	L	L	-	H
L	L	-	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**28.9917**	**0.3656**	**28.9917**
A to ZN	**fall**	0.01	0.02	-0.00	0.04
	rise	0.03	0.08	0.06	0.12
B to ZN	**fall**	0.01	0.02	-0.01	0.03
	rise	0.02	0.08	0.07	0.13
C1 to ZN	**fall**	0.01	0.03	-0.01	0.05
	rise	0.01	0.07	0.06	0.13
C2 to ZN	**fall**	0.01	0.03	-0.01	0.04
	rise	0.02	0.07	0.07	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**28.9917**	**0.3656**	**28.9917**
A to ZN	**fall**	0.01	0.02	0.04	0.05
	rise	0.01	0.07	0.03	0.07
B to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.07	0.03	0.07
C1 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.06	0.03	0.08
C2 to ZN	**fall**	0.00	0.02	0.02	0.04
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A	3.1820
B	3.5341
C1	3.2463
C2	3.5674

Leakage [nW]
204.47

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	28.9917	0.3656	28.9917
A to ZN	fall	2.93	3.74	17.33	15.88
	rise	13.45	13.62	26.36	22.07
B to ZN	fall	2.43	2.82	14.75	14.03
	rise	11.49	11.60	23.71	20.17
C1 to ZN	fall	0.71	0.76	12.09	10.46
	rise	7.40	7.73	22.64	18.60
C2 to ZN	fall	0.69	0.76	9.47	8.83
	rise	9.39	9.56	24.54	21.12

AOI211_X4

Databook Build Date: Thursday Feb 17 15:07 2011

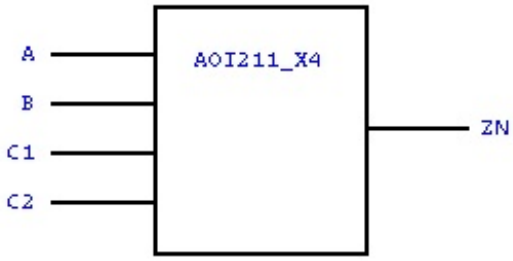
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	ZN = "!(!(!(((C1 & C2) | B) | A)))"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	L	L	-	H
L	L	-	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.02	0.08	0.03	0.09
	rise	0.05	0.14	0.09	0.18
B to ZN	**fall**	0.02	0.08	0.02	0.09
	rise	0.05	0.14	0.10	0.18
C1 to ZN	**fall**	0.02	0.09	0.03	0.09
	rise	0.04	0.12	0.09	0.18
C2 to ZN	**fall**	0.03	0.09	0.02	0.09
	rise	0.04	0.13	0.10	0.18

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
B to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
C1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
C2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08

Capacitance [fF]	
A	1.7081
B	1.7525
C1	1.6885
C2	1.8089

Leakage [nW]
439.08

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	fall	15.72	17.78	29.61	32.11
	rise	23.74	25.58	32.29	34.02
B to ZN	fall	15.21	17.07	27.88	30.35
	rise	22.76	24.12	31.64	33.39
C1 to ZN	fall	14.27	15.94	26.92	29.49
	rise	20.85	21.67	32.92	34.96
C2 to ZN	fall	14.21	15.95	23.44	26.58
	rise	22.33	23.74	34.02	35.29

AOI21_X1

Databook Build Date: Thursday Feb 17 15:07 2011

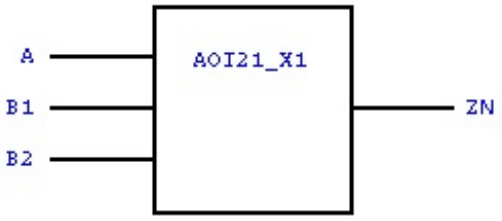
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!(A | (B1 & B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	L	-	H
L	-	L	H
H	-	-	L
-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to ZN	**fall**	0.01	0.03	-0.00	0.05
	rise	0.02	0.08	0.05	0.13
B1 to ZN	**fall**	0.01	0.05	-0.00	0.08
	rise	0.01	0.07	0.04	0.13
B2 to ZN	**fall**	0.01	0.05	-0.01	0.07
	rise	0.01	0.08	0.05	0.14

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to ZN	**fall**	0.00	0.03	0.03	0.06
	rise	0.01	0.07	0.03	0.08
B1 to ZN	**fall**	0.00	0.04	0.03	0.06
	rise	0.01	0.07	0.03	0.08
B2 to ZN	**fall**	0.00	0.04	0.02	0.05
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A	1.6617
B1	1.6720
B2	1.7195

Leakage [nW]
87.02

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to ZN	**fall**	1.17	1.42	9.67	7.66
	rise	5.24	5.27	13.56	10.24
B1 to ZN	**fall**	0.31	0.38	7.56	4.96
	rise	3.27	3.39	12.36	9.21
B2 to ZN	**fall**	0.31	0.38	6.11	4.22
	rise	4.21	4.14	13.46	10.61

AOI21_X2

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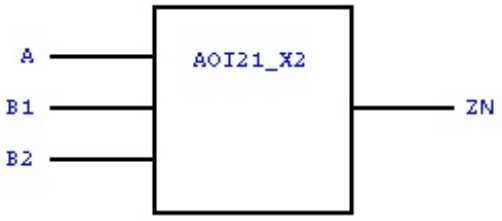
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!(A | (B1 & B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	L	-	H
L	-	L	H
H	-	-	L
-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	50.6592	0.3656	50.6592
A to ZN	**fall**	0.01	0.03	-0.00	0.05
	rise	0.01	0.08	0.05	0.13
B1 to ZN	**fall**	0.01	0.05	-0.00	0.08
	rise	0.01	0.07	0.04	0.13
B2 to ZN	**fall**	0.01	0.05	-0.01	0.07
	rise	0.01	0.08	0.05	0.14

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	50.6592	0.3656	50.6592
A to ZN	**fall**	0.00	0.03	0.03	0.06
	rise	0.01	0.07	0.03	0.08
B1 to ZN	**fall**	0.00	0.04	0.03	0.06
	rise	0.01	0.07	0.03	0.08
B2 to ZN	**fall**	0.00	0.04	0.02	0.05
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A	3.1946
B1	3.1775
B2	3.5898

Leakage [nW]
174.04

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	50.6592	0.3656	50.6592
A to ZN	**fall**	2.22	2.69	19.21	15.21
	rise	10.24	10.33	27.08	20.24
B1 to ZN	**fall**	0.61	0.76	15.19	9.91
	rise	6.44	6.79	24.76	18.29
B2 to ZN	**fall**	0.60	0.76	12.08	8.44
	rise	8.42	8.34	26.88	21.15

AOI21_X4

Databook Build Date: Thursday Feb 17 15:07 2011

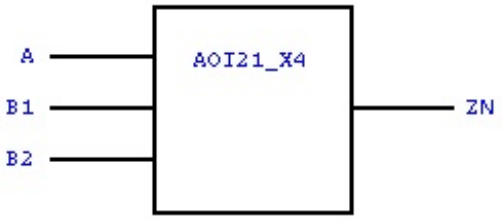
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(A | (B1 & B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	L	-	H
L	-	L	H
H	-	-	L
-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	101.013	0.3656	101.013
A to ZN	**fall**	0.01	0.03	-0.00	0.05
	rise	0.01	0.08	0.05	0.13
B1 to ZN	**fall**	0.01	0.05	-0.00	0.08
	rise	0.01	0.07	0.04	0.13
B2 to ZN	**fall**	0.01	0.05	-0.01	0.07
	rise	0.01	0.08	0.05	0.14

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	101.013	0.3656	101.013
A to ZN	**fall**	0.00	0.03	0.03	0.06
	rise	0.01	0.07	0.03	0.08
B1 to ZN	**fall**	0.00	0.04	0.03	0.06
	rise	0.01	0.07	0.03	0.08
B2 to ZN	**fall**	0.00	0.04	0.02	0.05
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A	6.2240
B1	6.5175
B2	6.8825

Leakage [nW]
348.08

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	101.013	0.3656	101.013
A to ZN	**fall**	4.41	5.36	38.35	30.26
	rise	20.91	21.12	54.58	41.18
B1 to ZN	**fall**	1.20	1.51	30.09	19.71
	rise	12.98	13.02	49.52	36.06
B2 to ZN	**fall**	1.22	1.51	24.18	16.80
	rise	16.94	17.24	53.73	41.77

AOI221_X1

Databook Build Date: Thursday Feb 17 15:07 2011

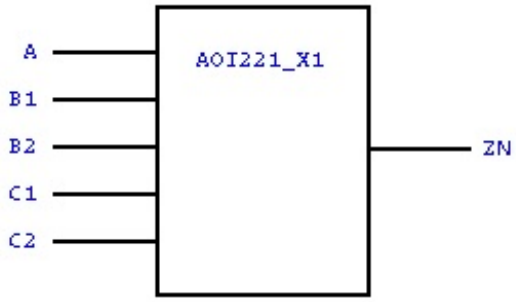
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "!(((C1 & C2) | A) | (B1 & B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	L	-	H
L	L	-	-	L	H
L	-	L	L	-	H
L	-	L	-	L	H
H	-	-	-	-	L
-	H	H	-	-	L
-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	13.8092	0.3656	13.8092
A to ZN	**fall**	0.01	0.02	-0.00	0.03
	rise	0.03	0.08	0.08	0.13
B1 to ZN	**fall**	0.01	0.03	0.02	0.06
	rise	0.03	0.08	0.06	0.11
B2 to ZN	**fall**	0.01	0.04	0.01	0.05
	rise	0.03	0.08	0.07	0.12
C1 to ZN	**fall**	0.01	0.03	-0.00	0.05
	rise	0.02	0.07	0.06	0.13
C2 to ZN	**fall**	0.01	0.03	-0.01	0.04
	rise	0.02	0.07	0.07	0.14

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	13.8092	0.3656	13.8092
A to ZN	**fall**	0.01	0.02	0.04	0.05
	rise	0.02	0.07	0.03	0.07
B1 to ZN	**fall**	0.01	0.03	0.04	0.06
	rise	0.01	0.06	0.03	0.07
B2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.02	0.07	0.03	0.07
C1 to ZN	**fall**	0.01	0.02	0.04	0.05
	rise	0.01	0.06	0.03	0.08
C2 to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.02	0.07	0.03	0.08

Capacitance [fF]	
A	1.7206
B1	1.6367
B2	1.6536
C1	1.6725
C2	1.7648

Leakage [nW]
123.72

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	13.8092	0.3656	13.8092
A to ZN	fall	1.15	1.30	7.73	7.26
	rise	6.03	6.06	12.38	10.69
B1 to ZN	fall	1.91	2.25	8.54	7.16
	rise	7.14	7.23	12.29	10.49
B2 to ZN	fall	1.91	2.25	7.12	6.24
	rise	8.09	8.14	13.29	11.73
C1 to ZN	fall	0.34	0.38	6.35	5.45
	rise	4.05	4.14	11.74	9.91
C2 to ZN	fall	0.35	0.38	5.00	4.59
	rise	5.01	5.05	12.77	11.19

AOI221_X2

Databook Build Date: Thursday Feb 17 15:07 2011

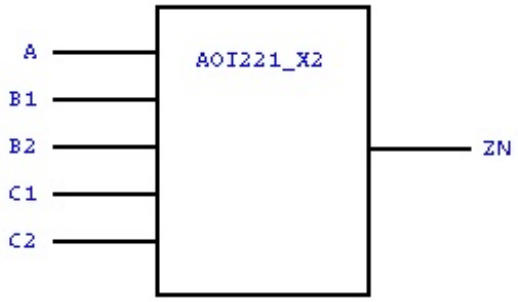
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.926 um^2
Equation	ZN = "!(((C1 & C2) | A) | (B1 & B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	L	-	H
L	L	-	-	L	H
L	-	L	L	-	H
L	-	L	-	L	H
H	-	-	-	-	L
-	H	H	-	-	L
-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	27.6184	0.3656	27.6184
A to ZN	**fall**	0.01	0.02	-0.01	0.03
	rise	0.03	0.08	0.07	0.13
B1 to ZN	**fall**	0.01	0.03	0.02	0.06
	rise	0.03	0.08	0.06	0.11
B2 to ZN	**fall**	0.01	0.04	0.01	0.05
	rise	0.03	0.08	0.06	0.12
C1 to ZN	**fall**	0.01	0.03	-0.01	0.05
	rise	0.02	0.07	0.06	0.13
C2 to ZN	**fall**	0.01	0.03	-0.01	0.04
	rise	0.02	0.07	0.07	0.14

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	27.6184	0.3656	27.6184
A to ZN	**fall**	0.01	0.02	0.04	0.05
	rise	0.02	0.07	0.03	0.07
B1 to ZN	**fall**	0.01	0.03	0.04	0.06
	rise	0.01	0.06	0.03	0.07
B2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.02	0.07	0.03	0.07
C1 to ZN	**fall**	0.00	0.02	0.04	0.05
	rise	0.01	0.06	0.03	0.08
C2 to ZN	**fall**	0.00	0.02	0.03	0.04
	rise	0.02	0.07	0.03	0.08

Capacitance [fF]	
A	3.5993
B1	3.3384
B2	3.2042
C1	3.2181
C2	3.6263

Leakage [nW]
247.44

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	27.6184	0.3656	27.6184
A to ZN	fall	2.43	2.76	15.55	14.63
	rise	12.10	12.25	24.79	21.36
B1 to ZN	fall	3.84	4.52	17.11	14.33
	rise	14.20	14.40	24.65	20.97
B2 to ZN	fall	3.83	4.52	14.27	12.52
	rise	16.09	16.21	26.61	23.59
C1 to ZN	fall	0.68	0.76	12.73	10.87
	rise	8.12	8.42	23.63	19.77
C2 to ZN	fall	0.68	0.76	9.98	9.19
	rise	10.11	10.25	25.64	22.36

AOI221_X4

Databook Build Date: Thursday Feb 17 15:07 2011

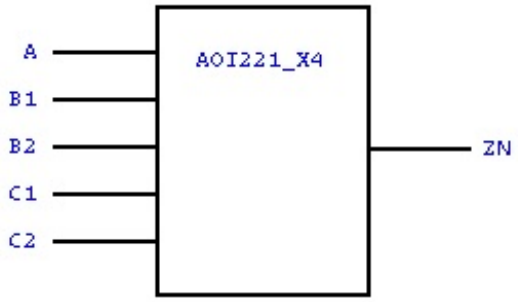
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(!(!(((C1 & C2) | A) | (B1 & B2))))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	L	-	H
L	L	-	-	L	H
L	-	L	L	-	H
L	-	L	-	L	H
H	-	-	-	-	L
-	H	H	-	-	L
-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.02	0.08	0.03	0.09
	rise	0.06	0.14	0.11	0.19
B1 to ZN	**fall**	0.03	0.09	0.05	0.11
	rise	0.06	0.14	0.09	0.17
B2 to ZN	**fall**	0.03	0.10	0.04	0.10
	rise	0.06	0.14	0.10	0.18
C1 to ZN	**fall**	0.03	0.09	0.03	0.10
	rise	0.04	0.13	0.10	0.18
C2 to ZN	**fall**	0.03	0.09	0.03	0.09
	rise	0.05	0.13	0.11	0.19

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
B1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
B2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
C1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
C2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08

Capacitance [fF]	
A	1.6899
B1	1.6112
B2	1.6798
C1	1.6484
C2	1.7443

Leakage [nW]
455.02

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	fall	15.25	17.10	28.51	30.92
	rise	23.07	24.58	32.26	34.02
B1 to ZN	fall	17.07	19.09	30.64	32.14
	rise	24.34	26.01	32.07	34.77
B2 to ZN	fall	17.06	19.08	26.92	29.01
	rise	25.86	28.02	33.48	35.76
C1 to ZN	fall	14.32	16.03	27.49	30.01
	rise	21.27	22.34	33.47	35.33
C2 to ZN	fall	14.39	16.04	23.93	27.03
	rise	22.74	23.92	34.75	35.47

AOI222_X1

Databook Build Date: Thursday Feb 17 15:07 2011

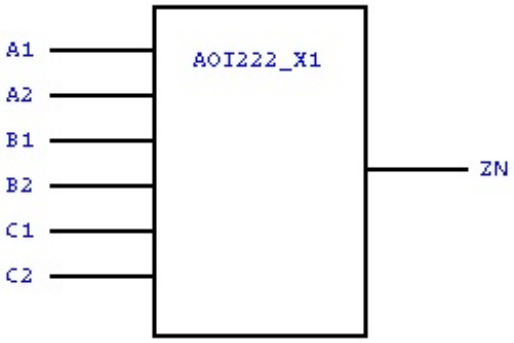
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.128 um^2
Equation	ZN = "!(((A1 & A2) | (B1 & B2)) | (C1 & C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	L	-	L	-	H
L	-	L	-	-	L	H
L	-	-	L	L	-	H
L	-	-	L	-	L	H
H	H	-	-	-	-	L
-	L	L	-	L	-	H
-	L	L	-	-	L	H
-	L	-	L	L	-	H
-	L	-	L	-	L	H
-	-	H	H	-	-	L
-	-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**13.0081**	**0.3656**	**13.0081**
A1 to ZN	**fall**	0.01	0.03	-0.00	0.05
	rise	0.02	0.07	0.07	0.13
A2 to ZN	**fall**	0.01	0.03	-0.00	0.04
	rise	0.02	0.07	0.08	0.14
B1 to ZN	**fall**	0.01	0.03	0.02	0.06
	rise	0.03	0.08	0.07	0.13
B2 to ZN	**fall**	0.01	0.03	0.01	0.05
	rise	0.03	0.08	0.08	0.13
C1 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.04	0.09	0.06	0.12
C2 to ZN	**fall**	0.01	0.04	0.02	0.05
	rise	0.04	0.09	0.07	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**13.0081**	**0.3656**	**13.0081**
A1 to ZN	**fall**	0.01	0.02	0.04	0.06
	rise	0.02	0.06	0.03	0.08
A2 to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.02	0.07	0.03	0.08
B1 to ZN	**fall**	0.01	0.03	0.04	0.06
	rise	0.02	0.06	0.03	0.07
B2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.02	0.07	0.03	0.07
C1 to ZN	**fall**	0.01	0.03	0.05	0.06
	rise	0.02	0.06	0.03	0.07
C2 to ZN	**fall**	0.01	0.03	0.04	0.05
	rise	0.02	0.07	0.03	0.07

Capacitance [fF]	
A1	1.6742
A2	1.7488
B1	1.6051
B2	1.6718
C1	1.5950
C2	1.6131

Leakage [nW]
140.87

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	13.0081	0.3656	13.0081
A1 to ZN	fall	0.34	0.36	6.64	5.68
	rise	4.33	4.46	12.28	10.45
A2 to ZN	fall	0.34	0.36	5.24	4.79
	rise	5.29	5.38	13.35	11.88
B1 to ZN	fall	1.58	1.73	7.81	6.72
	rise	6.51	6.60	11.77	10.18
B2 to ZN	fall	1.59	1.73	6.41	5.81
	rise	7.45	7.52	12.80	11.51
C1 to ZN	fall	2.44	2.86	9.06	7.75
	rise	8.28	8.36	12.89	11.31
C2 to ZN	fall	2.44	2.86	7.64	6.83
	rise	9.24	9.30	13.91	12.59

AOI222_X2

Databook Build Date: Thursday Feb 17 15:07 2011

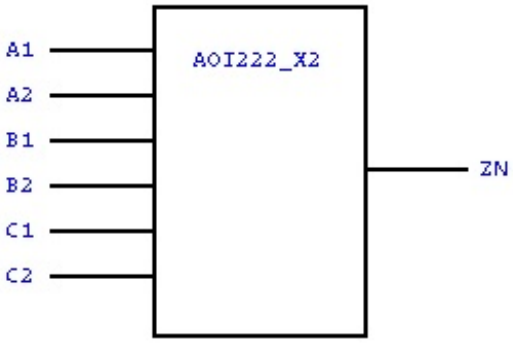
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	3.724 um^2
Equation	ZN = "!(((A1 & A2) | (B1 & B2)) | (C1 & C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	L	-	L	-	H
L	-	L	-	-	L	H
L	-	-	L	L	-	H
L	-	-	L	-	L	H
H	H	-	-	-	-	L
-	L	L	-	L	-	H
-	L	L	-	-	L	H
-	L	-	L	L	-	H
-	L	-	L	-	L	H
-	-	H	H	-	-	L
-	-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**25.6348**	**0.3656**	**25.6348**
A1 to ZN	**fall**	0.01	0.03	-0.00	0.05
	rise	0.02	0.07	0.07	0.13
A2 to ZN	**fall**	0.01	0.03	-0.00	0.04
	rise	0.02	0.07	0.08	0.14
B1 to ZN	**fall**	0.01	0.03	0.01	0.06
	rise	0.03	0.08	0.07	0.13
B2 to ZN	**fall**	0.01	0.03	0.01	0.05
	rise	0.03	0.08	0.08	0.13
C1 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.04	0.09	0.06	0.12
C2 to ZN	**fall**	0.01	0.04	0.02	0.05
	rise	0.04	0.09	0.07	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**25.6348**	**0.3656**	**25.6348**
A1 to ZN	**fall**	0.01	0.02	0.04	0.06
	rise	0.02	0.06	0.03	0.08
A2 to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.02	0.07	0.03	0.08
B1 to ZN	**fall**	0.01	0.03	0.04	0.06
	rise	0.02	0.06	0.03	0.07
B2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.02	0.07	0.03	0.07
C1 to ZN	**fall**	0.01	0.03	0.05	0.06
	rise	0.02	0.06	0.03	0.07
C2 to ZN	**fall**	0.01	0.03	0.04	0.05
	rise	0.02	0.07	0.03	0.07

Capacitance [fF]	
A1	3.2309
A2	3.3243
B1	3.0719
B2	3.5056
C1	3.0582
C2	3.3584

Leakage [nW]
281.75

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	25.6348	0.3656	25.6348
A1 to ZN	fall	0.70	0.71	13.28	11.35
	rise	8.91	9.17	24.85	21.18
A2 to ZN	fall	0.69	0.72	10.40	9.51
	rise	11.09	11.30	27.12	24.25
B1 to ZN	fall	3.16	3.47	15.66	13.45
	rise	13.30	13.54	23.88	20.72
B2 to ZN	fall	3.15	3.47	12.83	11.63
	rise	15.18	15.35	25.90	23.35
C1 to ZN	fall	4.89	5.72	18.13	15.50
	rise	16.97	17.19	26.17	23.00
C2 to ZN	fall	4.87	5.72	15.27	13.65
	rise	18.88	19.03	28.20	25.62

AOI222_X4

Databook Build Date: Thursday Feb 17 15:07 2011

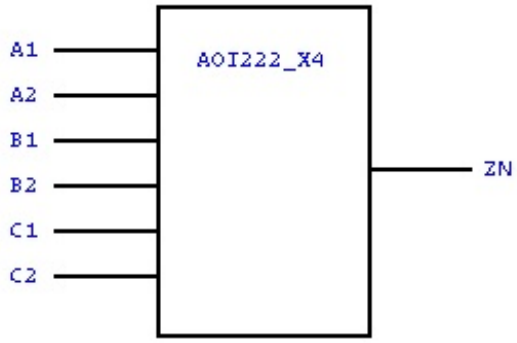
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.724 um^2
Equation	ZN = "!(!(!(((A1 & A2) | (B1 & B2)) | (C1 & C2))))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	L	-	L	-	H
L	-	L	-	-	L	H
L	-	-	L	L	-	H
L	-	-	L	-	L	H
H	H	-	-	-	-	L
-	L	L	-	L	-	H
-	L	L	-	-	L	H
-	L	-	L	L	-	H
-	L	-	L	-	L	H
-	-	H	H	-	-	L
-	-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.03	0.09	0.04	0.10
	rise	0.05	0.13	0.10	0.19
A2 to ZN	**fall**	0.03	0.09	0.03	0.09
	rise	0.05	0.13	0.11	0.20
B1 to ZN	**fall**	0.03	0.09	0.05	0.11
	rise	0.06	0.14	0.10	0.19
B2 to ZN	**fall**	0.03	0.09	0.04	0.10
	rise	0.06	0.14	0.11	0.19
C1 to ZN	**fall**	0.04	0.10	0.06	0.12
	rise	0.06	0.15	0.09	0.18
C2 to ZN	**fall**	0.04	0.10	0.05	0.11
	rise	0.07	0.15	0.10	0.18

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
A2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
B1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
B2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
C1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
C2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08

Capacitance [fF]	
A1	1.6321
A2	1.7627
B1	1.6256
B2	1.7235
C1	1.6239
C2	1.6796

Leakage [nW]
463.84

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	fall	14.44	16.08	28.00	30.42
	rise	21.42	22.66	33.89	35.38
A2 to ZN	fall	14.40	16.09	24.32	27.41
	rise	22.85	24.45	35.19	36.58
B1 to ZN	fall	16.27	18.11	29.48	31.01
	rise	23.45	24.96	32.15	34.70
B2 to ZN	fall	16.30	18.11	25.90	27.92
	rise	24.90	26.31	33.53	35.39
C1 to ZN	fall	17.82	19.91	31.31	32.86
	rise	25.18	26.44	32.60	34.91
C2 to ZN	fall	17.80	19.91	27.61	29.69
	rise	26.61	28.40	34.03	36.52

AOI22_X1

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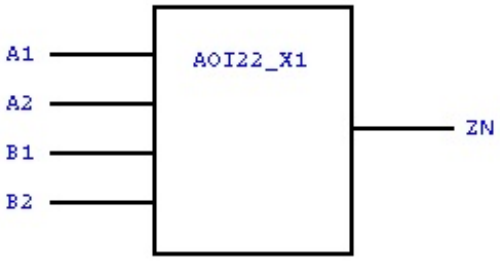
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!((A1 & A2) | (B1 & B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**24.6048**	**0.3656**	**24.6048**
A1 to ZN	**fall**	0.01	0.04	0.00	0.08
	rise	0.01	0.07	0.05	0.13
A2 to ZN	**fall**	0.01	0.05	-0.00	0.07
	rise	0.01	0.08	0.06	0.14
B1 to ZN	**fall**	0.01	0.05	0.02	0.08
	rise	0.02	0.08	0.05	0.13
B2 to ZN	**fall**	0.01	0.05	0.01	0.07
	rise	0.02	0.08	0.05	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**24.6048**	**0.3656**	**24.6048**
A1 to ZN	**fall**	0.00	0.04	0.03	0.07
	rise	0.01	0.07	0.03	0.08
A2 to ZN	**fall**	0.00	0.04	0.03	0.05
	rise	0.01	0.07	0.03	0.08
B1 to ZN	**fall**	0.01	0.04	0.04	0.07
	rise	0.01	0.07	0.03	0.08
B2 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A1	1.7111
A2	1.7350
B1	1.6133
B2	1.6528

Leakage [nW]
101.18

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	24.6048	0.3656	24.6048
A1 to ZN	fall	0.29	0.36	8.10	5.20
	rise	3.59	3.74	12.94	9.81
A2 to ZN	fall	0.30	0.36	6.52	4.43
	rise	4.57	4.54	14.15	11.34
B1 to ZN	fall	1.50	1.72	9.33	6.50
	rise	5.47	5.46	12.25	9.37
B2 to ZN	fall	1.50	1.72	7.75	5.64
	rise	6.40	6.37	13.30	10.76

AOI22_X2

Databook Build Date: Thursday Feb 17 15:07 2011

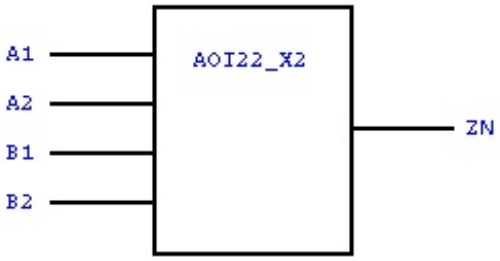
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!((A1 & A2) | (B1 & B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**49.1333**	**0.3656**	**49.1333**
A1 to ZN	**fall**	0.01	0.04	0.00	0.08
	rise	0.01	0.07	0.05	0.13
A2 to ZN	**fall**	0.01	0.05	-0.00	0.07
	rise	0.01	0.08	0.06	0.14
B1 to ZN	**fall**	0.01	0.05	0.02	0.08
	rise	0.02	0.08	0.04	0.13
B2 to ZN	**fall**	0.01	0.05	0.01	0.07
	rise	0.02	0.08	0.05	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**49.1333**	**0.3656**	**49.1333**
A1 to ZN	**fall**	0.00	0.04	0.03	0.07
	rise	0.01	0.07	0.03	0.08
A2 to ZN	**fall**	0.00	0.04	0.03	0.05
	rise	0.01	0.07	0.03	0.08
B1 to ZN	**fall**	0.01	0.04	0.04	0.07
	rise	0.01	0.07	0.03	0.08
B2 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A1	3.1849
A2	3.5654
B1	3.0519
B2	3.4907

Leakage [nW]
202.35

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	49.1333	0.3656	49.1333
A1 to ZN	fall	0.57	0.73	16.11	10.38
	rise	7.26	7.41	26.03	19.79
A2 to ZN	fall	0.57	0.73	12.96	8.84
	rise	9.13	9.31	28.33	22.65
B1 to ZN	fall	2.97	3.43	18.62	12.92
	rise	11.35	11.39	24.85	19.17
B2 to ZN	fall	2.97	3.43	15.41	11.22
	rise	13.19	13.32	26.99	21.84

AOI22_X4

Databook Build Date: Thursday Feb 17 15:07 2011

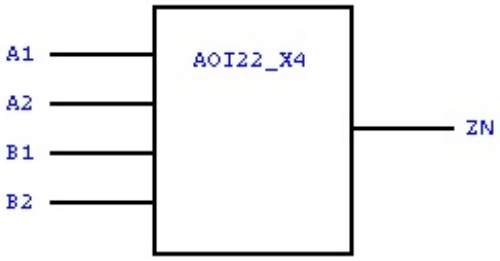
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Equation	ZN = "!((A1 & A2) | (B1 & B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**97.9614**	**0.3656**	**97.9614**
A1 to ZN	**fall**	0.01	0.04	0.00	0.08
	rise	0.01	0.08	0.05	0.13
A2 to ZN	**fall**	0.01	0.05	-0.00	0.07
	rise	0.01	0.08	0.06	0.14
B1 to ZN	**fall**	0.01	0.05	0.02	0.08
	rise	0.02	0.08	0.04	0.13
B2 to ZN	**fall**	0.01	0.05	0.01	0.07
	rise	0.02	0.08	0.05	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**97.9614**	**0.3656**	**97.9614**
A1 to ZN	**fall**	0.00	0.04	0.03	0.07
	rise	0.01	0.07	0.03	0.08
A2 to ZN	**fall**	0.00	0.04	0.03	0.05
	rise	0.01	0.07	0.03	0.08
B1 to ZN	**fall**	0.01	0.04	0.04	0.07
	rise	0.01	0.07	0.03	0.08
B2 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A1	6.4910
A2	6.9613
B1	6.2169
B2	6.7269

Leakage [nW]
404.71

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	97.9614	0.3656	97.9614
A1 to ZN	fall	1.13	1.45	32.04	20.64
	rise	14.43	15.05	51.87	38.81
A2 to ZN	fall	1.17	1.45	25.93	17.60
	rise	18.33	18.61	56.53	44.71
B1 to ZN	fall	5.95	6.86	37.15	25.80
	rise	22.33	22.50	49.57	37.94
B2 to ZN	fall	5.97	6.87	30.86	22.42
	rise	26.14	26.26	53.77	43.43

BUF_X16

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	6.650 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**965.576**	**0.3656**	**965.576**
A to Z	**fall**	0.01	0.07	0.04	0.11
	rise	0.01	0.09	0.01	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**965.576**	**0.3656**	**965.576**
A to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A	12.3998

Leakage [nW]
1272.69

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**965.576**	**0.3656**	**965.576**
A to Z	**fall**	52.10	59.95	185.27	187.22
	rise	32.82	30.18	164.13	164.04

BUF_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.6537**	**0.3656**	**60.6537**
A to Z	**fall**	0.01	0.07	0.05	0.11
	rise	0.01	0.09	0.01	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.6537**	**0.3656**	**60.6537**
A to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A	0.9832

Leakage [nW]
79.18

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.6537**	**0.3656**	**60.6537**
A to Z	**fall**	3.61	4.15	11.72	11.87
	rise	2.30	2.24	10.22	10.21

BUF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.064 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
A to Z	**fall**	0.01	0.07	0.04	0.11
	rise	0.01	0.09	0.01	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
A to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A	1.8114

Leakage [nW]
159.08

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
A to Z	**fall**	6.93	7.92	23.36	23.68
	rise	4.53	4.86	20.72	20.84

BUF_X32

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	32
Cell Area	13.034 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**1904.30**	**0.3656**	**1904.30**
A to Z	**fall**	0.01	0.07	0.04	0.11
	rise	0.01	0.10	0.01	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**1904.30**	**0.3656**	**1904.30**
A to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A	26.9452

Leakage [nW]
2545.40

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**1904.30**	**0.3656**	**1904.30**
A to Z	**fall**	103.16	117.85	364.85	366.71
	rise	64.45	56.89	319.66	318.62

BUF_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	1.862 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to Z	**fall**	0.01	0.07	0.04	0.11
	rise	0.01	0.09	0.01	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A	3.4463

Leakage [nW]
318.16

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to Z	**fall**	13.12	15.05	46.26	46.87
	rise	8.46	8.73	41.20	41.74

BUF_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	3.458 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	484.009	0.3656	484.009
A to Z	**fall**	0.01	0.07	0.04	0.11
	rise	0.01	0.09	0.01	0.10

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	484.009	0.3656	484.009
A to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A	6.6354

Leakage [nW]
636.35

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	484.009	0.3656	484.009
A to Z	**fall**	26.07	29.81	92.86	93.67
	rise	16.56	18.52	82.99	82.65

CLKBUF_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
A to Z	**fall**	0.01	0.14	0.03	0.16
	rise	0.01	0.10	0.04	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
A to Z	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A	0.7826

Leakage [nW]
46.91

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
A to Z	**fall**	3.01	3.35	8.27	8.44
	rise	1.93	2.11	7.63	6.55

CLKBUF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.064 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
A to Z	**fall**	0.01	0.14	0.03	0.16
	rise	0.01	0.10	0.04	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
A to Z	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A	1.4126

Leakage [nW]
95.50

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
A to Z	**fall**	5.72	6.39	16.61	17.16
	rise	3.66	3.35	15.44	14.11

CLKBUF_X3

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	3
Cell Area	1.330 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**181.885**	**0.3656**	**181.885**
A to Z	**fall**	0.01	0.14	0.04	0.17
	rise	0.02	0.10	0.04	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**181.885**	**0.3656**	**181.885**
A to Z	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A	1.4219

Leakage [nW]
127.35

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**181.885**	**0.3656**	**181.885**
A to Z	**fall**	7.21	8.15	20.92	18.65
	rise	5.70	5.04	20.33	15.50

NANGATE CLKGATETST_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	3.990 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to GCK	**fall**	0.01	0.14	0.04	0.17
	rise	0.01	0.10	0.02	0.11

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.08	0.01	0.08

Constraints Time [ns]		
Setup CK to E	**fall**	0.09
	rise	0.07
Hold CK to E	**fall**	0.10
	rise	0.07
Setup CK to SE	**fall**	0.10
	rise	0.07
Hold CK to SE	**fall**	0.11
	rise	0.06

Capacitance [fF]	
CK	1.8235
E	0.8837
SE	0.7872

Leakage [nW]
211.61

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	8341.65	17411.67
	rise	2571.97	11187.46
E	fall	2739.39	5054.47
	rise	1316.46	3900.07
SE	fall	2235.59	5011.09
	rise	1253.63	4148.90

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to GCK	fall	7.00	8.76	19.44	23.74
	rise	4.28	5.26	18.22	19.74

NANGATE CLKGATETST_X2

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	4.256 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
CK to GCK	**fall**	0.01	0.14	0.04	0.17
	rise	0.01	0.10	0.02	0.11

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.08	0.01	0.08

Constraints Time [ns]		
Setup CK to E	**fall**	0.09
	rise	0.07
Hold CK to E	**fall**	0.10
	rise	0.07
Setup CK to SE	**fall**	0.09
	rise	0.07
Hold CK to SE	**fall**	0.11
	rise	0.06

Capacitance [fF]	
CK	2.8419
E	0.8799
SE	0.8214

Leakage [nW]
283.68

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	9655.99	19097.58
	rise	1546.20	10147.10
E	fall	2739.13	5086.99
	rise	1267.72	3869.53
SE	fall	2234.94	5040.15
	rise	1202.86	4127.24

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
CK to GCK	fall	10.06	12.62	28.66	34.07
	rise	5.70	6.30	25.97	28.29

NANGATE CLKGATETST_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	5.320 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.92**	**0.3656**	**242.92**
CK to GCK	**fall**	0.01	0.14	0.05	0.18
	rise	0.01	0.10	0.02	0.11

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.92**	**0.3656**	**242.92**
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.08	0.01	0.08

Constraints Time [ns]		
Setup CK to E	**fall**	0.10
	rise	0.10
Hold CK to E	**fall**	0.08
	rise	0.06
Setup CK to SE	**fall**	0.11
	rise	0.10
Hold CK to SE	**fall**	0.09
	rise	0.06

Capacitance [fF]	
CK	4.5138
E	0.9399
SE	0.8241

Leakage [nW]
466.38

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	14238.98	23158.94
	rise	157.01	8712.69
E	fall	3597.75	5863.65
	rise	1947.18	4482.78
SE	fall	3094.91	5817.82
	rise	1884.45	4725.02

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	242.92	0.3656	242.92
CK to GCK	fall	17.38	21.04	52.28	55.52
	rise	10.57	11.93	36.88	43.60

NANGATE CLKGATETST_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	7.714 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	484.619	0.3656	484.619
CK to GCK	**fall**	0.01	0.14	0.05	0.18
	rise	0.01	0.10	0.02	0.11

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	484.619	0.3656	484.619
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.08	0.01	0.08

Constraints Time [ns]		
Setup CK to E	**fall**	0.11
	rise	0.15
Hold CK to E	**fall**	0.04
	rise	0.05
Setup CK to SE	**fall**	0.12
	rise	0.14
Hold CK to SE	**fall**	0.05
	rise	0.05

Capacitance [fF]	
CK	8.0886
E	0.9103
SE	0.8147

Leakage [nW]
821.18

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	25912.22	34647.11
	rise	-2532.99	5986.01
E	fall	5061.78	7343.94
	rise	3748.07	6275.98
SE	fall	4558.79	7288.82
	rise	3684.99	6526.92

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	484.619	0.3656	484.619
CK to GCK	fall	31.11	38.00	95.27	98.88
	rise	19.14	21.38	65.70	75.19

NANGATE CLKGATE_X1

Databook Build Date: Thursday Feb 17 15:07 2011

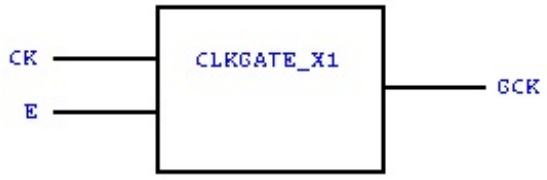
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	3.458 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	0.01	0.14	0.04	0.18
	rise	0.01	0.10	0.02	0.11

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.08	0.01	0.08

Constraints Time [ns]		
Setup CK to E	**fall**	0.06
	rise	0.06
Hold CK to E	**fall**	0.11
	rise	0.09

Capacitance [fF]	
CK	1.8523
E	0.9322

Leakage [nW]
175.94

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1462**
CK	**fall**	6470.40	15953.01
	rise	1688.23	10510.31
E	**fall**	2094.26	3641.65
	rise	882.25	2408.53

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	6.86	9.16	19.43	24.18
	rise	4.08	4.36	17.71	19.80

NANGATE CLKGATE_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	3.724 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.307**	**0.3656**	**121.307**
CK to GCK	**fall**	0.01	0.14	0.04	0.17
	rise	0.01	0.10	0.02	0.11

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.307**	**0.3656**	**121.307**
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.08	0.01	0.08

Constraints Time [ns]		
Setup CK to E	**fall**	0.07
	rise	0.06
Hold CK to E	**fall**	0.10
	rise	0.09

Capacitance [fF]	
CK	2.5719
E	0.9026

Leakage [nW]
245.04

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1462**
CK	**fall**	8053.38	17720.54
	rise	1379.90	10141.70
E	**fall**	2329.43	3812.57
	rise	1367.86	2855.17

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.307**	**0.3656**	**121.307**
CK to GCK	**fall**	10.00	13.15	28.81	34.54
	rise	5.99	6.86	26.22	28.81

NANGATE CLKGATE_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
CK to GCK	**fall**	0.01	0.14	0.04	0.17
	rise	0.01	0.10	0.02	0.11

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.08	0.01	0.08

Constraints Time [ns]		
Setup CK to E	**fall**	0.07
	rise	0.07
Hold CK to E	**fall**	0.10
	rise	0.08

Capacitance [fF]	
CK	4.2618
E	0.9003

Leakage [nW]
409.31

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1462**
CK	**fall**	10840.98	21121.08
	rise	301.75	9062.56
E	**fall**	3392.36	5136.73
	rise	2175.03	3953.90

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
CK to GCK	**fall**	15.77	21.24	47.50	55.55
	rise	9.71	11.52	43.22	43.89

NANGATE CLKGATE_X8

Databook Build Date: Thursday Feb 17 15:07 2011

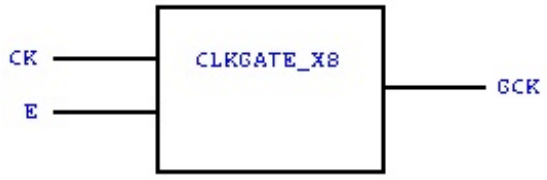
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	6.916 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	0.01	0.14	0.04	0.17
	rise	0.01	0.10	0.02	0.11

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.08	0.01	0.08

Constraints Time [ns]		
Setup CK to E	**fall**	0.08
	rise	0.07
Hold CK to E	**fall**	0.10
	rise	0.08

Capacitance [fF]	
CK	7.7165
E	1.1820

Leakage [nW]
744.22

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1462**
CK	**fall**	17050.05	28124.79
	rise	-1420.70	7079.18
E	**fall**	6030.35	8846.41
	rise	3581.40	6480.94

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	27.86	38.89	85.62	98.81
	rise	17.66	20.30	77.01	75.73

DFFRS_X1

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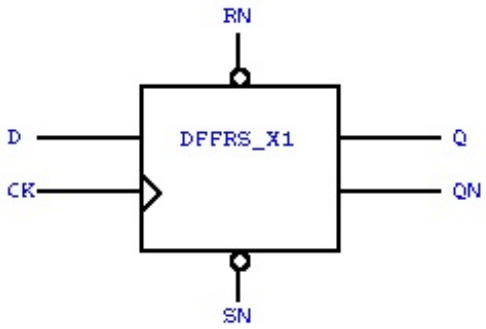
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, and drive strength X1
Strength	1
Cell Area	6.384 um^2
Equation	Q = "(!(!RN)*((D)+(!SN)))" QN = "(!(!SN)*(!(D)+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	RN	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	H	-	-	L	H
R	H	H	H	-	-	H	L
F	-	H	H	L	H	L	H
F	-	H	H	H	L	H	L
-	-	L	L	-	-	L	L
-	-	L	H	-	-	L	H
-	-	H	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.05	0.11	0.06	0.12
	rise	0.05	0.13	0.05	0.13
CK to QN	fall	0.06	*0.12*	0.06	*0.12*
	rise	0.07	*0.15*	0.07	*0.16*
RN to Q	fall	0.02	0.08	0.05	0.12
	rise	0.09	0.17	0.10	0.19
RN to QN	rise	0.03	*0.12*	0.07	*0.15*
SN to Q	rise	0.10	0.18	0.14	0.22
SN to QN	fall	0.01	*0.08*	0.05	*0.12*
	rise	0.02	*0.10*	0.03	*0.11*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
CK to QN	fall	0.00	*0.05*	0.00	*0.05*
	rise	0.00	*0.08*	0.00	*0.08*
RN to Q	fall	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
RN to QN	rise	0.00	*0.08*	0.00	*0.08*
SN to Q	rise	0.00	0.08	0.01	0.08
SN to QN	fall	0.00	*0.05*	0.01	*0.05*
	rise	0.00	*0.08*	0.01	*0.08*

Constraints Time [ns]		
Setup CK to D	fall	0.06
	rise	0.04
Hold CK to D	fall	0.12
	rise	0.10
Recovery CK to RN	rise	0.07
Removal CK to RN	rise	0.14
Recovery CK to SN	rise	-0.02
Removal CK to SN	rise	0.30

Capacitance [fF]	
CK	0.9792
D	1.1678
RN	1.4460
SN	2.2234

Leakage [nW]
344.76

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	7390.26	15711.72
	rise	9618.67	18055.31
D	fall	3252.55	5908.72
	rise	1681.53	4358.20
RN	fall	906.44	909.10
	rise	-610.90	-629.04
SN	fall	2104.66	3701.52
	rise	-951.66	601.18

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	8.19	7.93	12.21	12.02
	rise	8.00	7.98	12.05	12.31
CK to QN	fall	8.00	*7.98*	12.05	*12.31*
	rise	8.19	*7.80*	12.21	*12.07*
RN to Q	fall	5.06	7.96	11.23	13.16
	rise	6.00	6.21	10.58	9.57
RN to QN	rise	5.30	*7.54*	10.44	*11.48*
SN to Q	rise	7.90	7.92	15.16	13.94
SN to QN	fall	7.29	*7.93*	16.83	*16.98*
	rise	3.50	*4.11*	13.31	*15.03*

DFFRS_X2

Databook Build Date: Thursday Feb 17 15:07 2011

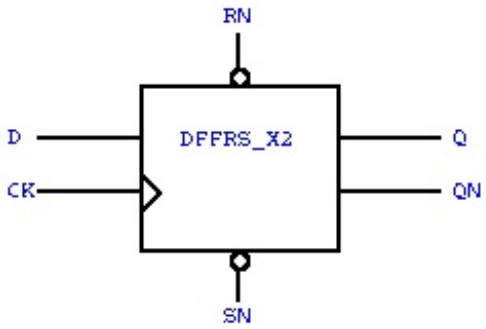
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, and drive strength X2
Strength	2
Cell Area	6.916 um^2
Equation	Q = "(!(!RN)*((D)+(!SN)))" QN = "(!(!SN)*(!(D)+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	RN	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	H	-	-	L	H
R	H	H	H	-	-	H	L
F	-	H	H	L	H	L	H
F	-	H	H	H	L	H	L
-	-	L	L	-	-	L	L
-	-	L	H	-	-	L	H
-	-	H	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.05	0.11	0.06	0.12
	rise	0.05	0.13	0.05	0.14
CK to QN	fall	0.06	0.12	0.07	0.13
	rise	0.07	0.15	0.07	0.16
RN to Q	fall	0.02	0.08	0.05	0.12
	rise	0.09	0.18	0.10	0.19
RN to QN	rise	0.04	0.12	0.08	0.16
SN to Q	rise	0.10	0.19	0.15	0.23
SN to QN	fall	0.01	0.08	0.05	0.12
	rise	0.02	0.10	0.03	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
RN to Q	fall	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
RN to QN	rise	0.00	0.08	0.00	0.08
SN to Q	rise	0.01	0.08	0.01	0.08
SN to QN	fall	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Constraints Time [ns]		
Setup CK to D	fall	0.06
	rise	0.04
Hold CK to D	fall	0.12
	rise	0.09
Recovery CK to RN	rise	0.02
Removal CK to RN	rise	0.14
Recovery CK to SN	rise	-0.02
Removal CK to SN	rise	0.31

Capacitance [fF]	
CK	0.9574
D	1.1725
RN	1.8528
SN	2.6271

Leakage [nW]
498.03

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	7282.31	15626.89
	rise	10737.37	19185.03
D	fall	3290.29	5944.18
	rise	1639.41	4314.76
RN	fall	1336.32	1251.01
	rise	-1127.84	-1146.10
SN	fall	2477.93	4054.46
	rise	-1305.14	260.75

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	11.46	10.57	15.51	14.64
	rise	11.82	11.22	16.07	15.45
CK to QN	fall	11.82	11.22	16.07	15.45
	rise	11.46	10.57	15.51	14.64
RN to Q	fall	8.76	11.47	19.90	19.59
	rise	9.72	9.58	15.65	14.60
RN to QN	rise	9.18	10.89	18.13	16.78
SN to Q	rise	12.35	11.55	23.20	19.41
SN to QN	fall	10.91	11.26	25.13	22.94
	rise	6.14	6.42	20.94	19.77

DFFR_X1

Databook Build Date: Thursday Feb 17 15:07 2011

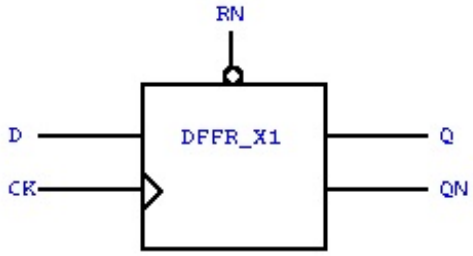
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, and drive strength X1
Strength	1
Cell Area	5.320 um^2
Equation	Q = "(!(!RN)*(D))" QN = "((!RN)+!(D))"
Clock	CK
Reset	RN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	RN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.05	0.11	0.06	0.12
	rise	0.06	0.14	0.06	0.14
CK to QN	fall	0.03	*0.10*	0.04	*0.11*
	rise	0.04	*0.12*	0.05	*0.13*
RN to Q	fall	0.01	0.08	0.05	0.12
RN to QN	rise	0.11	*0.19*	0.16	*0.24*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
CK to QN	fall	0.01	*0.05*	0.01	*0.05*
	rise	0.00	*0.08*	0.00	*0.08*
RN to Q	fall	0.00	0.05	0.01	0.05
RN to QN	rise	0.01	*0.08*	0.01	*0.08*

Constraints Time [ns]		
Setup CK to D	fall	0.06
	rise	0.04
Hold CK to D	fall	0.12
	rise	0.09
Recovery CK to RN	rise	0.00
Removal CK to RN	rise	0.29

Capacitance [fF]	
CK	0.9910
D	1.1485
RN	1.7893

Leakage [nW]
297.24

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	7144.68	15533.92
	rise	9507.71	18065.61
D	fall	3435.55	6399.92
	rise	1781.62	4750.54
RN	fall	926.20	870.10
	rise	-758.95	-776.40

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	6.42	6.33	10.28	10.49
	rise	7.38	6.64	11.73	11.09
CK to QN	fall	7.38	*6.52*	11.73	*10.94*
	rise	6.42	*6.37*	10.28	*10.57*
RN to Q	fall	7.34	7.00	12.29	10.42
RN to QN	rise	7.34	*6.84*	12.29	*10.45*

DFFR_X2

Databook Build Date: Thursday Feb 17 15:07 2011

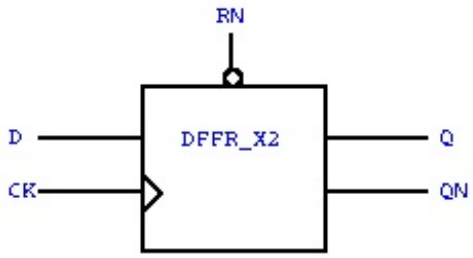
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, and drive strength X2
Strength	2
Cell Area	5.852 um^2
Equation	Q = "(!(!RN)*(D))" QN = "((!RN)+!(D))"
Clock	CK
Reset	RN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	RN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.06	0.12	0.07	0.12
	rise	0.07	0.15	0.08	0.15
CK to QN	fall	0.05	*0.12*	0.05	*0.13*
	rise	0.05	*0.13*	0.05	*0.14*
RN to Q	fall	0.01	0.07	0.05	0.11
RN to QN	rise	0.11	*0.20*	0.17	*0.25*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
CK to QN	fall	0.01	*0.06*	0.01	*0.06*
	rise	0.01	*0.08*	0.01	*0.08*
RN to Q	fall	0.00	0.05	0.01	0.05
RN to QN	rise	0.01	*0.08*	0.01	*0.08*

Constraints Time [ns]		
Setup CK to D	fall	0.06
	rise	0.04
Hold CK to D	fall	0.12
	rise	0.09
Recovery CK to RN	rise	-0.02
Removal CK to RN	rise	0.29

Capacitance [fF]	
CK	0.9802
D	1.1434
RN	2.4834

Leakage [nW]
441.53

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	7059.98	15472.86
	rise	11647.22	21022.26
D	fall	3414.59	6387.00
	rise	1735.98	4714.04
RN	fall	1699.15	1577.85
	rise	-1412.98	-1432.66

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	10.81	9.49	14.76	13.68
	rise	14.15	10.51	18.70	14.75
CK to QN	fall	14.15	*10.48*	18.70	*14.80*
	rise	10.81	*9.57*	14.76	*13.75*
RN to Q	fall	12.88	10.65	22.60	17.33
RN to QN	rise	12.88	*10.73*	22.60	*17.39*

DFFS_X1

Databook Build Date: Thursday Feb 17 15:07 2011

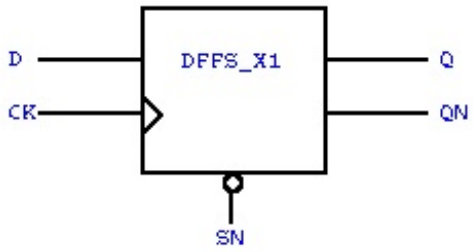
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, and drive strength X1
Strength	1
Cell Area	5.320 um^2
Equation	Q = "((!SN)+(D))" QN = "(!((!SN)+(D)))"
Clock	CK
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.05	0.11	0.06	0.12
	rise	0.05	0.13	0.06	0.14
CK to QN	fall	0.04	*0.10*	0.04	*0.11*
	rise	0.04	*0.12*	0.05	*0.13*
SN to Q	rise	0.11	0.18	0.16	0.24
SN to QN	fall	0.09	*0.17*	0.14	*0.22*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
CK to QN	fall	0.01	*0.05*	0.01	*0.05*
	rise	0.00	*0.08*	0.00	*0.08*
SN to Q	rise	0.01	0.08	0.01	0.08
SN to QN	fall	0.02	*0.06*	0.02	*0.06*

Constraints Time [ns]		
Setup CK to D	fall	0.06
	rise	0.04
Hold CK to D	fall	0.12
	rise	0.10
Recovery CK to SN	rise	-0.03
Removal CK to SN	rise	0.27

Capacitance [fF]	
CK	0.9863
D	1.1879
SN	1.3628

Leakage [nW]
292.60

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	8113.44	16324.00
	rise	9243.85	17417.22
D	fall	3176.68	5959.77
	rise	1363.79	4089.63
SN	fall	1277.89	2911.02
	rise	-421.92	1059.84

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	5.88	5.90	9.62	10.00
	rise	6.63	5.97	10.89	10.32
CK to QN	fall	6.63	*6.07*	10.89	*10.40*
	rise	5.88	*5.80*	9.62	*10.02*
SN to Q	rise	9.05	7.08	11.91	9.81
SN to QN	fall	9.04	*7.10*	11.93	*9.78*

DFFS_X2

Databook Build Date: Thursday Feb 17 15:07 2011

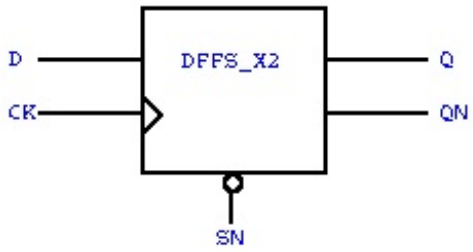
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, and drive strength X2
Strength	2
Cell Area	5.586 um^2
Equation	Q = "((!SN)+(D))" QN = "(!((!SN)+(D)))"
Clock	CK
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.06	0.12	0.07	0.13
	rise	0.06	0.14	0.07	0.15
CK to QN	fall	0.05	*0.12*	0.06	*0.13*
	rise	0.05	*0.13*	0.06	*0.14*
SN to Q	rise	0.16	0.23	0.21	0.28
SN to QN	fall	0.13	*0.23*	0.19	*0.28*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
CK to QN	fall	0.01	*0.06*	0.01	*0.06*
	rise	0.01	*0.08*	0.01	*0.08*
SN to Q	rise	0.01	0.08	0.01	0.08
SN to QN	fall	0.03	*0.07*	0.03	*0.07*

Constraints Time [ns]		
Setup CK to D	fall	0.06
	rise	0.04
Hold CK to D	fall	0.12
	rise	0.10
Recovery CK to SN	rise	-0.03
Removal CK to SN	rise	0.26

Capacitance [fF]	
CK	0.9854
D	1.1857
SN	1.3416

Leakage [nW]
428.68

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	8242.67	16429.12
	rise	10065.87	18283.59
D	fall	3239.20	6015.01
	rise	1352.08	4069.11
SN	fall	1267.37	2903.09
	rise	-419.96	1065.20

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	10.39	9.19	14.25	13.44
	rise	13.42	10.08	17.87	14.49
CK to QN	fall	13.42	*10.10*	17.87	*14.58*
	rise	10.39	*9.17*	14.25	*13.16*
SN to Q	rise	21.08	14.35	23.90	17.13
SN to QN	fall	21.08	*14.41*	23.90	*17.22*

DFF_X1

Databook Build Date: Thursday Feb 17 15:07 2011

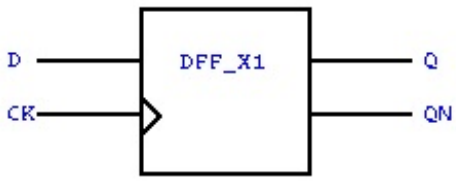
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with drive strength X1
Strength	1
Cell Area	4.522 um^2
Equation	Q = "(D)" QN = "(!(D))"
Clock	CK
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
CK	D	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	-	-	L	H
R	H	-	-	H	L
F	-	L	H	L	H
F	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to Q	fall	0.05	0.11	0.06	0.12
	rise	0.05	0.13	0.06	0.14
CK to QN	fall	0.03	*0.10*	0.04	*0.11*
	rise	0.04	*0.12*	0.05	*0.13*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
CK to QN	fall	0.01	*0.05*	0.01	*0.05*
	rise	0.00	*0.08*	0.00	*0.08*

Constraints Time [ns]		
Setup CK to D	fall	0.06
	rise	0.04
Hold CK to D	fall	0.12
	rise	0.10

Capacitance [fF]	
CK	0.9667
D	1.1576

Leakage [nW]
276.85

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	7848.71	16187.68
	rise	3339.73	11564.31
D	fall	3327.60	6319.14
	rise	1750.42	4714.60

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to Q	fall	5.79	5.69	9.59	9.84
	rise	6.51	5.84	10.82	10.36
CK to QN	fall	6.51	*5.87*	10.82	*10.22*
	rise	5.79	*5.78*	9.58	*9.91*

DFF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

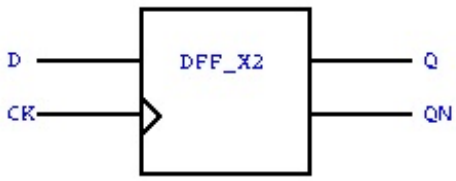
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with drive strength X2
Strength	2
Cell Area	5.054 um^2
Equation	Q = "(D)" QN = "(!(D))"
Clock	CK
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
CK	**D**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**	**QN**
R	L	-	-	L	H
R	H	-	-	H	L
F	-	L	H	L	H
F	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
CK to Q	**fall**	0.06	0.12	0.06	0.12
	rise	0.06	0.14	0.07	0.15
CK to QN	**fall**	0.05	*0.12*	0.05	*0.13*
	rise	0.05	*0.13*	0.05	*0.14*

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
CK to Q	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
CK to QN	**fall**	0.01	*0.06*	0.01	*0.06*
	rise	0.01	*0.08*	0.01	*0.08*

Constraints Time [ns]		
Setup CK to D	**fall**	0.06
	rise	0.04
Hold CK to D	**fall**	0.12
	rise	0.09

Capacitance [fF]	
CK	0.9463
D	1.1478

Leakage [nW]
410.32

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1462**
CK	**fall**	7914.18	16245.65
	rise	3360.70	11582.03
D	**fall**	3372.06	6351.28
	rise	1808.43	4758.51

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
CK to Q	**fall**	10.32	9.10	14.21	13.40
	rise	13.24	10.15	17.79	14.45
CK to QN	**fall**	13.24	*9.79*	17.79	*14.55*
	rise	10.32	*9.12*	14.21	*13.14*

DLH_X1

Databook Build Date: Thursday Feb 17 15:07 2011

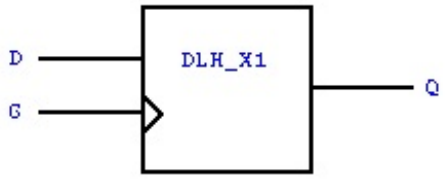
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	High enable Latch with drive strength X1
Strength	1
Cell Area	2.660 um^2
Equation	Q = "(D)"
Enable	G
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**G**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	H	-	-	L
H	H	-	-	H
-	L	L	H	L
-	L	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
D to Q	**fall**	0.03	0.09	0.07	0.14
	rise	0.02	0.10	0.03	0.11
G to Q	**fall**	0.03	0.10	0.03	0.10
	rise	0.03	0.12	0.03	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
D to Q	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
G to Q	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.08	0.00	0.08

Constraints Time [ns]		
Setup G to D	**fall**	0.07
	rise	0.02
Hold G to D	**fall**	0.16
	rise	0.09

Capacitance [fF]	
D	0.9374
G	1.0002

Leakage [nW]
144.20

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1462**
D	**fall**	679.26	660.78
	rise	-506.15	-592.14
G	**fall**	3966.86	12527.80
	rise	2925.21	11601.48

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
D to Q	**fall**	5.66	6.14	10.74	10.31
	rise	3.61	4.11	9.23	8.48
G to Q	**fall**	7.35	7.84	15.35	16.99
	rise	6.62	6.96	13.86	15.72

DLH_X2

Databook Build Date: Thursday Feb 17 15:07 2011

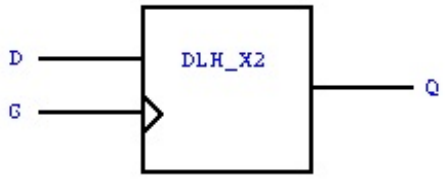
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	High enable Latch with drive strength X2
Strength	2
Cell Area	2.926 um^2
Equation	Q = "(D)"
Enable	G
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**G**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	H	-	-	L
H	H	-	-	H
-	L	L	H	L
-	L	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.697**	**0.3656**	**120.697**
D to Q	**fall**	0.03	0.10	0.07	0.14
	rise	0.02	0.11	0.03	0.12
G to Q	**fall**	0.03	0.10	0.04	0.10
	rise	0.03	0.12	0.04	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.697**	**0.3656**	**120.697**
D to Q	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
G to Q	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.08	0.00	0.08

Constraints Time [ns]		
Setup G to D	**fall**	0.07
	rise	0.03
Hold G to D	**fall**	0.16
	rise	0.09

Capacitance [fF]	
D	1.1861
G	0.9998

Leakage [nW]
204.47

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1462**
D	**fall**	914.38	888.80
	rise	-681.37	-795.56
G	**fall**	3865.28	12449.42
	rise	2938.91	11613.92

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.697**	**0.3656**	**120.697**
D to Q	**fall**	9.95	9.43	17.95	15.02
	rise	6.63	5.91	16.27	12.95
G to Q	**fall**	11.26	10.72	19.98	19.92
	rise	9.72	9.76	16.88	17.87

DLL_X1

Databook Build Date: Thursday Feb 17 15:07 2011

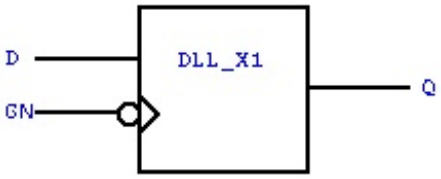
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Low enable Latch with drive strength X1
Strength	1
Cell Area	2.660 um^2
Equation	Q = "(D)"
Enable	GN
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**GN**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	L	-	-	L
H	L	-	-	H
-	H	L	H	L
-	H	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
D to Q	**fall**	0.03	0.09	0.07	0.14
	rise	0.02	0.10	0.03	0.11
GN to Q	**fall**	0.04	0.10	0.07	0.14
	rise	0.03	0.11	0.06	0.15

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
D to Q	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
GN to Q	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Constraints Time [ns]		
Setup GN to D	**fall**	0.06
	rise	0.07
Hold GN to D	**fall**	0.11
	rise	0.09

Capacitance [fF]	
D	0.9067
GN	1.0000

Leakage [nW]
144.20

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1462**
D	**fall**	678.93	660.62
	rise	-505.88	-592.33
GN	**fall**	4264.53	12955.38
	rise	2336.23	11066.07

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
D to Q	**fall**	5.74	6.07	10.79	10.27
	rise	3.74	4.09	9.35	8.06
GN to Q	**fall**	8.79	9.22	16.57	17.87
	rise	8.05	8.49	16.42	17.32

DLL_X2

Databook Build Date: Thursday Feb 17 15:07 2011

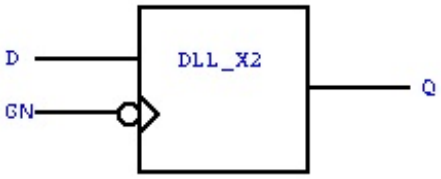
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Low enable Latch with drive strength X2
Strength	2
Cell Area	2.926 um^2
Equation	Q = "(D)"
Enable	GN
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**GN**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	L	-	-	L
H	L	-	-	H
-	H	L	H	L
-	H	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
D to Q	**fall**	0.03	0.10	0.07	0.14
	rise	0.02	0.11	0.03	0.12
GN to Q	**fall**	0.04	0.10	0.07	0.14
	rise	0.03	0.11	0.07	0.15

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
D to Q	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
GN to Q	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Constraints Time [ns]		
Setup GN to D	**fall**	0.07
	rise	0.08
Hold GN to D	**fall**	0.11
	rise	0.09

Capacitance [fF]	
D	1.1590
GN	0.9952

Leakage [nW]
204.47

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1462**
D	**fall**	915.40	888.61
	rise	-682.48	-795.83
GN	**fall**	4437.30	13135.00
	rise	2219.28	10932.92

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
D to Q	**fall**	9.73	9.09	17.75	14.72
	rise	6.52	6.15	16.23	11.75
GN to Q	**fall**	12.49	11.95	20.34	20.62
	rise	11.16	11.25	20.50	20.13

FA_X1

Databook Build Date: Thursday Feb 17 15:07 2011

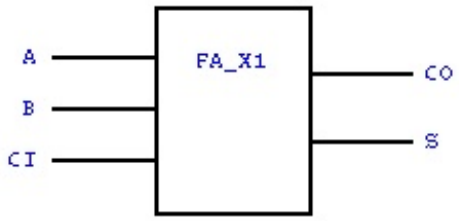
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	4.256 um^2
Equation	CO = "((A & B) | (CI & (A | B)))" S = "(CI ^ (A ^ B))"
Type	Combinational
Input	A, B, CI
Output	CO, S
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**CI**	**CO**	**S**
L	L	L	L	L
L	L	H	L	H
L	H	L	L	H
L	H	H	H	L
H	L	L	L	H
H	L	H	H	L
H	H	L	H	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A to CO	**fall**	0.03	*0.10*	0.09	*0.16*
	rise	0.03	*0.11*	0.05	*0.13*
A to S	**fall**	0.05	0.11	0.08	0.15
	rise	0.06	0.14	0.11	0.19
B to CO	**fall**	0.03	*0.10*	0.08	*0.15*
	rise	0.03	*0.11*	0.04	*0.12*
B to S	**fall**	0.05	0.12	0.09	0.16
	rise	0.06	0.14	0.11	0.19
CI to CO	**fall**	0.03	*0.10*	0.08	*0.15*
	rise	0.02	*0.11*	0.04	*0.13*
CI to S	**fall**	0.05	0.11	0.10	0.17
	rise	0.06	0.14	0.11	0.19

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A to CO	**fall**	0.01	*0.05*	0.01	*0.06*
	rise	0.00	*0.08*	0.01	*0.08*
A to S	**fall**	0.01	0.05	0.01	0.06
	rise	0.01	0.08	0.01	0.08
B to CO	**fall**	0.01	*0.05*	0.01	*0.05*
	rise	0.00	*0.08*	0.01	*0.08*
B to S	**fall**	0.01	0.05	0.01	0.06
	rise	0.01	0.08	0.01	0.08
CI to CO	**fall**	0.01	*0.05*	0.01	*0.06*
	rise	0.00	*0.08*	0.01	*0.08*
CI to S	**fall**	0.01	0.05	0.01	0.06
	rise	0.01	0.08	0.01	0.08

Capacitance [fF]	
A	3.7875
B	3.4962
CI	2.7735

Leakage [nW]
248.53

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.2722	0.3656	60.2722
A to CO	fall	7.59	*6.73*	10.60	*8.98*
	rise	3.71	*3.55*	7.50	*6.05*
A to S	fall	6.14	6.10	9.69	8.91
	rise	5.01	4.39	9.01	7.19
B to CO	fall	7.87	*6.95*	10.43	*8.84*
	rise	3.93	*3.77*	7.22	*5.91*
B to S	fall	5.99	5.96	9.29	8.49
	rise	5.03	4.51	8.84	7.04
CI to CO	fall	7.17	*6.26*	10.86	*8.84*
	rise	4.24	*3.99*	8.49	*6.90*
CI to S	fall	5.58	5.62	9.72	8.73
	rise	5.05	4.49	9.54	7.66

FILLCELL_X16

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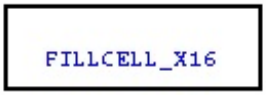
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	4.256 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X1

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.266 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X2

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	0.266 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X32

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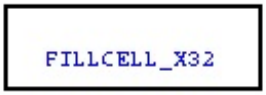
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	32
Cell Area	8.512 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X4

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	1.064 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X8

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].
Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.
Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	2.128 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



HA_X1

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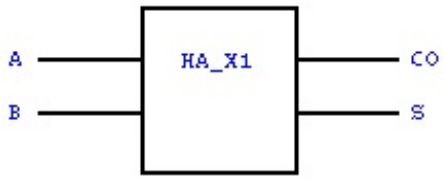
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.660 um^2
Equation	CO = "(A & B)" S = "(A ^ B)"
Type	Combinational
Input	A, B
Output	CO, S
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B**	**CO**	**S**
L	L	L	L
L	H	L	H
H	L	L	H
H	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A to CO	**fall**	0.01	0.07	0.04	0.11
	rise	0.02	0.10	0.03	0.12
A to S	**fall**	0.02	*0.10*	0.07	*0.17*
	rise	0.02	*0.18*	0.04	*0.24*
B to CO	**fall**	0.01	0.07	0.05	0.11
	rise	0.02	0.10	0.02	0.11
B to S	**fall**	0.03	*0.10*	0.07	*0.15*
	rise	0.02	*0.18*	0.05	*0.24*

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A to CO	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
A to S	**fall**	0.01	*0.09*	0.03	*0.11*
	rise	0.01	*0.15*	0.03	*0.15*
B to CO	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
B to S	**fall**	0.01	*0.09*	0.02	*0.09*
	rise	0.01	*0.15*	0.03	*0.16*

Capacitance [fF]	
A	3.1665
B	3.4702

Leakage [nW]
211.09

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A to CO	fall	3.95	3.98	13.01	9.34
	rise	1.39	1.38	9.47	6.11
A to S	fall	4.41	*4.21*	10.93	*8.82*
	rise	4.25	*3.66*	11.09	*8.37*
B to CO	fall	4.92	4.89	13.95	10.40
	rise	1.23	1.24	7.65	5.18
B to S	fall	4.56	*4.49*	9.47	*8.51*
	rise	4.32	*4.43*	11.70	*8.67*

INV_X16

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	4.522 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**969.238**	**0.3656**	**969.238**
A to ZN	**fall**	0.00	0.06	-0.01	0.10
	rise	0.00	0.09	0.03	0.15

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**969.238**	**0.3656**	**969.238**
A to ZN	**fall**	0.00	0.05	0.02	0.08
	rise	0.00	0.08	0.02	0.09

Capacitance [fF]	
A	25.4110

Leakage [nW]
848.13

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**969.238**	**0.3656**	**969.238**
A to ZN	**fall**	-0.00	-0.06	171.28	75.94
	rise	35.99	39.49	268.72	150.02

INV_X1

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.532 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
A to ZN	**fall**	0.00	0.06	-0.01	0.10
	rise	0.00	0.09	0.03	0.15

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
A to ZN	**fall**	0.00	0.05	0.03	0.08
	rise	0.00	0.08	0.02	0.09

Capacitance [fF]	
A	1.7196

Leakage [nW]
53.01

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
A to ZN	**fall**	-0.00	-0.00	10.61	4.61
	rise	2.37	2.33	16.76	9.84

INV_X2

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	0.798 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
A to ZN	**fall**	0.00	0.06	-0.01	0.10
	rise	0.00	0.09	0.03	0.15

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
A to ZN	**fall**	0.00	0.05	0.02	0.08
	rise	0.00	0.08	0.02	0.09

Capacitance [fF]	
A	3.2774

Leakage [nW]
106.01

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
A to ZN	**fall**	-0.00	-0.01	21.33	9.42
	rise	4.61	4.64	33.60	19.38

INV_X32

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	32
Cell Area	8.778 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**1923.83**	**0.3656**	**1923.83**
A to ZN	**fall**	0.00	0.06	-0.01	0.10
	rise	0.00	0.09	0.03	0.16

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**1923.83**	**0.3656**	**1923.83**
A to ZN	**fall**	0.00	0.05	0.02	0.08
	rise	0.00	0.08	0.02	0.09

Capacitance [fF]	
A	49.4203

Leakage [nW]
1696.29

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**1923.83**	**0.3656**	**1923.83**
A to ZN	**fall**	-0.00	-0.12	337.88	154.64
	rise	70.25	69.46	532.08	311.62

INV_X4

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	1.330 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.92**	**0.3656**	**242.92**
A to ZN	**fall**	0.00	0.06	-0.01	0.10
	rise	0.00	0.09	0.03	0.15

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.92**	**0.3656**	**242.92**
A to ZN	**fall**	0.00	0.05	0.02	0.08
	rise	0.00	0.08	0.02	0.09

Capacitance [fF]	
A	6.2972

Leakage [nW]
212.03

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.92**	**0.3656**	**242.92**
A to ZN	**fall**	-0.00	-0.01	42.80	18.87
	rise	9.00	9.23	67.29	38.45

INV_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	2.394 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**485.229**	**0.3656**	**485.229**
A to ZN	**fall**	0.00	0.06	-0.01	0.10
	rise	0.00	0.09	0.03	0.15

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**485.229**	**0.3656**	**485.229**
A to ZN	**fall**	0.00	0.05	0.02	0.08
	rise	0.00	0.08	0.02	0.09

Capacitance [fF]	
A	11.7519

Leakage [nW]
424.10

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**485.229**	**0.3656**	**485.229**
A to ZN	**fall**	-0.00	-0.03	86.27	37.87
	rise	17.56	15.12	135.19	76.60

LOGIC0_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].
Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.
Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.532 um^2
Equation	Z = "0"
Type	Constant
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)

LOGIC1_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].
Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.
Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.532 um^2
Equation	Z = "1"
Type	Constant
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)

MUX2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

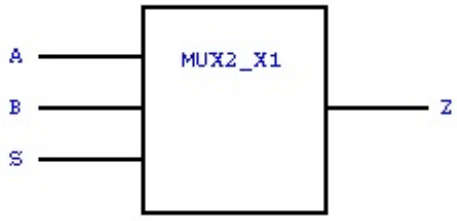
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.862 um^2
Equation	Z = "((S & B) | (A & !S))"
Type	Combinational
Input	A, B, S
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B**	**S**	**Z**
L	-	L	L
H	-	L	H
-	L	H	L
-	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5011	0.3656	60.5011
A to Z	**fall**	0.03	0.09	0.07	0.14
	rise	0.02	0.10	0.03	0.11
B to Z	**fall**	0.03	0.09	0.07	0.14
	rise	0.02	0.10	0.03	0.11
S to Z	**fall**	0.03	0.09	0.06	0.13
	rise	0.03	0.11	0.06	0.15

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5011	0.3656	60.5011
A to Z	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
B to Z	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
S to Z	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A	0.9691
B	0.9681
S	1.9210

Leakage [nW]
123.33

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5011	0.3656	60.5011
A to Z	**fall**	5.07	5.31	9.83	9.21
	rise	3.43	3.62	8.71	7.73
B to Z	**fall**	5.40	5.77	10.15	9.66
	rise	3.08	2.94	8.40	7.26
S to Z	**fall**	5.60	5.92	15.61	16.46
	rise	4.88	4.89	15.15	15.69

MUX2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

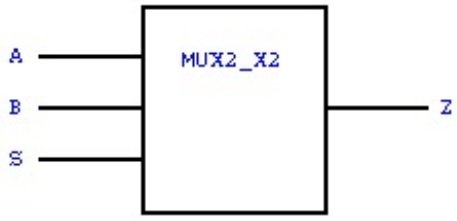
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	Z = "((S & B) | (A & !S))"
Type	Combinational
Input	A, B, S
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B**	**S**	**Z**
L	-	L	L
H	-	L	H
-	L	H	L
-	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.85**	**0.3656**	**120.85**
A to Z	**fall**	0.02	0.09	0.06	0.13
	rise	0.02	0.11	0.04	0.13
B to Z	**fall**	0.03	0.09	0.07	0.14
	rise	0.02	0.10	0.02	0.11
S to Z	**fall**	0.04	0.10	0.05	0.13
	rise	0.03	0.12	0.07	0.15

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.85**	**0.3656**	**120.85**
A to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
B to Z	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
S to Z	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A	1.6100
B	1.7885
S	2.6091

Leakage [nW]
236.62

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.85**	**0.3656**	**120.85**
A to Z	**fall**	9.93	11.16	18.86	18.76
	rise	6.96	6.85	20.86	16.29
B to Z	**fall**	9.92	10.93	23.28	21.48
	rise	5.21	5.32	16.01	14.58
S to Z	**fall**	11.38	12.35	24.71	25.34
	rise	9.37	9.18	22.78	22.50

NANGATE NAND2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

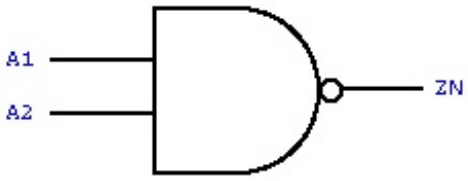
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	ZN = "!(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	59.3567	0.3656	59.3567
A1 to ZN	**fall**	0.00	0.10	0.01	0.14
	rise	0.00	0.08	0.02	0.15
A2 to ZN	**fall**	0.01	0.10	0.00	0.13
	rise	0.01	0.09	0.03	0.15

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	59.3567	0.3656	59.3567
A1 to ZN	**fall**	0.00	0.08	0.03	0.10
	rise	0.00	0.08	0.03	0.09
A2 to ZN	**fall**	0.00	0.08	0.02	0.09
	rise	0.00	0.08	0.02	0.09

Capacitance [fF]	
A1	1.6001
A2	1.6936

Leakage [nW]
59.12

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	59.3567	0.3656	59.3567
A1 to ZN	**fall**	0.22	0.31	10.34	3.97
	rise	2.82	2.47	14.06	8.36
A2 to ZN	**fall**	0.23	0.30	8.85	3.17
	rise	3.75	3.82	15.53	9.83

NANGATE NAND2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

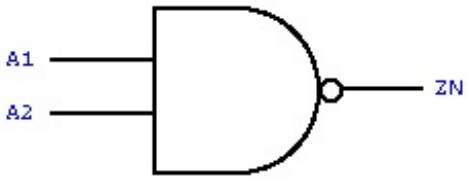
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "!(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**118.713**	**0.3656**	**118.713**
A1 to ZN	**fall**	0.00	0.10	0.01	0.14
	rise	0.00	0.08	0.02	0.15
A2 to ZN	**fall**	0.01	0.10	-0.00	0.13
	rise	0.01	0.09	0.03	0.15

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**118.713**	**0.3656**	**118.713**
A1 to ZN	**fall**	0.00	0.08	0.03	0.10
	rise	0.00	0.08	0.03	0.09
A2 to ZN	**fall**	0.00	0.08	0.02	0.09
	rise	0.00	0.08	0.02	0.09

Capacitance [fF]	
A1	3.0603
A2	3.5091

Leakage [nW]
118.23

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**118.713**	**0.3656**	**118.713**
A1 to ZN	**fall**	0.43	0.61	20.76	7.94
	rise	5.75	5.79	28.23	16.71
A2 to ZN	**fall**	0.45	0.62	17.72	6.36
	rise	7.60	7.49	31.14	20.21

NANGATE NAND2_X4

Databook Build Date: Thursday Feb 17 15:07 2011

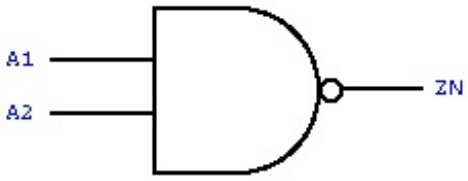
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "!(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	237.427	0.3656	237.427
A1 to ZN	**fall**	0.00	0.10	0.01	0.14
	rise	0.00	0.08	0.02	0.15
A2 to ZN	**fall**	0.01	0.10	-0.00	0.13
	rise	0.01	0.09	0.03	0.15

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	237.427	0.3656	237.427
A1 to ZN	**fall**	0.00	0.08	0.03	0.10
	rise	0.00	0.08	0.03	0.09
A2 to ZN	**fall**	0.00	0.08	0.02	0.09
	rise	0.00	0.08	0.02	0.09

Capacitance [fF]	
A1	5.9388
A2	6.2651

Leakage [nW]
236.47

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	237.427	0.3656	237.427
A1 to ZN	**fall**	0.80	1.23	41.70	15.81
	rise	10.52	10.22	55.52	32.54
A2 to ZN	**fall**	0.82	1.21	35.63	12.56
	rise	14.68	13.45	61.60	39.44

NANGATE NAND3_X1

Databook Build Date: Thursday Feb 17 15:07 2011

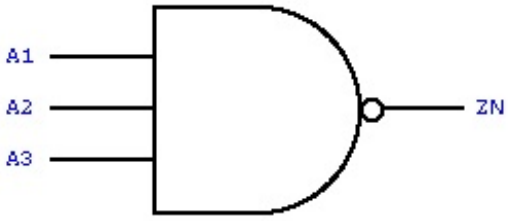
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	H
H	H	H	L
-	L	-	H
-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	58.3649	0.3656	58.3649
A1 to ZN	**fall**	0.01	0.13	0.02	0.18
	rise	0.00	0.08	0.02	0.14
A2 to ZN	**fall**	0.01	0.13	0.02	0.17
	rise	0.01	0.09	0.03	0.15
A3 to ZN	**fall**	0.01	0.14	0.01	0.15
	rise	0.01	0.09	0.04	0.15

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	58.3649	0.3656	58.3649
A1 to ZN	**fall**	0.01	0.11	0.03	0.12
	rise	0.00	0.08	0.03	0.09
A2 to ZN	**fall**	0.01	0.11	0.03	0.12
	rise	0.00	0.08	0.03	0.09
A3 to ZN	**fall**	0.01	0.11	0.02	0.12
	rise	0.01	0.08	0.03	0.09

Capacitance [fF]	
A1	1.6385
A2	1.6255
A3	1.6794

Leakage [nW]
57.58

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	58.3649	0.3656	58.3649
A1 to ZN	**fall**	0.62	0.68	9.85	3.68
	rise	3.19	2.85	12.37	7.47
A2 to ZN	**fall**	0.62	0.68	8.47	2.86
	rise	4.18	3.90	13.73	9.09
A3 to ZN	**fall**	0.63	0.68	9.11	3.11
	rise	4.94	4.99	15.81	10.44

NANGATE NAND3_X2

Databook Build Date: Thursday Feb 17 15:07 2011

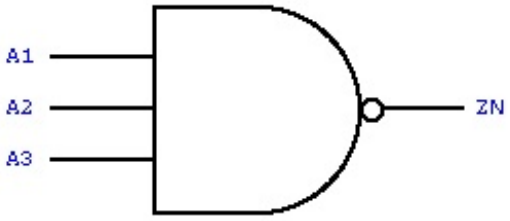
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	H
H	H	H	L
-	L	-	H
-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	116.272	0.3656	116.272
A1 to ZN	**fall**	0.01	0.13	0.02	0.18
	rise	0.01	0.08	0.02	0.14
A2 to ZN	**fall**	0.01	0.13	0.02	0.17
	rise	0.01	0.09	0.03	0.15
A3 to ZN	**fall**	0.01	0.14	0.01	0.15
	rise	0.01	0.09	0.04	0.15

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	116.272	0.3656	116.272
A1 to ZN	**fall**	0.00	0.11	0.03	0.12
	rise	0.00	0.08	0.03	0.09
A2 to ZN	**fall**	0.00	0.11	0.03	0.12
	rise	0.00	0.08	0.03	0.09
A3 to ZN	**fall**	0.00	0.11	0.02	0.12
	rise	0.01	0.08	0.03	0.09

Capacitance [fF]	
A1	3.0774
A2	3.2880
A3	3.6107

Leakage [nW]
115.15

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	116.272	0.3656	116.272
A1 to ZN	**fall**	1.46	1.58	19.92	7.57
	rise	6.62	6.83	24.99	15.30
A2 to ZN	**fall**	1.45	1.57	17.18	5.94
	rise	8.62	8.84	27.55	18.32
A3 to ZN	**fall**	1.48	1.58	18.47	6.44
	rise	10.09	10.71	31.88	20.53

NANGATE NAND3_X4

Databook Build Date: Thursday Feb 17 15:07 2011

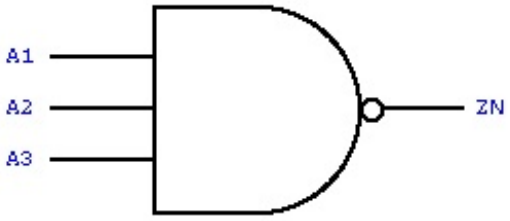
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	H
H	H	H	L
-	L	-	H
-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	233.154	0.3656	233.154
A1 to ZN	**fall**	0.01	0.13	0.02	0.18
	rise	0.00	0.08	0.02	0.14
A2 to ZN	**fall**	0.01	0.13	0.02	0.17
	rise	0.01	0.09	0.03	0.15
A3 to ZN	**fall**	0.01	0.14	0.01	0.15
	rise	0.01	0.09	0.03	0.15

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	233.154	0.3656	233.154
A1 to ZN	**fall**	0.00	0.11	0.03	0.12
	rise	0.00	0.08	0.03	0.09
A2 to ZN	**fall**	0.00	0.11	0.03	0.12
	rise	0.00	0.08	0.03	0.09
A3 to ZN	**fall**	0.00	0.11	0.02	0.12
	rise	0.00	0.08	0.03	0.09

Capacitance [fF]	
A1	6.4603
A2	6.9305
A3	7.2361

Leakage [nW]
230.31

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	233.154	0.3656	233.154
A1 to ZN	**fall**	2.92	3.17	39.95	15.19
	rise	12.89	13.04	49.88	29.93
A2 to ZN	**fall**	2.94	3.16	34.52	11.92
	rise	16.85	17.21	54.72	36.17
A3 to ZN	**fall**	2.95	3.17	36.95	12.93
	rise	19.91	19.77	63.49	41.56

NANGATE NAND4_X1

Databook Build Date: Thursday Feb 17 15:07 2011

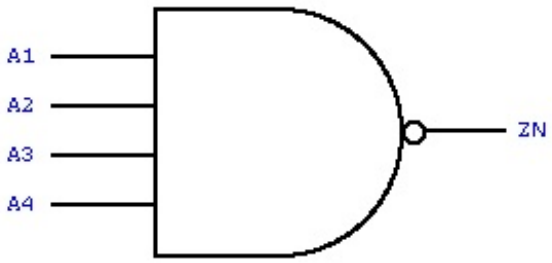
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	H
H	H	H	H	L
-	L	-	-	H
-	-	L	-	H
-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**55.9998**	**0.3656**	**55.9998**
A1 to ZN	**fall**	0.01	0.16	0.03	0.21
	rise	0.01	0.08	0.01	0.14
A2 to ZN	**fall**	0.01	0.17	0.03	0.20
	rise	0.01	0.08	0.02	0.14
A3 to ZN	**fall**	0.02	0.17	0.02	0.19
	rise	0.01	0.08	0.03	0.14
A4 to ZN	**fall**	0.02	0.17	0.01	0.17
	rise	0.01	0.09	0.04	0.15

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**55.9998**	**0.3656**	**55.9998**
A1 to ZN	**fall**	0.01	0.14	0.03	0.14
	rise	0.00	0.07	0.03	0.09
A2 to ZN	**fall**	0.01	0.14	0.03	0.14
	rise	0.00	0.08	0.03	0.09
A3 to ZN	**fall**	0.01	0.14	0.03	0.14
	rise	0.01	0.08	0.03	0.09
A4 to ZN	**fall**	0.01	0.14	0.03	0.14
	rise	0.01	0.08	0.03	0.09

Capacitance [fF]	
A1	1.6054
A2	1.6070
A3	1.6402
A4	1.6923

Leakage [nW]
54.73

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	55.9998	0.3656	55.9998
A1 to ZN	fall	1.24	1.27	9.50	3.88
	rise	3.55	3.26	11.39	7.32
A2 to ZN	fall	1.25	1.27	8.11	3.08
	rise	4.55	4.46	12.53	8.67
A3 to ZN	fall	1.24	1.27	8.50	3.13
	rise	5.40	5.41	14.20	9.72
A4 to ZN	fall	1.24	1.27	9.20	3.37
	rise	6.04	6.34	16.10	10.98

NANGATE NAND4_X2

Databook Build Date: Thursday Feb 17 15:07 2011

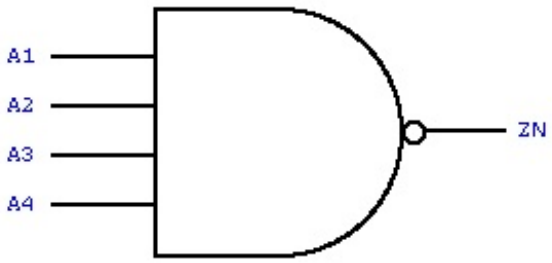
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	H
H	H	H	H	L
-	L	-	-	H
-	-	L	-	H
-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**111.542**	**0.3656**	**111.542**
A1 to ZN	**fall**	0.01	0.16	0.03	0.21
	rise	0.01	0.08	0.01	0.14
A2 to ZN	**fall**	0.01	0.17	0.03	0.20
	rise	0.01	0.08	0.02	0.14
A3 to ZN	**fall**	0.02	0.17	0.02	0.19
	rise	0.01	0.08	0.03	0.14
A4 to ZN	**fall**	0.02	0.17	0.01	0.17
	rise	0.01	0.09	0.04	0.15

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**111.542**	**0.3656**	**111.542**
A1 to ZN	**fall**	0.01	0.14	0.03	0.14
	rise	0.00	0.07	0.03	0.09
A2 to ZN	**fall**	0.01	0.14	0.03	0.14
	rise	0.01	0.08	0.03	0.09
A3 to ZN	**fall**	0.01	0.14	0.03	0.14
	rise	0.01	0.08	0.03	0.09
A4 to ZN	**fall**	0.01	0.14	0.03	0.14
	rise	0.01	0.08	0.03	0.09

Capacitance [fF]	
A1	3.0844
A2	3.2981
A3	3.4836
A4	3.8773

Leakage [nW]
109.47

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	111.542	0.3656	111.542
A1 to ZN	fall	2.47	2.54	18.90	7.74
	rise	7.71	7.72	23.39	15.46
A2 to ZN	fall	2.49	2.54	16.26	6.16
	rise	9.66	9.95	25.63	18.31
A3 to ZN	fall	2.51	2.54	16.94	6.25
	rise	11.45	11.72	28.95	19.97
A4 to ZN	fall	2.49	2.54	18.46	6.73
	rise	12.80	13.29	32.76	22.64

NANGATE NAND4_X4

Databook Build Date: Thursday Feb 17 15:07 2011

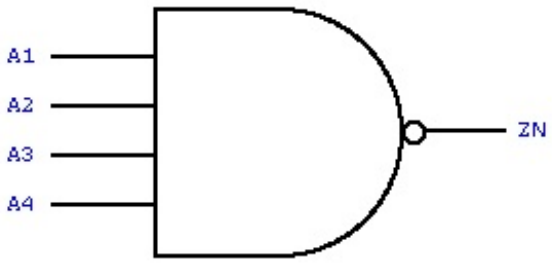
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.788 um^2
Equation	ZN = "!(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	H
H	H	H	H	L
-	L	-	-	H
-	-	L	-	H
-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**222.778**	**0.3656**	**222.778**
A1 to ZN	**fall**	0.01	0.16	0.03	0.21
	rise	0.01	0.08	0.01	0.14
A2 to ZN	**fall**	0.01	0.17	0.03	0.21
	rise	0.01	0.08	0.02	0.14
A3 to ZN	**fall**	0.02	0.17	0.02	0.19
	rise	0.01	0.08	0.03	0.14
A4 to ZN	**fall**	0.02	0.17	0.01	0.17
	rise	0.01	0.09	0.04	0.15

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**222.778**	**0.3656**	**222.778**
A1 to ZN	**fall**	0.01	0.14	0.03	0.15
	rise	0.00	0.07	0.03	0.09
A2 to ZN	**fall**	0.01	0.14	0.03	0.14
	rise	0.00	0.08	0.03	0.09
A3 to ZN	**fall**	0.01	0.14	0.03	0.14
	rise	0.01	0.08	0.03	0.09
A4 to ZN	**fall**	0.01	0.14	0.03	0.14
	rise	0.01	0.08	0.03	0.09

Capacitance [fF]	
A1	5.9599
A2	5.8398
A3	5.9008
A4	6.2153

Leakage [nW]
218.93

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	222.778	0.3656	222.778
A1 to ZN	fall	4.88	5.08	38.30	15.54
	rise	13.47	13.09	44.55	29.74
A2 to ZN	fall	4.90	5.10	32.61	12.18
	rise	18.11	17.67	49.56	35.11
A3 to ZN	fall	4.93	5.08	33.93	12.25
	rise	21.79	23.06	56.69	39.32
A4 to ZN	fall	4.88	5.08	36.69	13.11
	rise	24.63	26.18	64.56	43.99

NOR2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

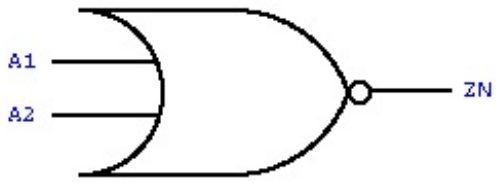
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	ZN = "!(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**26.7029**	**0.3656**	**26.7029**
A1 to ZN	**fall**	0.00	0.03	-0.02	0.05
	rise	0.01	0.08	0.05	0.14
A2 to ZN	**fall**	0.00	0.03	-0.01	0.05
	rise	0.01	0.08	0.04	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**26.7029**	**0.3656**	**26.7029**
A1 to ZN	**fall**	0.00	0.02	0.03	0.06
	rise	0.00	0.07	0.02	0.08
A2 to ZN	**fall**	0.00	0.02	0.03	0.06
	rise	0.00	0.07	0.02	0.08

Capacitance [fF]	
A1	1.7638
A2	1.6823

Leakage [nW]
68.18

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**26.7029**	**0.3656**	**26.7029**
A1 to ZN	**fall**	-0.00	-0.00	7.15	5.67
	rise	2.93	3.08	13.65	10.12
A2 to ZN	**fall**	0.17	0.36	8.50	6.54
	rise	3.98	4.01	12.90	9.16

NOR2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

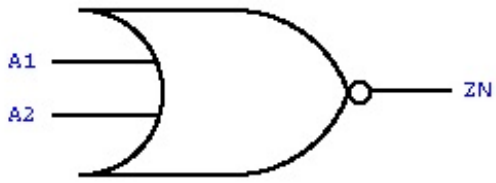
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "!(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	53.4058	0.3656	53.4058
A1 to ZN	**fall**	0.00	0.03	-0.02	0.05
	rise	0.01	0.08	0.05	0.14
A2 to ZN	**fall**	0.00	0.03	-0.01	0.05
	rise	0.01	0.08	0.04	0.13

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	53.4058	0.3656	53.4058
A1 to ZN	**fall**	0.00	0.02	0.03	0.06
	rise	0.00	0.07	0.02	0.08
A2 to ZN	**fall**	0.00	0.02	0.03	0.06
	rise	0.00	0.07	0.02	0.08

Capacitance [fF]	
A1	3.3796
A2	3.4123

Leakage [nW]
136.36

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	53.4058	0.3656	53.4058
A1 to ZN	**fall**	-0.00	-0.00	14.52	11.56
	rise	5.56	5.96	27.20	19.61
A2 to ZN	**fall**	0.57	0.95	17.10	13.29
	rise	7.73	7.75	25.68	18.10

NOR2_X4

Databook Build Date: Thursday Feb 17 15:07 2011

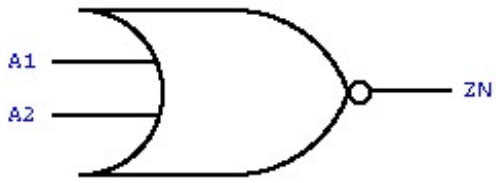
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "!(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	106.811	0.3656	106.811
A1 to ZN	**fall**	0.00	0.03	-0.02	0.05
	rise	0.01	0.08	0.04	0.14
A2 to ZN	**fall**	0.00	0.03	-0.01	0.05
	rise	0.01	0.08	0.04	0.13

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	106.811	0.3656	106.811
A1 to ZN	**fall**	0.00	0.02	0.03	0.06
	rise	0.00	0.07	0.02	0.08
A2 to ZN	**fall**	0.00	0.02	0.03	0.06
	rise	0.00	0.07	0.02	0.08

Capacitance [fF]	
A1	6.9485
A2	6.7881

Leakage [nW]
272.71

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	106.811	0.3656	106.811
A1 to ZN	**fall**	-0.00	-0.01	28.87	23.17
	rise	10.83	11.30	54.53	38.90
A2 to ZN	**fall**	1.10	1.90	34.02	26.65
	rise	15.35	15.74	51.22	36.04

NANGATE NOR3_X1

Databook Build Date: Thursday Feb 17 15:07 2011

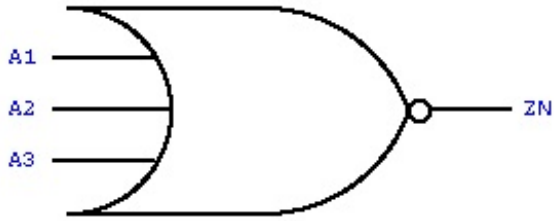
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	H
H	-	-	L
-	H	-	L
-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	16.0217	0.3656	16.0217
A1 to ZN	**fall**	0.00	0.02	-0.03	0.02
	rise	0.01	0.07	0.06	0.14
A2 to ZN	**fall**	0.00	0.02	-0.02	0.03
	rise	0.02	0.08	0.06	0.13
A3 to ZN	**fall**	0.00	0.02	-0.01	0.03
	rise	0.02	0.08	0.05	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	16.0217	0.3656	16.0217
A1 to ZN	**fall**	0.00	0.01	0.03	0.05
	rise	0.01	0.06	0.03	0.08
A2 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.06	0.03	0.07
A3 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.06	0.03	0.07

Capacitance [fF]	
A1	1.8198
A2	1.7160
A3	1.6547

Leakage [nW]
79.98

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	16.0217	0.3656	16.0217
A1 to ZN	**fall**	-0.00	-0.00	5.40	5.32
	rise	3.24	3.33	11.82	9.71
A2 to ZN	**fall**	0.29	0.44	6.34	6.04
	rise	4.35	4.46	10.97	8.88
A3 to ZN	**fall**	0.47	0.94	7.60	6.92
	rise	5.33	5.38	12.35	9.75

NOR3_X2

Databook Build Date: Thursday Feb 17 15:07 2011

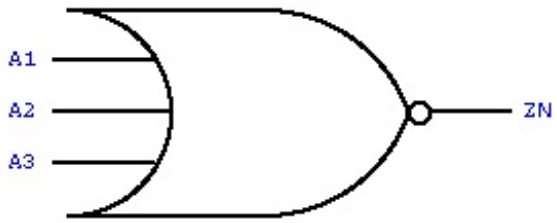
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	H
H	-	-	L
-	H	-	L
-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	31.7383	0.3656	31.7383
A1 to ZN	**fall**	0.00	0.02	-0.03	0.02
	rise	0.01	0.07	0.06	0.14
A2 to ZN	**fall**	0.00	0.02	-0.02	0.03
	rise	0.02	0.08	0.06	0.13
A3 to ZN	**fall**	0.00	0.02	-0.01	0.03
	rise	0.02	0.08	0.05	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	31.7383	0.3656	31.7383
A1 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.06	0.03	0.08
A2 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.06	0.03	0.07
A3 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.06	0.03	0.07

Capacitance [fF]	
A1	3.4755
A2	3.5254
A3	3.5246

Leakage [nW]
159.95

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	31.7383	0.3656	31.7383
A1 to ZN	**fall**	-0.00	-0.00	11.06	10.85
	rise	6.65	6.79	23.86	19.54
A2 to ZN	**fall**	0.81	1.10	12.82	12.28
	rise	8.81	8.95	22.12	17.94
A3 to ZN	**fall**	1.21	2.10	15.38	14.06
	rise	10.75	10.79	24.92	19.82

NOR3_X4

Databook Build Date: Thursday Feb 17 15:07 2011

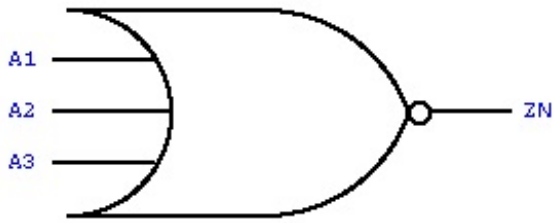
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.724 um^2
Equation	ZN = "!((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	H
H	-	-	L
-	H	-	L
-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**63.324**	**0.3656**	**63.324**
A1 to ZN	**fall**	0.00	0.02	-0.03	0.02
	rise	0.01	0.07	0.06	0.14
A2 to ZN	**fall**	0.00	0.02	-0.02	0.03
	rise	0.02	0.08	0.06	0.13
A3 to ZN	**fall**	0.00	0.02	-0.01	0.03
	rise	0.02	0.08	0.05	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**63.324**	**0.3656**	**63.324**
A1 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.07	0.03	0.08
A2 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.07	0.03	0.07
A3 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.07	0.03	0.07

Capacitance [fF]	
A1	6.7201
A2	6.3502
A3	6.2025

Leakage [nW]
319.90

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**63.324**	**0.3656**	**63.324**
A1 to ZN	**fall**	-0.00	-0.00	21.71	21.51
	rise	13.15	13.64	47.60	38.45
A2 to ZN	**fall**	1.56	2.20	25.37	24.32
	rise	18.38	18.21	44.59	35.79
A3 to ZN	**fall**	2.29	4.19	30.40	27.84
	rise	22.72	23.00	50.56	40.43

NOR4_X1

Databook Build Date: Thursday Feb 17 15:07 2011

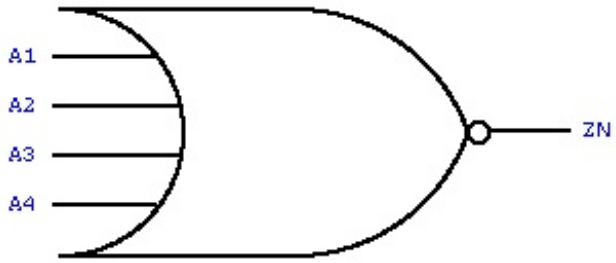
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	-	L
-	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**10.4713**	**0.3656**	**10.4713**
A1 to ZN	**fall**	0.00	0.01	-0.03	0.01
	rise	0.01	0.07	0.07	0.13
A2 to ZN	**fall**	0.00	0.02	-0.02	0.01
	rise	0.02	0.08	0.07	0.13
A3 to ZN	**fall**	0.00	0.02	-0.01	0.02
	rise	0.03	0.08	0.07	0.12
A4 to ZN	**fall**	0.00	0.02	-0.01	0.02
	rise	0.03	0.08	0.06	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**10.4713**	**0.3656**	**10.4713**
A1 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.01	0.06	0.03	0.07
A2 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.02	0.06	0.03	0.07
A3 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.02	0.06	0.03	0.07
A4 to ZN	**fall**	0.00	0.01	0.03	0.05
	rise	0.02	0.06	0.03	0.07

Capacitance [fF]	
A1	1.8046
A2	1.7364
A3	1.6924
A4	1.6419

Leakage [nW]
93.09

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	10.4713	0.3656	10.4713
A1 to ZN	fall	-0.00	-0.00	4.35	4.62
	rise	3.91	3.95	10.99	9.67
A2 to ZN	fall	0.35	0.43	5.12	5.27
	rise	5.00	5.08	10.18	8.97
A3 to ZN	fall	0.63	0.98	6.00	6.02
	rise	5.98	6.04	11.02	9.58
A4 to ZN	fall	0.63	1.29	6.82	6.73
	rise	6.94	7.01	12.21	10.64

NOR4_X2

Databook Build Date: Thursday Feb 17 15:07 2011

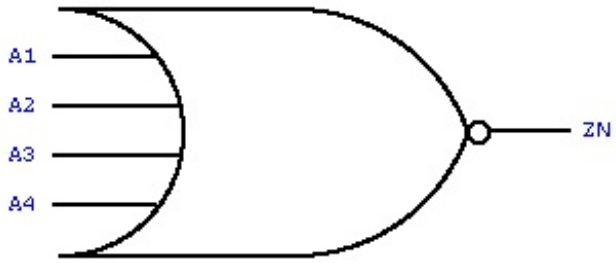
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	-	L
-	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**20.9045**	**0.3656**	**20.9045**
A1 to ZN	**fall**	0.00	0.01	-0.03	0.01
	rise	0.01	0.07	0.07	0.13
A2 to ZN	**fall**	0.00	0.02	-0.02	0.01
	rise	0.02	0.08	0.07	0.13
A3 to ZN	**fall**	0.00	0.02	-0.02	0.02
	rise	0.03	0.08	0.07	0.12
A4 to ZN	**fall**	0.00	0.02	-0.01	0.02
	rise	0.03	0.08	0.06	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**20.9045**	**0.3656**	**20.9045**
A1 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.01	0.06	0.03	0.07
A2 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.01	0.06	0.03	0.07
A3 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.01	0.06	0.03	0.07
A4 to ZN	**fall**	0.00	0.01	0.03	0.05
	rise	0.01	0.06	0.03	0.07

Capacitance [fF]	
A1	3.5116
A2	3.5307
A3	3.6096
A4	3.6855

Leakage [nW]
186.18

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	20.9045	0.3656	20.9045
A1 to ZN	fall	0.02	-0.00	8.90	9.45
	rise	7.63	7.80	21.95	19.25
A2 to ZN	fall	0.93	1.09	10.39	10.76
	rise	9.82	10.00	20.24	17.73
A3 to ZN	fall	1.44	2.19	12.20	12.26
	rise	11.75	11.89	21.91	19.18
A4 to ZN	fall	1.50	2.82	13.76	13.67
	rise	13.68	13.88	24.29	21.10

NOR4_X4

Databook Build Date: Thursday Feb 17 15:07 2011

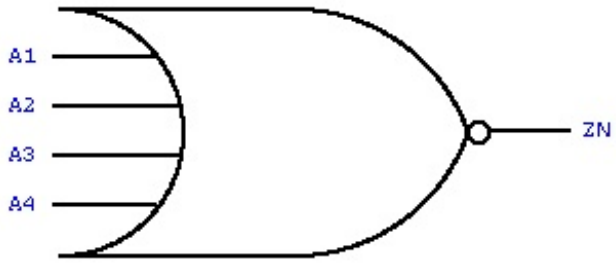
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.788 um^2
Equation	ZN = "!(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	-	L
-	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**41.5039**	**0.3656**	**41.5039**
A1 to ZN	**fall**	0.00	0.01	-0.03	0.01
	rise	0.01	0.07	0.07	0.13
A2 to ZN	**fall**	0.00	0.02	-0.02	0.01
	rise	0.02	0.08	0.07	0.13
A3 to ZN	**fall**	0.00	0.02	-0.01	0.02
	rise	0.03	0.08	0.07	0.12
A4 to ZN	**fall**	0.00	0.02	-0.01	0.02
	rise	0.03	0.08	0.06	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**41.5039**	**0.3656**	**41.5039**
A1 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.01	0.06	0.03	0.07
A2 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.01	0.06	0.03	0.07
A3 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.01	0.06	0.03	0.07
A4 to ZN	**fall**	0.00	0.01	0.03	0.05
	rise	0.01	0.06	0.03	0.07

Capacitance [fF]	
A1	6.7997
A2	6.4032
A3	6.2535
A4	6.1732

Leakage [nW]
372.36

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	41.5039	0.3656	41.5039
A1 to ZN	fall	0.16	-0.00	17.67	18.68
	rise	15.10	15.40	43.77	38.28
A2 to ZN	fall	1.89	2.16	20.47	21.25
	rise	20.21	20.22	40.85	35.51
A3 to ZN	fall	2.88	4.36	23.95	24.19
	rise	24.92	24.86	44.66	38.98
A4 to ZN	fall	2.99	5.56	27.24	26.92
	rise	29.28	29.48	49.67	42.75

OAI211_X1

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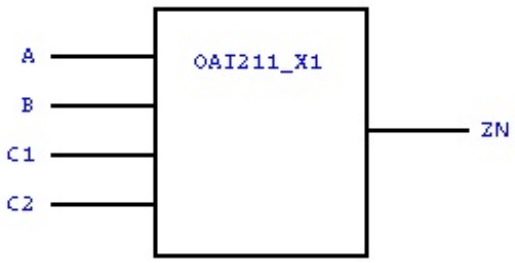
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((C1 | C2) & A) & B)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	-	-	-	H
H	H	H	-	L
H	H	-	H	L
-	L	-	-	H
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**25.5585**	**0.3656**	**25.5585**
A to ZN	**fall**	0.01	0.07	0.03	0.10
	rise	0.01	0.04	0.03	0.09
B to ZN	**fall**	0.01	0.07	0.02	0.08
	rise	0.01	0.04	0.04	0.10
C1 to ZN	**fall**	0.01	0.06	0.01	0.10
	rise	0.01	0.08	0.04	0.13
C2 to ZN	**fall**	0.01	0.07	0.02	0.11
	rise	0.01	0.08	0.03	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**25.5585**	**0.3656**	**25.5585**
A to ZN	**fall**	0.01	0.06	0.03	0.07
	rise	0.01	0.04	0.04	0.07
B to ZN	**fall**	0.01	0.06	0.03	0.06
	rise	0.01	0.04	0.03	0.07
C1 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.07	0.03	0.08
C2 to ZN	**fall**	0.01	0.06	0.03	0.08
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A	1.6162
B	1.6972
C1	1.6115
C2	1.6156

Leakage [nW]
70.26

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	25.5585	0.3656	25.5585
A to ZN	fall	0.87	0.94	8.35	4.36
	rise	5.71	5.80	15.29	12.62
B to ZN	fall	0.87	0.94	8.87	4.83
	rise	6.46	6.57	17.36	14.27
C1 to ZN	fall	0.72	0.78	7.77	4.44
	rise	3.59	3.73	11.47	8.51
C2 to ZN	fall	1.30	1.37	8.57	5.23
	rise	4.68	4.78	10.75	8.16

OAI211_X2

Databook Build Date: Thursday Feb 17 15:07 2011

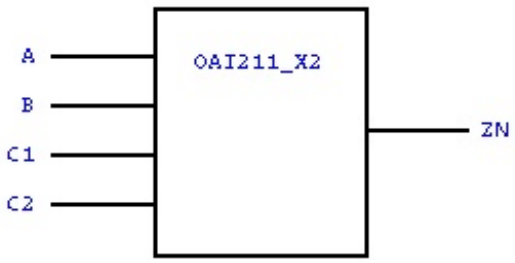
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((C1 | C2) & A) & B)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	-	-	-	H
H	H	H	-	L
H	H	-	H	L
-	L	-	-	H
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**50.8118**	**0.3656**	**50.8118**
A to ZN	**fall**	0.01	0.07	0.03	0.10
	rise	0.01	0.04	0.03	0.09
B to ZN	**fall**	0.01	0.07	0.02	0.08
	rise	0.01	0.04	0.04	0.10
C1 to ZN	**fall**	0.01	0.06	0.01	0.10
	rise	0.01	0.08	0.04	0.13
C2 to ZN	**fall**	0.01	0.07	0.02	0.11
	rise	0.01	0.08	0.03	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**50.8118**	**0.3656**	**50.8118**
A to ZN	**fall**	0.01	0.06	0.03	0.07
	rise	0.01	0.04	0.04	0.07
B to ZN	**fall**	0.01	0.06	0.03	0.06
	rise	0.01	0.04	0.03	0.07
C1 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.07	0.03	0.08
C2 to ZN	**fall**	0.01	0.06	0.03	0.08
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A	3.3328
B	3.2744
C1	3.0130
C2	3.3515

Leakage [nW]
140.52

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	50.8118	0.3656	50.8118
A to ZN	fall	1.95	2.11	16.99	8.97
	rise	11.47	11.63	30.64	25.18
B to ZN	fall	1.95	2.12	18.03	9.91
	rise	12.93	13.16	34.78	28.58
C1 to ZN	fall	1.67	1.78	15.76	9.07
	rise	7.38	7.22	23.11	17.15
C2 to ZN	fall	2.82	2.98	17.37	10.66
	rise	9.49	9.50	21.68	16.59

OAI211_X4

Databook Build Date: Thursday Feb 17 15:07 2011

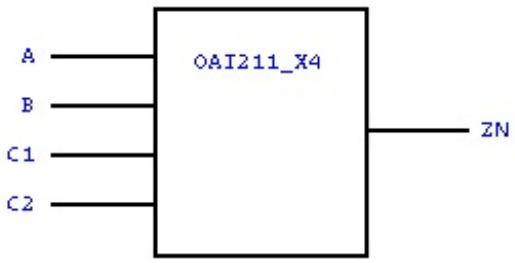
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Equation	ZN = "!(((C1 | C2) & A) & B)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	-	-	-	H
H	H	H	-	L
H	H	-	H	L
-	L	-	-	H
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**101.624**	**0.3656**	**101.624**
A to ZN	**fall**	0.01	0.07	0.03	0.10
	rise	0.01	0.04	0.03	0.09
B to ZN	**fall**	0.01	0.07	0.02	0.08
	rise	0.01	0.04	0.04	0.10
C1 to ZN	**fall**	0.01	0.06	0.01	0.10
	rise	0.01	0.08	0.04	0.13
C2 to ZN	**fall**	0.01	0.07	0.02	0.11
	rise	0.01	0.08	0.03	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**101.624**	**0.3656**	**101.624**
A to ZN	**fall**	0.01	0.06	0.03	0.07
	rise	0.01	0.04	0.04	0.07
B to ZN	**fall**	0.01	0.06	0.03	0.06
	rise	0.01	0.04	0.03	0.07
C1 to ZN	**fall**	0.01	0.05	0.03	0.08
	rise	0.01	0.07	0.03	0.09
C2 to ZN	**fall**	0.01	0.06	0.03	0.08
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A	6.3661
B	6.6701
C1	6.2392
C2	6.6787

Leakage [nW]
281.05

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	101.624	0.3656	101.624
A to ZN	fall	3.90	4.19	33.92	17.75
	rise	23.20	23.47	61.34	50.90
B to ZN	fall	3.91	4.19	36.00	19.63
	rise	26.22	27.08	69.60	56.75
C1 to ZN	fall	3.33	3.57	31.43	18.00
	rise	15.00	15.32	46.40	34.04
C2 to ZN	fall	5.61	5.94	34.64	21.18
	rise	19.23	19.33	43.33	33.24

NANGATE OAI21_X1

Databook Build Date: Thursday Feb 17 15:07 2011

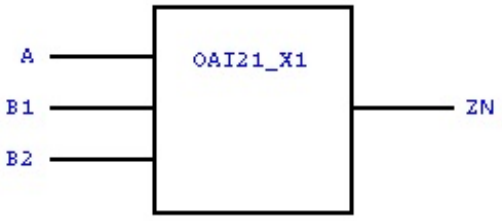
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!(A & (B1 | B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	-	-	H
H	H	-	L
H	-	H	L
-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	26.0544	0.3656	26.0544
A to ZN	**fall**	0.01	0.05	0.01	0.07
	rise	0.01	0.04	0.04	0.10
B1 to ZN	**fall**	0.01	0.05	-0.00	0.08
	rise	0.01	0.08	0.04	0.13
B2 to ZN	**fall**	0.01	0.05	0.01	0.08
	rise	0.01	0.08	0.04	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	26.0544	0.3656	26.0544
A to ZN	**fall**	0.00	0.04	0.03	0.05
	rise	0.01	0.04	0.03	0.06
B1 to ZN	**fall**	0.00	0.04	0.03	0.07
	rise	0.01	0.07	0.03	0.08
B2 to ZN	**fall**	0.00	0.04	0.03	0.07
	rise	0.01	0.07	0.02	0.08

Capacitance [fF]	
A	1.7017
B1	1.6940
B2	1.6165

Leakage [nW]
74.32

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	26.0544	0.3656	26.0544
A to ZN	**fall**	0.46	0.56	8.79	5.26
	rise	5.32	5.41	17.20	13.92
B1 to ZN	**fall**	0.33	0.40	7.68	4.89
	rise	3.28	3.35	12.37	9.06
B2 to ZN	**fall**	0.88	0.99	8.73	5.72
	rise	4.39	4.46	11.56	8.34

OAI21_X2

Databook Build Date: Thursday Feb 17 15:07 2011

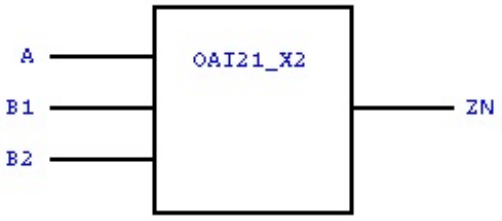
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!(A & (B1 | B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	-	-	H
H	H	-	L
H	-	H	L
-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	52.1088	0.3656	52.1088
A to ZN	**fall**	0.01	0.05	0.01	0.07
	rise	0.01	0.04	0.04	0.10
B1 to ZN	**fall**	0.01	0.05	-0.00	0.08
	rise	0.01	0.08	0.04	0.13
B2 to ZN	**fall**	0.01	0.05	0.01	0.08
	rise	0.01	0.08	0.04	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	52.1088	0.3656	52.1088
A to ZN	**fall**	0.00	0.04	0.02	0.05
	rise	0.01	0.04	0.03	0.06
B1 to ZN	**fall**	0.00	0.04	0.03	0.07
	rise	0.01	0.07	0.03	0.08
B2 to ZN	**fall**	0.00	0.04	0.03	0.07
	rise	0.01	0.07	0.02	0.08

Capacitance [fF]	
A	3.2299
B1	3.1473
B2	3.4280

Leakage [nW]
148.64

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	52.1088	0.3656	52.1088
A to ZN	**fall**	0.92	1.12	17.59	10.53
	rise	10.48	10.64	34.24	27.78
B1 to ZN	**fall**	0.63	0.80	15.20	9.74
	rise	6.60	6.40	24.81	17.82
B2 to ZN	**fall**	1.73	1.97	17.48	11.41
	rise	8.75	8.67	23.20	16.86

OAI21_X4

Databook Build Date: Thursday Feb 17 15:07 2011

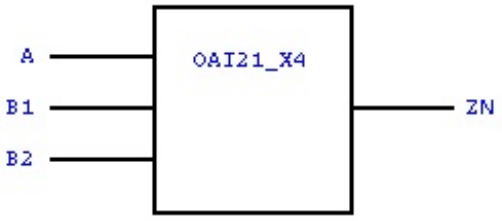
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(A & (B1 | B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	-	-	H
H	H	-	L
H	-	H	L
-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	104.065	0.3656	104.065
A to ZN	**fall**	0.01	0.05	0.01	0.07
	rise	0.01	0.04	0.04	0.10
B1 to ZN	**fall**	0.01	0.05	-0.00	0.08
	rise	0.01	0.08	0.04	0.13
B2 to ZN	**fall**	0.01	0.05	0.01	0.08
	rise	0.01	0.08	0.04	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	104.065	0.3656	104.065
A to ZN	**fall**	0.00	0.04	0.02	0.05
	rise	0.01	0.04	0.03	0.06
B1 to ZN	**fall**	0.00	0.04	0.03	0.07
	rise	0.01	0.07	0.03	0.08
B2 to ZN	**fall**	0.00	0.04	0.03	0.07
	rise	0.01	0.07	0.02	0.08

Capacitance [fF]	
A	6.2869
B1	6.4403
B2	6.6898

Leakage [nW]
297.28

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	104.065	0.3656	104.065
A to ZN	**fall**	1.79	2.22	35.38	20.91
	rise	21.01	20.85	68.29	55.66
B1 to ZN	**fall**	1.29	1.60	30.33	19.34
	rise	13.22	13.78	49.45	36.11
B2 to ZN	**fall**	3.48	3.95	34.87	22.65
	rise	17.50	17.40	46.46	33.60

OAI221_X1

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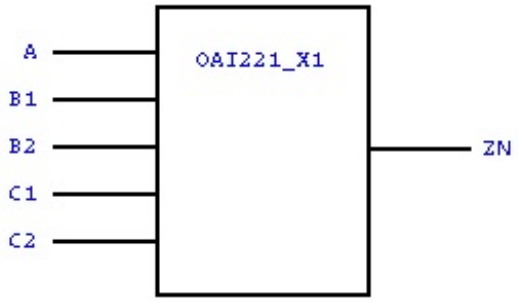
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "!(((C1 | C2) & A) & (B1 | B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	-	-	-	H
H	H	-	H	-	L
H	H	-	-	H	L
H	-	H	H	-	L
H	-	H	-	H	L
-	L	L	-	-	H
-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	22.1634	0.3656	22.1634
A to ZN	**fall**	0.02	0.06	0.03	0.10
	rise	0.01	0.04	0.04	0.09
B1 to ZN	**fall**	0.02	0.06	0.01	0.07
	rise	0.02	0.08	0.07	0.14
B2 to ZN	**fall**	0.02	0.07	0.02	0.08
	rise	0.02	0.08	0.06	0.13
C1 to ZN	**fall**	0.01	0.06	0.02	0.10
	rise	0.01	0.07	0.04	0.12
C2 to ZN	**fall**	0.01	0.06	0.03	0.10
	rise	0.01	0.07	0.04	0.11

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	22.1634	0.3656	22.1634
A to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.01	0.04	0.04	0.07
B1 to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.02	0.07	0.04	0.08
B2 to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.02	0.07	0.03	0.08
C1 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.06	0.04	0.08
C2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.06	0.03	0.08

Capacitance [fF]	
A	1.6389
B1	1.7087
B2	1.6405
C1	1.5787
C2	1.6476

Leakage [nW]
104.27

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	22.1634	0.3656	22.1634
A to ZN	fall	1.11	1.20	8.63	5.10
	rise	5.87	5.97	15.93	13.44
B1 to ZN	fall	0.96	1.04	6.36	4.26
	rise	6.87	6.98	15.26	12.79
B2 to ZN	fall	1.54	1.63	7.19	4.99
	rise	7.92	8.11	14.41	12.32
C1 to ZN	fall	0.96	1.04	7.95	5.06
	rise	3.90	4.00	11.99	9.17
C2 to ZN	fall	1.53	1.63	8.81	5.86
	rise	4.96	5.01	11.26	8.72

OAI221_X2

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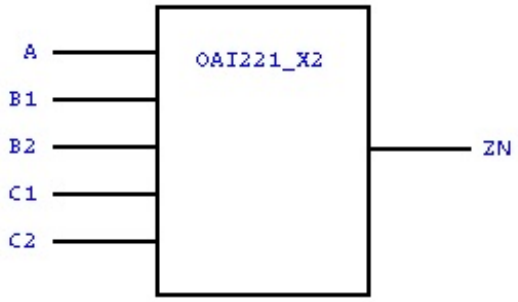
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.926 um^2
Equation	ZN = "!(((C1 | C2) & A) & (B1 | B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	-	-	-	H
H	H	-	H	-	L
H	H	-	-	H	L
H	-	H	H	-	L
H	-	H	-	H	L
-	L	L	-	-	H
-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	43.869	0.3656	43.869
A to ZN	**fall**	0.02	0.07	0.03	0.10
	rise	0.01	0.04	0.04	0.09
B1 to ZN	**fall**	0.02	0.06	0.01	0.07
	rise	0.02	0.08	0.07	0.14
B2 to ZN	**fall**	0.02	0.07	0.02	0.08
	rise	0.02	0.08	0.06	0.13
C1 to ZN	**fall**	0.01	0.06	0.02	0.10
	rise	0.01	0.07	0.04	0.12
C2 to ZN	**fall**	0.01	0.06	0.03	0.10
	rise	0.01	0.07	0.04	0.11

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	43.869	0.3656	43.869
A to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.01	0.04	0.04	0.07
B1 to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.02	0.07	0.04	0.08
B2 to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.02	0.07	0.03	0.08
C1 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.06	0.04	0.08
C2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.06	0.03	0.08

Capacitance [fF]	
A	3.5527
B1	3.5706
B2	3.2024
C1	3.0204
C2	3.3723

Leakage [nW]
208.53

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	43.869	0.3656	43.869
A to ZN	fall	2.44	2.62	17.38	10.34
	rise	12.24	12.41	32.30	27.38
B1 to ZN	fall	2.15	2.30	12.95	8.72
	rise	14.12	14.50	30.84	25.88
B2 to ZN	fall	3.31	3.49	14.57	10.17
	rise	16.18	16.47	29.24	24.88
C1 to ZN	fall	2.14	2.30	16.11	10.34
	rise	8.11	8.15	24.26	18.60
C2 to ZN	fall	3.29	3.48	17.82	11.94
	rise	10.24	10.37	22.83	17.78

NANGATE OAI221_X4

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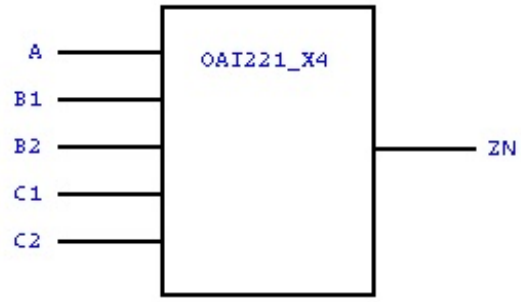
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(!(!(((C1 | C2) & A) & (B1 | B2))))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	-	-	-	H
H	H	-	H	-	L
H	H	-	-	H	L
H	-	H	H	-	L
H	-	H	-	H	L
-	L	L	-	-	H
-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.04	0.10	0.07	0.13
	rise	0.03	0.11	0.06	0.14
B1 to ZN	**fall**	0.04	0.10	0.04	0.10
	rise	0.04	0.12	0.09	0.18
B2 to ZN	**fall**	0.04	0.10	0.05	0.11
	rise	0.04	0.13	0.09	0.17
C1 to ZN	**fall**	0.03	0.10	0.06	0.12
	rise	0.03	0.12	0.08	0.16
C2 to ZN	**fall**	0.04	0.10	0.07	0.13
	rise	0.04	0.12	0.07	0.15

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
B1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
B2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
C1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
C2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08

Capacitance [fF]	
A	1.6548
B1	1.7192
B2	1.6930
C1	1.5635
C2	1.6426

Leakage [nW]
409.52

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	fall	15.96	17.72	27.66	29.94
	rise	21.06	22.17	37.09	38.65
B1 to ZN	fall	15.84	17.53	25.12	27.99
	rise	23.99	25.01	37.50	38.90
B2 to ZN	fall	16.78	18.54	26.32	28.23
	rise	24.95	25.65	34.96	36.77
C1 to ZN	fall	15.79	17.50	29.14	31.14
	rise	19.90	20.26	33.90	35.48
C2 to ZN	fall	16.79	18.51	30.06	31.42
	rise	20.92	21.67	31.05	33.20

OAI222_X1

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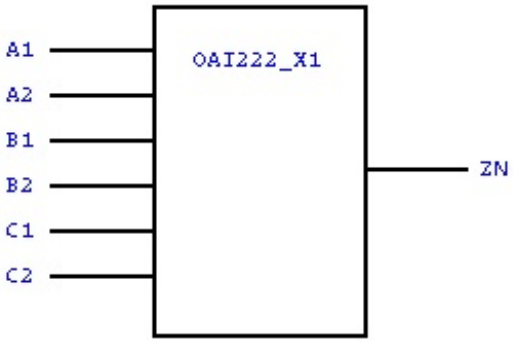
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.128 um^2
Equation	ZN = "!(((A1 | A2) & (B1 | B2)) & (C1 | C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	-	-	-	H
H	-	H	-	H	-	L
H	-	H	-	-	H	L
H	-	-	H	H	-	L
H	-	-	H	-	H	L
-	H	H	-	H	-	L
-	H	H	-	-	H	L
-	H	-	H	H	-	L
-	H	-	H	-	H	L
-	-	L	L	-	-	H
-	-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**20.0653**	**0.3656**	**20.0653**
A1 to ZN	**fall**	0.01	0.06	0.03	0.09
	rise	0.01	0.06	0.05	0.12
A2 to ZN	**fall**	0.02	0.06	0.04	0.10
	rise	0.01	0.07	0.04	0.11
B1 to ZN	**fall**	0.02	0.06	0.03	0.09
	rise	0.02	0.07	0.06	0.13
B2 to ZN	**fall**	0.02	0.06	0.04	0.09
	rise	0.02	0.07	0.05	0.12
C1 to ZN	**fall**	0.02	0.06	0.02	0.07
	rise	0.02	0.07	0.07	0.13
C2 to ZN	**fall**	0.02	0.07	0.02	0.08
	rise	0.02	0.08	0.06	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**20.0653**	**0.3656**	**20.0653**
A1 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.06	0.05	0.08
A2 to ZN	**fall**	0.01	0.05	0.04	0.07
	rise	0.02	0.06	0.04	0.08
B1 to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.02	0.07	0.05	0.08
B2 to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.02	0.07	0.04	0.08
C1 to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.02	0.07	0.04	0.09
C2 to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.02	0.07	0.04	0.08

Capacitance [fF]	
A1	1.5991
A2	1.6566
B1	1.6468
B2	1.6531
C1	1.6469
C2	1.6015

Leakage [nW]
130.26

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	20.0653	0.3656	20.0653
A1 to ZN	fall	1.20	1.29	8.16	5.63
	rise	4.17	4.22	12.50	9.77
A2 to ZN	fall	1.77	1.88	9.06	6.44
	rise	5.25	5.29	11.75	9.28
B1 to ZN	fall	1.20	1.29	6.66	4.43
	rise	6.28	6.34	14.30	12.11
B2 to ZN	fall	1.78	1.88	7.43	5.16
	rise	7.35	7.37	13.47	11.55
C1 to ZN	fall	1.20	1.29	6.35	4.53
	rise	8.01	8.22	16.30	14.11
C2 to ZN	fall	1.78	1.88	7.16	5.23
	rise	9.05	9.26	15.48	13.55

OAI222_X2

Databook Build Date: Thursday Feb 17 15:07 2011

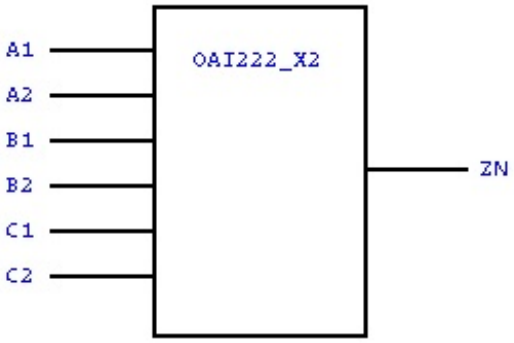
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	3.724 um^2
Equation	ZN = "!(((A1 | A2) & (B1 | B2)) & (C1 | C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	-	-	-	H
H	-	H	-	H	-	L
H	-	H	-	-	H	L
H	-	-	H	H	-	L
H	-	-	H	-	H	L
-	H	H	-	H	-	L
-	H	H	-	-	H	L
-	H	-	H	H	-	L
-	H	-	H	-	H	L
-	-	L	L	-	-	H
-	-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**39.5966**	**0.3656**	**39.5966**
A1 to ZN	**fall**	0.01	0.06	0.03	0.09
	rise	0.01	0.06	0.05	0.12
A2 to ZN	**fall**	0.02	0.06	0.04	0.10
	rise	0.01	0.07	0.04	0.11
B1 to ZN	**fall**	0.02	0.06	0.03	0.09
	rise	0.02	0.07	0.06	0.13
B2 to ZN	**fall**	0.02	0.06	0.04	0.10
	rise	0.02	0.07	0.05	0.12
C1 to ZN	**fall**	0.02	0.06	0.02	0.07
	rise	0.02	0.07	0.07	0.13
C2 to ZN	**fall**	0.02	0.07	0.02	0.08
	rise	0.02	0.08	0.06	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**39.5966**	**0.3656**	**39.5966**
A1 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.06	0.05	0.08
A2 to ZN	**fall**	0.01	0.05	0.04	0.07
	rise	0.02	0.06	0.04	0.08
B1 to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.02	0.07	0.05	0.08
B2 to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.02	0.07	0.04	0.08
C1 to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.02	0.07	0.05	0.09
C2 to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.02	0.07	0.04	0.08

Capacitance [fF]	
A1	3.2660
A2	3.1183
B1	3.3625
B2	3.0580
C1	3.4738
C2	3.1369

Leakage [nW]
260.53

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	39.5966	0.3656	39.5966
A1 to ZN	fall	2.63	2.82	16.51	11.46
	rise	8.79	8.92	25.40	20.03
A2 to ZN	fall	3.76	3.99	18.30	13.05
	rise	10.97	11.04	23.88	18.98
B1 to ZN	fall	2.64	2.82	13.52	9.08
	rise	12.97	13.21	28.93	24.51
B2 to ZN	fall	3.78	3.99	15.09	10.52
	rise	15.09	15.31	27.28	23.52
C1 to ZN	fall	2.63	2.81	12.90	9.26
	rise	16.44	16.88	32.99	28.70
C2 to ZN	fall	3.77	3.99	14.50	10.67
	rise	18.52	18.95	31.38	27.60

OAI222_X4

Databook Build Date: Thursday Feb 17 15:07 2011

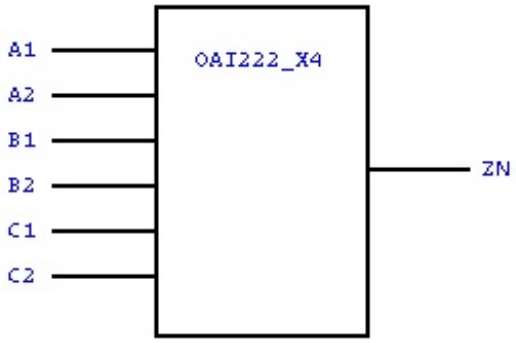
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.724 um^2
Equation	ZN = "!(!(!(((A1 | A2) & (B1 | B2)) & (C1 | C2))))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	-	-	-	H
H	-	H	-	H	-	L
H	-	H	-	-	H	L
H	-	-	H	H	-	L
H	-	-	H	-	H	L
-	H	H	-	H	-	L
-	H	H	-	-	H	L
-	H	-	H	H	-	L
-	H	-	H	-	H	L
-	-	L	L	-	-	H
-	-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.04	0.10	0.06	0.13
	rise	0.04	0.12	0.08	0.16
A2 to ZN	**fall**	0.04	0.10	0.07	0.13
	rise	0.04	0.12	0.07	0.16
B1 to ZN	**fall**	0.04	0.10	0.06	0.12
	rise	0.04	0.12	0.09	0.17
B2 to ZN	**fall**	0.05	0.11	0.07	0.13
	rise	0.04	0.13	0.08	0.17
C1 to ZN	**fall**	0.04	0.11	0.05	0.11
	rise	0.05	0.13	0.10	0.18
C2 to ZN	**fall**	0.05	0.11	0.06	0.12
	rise	0.05	0.13	0.09	0.17

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
B1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.01	0.08
B2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
C1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.01	0.08
C2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08

Capacitance [fF]	
A1	1.5994
A2	1.6425
B1	1.6579
B2	1.6437
C1	1.7013
C2	1.6809

Leakage [nW]
443.92

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	fall	16.10	17.83	29.38	31.40
	rise	20.46	20.83	34.86	36.16
A2 to ZN	fall	17.05	18.82	30.36	31.74
	rise	21.52	22.94	31.92	33.87
B1 to ZN	fall	16.13	17.86	26.44	28.61
	rise	23.34	24.19	37.08	38.00
B2 to ZN	fall	17.08	18.84	27.45	28.92
	rise	24.39	25.71	34.31	36.05
C1 to ZN	fall	16.10	17.84	25.21	27.91
	rise	25.78	27.42	39.28	40.71
C2 to ZN	fall	17.05	18.83	26.33	28.18
	rise	26.82	28.46	36.69	38.53

OAI22_X1

Databook Build Date: Thursday Feb 17 15:07 2011

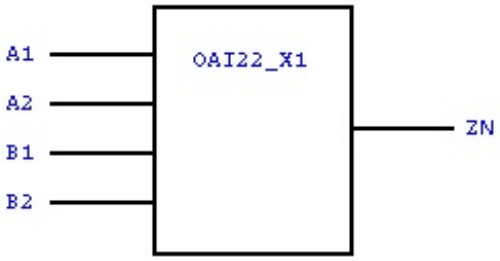
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!((A1 | A2) & (B1 | B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	L	-	-	H
H	-	H	-	L
H	-	-	H	L
-	H	H	-	L
-	H	-	H	L
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**23.2315**	**0.3656**	**23.2315**
A1 to ZN	**fall**	0.01	0.04	0.00	0.07
	rise	0.01	0.07	0.05	0.13
A2 to ZN	**fall**	0.01	0.05	0.02	0.08
	rise	0.01	0.07	0.04	0.12
B1 to ZN	**fall**	0.01	0.05	0.00	0.07
	rise	0.01	0.07	0.06	0.14
B2 to ZN	**fall**	0.01	0.05	0.01	0.07
	rise	0.02	0.08	0.05	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**23.2315**	**0.3656**	**23.2315**
A1 to ZN	**fall**	0.00	0.04	0.03	0.06
	rise	0.01	0.07	0.04	0.08
A2 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.07	0.03	0.08
B1 to ZN	**fall**	0.00	0.04	0.03	0.05
	rise	0.01	0.07	0.03	0.08
B2 to ZN	**fall**	0.01	0.04	0.03	0.05
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A1	1.7009
A2	1.6214
B1	1.7161
B2	1.6472

Leakage [nW]
105.07

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	23.2315	0.3656	23.2315
A1 to ZN	fall	0.55	0.66	7.82	5.60
	rise	3.51	3.61	12.89	9.69
A2 to ZN	fall	1.10	1.23	8.98	6.42
	rise	4.60	4.53	12.08	9.06
B1 to ZN	fall	0.56	0.66	6.25	4.50
	rise	5.51	5.59	14.60	12.01
B2 to ZN	fall	1.11	1.23	7.24	5.22
	rise	6.57	6.62	13.72	11.36

OAI22_X2

Databook Build Date: Thursday Feb 17 15:07 2011

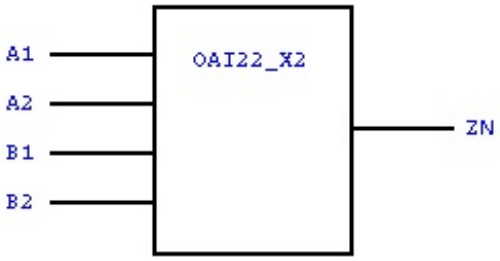
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!((A1 | A2) & (B1 | B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	L	-	-	H
H	-	H	-	L
H	-	-	H	L
-	H	H	-	L
-	H	-	H	L
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**46.3104**	**0.3656**	**46.3104**
A1 to ZN	**fall**	0.01	0.04	0.00	0.07
	rise	0.01	0.07	0.04	0.13
A2 to ZN	**fall**	0.01	0.05	0.01	0.08
	rise	0.01	0.07	0.04	0.12
B1 to ZN	**fall**	0.01	0.05	0.00	0.07
	rise	0.01	0.07	0.06	0.14
B2 to ZN	**fall**	0.01	0.05	0.01	0.07
	rise	0.02	0.08	0.05	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**46.3104**	**0.3656**	**46.3104**
A1 to ZN	**fall**	0.00	0.04	0.03	0.06
	rise	0.01	0.07	0.04	0.08
A2 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.07	0.03	0.08
B1 to ZN	**fall**	0.00	0.04	0.03	0.05
	rise	0.01	0.07	0.03	0.08
B2 to ZN	**fall**	0.01	0.04	0.03	0.05
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A1	3.1879
A2	3.4765
B1	3.2485
B2	3.4137

Leakage [nW]
210.14

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	46.3104	0.3656	46.3104
A1 to ZN	fall	1.09	1.31	15.67	11.13
	rise	7.19	7.52	25.96	19.69
A2 to ZN	fall	2.19	2.47	17.93	12.81
	rise	9.37	9.24	24.37	18.29
B1 to ZN	fall	1.10	1.31	12.49	8.97
	rise	11.23	11.42	29.39	24.32
B2 to ZN	fall	2.21	2.47	14.54	10.41
	rise	13.32	13.61	27.63	22.98

OAI22_X4

Databook Build Date: Thursday Feb 17 15:07 2011

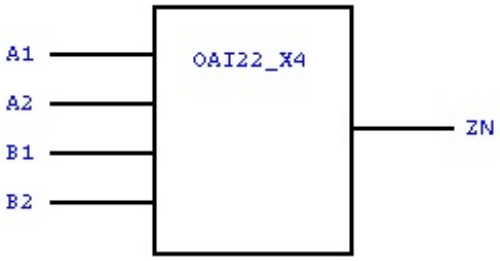
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Equation	ZN = "!((A1 | A2) & (B1 | B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	L	-	-	H
H	-	H	-	L
H	-	-	H	L
-	H	H	-	L
-	H	-	H	L
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**92.4683**	**0.3656**	**92.4683**
A1 to ZN	**fall**	0.01	0.04	0.00	0.08
	rise	0.01	0.07	0.04	0.13
A2 to ZN	**fall**	0.01	0.05	0.02	0.08
	rise	0.01	0.07	0.04	0.12
B1 to ZN	**fall**	0.01	0.05	0.00	0.07
	rise	0.01	0.08	0.06	0.14
B2 to ZN	**fall**	0.01	0.05	0.01	0.07
	rise	0.02	0.08	0.05	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**92.4683**	**0.3656**	**92.4683**
A1 to ZN	**fall**	0.00	0.04	0.03	0.06
	rise	0.01	0.07	0.04	0.08
A2 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.07	0.03	0.08
B1 to ZN	**fall**	0.00	0.04	0.03	0.05
	rise	0.01	0.07	0.03	0.08
B2 to ZN	**fall**	0.01	0.04	0.03	0.05
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A1	6.5496
A2	6.6777
B1	6.6978
B2	6.6580

Leakage [nW]
420.27

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	92.4683	0.3656	92.4683
A1 to ZN	fall	2.18	2.62	31.24	22.07
	rise	14.55	14.88	51.84	38.95
A2 to ZN	fall	4.38	4.95	35.75	25.38
	rise	18.85	18.57	48.75	36.71
B1 to ZN	fall	2.19	2.59	24.92	17.83
	rise	22.63	22.97	58.79	48.37
B2 to ZN	fall	4.44	4.94	28.87	20.69
	rise	26.83	26.82	55.33	45.58

OAI33_X1

Databook Build Date: Thursday Feb 17 15:07 2011

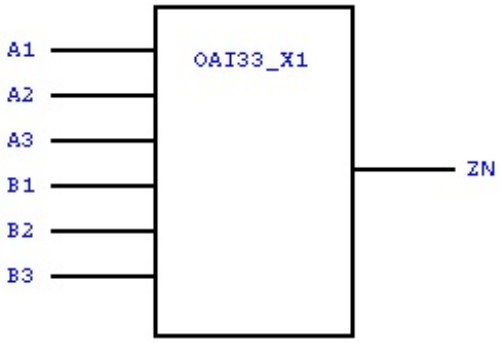
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.862 um^2
Equation	ZN = "!(((A1 | A2) | A3) & ((B1 | B2) | B3))"
Type	Combinational
Input	A1, A2, A3, B1, B2, B3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**A3**	**B1**	**B2**	**B3**	**ZN**
L	L	L	-	-	-	H
H	-	-	H	-	-	L
H	-	-	-	H	-	L
H	-	-	-	-	H	L
-	H	-	H	-	-	L
-	H	-	-	H	-	L
-	H	-	-	-	H	L
-	-	H	H	-	-	L
-	-	H	-	H	-	L
-	-	H	-	-	H	L
-	-	-	L	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**11.4822**	**0.3656**	**11.4822**
A1 to ZN	**fall**	0.01	0.03	0.01	0.05
	rise	0.01	0.06	0.06	0.12
A2 to ZN	**fall**	0.01	0.03	0.01	0.05
	rise	0.02	0.06	0.06	0.11
A3 to ZN	**fall**	0.01	0.03	0.02	0.06
	rise	0.02	0.07	0.05	0.10
B1 to ZN	**fall**	0.01	0.03	0.01	0.04
	rise	0.03	0.07	0.08	0.13
B2 to ZN	**fall**	0.01	0.03	0.01	0.05
	rise	0.03	0.08	0.08	0.13
B3 to ZN	**fall**	0.02	0.03	0.02	0.05
	rise	0.03	0.08	0.07	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**11.4822**	**0.3656**	**11.4822**
A1 to ZN	**fall**	0.01	0.02	0.03	0.05
	rise	0.02	0.06	0.05	0.08
A2 to ZN	**fall**	0.01	0.02	0.03	0.05
	rise	0.02	0.06	0.04	0.07
A3 to ZN	**fall**	0.01	0.03	0.04	0.05
	rise	0.02	0.06	0.04	0.07
B1 to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.03	0.07	0.05	0.08
B2 to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.03	0.07	0.04	0.08
B3 to ZN	**fall**	0.01	0.03	0.03	0.04
	rise	0.03	0.07	0.04	0.08

Capacitance [fF]	
A1	1.7318
A2	1.6535
A3	1.6271
B1	1.7222
B2	1.6649
B3	1.6071

Leakage [nW]
140.80

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	11.4822	0.3656	11.4822
A1 to ZN	fall	0.72	0.81	6.42	6.07
	rise	4.18	4.22	11.95	10.24
A2 to ZN	fall	1.34	1.46	7.28	6.74
	rise	5.26	5.25	11.14	9.52
A3 to ZN	fall	1.72	1.98	8.42	7.55
	rise	6.23	6.24	12.33	10.55
B1 to ZN	fall	0.72	0.82	5.06	4.75
	rise	7.08	7.17	14.30	13.22
B2 to ZN	fall	1.36	1.46	5.80	5.36
	rise	8.13	8.21	13.46	12.45
B3 to ZN	fall	1.75	1.99	6.82	6.09
	rise	9.10	9.16	14.59	13.47

OR2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

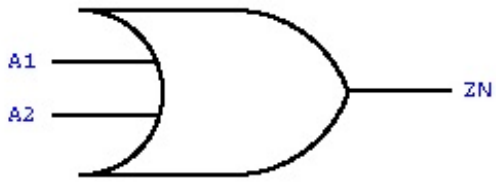
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	L
H	-	H
-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.02	0.08	0.07	0.14
	rise	0.01	0.09	0.01	0.09
A2 to ZN	**fall**	0.02	0.09	0.06	0.13
	rise	0.01	0.09	0.01	0.10

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A1	0.9729
A2	0.9609

Leakage [nW]
79.26

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	4.06	4.43	10.61	9.86
	rise	2.38	2.45	8.94	8.40
A2 to ZN	**fall**	4.55	4.95	9.55	9.41
	rise	2.72	2.80	9.78	8.69

NANGATE OR2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

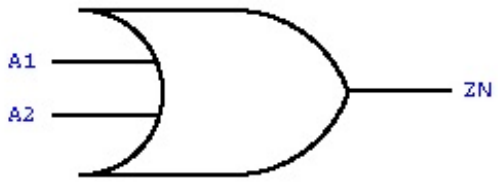
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	L
H	-	H
-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
A1 to ZN	**fall**	0.02	0.08	0.07	0.13
	rise	0.01	0.09	0.00	0.09
A2 to ZN	**fall**	0.02	0.09	0.06	0.13
	rise	0.01	0.09	0.01	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A1	1.7952
A2	1.7220

Leakage [nW]
159.40

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
A1 to ZN	**fall**	7.79	8.45	21.25	19.57
	rise	4.67	4.48	17.89	16.74
A2 to ZN	**fall**	8.86	9.52	18.95	18.69
	rise	5.28	5.56	19.74	17.07

NANGATE OR2_X4

Databook Build Date: Thursday Feb 17 15:07 2011

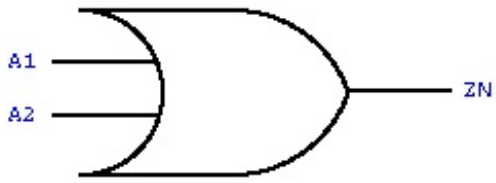
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	L
H	-	H
-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.02	0.08	0.07	0.13
	rise	0.01	0.09	0.00	0.09
A2 to ZN	**fall**	0.02	0.08	0.06	0.13
	rise	0.01	0.09	0.01	0.10

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A1	3.4801
A2	3.5116

Leakage [nW]
318.80

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	15.19	16.50	42.26	38.89
	rise	8.68	8.47	35.73	32.89
A2 to ZN	**fall**	17.37	18.64	37.89	37.18
	rise	9.95	10.53	39.10	33.85

OR3_X1

Databook Build Date: Thursday Feb 17 15:07 2011

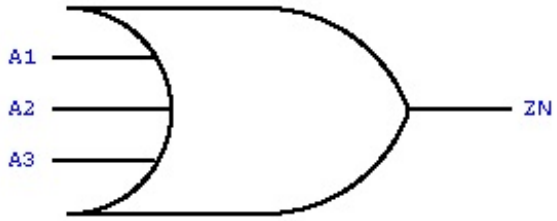
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	L
H	-	-	H
-	H	-	H
-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.03	0.10	0.09	0.16
	rise	0.01	0.09	0.00	0.09
A2 to ZN	**fall**	0.03	0.10	0.08	0.16
	rise	0.01	0.10	0.01	0.09
A3 to ZN	**fall**	0.04	0.11	0.07	0.15
	rise	0.01	0.10	0.01	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.01	0.05	0.01	0.06
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.01	0.05	0.01	0.06
	rise	0.00	0.08	0.01	0.08
A3 to ZN	**fall**	0.01	0.05	0.01	0.06
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A1	0.9898
A2	0.9704
A3	0.9441

Leakage [nW]
81.37

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	4.93	4.99	10.35	9.21
	rise	2.40	2.56	8.26	7.43
A2 to ZN	**fall**	5.43	5.51	9.45	8.71
	rise	2.79	2.79	8.85	7.39
A3 to ZN	**fall**	5.92	5.99	9.85	9.36
	rise	3.01	2.85	9.74	8.26

OR3_X2

Databook Build Date: Thursday Feb 17 15:07 2011

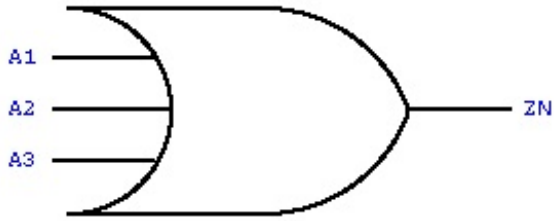
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.596 um^2
Equation	ZN = "((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	L
H	-	-	H
-	H	-	H
-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
A1 to ZN	**fall**	0.03	0.10	0.08	0.15
	rise	0.01	0.09	0.00	0.09
A2 to ZN	**fall**	0.03	0.10	0.08	0.15
	rise	0.01	0.10	0.01	0.09
A3 to ZN	**fall**	0.03	0.10	0.07	0.14
	rise	0.01	0.10	0.01	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
A1 to ZN	**fall**	0.01	0.05	0.01	0.06
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.01	0.05	0.01	0.06
	rise	0.00	0.08	0.01	0.08
A3 to ZN	**fall**	0.01	0.05	0.01	0.06
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A1	1.8259
A2	1.7659
A3	1.7103

Leakage [nW]
163.76

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
A1 to ZN	**fall**	9.32	9.39	20.51	18.02
	rise	4.65	5.10	16.33	15.26
A2 to ZN	**fall**	10.35	10.46	18.68	17.04
	rise	5.40	5.35	17.72	13.98
A3 to ZN	**fall**	11.32	11.42	19.50	18.39
	rise	5.87	5.72	19.35	16.63

NANGATE OR3_X4

Databook Build Date: Thursday Feb 17 15:07 2011

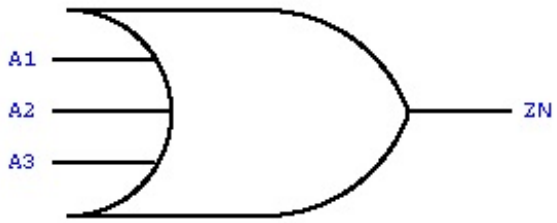
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	ZN = "((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	L
H	-	-	H
-	H	-	H
-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.03	0.09	0.08	0.15
	rise	0.01	0.09	-0.00	0.09
A2 to ZN	**fall**	0.03	0.10	0.08	0.15
	rise	0.01	0.10	0.01	0.09
A3 to ZN	**fall**	0.03	0.10	0.07	0.14
	rise	0.01	0.10	0.01	0.10

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.01	0.05	0.01	0.06
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.01	0.05	0.01	0.06
	rise	0.00	0.08	0.01	0.08
A3 to ZN	**fall**	0.01	0.05	0.01	0.06
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A1	3.4909
A2	3.4790
A3	3.5032

Leakage [nW]
327.52

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	17.81	17.94	40.54	35.39
	rise	8.74	9.66	32.59	29.36
A2 to ZN	**fall**	20.09	20.08	36.72	33.45
	rise	10.24	10.21	35.12	27.94
A3 to ZN	**fall**	22.00	22.03	38.44	36.20
	rise	11.17	10.86	38.77	32.11

OR4_X1

Databook Build Date: Thursday Feb 17 15:07 2011

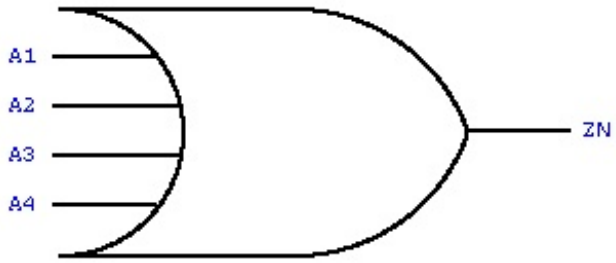
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	L
H	-	-	-	H
-	H	-	-	H
-	-	H	-	H
-	-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.04	0.11	0.10	0.18
	rise	0.01	0.09	0.00	0.09
A2 to ZN	**fall**	0.05	0.12	0.10	0.18
	rise	0.01	0.10	0.01	0.09
A3 to ZN	**fall**	0.05	0.13	0.09	0.17
	rise	0.01	0.10	0.01	0.09
A4 to ZN	**fall**	0.05	0.13	0.09	0.16
	rise	0.01	0.10	0.01	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.01	0.06	0.01	0.06
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.01	0.06	0.01	0.06
	rise	0.00	0.08	0.01	0.08
A3 to ZN	**fall**	0.01	0.06	0.01	0.06
	rise	0.00	0.08	0.01	0.08
A4 to ZN	**fall**	0.01	0.06	0.01	0.06
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A1	0.9789
A2	0.9692
A3	0.9500
A4	0.9316

Leakage [nW]
85.99

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	fall	5.75	5.39	10.41	8.93
	rise	2.43	2.49	7.74	6.28
A2 to ZN	fall	6.27	5.91	9.59	8.48
	rise	2.80	2.85	8.23	6.81
A3 to ZN	fall	6.70	6.39	9.77	8.84
	rise	3.09	3.00	8.89	7.33
A4 to ZN	fall	7.17	6.86	10.20	9.38
	rise	3.18	3.36	9.52	7.88

OR4_X2

Databook Build Date: Thursday Feb 17 15:07 2011

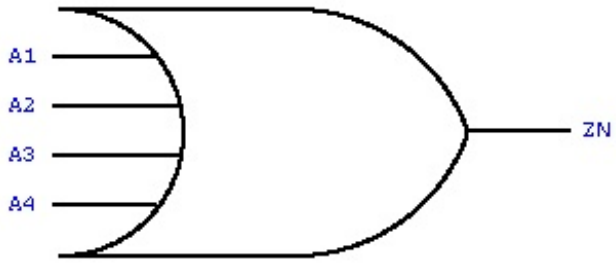
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	L
H	-	-	-	H
-	H	-	-	H
-	-	H	-	H
-	-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.85**	**0.3656**	**120.85**
A1 to ZN	**fall**	0.03	0.11	0.10	0.17
	rise	0.01	0.09	-0.00	0.08
A2 to ZN	**fall**	0.04	0.12	0.10	0.17
	rise	0.01	0.10	0.00	0.09
A3 to ZN	**fall**	0.05	0.12	0.09	0.17
	rise	0.01	0.10	0.01	0.09
A4 to ZN	**fall**	0.05	0.12	0.08	0.16
	rise	0.01	0.10	0.01	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**120.85**	**0.3656**	**120.85**
A1 to ZN	**fall**	0.01	0.06	0.01	0.06
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.01	0.06	0.01	0.06
	rise	0.00	0.08	0.01	0.08
A3 to ZN	**fall**	0.01	0.06	0.01	0.06
	rise	0.00	0.08	0.01	0.08
A4 to ZN	**fall**	0.01	0.06	0.01	0.06
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A1	1.8072
A2	1.7466
A3	1.7107
A4	1.6721

Leakage [nW]
173.15

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	fall	10.70	10.07	20.34	17.29
	rise	4.74	5.12	15.46	13.36
A2 to ZN	fall	11.91	11.14	18.72	16.40
	rise	5.46	5.48	16.36	13.21
A3 to ZN	fall	12.81	12.11	19.24	17.19
	rise	6.02	6.09	17.78	14.70
A4 to ZN	fall	13.76	13.07	20.19	18.29
	rise	6.17	6.70	19.18	15.75

OR4_X4

Databook Build Date: Thursday Feb 17 15:07 2011

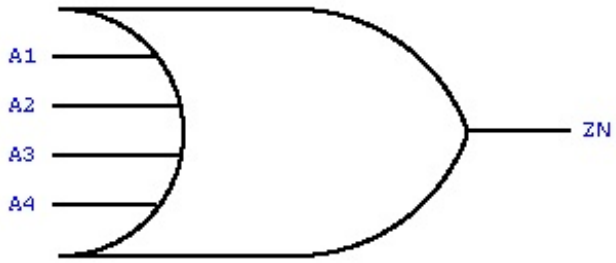
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	L
H	-	-	-	H
-	H	-	-	H
-	-	H	-	H
-	-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**241.699**	**0.3656**	**241.699**
A1 to ZN	**fall**	0.03	0.11	0.09	0.17
	rise	0.01	0.09	-0.00	0.08
A2 to ZN	**fall**	0.04	0.12	0.09	0.17
	rise	0.01	0.10	0.00	0.09
A3 to ZN	**fall**	0.05	0.12	0.09	0.16
	rise	0.01	0.10	0.01	0.09
A4 to ZN	**fall**	0.05	0.12	0.08	0.16
	rise	0.01	0.10	0.01	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**241.699**	**0.3656**	**241.699**
A1 to ZN	**fall**	0.01	0.06	0.01	0.06
	rise	0.00	0.08	0.01	0.08
A2 to ZN	**fall**	0.01	0.06	0.01	0.06
	rise	0.00	0.08	0.01	0.08
A3 to ZN	**fall**	0.01	0.06	0.01	0.06
	rise	0.00	0.08	0.01	0.08
A4 to ZN	**fall**	0.01	0.06	0.01	0.06
	rise	0.00	0.08	0.01	0.08

Capacitance [fF]	
A1	3.4900
A2	3.4856
A3	3.6131
A4	3.7140

Leakage [nW]
346.30

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	fall	21.16	19.70	40.56	34.24
	rise	9.02	9.62	30.73	26.12
A2 to ZN	fall	23.32	21.84	37.37	32.45
	rise	10.51	11.04	32.77	26.21
A3 to ZN	fall	25.24	23.78	38.06	34.01
	rise	11.60	11.63	35.28	28.88
A4 to ZN	fall	27.16	25.71	39.88	36.25
	rise	11.86	13.34	38.16	30.34

SDFFRS_X1

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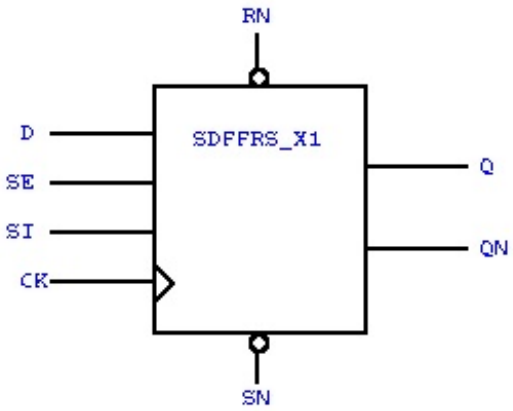
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, scan, and drive strength X1
Strength	1
Cell Area	7.714 um^2
Equation	Q = "(!(!RN)*((((SE * SI) + (D * !SE)))+(!SN)))" QN = "(!(!SN)*(!((((SE * SI) + (D * !SE)))+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table									
CK	D	RN	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	H	-	-	L	H
R	H	H	L	-	H	-	-	H	L
R	-	H	H	L	H	-	-	L	H
R	-	H	H	H	H	-	-	H	L
F	-	H	-	-	H	L	H	L	H
F	-	H	-	-	H	H	L	H	L
-	-	L	-	-	L	-	-	L	L
-	-	L	-	-	H	-	-	L	H
-	-	H	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.06	*0.12*	0.06	*0.12*
	rise	0.07	*0.15*	0.07	*0.16*
CK to QN	fall	0.05	0.11	0.06	0.12
	rise	0.05	0.13	0.05	0.14
RN to Q	fall	0.01	*0.08*	0.05	*0.12*
	rise	0.02	*0.10*	0.03	*0.11*
RN to QN	rise	0.10	0.18	0.14	0.22
SN to Q	rise	0.04	*0.12*	0.08	*0.16*
SN to QN	fall	0.02	0.08	0.06	0.13
	rise	0.09	0.18	0.10	0.18

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.00	*0.05*	0.00	*0.05*
	rise	0.00	*0.08*	0.00	*0.08*
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
RN to Q	fall	0.00	*0.05*	0.01	*0.05*
	rise	0.00	*0.08*	0.01	*0.08*
RN to QN	rise	0.01	0.08	0.01	0.08
SN to Q	rise	0.00	*0.08*	0.00	*0.08*
SN to QN	fall	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Constraints Time [ns]		
Setup CK to D	fall	0.09
	rise	0.04
Hold CK to D	fall	0.11
	rise	0.07
Recovery CK to RN	rise	-0.02
Removal CK to RN	rise	0.30
Setup CK to SE	fall	0.08
	rise	0.07
Hold CK to SE	fall	0.10
	rise	0.08
Setup CK to SI	fall	0.10
	rise	0.04
Hold CK to SI	fall	0.11
	rise	0.06
Recovery CK to SN	rise	0.03
Removal CK to SN	rise	0.14

Capacitance [fF]	
CK	0.9694
D	1.1592
RN	2.2576
SE	2.1542
SI	0.8781
SN	1.5338

Leakage [nW]
418.20

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	7403.12	15627.19
	rise	9580.35	17967.82
D	fall	4006.65	7050.70
	rise	2261.67	5166.66
RN	fall	2084.87	3687.48
	rise	-965.57	595.92
SE	fall	5273.75	15440.06
	rise	2398.50	12297.03
SI	fall	3589.70	5446.91
	rise	1811.40	3586.29
SN	fall	928.36	871.25
	rise	-757.65	-775.28

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	8.14	*8.07*	12.23	*12.44*
	rise	8.06	*7.83*	12.06	*11.98*
CK to QN	fall	8.07	7.82	12.06	11.98
	rise	8.14	8.10	12.23	12.44
RN to Q	fall	7.36	*7.94*	16.89	*16.91*
	rise	3.53	*4.13*	13.43	*15.06*
RN to QN	rise	8.00	7.92	15.29	13.86
SN to Q	rise	5.69	*7.96*	10.80	*11.65*
SN to QN	fall	5.52	8.42	11.97	13.69
	rise	6.13	6.27	9.75	9.30

SDFFRS_X2

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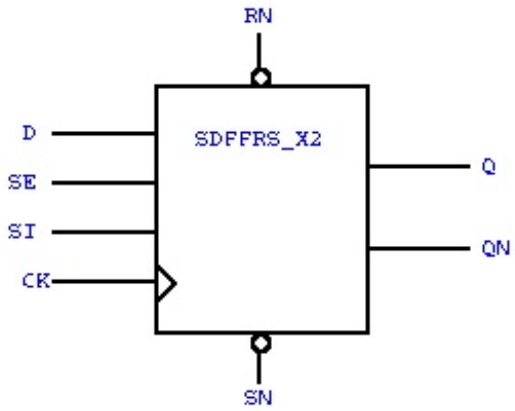
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, scan, and drive strength X2
Strength	2
Cell Area	8.246 um^2
Equation	Q = "(!(!RN)*((((SE * SI) + (D * !SE)))+(!SN)))" QN = "(!(!SN)*(!((((SE * SI) + (D * !SE)))+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table									
CK	D	RN	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	H	-	-	L	H
R	H	H	L	-	H	-	-	H	L
R	-	H	H	L	H	-	-	L	H
R	-	H	H	H	H	-	-	H	L
F	-	H	-	-	H	L	H	L	H
F	-	H	-	-	H	H	L	H	L
-	-	L	-	-	L	-	-	L	L
-	-	L	-	-	H	-	-	L	H
-	-	H	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.06	0.12	0.07	0.13
	rise	0.07	0.15	0.08	0.16
CK to QN	fall	0.05	0.12	0.06	0.12
	rise	0.05	0.13	0.05	0.14
RN to Q	fall	0.01	0.08	0.04	0.11
	rise	0.02	0.10	0.04	0.12
RN to QN	rise	0.10	0.19	0.15	0.23
SN to Q	rise	0.03	0.12	0.07	0.15
SN to QN	fall	0.02	0.08	0.05	0.12
	rise	0.09	0.18	0.10	0.19

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
RN to Q	fall	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08
RN to QN	rise	0.01	0.08	0.01	0.08
SN to Q	rise	0.00	0.08	0.00	0.08
SN to QN	fall	0.00	0.05	0.01	0.05
	rise	0.00	0.08	0.01	0.08

Constraints Time [ns]		
Setup CK to D	fall	0.09
	rise	0.04
Hold CK to D	fall	0.11
	rise	0.07
Recovery CK to RN	rise	0.01
Removal CK to RN	rise	0.30
Setup CK to SE	fall	0.08
	rise	0.07
Hold CK to SE	fall	0.10
	rise	0.08
Setup CK to SI	fall	0.10
	rise	0.04
Hold CK to SI	fall	0.11
	rise	0.06
Recovery CK to SN	rise	0.07
Removal CK to SN	rise	0.14

Capacitance [fF]	
CK	0.9525
D	1.1389
RN	2.6298
SE	2.0426
SI	0.8797
SN	1.8882

Leakage [nW]
563.33

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	7574.74	15792.36
	rise	10727.89	19101.58
D	fall	4079.66	7108.42
	rise	2204.67	5099.79
RN	fall	2458.62	4121.83
	rise	-1097.33	465.47
SE	fall	5242.37	15421.14
	rise	2396.81	12292.86
SI	fall	3649.25	5498.86
	rise	1754.53	3521.24
SN	fall	1303.10	1308.85
	rise	-873.55	-892.33

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	12.10	11.43	16.27	15.75
	rise	12.19	11.34	16.20	15.48
CK to QN	fall	12.19	11.34	16.20	15.48
	rise	12.10	11.43	16.27	15.75
RN to Q	fall	10.51	10.98	24.14	22.52
	rise	5.77	5.94	24.12	21.22
RN to QN	rise	11.80	11.03	22.77	18.87
SN to Q	rise	9.07	10.97	17.61	16.58
SN to QN	fall	8.79	11.61	19.29	19.41
	rise	9.39	9.32	17.22	14.74

SDFFR_X1

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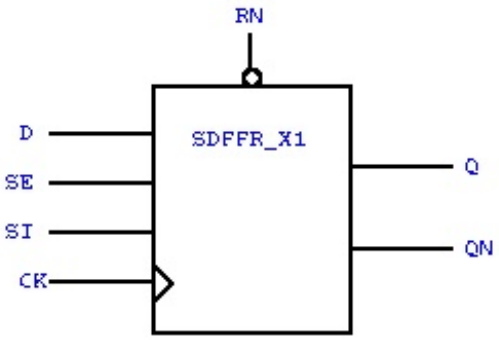
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, scan, and drive strength X1
Strength	1
Cell Area	6.650 um^2
Equation	Q = "(!(!RN)*(((SE * SI) + (D * !SE))))" QN = "((!RN)+!(((SE * SI) + (D * !SE))))"
Clock	CK
Reset	RN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	RN	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	-	-	L	H
R	H	H	L	-	-	-	H	L
R	-	H	H	L	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	H	-	-	L	H	L	H
F	-	H	-	-	H	L	H	L
-	-	L	-	-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
CK to Q	fall	0.04	0.10	0.04	0.11
	rise	0.04	0.13	0.05	0.13
CK to QN	fall	0.06	0.12	0.06	0.13
	rise	0.05	0.13	0.06	0.14
RN to Q	fall	0.09	0.15	0.14	0.21
RN to QN	rise	0.10	0.18	0.16	0.24

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
CK to Q	fall	0.01	0.05	0.01	0.05
	rise	0.00	0.08	0.00	0.08
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
RN to Q	fall	0.01	0.05	0.01	0.05
RN to QN	rise	0.00	0.08	0.00	0.08

Constraints Time [ns]		
Setup CK to D	fall	0.09
	rise	0.04
Hold CK to D	fall	0.11
	rise	0.07
Recovery CK to RN	rise	-0.02
Removal CK to RN	rise	0.26
Setup CK to SE	fall	0.08
	rise	0.06
Hold CK to SE	fall	0.10
	rise	0.08
Setup CK to SI	fall	0.10
	rise	0.05
Hold CK to SI	fall	0.11
	rise	0.06

Capacitance [fF]	
CK	1.0449
D	1.1692
RN	1.4973
SE	1.9943
SI	0.8939

Leakage [nW]
355.94

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	7477.19	15684.07
	rise	7838.88	16211.65
D	fall	3531.02	6198.17
	rise	1605.96	4145.96
RN	fall	1412.54	1836.86
	rise	-646.07	-294.12
SE	fall	5005.69	14717.03
	rise	2024.50	11454.85
SI	fall	3393.97	5336.06
	rise	1393.89	3245.37

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
CK to Q	fall	7.62	6.87	11.95	11.19
	rise	6.93	6.80	10.82	11.04
CK to QN	fall	6.93	6.80	10.82	11.04
	rise	7.62	6.87	11.95	11.19
RN to Q	fall	7.82	6.94	9.75	8.58
RN to QN	rise	7.82	6.94	9.75	8.58

SDFFR_X2

Databook Build Date: Thursday Feb 17 15:07 2011

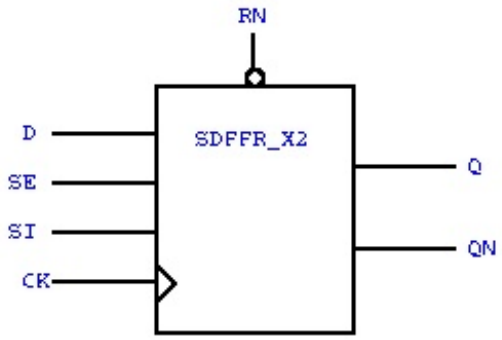
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, scan, and drive strength X2
Strength	2
Cell Area	6.916 um^2
Equation	Q = "(!(!RN)*(((SE * SI) + (D * !SE))))" QN = "((!RN)+!(((SE * SI) + (D * !SE))))"
Clock	CK
Reset	RN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	RN	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	-	-	L	H
R	H	H	L	-	-	-	H	L
R	-	H	H	L	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	H	-	-	L	H	L	H
F	-	H	-	-	H	L	H	L
-	-	L	-	-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.05	*0.12*	0.06	*0.13*
	rise	0.05	*0.14*	0.06	*0.15*
CK to QN	fall	0.06	0.12	0.07	0.13
	rise	0.07	0.14	0.07	0.15
RN to Q	fall	0.09	*0.17*	0.15	*0.22*
RN to QN	rise	0.11	0.19	0.16	0.24

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.01	*0.06*	0.01	*0.06*
	rise	0.01	*0.08*	0.01	*0.08*
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
RN to Q	fall	0.01	*0.06*	0.01	*0.06*
RN to QN	rise	0.00	0.08	0.00	0.08

Constraints Time [ns]		
Setup CK to D	fall	0.09
	rise	0.04
Hold CK to D	fall	0.11
	rise	0.07
Recovery CK to RN	rise	-0.02
Removal CK to RN	rise	0.05
Setup CK to SE	fall	0.08
	rise	0.06
Hold CK to SE	fall	0.10
	rise	0.08
Setup CK to SI	fall	0.10
	rise	0.05
Hold CK to SI	fall	0.11
	rise	0.06

Capacitance [fF]	
CK	0.9803
D	1.1562
RN	1.5366
SE	1.9233
SI	0.8970

Leakage [nW]
491.67

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	7318.13	15532.96
	rise	8212.25	16605.14
D	fall	3484.75	6155.42
	rise	1571.60	4116.94
RN	fall	1362.92	1794.45
	rise	-643.69	-281.77
SE	fall	4975.09	14691.76
	rise	1985.20	11413.34
SI	fall	3347.67	5294.30
	rise	1359.69	3215.06

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	14.56	*11.16*	19.08	*15.31*
	rise	11.50	*10.41*	15.47	*14.33*
CK to QN	fall	11.51	10.42	15.47	14.48
	rise	14.57	11.23	19.08	15.61
RN to Q	fall	14.67	*11.18*	17.27	*12.93*
RN to QN	rise	14.67	11.17	17.27	13.05

SDFFS_X1

Databook Build Date: Thursday Feb 17 15:07 2011

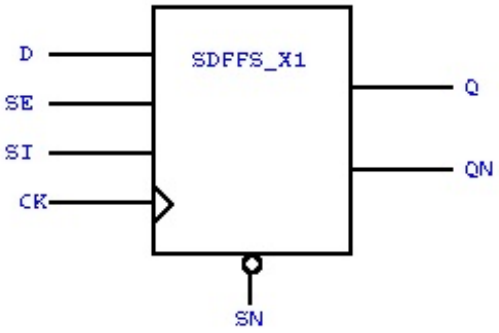
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, scan, and drive strength X1
Strength	1
Cell Area	6.650 um^2
Equation	Q = "((!SN)+(((SE * SI) + (D * !SE))))" QN = "(!((!SN)+(((SE * SI) + (D * !SE)))))"
Clock	CK
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	H	-	-	L	H
R	H	L	-	H	-	-	H	L
R	-	H	L	H	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	-	-	H	L	H	L	H
F	-	-	-	H	H	L	H	L
-	-	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.03	*0.10*	0.04	*0.11*
	rise	0.04	*0.12*	0.05	*0.13*
CK to QN	fall	0.05	0.11	0.06	0.12
	rise	0.06	0.14	0.06	0.14
SN to Q	rise	0.10	*0.19*	0.16	*0.24*
SN to QN	fall	0.02	0.08	0.05	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.01	*0.05*	0.01	*0.05*
	rise	0.00	*0.08*	0.00	*0.08*
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
SN to Q	rise	0.01	*0.08*	0.01	*0.08*
SN to QN	fall	0.00	0.05	0.01	0.05

Constraints Time [ns]		
Setup CK to D	fall	0.10
	rise	0.04
Hold CK to D	fall	0.11
	rise	0.06
Setup CK to SE	fall	0.09
	rise	0.07
Hold CK to SE	fall	0.09
	rise	0.07
Setup CK to SI	fall	0.10
	rise	0.05
Hold CK to SI	fall	0.11
	rise	0.06
Recovery CK to SN	rise	0.00
Removal CK to SN	rise	0.29

Capacitance [fF]	
CK	0.9964
D	1.1684
SE	2.0016
SI	0.9109
SN	1.3470

Leakage [nW]
367.38

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	7427.23	15715.06
	rise	9613.27	18118.27
D	fall	3885.30	6562.46
	rise	1869.63	4413.46
SE	fall	5542.68	15134.05
	rise	2470.52	11773.75
SI	fall	3752.57	5703.88
	rise	1653.93	3511.08
SN	fall	928.64	870.22
	rise	-757.88	-775.93

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	7.52	*6.75*	11.86	*11.19*
	rise	6.63	*6.63*	10.48	*10.83*
CK to QN	fall	6.63	6.68	10.48	10.82
	rise	7.52	6.71	11.86	11.18
SN to Q	rise	7.42	*6.94*	12.39	*10.56*
SN to QN	fall	7.43	6.96	12.39	10.45

SDFFS_X2

Databook Build Date: Thursday Feb 17 15:07 2011

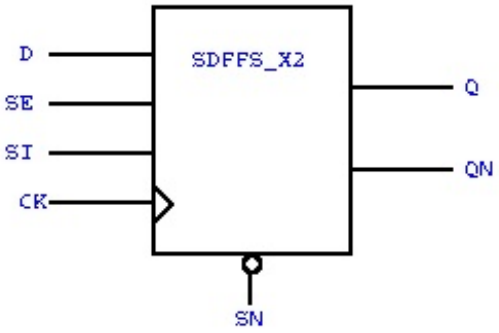
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, scan, and drive strength X2
Strength	2
Cell Area	7.182 um^2
Equation	Q = "((!SN)+(((SE * SI) + (D * !SE))))" QN = "(!((!SN)+(((SE * SI) + (D * !SE)))))"
Clock	CK
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	H	-	-	L	H
R	H	L	-	H	-	-	H	L
R	-	H	L	H	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	-	-	H	L	H	L	H
F	-	-	-	H	H	L	H	L
-	-	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.05	*0.12*	0.05	*0.13*
	rise	0.05	*0.14*	0.05	*0.14*
CK to QN	fall	0.06	0.12	0.07	0.13
	rise	0.07	0.15	0.08	0.16
SN to Q	rise	0.12	*0.21*	0.17	*0.26*
SN to QN	fall	0.01	0.07	0.04	0.11

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.01	*0.06*	0.01	*0.06*
	rise	0.01	*0.08*	0.01	*0.08*
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08
SN to Q	rise	0.01	*0.08*	0.01	*0.08*
SN to QN	fall	0.00	0.05	0.01	0.05

Constraints Time [ns]		
Setup CK to D	fall	0.10
	rise	0.05
Hold CK to D	fall	0.11
	rise	0.06
Setup CK to SE	fall	0.09
	rise	0.07
Hold CK to SE	fall	0.09
	rise	0.07
Setup CK to SI	fall	0.10
	rise	0.05
Hold CK to SI	fall	0.10
	rise	0.06
Recovery CK to SN	rise	0.00
Removal CK to SN	rise	0.29

Capacitance [fF]	
CK	0.9763
D	1.1411
SE	1.9038
SI	0.9233
SN	2.2508

Leakage [nW]
489.73

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	7560.33	15877.87
	rise	12182.90	21325.10
D	fall	4105.58	6778.25
	rise	1931.53	4467.61
SE	fall	5631.94	15239.94
	rise	2609.70	11921.78
SI	fall	3972.90	5918.51
	rise	1714.40	3565.22
SN	fall	1653.31	1656.45
	rise	-1098.75	-1115.68

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	14.12	*10.67*	18.58	*14.92*
	rise	11.88	*10.69*	15.90	*14.98*
CK to QN	fall	11.88	10.68	15.90	14.99
	rise	14.12	10.68	18.58	15.00
SN to Q	rise	13.17	*10.72*	23.22	*17.57*
SN to QN	fall	13.17	10.65	23.23	17.84

SDFF_X1

Databook Build Date: Thursday Feb 17 15:07 2011

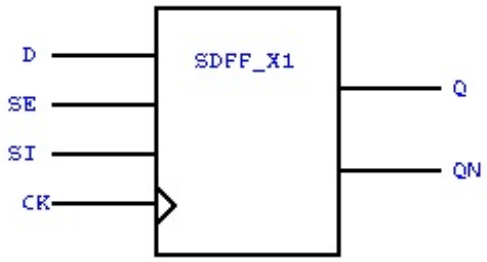
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with scan, and drive strength X1
Strength	1
Cell Area	6.118 um^2
Equation	Q = "(((SE * SI) + (D * !SE)))" QN = "(!(((SE * SI) + (D * !SE))))"
Clock	CK
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	-	-	L	H
R	H	L	-	-	-	H	L
R	-	H	L	-	-	L	H
R	-	H	H	-	-	H	L
F	-	-	-	L	H	L	H
F	-	-	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
CK to Q	**fall**	0.03	0.10	0.04	0.11
	rise	0.04	0.12	0.04	0.13
CK to QN	**fall**	0.05	*0.11*	0.05	*0.11*
	rise	0.05	*0.13*	0.05	*0.13*

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1462**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
CK to Q	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.08	0.00	0.08
CK to QN	**fall**	0.00	*0.05*	0.00	*0.05*
	rise	0.00	*0.08*	0.00	*0.08*

Constraints Time [ns]		
Setup CK to D	fall	0.10
	rise	0.05
Hold CK to D	fall	0.11
	rise	0.06
Setup CK to SE	fall	0.09
	rise	0.07
Hold CK to SE	fall	0.09
	rise	0.07
Setup CK to SI	fall	0.10
	rise	0.05
Hold CK to SI	fall	0.11
	rise	0.06

Capacitance [fF]	
CK	0.9747
D	1.1321
SE	1.8753
SI	0.9281

Leakage [nW]
347.11

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	7015.23	15468.48
	rise	3513.05	11909.14
D	fall	3853.01	6528.00
	rise	1910.90	4456.72
SE	fall	5250.46	15001.61
	rise	2344.65	11812.02
SI	fall	3718.78	5670.77
	rise	1697.23	3555.61

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	6.85	6.17	11.21	10.64
	rise	6.16	6.16	10.01	10.37
CK to QN	fall	6.16	*6.04*	10.02	*10.39*
	rise	6.85	*6.06*	11.21	*10.65*

SDFF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

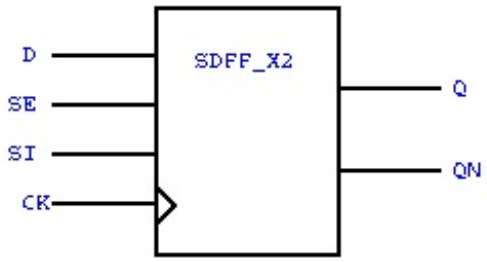
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with scan, and drive strength X2
Strength	2
Cell Area	6.384 um^2
Equation	Q = "(((SE * SI) + (D * !SE)))" QN = "(!(((SE * SI) + (D * !SE))))"
Clock	CK
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table

CK	D	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	-	-	L	H
R	H	L	-	-	-	H	L
R	-	H	L	-	-	L	H
R	-	H	H	-	-	H	L
F	-	-	-	L	H	L	H
F	-	-	-	H	L	H	L

Propagation Delay [ns]

Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.05	*0.12*	0.05	*0.13*
	rise	0.04	*0.13*	0.05	*0.14*
CK to QN	fall	0.06	0.12	0.06	0.12
	rise	0.06	0.14	0.07	0.15

Output Transition [ns]

Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.01	*0.06*	0.01	*0.06*
	rise	0.01	*0.08*	0.01	*0.08*
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.08	0.00	0.08

Constraints Time [ns]		
Setup CK to D	fall	0.10
	rise	0.05
Hold CK to D	fall	0.11
	rise	0.06
Setup CK to SE	fall	0.09
	rise	0.07
Hold CK to SE	fall	0.09
	rise	0.07
Setup CK to SI	fall	0.10
	rise	0.05
Hold CK to SI	fall	0.11
	rise	0.06

Capacitance [fF]	
CK	0.9994
D	1.1457
SE	1.8559
SI	0.9120

Leakage [nW]
480.58

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
CK	fall	7110.17	15520.33
	rise	3614.25	11972.12
D	fall	3863.49	6532.49
	rise	1918.12	4458.07
SE	fall	5239.03	14992.71
	rise	2352.84	11821.57
SI	fall	3729.28	5675.58
	rise	1704.28	3558.45

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	13.59	*10.37*	18.20	*14.87*
	rise	10.61	*9.51*	14.55	*13.71*
CK to QN	fall	10.61	9.51	14.55	13.69
	rise	13.59	10.46	18.21	14.80

TBUF_X16

Databook Build Date: Thursday Feb 17 15:07 2011

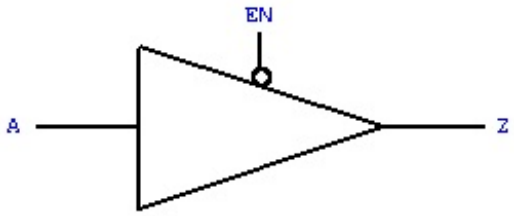
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	6.916 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		13.129	833.1075	13.129	833.1075
A to Z	**fall**	*0.02*	0.09	*0.06*	0.14
	rise	0.02	*0.11*	0.03	*0.13*
EN to Z	**fall**	*0.02*	0.09	*0.05*	0.13
	rise	0.03	*0.12*	0.06	*0.15*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		13.129	833.1075	13.129	833.1075
A to Z	**fall**	*0.00*	0.05	*0.01*	0.06
	rise	0.00	*0.08*	0.01	*0.08*
EN to Z	**fall**	*0.00*	0.05	*0.01*	0.06
	rise	0.00	*0.08*	0.01	*0.08*

Capacitance [fF]	
A	6.6209
EN	5.0364
Z	12.7955

Leakage [nW]
690.06

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		13.129	833.1075	13.129	833.1075
A to Z	**fall**	*45.60*	48.19	*90.40*	90.89
	rise	23.90	*34.13*	60.08	*75.70*
EN to Z	**fall**	*29.01*	39.41	*53.09*	71.74
	rise	32.37	*40.11*	45.04	*55.52*

TBUF_X1

Databook Build Date: Thursday Feb 17 15:07 2011

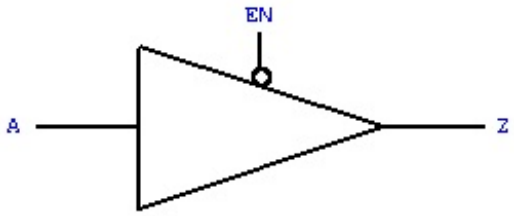
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.128 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		1.3987	52.6086	1.3987	52.6086
A to Z	**fall**	*0.02*	0.08	*0.05*	0.12
	rise	0.01	*0.09*	0.02	*0.11*
EN to Z	**fall**	*0.01*	0.08	*0.05*	0.12
	rise	0.02	*0.10*	0.05	*0.14*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		1.3987	52.6086	1.3987	52.6086
A to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.08*	0.01	*0.08*
EN to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.08*	0.00	*0.08*

Capacitance [fF]	
A	1.8892
EN	1.7641
Z	1.0339

Leakage [nW]
112.88

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		1.3987	52.6086	1.3987	52.6086
A to Z	**fall**	*5.54*	6.61	*13.55*	17.77
	rise	2.45	*2.87*	11.09	*13.95*
EN to Z	**fall**	*5.14*	6.00	*15.47*	19.55
	rise	5.32	*5.46*	11.60	*13.92*

TBUF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

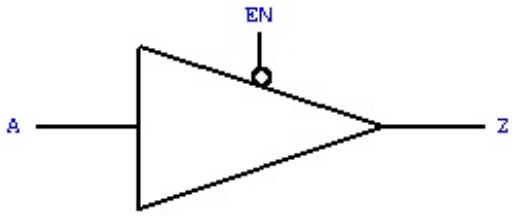
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		1.9666	105.2095	1.9666	105.2095
A to Z	**fall**	*0.01*	0.08	*0.04*	0.12
	rise	0.01	*0.09*	0.01	*0.11*
EN to Z	**fall**	*0.01*	0.07	*0.05*	0.12
	rise	0.02	*0.10*	0.06	*0.14*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		1.9666	105.2095	1.9666	105.2095
A to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.08*	0.01	*0.08*
EN to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.08*	0.00	*0.08*

Capacitance [fF]	
A	3.3322
EN	2.7781
Z	1.6025

Leakage [nW]
201.28

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		1.9666	105.2095	1.9666	105.2095
A to Z	**fall**	*9.80*	11.89	*25.98*	34.73
	rise	4.40	*5.34*	21.48	*28.78*
EN to Z	**fall**	*7.14*	9.00	*21.13*	28.15
	rise	8.04	*8.59*	14.43	*17.55*

TBUF_X4

Databook Build Date: Thursday Feb 17 15:07 2011

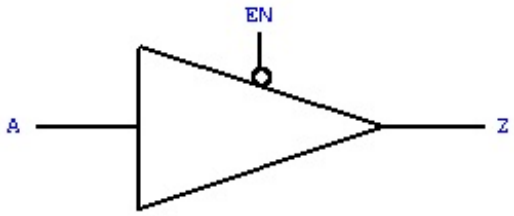
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		3.516	210.0645	3.516	210.0645
A to Z	**fall**	*0.01*	0.08	*0.05*	0.13
	rise	0.01	*0.10*	0.02	*0.12*
EN to Z	**fall**	*0.01*	0.08	*0.05*	0.12
	rise	0.02	*0.11*	0.06	*0.14*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		3.516	210.0645	3.516	210.0645
A to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.08*	0.01	*0.08*
EN to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.08*	0.00	*0.08*

Capacitance [fF]	
A	3.4442
EN	2.4866
Z	3.1555

Leakage [nW]
249.09

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		3.516	210.0645	3.516	210.0645
A to Z	**fall**	*11.55*	14.75	*30.78*	38.15
	rise	6.56	*8.32*	22.84	*30.68*
EN to Z	**fall**	*10.44*	13.24	*21.70*	30.13
	rise	11.40	*12.55*	17.48	*21.52*

TBUF_X8

Databook Build Date: Thursday Feb 17 15:07 2011

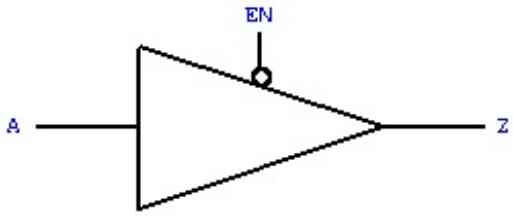
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	4.788 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		6.9635	419.2074	6.9635	419.2074
A to Z	**fall**	*0.01*	0.08	*0.05*	0.13
	rise	0.01	*0.10*	0.02	*0.12*
EN to Z	**fall**	*0.01*	0.08	*0.05*	0.12
	rise	0.02	*0.11*	0.06	*0.15*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		6.9635	419.2074	6.9635	419.2074
A to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.08*	0.01	*0.08*
EN to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.08*	0.00	*0.08*

Capacitance [fF]	
A	6.8548
EN	5.0568
Z	6.6094

Leakage [nW]
498.90

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		6.9635	419.2074	6.9635	419.2074
A to Z	**fall**	*24.06*	30.06	*62.61*	76.73
	rise	13.25	*17.93*	45.97	*63.43*
EN to Z	**fall**	*21.14*	26.93	*43.84*	60.83
	rise	23.39	*24.97*	35.83	*43.69*

TINV_X1

Databook Build Date: Thursday Feb 17 15:07 2011

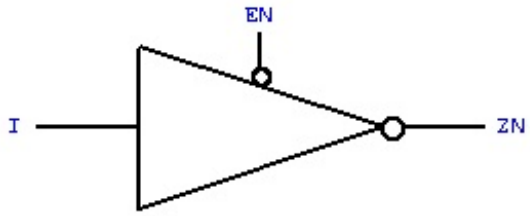
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!I", when !(EN)
Type	Tristate
Input	EN, I
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
I	**EN**	**ZN**
L	L	H
H	L	L
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		1.1499	23.4061	1.1499	23.4061
EN to ZN	**fall**	0.01	*0.05*	0.03	*0.09*
	rise	*0.00*	0.07	*-0.05*	0.11
I to ZN	**fall**	0.01	*0.05*	-0.00	*0.07*
	rise	*0.01*	0.08	*0.04*	0.13

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		1.1499	23.4061	1.1499	23.4061
EN to ZN	**fall**	0.00	*0.04*	0.01	*0.04*
	rise	*0.00*	0.07	*0.03*	0.09
I to ZN	**fall**	0.00	*0.04*	0.02	*0.05*
	rise	*0.01*	0.07	*0.02*	0.08

Capacitance [fF]	
EN	1.7653
I	1.4810
ZN	0.7849

Leakage [nW]
56.51

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		1.1499	23.4061	1.1499	23.4061
EN to ZN	**fall**	3.35	*3.58*	8.15	*11.44*
	rise	*3.91*	3.92	*4.07*	12.14
I to ZN	**fall**	0.65	*0.84*	6.06	*4.24*
	rise	*3.29*	3.28	*9.61*	7.04

TLAT_X1

Databook Build Date: Thursday Feb 17 15:07 2011

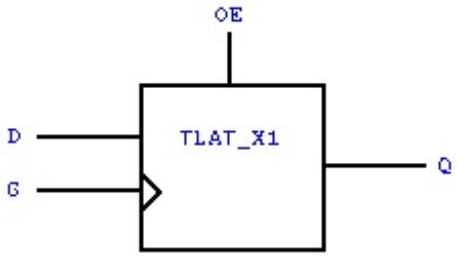
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	High enable Latch with drive strength X1
Strength	1
Cell Area	3.458 um^2
Equation	Q = "(D)", when !(!OE)
Enable	G
Type	Sequential
Input	D, OE
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
OE	**D**	**G**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	-	-	-	-	Z
H	L	H	-	-	L
H	H	H	-	-	H
H	-	L	L	H	L
H	-	L	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		1.1424	23.3612	1.1424	23.3612
D to Q	**fall**	*0.03*	0.07	*0.07*	0.11
	rise	0.03	*0.09*	0.03	*0.10*
G to Q	**fall**	*0.03*	0.07	*0.03*	0.08
	rise	0.03	*0.10*	0.04	*0.11*
OE to Q	**fall**	*0.00*	0.04	*-0.06*	0.06
	rise	0.01	*0.08*	-0.00	*0.08*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		1.1424	23.3612	1.1424	23.3612
D to Q	**fall**	*0.01*	0.04	*0.01*	0.04
	rise	0.01	*0.07*	0.01	*0.07*
G to Q	**fall**	*0.00*	0.04	*0.01*	0.04
	rise	0.01	*0.07*	0.01	*0.07*
OE to Q	**fall**	*0.00*	0.04	*0.03*	0.07
	rise	0.01	*0.07*	0.01	*0.07*

Constraints Time [ns]		
Setup G to D	**fall**	0.06
	rise	0.02
Hold G to D	**fall**	0.17
	rise	0.10

Capacitance [fF]	
D	1.1582
G	1.0281
OE	1.4980
Q	0.7782

Leakage [nW]
158.60

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1462
D	fall	1946.45	3260.26
	rise	263.57	1527.30
G	fall	3721.08	12367.59
	rise	3708.99	12459.78
OE	fall	2161.31	10069.56
	rise	1119.58	8978.97

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		1.1424	23.3612	1.1424	23.3612
D to Q	fall	*5.82*	6.53	*10.69*	11.94
	rise	5.47	*5.85*	10.34	*11.20*
G to Q	fall	*7.02*	7.73	*14.70*	17.06
	rise	8.35	*8.74*	15.93	*17.47*
OE to Q	fall	*0.70*	1.12	*2.29*	9.02
	rise	1.91	*2.00*	6.19	*9.68*

NANGATE XNOR2_X1

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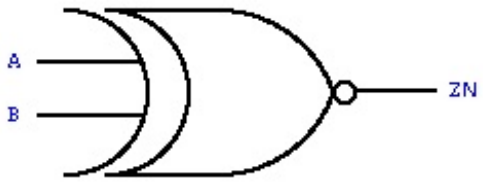
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "!(A ^ B)"
Type	Combinational
Input	A, B
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**ZN**
L	L	H
L	H	L
H	L	L
H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	26.0162	0.3656	26.0162
A to ZN	**fall**	0.02	0.06	0.05	0.09
	rise	0.02	0.08	0.04	0.13
B to ZN	**fall**	0.02	0.06	0.05	0.10
	rise	0.02	0.08	0.04	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	26.0162	0.3656	26.0162
A to ZN	**fall**	0.01	0.04	0.03	0.07
	rise	0.01	0.07	0.03	0.08
B to ZN	**fall**	0.00	0.04	0.03	0.07
	rise	0.01	0.07	0.02	0.08

Capacitance [fF]	
A	2.2244
B	2.5706

Leakage [nW]
124.47

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	26.0162	0.3656	26.0162
A to ZN	**fall**	3.20	3.04	9.52	8.53
	rise	4.48	4.34	12.26	10.45
B to ZN	**fall**	3.42	3.63	10.46	9.16
	rise	4.64	4.49	10.77	9.40

NANGATE XNOR2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

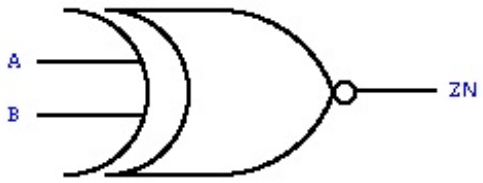
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.660 um^2
Equation	ZN = "!(A ^ B)"
Type	Combinational
Input	A, B
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**ZN**
L	L	H
L	H	L
H	L	L
H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	52.0325	0.3656	52.0325
A to ZN	**fall**	0.02	0.06	0.04	0.09
	rise	0.02	0.08	0.04	0.14
B to ZN	**fall**	0.02	0.06	0.05	0.10
	rise	0.02	0.08	0.04	0.13

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	52.0325	0.3656	52.0325
A to ZN	**fall**	0.01	0.04	0.03	0.07
	rise	0.01	0.07	0.03	0.08
B to ZN	**fall**	0.00	0.04	0.03	0.07
	rise	0.01	0.07	0.02	0.08

Capacitance [fF]	
A	3.9704
B	4.8318

Leakage [nW]
249.72

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	52.0325	0.3656	52.0325
A to ZN	**fall**	5.97	5.64	18.66	16.72
	rise	8.91	8.73	24.56	21.31
B to ZN	**fall**	6.45	6.85	20.59	17.96
	rise	9.33	9.06	21.64	19.33

XOR2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

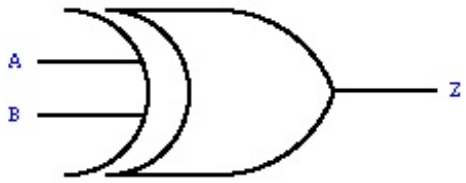
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	Z = "(A ^ B)"
Type	Combinational
Input	A, B
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**Z**
L	L	L
L	H	H
H	L	H
H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	25.3296	0.3656	25.3296
A to Z	**fall**	0.02	0.05	0.07	0.10
	rise	0.02	0.09	0.04	0.13
B to Z	**fall**	0.03	0.05	0.07	0.10
	rise	0.02	0.09	0.05	0.14

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	25.3296	0.3656	25.3296
A to Z	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.07	0.03	0.08
B to Z	**fall**	0.01	0.04	0.02	0.05
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A	2.2041
B	2.4500

Leakage [nW]
121.45

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	25.3296	0.3656	25.3296
A to Z	**fall**	3.33	3.32	9.65	8.39
	rise	4.42	4.19	11.13	9.96
B to Z	**fall**	3.48	3.47	8.20	7.51
	rise	4.68	4.75	11.96	10.55

NANGATE XOR2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

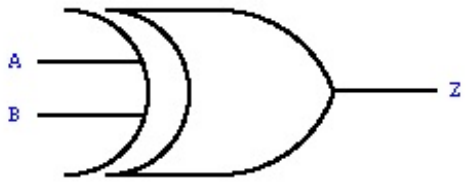
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_fast_fast**: Vdd= **1.25**V, Tj= **.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	Z = "(A ^ B)"
Type	Combinational
Input	A, B
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**Z**
L	L	L
L	H	H
H	L	H
H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	50.5066	0.3656	50.5066
A to Z	**fall**	0.02	0.05	0.07	0.10
	rise	0.02	0.09	0.04	0.13
B to Z	**fall**	0.03	0.05	0.07	0.10
	rise	0.02	0.09	0.05	0.14

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	50.5066	0.3656	50.5066
A to Z	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.07	0.03	0.08
B to Z	**fall**	0.01	0.04	0.02	0.05
	rise	0.01	0.07	0.03	0.08

Capacitance [fF]	
A	4.2679
B	4.5584

Leakage [nW]
243.78

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1462	
Load Capacitance [fF]		0.3656	50.5066	0.3656	50.5066
A to Z	**fall**	6.88	6.90	19.45	17.06
	rise	8.96	8.77	22.46	20.19
B to Z	**fall**	7.17	7.20	16.58	15.29
	rise	9.51	9.60	24.17	21.27

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Library

NangateOpenCellLibrary_worst_low

Corner

worst_low

AND2_X1

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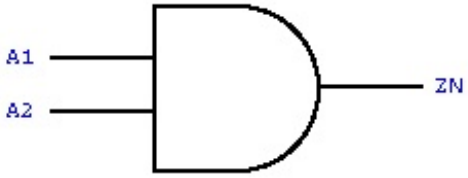
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	L
H	H	H
-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.02	0.10	0.09	0.18
	rise	0.02	0.14	0.07	0.19
A2 to ZN	**fall**	0.02	0.10	0.10	0.19
	rise	0.02	0.14	0.08	0.19

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A1	0.8631
A2	0.9048

Leakage [nW]
5.61

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	2.04	2.50	1.84	2.60
	rise	1.27	1.50	1.31	1.41
A2 to ZN	**fall**	2.24	2.75	2.09	2.86
	rise	1.28	1.45	1.20	1.28

AND2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

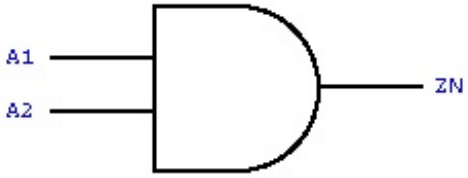
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	L
H	H	H
-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	**fall**	0.02	0.10	0.09	0.18
	rise	0.02	0.14	0.07	0.19
A2 to ZN	**fall**	0.02	0.10	0.09	0.18
	rise	0.02	0.14	0.07	0.19

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A1	1.5485
A2	1.5990

Leakage [nW]
11.26

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	**fall**	3.91	4.77	3.57	4.97
	rise	2.49	3.12	2.60	2.59
A2 to ZN	**fall**	4.29	5.26	4.03	5.49
	rise	2.49	3.00	2.37	2.68

AND2_X4

Databook Build Date: Thursday Feb 17 15:07 2011

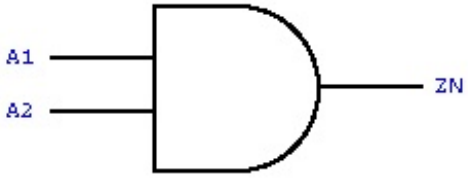
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	L
H	H	H
-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	**fall**	0.02	0.10	0.09	0.18
	rise	0.02	0.14	0.06	0.19
A2 to ZN	**fall**	0.02	0.10	0.09	0.18
	rise	0.02	0.14	0.07	0.19

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A1	2.9866
A2	3.2793

Leakage [nW]
22.52

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	**fall**	7.48	9.21	6.91	9.65
	rise	4.86	5.85	5.08	5.39
A2 to ZN	**fall**	8.30	10.21	7.84	10.69
	rise	4.86	5.70	4.58	5.54

AND3_X1

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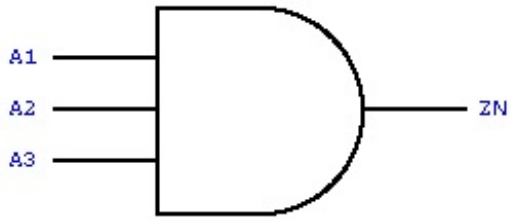
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	L
H	H	H	H
-	L	-	L
-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.4248**	**0.3656**	**60.4248**
A1 to ZN	**fall**	0.02	0.10	0.10	0.19
	rise	0.03	0.15	0.09	0.21
A2 to ZN	**fall**	0.02	0.11	0.10	0.19
	rise	0.03	0.15	0.09	0.21
A3 to ZN	**fall**	0.02	0.11	0.11	0.20
	rise	0.04	0.15	0.09	0.21

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.4248**	**0.3656**	**60.4248**
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
A3 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10

Capacitance [fF]	
A1	0.8448
A2	0.8830
A3	0.8967

Leakage [nW]
6.34

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.4248**	**0.3656**	**60.4248**
A1 to ZN	**fall**	2.15	2.67	1.92	2.73
	rise	1.35	1.69	1.41	1.66
A2 to ZN	**fall**	2.36	2.93	2.16	2.99
	rise	1.34	1.66	1.32	1.58
A3 to ZN	**fall**	2.52	3.15	2.38	3.22
	rise	1.36	1.62	1.31	1.31

AND3_X2

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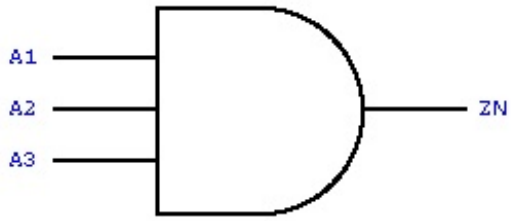
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.596 um^2
Equation	ZN = "((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	L
H	H	H	H
-	L	-	L
-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
A1 to ZN	**fall**	0.02	0.10	0.09	0.18
	rise	0.03	0.15	0.08	0.21
A2 to ZN	**fall**	0.02	0.10	0.10	0.19
	rise	0.03	0.15	0.09	0.21
A3 to ZN	**fall**	0.02	0.11	0.10	0.19
	rise	0.03	0.15	0.08	0.20

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
A3 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10

Capacitance [fF]	
A1	1.5158
A2	1.5662
A3	1.5772

Leakage [nW]
12.73

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
A1 to ZN	**fall**	4.08	5.05	3.70	5.20
	rise	2.70	3.21	2.81	3.18
A2 to ZN	**fall**	4.51	5.58	4.19	5.72
	rise	2.68	3.34	2.63	3.14
A3 to ZN	**fall**	4.86	6.01	4.61	6.18
	rise	2.73	3.44	2.60	3.21

AND3_X4

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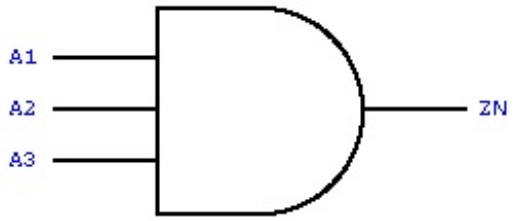
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Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	ZN = "((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	L
H	H	H	H
-	L	-	L
-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.02	0.10	0.09	0.18
	rise	0.03	0.15	0.08	0.20
A2 to ZN	**fall**	0.02	0.10	0.10	0.19
	rise	0.03	0.15	0.09	0.21
A3 to ZN	**fall**	0.02	0.11	0.10	0.19
	rise	0.03	0.15	0.08	0.20

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
A3 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10

Capacitance [fF]	
A1	2.9143
A2	3.1351
A3	3.3024

Leakage [nW]
25.45

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	7.86	9.80	7.18	10.12
	rise	5.11	6.36	5.41	6.70
A2 to ZN	**fall**	8.74	10.86	8.15	11.16
	rise	5.16	5.86	5.00	6.33
A3 to ZN	**fall**	9.43	11.74	9.00	12.09
	rise	5.21	6.05	5.00	6.10

AND4_X1

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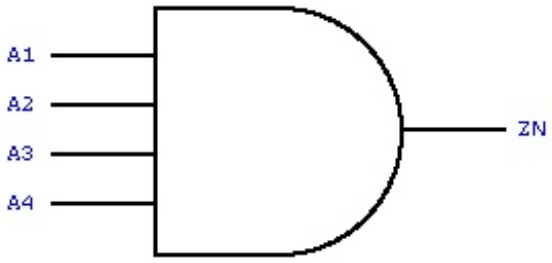
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Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	L
H	H	H	H	H
-	L	-	-	L
-	-	L	-	L
-	-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A1 to ZN	**fall**	0.02	0.11	0.10	0.19
	rise	0.04	0.16	0.10	0.23
A2 to ZN	**fall**	0.02	0.11	0.10	0.20
	rise	0.04	0.17	0.11	0.23
A3 to ZN	**fall**	0.03	0.11	0.11	0.20
	rise	0.05	0.17	0.10	0.23
A4 to ZN	**fall**	0.03	0.11	0.11	0.20
	rise	0.05	0.17	0.10	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
A3 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
A4 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10

Capacitance [fF]	
A1	0.8250
A2	0.8777
A3	0.8801
A4	0.8785

Leakage [nW]
6.70

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.2722	0.3656	60.2722
A1 to ZN	fall	2.26	2.83	2.03	2.88
	rise	1.46	1.78	1.51	1.67
A2 to ZN	fall	2.48	3.10	2.27	3.14
	rise	1.47	1.60	1.44	1.74
A3 to ZN	fall	2.67	3.33	2.48	3.37
	rise	1.44	1.72	1.42	1.75
A4 to ZN	fall	2.82	3.54	2.69	3.58
	rise	1.44	1.70	1.42	1.56

AND4_X2

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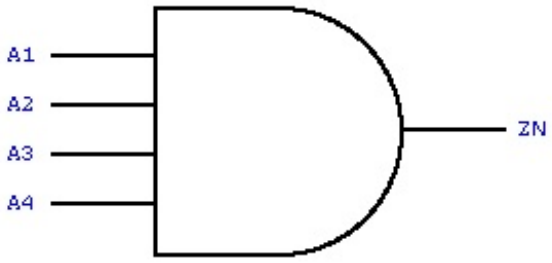
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	L
H	H	H	H	H
-	L	-	-	L
-	-	L	-	L
-	-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.392	0.3656	120.392
A1 to ZN	**fall**	0.02	0.10	0.09	0.19
	rise	0.04	0.16	0.10	0.22
A2 to ZN	**fall**	0.02	0.11	0.10	0.19
	rise	0.04	0.16	0.10	0.23
A3 to ZN	**fall**	0.02	0.11	0.10	0.20
	rise	0.04	0.17	0.10	0.22
A4 to ZN	**fall**	0.03	0.11	0.11	0.20
	rise	0.04	0.17	0.09	0.22

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.392	0.3656	120.392
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
A3 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
A4 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10

Capacitance [fF]	
A1	1.4826
A2	1.5568
A3	1.5630
A4	1.5675

Leakage [nW]
13.47

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.392	0.3656	120.392
A1 to ZN	fall	4.22	5.26	3.82	5.37
	rise	2.82	3.51	2.97	3.08
A2 to ZN	fall	4.65	5.79	4.29	5.90
	rise	2.83	3.42	2.75	3.46
A3 to ZN	fall	5.01	6.26	4.72	6.35
	rise	2.83	3.47	2.73	3.37
A4 to ZN	fall	5.33	6.67	5.11	6.78
	rise	2.82	3.68	2.73	3.41

AND4_X4

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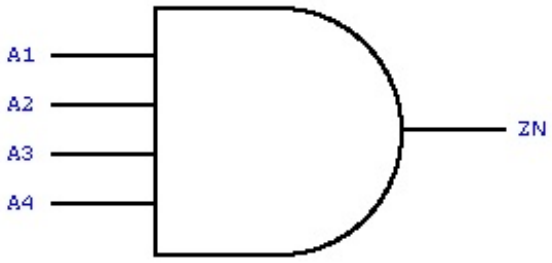
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	L
H	H	H	H	H
-	L	-	-	L
-	-	L	-	L
-	-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.02	0.10	0.09	0.18
	rise	0.04	0.16	0.09	0.22
A2 to ZN	**fall**	0.02	0.11	0.10	0.19
	rise	0.04	0.16	0.10	0.23
A3 to ZN	**fall**	0.02	0.11	0.10	0.19
	rise	0.04	0.16	0.10	0.22
A4 to ZN	**fall**	0.02	0.11	0.11	0.20
	rise	0.04	0.17	0.09	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
A3 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
A4 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10

Capacitance [fF]	
A1	2.8724
A2	3.1337
A3	3.2925
A4	3.4969

Leakage [nW]
26.93

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	241.089	0.3656	241.089
A1 to ZN	fall	7.94	10.04	7.29	10.30
	rise	5.39	7.19	5.73	6.72
A2 to ZN	fall	8.84	11.11	8.22	11.34
	rise	5.35	7.04	5.33	6.14
A3 to ZN	fall	9.60	12.04	9.06	12.26
	rise	5.44	7.02	5.25	6.12
A4 to ZN	fall	10.17	12.88	9.84	13.14
	rise	5.44	7.21	5.23	6.20

ANTENNA_X1

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Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.266 um^2
Type	Physical
Input	A
PG Pins	VDD (primary_power), VSS (primary_ground)



Capacitance [fF]	
A	0.0236

Leakage [nW]
0.00

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1792**
A	**fall**	0.00	0.00
	rise	0.00	0.00

AOI211_X1

Databook Build Date: Thursday Feb 17 15:07 2011

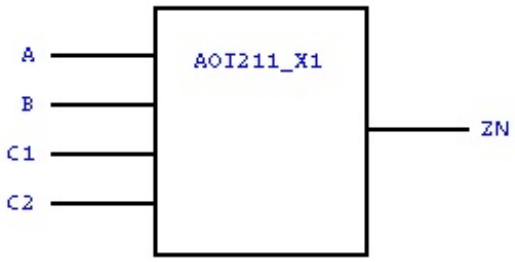
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((C1 & C2) | B) | A)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	L	L	-	H
L	L	-	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**14.4959**	**0.3656**	**14.4959**
A to ZN	**fall**	0.01	0.03	0.05	0.10
	rise	0.04	0.12	0.12	0.21
B to ZN	**fall**	0.01	0.03	0.04	0.10
	rise	0.04	0.12	0.11	0.22
C1 to ZN	**fall**	0.01	0.04	0.03	0.11
	rise	0.02	0.10	0.08	0.20
C2 to ZN	**fall**	0.01	0.04	0.04	0.11
	rise	0.02	0.11	0.09	0.20

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**14.4959**	**0.3656**	**14.4959**
A to ZN	**fall**	0.01	0.02	0.03	0.05
	rise	0.02	0.09	0.03	0.09
B to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.02	0.09	0.03	0.09
C1 to ZN	**fall**	0.00	0.03	0.03	0.06
	rise	0.01	0.08	0.04	0.10
C2 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.02	0.09	0.04	0.10

Capacitance [fF]	
A	1.5124
B	1.5394
C1	1.5413
C2	1.5494

Leakage [nW]
9.22

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	14.4959	0.3656	14.4959
A to ZN	fall	0.70	0.90	1.10	1.01
	rise	3.70	3.69	3.41	3.55
B to ZN	fall	0.60	0.70	0.89	0.78
	rise	3.17	3.21	2.93	3.06
C1 to ZN	fall	0.18	0.21	0.47	0.26
	rise	2.12	2.21	2.22	2.19
C2 to ZN	fall	0.19	0.20	0.24	0.17
	rise	2.60	2.59	2.72	2.78

AOI211_X2

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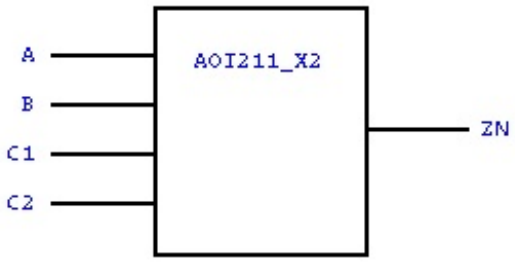
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((C1 & C2) | B) | A)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	L	L	-	H
L	L	-	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	28.9917	0.3656	28.9917
A to ZN	**fall**	0.01	0.03	0.05	0.10
	rise	0.04	0.12	0.11	0.21
B to ZN	**fall**	0.01	0.03	0.04	0.10
	rise	0.04	0.12	0.11	0.22
C1 to ZN	**fall**	0.01	0.04	0.03	0.11
	rise	0.02	0.10	0.08	0.20
C2 to ZN	**fall**	0.01	0.04	0.04	0.11
	rise	0.02	0.11	0.09	0.20

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	28.9917	0.3656	28.9917
A to ZN	**fall**	0.01	0.02	0.03	0.05
	rise	0.02	0.09	0.03	0.09
B to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.02	0.09	0.03	0.09
C1 to ZN	**fall**	0.00	0.03	0.03	0.06
	rise	0.01	0.08	0.04	0.10
C2 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.02	0.09	0.04	0.10

Capacitance [fF]	
A	2.8853
B	3.1886
C1	2.9273
C2	3.1781

Leakage [nW]
18.45

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	28.9917	0.3656	28.9917
A to ZN	fall	1.34	1.75	2.14	1.96
	rise	7.47	7.47	6.88	7.27
B to ZN	fall	1.14	1.33	1.72	1.48
	rise	6.44	6.49	5.94	6.28
C1 to ZN	fall	0.38	0.41	0.94	0.51
	rise	4.23	4.31	4.45	4.35
C2 to ZN	fall	0.38	0.41	0.49	0.34
	rise	5.21	5.16	5.41	5.45

AOI211_X4

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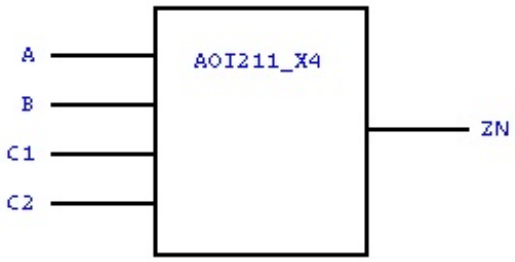
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	ZN = "!(!(!(((C1 & C2) | B) | A)))"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	L	L	-	H
L	L	-	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.03	0.12	0.09	0.18
	rise	0.09	0.20	0.17	0.29
B to ZN	**fall**	0.03	0.12	0.09	0.17
	rise	0.08	0.20	0.17	0.29
C1 to ZN	**fall**	0.04	0.12	0.09	0.17
	rise	0.07	0.18	0.15	0.27
C2 to ZN	**fall**	0.04	0.12	0.09	0.18
	rise	0.07	0.19	0.16	0.27

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.00	0.10
B to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
C1 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
C2 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A	1.5620
B	1.5781
C1	1.5269
C2	1.6090

Leakage [nW]
28.10

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	fall	7.69	9.51	7.69	9.54
	rise	10.57	12.04	10.11	11.70
B to ZN	fall	7.51	9.24	7.46	9.26
	rise	10.06	11.42	9.62	11.62
C1 to ZN	fall	6.92	8.62	6.84	8.75
	rise	9.05	10.95	8.89	11.20
C2 to ZN	fall	6.91	8.62	6.75	8.53
	rise	9.48	11.38	9.38	11.37

AOI21_X1

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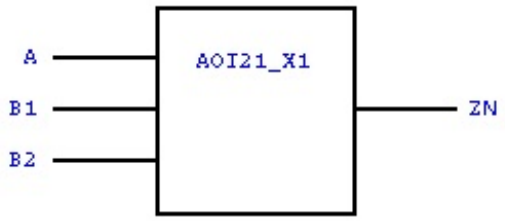
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!(A | (B1 & B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	L	-	H
L	-	L	H
H	-	-	L
-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to ZN	**fall**	0.01	0.04	0.04	0.12
	rise	0.02	0.12	0.09	0.22
B1 to ZN	**fall**	0.01	0.06	0.03	0.15
	rise	0.01	0.11	0.06	0.21
B2 to ZN	**fall**	0.01	0.06	0.04	0.14
	rise	0.02	0.11	0.08	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to ZN	**fall**	0.00	0.03	0.03	0.06
	rise	0.01	0.09	0.03	0.10
B1 to ZN	**fall**	0.00	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.10
B2 to ZN	**fall**	0.00	0.05	0.02	0.06
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A	1.5136
B1	1.5444
B2	1.5545

Leakage [nW]
6.84

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to ZN	**fall**	0.56	0.68	0.91	0.72
	rise	2.92	2.94	2.62	2.83
B1 to ZN	**fall**	0.17	0.21	0.50	0.19
	rise	1.85	1.92	1.97	1.88
B2 to ZN	**fall**	0.18	0.20	0.25	0.13
	rise	2.30	2.39	2.45	2.41

AOI21_X2

Databook Build Date: Thursday Feb 17 15:07 2011

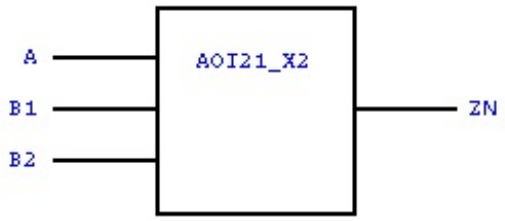
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!(A | (B1 & B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	L	-	H
L	-	L	H
H	-	-	L
-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**50.6592**	**0.3656**	**50.6592**
A to ZN	**fall**	0.01	0.04	0.04	0.12
	rise	0.02	0.12	0.09	0.22
B1 to ZN	**fall**	0.01	0.06	0.03	0.15
	rise	0.01	0.11	0.06	0.21
B2 to ZN	**fall**	0.01	0.06	0.04	0.14
	rise	0.02	0.11	0.07	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**50.6592**	**0.3656**	**50.6592**
A to ZN	**fall**	0.00	0.03	0.03	0.06
	rise	0.01	0.09	0.03	0.10
B1 to ZN	**fall**	0.00	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.10
B2 to ZN	**fall**	0.00	0.05	0.02	0.06
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A	2.8982
B1	2.9226
B2	3.2445

Leakage [nW]
13.68

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**50.6592**	**0.3656**	**50.6592**
A to ZN	**fall**	1.05	1.30	1.77	1.37
	rise	5.69	5.76	5.08	5.57
B1 to ZN	**fall**	0.34	0.42	1.01	0.38
	rise	3.70	3.67	3.92	3.69
B2 to ZN	**fall**	0.35	0.42	0.51	0.26
	rise	4.63	4.77	4.89	4.84

AOI21_X4

Databook Build Date: Thursday Feb 17 15:07 2011

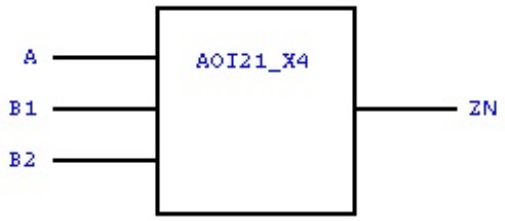
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(A | (B1 & B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	L	-	H
L	-	L	H
H	-	-	L
-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**101.013**	**0.3656**	**101.013**
A to ZN	**fall**	0.01	0.04	0.04	0.12
	rise	0.02	0.12	0.09	0.22
B1 to ZN	**fall**	0.01	0.06	0.03	0.15
	rise	0.01	0.11	0.06	0.21
B2 to ZN	**fall**	0.01	0.06	0.04	0.14
	rise	0.01	0.11	0.07	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**101.013**	**0.3656**	**101.013**
A to ZN	**fall**	0.00	0.03	0.03	0.06
	rise	0.01	0.09	0.02	0.10
B1 to ZN	**fall**	0.00	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.10
B2 to ZN	**fall**	0.00	0.05	0.02	0.06
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A	5.6692
B1	5.9917
B2	6.2276

Leakage [nW]
27.36

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**101.013**	**0.3656**	**101.013**
A to ZN	**fall**	2.08	2.59	3.53	2.72
	rise	11.62	11.47	10.44	11.15
B1 to ZN	**fall**	0.68	0.83	2.04	0.76
	rise	7.45	7.32	7.90	7.60
B2 to ZN	**fall**	0.71	0.83	1.02	0.52
	rise	9.27	9.51	9.81	9.55

AOI221_X1

Databook Build Date: Thursday Feb 17 15:07 2011

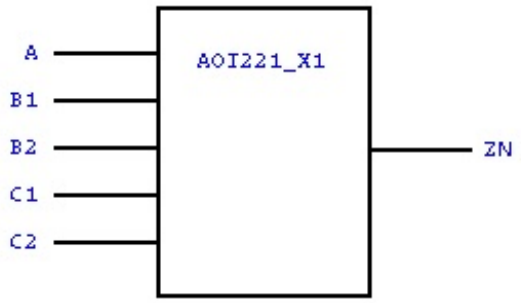
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "!(((C1 & C2) | A) | (B1 & B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	L	-	H
L	L	-	-	L	H
L	-	L	L	-	H
L	-	L	-	L	H
H	-	-	-	-	L
-	H	H	-	-	L
-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	13.8092	0.3656	13.8092
A to ZN	**fall**	0.01	0.03	0.04	0.10
	rise	0.04	0.12	0.12	0.22
B1 to ZN	**fall**	0.01	0.05	0.06	0.12
	rise	0.04	0.12	0.12	0.21
B2 to ZN	**fall**	0.02	0.05	0.07	0.12
	rise	0.05	0.12	0.13	0.22
C1 to ZN	**fall**	0.01	0.04	0.03	0.11
	rise	0.02	0.10	0.09	0.20
C2 to ZN	**fall**	0.01	0.04	0.04	0.11
	rise	0.03	0.11	0.10	0.21

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	13.8092	0.3656	13.8092
A to ZN	**fall**	0.01	0.02	0.04	0.05
	rise	0.02	0.09	0.03	0.09
B1 to ZN	**fall**	0.01	0.03	0.04	0.06
	rise	0.02	0.08	0.03	0.09
B2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.02	0.09	0.03	0.09
C1 to ZN	**fall**	0.01	0.03	0.04	0.06
	rise	0.02	0.08	0.04	0.09
C2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.02	0.09	0.04	0.10

Capacitance [fF]	
A	1.5518
B1	1.5116
B2	1.5472
C1	1.5206
C2	1.5766

Leakage [nW]
11.09

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	13.8092	0.3656	13.8092
A to ZN	fall	0.55	0.62	0.84	0.71
	rise	3.34	3.35	3.11	3.27
B1 to ZN	fall	0.85	1.03	1.15	1.07
	rise	3.96	3.97	3.62	3.78
B2 to ZN	fall	0.84	1.03	0.98	1.00
	rise	4.41	4.47	4.15	4.33
C1 to ZN	fall	0.19	0.21	0.47	0.26
	rise	2.29	2.32	2.38	2.36
C2 to ZN	fall	0.19	0.21	0.24	0.17
	rise	2.76	2.80	2.87	2.87

AOI221_X2

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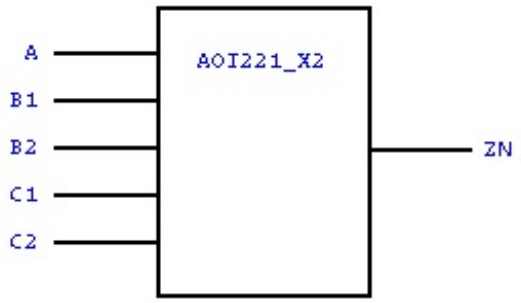
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.926 um^2
Equation	ZN = "!(((C1 & C2) | A) | (B1 & B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	L	-	H
L	L	-	-	L	H
L	-	L	L	-	H
L	-	L	-	L	H
H	-	-	-	-	L
-	H	H	-	-	L
-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**27.6184**	**0.3656**	**27.6184**
A to ZN	**fall**	0.01	0.03	0.04	0.10
	rise	0.04	0.12	0.12	0.22
B1 to ZN	**fall**	0.01	0.05	0.06	0.12
	rise	0.04	0.12	0.12	0.21
B2 to ZN	**fall**	0.02	0.05	0.07	0.12
	rise	0.05	0.12	0.13	0.22
C1 to ZN	**fall**	0.01	0.04	0.03	0.11
	rise	0.02	0.10	0.09	0.20
C2 to ZN	**fall**	0.01	0.04	0.04	0.11
	rise	0.03	0.11	0.10	0.21

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**27.6184**	**0.3656**	**27.6184**
A to ZN	**fall**	0.00	0.02	0.04	0.05
	rise	0.02	0.09	0.03	0.09
B1 to ZN	**fall**	0.01	0.03	0.04	0.06
	rise	0.02	0.08	0.03	0.09
B2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.02	0.09	0.03	0.09
C1 to ZN	**fall**	0.01	0.03	0.04	0.06
	rise	0.02	0.08	0.04	0.09
C2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.02	0.09	0.04	0.10

Capacitance [fF]	
A	3.2635
B1	3.0816
B2	2.9689
C1	2.9126
C2	3.2399

Leakage [nW]
22.18

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	27.6184	0.3656	27.6184
A to ZN	fall	1.14	1.31	1.73	1.49
	rise	6.70	6.72	6.23	6.56
B1 to ZN	fall	1.70	2.07	2.31	2.15
	rise	7.87	7.84	7.21	7.59
B2 to ZN	fall	1.70	2.07	1.98	2.00
	rise	8.78	8.78	8.27	8.57
C1 to ZN	fall	0.37	0.41	0.94	0.51
	rise	4.59	4.76	4.79	4.84
C2 to ZN	fall	0.38	0.41	0.49	0.33
	rise	5.54	5.59	5.77	5.82

AOI221_X4

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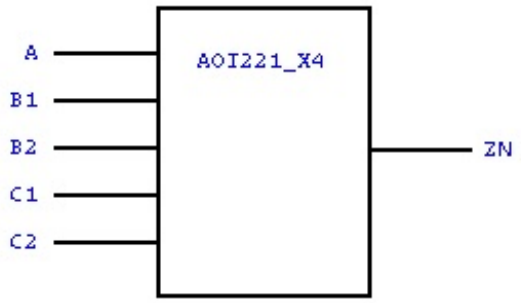
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(!(!(((C1 & C2) | A) | (B1 & B2))))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	L	-	H
L	L	-	-	L	H
L	-	L	L	-	H
L	-	L	-	L	H
H	-	-	-	-	L
-	H	H	-	-	L
-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	**fall**	0.03	0.12	0.09	0.18
	rise	0.09	0.21	0.18	0.30
B1 to ZN	**fall**	0.05	0.13	0.11	0.20
	rise	0.09	0.20	0.18	0.29
B2 to ZN	**fall**	0.05	0.13	0.11	0.20
	rise	0.09	0.21	0.18	0.30
C1 to ZN	**fall**	0.04	0.12	0.10	0.18
	rise	0.07	0.19	0.16	0.28
C2 to ZN	**fall**	0.04	0.12	0.10	0.18
	rise	0.08	0.19	0.17	0.28

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
B1 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.00	0.10
B2 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.00	0.10
C1 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
C2 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A	1.5257
B1	1.4837
B2	1.5714
C1	1.4978
C2	1.5480

Leakage [nW]
29.30

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	fall	7.54	9.25	7.47	9.28
	rise	10.25	11.87	9.81	11.64
B1 to ZN	fall	7.70	9.67	7.71	9.72
	rise	10.87	12.78	10.39	12.31
B2 to ZN	fall	7.70	9.67	7.63	9.58
	rise	11.30	13.03	10.95	12.97
C1 to ZN	fall	6.92	8.63	6.83	8.76
	rise	9.31	10.80	9.10	10.79
C2 to ZN	fall	6.92	8.64	6.74	8.54
	rise	9.70	11.34	9.66	11.28

AOI222_X1

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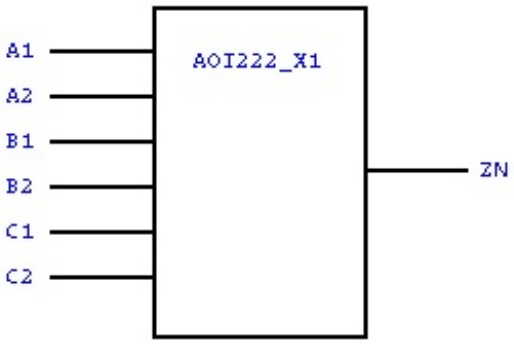
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.128 um^2
Equation	ZN = "!(((A1 & A2) | (B1 & B2)) | (C1 & C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	L	-	L	-	H
L	-	L	-	-	L	H
L	-	-	L	L	-	H
L	-	-	L	-	L	H
H	H	-	-	-	-	L
-	L	L	-	L	-	H
-	L	L	-	-	L	H
-	L	-	L	L	-	H
-	L	-	L	-	L	H
-	-	H	H	-	-	L
-	-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**13.0081**	**0.3656**	**13.0081**
A1 to ZN	**fall**	0.01	0.04	0.03	0.11
	rise	0.03	0.10	0.10	0.20
A2 to ZN	**fall**	0.01	0.04	0.04	0.11
	rise	0.03	0.11	0.11	0.21
B1 to ZN	**fall**	0.01	0.04	0.06	0.12
	rise	0.05	0.12	0.13	0.22
B2 to ZN	**fall**	0.02	0.04	0.06	0.12
	rise	0.05	0.12	0.14	0.22
C1 to ZN	**fall**	0.02	0.05	0.07	0.13
	rise	0.05	0.12	0.14	0.22
C2 to ZN	**fall**	0.02	0.05	0.07	0.12
	rise	0.06	0.13	0.14	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**13.0081**	**0.3656**	**13.0081**
A1 to ZN	**fall**	0.01	0.03	0.04	0.06
	rise	0.02	0.08	0.04	0.09
A2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.02	0.09	0.04	0.09
B1 to ZN	**fall**	0.01	0.03	0.04	0.06
	rise	0.02	0.08	0.03	0.09
B2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.02	0.09	0.03	0.09
C1 to ZN	**fall**	0.01	0.04	0.04	0.06
	rise	0.02	0.08	0.03	0.09
C2 to ZN	**fall**	0.01	0.04	0.03	0.05
	rise	0.02	0.09	0.03	0.09

Capacitance [fF]	
A1	1.5252
A2	1.5663
B1	1.5006
B2	1.5324
C1	1.4790
C2	1.5098

Leakage [nW]
12.51

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	13.0081	0.3656	13.0081
A1 to ZN	fall	0.19	0.20	0.47	0.26
	rise	2.42	2.49	2.53	2.56
A2 to ZN	fall	0.19	0.20	0.24	0.16
	rise	2.89	2.96	3.03	3.09
B1 to ZN	fall	0.72	0.81	0.96	0.85
	rise	3.60	3.66	3.32	3.48
B2 to ZN	fall	0.72	0.81	0.78	0.77
	rise	4.05	4.08	3.85	4.01
C1 to ZN	fall	1.05	1.29	1.37	1.34
	rise	4.56	4.62	4.25	4.41
C2 to ZN	fall	1.05	1.28	1.21	1.25
	rise	5.01	5.07	4.78	4.95

AOI222_X2

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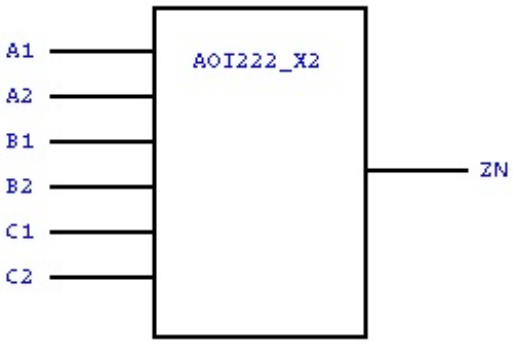
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	3.724 um^2
Equation	ZN = "!(((A1 & A2) | (B1 & B2)) | (C1 & C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	L	-	L	-	H
L	-	L	-	-	L	H
L	-	-	L	L	-	H
L	-	-	L	-	L	H
H	H	-	-	-	-	L
-	L	L	-	L	-	H
-	L	L	-	-	L	H
-	L	-	L	L	-	H
-	L	-	L	-	L	H
-	-	H	H	-	-	L
-	-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**25.6348**	**0.3656**	**25.6348**
A1 to ZN	**fall**	0.01	0.04	0.03	0.11
	rise	0.03	0.10	0.10	0.20
A2 to ZN	**fall**	0.01	0.04	0.04	0.11
	rise	0.03	0.11	0.11	0.21
B1 to ZN	**fall**	0.01	0.04	0.06	0.12
	rise	0.05	0.12	0.13	0.22
B2 to ZN	**fall**	0.02	0.04	0.06	0.12
	rise	0.05	0.12	0.14	0.22
C1 to ZN	**fall**	0.02	0.05	0.07	0.13
	rise	0.05	0.13	0.14	0.22
C2 to ZN	**fall**	0.02	0.05	0.07	0.12
	rise	0.06	0.13	0.14	0.23

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**25.6348**	**0.3656**	**25.6348**
A1 to ZN	**fall**	0.01	0.03	0.04	0.06
	rise	0.02	0.08	0.04	0.09
A2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.02	0.09	0.04	0.09
B1 to ZN	**fall**	0.01	0.03	0.04	0.06
	rise	0.02	0.08	0.03	0.09
B2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.02	0.09	0.03	0.09
C1 to ZN	**fall**	0.01	0.04	0.04	0.06
	rise	0.02	0.08	0.03	0.09
C2 to ZN	**fall**	0.01	0.04	0.03	0.05
	rise	0.02	0.09	0.03	0.09

Capacitance [fF]	
A1	2.9373
A2	2.9515
B1	2.8627
B2	3.2254
C1	2.8279
C2	3.1346

Leakage [nW]
25.01

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	25.6348	0.3656	25.6348
A1 to ZN	fall	0.37	0.40	0.94	0.52
	rise	4.99	5.13	5.22	5.27
A2 to ZN	fall	0.38	0.40	0.46	0.32
	rise	6.05	6.18	6.33	6.38
B1 to ZN	fall	1.44	1.62	1.92	1.70
	rise	7.35	7.42	6.79	7.11
B2 to ZN	fall	1.44	1.62	1.57	1.54
	rise	8.25	8.37	7.84	8.17
C1 to ZN	fall	2.10	2.57	2.75	2.67
	rise	9.33	9.43	8.70	9.04
C2 to ZN	fall	2.10	2.58	2.43	2.51
	rise	10.23	10.34	9.77	10.10

AOI222_X4

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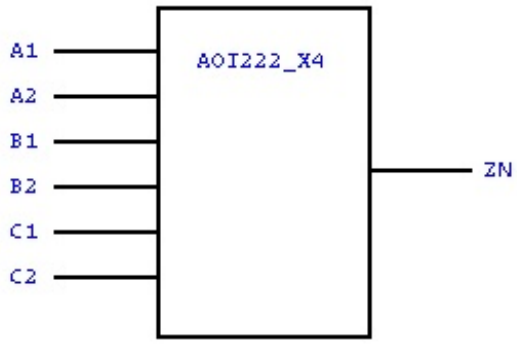
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.724 um^2
Equation	ZN = "!(!(!(((A1 & A2) | (B1 & B2)) | (C1 & C2))))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	L	-	L	-	H
L	-	L	-	-	L	H
L	-	-	L	L	-	H
L	-	-	L	-	L	H
H	H	-	-	-	-	L
-	L	L	-	L	-	H
-	L	L	-	-	L	H
-	L	-	L	L	-	H
-	L	-	L	-	L	H
-	-	H	H	-	-	L
-	-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.04	0.12	0.10	0.18
	rise	0.08	0.19	0.17	0.28
A2 to ZN	**fall**	0.04	0.13	0.10	0.18
	rise	0.08	0.20	0.18	0.29
B1 to ZN	**fall**	0.05	0.13	0.11	0.20
	rise	0.09	0.21	0.19	0.30
B2 to ZN	**fall**	0.05	0.13	0.11	0.20
	rise	0.10	0.21	0.19	0.31
C1 to ZN	**fall**	0.05	0.13	0.12	0.20
	rise	0.10	0.21	0.19	0.30
C2 to ZN	**fall**	0.05	0.13	0.12	0.20
	rise	0.10	0.22	0.20	0.31

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
A2 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
B1 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
B2 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
C1 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.00	0.10
C2 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.00	0.10

Capacitance [fF]	
A1	1.4938
A2	1.5760
B1	1.5190
B2	1.5887
C1	1.5117
C2	1.5748

Leakage [nW]
29.71

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	fall	6.88	8.62	6.78	8.75
	rise	9.48	11.13	9.28	11.16
A2 to ZN	fall	6.88	8.62	6.70	8.53
	rise	9.87	11.68	9.78	11.59
B1 to ZN	fall	7.46	9.33	7.42	9.39
	rise	10.43	11.96	9.93	11.75
B2 to ZN	fall	7.45	9.32	7.30	9.22
	rise	10.88	12.60	10.50	12.40
C1 to ZN	fall	7.83	9.90	7.87	9.94
	rise	11.39	13.14	10.90	12.88
C2 to ZN	fall	7.84	9.90	7.78	9.81
	rise	11.77	13.73	11.43	13.39

AOI22_X1

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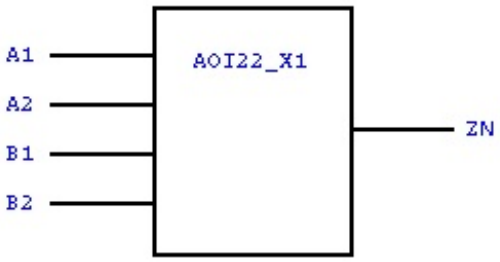
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!((A1 & A2) | (B1 & B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**24.6048**	**0.3656**	**24.6048**
A1 to ZN	**fall**	0.01	0.06	0.03	0.14
	rise	0.02	0.11	0.08	0.21
A2 to ZN	**fall**	0.01	0.06	0.04	0.14
	rise	0.02	0.11	0.09	0.22
B1 to ZN	**fall**	0.01	0.06	0.05	0.15
	rise	0.02	0.11	0.10	0.22
B2 to ZN	**fall**	0.01	0.07	0.06	0.15
	rise	0.03	0.12	0.10	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**24.6048**	**0.3656**	**24.6048**
A1 to ZN	**fall**	0.00	0.05	0.04	0.07
	rise	0.01	0.09	0.03	0.10
A2 to ZN	**fall**	0.00	0.05	0.03	0.06
	rise	0.01	0.09	0.03	0.10
B1 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.09
B2 to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A1	1.5867
A2	1.5681
B1	1.5031
B2	1.5336

Leakage [nW]
8.13

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	24.6048	0.3656	24.6048
A1 to ZN	fall	0.16	0.20	0.50	0.19
	rise	2.02	2.10	2.13	2.08
A2 to ZN	fall	0.18	0.20	0.25	0.12
	rise	2.49	2.56	2.61	2.63
B1 to ZN	fall	0.69	0.81	0.97	0.80
	rise	3.04	3.06	2.68	2.92
B2 to ZN	fall	0.69	0.82	0.78	0.74
	rise	3.47	3.51	3.20	3.42

AOI22_X2

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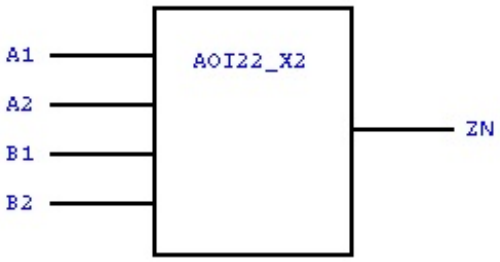
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!((A1 & A2) | (B1 & B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**49.1333**	**0.3656**	**49.1333**
A1 to ZN	**fall**	0.01	0.06	0.03	0.14
	rise	0.02	0.11	0.07	0.21
A2 to ZN	**fall**	0.01	0.06	0.04	0.14
	rise	0.02	0.11	0.08	0.22
B1 to ZN	**fall**	0.01	0.07	0.05	0.15
	rise	0.02	0.12	0.10	0.22
B2 to ZN	**fall**	0.01	0.07	0.06	0.15
	rise	0.03	0.12	0.10	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**49.1333**	**0.3656**	**49.1333**
A1 to ZN	**fall**	0.00	0.05	0.04	0.07
	rise	0.01	0.09	0.03	0.10
A2 to ZN	**fall**	0.00	0.05	0.03	0.06
	rise	0.01	0.09	0.03	0.10
B1 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.02	0.09
B2 to ZN	**fall**	0.01	0.05	0.03	0.06
	rise	0.01	0.09	0.02	0.10

Capacitance [fF]	
A1	2.9409
A2	3.2363
B1	2.8359
B2	3.2480

Leakage [nW]
16.26

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	49.1333	0.3656	49.1333
A1 to ZN	fall	0.33	0.40	1.02	0.38
	rise	4.07	4.15	4.31	4.16
A2 to ZN	fall	0.34	0.40	0.50	0.25
	rise	4.99	5.08	5.26	5.27
B1 to ZN	fall	1.36	1.62	1.94	1.58
	rise	6.30	6.38	5.58	6.01
B2 to ZN	fall	1.37	1.63	1.55	1.47
	rise	7.18	7.21	6.62	7.10

AOI22_X4

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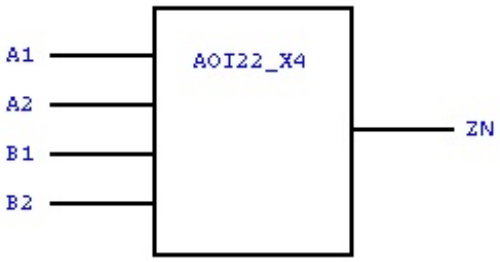
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Equation	ZN = "!((A1 & A2) | (B1 & B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**97.9614**	**0.3656**	**97.9614**
A1 to ZN	**fall**	0.01	0.06	0.03	0.14
	rise	0.02	0.11	0.07	0.22
A2 to ZN	**fall**	0.01	0.06	0.04	0.14
	rise	0.02	0.11	0.08	0.22
B1 to ZN	**fall**	0.01	0.06	0.05	0.15
	rise	0.02	0.12	0.09	0.22
B2 to ZN	**fall**	0.01	0.07	0.06	0.15
	rise	0.03	0.12	0.10	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**97.9614**	**0.3656**	**97.9614**
A1 to ZN	**fall**	0.00	0.05	0.04	0.07
	rise	0.01	0.09	0.03	0.10
A2 to ZN	**fall**	0.00	0.05	0.03	0.06
	rise	0.01	0.09	0.03	0.10
B1 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.02	0.09
B2 to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.01	0.09	0.02	0.10

Capacitance [fF]	
A1	6.0125
A2	6.2927
B1	5.7848
B2	6.2205

Leakage [nW]
32.51

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	97.9614	0.3656	97.9614
A1 to ZN	fall	0.66	0.80	2.05	0.75
	rise	8.17	8.12	8.59	8.17
A2 to ZN	fall	0.69	0.80	1.02	0.50
	rise	10.00	10.06	10.54	10.60
B1 to ZN	fall	2.72	3.25	3.90	3.15
	rise	12.38	12.41	11.01	11.74
B2 to ZN	fall	2.73	3.26	3.11	2.94
	rise	14.14	14.38	13.09	13.64

BUF_X16

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	6.650 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	965.576	0.3656	965.576
A to Z	**fall**	0.02	0.10	0.08	0.17
	rise	0.01	0.13	0.05	0.17

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	965.576	0.3656	965.576
A to Z	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.11	0.01	0.11

Capacitance [fF]	
A	11.2325

Leakage [nW]
66.61

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	965.576	0.3656	965.576
A to Z	**fall**	27.63	34.08	26.41	36.81
	rise	17.60	17.85	18.20	21.10

BUF_X1

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
A to Z	**fall**	0.02	0.10	0.09	0.18
	rise	0.02	0.13	0.06	0.17

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
A to Z	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A	0.8929

Leakage [nW]
4.15

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
A to Z	**fall**	1.94	2.36	1.78	2.50
	rise	1.22	1.22	1.23	1.14

BUF_X2

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.064 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A to Z	**fall**	0.02	0.10	0.08	0.17
	rise	0.01	0.13	0.05	0.17

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A to Z	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A	1.6267

Leakage [nW]
8.33

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A to Z	**fall**	3.71	4.51	3.46	4.80
	rise	2.43	2.69	2.42	2.91

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BUF_X32

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	32
Cell Area	13.034 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	1904.30	0.3656	1904.30
A to Z	**fall**	0.02	0.10	0.08	0.17
	rise	0.01	0.14	0.05	0.18

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	1904.30	0.3656	1904.30
A to Z	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.11	0.01	0.11

Capacitance [fF]	
A	24.1263

Leakage [nW]
133.22

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	1904.30	0.3656	1904.30
A to Z	**fall**	54.37	66.93	52.56	72.81
	rise	34.36	30.43	36.49	38.49

BUF_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	1.862 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to Z	**fall**	0.02	0.10	0.08	0.17
	rise	0.01	0.13	0.05	0.17

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to Z	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A	3.0762

Leakage [nW]
16.65

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to Z	**fall**	7.01	8.56	6.61	9.21
	rise	4.54	5.50	4.68	5.72

BUF_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	3.458 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	484.009	0.3656	484.009
A to Z	**fall**	0.02	0.10	0.08	0.17
	rise	0.01	0.13	0.05	0.17

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	484.009	0.3656	484.009
A to Z	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A	5.9752

Leakage [nW]
33.30

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	484.009	0.3656	484.009
A to Z	**fall**	13.80	16.94	13.20	18.32
	rise	8.86	9.94	9.28	10.20

CLKBUF_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
A to Z	**fall**	0.02	0.19	0.08	0.26
	rise	0.02	0.14	0.08	0.20

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
A to Z	**fall**	0.00	0.15	0.01	0.15
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A	0.7348

Leakage [nW]
1.90

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
A to Z	**fall**	1.69	1.91	1.47	1.98
	rise	0.95	1.04	0.93	1.06

CLKBUF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.064 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to Z	**fall**	0.02	0.19	0.08	0.26
	rise	0.02	0.14	0.07	0.19

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to Z	**fall**	0.00	0.15	0.01	0.15
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A	1.3095

Leakage [nW]
3.89

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to Z	**fall**	3.21	3.64	2.83	3.80
	rise	1.90	2.11	1.83	1.68

CLKBUF_X3

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	3
Cell Area	1.330 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	181.885	0.3656	181.885
A to Z	**fall**	0.02	0.19	0.09	0.27
	rise	0.02	0.14	0.08	0.21

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	181.885	0.3656	181.885
A to Z	**fall**	0.00	0.15	0.01	0.15
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A	1.3308

Leakage [nW]
5.19

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	181.885	0.3656	181.885
A to Z	**fall**	3.86	4.64	3.49	4.73
	rise	2.51	3.23	2.52	2.92

NANGATE CLKGATETST_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	3.990 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	0.02	0.19	0.09	0.27
	rise	0.02	0.13	0.06	0.18

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	0.00	0.15	0.01	0.15
	rise	0.00	0.10	0.01	0.10

Constraints Time [ns]		
Setup CK to E	**fall**	0.13
	rise	0.11
Hold CK to E	**fall**	0.09
	rise	0.05
Setup CK to SE	**fall**	0.13
	rise	0.10
Hold CK to SE	**fall**	0.09
	rise	0.06

Capacitance [fF]	
CK	1.6692
E	0.8354
SE	0.7366

Leakage [nW]
12.53

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	4554.38	4723.84
	rise	1476.19	1581.35
E	fall	1467.04	1463.16
	rise	740.71	771.68
SE	fall	1223.48	1301.40
	rise	654.14	692.44

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to GCK	fall	3.93	4.92	3.63	5.19
	rise	2.38	2.76	2.68	3.18

NANGATE CLKGATETST_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	4.256 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
CK to GCK	**fall**	0.02	0.19	0.09	0.26
	rise	0.02	0.13	0.06	0.18

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
CK to GCK	**fall**	0.00	0.15	0.01	0.15
	rise	0.00	0.10	0.01	0.10

Constraints Time [ns]		
Setup CK to E	**fall**	0.13
	rise	0.11
Hold CK to E	**fall**	0.08
	rise	0.06
Setup CK to SE	**fall**	0.12
	rise	0.10
Hold CK to SE	**fall**	0.09
	rise	0.07

Capacitance [fF]	
CK	2.6160
E	0.8306
SE	0.7752

Leakage [nW]
16.41

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	5207.17	5377.03
	rise	885.41	991.36
E	fall	1454.85	1451.97
	rise	673.28	705.34
SE	fall	1210.92	1291.65
	rise	597.16	639.99

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
CK to GCK	fall	5.61	7.09	5.02	7.48
	rise	3.13	3.89	3.22	4.14

NANGATE CLKGATETST_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	5.320 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.92	0.3656	242.92
CK to GCK	**fall**	0.02	0.19	0.09	0.27
	rise	0.02	0.14	0.06	0.18

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.92	0.3656	242.92
CK to GCK	**fall**	0.00	0.15	0.01	0.15
	rise	0.00	0.10	0.01	0.10

Constraints Time [ns]		
Setup CK to E	**fall**	0.14
	rise	0.15
Hold CK to E	**fall**	0.06
	rise	0.04
Setup CK to SE	**fall**	0.14
	rise	0.14
Hold CK to SE	**fall**	0.07
	rise	0.05

Capacitance [fF]	
CK	4.0635
E	0.8886
SE	0.7749

Leakage [nW]
26.54

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	7049.80	7123.18
	rise	101.65	201.08
E	fall	1868.07	1863.77
	rise	858.31	884.98
SE	fall	1625.12	1700.05
	rise	814.64	849.27

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.92	0.3656	242.92
CK to GCK	fall	9.56	11.70	8.93	12.32
	rise	5.68	6.53	5.57	6.00

NANGATE CLKGATETST_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	7.714 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	484.619	0.3656	484.619
CK to GCK	**fall**	0.02	0.20	0.09	0.27
	rise	0.02	0.14	0.06	0.19

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	484.619	0.3656	484.619
CK to GCK	**fall**	0.00	0.15	0.01	0.15
	rise	0.00	0.10	0.01	0.11

Constraints Time [ns]		
Setup CK to E	**fall**	0.15
	rise	0.20
Hold CK to E	**fall**	0.02
	rise	0.04
Setup CK to SE	**fall**	0.15
	rise	0.19
Hold CK to SE	**fall**	0.03
	rise	0.04

Capacitance [fF]	
CK	7.2988
E	0.8584
SE	0.7653

Leakage [nW]
45.89

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	10868.49	10837.22
	rise	-353.03	-366.95
E	fall	2362.54	2360.05
	rise	1216.04	1214.15
SE	fall	2119.46	2196.71
	rise	1202.97	1239.08

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	484.619	0.3656	484.619
CK to GCK	fall	17.02	20.76	15.93	21.89
	rise	10.06	10.78	9.69	10.16

NANGATE CLKGATE_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	3.458 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	0.02	0.19	0.09	0.27
	rise	0.02	0.14	0.06	0.18

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	0.00	0.15	0.01	0.15
	rise	0.00	0.10	0.01	0.10

Constraints Time [ns]		
Setup CK to E	**fall**	0.11
	rise	0.10
Hold CK to E	**fall**	0.09
	rise	0.10

Capacitance [fF]	
CK	1.7048
E	0.8707

Leakage [nW]
10.20

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1792**
CK	**fall**	3586.82	3767.30
	rise	962.63	1079.93
E	**fall**	1093.68	1088.94
	rise	601.01	588.31

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	3.85	5.11	3.60	5.40
	rise	2.29	2.56	2.46	2.61

NANGATE CLKGATE_X2

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	3.724 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**121.307**	**0.3656**	**121.307**
CK to GCK	**fall**	0.02	0.19	0.09	0.27
	rise	0.02	0.13	0.06	0.18

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**121.307**	**0.3656**	**121.307**
CK to GCK	**fall**	0.00	0.15	0.01	0.15
	rise	0.00	0.10	0.01	0.10

Constraints Time [ns]		
Setup CK to E	**fall**	0.12
	rise	0.11
Hold CK to E	**fall**	0.09
	rise	0.09

Capacitance [fF]	
CK	2.3652
E	0.8500

Leakage [nW]
13.89

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1792**
CK	**fall**	4504.06	4676.05
	rise	801.41	912.95
E	**fall**	1232.01	1223.41
	rise	865.35	847.97

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**121.307**	**0.3656**	**121.307**
CK to GCK	**fall**	5.56	7.38	5.11	7.79
	rise	3.37	3.52	3.56	4.29

NANGATE CLKGATE_X4

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
CK to GCK	**fall**	0.02	0.19	0.09	0.26
	rise	0.02	0.13	0.06	0.18

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
CK to GCK	**fall**	0.00	0.15	0.01	0.15
	rise	0.00	0.10	0.01	0.10

Constraints Time [ns]		
Setup CK to E	**fall**	0.13
	rise	0.12
Hold CK to E	**fall**	0.08
	rise	0.09

Capacitance [fF]	
CK	3.9448
E	0.8421

Leakage [nW]
22.62

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1792**
CK	**fall**	5996.77	6179.27
	rise	204.20	310.46
E	**fall**	1724.36	1714.24
	rise	1217.76	1199.92

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
CK to GCK	**fall**	8.71	11.82	7.96	12.51
	rise	5.44	6.11	5.53	6.41

NANGATE CLKGATE_X8

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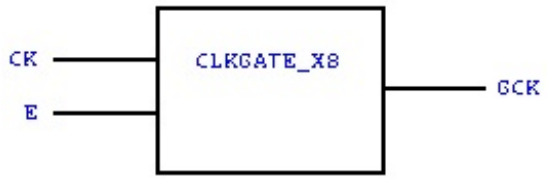
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	6.916 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	0.02	0.19	0.09	0.27
	rise	0.02	0.14	0.06	0.18

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	0.00	0.15	0.01	0.15
	rise	0.00	0.10	0.01	0.11

Constraints Time [ns]		
Setup CK to E	**fall**	0.13
	rise	0.12
Hold CK to E	**fall**	0.07
	rise	0.09

Capacitance [fF]	
CK	7.1487
E	1.1091

Leakage [nW]
40.60

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1792**
CK	**fall**	9275.99	9461.97
	rise	-398.35	-412.28
E	**fall**	2860.33	2852.29
	rise	1819.66	1803.59

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	15.41	21.35	14.03	22.56
	rise	9.43	10.12	9.50	10.65

DFFRS_X1

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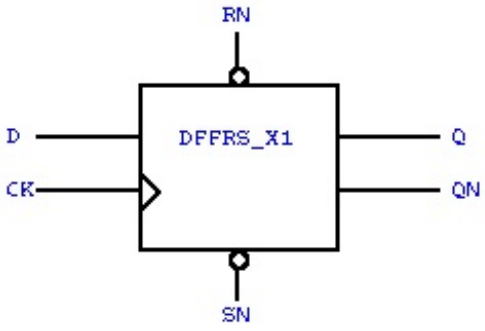
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, and drive strength X1
Strength	1
Cell Area	6.384 um^2
Equation	Q = "(!(!RN)*((D)+(!SN)))" QN = "(!(!SN)*(!(D)+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	RN	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	H	-	-	L	H
R	H	H	H	-	-	H	L
F	-	H	H	L	H	L	H
F	-	H	H	H	L	H	L
-	-	L	L	-	-	L	L
-	-	L	H	-	-	L	H
-	-	H	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.08	0.16	0.13	0.21
	rise	0.07	0.19	0.12	0.24
CK to QN	fall	0.09	*0.17*	0.14	*0.22*
	rise	0.10	*0.22*	0.15	*0.26*
RN to Q	fall	0.02	0.11	0.10	0.20
	rise	0.14	0.25	0.21	0.32
RN to QN	rise	0.05	*0.16*	0.14	*0.25*
SN to Q	rise	0.15	0.27	0.24	0.36
SN to QN	fall	0.02	*0.10*	0.10	*0.19*
	rise	0.03	*0.14*	0.08	*0.20*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.00	0.07	0.00	0.07
	rise	0.01	0.10	0.01	0.10
CK to QN	fall	0.00	*0.07*	0.00	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*
RN to Q	fall	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
RN to QN	rise	0.01	*0.10*	0.01	*0.10*
SN to Q	rise	0.01	0.10	0.01	0.10
SN to QN	fall	0.00	*0.07*	0.01	*0.07*
	rise	0.00	*0.10*	0.01	*0.10*

Constraints Time [ns]		
Setup CK to D	fall	0.11
	rise	0.09
Hold CK to D	fall	0.11
	rise	0.12
Recovery CK to RN	rise	0.09
Removal CK to RN	rise	0.24
Recovery CK to SN	rise	0.00
Removal CK to SN	rise	0.44

Capacitance [fF]	
CK	0.8836
D	1.0900
RN	1.3555
SN	2.1008

Leakage [nW]
22.66

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	4132.66	4284.56
	rise	4871.18	4943.94
D	fall	1752.62	1741.43
	rise	988.60	961.27
RN	fall	491.11	502.86
	rise	-214.02	-221.49
SN	fall	1143.90	1173.59
	rise	-116.36	-94.65

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	4.12	4.34	4.18	4.36
	rise	4.11	4.41	4.15	4.45
CK to QN	fall	4.11	*4.39*	4.15	*4.40*
	rise	4.12	*4.36*	4.18	*4.42*
RN to Q	fall	2.56	4.01	2.40	3.97
	rise	2.97	3.20	2.95	3.16
RN to QN	rise	2.61	*3.71*	2.54	*3.71*
SN to Q	rise	3.93	4.16	4.07	4.31
SN to QN	fall	3.79	*4.39*	3.74	*4.53*
	rise	1.83	*2.12*	1.89	*2.43*

DFFRS_X2

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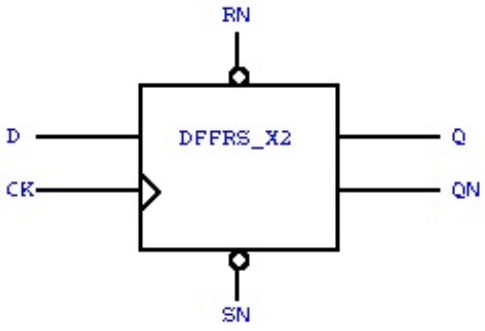
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, and drive strength X2
Strength	2
Cell Area	6.916 um^2
Equation	Q = "(!(!RN)*((D)+(!SN)))" QN = "(!(!SN)*(!(D)+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	RN	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	H	-	-	L	H
R	H	H	H	-	-	H	L
F	-	H	H	L	H	L	H
F	-	H	H	H	L	H	L
-	-	L	L	-	-	L	L
-	-	L	H	-	-	L	H
-	-	H	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.08	0.16	0.13	0.21
	rise	0.07	0.19	0.12	0.24
CK to QN	fall	0.10	0.18	0.15	0.23
	rise	0.10	0.22	0.15	0.27
RN to Q	fall	0.03	0.11	0.11	0.20
	rise	0.14	0.26	0.21	0.33
RN to QN	rise	0.05	0.17	0.15	0.26
SN to Q	rise	0.15	0.27	0.25	0.37
SN to QN	fall	0.02	0.11	0.10	0.19
	rise	0.03	0.14	0.08	0.20

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.00	0.07	0.00	0.07
	rise	0.01	0.10	0.01	0.10
CK to QN	fall	0.00	0.07	0.00	0.07
	rise	0.01	0.10	0.01	0.10
RN to Q	fall	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
RN to QN	rise	0.01	0.10	0.01	0.10
SN to Q	rise	0.01	0.10	0.01	0.10
SN to QN	fall	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Constraints Time [ns]		
Setup CK to D	fall	0.12
	rise	0.09
Hold CK to D	fall	0.11
	rise	0.12
Recovery CK to RN	rise	0.06
Removal CK to RN	rise	0.25
Recovery CK to SN	rise	0.00
Removal CK to SN	rise	0.45

Capacitance [fF]	
CK	0.8594
D	1.1029
RN	1.7669
SN	2.4719

Leakage [nW]
31.65

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	4070.48	4225.33
	rise	5292.33	5365.91
D	fall	1772.35	1761.19
	rise	918.20	889.60
RN	fall	679.98	654.97
	rise	-294.79	-305.41
SN	fall	1335.54	1354.40
	rise	-229.18	-208.42

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	5.41	5.87	5.47	5.90
	rise	5.47	6.15	5.52	6.06
CK to QN	fall	5.47	6.15	5.52	6.06
	rise	5.41	5.87	5.47	5.90
RN to Q	fall	4.07	5.98	3.96	6.00
	rise	4.32	5.12	4.30	4.79
RN to QN	rise	4.16	5.61	4.17	5.61
SN to Q	rise	5.39	6.02	5.63	6.06
SN to QN	fall	5.10	6.09	5.11	6.24
	rise	2.89	3.80	2.93	3.90

DFFR_X1

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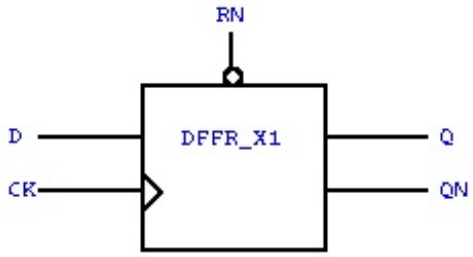
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, and drive strength X1
Strength	1
Cell Area	5.320 um^2
Equation	Q = "(!(!RN)*(D))" QN = "((!RN)+!(D))"
Clock	CK
Reset	RN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	RN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.08	0.16	0.13	0.21
	rise	0.09	0.20	0.14	0.25
CK to QN	fall	0.05	*0.15*	0.10	*0.20*
	rise	0.06	*0.18*	0.11	*0.23*
RN to Q	fall	0.02	0.11	0.10	0.19
RN to QN	rise	0.17	*0.29*	0.28	*0.39*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.00	0.07	0.00	0.07
	rise	0.01	0.10	0.01	0.10
CK to QN	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*
RN to Q	fall	0.00	0.07	0.01	0.07
RN to QN	rise	0.01	*0.11*	0.01	*0.11*

Constraints Time [ns]		
Setup CK to D	fall	0.11
	rise	0.09
Hold CK to D	fall	0.12
	rise	0.12
Recovery CK to RN	rise	0.02
Removal CK to RN	rise	0.42

Capacitance [fF]	
CK	0.8969
D	1.0692
RN	1.7340

Leakage [nW]
18.97

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	3957.88	4092.88
	rise	3862.92	3919.67
D	fall	1770.38	1761.49
	rise	1049.44	1027.35
RN	fall	467.02	454.07
	rise	-184.70	-191.12

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	3.19	3.56	3.26	3.58
	rise	3.33	3.58	3.37	3.57
CK to QN	fall	3.33	*3.59*	3.37	*3.62*
	rise	3.20	*3.53*	3.26	*3.57*
RN to Q	fall	2.94	3.19	3.02	3.30
RN to QN	rise	2.94	*3.25*	3.02	*3.25*

DFFR_X2

Databook Build Date: Thursday Feb 17 15:07 2011

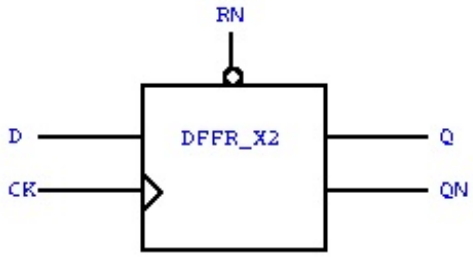
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, and drive strength X2
Strength	2
Cell Area	5.852 um^2
Equation	Q = "(!(!RN)*(D))" QN = "((!RN)+!(D))"
Clock	CK
Reset	RN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	RN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.09	0.17	0.14	0.22
	rise	0.11	0.22	0.16	0.27
CK to QN	fall	0.07	*0.17*	0.12	*0.22*
	rise	0.07	*0.19*	0.12	*0.24*
RN to Q	fall	0.02	0.10	0.09	0.19
RN to QN	rise	0.18	*0.30*	0.29	*0.41*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.00	0.07	0.00	0.07
	rise	0.01	0.10	0.01	0.10
CK to QN	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*
RN to Q	fall	0.00	0.07	0.01	0.07
RN to QN	rise	0.01	*0.11*	0.01	*0.11*

Constraints Time [ns]		
Setup CK to D	fall	0.11
	rise	0.09
Hold CK to D	fall	0.12
	rise	0.12
Recovery CK to RN	rise	-0.02
Removal CK to RN	rise	0.33

Capacitance [fF]	
CK	0.8841
D	1.0691
RN	2.4068

Leakage [nW]
27.13

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	3923.53	4058.84
	rise	3665.56	3678.43
D	fall	1757.30	1748.09
	rise	981.50	958.38
RN	fall	866.73	827.09
	rise	-257.37	-266.45

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	4.48	5.09	4.54	5.13
	rise	4.67	5.30	4.69	5.25
CK to QN	fall	4.67	*5.31*	4.69	*5.31*
	rise	4.48	*5.10*	4.54	*5.18*
RN to Q	fall	4.60	5.10	4.78	5.10
RN to QN	rise	4.60	*5.06*	4.78	*5.15*

DFFS_X1

Databook Build Date: Thursday Feb 17 15:07 2011

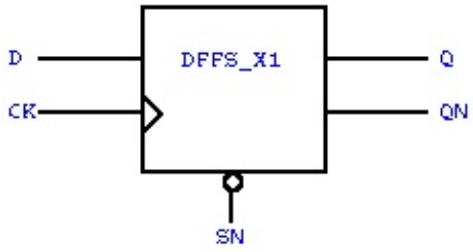
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, and drive strength X1
Strength	1
Cell Area	5.320 um^2
Equation	Q = "((!SN)+(D))" QN = "(!((!SN)+(D)))"
Clock	CK
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.08	0.16	0.13	0.21
	rise	0.08	0.19	0.13	0.24
CK to QN	fall	0.06	*0.15*	0.11	*0.20*
	rise	0.06	*0.18*	0.11	*0.23*
SN to Q	rise	0.18	0.28	0.29	0.39
SN to QN	fall	0.13	*0.26*	0.24	*0.36*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.00	0.10
CK to QN	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*
SN to Q	rise	0.01	0.10	0.01	0.10
SN to QN	fall	0.02	*0.08*	0.02	*0.08*

Constraints Time [ns]		
Setup CK to D	fall	0.11
	rise	0.08
Hold CK to D	fall	0.13
	rise	0.12
Recovery CK to SN	rise	-0.02
Removal CK to SN	rise	0.38

Capacitance [fF]	
CK	0.8911
D	1.1131
SN	1.3097

Leakage [nW]
18.38

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	4487.26	4630.49
	rise	3261.53	3300.22
D	fall	1694.15	1690.67
	rise	796.65	772.48
SN	fall	682.30	718.18
	rise	-15.87	-21.51

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	2.89	3.21	2.95	3.24
	rise	2.96	3.27	3.00	3.27
CK to QN	fall	2.96	*3.28*	3.00	*3.25*
	rise	2.89	*3.24*	2.95	*3.27*
SN to Q	rise	2.98	3.23	3.02	3.31
SN to QN	fall	2.98	*3.26*	3.02	*3.27*

DFFS_X2

Databook Build Date: Thursday Feb 17 15:07 2011

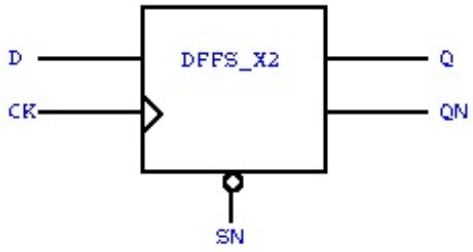
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, and drive strength X2
Strength	2
Cell Area	5.586 um^2
Equation	Q = "((!SN)+(D))" QN = "(!((!SN)+(D)))"
Clock	CK
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.09	0.17	0.15	0.22
	rise	0.10	0.21	0.16	0.26
CK to QN	fall	0.07	*0.18*	0.13	*0.23*
	rise	0.07	*0.19*	0.12	*0.25*
SN to Q	rise	0.26	0.35	0.37	0.46
SN to QN	fall	0.20	*0.35*	0.31	*0.46*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.00	0.07	0.00	0.07
	rise	0.01	0.10	0.01	0.10
CK to QN	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*
SN to Q	rise	0.01	0.10	0.01	0.10
SN to QN	fall	0.04	*0.09*	0.04	*0.09*

Constraints Time [ns]		
Setup CK to D	fall	0.11
	rise	0.08
Hold CK to D	fall	0.13
	rise	0.12
Recovery CK to SN	rise	-0.01
Removal CK to SN	rise	0.10

Capacitance [fF]	
CK	0.8923
D	1.1124
SN	1.2835

Leakage [nW]
25.63

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	4576.11	4716.58
	rise	3013.06	3052.78
D	fall	1727.25	1723.68
	rise	743.99	718.48
SN	fall	674.86	711.80
	rise	-86.01	-94.44

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	4.17	4.77	4.21	4.90
	rise	4.22	4.90	4.27	4.81
CK to QN	fall	4.22	*4.93*	4.27	*4.82*
	rise	4.17	*4.87*	4.21	*4.89*
SN to Q	rise	4.59	4.85	4.63	4.86
SN to QN	fall	4.60	*4.86*	4.63	*4.87*

DFF_X1

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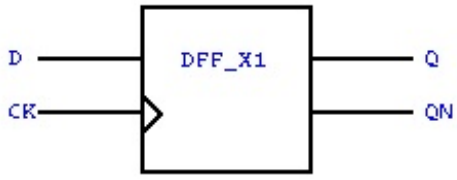
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with drive strength X1
Strength	1
Cell Area	4.522 um^2
Equation	Q = "(D)" QN = "(!(D))"
Clock	CK
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
CK	**D**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**	**QN**
R	L	-	-	L	H
R	H	-	-	H	L
F	-	L	H	L	H
F	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to Q	**fall**	0.07	0.15	0.13	0.21
	rise	0.08	0.19	0.13	0.24
CK to QN	**fall**	0.05	*0.14*	0.11	*0.20*
	rise	0.06	*0.17*	0.11	*0.23*

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to Q	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.00	0.10
CK to QN	**fall**	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*

Constraints Time [ns]		
Setup CK to D	**fall**	0.11
	rise	0.09
Hold CK to D	**fall**	0.12
	rise	0.12

Capacitance [fF]	
CK	0.8738
D	1.0799

Leakage [nW]
16.70

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	4317.07	4449.51
	rise	1858.61	1942.51
D	fall	1706.19	1701.98
	rise	1002.95	981.97

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to Q	fall	2.84	3.16	2.90	3.14
	rise	2.89	3.08	2.93	3.23
CK to QN	fall	2.89	*3.17*	2.93	*3.18*
	rise	2.84	*3.18*	2.90	*3.16*

DFF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

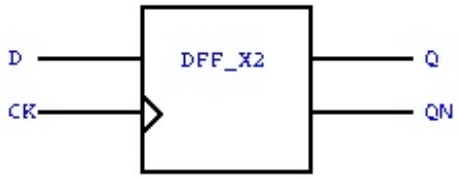
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with drive strength X2
Strength	2
Cell Area	5.054 um^2
Equation	Q = "(D)" QN = "(!(D))"
Clock	CK
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
CK	**D**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**	**QN**
R	L	-	-	L	H
R	H	-	-	H	L
F	-	L	H	L	H
F	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
CK to Q	**fall**	0.09	0.17	0.14	0.22
	rise	0.10	0.21	0.15	0.26
CK to QN	**fall**	0.07	*0.17*	0.12	*0.23*
	rise	0.07	*0.19*	0.12	*0.24*

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
CK to Q	**fall**	0.00	0.07	0.00	0.07
	rise	0.01	0.10	0.01	0.10
CK to QN	**fall**	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*

Constraints Time [ns]		
Setup CK to D	**fall**	0.11
	rise	0.09
Hold CK to D	**fall**	0.12
	rise	0.12

Capacitance [fF]	
CK	0.8538
D	1.0683

Leakage [nW]
23.66

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	4375.77	4505.77
	rise	1870.49	1953.99
D	fall	1733.61	1728.45
	rise	993.39	970.62

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	4.17	4.76	4.21	4.77
	rise	4.20	4.83	4.24	4.73
CK to QN	fall	4.20	*4.84*	4.24	*4.83*
	rise	4.17	*4.78*	4.21	*4.85*

DLH_X1

Databook Build Date: Thursday Feb 17 15:07 2011

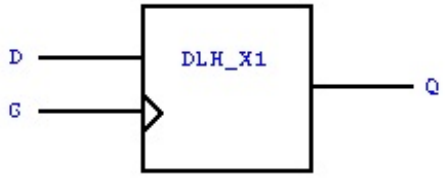
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	High enable Latch with drive strength X1
Strength	1
Cell Area	2.660 um^2
Equation	Q = "(D)"
Enable	G
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**G**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	H	-	-	L
H	H	-	-	H
-	L	L	H	L
-	L	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
D to Q	**fall**	0.04	0.13	0.13	0.22
	rise	0.03	0.15	0.09	0.21
G to Q	**fall**	0.05	0.14	0.09	0.18
	rise	0.05	0.16	0.09	0.21

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
D to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
G to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.00	0.10

Constraints Time [ns]		
Setup G to D	**fall**	0.13
	rise	0.07
Hold G to D	**fall**	0.18
	rise	0.11

Capacitance [fF]	
D	0.8668
G	0.9085

Leakage [nW]
8.51

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1792**
D	**fall**	348.85	348.57
	rise	-91.24	-93.85
G	**fall**	2232.33	2396.30
	rise	1633.66	1745.05

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
D to Q	**fall**	2.42	3.31	2.26	3.26
	rise	1.55	1.92	1.53	2.07
G to Q	**fall**	3.37	4.28	3.50	4.35
	rise	3.25	3.95	3.44	4.08

DLH_X2

Databook Build Date: Thursday Feb 17 15:07 2011

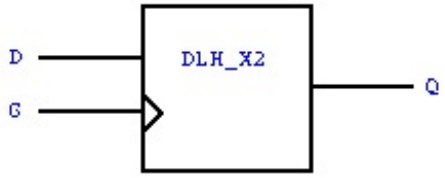
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	High enable Latch with drive strength X2
Strength	2
Cell Area	2.926 um^2
Equation	Q = "(D)"
Enable	G
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**G**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	H	-	-	L
H	H	-	-	H
-	L	L	H	L
-	L	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**120.697**	**0.3656**	**120.697**
D to Q	**fall**	0.05	0.14	0.13	0.23
	rise	0.03	0.15	0.09	0.21
G to Q	**fall**	0.05	0.14	0.10	0.19
	rise	0.05	0.17	0.09	0.21

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**120.697**	**0.3656**	**120.697**
D to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
G to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10

Constraints Time [ns]		
Setup G to D	**fall**	0.14
	rise	0.08
Hold G to D	**fall**	0.17
	rise	0.11

Capacitance [fF]	
D	1.1032
G	0.9085

Leakage [nW]
11.82

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1792**
D	**fall**	471.48	468.62
	rise	-126.77	-130.43
G	**fall**	2161.61	2326.35
	rise	1641.46	1751.93

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**120.697**	**0.3656**	**120.697**
D to Q	**fall**	3.67	5.06	3.56	4.97
	rise	2.63	3.26	2.55	3.18
G to Q	**fall**	4.40	5.81	4.52	5.87
	rise	4.32	5.13	4.49	5.30

DLL_X1

Databook Build Date: Thursday Feb 17 15:07 2011

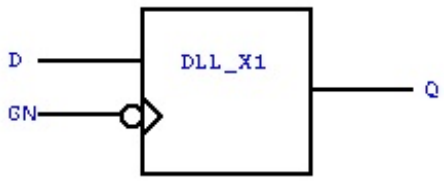
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Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Low enable Latch with drive strength X1
Strength	1
Cell Area	2.660 um^2
Equation	Q = "(D)"
Enable	GN
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**GN**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	L	-	-	L
H	L	-	-	H
-	H	L	H	L
-	H	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
D to Q	**fall**	0.04	0.13	0.13	0.22
	rise	0.03	0.14	0.08	0.20
GN to Q	**fall**	0.06	0.15	0.13	0.22
	rise	0.04	0.16	0.11	0.23

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
D to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
GN to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10

Constraints Time [ns]		
Setup GN to D	**fall**	0.12
	rise	0.09
Hold GN to D	**fall**	0.11
	rise	0.11

Capacitance [fF]	
D	0.8333
GN	0.9087

Leakage [nW]
8.51

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1792**
D	**fall**	348.83	348.53
	rise	-64.64	-66.25
GN	**fall**	2402.84	2577.31
	rise	1316.56	1433.05

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
D to Q	**fall**	2.49	3.29	2.31	3.24
	rise	1.73	2.16	1.67	2.15
GN to Q	**fall**	4.30	5.10	4.35	5.26
	rise	4.11	4.63	4.13	4.95

DLL_X2

Databook Build Date: Thursday Feb 17 15:07 2011

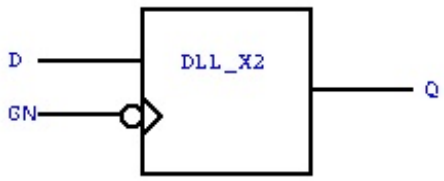
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Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Low enable Latch with drive strength X2
Strength	2
Cell Area	2.926 um^2
Equation	Q = "(D)"
Enable	GN
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**GN**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	L	-	-	L
H	L	-	-	H
-	H	L	H	L
-	H	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
D to Q	**fall**	0.05	0.13	0.13	0.23
	rise	0.03	0.15	0.09	0.21
GN to Q	**fall**	0.06	0.15	0.13	0.22
	rise	0.04	0.16	0.12	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
D to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
GN to Q	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10

Constraints Time [ns]		
Setup GN to D	**fall**	0.13
	rise	0.09
Hold GN to D	**fall**	0.10
	rise	0.12

Capacitance [fF]	
D	1.0711
GN	0.9030

Leakage [nW]
11.82

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0012**	**0.1792**
D	**fall**	471.52	468.57
	rise	-100.18	-102.86
GN	**fall**	2498.86	2675.59
	rise	1250.21	1365.42

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
D to Q	**fall**	3.57	4.88	3.41	4.79
	rise	2.64	3.48	2.52	3.57
GN to Q	**fall**	5.23	6.54	5.30	6.71
	rise	5.16	6.06	5.17	6.25

FA_X1

Databook Build Date: Thursday Feb 17 15:07 2011

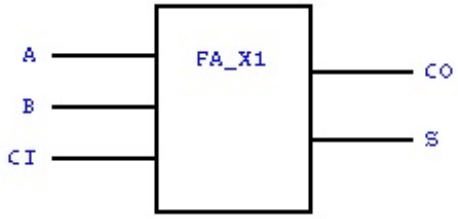
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	4.256 um^2
Equation	CO = "((A & B) | (CI & (A | B)))" S = "(CI ^ (A ^ B))"
Type	Combinational
Input	A, B, CI
Output	CO, S
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**CI**	**CO**	**S**
L	L	L	L	L
L	L	H	L	H
L	H	L	L	H
L	H	H	H	L
H	L	L	L	H
H	L	H	H	L
H	H	L	H	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A to CO	**fall**	0.05	*0.15*	0.14	*0.24*
	rise	0.04	*0.16*	0.10	*0.22*
A to S	**fall**	0.07	0.17	0.15	0.25
	rise	0.09	0.20	0.18	0.29
B to CO	**fall**	0.05	*0.15*	0.15	*0.25*
	rise	0.04	*0.16*	0.10	*0.22*
B to S	**fall**	0.08	0.16	0.16	0.26
	rise	0.09	0.21	0.19	0.30
CI to CO	**fall**	0.05	*0.14*	0.14	*0.24*
	rise	0.03	*0.15*	0.10	*0.22*
CI to S	**fall**	0.07	0.16	0.15	0.25
	rise	0.09	0.20	0.18	0.29

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A to CO	**fall**	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*
A to S	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
B to CO	**fall**	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*
B to S	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
CI to CO	**fall**	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*
CI to S	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10

Capacitance [fF]	
A	3.6381
B	3.3871
CI	2.6926

Leakage [nW]
17.74

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.2722	0.3656	60.2722
A to CO	fall	3.27	*3.56*	3.16	*3.52*
	rise	1.40	*1.89*	1.44	*1.78*
A to S	fall	2.60	3.28	2.45	3.21
	rise	2.00	2.28	2.00	2.15
B to CO	fall	3.39	*3.63*	3.27	*3.67*
	rise	1.47	*1.98*	1.50	*1.84*
B to S	fall	2.52	3.20	2.37	3.08
	rise	2.00	2.23	2.00	2.13
CI to CO	fall	3.02	*3.34*	2.95	*3.34*
	rise	1.68	*2.13*	1.72	*2.08*
CI to S	fall	2.33	3.03	2.23	2.95
	rise	2.05	2.30	2.01	2.24

FILLCELL_X16

Databook Build Date: Thursday Feb 17 15:07 2011

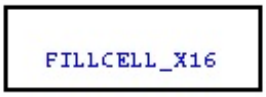
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	4.256 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X1

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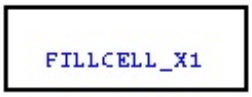
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.266 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X2

Databook Build Date: Thursday Feb 17 15:07 2011

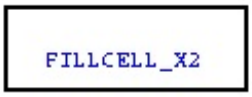
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	0.266 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



NANGATE FILLCELL_X32

Databook Build Date: Thursday Feb 17 15:07 2011

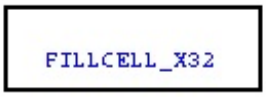
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	32
Cell Area	8.512 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X4

Databook Build Date: Thursday Feb 17 15:07 2011

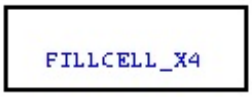
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	1.064 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X8

Databook Build Date: Thursday Feb 17 15:07 2011

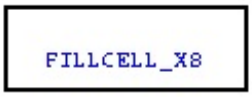
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	2.128 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



HA_X1

Databook Build Date: Thursday Feb 17 15:07 2011

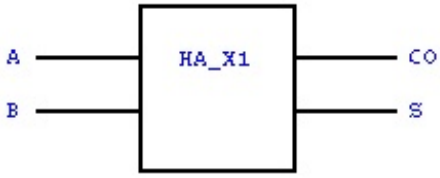
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.660 um^2
Equation	CO = "(A & B)" S = "(A ^ B)"
Type	Combinational
Input	A, B
Output	CO, S
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B**	**CO**	**S**
L	L	L	L
L	H	L	H
H	L	L	H
H	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A to CO	**fall**	0.02	0.10	0.09	0.18
	rise	0.02	0.14	0.07	0.19
A to S	**fall**	0.04	*0.13*	0.12	*0.27*
	rise	0.03	*0.25*	0.08	*0.37*
B to CO	**fall**	0.02	0.10	0.10	0.19
	rise	0.02	0.14	0.07	0.19
B to S	**fall**	0.04	*0.14*	0.12	*0.25*
	rise	0.03	*0.26*	0.08	*0.37*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A to CO	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A to S	**fall**	0.01	*0.11*	0.03	*0.12*
	rise	0.01	*0.21*	0.04	*0.21*
B to CO	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
B to S	**fall**	0.01	*0.11*	0.02	*0.11*
	rise	0.01	*0.21*	0.03	*0.20*

Capacitance [fF]	
A	3.0898
B	3.3022

Leakage [nW]
13.43

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A to CO	fall	2.11	2.31	1.95	2.28
	rise	0.68	0.77	0.88	0.78
A to S	fall	2.35	*2.37*	2.19	*2.59*
	rise	2.28	*2.08*	2.16	*2.22*
B to CO	fall	2.57	2.77	2.43	2.77
	rise	0.59	0.62	0.61	0.63
B to S	fall	2.40	*2.39*	2.07	*2.55*
	rise	2.32	*2.35*	2.38	*2.36*

INV_X16

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	4.522 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	969.238	0.3656	969.238
A to ZN	**fall**	0.00	0.08	0.01	0.18
	rise	0.00	0.12	0.04	0.24

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	969.238	0.3656	969.238
A to ZN	**fall**	0.00	0.07	0.02	0.09
	rise	0.00	0.10	0.02	0.11

Capacitance [fF]	
A	23.3571

Leakage [nW]
44.41

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	969.238	0.3656	969.238
A to ZN	**fall**	-0.00	-0.00	5.82	-0.00
	rise	20.33	22.55	24.24	18.74

INV_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.532 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
A to ZN	**fall**	0.00	0.08	0.01	0.18
	rise	0.00	0.12	0.04	0.24

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
A to ZN	**fall**	0.00	0.07	0.02	0.09
	rise	0.00	0.10	0.03	0.11

Capacitance [fF]	
A	1.5715

Leakage [nW]
2.78

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
A to ZN	**fall**	-0.00	-0.00	0.28	-0.00
	rise	1.34	1.36	1.56	1.08

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INV_X2

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	0.798 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to ZN	**fall**	0.00	0.08	0.01	0.18
	rise	0.00	0.12	0.04	0.24

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to ZN	**fall**	0.00	0.07	0.02	0.09
	rise	0.00	0.10	0.02	0.11

Capacitance [fF]	
A	2.9933

Leakage [nW]
5.55

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to ZN	**fall**	-0.00	-0.00	0.70	-0.00
	rise	2.65	2.56	3.09	2.88

INV_X32

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	32
Cell Area	8.778 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**1923.83**	**0.3656**	**1923.83**
A to ZN	**fall**	0.00	0.08	0.01	0.18
	rise	0.00	0.12	0.04	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**1923.83**	**0.3656**	**1923.83**
A to ZN	**fall**	0.00	0.07	0.02	0.09
	rise	0.00	0.11	0.02	0.11

Capacitance [fF]	
A	45.5977

Leakage [nW]
88.81

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**1923.83**	**0.3656**	**1923.83**
A to ZN	**fall**	-0.00	-0.00	11.77	-0.01
	rise	40.29	40.24	48.31	44.93

INV_X4

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	1.330 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.92	0.3656	242.92
A to ZN	**fall**	0.00	0.08	0.01	0.18
	rise	0.00	0.12	0.04	0.24

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.92	0.3656	242.92
A to ZN	**fall**	0.00	0.07	0.02	0.09
	rise	0.00	0.10	0.02	0.11

Capacitance [fF]	
A	5.7773

Leakage [nW]
11.10

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.92	0.3656	242.92
A to ZN	**fall**	-0.00	-0.00	1.45	-0.00
	rise	5.10	5.68	6.06	5.56

INV_X8

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	2.394 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	485.229	0.3656	485.229
A to ZN	**fall**	0.00	0.08	0.01	0.18
	rise	0.01	0.12	0.04	0.24

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	485.229	0.3656	485.229
A to ZN	**fall**	0.00	0.07	0.02	0.09
	rise	0.00	0.10	0.02	0.11

Capacitance [fF]	
A	10.9132

Leakage [nW]
22.20

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	485.229	0.3656	485.229
A to ZN	**fall**	-0.00	-0.00	2.98	-0.00
	rise	9.88	9.94	12.18	10.21

LOGIC0_X1

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.532 um^2
Equation	Z = "0"
Type	Constant
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)

LOGIC1_X1

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.532 um^2
Equation	Z = "1"
Type	Constant
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)

MUX2_X1

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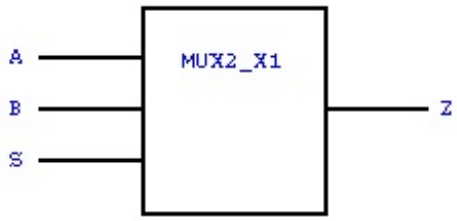
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.862 um^2
Equation	Z = "((S & B) | (A & !S))"
Type	Combinational
Input	A, B, S
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B**	**S**	**Z**
L	-	L	L
H	-	L	H
-	L	H	L
-	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.5011**	**0.3656**	**60.5011**
A to Z	**fall**	0.04	0.13	0.13	0.22
	rise	0.03	0.15	0.08	0.20
B to Z	**fall**	0.04	0.13	0.13	0.22
	rise	0.03	0.14	0.08	0.20
S to Z	**fall**	0.05	0.13	0.11	0.21
	rise	0.04	0.15	0.11	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.5011**	**0.3656**	**60.5011**
A to Z	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
B to Z	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
S to Z	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A	0.9015
B	0.8938
S	1.8386

Leakage [nW]
7.78

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.5011**	**0.3656**	**60.5011**
A to Z	**fall**	2.25	2.93	2.07	2.87
	rise	1.61	1.75	1.56	1.59
B to Z	**fall**	2.43	3.18	2.25	3.12
	rise	1.40	1.50	1.36	1.46
S to Z	**fall**	2.58	3.30	2.70	3.46
	rise	2.44	2.61	2.55	2.67

MUX2_X2

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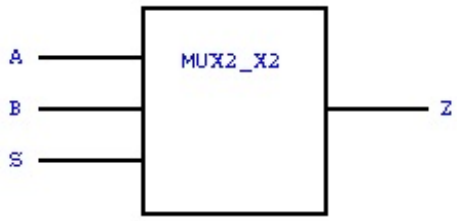
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	Z = "((S & B) | (A & !S))"
Type	Combinational
Input	A, B, S
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B**	**S**	**Z**
L	-	L	L
H	-	L	H
-	L	H	L
-	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A to Z	**fall**	0.04	0.13	0.12	0.22
	rise	0.03	0.15	0.09	0.21
B to Z	**fall**	0.04	0.13	0.12	0.22
	rise	0.03	0.14	0.08	0.20
S to Z	**fall**	0.06	0.15	0.10	0.20
	rise	0.05	0.16	0.12	0.24

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A to Z	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
B to Z	**fall**	0.01	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
S to Z	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A	1.5126
B	1.6049
S	2.5218

Leakage [nW]
14.96

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A to Z	**fall**	4.74	6.24	4.19	6.04
	rise	3.01	3.72	3.09	3.37
B to Z	**fall**	4.50	6.02	4.53	6.19
	rise	2.38	2.89	2.27	2.42
S to Z	**fall**	5.30	6.85	5.32	7.00
	rise	4.55	4.96	4.61	5.26

NANGATE NAND2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

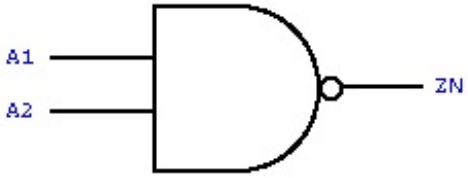
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	ZN = "!(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**59.3567**	**0.3656**	**59.3567**
A1 to ZN	**fall**	0.01	0.13	0.03	0.23
	rise	0.01	0.12	0.05	0.24
A2 to ZN	**fall**	0.01	0.13	0.03	0.22
	rise	0.01	0.12	0.06	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**59.3567**	**0.3656**	**59.3567**
A1 to ZN	**fall**	0.00	0.11	0.03	0.11
	rise	0.00	0.10	0.03	0.11
A2 to ZN	**fall**	0.00	0.11	0.02	0.11
	rise	0.00	0.10	0.03	0.11

Capacitance [fF]	
A1	1.5118
A2	1.5510

Leakage [nW]
3.93

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**59.3567**	**0.3656**	**59.3567**
A1 to ZN	**fall**	0.14	0.18	0.54	0.14
	rise	1.60	1.38	1.70	1.55
A2 to ZN	**fall**	0.15	0.18	0.26	0.10
	rise	2.04	2.12	2.17	1.94

NANGATE NAND2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

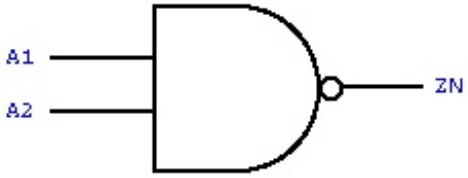
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "!(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	118.713	0.3656	118.713
A1 to ZN	**fall**	0.01	0.13	0.02	0.23
	rise	0.01	0.12	0.05	0.24
A2 to ZN	**fall**	0.01	0.13	0.03	0.22
	rise	0.01	0.12	0.06	0.24

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	118.713	0.3656	118.713
A1 to ZN	**fall**	0.00	0.11	0.03	0.11
	rise	0.00	0.10	0.03	0.11
A2 to ZN	**fall**	0.00	0.11	0.02	0.11
	rise	0.00	0.10	0.03	0.11

Capacitance [fF]	
A1	2.8904
A2	3.2319

Leakage [nW]
7.86

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	118.713	0.3656	118.713
A1 to ZN	**fall**	0.26	0.36	1.09	0.28
	rise	3.21	3.05	3.49	3.21
A2 to ZN	**fall**	0.29	0.37	0.53	0.21
	rise	4.12	4.23	4.37	3.72

NANGATE NAND2_X4

Databook Build Date: Thursday Feb 17 15:07 2011

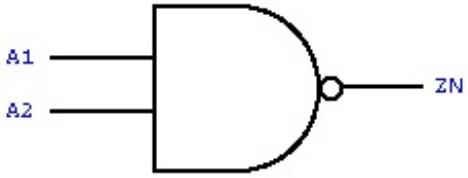
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "!(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**237.427**	**0.3656**	**237.427**
A1 to ZN	**fall**	0.01	0.13	0.02	0.23
	rise	0.01	0.12	0.04	0.24
A2 to ZN	**fall**	0.01	0.13	0.03	0.22
	rise	0.01	0.12	0.06	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**237.427**	**0.3656**	**237.427**
A1 to ZN	**fall**	0.00	0.11	0.03	0.11
	rise	0.00	0.10	0.03	0.11
A2 to ZN	**fall**	0.00	0.11	0.02	0.11
	rise	0.00	0.10	0.03	0.11

Capacitance [fF]	
A1	5.6249
A2	5.7640

Leakage [nW]
15.72

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**237.427**	**0.3656**	**237.427**
A1 to ZN	**fall**	0.52	0.73	2.26	0.57
	rise	5.96	5.38	6.53	5.63
A2 to ZN	**fall**	0.55	0.72	1.06	0.42
	rise	7.98	8.36	8.49	8.40

NANGATE NAND3_X1

Databook Build Date: Thursday Feb 17 15:07 2011

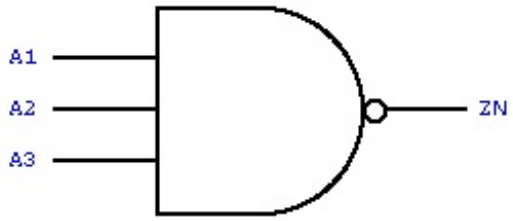
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	H
H	H	H	L
-	L	-	H
-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**58.3649**	**0.3656**	**58.3649**
A1 to ZN	**fall**	0.01	0.18	0.04	0.27
	rise	0.01	0.12	0.05	0.24
A2 to ZN	**fall**	0.01	0.18	0.05	0.27
	rise	0.01	0.12	0.06	0.24
A3 to ZN	**fall**	0.01	0.18	0.05	0.25
	rise	0.01	0.12	0.07	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**58.3649**	**0.3656**	**58.3649**
A1 to ZN	**fall**	0.01	0.14	0.03	0.15
	rise	0.00	0.10	0.03	0.11
A2 to ZN	**fall**	0.01	0.14	0.03	0.15
	rise	0.01	0.10	0.03	0.11
A3 to ZN	**fall**	0.01	0.14	0.02	0.15
	rise	0.01	0.10	0.03	0.11

Capacitance [fF]	
A1	1.5235
A2	1.5509
A3	1.5396

Leakage [nW]
4.50

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**58.3649**	**0.3656**	**58.3649**
A1 to ZN	**fall**	0.36	0.40	0.69	0.35
	rise	1.79	1.76	1.86	1.87
A2 to ZN	**fall**	0.37	0.40	0.44	0.32
	rise	2.29	2.12	2.33	2.21
A3 to ZN	**fall**	0.38	0.40	0.38	0.31
	rise	2.62	2.76	2.76	2.81

NANGATE NAND3_X2

Databook Build Date: Thursday Feb 17 15:07 2011

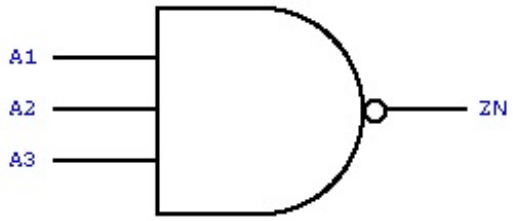
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	H
H	H	H	L
-	L	-	H
-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**116.272**	**0.3656**	**116.272**
A1 to ZN	**fall**	0.01	0.18	0.04	0.27
	rise	0.01	0.12	0.05	0.24
A2 to ZN	**fall**	0.01	0.18	0.05	0.27
	rise	0.01	0.12	0.06	0.24
A3 to ZN	**fall**	0.01	0.18	0.05	0.25
	rise	0.01	0.12	0.07	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**116.272**	**0.3656**	**116.272**
A1 to ZN	**fall**	0.01	0.14	0.03	0.15
	rise	0.00	0.10	0.03	0.11
A2 to ZN	**fall**	0.01	0.14	0.03	0.15
	rise	0.00	0.10	0.03	0.11
A3 to ZN	**fall**	0.01	0.14	0.02	0.15
	rise	0.01	0.10	0.03	0.11

Capacitance [fF]	
A1	2.8545
A2	3.1346
A3	3.3360

Leakage [nW]
9.01

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**116.272**	**0.3656**	**116.272**
A1 to ZN	**fall**	0.85	0.93	1.52	0.84
	rise	3.77	3.56	3.87	3.84
A2 to ZN	**fall**	0.87	0.93	1.02	0.77
	rise	4.72	4.70	4.81	4.56
A3 to ZN	**fall**	0.88	0.93	0.91	0.76
	rise	5.38	5.64	5.65	5.47

NANGATE NAND3_X4

Databook Build Date: Thursday Feb 17 15:07 2011

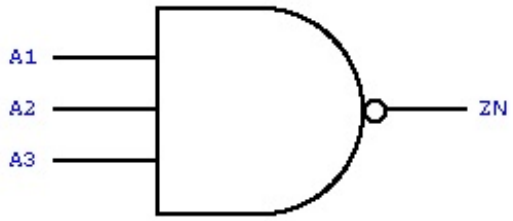
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	H
H	H	H	L
-	L	-	H
-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**233.154**	**0.3656**	**233.154**
A1 to ZN	**fall**	0.01	0.18	0.03	0.27
	rise	0.01	0.12	0.05	0.24
A2 to ZN	**fall**	0.01	0.18	0.04	0.27
	rise	0.01	0.12	0.06	0.24
A3 to ZN	**fall**	0.01	0.18	0.05	0.25
	rise	0.01	0.12	0.07	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**233.154**	**0.3656**	**233.154**
A1 to ZN	**fall**	0.01	0.14	0.03	0.15
	rise	0.00	0.10	0.03	0.11
A2 to ZN	**fall**	0.01	0.14	0.02	0.15
	rise	0.00	0.10	0.03	0.11
A3 to ZN	**fall**	0.01	0.14	0.02	0.15
	rise	0.01	0.10	0.03	0.11

Capacitance [fF]	
A1	5.9991
A2	6.6340
A3	6.7204

Leakage [nW]
18.02

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**233.154**	**0.3656**	**233.154**
A1 to ZN	**fall**	1.68	1.86	3.10	1.68
	rise	7.34	6.81	7.57	7.39
A2 to ZN	**fall**	1.76	1.86	2.05	1.54
	rise	9.23	8.72	9.45	9.14
A3 to ZN	**fall**	1.74	1.87	1.80	1.52
	rise	10.48	10.89	11.09	10.58

NANGATE NAND4_X1

Databook Build Date: Thursday Feb 17 15:07 2011

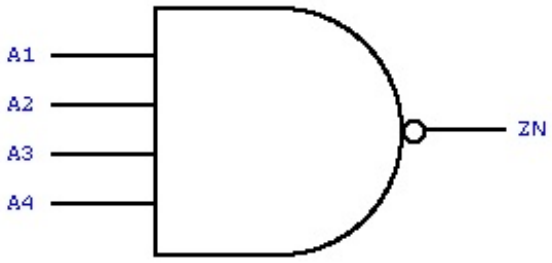
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	H
H	H	H	H	L
-	L	-	-	H
-	-	L	-	H
-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**55.9998**	**0.3656**	**55.9998**
A1 to ZN	**fall**	0.02	0.22	0.05	0.30
	rise	0.01	0.11	0.06	0.23
A2 to ZN	**fall**	0.02	0.22	0.06	0.31
	rise	0.01	0.11	0.07	0.24
A3 to ZN	**fall**	0.02	0.22	0.06	0.30
	rise	0.01	0.12	0.07	0.24
A4 to ZN	**fall**	0.02	0.22	0.06	0.28
	rise	0.01	0.12	0.08	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**55.9998**	**0.3656**	**55.9998**
A1 to ZN	**fall**	0.01	0.18	0.04	0.18
	rise	0.01	0.10	0.03	0.10
A2 to ZN	**fall**	0.01	0.18	0.03	0.18
	rise	0.01	0.10	0.03	0.10
A3 to ZN	**fall**	0.01	0.18	0.03	0.18
	rise	0.01	0.10	0.03	0.10
A4 to ZN	**fall**	0.01	0.18	0.02	0.18
	rise	0.01	0.10	0.03	0.11

Capacitance [fF]	
A1	1.4692
A2	1.5415
A3	1.5624
A4	1.5469

Leakage [nW]
4.80

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	55.9998	0.3656	55.9998
A1 to ZN	fall	0.72	0.75	0.99	0.70
	rise	2.01	1.79	2.04	1.82
A2 to ZN	fall	0.73	0.75	0.76	0.67
	rise	2.48	2.42	2.51	2.57
A3 to ZN	fall	0.73	0.75	0.71	0.66
	rise	2.87	2.69	2.94	2.84
A4 to ZN	fall	0.72	0.75	0.69	0.66
	rise	3.16	3.38	3.33	3.15

NANGATE NAND4_X2

Databook Build Date: Thursday Feb 17 15:07 2011

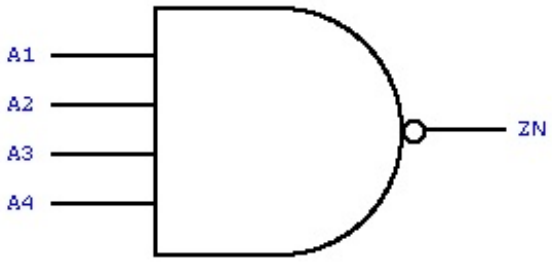
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	H
H	H	H	H	L
-	L	-	-	H
-	-	L	-	H
-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**111.542**	**0.3656**	**111.542**
A1 to ZN	**fall**	0.02	0.22	0.05	0.30
	rise	0.01	0.11	0.06	0.23
A2 to ZN	**fall**	0.02	0.22	0.06	0.31
	rise	0.01	0.11	0.07	0.24
A3 to ZN	**fall**	0.02	0.22	0.06	0.30
	rise	0.01	0.12	0.07	0.24
A4 to ZN	**fall**	0.02	0.22	0.06	0.28
	rise	0.01	0.12	0.08	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**111.542**	**0.3656**	**111.542**
A1 to ZN	**fall**	0.01	0.18	0.04	0.18
	rise	0.01	0.10	0.03	0.10
A2 to ZN	**fall**	0.01	0.18	0.03	0.18
	rise	0.01	0.10	0.03	0.10
A3 to ZN	**fall**	0.01	0.18	0.03	0.18
	rise	0.01	0.10	0.03	0.10
A4 to ZN	**fall**	0.01	0.18	0.02	0.18
	rise	0.01	0.10	0.03	0.11

Capacitance [fF]	
A1	2.8264
A2	3.1762
A3	3.3337
A4	3.6014

Leakage [nW]
9.60

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	111.542	0.3656	111.542
A1 to ZN	fall	1.45	1.50	1.97	1.40
	rise	4.41	4.41	4.42	4.14
A2 to ZN	fall	1.45	1.50	1.51	1.33
	rise	5.30	5.11	5.39	5.42
A3 to ZN	fall	1.46	1.49	1.42	1.32
	rise	6.12	6.26	6.23	6.01
A4 to ZN	fall	1.46	1.50	1.39	1.31
	rise	6.74	7.12	7.04	6.94

NANGATE NAND4_X4

Databook Build Date: Thursday Feb 17 15:07 2011

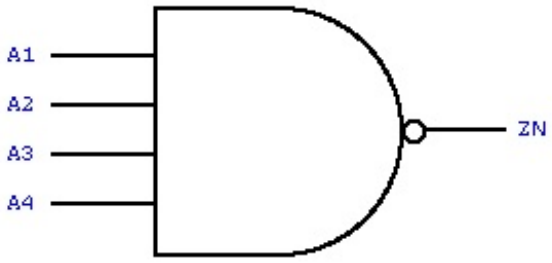
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.788 um^2
Equation	ZN = "!(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	H
H	H	H	H	L
-	L	-	-	H
-	-	L	-	H
-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**222.778**	**0.3656**	**222.778**
A1 to ZN	**fall**	0.01	0.22	0.05	0.31
	rise	0.01	0.11	0.06	0.23
A2 to ZN	**fall**	0.02	0.22	0.06	0.31
	rise	0.01	0.12	0.07	0.24
A3 to ZN	**fall**	0.02	0.22	0.06	0.30
	rise	0.01	0.12	0.07	0.24
A4 to ZN	**fall**	0.02	0.23	0.06	0.29
	rise	0.01	0.12	0.08	0.24

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**222.778**	**0.3656**	**222.778**
A1 to ZN	**fall**	0.01	0.18	0.04	0.18
	rise	0.00	0.10	0.03	0.10
A2 to ZN	**fall**	0.01	0.18	0.03	0.18
	rise	0.01	0.10	0.03	0.10
A3 to ZN	**fall**	0.01	0.18	0.02	0.18
	rise	0.01	0.10	0.03	0.10
A4 to ZN	**fall**	0.01	0.18	0.02	0.18
	rise	0.01	0.10	0.03	0.11

Capacitance [fF]	
A1	5.4411
A2	5.5945
A3	5.6278
A4	5.6841

Leakage [nW]
19.20

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	222.778	0.3656	222.778
A1 to ZN	fall	2.85	3.00	4.01	2.80
	rise	7.69	6.50	7.80	8.01
A2 to ZN	fall	2.88	3.00	3.01	2.66
	rise	9.89	9.88	9.97	10.21
A3 to ZN	fall	2.92	3.01	2.79	2.63
	rise	11.49	10.89	11.77	12.04
A4 to ZN	fall	2.92	3.00	2.75	2.63
	rise	12.80	13.68	13.54	12.81

NANGATE NOR2_X1

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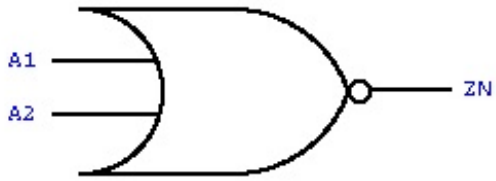
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	ZN = "!(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	26.7029	0.3656	26.7029
A1 to ZN	**fall**	0.00	0.04	0.02	0.12
	rise	0.01	0.11	0.06	0.22
A2 to ZN	**fall**	0.01	0.04	0.03	0.12
	rise	0.01	0.11	0.07	0.22

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	26.7029	0.3656	26.7029
A1 to ZN	**fall**	0.00	0.03	0.03	0.06
	rise	0.01	0.09	0.03	0.10
A2 to ZN	**fall**	0.00	0.03	0.02	0.06
	rise	0.01	0.09	0.02	0.10

Capacitance [fF]	
A1	1.5732
A2	1.5342

Leakage [nW]
5.01

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	26.7029	0.3656	26.7029
A1 to ZN	**fall**	-0.00	-0.00	0.22	-0.00
	rise	1.68	1.64	1.85	1.72
A2 to ZN	**fall**	0.04	0.14	0.43	0.17
	rise	2.26	2.27	1.99	2.22

NOR2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

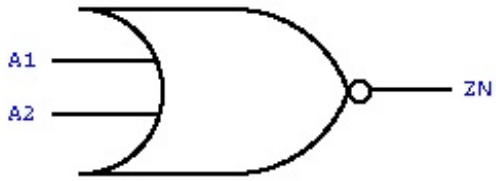
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "!(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	53.4058	0.3656	53.4058
A1 to ZN	**fall**	0.00	0.04	0.01	0.12
	rise	0.01	0.11	0.05	0.22
A2 to ZN	**fall**	0.01	0.04	0.03	0.12
	rise	0.01	0.11	0.07	0.22

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	53.4058	0.3656	53.4058
A1 to ZN	**fall**	0.00	0.03	0.02	0.06
	rise	0.01	0.09	0.03	0.10
A2 to ZN	**fall**	0.00	0.03	0.02	0.06
	rise	0.01	0.09	0.02	0.10

Capacitance [fF]	
A1	3.0079
A2	3.1209

Leakage [nW]
10.02

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	53.4058	0.3656	53.4058
A1 to ZN	**fall**	-0.00	-0.00	0.58	-0.00
	rise	3.18	3.12	3.57	3.28
A2 to ZN	**fall**	0.19	0.40	1.00	0.48
	rise	4.39	4.40	3.82	4.25

NANGATE NOR2_X4

Databook Build Date: Thursday Feb 17 15:07 2011

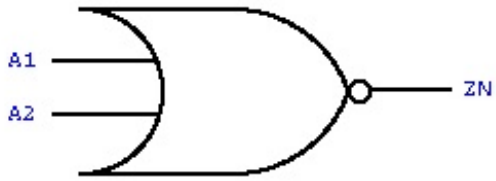
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "!(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	106.811	0.3656	106.811
A1 to ZN	**fall**	0.00	0.04	0.01	0.12
	rise	0.01	0.11	0.05	0.22
A2 to ZN	**fall**	0.01	0.04	0.02	0.12
	rise	0.01	0.11	0.07	0.22

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	106.811	0.3656	106.811
A1 to ZN	**fall**	0.00	0.03	0.02	0.06
	rise	0.01	0.09	0.03	0.10
A2 to ZN	**fall**	0.00	0.03	0.02	0.06
	rise	0.01	0.09	0.02	0.10

Capacitance [fF]	
A1	6.2136
A2	6.2043

Leakage [nW]
20.05

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	106.811	0.3656	106.811
A1 to ZN	**fall**	-0.00	-0.00	1.19	-0.00
	rise	6.25	6.31	7.04	6.76
A2 to ZN	**fall**	0.36	0.81	2.01	0.96
	rise	8.64	8.66	7.46	8.45

NANGATE NOR3_X1

Databook Build Date: Thursday Feb 17 15:07 2011

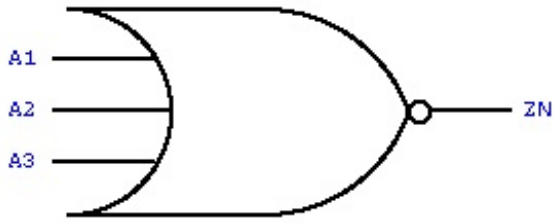
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	H
H	-	-	L
-	H	-	L
-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	16.0217	0.3656	16.0217
A1 to ZN	**fall**	0.00	0.03	0.02	0.09
	rise	0.02	0.11	0.07	0.20
A2 to ZN	**fall**	0.01	0.03	0.03	0.10
	rise	0.02	0.11	0.09	0.21
A3 to ZN	**fall**	0.01	0.03	0.04	0.10
	rise	0.03	0.12	0.09	0.21

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	16.0217	0.3656	16.0217
A1 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.09	0.04	0.10
A2 to ZN	**fall**	0.00	0.02	0.02	0.05
	rise	0.01	0.09	0.03	0.09
A3 to ZN	**fall**	0.00	0.02	0.02	0.05
	rise	0.01	0.09	0.02	0.09

Capacitance [fF]	
A1	1.6055
A2	1.5476
A3	1.5039

Leakage [nW]
7.12

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	16.0217	0.3656	16.0217
A1 to ZN	**fall**	-0.00	-0.00	0.18	-0.00
	rise	1.88	1.95	2.02	2.02
A2 to ZN	**fall**	0.08	0.16	0.40	0.24
	rise	2.46	2.44	2.21	2.37
A3 to ZN	**fall**	0.17	0.38	0.61	0.48
	rise	2.95	3.02	2.64	2.80

NANGATE NOR3_X2

Databook Build Date: Thursday Feb 17 15:07 2011

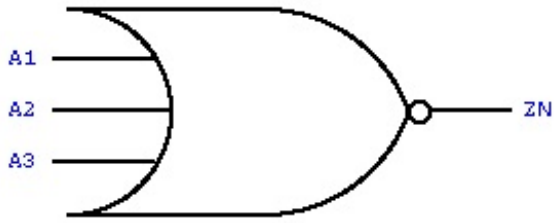
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	H
H	-	-	L
-	H	-	L
-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	31.7383	0.3656	31.7383
A1 to ZN	**fall**	0.00	0.03	0.02	0.09
	rise	0.02	0.11	0.07	0.20
A2 to ZN	**fall**	0.01	0.03	0.03	0.10
	rise	0.02	0.11	0.09	0.21
A3 to ZN	**fall**	0.01	0.03	0.03	0.10
	rise	0.03	0.12	0.09	0.21

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	31.7383	0.3656	31.7383
A1 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.09	0.04	0.10
A2 to ZN	**fall**	0.00	0.02	0.02	0.05
	rise	0.01	0.09	0.03	0.09
A3 to ZN	**fall**	0.00	0.02	0.02	0.05
	rise	0.01	0.09	0.02	0.09

Capacitance [fF]	
A1	3.0475
A2	3.1804
A3	3.2152

Leakage [nW]
14.25

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	31.7383	0.3656	31.7383
A1 to ZN	**fall**	-0.00	-0.00	0.49	0.02
	rise	3.82	3.99	4.13	4.12
A2 to ZN	**fall**	0.30	0.45	0.94	0.60
	rise	4.98	4.97	4.49	4.74
A3 to ZN	**fall**	0.45	0.90	1.35	1.08
	rise	6.02	6.13	5.36	5.70

NANGATE NOR3_X4

Databook Build Date: Thursday Feb 17 15:07 2011

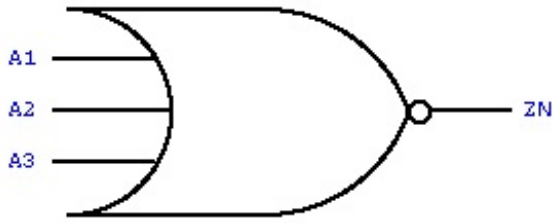
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.724 um^2
Equation	ZN = "!((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	H
H	-	-	L
-	H	-	L
-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	63.324	0.3656	63.324
A1 to ZN	**fall**	0.00	0.03	0.02	0.09
	rise	0.02	0.11	0.07	0.20
A2 to ZN	**fall**	0.01	0.03	0.03	0.10
	rise	0.02	0.12	0.09	0.21
A3 to ZN	**fall**	0.01	0.03	0.04	0.10
	rise	0.03	0.12	0.09	0.21

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	63.324	0.3656	63.324
A1 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.09	0.03	0.10
A2 to ZN	**fall**	0.00	0.02	0.02	0.05
	rise	0.01	0.09	0.03	0.09
A3 to ZN	**fall**	0.00	0.02	0.02	0.05
	rise	0.01	0.09	0.02	0.09

Capacitance [fF]	
A1	5.8988
A2	5.6928
A3	5.6473

Leakage [nW]
28.50

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	63.324	0.3656	63.324
A1 to ZN	**fall**	-0.00	-0.00	0.98	0.03
	rise	7.58	7.70	8.19	7.95
A2 to ZN	**fall**	0.57	0.90	1.87	1.19
	rise	10.41	10.57	9.38	10.24
A3 to ZN	**fall**	0.88	1.78	2.65	2.14
	rise	12.73	12.98	11.42	12.26

NOR4_X1

Databook Build Date: Thursday Feb 17 15:07 2011

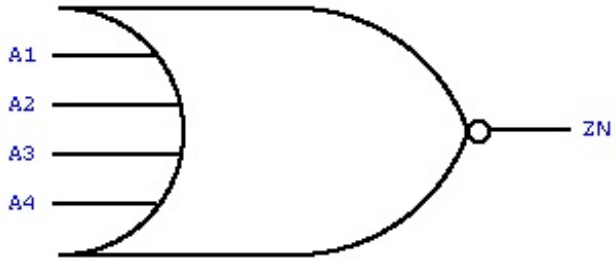
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	-	L
-	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**10.4713**	**0.3656**	**10.4713**
A1 to ZN	**fall**	0.01	0.02	0.02	0.08
	rise	0.02	0.10	0.09	0.19
A2 to ZN	**fall**	0.01	0.02	0.03	0.08
	rise	0.04	0.11	0.10	0.21
A3 to ZN	**fall**	0.01	0.02	0.04	0.08
	rise	0.04	0.12	0.11	0.21
A4 to ZN	**fall**	0.01	0.02	0.04	0.09
	rise	0.05	0.12	0.12	0.21

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**10.4713**	**0.3656**	**10.4713**
A1 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.02	0.09	0.04	0.10
A2 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.02	0.09	0.04	0.09
A3 to ZN	**fall**	0.00	0.02	0.03	0.04
	rise	0.02	0.09	0.03	0.09
A4 to ZN	**fall**	0.00	0.02	0.03	0.04
	rise	0.02	0.09	0.03	0.09

Capacitance [fF]	
A1	1.5659
A2	1.5494
A3	1.5315
A4	1.4949

Leakage [nW]
9.24

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	10.4713	0.3656	10.4713
A1 to ZN	fall	-0.00	-0.00	0.14	-0.00
	rise	2.27	2.33	2.40	2.44
A2 to ZN	fall	0.11	0.15	0.38	0.28
	rise	2.84	2.84	2.63	2.71
A3 to ZN	fall	0.22	0.39	0.59	0.52
	rise	3.36	3.32	3.10	3.19
A4 to ZN	fall	0.23	0.53	0.75	0.71
	rise	3.85	3.87	3.60	3.73

NOR4_X2

Databook Build Date: Thursday Feb 17 15:07 2011

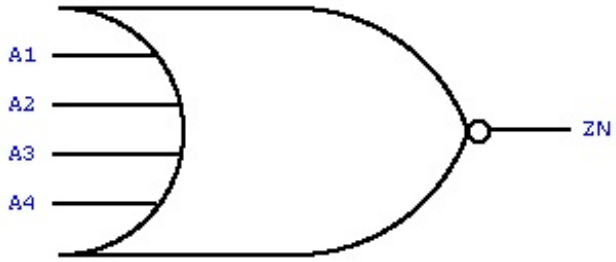
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	-	L
-	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**20.9045**	**0.3656**	**20.9045**
A1 to ZN	**fall**	0.01	0.02	0.02	0.08
	rise	0.02	0.10	0.08	0.19
A2 to ZN	**fall**	0.01	0.02	0.03	0.08
	rise	0.03	0.11	0.10	0.21
A3 to ZN	**fall**	0.01	0.02	0.04	0.08
	rise	0.04	0.12	0.11	0.21
A4 to ZN	**fall**	0.01	0.02	0.04	0.09
	rise	0.04	0.12	0.12	0.21

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**20.9045**	**0.3656**	**20.9045**
A1 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.02	0.09	0.04	0.10
A2 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.02	0.09	0.04	0.09
A3 to ZN	**fall**	0.00	0.02	0.03	0.04
	rise	0.02	0.09	0.03	0.09
A4 to ZN	**fall**	0.00	0.02	0.03	0.04
	rise	0.02	0.09	0.03	0.09

Capacitance [fF]	
A1	3.0539
A2	3.1508
A3	3.2952
A4	3.3854

Leakage [nW]
18.47

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	20.9045	0.3656	20.9045
A1 to ZN	fall	-0.00	-0.00	0.42	0.10
	rise	4.42	4.55	4.69	4.80
A2 to ZN	fall	0.33	0.44	0.90	0.69
	rise	5.56	5.57	5.15	5.31
A3 to ZN	fall	0.55	0.92	1.33	1.17
	rise	6.59	6.57	6.08	6.30
A4 to ZN	fall	0.57	1.18	1.65	1.56
	rise	7.62	7.64	7.06	7.32

NOR4_X4

Databook Build Date: Thursday Feb 17 15:07 2011

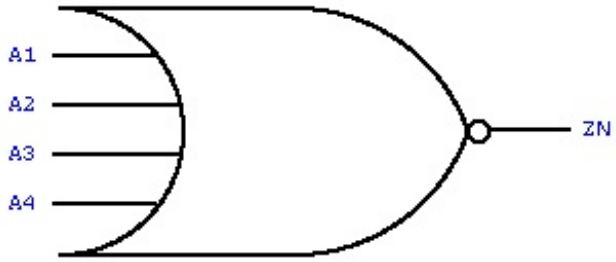
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.788 um^2
Equation	ZN = "!(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	-	L
-	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**41.5039**	**0.3656**	**41.5039**
A1 to ZN	**fall**	0.01	0.02	0.02	0.08
	rise	0.02	0.10	0.08	0.19
A2 to ZN	**fall**	0.01	0.02	0.03	0.08
	rise	0.04	0.12	0.10	0.21
A3 to ZN	**fall**	0.01	0.02	0.04	0.08
	rise	0.04	0.12	0.11	0.21
A4 to ZN	**fall**	0.01	0.02	0.04	0.09
	rise	0.05	0.13	0.12	0.21

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**41.5039**	**0.3656**	**41.5039**
A1 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.02	0.09	0.04	0.10
A2 to ZN	**fall**	0.00	0.02	0.03	0.04
	rise	0.02	0.09	0.03	0.09
A3 to ZN	**fall**	0.00	0.02	0.03	0.04
	rise	0.02	0.09	0.03	0.09
A4 to ZN	**fall**	0.00	0.02	0.03	0.04
	rise	0.02	0.09	0.02	0.09

Capacitance [fF]	
A1	5.8770
A2	5.6744
A3	5.6148
A4	5.5881

Leakage [nW]
36.95

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	41.5039	0.3656	41.5039
A1 to ZN	fall	-0.00	-0.00	0.84	0.20
	rise	8.74	8.93	9.33	9.39
A2 to ZN	fall	0.70	0.87	1.78	1.36
	rise	11.46	11.68	10.65	11.15
A3 to ZN	fall	1.09	1.82	2.59	2.30
	rise	13.97	14.00	12.97	13.47
A4 to ZN	fall	1.06	2.33	3.23	3.05
	rise	16.30	16.43	15.21	15.91

NANGATE OAI211_X1

Databook Build Date: Thursday Feb 17 15:07 2011

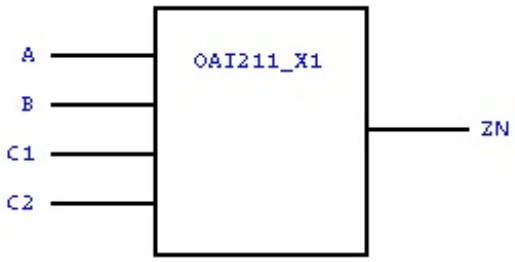
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((C1 | C2) & A) & B)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	-	-	-	H
H	H	H	-	L
H	H	-	H	L
-	L	-	-	H
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**25.5585**	**0.3656**	**25.5585**
A to ZN	**fall**	0.02	0.09	0.07	0.18
	rise	0.01	0.06	0.07	0.17
B to ZN	**fall**	0.02	0.09	0.06	0.16
	rise	0.01	0.06	0.08	0.18
C1 to ZN	**fall**	0.01	0.08	0.04	0.17
	rise	0.01	0.11	0.07	0.22
C2 to ZN	**fall**	0.01	0.09	0.05	0.18
	rise	0.02	0.11	0.08	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**25.5585**	**0.3656**	**25.5585**
A to ZN	**fall**	0.01	0.07	0.03	0.08
	rise	0.01	0.05	0.04	0.07
B to ZN	**fall**	0.01	0.07	0.02	0.07
	rise	0.01	0.05	0.03	0.07
C1 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.01	0.09	0.04	0.10
C2 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A	1.5344
B	1.5420
C1	1.5213
C2	1.4684

Leakage [nW]
5.48

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	25.5585	0.3656	25.5585
A to ZN	fall	0.48	0.52	0.50	0.41
	rise	3.06	3.05	3.12	3.04
B to ZN	fall	0.48	0.52	0.47	0.40
	rise	3.40	3.45	3.55	3.60
C1 to ZN	fall	0.40	0.43	0.66	0.40
	rise	2.06	2.01	2.12	2.11
C2 to ZN	fall	0.66	0.72	0.92	0.70
	rise	2.64	2.69	2.32	2.42

OAI211_X2

Databook Build Date: Thursday Feb 17 15:07 2011

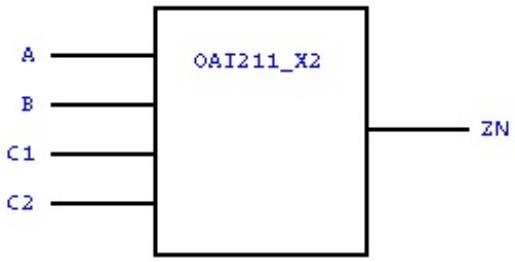
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((C1 | C2) & A) & B)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	-	-	-	H
H	H	H	-	L
H	H	-	H	L
-	L	-	-	H
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	50.8118	0.3656	50.8118
A to ZN	**fall**	0.02	0.09	0.06	0.18
	rise	0.01	0.06	0.07	0.17
B to ZN	**fall**	0.02	0.09	0.06	0.16
	rise	0.01	0.06	0.08	0.18
C1 to ZN	**fall**	0.01	0.08	0.04	0.17
	rise	0.01	0.11	0.07	0.22
C2 to ZN	**fall**	0.01	0.09	0.05	0.18
	rise	0.02	0.11	0.08	0.22

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	50.8118	0.3656	50.8118
A to ZN	**fall**	0.01	0.07	0.03	0.08
	rise	0.01	0.05	0.04	0.07
B to ZN	**fall**	0.01	0.07	0.02	0.07
	rise	0.01	0.05	0.03	0.07
C1 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.01	0.09	0.04	0.10
C2 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A	3.1689
B	2.9711
C1	2.8459
C2	3.0494

Leakage [nW]
10.95

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	50.8118	0.3656	50.8118
A to ZN	fall	1.09	1.17	1.13	0.95
	rise	6.17	6.24	6.28	6.36
B to ZN	fall	1.08	1.17	1.06	0.92
	rise	6.83	7.07	7.11	7.27
C1 to ZN	fall	0.92	1.00	1.46	0.91
	rise	4.22	4.11	4.36	4.17
C2 to ZN	fall	1.46	1.58	1.96	1.54
	rise	5.35	5.36	4.71	4.91

OAI211_X4

Databook Build Date: Thursday Feb 17 15:07 2011

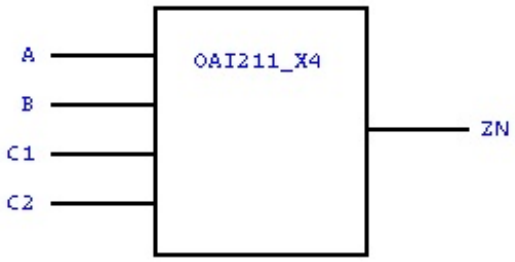
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Equation	ZN = "!(((C1 | C2) & A) & B)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	-	-	-	H
H	H	H	-	L
H	H	-	H	L
-	L	-	-	H
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	101.624	0.3656	101.624
A to ZN	**fall**	0.02	0.09	0.06	0.18
	rise	0.01	0.06	0.07	0.17
B to ZN	**fall**	0.02	0.09	0.06	0.16
	rise	0.01	0.06	0.08	0.18
C1 to ZN	**fall**	0.01	0.08	0.04	0.17
	rise	0.01	0.11	0.07	0.22
C2 to ZN	**fall**	0.01	0.09	0.05	0.18
	rise	0.02	0.11	0.08	0.22

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	101.624	0.3656	101.624
A to ZN	**fall**	0.01	0.07	0.03	0.08
	rise	0.01	0.05	0.04	0.07
B to ZN	**fall**	0.01	0.07	0.02	0.08
	rise	0.01	0.05	0.03	0.07
C1 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.01	0.09	0.04	0.10
C2 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A	6.0535
B	6.0744
C1	5.9031
C2	6.0765

Leakage [nW]
21.90

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	101.624	0.3656	101.624
A to ZN	fall	2.17	2.32	2.26	1.89
	rise	12.53	12.73	12.74	12.69
B to ZN	fall	2.17	2.31	2.13	1.85
	rise	13.87	14.13	14.43	14.67
C1 to ZN	fall	1.88	1.99	2.93	1.84
	rise	8.52	8.54	8.83	8.48
C2 to ZN	fall	2.93	3.15	3.93	3.06
	rise	10.87	10.91	9.60	10.48

OAI21_X1

Databook Build Date: Thursday Feb 17 15:07 2011

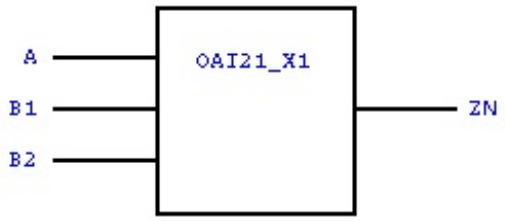
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!(A & (B1 | B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	-	-	H
H	H	-	L
H	-	H	L
-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**26.0544**	**0.3656**	**26.0544**
A to ZN	**fall**	0.01	0.07	0.05	0.15
	rise	0.01	0.06	0.07	0.17
B1 to ZN	**fall**	0.01	0.06	0.03	0.15
	rise	0.01	0.11	0.06	0.22
B2 to ZN	**fall**	0.01	0.06	0.04	0.15
	rise	0.02	0.11	0.08	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**26.0544**	**0.3656**	**26.0544**
A to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.01	0.05	0.03	0.07
B1 to ZN	**fall**	0.00	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.10
B2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A	1.5491
B1	1.5619
B2	1.4892

Leakage [nW]
5.33

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**26.0544**	**0.3656**	**26.0544**
A to ZN	**fall**	0.25	0.30	0.31	0.20
	rise	2.84	2.79	2.98	2.94
B1 to ZN	**fall**	0.18	0.22	0.50	0.19
	rise	1.87	1.83	1.97	1.88
B2 to ZN	**fall**	0.43	0.49	0.74	0.49
	rise	2.46	2.46	2.14	2.39

OAI21_X2

Databook Build Date: Thursday Feb 17 15:07 2011

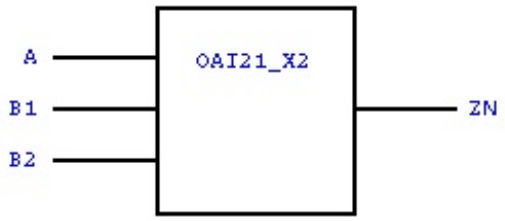
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!(A & (B1 | B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	-	-	H
H	H	-	L
H	-	H	L
-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**52.1088**	**0.3656**	**52.1088**
A to ZN	**fall**	0.01	0.07	0.05	0.15
	rise	0.01	0.06	0.07	0.17
B1 to ZN	**fall**	0.01	0.06	0.03	0.15
	rise	0.01	0.11	0.06	0.22
B2 to ZN	**fall**	0.01	0.06	0.04	0.15
	rise	0.02	0.11	0.08	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**52.1088**	**0.3656**	**52.1088**
A to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.01	0.05	0.03	0.07
B1 to ZN	**fall**	0.00	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.10
B2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A	2.9334
B1	2.8973
B2	3.1710

Leakage [nW]
10.67

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**52.1088**	**0.3656**	**52.1088**
A to ZN	**fall**	0.49	0.59	0.63	0.41
	rise	5.62	5.71	5.89	5.85
B1 to ZN	**fall**	0.36	0.43	1.00	0.37
	rise	3.77	3.95	3.96	3.73
B2 to ZN	**fall**	0.84	0.99	1.48	0.97
	rise	4.90	4.91	4.26	4.45

OAI21_X4

Databook Build Date: Thursday Feb 17 15:07 2011

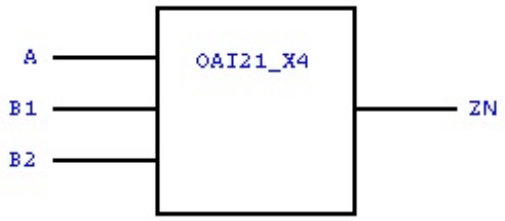
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(A & (B1 | B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	-	-	H
H	H	-	L
H	-	H	L
-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**104.065**	**0.3656**	**104.065**
A to ZN	**fall**	0.01	0.07	0.05	0.15
	rise	0.01	0.06	0.07	0.17
B1 to ZN	**fall**	0.01	0.06	0.03	0.15
	rise	0.01	0.11	0.06	0.22
B2 to ZN	**fall**	0.01	0.07	0.04	0.15
	rise	0.02	0.11	0.08	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**104.065**	**0.3656**	**104.065**
A to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.01	0.05	0.03	0.07
B1 to ZN	**fall**	0.00	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.10
B2 to ZN	**fall**	0.01	0.05	0.03	0.08
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A	5.7223
B1	5.9421
B2	6.1758

Leakage [nW]
21.33

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**104.065**	**0.3656**	**104.065**
A to ZN	**fall**	0.98	1.17	1.27	0.81
	rise	11.27	11.48	11.81	11.82
B1 to ZN	**fall**	0.70	0.86	2.03	0.74
	rise	7.52	7.60	7.97	7.43
B2 to ZN	**fall**	1.69	1.99	2.96	1.92
	rise	9.83	9.79	8.56	9.51

OAI221_X1

Databook Build Date: Thursday Feb 17 15:07 2011

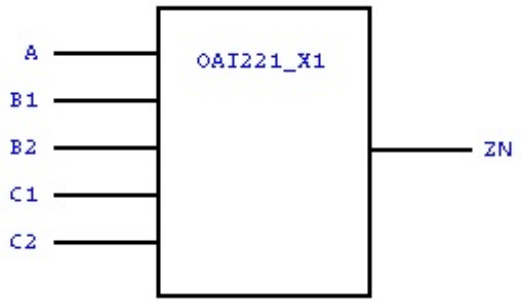
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "!(((C1 | C2) & A) & (B1 | B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	-	-	-	H
H	H	-	H	-	L
H	H	-	-	H	L
H	-	H	H	-	L
H	-	H	-	H	L
-	L	L	-	-	H
-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	22.1634	0.3656	22.1634
A to ZN	**fall**	0.02	0.09	0.08	0.17
	rise	0.01	0.05	0.07	0.16
B1 to ZN	**fall**	0.02	0.08	0.06	0.15
	rise	0.03	0.11	0.10	0.22
B2 to ZN	**fall**	0.02	0.09	0.07	0.16
	rise	0.03	0.11	0.11	0.22
C1 to ZN	**fall**	0.01	0.08	0.05	0.17
	rise	0.02	0.10	0.07	0.20
C2 to ZN	**fall**	0.02	0.08	0.06	0.17
	rise	0.02	0.10	0.08	0.20

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	22.1634	0.3656	22.1634
A to ZN	**fall**	0.01	0.06	0.03	0.07
	rise	0.01	0.05	0.05	0.07
B1 to ZN	**fall**	0.01	0.06	0.02	0.07
	rise	0.02	0.09	0.04	0.10
B2 to ZN	**fall**	0.01	0.06	0.02	0.07
	rise	0.02	0.09	0.03	0.09
C1 to ZN	**fall**	0.01	0.06	0.03	0.08
	rise	0.01	0.09	0.05	0.10
C2 to ZN	**fall**	0.01	0.06	0.03	0.08
	rise	0.01	0.09	0.03	0.09

Capacitance [fF]	
A	1.5454
B1	1.5480
B2	1.5247
C1	1.4917
C2	1.5001

Leakage [nW]
8.80

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	22.1634	0.3656	22.1634
A to ZN	fall	0.58	0.63	0.61	0.53
	rise	3.16	3.18	3.24	3.22
B1 to ZN	fall	0.51	0.55	0.46	0.42
	rise	3.61	3.72	3.71	3.70
B2 to ZN	fall	0.77	0.84	0.77	0.73
	rise	4.16	4.25	3.94	4.14
C1 to ZN	fall	0.50	0.55	0.76	0.52
	rise	2.22	2.27	2.30	2.30
C2 to ZN	fall	0.77	0.84	1.02	0.82
	rise	2.79	2.74	2.49	2.67

NANGATE OAI221_X2

Databook Build Date: Thursday Feb 17 15:07 2011

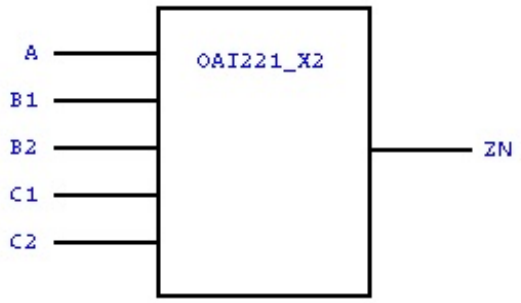
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.926 um^2
Equation	ZN = "!(((C1 | C2) & A) & (B1 | B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	-	-	-	H
H	H	-	H	-	L
H	H	-	-	H	L
H	-	H	H	-	L
H	-	H	-	H	L
-	L	L	-	-	H
-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	43.869	0.3656	43.869
A to ZN	**fall**	0.02	0.09	0.08	0.17
	rise	0.01	0.05	0.07	0.16
B1 to ZN	**fall**	0.02	0.08	0.06	0.15
	rise	0.03	0.11	0.10	0.22
B2 to ZN	**fall**	0.02	0.09	0.07	0.16
	rise	0.03	0.11	0.11	0.22
C1 to ZN	**fall**	0.01	0.08	0.05	0.17
	rise	0.02	0.10	0.07	0.20
C2 to ZN	**fall**	0.02	0.08	0.06	0.17
	rise	0.02	0.10	0.08	0.20

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	43.869	0.3656	43.869
A to ZN	**fall**	0.01	0.06	0.03	0.07
	rise	0.01	0.05	0.04	0.07
B1 to ZN	**fall**	0.01	0.06	0.02	0.07
	rise	0.02	0.09	0.04	0.10
B2 to ZN	**fall**	0.01	0.06	0.02	0.07
	rise	0.02	0.09	0.03	0.09
C1 to ZN	**fall**	0.01	0.06	0.03	0.08
	rise	0.01	0.09	0.05	0.10
C2 to ZN	**fall**	0.01	0.06	0.03	0.08
	rise	0.01	0.09	0.03	0.09

Capacitance [fF]	
A	3.3705
B1	3.2484
B2	2.9380
C1	2.8450
C2	3.0747

Leakage [nW]
17.60

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	43.869	0.3656	43.869
A to ZN	fall	1.29	1.40	1.33	1.18
	rise	6.58	6.65	6.74	6.78
B1 to ZN	fall	1.15	1.23	1.05	0.98
	rise	7.40	7.62	7.63	7.70
B2 to ZN	fall	1.67	1.80	1.66	1.60
	rise	8.51	8.76	8.12	8.66
C1 to ZN	fall	1.15	1.23	1.65	1.16
	rise	4.60	4.70	4.77	4.76
C2 to ZN	fall	1.66	1.81	2.16	1.77
	rise	5.76	5.68	5.14	5.55

NANGATE OAI221_X4

Databook Build Date: Thursday Feb 17 15:07 2011

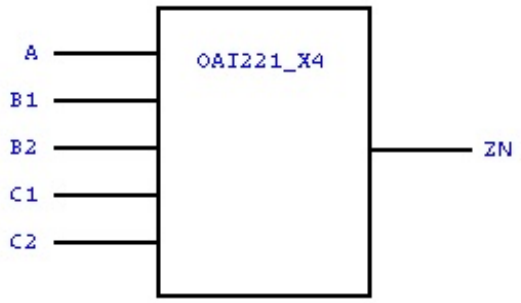
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(!(!(((C1 | C2) & A) & (B1 | B2))))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	-	-	-	H
H	H	-	H	-	L
H	H	-	-	H	L
H	-	H	H	-	L
H	-	H	-	H	L
-	L	L	-	-	H
-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	**fall**	0.06	0.14	0.13	0.21
	rise	0.04	0.15	0.13	0.24
B1 to ZN	**fall**	0.06	0.14	0.11	0.20
	rise	0.06	0.18	0.16	0.27
B2 to ZN	**fall**	0.06	0.14	0.12	0.20
	rise	0.07	0.18	0.16	0.28
C1 to ZN	**fall**	0.05	0.13	0.11	0.20
	rise	0.05	0.17	0.14	0.25
C2 to ZN	**fall**	0.06	0.14	0.12	0.21
	rise	0.06	0.17	0.14	0.26

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
B1 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
B2 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
C1 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
C2 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A	1.5572
B1	1.5550
B2	1.5533
C1	1.4746
C2	1.5017

Leakage [nW]
23.90

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	fall	7.32	9.21	7.13	9.12
	rise	10.80	11.79	10.08	11.82
B1 to ZN	fall	7.22	9.12	7.00	8.95
	rise	10.66	12.09	10.48	12.13
B2 to ZN	fall	7.48	9.43	7.31	9.31
	rise	11.23	12.90	10.81	12.83
C1 to ZN	fall	7.23	9.11	7.19	9.23
	rise	9.42	10.72	8.91	10.84
C2 to ZN	fall	7.49	9.43	7.52	9.55
	rise	9.95	10.77	9.26	11.01

OAI222_X1

Databook Build Date: Thursday Feb 17 15:07 2011

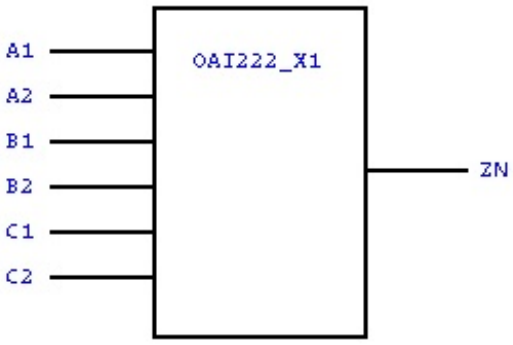
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.128 um^2
Equation	ZN = "!(((A1 | A2) & (B1 | B2)) & (C1 | C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	-	-	-	H
H	-	H	-	H	-	L
H	-	H	-	-	H	L
H	-	-	H	H	-	L
H	-	-	H	-	H	L
-	H	H	-	H	-	L
-	H	H	-	-	H	L
-	H	-	H	H	-	L
-	H	-	H	-	H	L
-	-	L	L	-	-	H
-	-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**20.0653**	**0.3656**	**20.0653**
A1 to ZN	**fall**	0.02	0.08	0.06	0.16
	rise	0.02	0.09	0.07	0.20
A2 to ZN	**fall**	0.02	0.08	0.07	0.17
	rise	0.02	0.09	0.09	0.20
B1 to ZN	**fall**	0.02	0.08	0.08	0.17
	rise	0.02	0.10	0.09	0.20
B2 to ZN	**fall**	0.03	0.08	0.09	0.17
	rise	0.03	0.10	0.10	0.20
C1 to ZN	**fall**	0.03	0.08	0.08	0.15
	rise	0.03	0.10	0.11	0.21
C2 to ZN	**fall**	0.03	0.09	0.08	0.16
	rise	0.03	0.11	0.12	0.21

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**20.0653**	**0.3656**	**20.0653**
A1 to ZN	**fall**	0.01	0.06	0.03	0.08
	rise	0.01	0.08	0.06	0.10
A2 to ZN	**fall**	0.01	0.06	0.03	0.08
	rise	0.01	0.08	0.04	0.09
B1 to ZN	**fall**	0.01	0.06	0.03	0.07
	rise	0.02	0.09	0.06	0.10
B2 to ZN	**fall**	0.01	0.06	0.03	0.07
	rise	0.02	0.09	0.04	0.09
C1 to ZN	**fall**	0.01	0.06	0.02	0.06
	rise	0.02	0.09	0.05	0.10
C2 to ZN	**fall**	0.01	0.06	0.02	0.06
	rise	0.02	0.09	0.04	0.09

Capacitance [fF]	
A1	1.4954
A2	1.5122
B1	1.5382
B2	1.5550
C1	1.4915
C2	1.4793

Leakage [nW]
11.30

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	20.0653	0.3656	20.0653
A1 to ZN	fall	0.61	0.67	0.85	0.64
	rise	2.36	2.33	2.47	2.37
A2 to ZN	fall	0.87	0.95	1.12	0.94
	rise	2.93	2.93	2.65	2.79
B1 to ZN	fall	0.62	0.67	0.60	0.55
	rise	3.36	3.37	3.44	3.41
B2 to ZN	fall	0.87	0.95	0.90	0.86
	rise	3.92	3.97	3.67	3.87
C1 to ZN	fall	0.62	0.67	0.56	0.54
	rise	4.12	4.23	4.24	4.30
C2 to ZN	fall	0.87	0.95	0.87	0.85
	rise	4.68	4.78	4.50	4.74

OAI222_X2

Databook Build Date: Thursday Feb 17 15:07 2011

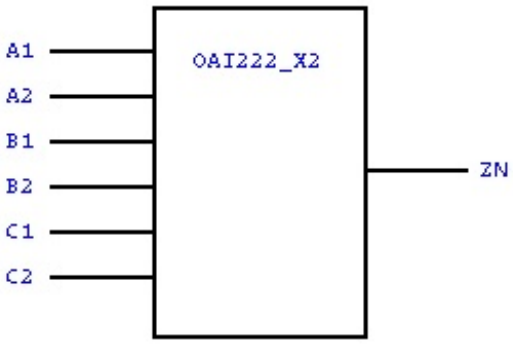
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	3.724 um^2
Equation	ZN = "!(((A1 | A2) & (B1 | B2)) & (C1 | C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	-	-	-	H
H	-	H	-	H	-	L
H	-	H	-	-	H	L
H	-	-	H	H	-	L
H	-	-	H	-	H	L
-	H	H	-	H	-	L
-	H	H	-	-	H	L
-	H	-	H	H	-	L
-	H	-	H	-	H	L
-	-	L	L	-	-	H
-	-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**39.5966**	**0.3656**	**39.5966**
A1 to ZN	**fall**	0.02	0.08	0.06	0.16
	rise	0.02	0.09	0.07	0.20
A2 to ZN	**fall**	0.02	0.08	0.07	0.17
	rise	0.02	0.09	0.09	0.20
B1 to ZN	**fall**	0.02	0.08	0.08	0.17
	rise	0.02	0.10	0.09	0.20
B2 to ZN	**fall**	0.03	0.08	0.09	0.17
	rise	0.03	0.10	0.10	0.20
C1 to ZN	**fall**	0.03	0.08	0.08	0.15
	rise	0.03	0.10	0.11	0.21
C2 to ZN	**fall**	0.03	0.09	0.08	0.16
	rise	0.03	0.11	0.12	0.21

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**39.5966**	**0.3656**	**39.5966**
A1 to ZN	**fall**	0.01	0.06	0.03	0.08
	rise	0.01	0.08	0.06	0.10
A2 to ZN	**fall**	0.01	0.06	0.03	0.08
	rise	0.01	0.08	0.04	0.09
B1 to ZN	**fall**	0.01	0.06	0.03	0.07
	rise	0.02	0.09	0.06	0.10
B2 to ZN	**fall**	0.01	0.06	0.03	0.07
	rise	0.02	0.09	0.04	0.09
C1 to ZN	**fall**	0.01	0.06	0.02	0.06
	rise	0.02	0.09	0.05	0.10
C2 to ZN	**fall**	0.01	0.06	0.02	0.07
	rise	0.02	0.09	0.04	0.09

Capacitance [fF]	
A1	3.0643
A2	2.8277
B1	3.1472
B2	2.8749
C1	3.1515
C2	2.8998

Leakage [nW]
22.60

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	39.5966	0.3656	39.5966
A1 to ZN	fall	1.36	1.47	1.83	1.41
	rise	5.00	4.98	5.19	5.09
A2 to ZN	fall	1.89	2.03	2.36	2.01
	rise	6.13	6.14	5.57	5.87
B1 to ZN	fall	1.37	1.47	1.33	1.23
	rise	6.95	6.97	7.10	7.06
B2 to ZN	fall	1.89	2.03	1.93	1.85
	rise	8.08	8.14	7.56	7.96
C1 to ZN	fall	1.37	1.46	1.25	1.21
	rise	8.50	8.81	8.74	8.93
C2 to ZN	fall	1.89	2.03	1.86	1.83
	rise	9.59	9.83	9.25	9.74

OAI222_X4

Databook Build Date: Thursday Feb 17 15:07 2011

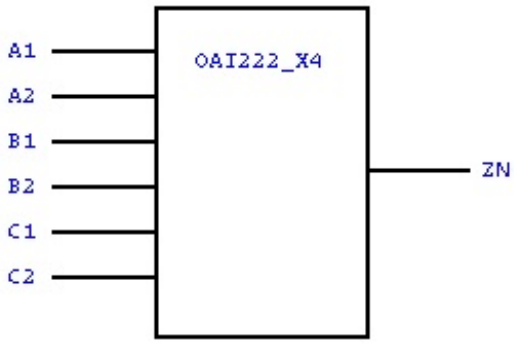
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.724 um^2
Equation	ZN = "!(!(!(((A1 | A2) & (B1 | B2)) & (C1 | C2))))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	-	-	-	H
H	-	H	-	H	-	L
H	-	H	-	-	H	L
H	-	-	H	H	-	L
H	-	-	H	-	H	L
-	H	H	-	H	-	L
-	H	H	-	-	H	L
-	H	-	H	H	-	L
-	H	-	H	-	H	L
-	-	L	L	-	-	H
-	-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.06	0.14	0.12	0.21
	rise	0.06	0.17	0.14	0.26
A2 to ZN	**fall**	0.06	0.14	0.13	0.21
	rise	0.06	0.18	0.15	0.26
B1 to ZN	**fall**	0.06	0.14	0.13	0.21
	rise	0.06	0.18	0.15	0.27
B2 to ZN	**fall**	0.06	0.15	0.14	0.22
	rise	0.07	0.18	0.16	0.27
C1 to ZN	**fall**	0.06	0.15	0.12	0.21
	rise	0.07	0.19	0.16	0.28
C2 to ZN	**fall**	0.07	0.15	0.13	0.21
	rise	0.07	0.19	0.17	0.28

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
A2 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
B1 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
B2 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
C1 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10
C2 to ZN	**fall**	0.00	0.07	0.00	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A1	1.4938
A2	1.4952
B1	1.5451
B2	1.5483
C1	1.5423
C2	1.5449

Leakage [nW]
27.39

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	fall	7.35	9.23	7.33	9.34
	rise	9.55	10.68	9.00	11.11
A2 to ZN	fall	7.59	9.55	7.64	9.66
	rise	10.11	11.26	9.41	11.38
B1 to ZN	fall	7.33	9.24	7.11	9.10
	rise	10.48	12.10	10.11	12.24
B2 to ZN	fall	7.60	9.55	7.43	9.46
	rise	11.05	12.51	10.50	12.49
C1 to ZN	fall	7.34	9.24	7.10	9.06
	rise	11.21	12.72	10.96	13.14
C2 to ZN	fall	7.59	9.55	7.42	9.43
	rise	11.77	13.31	11.36	13.23

OAI22_X1

Databook Build Date: Thursday Feb 17 15:07 2011

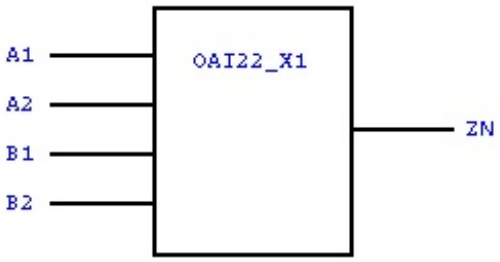
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!((A1 | A2) & (B1 | B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	L	-	-	H
H	-	H	-	L
H	-	-	H	L
-	H	H	-	L
-	H	-	H	L
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**23.2315**	**0.3656**	**23.2315**
A1 to ZN	**fall**	0.01	0.06	0.04	0.14
	rise	0.01	0.10	0.07	0.21
A2 to ZN	**fall**	0.01	0.06	0.05	0.15
	rise	0.02	0.10	0.08	0.21
B1 to ZN	**fall**	0.01	0.06	0.05	0.14
	rise	0.02	0.11	0.09	0.21
B2 to ZN	**fall**	0.02	0.06	0.06	0.15
	rise	0.02	0.11	0.10	0.21

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**23.2315**	**0.3656**	**23.2315**
A1 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.04	0.10
A2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.09
B1 to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.01	0.09	0.04	0.10
B2 to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A1	1.5600
A2	1.4995
B1	1.5469
B2	1.5248

Leakage [nW]
8.59

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	23.2315	0.3656	23.2315
A1 to ZN	fall	0.28	0.33	0.59	0.32
	rise	1.99	2.07	2.12	2.06
A2 to ZN	fall	0.52	0.61	0.83	0.62
	rise	2.57	2.58	2.27	2.48
B1 to ZN	fall	0.28	0.33	0.30	0.24
	rise	2.93	2.97	3.05	2.97
B2 to ZN	fall	0.53	0.61	0.59	0.54
	rise	3.50	3.52	3.25	3.48

OAI22_X2

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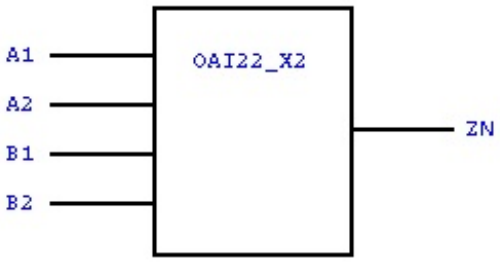
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!((A1 | A2) & (B1 | B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	L	-	-	H
H	-	H	-	L
H	-	-	H	L
-	H	H	-	L
-	H	-	H	L
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**46.3104**	**0.3656**	**46.3104**
A1 to ZN	**fall**	0.01	0.06	0.04	0.14
	rise	0.01	0.10	0.06	0.21
A2 to ZN	**fall**	0.01	0.06	0.05	0.15
	rise	0.02	0.10	0.08	0.21
B1 to ZN	**fall**	0.01	0.06	0.05	0.14
	rise	0.02	0.11	0.09	0.21
B2 to ZN	**fall**	0.01	0.06	0.06	0.15
	rise	0.02	0.11	0.10	0.21

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**46.3104**	**0.3656**	**46.3104**
A1 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.04	0.10
A2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.09
B1 to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.01	0.09	0.04	0.10
B2 to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A1	2.9096
A2	3.2276
B1	2.9202
B2	3.1499

Leakage [nW]
17.19

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	46.3104	0.3656	46.3104
A1 to ZN	fall	0.55	0.66	1.19	0.64
	rise	4.10	4.25	4.35	4.33
A2 to ZN	fall	1.05	1.21	1.67	1.23
	rise	5.24	5.20	4.66	5.07
B1 to ZN	fall	0.56	0.66	0.60	0.48
	rise	5.99	6.19	6.21	6.32
B2 to ZN	fall	1.06	1.22	1.18	1.07
	rise	7.11	7.21	6.62	7.16

OAI22_X4

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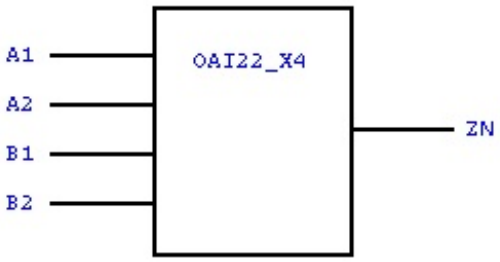
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Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Equation	ZN = "!((A1 | A2) & (B1 | B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	L	-	-	H
H	-	H	-	L
H	-	-	H	L
-	H	H	-	L
-	H	-	H	L
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**92.4683**	**0.3656**	**92.4683**
A1 to ZN	**fall**	0.01	0.06	0.04	0.15
	rise	0.01	0.10	0.06	0.21
A2 to ZN	**fall**	0.01	0.06	0.05	0.15
	rise	0.02	0.10	0.08	0.21
B1 to ZN	**fall**	0.01	0.06	0.05	0.14
	rise	0.02	0.11	0.09	0.21
B2 to ZN	**fall**	0.01	0.06	0.06	0.15
	rise	0.02	0.11	0.10	0.22

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**92.4683**	**0.3656**	**92.4683**
A1 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.04	0.10
A2 to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.09
B1 to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.01	0.09	0.04	0.10
B2 to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A1	6.0132
A2	6.1949
B1	6.0319
B2	6.1105

Leakage [nW]
34.37

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	92.4683	0.3656	92.4683
A1 to ZN	fall	1.10	1.32	2.38	1.27
	rise	8.24	8.46	8.76	8.68
A2 to ZN	fall	2.09	2.43	3.34	2.45
	rise	10.59	10.65	9.38	10.15
B1 to ZN	fall	1.12	1.31	1.21	0.95
	rise	12.05	12.37	12.49	12.63
B2 to ZN	fall	2.12	2.44	2.35	2.15
	rise	14.29	14.29	13.34	14.30

OAI33_X1

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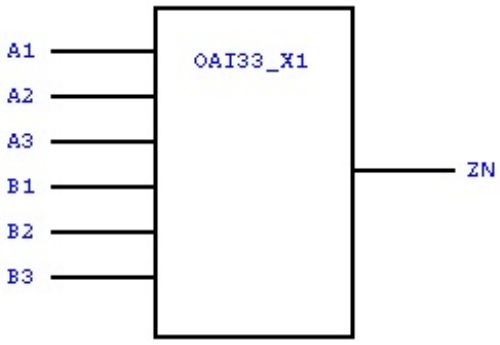
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Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.862 um^2
Equation	ZN = "!(((A1 | A2) | A3) & ((B1 | B2) | B3))"
Type	Combinational
Input	A1, A2, A3, B1, B2, B3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**A3**	**B1**	**B2**	**B3**	**ZN**
L	L	L	-	-	-	H
H	-	-	H	-	-	L
H	-	-	-	H	-	L
H	-	-	-	-	H	L
-	H	-	H	-	-	L
-	H	-	-	H	-	L
-	H	-	-	-	H	L
-	-	H	H	-	-	L
-	-	H	-	H	-	L
-	-	H	-	-	H	L
-	-	-	L	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**11.4822**	**0.3656**	**11.4822**
A1 to ZN	**fall**	0.01	0.04	0.05	0.11
	rise	0.02	0.09	0.08	0.18
A2 to ZN	**fall**	0.01	0.04	0.06	0.11
	rise	0.03	0.09	0.10	0.19
A3 to ZN	**fall**	0.02	0.04	0.07	0.12
	rise	0.03	0.10	0.10	0.18
B1 to ZN	**fall**	0.02	0.04	0.06	0.11
	rise	0.04	0.10	0.11	0.20
B2 to ZN	**fall**	0.02	0.04	0.07	0.12
	rise	0.04	0.11	0.13	0.21
B3 to ZN	**fall**	0.02	0.05	0.08	0.12
	rise	0.05	0.11	0.13	0.20

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**11.4822**	**0.3656**	**11.4822**
A1 to ZN	**fall**	0.01	0.03	0.03	0.06
	rise	0.02	0.08	0.05	0.09
A2 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.02	0.08	0.04	0.09
A3 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.02	0.08	0.03	0.08
B1 to ZN	**fall**	0.01	0.03	0.02	0.04
	rise	0.02	0.08	0.05	0.09
B2 to ZN	**fall**	0.01	0.03	0.02	0.04
	rise	0.02	0.08	0.04	0.09
B3 to ZN	**fall**	0.01	0.03	0.02	0.04
	rise	0.02	0.08	0.04	0.09

Capacitance [fF]	
A1	1.5519
A2	1.5242
A3	1.4920
B1	1.5296
B2	1.5234
B3	1.4957

Leakage [nW]
13.24

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	11.4822	0.3656	11.4822
A1 to ZN	fall	0.35	0.40	0.62	0.47
	rise	2.37	2.36	2.50	2.52
A2 to ZN	fall	0.62	0.69	0.88	0.77
	rise	2.95	2.96	2.71	2.81
A3 to ZN	fall	0.77	0.92	1.09	1.01
	rise	3.46	3.45	3.16	3.32
B1 to ZN	fall	0.35	0.40	0.35	0.33
	rise	3.75	3.80	3.85	3.88
B2 to ZN	fall	0.63	0.70	0.65	0.65
	rise	4.30	4.33	4.13	4.27
B3 to ZN	fall	0.79	0.93	0.89	0.90
	rise	4.82	4.86	4.60	4.77

NANGATE OR2_X1

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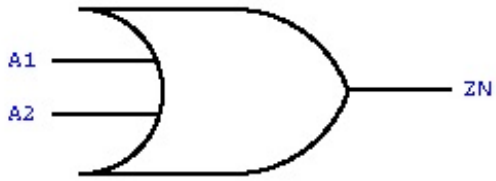
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	L
H	-	H
-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.03	0.12	0.11	0.20
	rise	0.02	0.13	0.06	0.18
A2 to ZN	**fall**	0.04	0.12	0.12	0.21
	rise	0.02	0.13	0.06	0.18

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A2 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A1	0.8679
A2	0.8826

Leakage [nW]
4.36

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	1.84	2.50	1.84	2.60
	rise	1.25	1.19	1.23	1.48
A2 to ZN	**fall**	2.12	2.78	1.91	2.71
	rise	1.38	1.63	1.37	1.39

OR2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

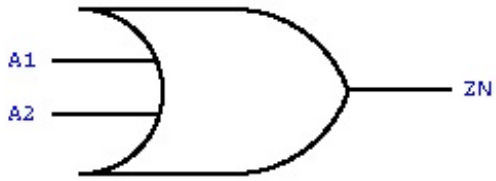
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	L
H	-	H
-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	0.03	0.12	0.10	0.20
	rise	0.02	0.13	0.05	0.17
A2 to ZN	**fall**	0.03	0.12	0.11	0.20
	rise	0.02	0.13	0.06	0.18

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A2 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A1	1.5767
A2	1.5710

Leakage [nW]
8.78

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	3.55	4.78	3.60	5.00
	rise	2.48	2.45	2.46	2.58
A2 to ZN	**fall**	4.13	5.34	3.68	5.23
	rise	2.71	3.19	2.72	3.15

NANGATE OR2_X4

Databook Build Date: Thursday Feb 17 15:07 2011

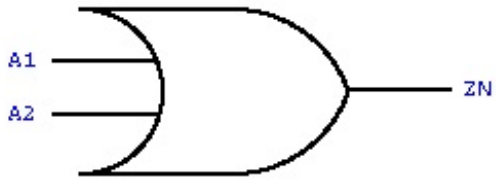
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	L
H	-	H
-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.03	0.12	0.10	0.19
	rise	0.01	0.13	0.05	0.17
A2 to ZN	**fall**	0.03	0.12	0.11	0.20
	rise	0.02	0.13	0.06	0.18

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A2 to ZN	**fall**	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A1	3.0511
A2	3.1822

Leakage [nW]
17.57

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	6.89	9.32	7.05	9.81
	rise	4.52	5.32	4.59	4.99
A2 to ZN	**fall**	8.03	10.46	7.17	10.26
	rise	5.03	6.21	5.11	5.32

NANGATE OR3_X1

Databook Build Date: Thursday Feb 17 15:07 2011

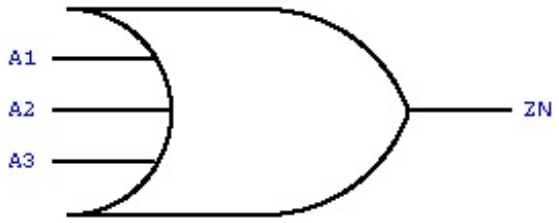
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	L
H	-	-	H
-	H	-	H
-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.05	0.14	0.13	0.23
	rise	0.02	0.13	0.06	0.18
A2 to ZN	**fall**	0.06	0.15	0.14	0.24
	rise	0.02	0.13	0.07	0.19
A3 to ZN	**fall**	0.06	0.15	0.14	0.24
	rise	0.02	0.14	0.07	0.19

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A2 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A3 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A1	0.8766
A2	0.8841
A3	0.8672

Leakage [nW]
4.96

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	1.91	2.76	2.00	2.85
	rise	1.26	1.11	1.23	1.37
A2 to ZN	**fall**	2.18	3.04	2.07	2.97
	rise	1.38	1.52	1.35	1.56
A3 to ZN	**fall**	2.44	3.29	2.27	3.21
	rise	1.48	1.60	1.48	1.65

NANGATE OR3_X2

Databook Build Date: Thursday Feb 17 15:07 2011

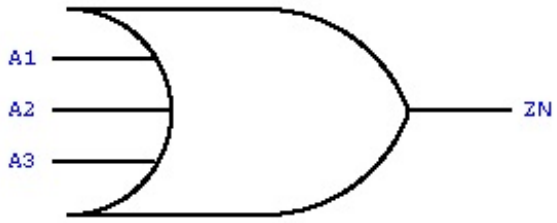
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Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.596 um^2
Equation	ZN = "((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	L
H	-	-	H
-	H	-	H
-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	0.05	0.14	0.12	0.22
	rise	0.02	0.13	0.06	0.18
A2 to ZN	**fall**	0.05	0.15	0.13	0.23
	rise	0.02	0.13	0.06	0.18
A3 to ZN	**fall**	0.06	0.15	0.13	0.23
	rise	0.02	0.13	0.07	0.19

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A2 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A3 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A1	1.5867
A2	1.5739
A3	1.5583

Leakage [nW]
10.00

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	3.62	5.22	3.85	5.40
	rise	2.43	2.25	2.42	2.68
A2 to ZN	**fall**	4.21	5.79	3.96	5.66
	rise	2.68	3.10	2.68	2.98
A3 to ZN	**fall**	4.71	6.30	4.35	6.16
	rise	2.85	3.48	2.89	3.21

NANGATE OR3_X4

Databook Build Date: Thursday Feb 17 15:07 2011

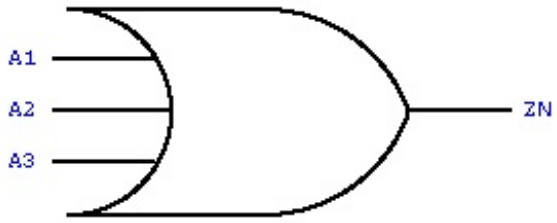
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Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	ZN = "((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	L
H	-	-	H
-	H	-	H
-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.04	0.14	0.12	0.22
	rise	0.02	0.13	0.06	0.18
A2 to ZN	**fall**	0.05	0.14	0.13	0.23
	rise	0.02	0.13	0.06	0.18
A3 to ZN	**fall**	0.05	0.15	0.13	0.23
	rise	0.02	0.13	0.07	0.19

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A2 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A3 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A1	3.0284
A2	3.1297
A3	3.1595

Leakage [nW]
20.01

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	6.91	10.00	7.42	10.38
	rise	4.52	5.17	4.53	5.13
A2 to ZN	**fall**	8.04	11.13	7.61	10.88
	rise	5.13	5.91	5.04	5.94
A3 to ZN	**fall**	9.08	12.16	8.41	11.88
	rise	5.37	6.03	5.57	5.95

OR4_X1

Databook Build Date: Thursday Feb 17 15:07 2011

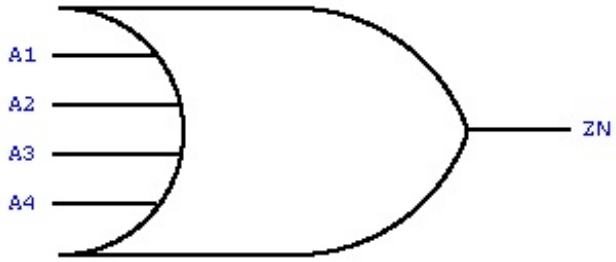
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Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	L
H	-	-	-	H
-	H	-	-	H
-	-	H	-	H
-	-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.06	0.17	0.15	0.25
	rise	0.02	0.13	0.06	0.18
A2 to ZN	**fall**	0.08	0.18	0.16	0.27
	rise	0.02	0.13	0.07	0.19
A3 to ZN	**fall**	0.08	0.19	0.17	0.27
	rise	0.02	0.14	0.07	0.19
A4 to ZN	**fall**	0.09	0.19	0.17	0.27
	rise	0.02	0.14	0.08	0.20

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A2 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A3 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A4 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A1	0.8577
A2	0.8785
A3	0.8721
A4	0.8620

Leakage [nW]
5.78

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	fall	1.93	2.91	2.06	2.97
	rise	1.24	1.44	1.23	1.16
A2 to ZN	fall	2.21	3.18	2.15	3.11
	rise	1.37	1.58	1.36	1.26
A3 to ZN	fall	2.46	3.44	2.36	3.36
	rise	1.48	1.51	1.48	1.46
A4 to ZN	fall	2.71	3.68	2.59	3.61
	rise	1.51	1.71	1.58	1.72

NANGATE OR4_X2

Databook Build Date: Thursday Feb 17 15:07 2011

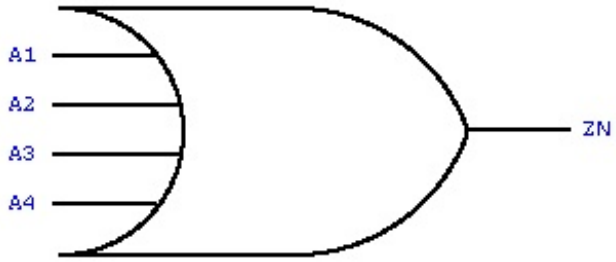
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	L
H	-	-	-	H
-	H	-	-	H
-	-	H	-	H
-	-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**120.85**	**0.3656**	**120.85**
A1 to ZN	**fall**	0.06	0.16	0.14	0.25
	rise	0.02	0.13	0.06	0.18
A2 to ZN	**fall**	0.07	0.17	0.15	0.26
	rise	0.02	0.13	0.07	0.18
A3 to ZN	**fall**	0.08	0.18	0.16	0.26
	rise	0.02	0.13	0.07	0.19
A4 to ZN	**fall**	0.08	0.18	0.16	0.26
	rise	0.02	0.14	0.07	0.19

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**120.85**	**0.3656**	**120.85**
A1 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A2 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A3 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A4 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A1	1.5590
A2	1.5520
A3	1.5402
A4	1.5266

Leakage [nW]
11.67

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	fall	3.64	5.47	3.94	5.62
	rise	2.48	2.93	2.42	2.66
A2 to ZN	fall	4.20	6.03	4.08	5.90
	rise	2.72	3.07	2.69	3.10
A3 to ZN	fall	4.74	6.55	4.52	6.40
	rise	2.90	3.55	2.89	3.47
A4 to ZN	fall	5.25	7.06	4.98	6.91
	rise	2.97	3.67	3.08	3.56

OR4_X4

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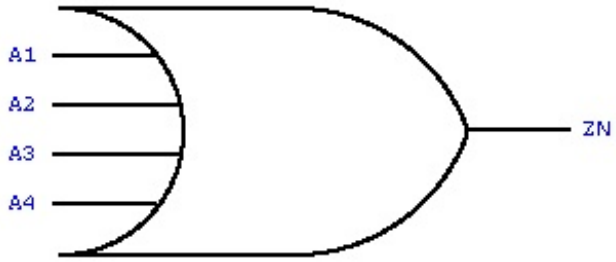
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Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	L
H	-	-	-	H
-	H	-	-	H
-	-	H	-	H
-	-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**241.699**	**0.3656**	**241.699**
A1 to ZN	**fall**	0.06	0.16	0.14	0.24
	rise	0.02	0.13	0.06	0.18
A2 to ZN	**fall**	0.07	0.17	0.15	0.26
	rise	0.02	0.13	0.06	0.18
A3 to ZN	**fall**	0.08	0.18	0.16	0.26
	rise	0.02	0.13	0.07	0.19
A4 to ZN	**fall**	0.08	0.18	0.16	0.26
	rise	0.02	0.14	0.07	0.19

Output Transition [ns]					
Input Transition [ns]		**0.0012**		**0.1792**	
Load Capacitance [fF]		**0.3656**	**241.699**	**0.3656**	**241.699**
A1 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A2 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A3 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
A4 to ZN	**fall**	0.01	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10

Capacitance [fF]	
A1	2.9927
A2	3.1203
A3	3.2604
A4	3.3970

Leakage [nW]
23.35

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	fall	7.14	10.71	7.74	11.04
	rise	4.62	5.46	4.61	5.23
A2 to ZN	fall	8.23	11.84	8.00	11.58
	rise	5.21	5.80	5.16	6.22
A3 to ZN	fall	9.30	12.88	8.86	12.58
	rise	5.54	6.38	5.61	5.85
A4 to ZN	fall	10.28	13.89	9.74	13.61
	rise	5.70	7.11	5.98	7.27

NANGATE SDFFRS_X1

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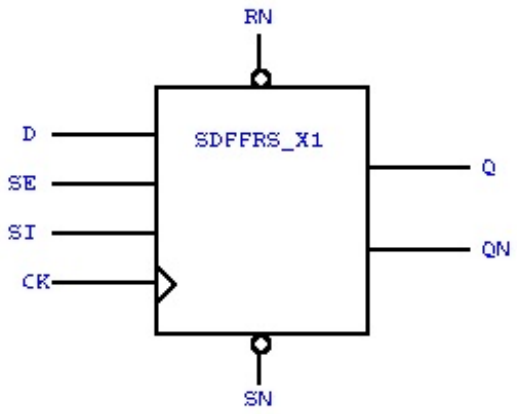
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Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, scan, and drive strength X1
Strength	1
Cell Area	7.714 um^2
Equation	Q = "(!(!RN)*((((SE * SI) + (D * !SE)))+(!SN)))" QN = "(!(!SN)*(!((((SE * SI) + (D * !SE)))+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table									
CK	D	RN	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	H	-	-	L	H
R	H	H	L	-	H	-	-	H	L
R	-	H	H	L	H	-	-	L	H
R	-	H	H	H	H	-	-	H	L
F	-	H	-	-	H	L	H	L	H
F	-	H	-	-	H	H	L	H	L
-	-	L	-	-	L	-	-	L	L
-	-	L	-	-	H	-	-	L	H
-	-	H	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.09	*0.17*	0.14	*0.22*
	rise	0.10	*0.21*	0.15	*0.27*
CK to QN	fall	0.08	0.16	0.13	0.21
	rise	0.07	0.19	0.12	0.24
RN to Q	fall	0.02	*0.10*	0.10	*0.19*
	rise	0.03	*0.14*	0.08	*0.20*
RN to QN	rise	0.15	0.27	0.24	0.36
SN to Q	rise	0.05	*0.17*	0.15	*0.26*
SN to QN	fall	0.03	0.11	0.11	0.21
	rise	0.14	0.25	0.20	0.32

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.00	*0.07*	0.00	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*
CK to QN	fall	0.00	0.07	0.00	0.07
	rise	0.01	0.10	0.01	0.10
RN to Q	fall	0.00	*0.07*	0.01	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*
RN to QN	rise	0.01	0.10	0.01	0.10
SN to Q	rise	0.01	*0.10*	0.01	*0.10*
SN to QN	fall	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10

Constraints Time [ns]		
Setup CK to D	fall	0.14
	rise	0.10
Hold CK to D	fall	0.13
	rise	0.05
Recovery CK to RN	rise	0.00
Removal CK to RN	rise	0.44
Setup CK to SE	fall	0.13
	rise	0.11
Hold CK to SE	fall	0.09
	rise	0.07
Setup CK to SI	fall	0.15
	rise	0.11
Hold CK to SI	fall	0.12
	rise	0.04
Recovery CK to SN	rise	0.06
Removal CK to SN	rise	0.25

Capacitance [fF]	
CK	0.8730
D	1.0942
RN	2.1353
SE	2.0222
SI	0.8260
SN	1.4743

Leakage [nW]
27.91

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	4134.71	4277.21
	rise	4824.26	4897.48
D	fall	2144.04	2139.35
	rise	1280.15	1248.40
RN	fall	1131.96	1162.27
	rise	-141.54	-119.56
SE	fall	2932.93	3262.03
	rise	1284.98	1442.21
SI	fall	1922.47	1914.06
	rise	1003.09	979.16
SN	fall	467.54	454.10
	rise	-237.83	-246.32

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	4.10	*4.43*	4.13	*4.40*
	rise	4.10	*4.29*	4.15	*4.34*
CK to QN	fall	4.10	4.35	4.15	4.37
	rise	4.10	4.44	4.13	4.37
RN to Q	fall	3.75	*4.29*	3.71	*4.49*
	rise	1.84	*2.13*	1.90	*2.41*
RN to QN	rise	3.86	4.13	4.00	4.29
SN to Q	rise	2.78	*3.82*	2.72	*3.91*
SN to QN	fall	2.76	4.25	2.61	4.28
	rise	2.94	3.25	2.90	3.31

SDFFRS_X2

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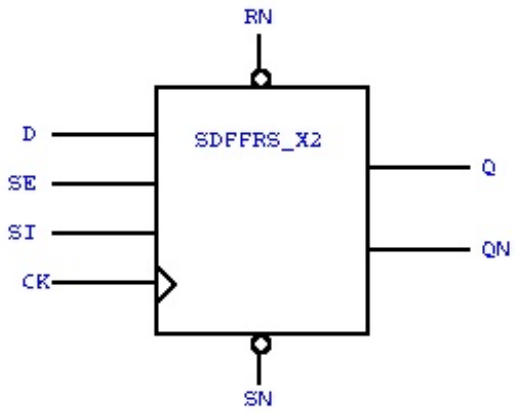
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, scan, and drive strength X2
Strength	2
Cell Area	8.246 um^2
Equation	Q = "(!(!RN)*((((SE * SI) + (D * !SE)))+(!SN)))" QN = "(!(!SN)*(!((((SE * SI) + (D * !SE)))+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table									
CK	D	RN	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	H	-	-	L	H
R	H	H	L	-	H	-	-	H	L
R	-	H	H	L	H	-	-	L	H
R	-	H	H	H	H	-	-	H	L
F	-	H	-	-	H	L	H	L	H
F	-	H	-	-	H	H	L	H	L
-	-	L	-	-	L	-	-	L	L
-	-	L	-	-	H	-	-	L	H
-	-	H	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.10	0.18	0.15	0.23
	rise	0.11	0.22	0.16	0.27
CK to QN	fall	0.08	0.16	0.13	0.22
	rise	0.08	0.20	0.13	0.25
RN to Q	fall	0.02	0.10	0.09	0.19
	rise	0.03	0.14	0.08	0.20
RN to QN	rise	0.15	0.27	0.26	0.38
SN to Q	rise	0.05	0.17	0.15	0.26
SN to QN	fall	0.03	0.11	0.11	0.20
	rise	0.14	0.26	0.21	0.33

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.00	0.07	0.00	0.07
	rise	0.01	0.10	0.01	0.10
CK to QN	fall	0.00	0.07	0.00	0.07
	rise	0.01	0.10	0.01	0.10
RN to Q	fall	0.00	0.07	0.01	0.07
	rise	0.00	0.10	0.01	0.10
RN to QN	rise	0.01	0.10	0.01	0.10
SN to Q	rise	0.00	0.10	0.01	0.10
SN to QN	fall	0.00	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10

Constraints Time [ns]		
Setup CK to D	fall	0.14
	rise	0.10
Hold CK to D	fall	0.13
	rise	0.05
Recovery CK to RN	rise	0.02
Removal CK to RN	rise	0.45
Setup CK to SE	fall	0.13
	rise	0.11
Hold CK to SE	fall	0.09
	rise	0.07
Setup CK to SI	fall	0.16
	rise	0.11
Hold CK to SI	fall	0.12
	rise	0.04
Recovery CK to SN	rise	0.09
Removal CK to SN	rise	0.24

Capacitance [fF]	
CK	0.8583
D	1.0717
RN	2.5163
SE	1.9208
SI	0.8268
SN	1.7638

Leakage [nW]
36.26

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	4229.73	4368.49
	rise	5211.91	5283.77
D	fall	2186.05	2180.09
	rise	1203.96	1166.62
RN	fall	1379.86	1434.65
	rise	-242.10	-222.69
SE	fall	2915.31	3246.84
	rise	1284.87	1443.06
SI	fall	1956.52	1947.75
	rise	940.61	905.14
SN	fall	716.05	728.64
	rise	-325.16	-336.51

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	5.68	6.32	5.69	6.34
	rise	5.79	6.31	5.84	6.18
CK to QN	fall	5.79	6.31	5.84	6.18
	rise	5.68	6.32	5.69	6.34
RN to Q	fall	5.10	6.05	5.06	6.11
	rise	2.74	3.77	2.95	3.20
RN to QN	rise	5.41	5.90	5.62	5.82
SN to Q	rise	4.27	5.51	4.20	5.62
SN to QN	fall	4.20	6.03	4.03	6.09
	rise	4.16	4.73	4.15	4.40

SDFFR_X1

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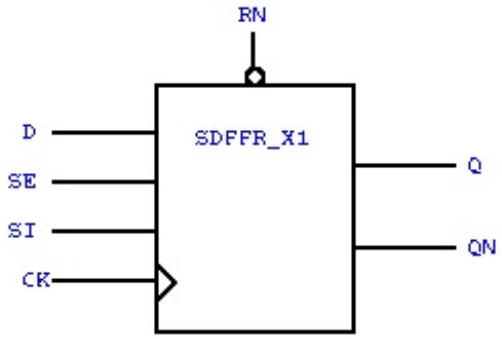
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, scan, and drive strength X1
Strength	1
Cell Area	6.650 um^2
Equation	Q = "(!(!RN)*(((SE * SI) + (D * !SE))))" QN = "((!RN)+!(((SE * SI) + (D * !SE))))"
Clock	CK
Reset	RN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	RN	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	-	-	L	H
R	H	H	L	-	-	-	H	L
R	-	H	H	L	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	H	-	-	L	H	L	H
F	-	H	-	-	H	L	H	L
-	-	L	-	-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
CK to Q	fall	0.06	0.15	0.11	0.20
	rise	0.06	0.18	0.11	0.23
CK to QN	fall	0.09	0.17	0.14	0.22
	rise	0.09	0.20	0.14	0.25
RN to Q	fall	0.13	0.23	0.24	0.33
RN to QN	rise	0.16	0.27	0.27	0.38

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
CK to Q	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
CK to QN	fall	0.00	0.07	0.00	0.07
	rise	0.01	0.10	0.01	0.10
RN to Q	fall	0.01	0.07	0.01	0.07
RN to QN	rise	0.01	0.10	0.01	0.10

Constraints Time [ns]		
Setup CK to D	fall	0.14
	rise	0.11
Hold CK to D	fall	0.13
	rise	0.05
Recovery CK to RN	rise	0.01
Removal CK to RN	rise	0.36
Setup CK to SE	fall	0.13
	rise	0.11
Hold CK to SE	fall	0.09
	rise	0.07
Setup CK to SI	fall	0.16
	rise	0.11
Hold CK to SI	fall	0.12
	rise	0.05

Capacitance [fF]	
CK	0.9484
D	1.1002
RN	1.4461
SE	1.9152
SI	0.8420

Leakage [nW]
23.55

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	4192.02	4336.07
	rise	3102.80	3185.14
D	fall	1904.28	1898.91
	rise	958.69	930.73
RN	fall	764.53	771.87
	rise	-166.20	-174.63
SE	fall	2807.77	3123.55
	rise	1100.30	1234.73
SI	fall	1841.25	1833.34
	rise	788.62	766.03

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
CK to Q	fall	3.39	3.68	3.42	3.72
	rise	3.40	3.79	3.45	3.74
CK to QN	fall	3.40	3.79	3.45	3.74
	rise	3.39	3.68	3.42	3.72
RN to Q	fall	3.40	3.67	3.38	3.65
RN to QN	rise	3.40	3.67	3.38	3.65

SDFFR_X2

Databook Build Date: Thursday Feb 17 15:07 2011

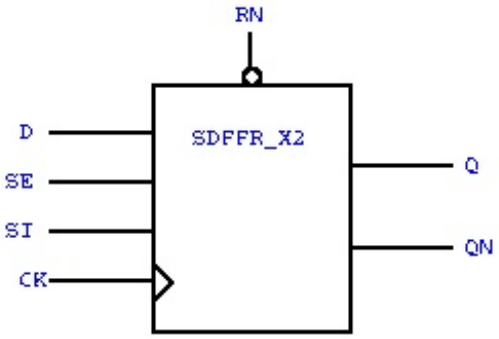
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, scan, and drive strength X2
Strength	2
Cell Area	6.916 um^2
Equation	Q = "(!(!RN)*(((SE * SI) + (D * !SE))))" QN = "((!RN)+!(((SE * SI) + (D * !SE))))"
Clock	CK
Reset	RN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	RN	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	-	-	L	H
R	H	H	L	-	-	-	H	L
R	-	H	H	L	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	H	-	-	L	H	L	H
F	-	H	-	-	H	L	H	L
-	-	L	-	-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.07	*0.18*	0.13	*0.23*
	rise	0.07	*0.20*	0.12	*0.25*
CK to QN	fall	0.10	0.18	0.15	0.23
	rise	0.11	0.21	0.16	0.26
RN to Q	fall	0.15	*0.25*	0.25	*0.35*
RN to QN	rise	0.18	0.29	0.28	0.39

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*
CK to QN	fall	0.00	0.07	0.00	0.07
	rise	0.01	0.10	0.01	0.10
RN to Q	fall	0.01	*0.07*	0.01	*0.07*
RN to QN	rise	0.01	0.10	0.01	0.10

Constraints Time [ns]		
Setup CK to D	fall	0.14
	rise	0.10
Hold CK to D	fall	0.13
	rise	0.05
Recovery CK to RN	rise	0.01
Removal CK to RN	rise	0.09
Setup CK to SE	fall	0.13
	rise	0.11
Hold CK to SE	fall	0.09
	rise	0.07
Setup CK to SI	fall	0.15
	rise	0.11
Hold CK to SI	fall	0.12
	rise	0.05

Capacitance [fF]	
CK	0.8836
D	1.0917
RN	1.4823
SE	1.8445
SI	0.8461

Leakage [nW]
30.81

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	4091.08	4235.24
	rise	3011.12	3085.62
D	fall	1877.67	1872.09
	rise	894.87	865.37
RN	fall	735.76	743.02
	rise	-236.57	-247.80
SE	fall	2789.32	3105.96
	rise	1077.10	1210.31
SI	fall	1814.28	1806.60
	rise	731.88	700.53

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	4.74	*5.38*	4.77	*5.39*
	rise	4.68	*5.44*	4.73	*5.41*
CK to QN	fall	4.68	5.42	4.73	5.33
	rise	4.74	5.42	4.77	5.47
RN to Q	fall	4.63	*5.32*	4.66	*5.21*
RN to QN	rise	4.63	5.22	4.66	5.26

SDFFS_X1

Databook Build Date: Thursday Feb 17 15:07 2011

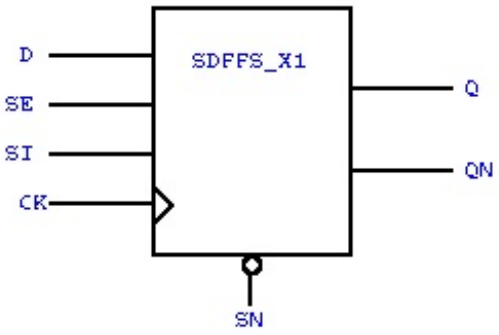
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, scan, and drive strength X1
Strength	1
Cell Area	6.650 um^2
Equation	Q = "((!SN)+(((SE * SI) + (D * !SE))))" QN = "(!((!SN)+(((SE * SI) + (D * !SE)))))"
Clock	CK
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	H	-	-	L	H
R	H	L	-	H	-	-	H	L
R	-	H	L	H	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	-	-	H	L	H	L	H
F	-	-	-	H	H	L	H	L
-	-	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.06	*0.15*	0.11	*0.20*
	rise	0.06	*0.18*	0.11	*0.23*
CK to QN	fall	0.08	0.16	0.13	0.21
	rise	0.09	0.20	0.14	0.25
SN to Q	rise	0.17	*0.28*	0.27	*0.39*
SN to QN	fall	0.02	0.11	0.10	0.19

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*
CK to QN	fall	0.00	0.07	0.00	0.07
	rise	0.01	0.10	0.01	0.10
SN to Q	rise	0.01	*0.11*	0.01	*0.11*
SN to QN	fall	0.00	0.07	0.01	0.07

Constraints Time [ns]		
Setup CK to D	fall	0.15
	rise	0.11
Hold CK to D	fall	0.12
	rise	0.05
Setup CK to SE	fall	0.13
	rise	0.11
Hold CK to SE	fall	0.08
	rise	0.07
Setup CK to SI	fall	0.16
	rise	0.11
Hold CK to SI	fall	0.12
	rise	0.04
Recovery CK to SN	rise	0.02
Removal CK to SN	rise	0.41

Capacitance [fF]	
CK	0.9030
D	1.1046
SE	1.9164
SI	0.8611
SN	1.2940

Leakage [nW]
23.97

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	4112.41	4243.84
	rise	3941.35	3992.76
D	fall	2078.58	2073.56
	rise	1039.98	1010.59
SE	fall	3068.61	3370.86
	rise	1302.02	1424.54
SI	fall	2016.11	2008.09
	rise	867.52	843.93
SN	fall	468.66	454.32
	rise	-210.61	-217.88

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	3.42	*3.74*	3.45	*3.65*
	rise	3.31	*3.67*	3.37	*3.72*
CK to QN	fall	3.31	3.67	3.37	3.73
	rise	3.42	3.65	3.45	3.72
SN to Q	rise	3.02	*3.35*	3.09	*3.32*
SN to QN	fall	3.02	3.32	3.09	3.30

SDFFS_X2

Databook Build Date: Thursday Feb 17 15:07 2011

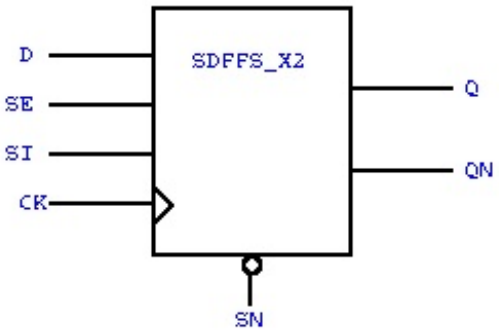
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, scan, and drive strength X2
Strength	2
Cell Area	7.182 um^2
Equation	Q = "((!SN)+(((SE * SI) + (D * !SE))))" QN = "(!((!SN)+(((SE * SI) + (D * !SE)))))"
Clock	CK
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	H	-	-	L	H
R	H	L	-	H	-	-	H	L
R	-	H	L	H	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	-	-	H	L	H	L	H
F	-	-	-	H	H	L	H	L
-	-	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.07	*0.18*	0.12	*0.23*
	rise	0.07	*0.19*	0.12	*0.24*
CK to QN	fall	0.10	0.18	0.15	0.23
	rise	0.11	0.22	0.16	0.27
SN to Q	rise	0.19	*0.31*	0.30	*0.42*
SN to QN	fall	0.02	0.10	0.09	0.18

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*
CK to QN	fall	0.00	0.07	0.00	0.07
	rise	0.01	0.10	0.01	0.10
SN to Q	rise	0.01	*0.11*	0.01	*0.11*
SN to QN	fall	0.00	0.07	0.01	0.07

Constraints Time [ns]		
Setup CK to D	fall	0.15
	rise	0.11
Hold CK to D	fall	0.12
	rise	0.04
Setup CK to SE	fall	0.14
	rise	0.12
Hold CK to SE	fall	0.08
	rise	0.06
Setup CK to SI	fall	0.16
	rise	0.12
Hold CK to SI	fall	0.11
	rise	0.04
Recovery CK to SN	rise	-0.00
Removal CK to SN	rise	0.33

Capacitance [fF]	
CK	0.8816
D	1.0694
SE	1.8131
SI	0.8732
SN	2.1672

Leakage [nW]
30.39

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	4197.27	4328.89
	rise	3894.50	3905.12
D	fall	2206.33	2201.66
	rise	1038.65	1007.79
SE	fall	3122.33	3426.55
	rise	1385.13	1509.02
SI	fall	2143.86	2136.05
	rise	874.34	841.74
SN	fall	913.33	925.17
	rise	-288.16	-298.35

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	4.82	*5.40*	4.87	*5.36*
	rise	4.96	*5.59*	5.02	*5.81*
CK to QN	fall	4.96	5.77	5.02	5.80
	rise	4.82	5.47	4.87	5.35
SN to Q	rise	4.67	*5.20*	4.84	*5.27*
SN to QN	fall	4.67	5.12	4.84	5.34

SDFF_X1

Databook Build Date: Thursday Feb 17 15:07 2011

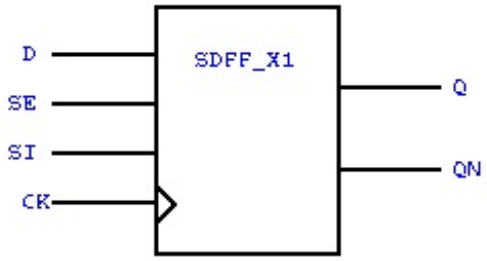
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with scan, and drive strength X1
Strength	1
Cell Area	6.118 um^2
Equation	Q = "(((SE * SI) + (D * !SE)))" QN = "(!(((SE * SI) + (D * !SE))))"
Clock	CK
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	-	-	L	H
R	H	L	-	-	-	H	L
R	-	H	L	-	-	L	H
R	-	H	H	-	-	H	L
F	-	-	-	L	H	L	H
F	-	-	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.05	0.14	0.10	0.19
	rise	0.05	0.17	0.10	0.22
CK to QN	fall	0.07	*0.15*	0.12	*0.20*
	rise	0.08	*0.19*	0.13	*0.24*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.01	0.07	0.01	0.07
	rise	0.01	0.10	0.01	0.10
CK to QN	fall	0.00	*0.07*	0.00	*0.07*
	rise	0.00	*0.10*	0.00	*0.10*

Constraints Time [ns]		
Setup CK to D	fall	0.15
	rise	0.11
Hold CK to D	fall	0.12
	rise	0.05
Setup CK to SE	fall	0.13
	rise	0.11
Hold CK to SE	fall	0.08
	rise	0.07
Setup CK to SI	fall	0.16
	rise	0.12
Hold CK to SI	fall	0.11
	rise	0.04

Capacitance [fF]	
CK	0.8787
D	1.0691
SE	1.7969
SI	0.8804

Leakage [nW]
21.71

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	3831.48	3974.52
	rise	1974.51	2062.31
D	fall	2051.81	2046.59
	rise	1054.97	1026.25
SE	fall	2893.29	3210.82
	rise	1224.27	1367.36
SI	fall	1988.46	1980.79
	rise	882.29	859.29

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	3.04	3.33	3.09	3.36
	rise	3.04	3.36	3.11	3.42
CK to QN	fall	3.03	*3.38*	3.11	*3.42*
	rise	3.03	*3.29*	3.09	*3.30*

SDFF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

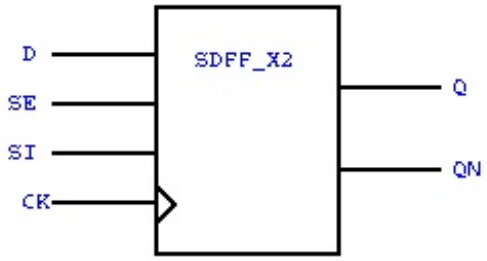
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with scan, and drive strength X2
Strength	2
Cell Area	6.384 um^2
Equation	Q = "(((SE * SI) + (D * !SE)))" QN = "(!(((SE * SI) + (D * !SE))))"
Clock	CK
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	-	-	L	H
R	H	L	-	-	-	H	L
R	-	H	L	-	-	L	H
R	-	H	H	-	-	H	L
F	-	-	-	L	H	L	H
F	-	-	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.07	*0.17*	0.12	*0.22*
	rise	0.06	*0.19*	0.11	*0.24*
CK to QN	fall	0.09	0.17	0.14	0.21
	rise	0.10	0.21	0.15	0.26

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.01	*0.07*	0.01	*0.07*
	rise	0.01	*0.10*	0.01	*0.10*
CK to QN	fall	0.00	0.07	0.00	0.07
	rise	0.01	0.10	0.01	0.10

Constraints Time [ns]		
Setup CK to D	fall	0.15
	rise	0.11
Hold CK to D	fall	0.12
	rise	0.05
Setup CK to SE	fall	0.13
	rise	0.11
Hold CK to SE	fall	0.08
	rise	0.07
Setup CK to SI	fall	0.16
	rise	0.12
Hold CK to SI	fall	0.11
	rise	0.04

Capacitance [fF]	
CK	0.9027
D	1.0774
SE	1.7705
SI	0.8620

Leakage [nW]
28.66

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
CK	fall	3881.96	4019.50
	rise	2030.37	2116.70
D	fall	2056.16	2050.57
	rise	1014.60	984.26
SE	fall	2884.88	3203.00
	rise	1226.46	1370.42
SI	fall	1993.06	1984.79
	rise	849.79	818.04

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	4.39	*5.06*	4.42	*5.07*
	rise	4.25	*4.97*	4.33	*5.04*
CK to QN	fall	4.25	4.96	4.33	5.00
	rise	4.39	5.06	4.42	4.95

TBUF_X16

Databook Build Date: Thursday Feb 17 15:07 2011

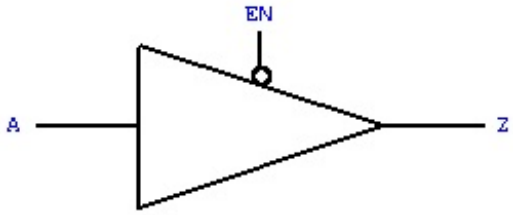
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	6.916 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		13.2634	833.2427	13.2634	833.2427
A to Z	**fall**	*0.03*	0.12	*0.11*	0.21
	rise	0.03	*0.15*	0.08	*0.21*
EN to Z	**fall**	*0.03*	0.12	*0.11*	0.21
	rise	0.04	*0.17*	0.12	*0.24*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		13.2634	833.2427	13.2634	833.2427
A to Z	**fall**	*0.01*	0.07	*0.01*	0.07
	rise	0.01	*0.11*	0.01	*0.11*
EN to Z	**fall**	*0.01*	0.07	*0.01*	0.07
	rise	0.01	*0.11*	0.01	*0.11*

Capacitance [fF]	
A	5.9783
EN	4.6000
Z	12.9307

Leakage [nW]
39.81

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		13.2634	833.2427	13.2634	833.2427
A to Z	**fall**	*20.02*	27.22	*23.86*	27.89
	rise	14.50	*18.37*	14.01	*19.14*
EN to Z	**fall**	*18.02*	22.57	*17.01*	22.88
	rise	19.88	*22.62*	19.47	*23.28*

TBUF_X1

Databook Build Date: Thursday Feb 17 15:07 2011

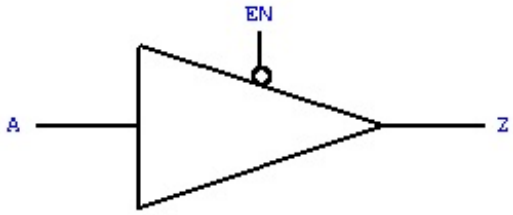
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.128 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		1.4062	52.6163	1.4062	52.6163
A to Z	**fall**	*0.03*	0.11	*0.10*	0.19
	rise	0.02	*0.13*	0.06	*0.17*
EN to Z	**fall**	*0.02*	0.11	*0.08*	0.18
	rise	0.03	*0.14*	0.09	*0.21*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		1.4062	52.6163	1.4062	52.6163
A to Z	**fall**	*0.00*	0.07	*0.01*	0.07
	rise	0.00	*0.10*	0.01	*0.10*
EN to Z	**fall**	*0.00*	0.07	*0.01*	0.07
	rise	0.00	*0.10*	0.00	*0.10*

Capacitance [fF]	
A	1.7551
EN	1.5978
Z	1.0416

Leakage [nW]
6.99

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		1.4062	52.6163	1.4062	52.6163
A to Z	**fall**	*3.34*	3.75	*2.90*	3.86
	rise	1.40	*1.61*	1.58	*1.75*
EN to Z	**fall**	*3.12*	3.45	*3.01*	3.85
	rise	3.08	*3.05*	2.97	*3.25*

TBUF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

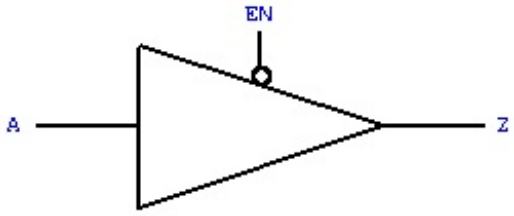
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		1.9821	105.2256	1.9821	105.2256
A to Z	**fall**	*0.02*	0.11	*0.09*	0.18
	rise	0.02	*0.13*	0.05	*0.17*
EN to Z	**fall**	*0.02*	0.10	*0.08*	0.17
	rise	0.03	*0.15*	0.11	*0.22*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		1.9821	105.2256	1.9821	105.2256
A to Z	**fall**	*0.00*	0.07	*0.01*	0.07
	rise	0.00	*0.10*	0.01	*0.10*
EN to Z	**fall**	*0.00*	0.07	*0.01*	0.07
	rise	0.00	*0.10*	0.00	*0.10*

Capacitance [fF]	
A	3.0563
EN	2.5658
Z	1.6186

Leakage [nW]
12.71

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		1.9821	105.2256	1.9821	105.2256
A to Z	**fall**	*5.88*	6.74	*5.01*	7.00
	rise	2.57	*3.11*	2.90	*3.39*
EN to Z	**fall**	*4.47*	5.16	*4.03*	5.75
	rise	4.73	*4.57*	4.38	*4.82*

TBUF_X4

Databook Build Date: Thursday Feb 17 15:07 2011

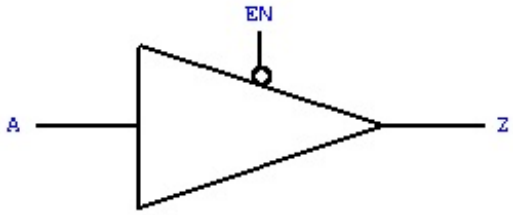
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		3.5482	210.0972	3.5482	210.0972
A to Z	**fall**	*0.02*	0.11	*0.09*	0.18
	rise	0.02	*0.14*	0.06	*0.18*
EN to Z	**fall**	*0.03*	0.11	*0.10*	0.19
	rise	0.04	*0.15*	0.11	*0.23*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		3.5482	210.0972	3.5482	210.0972
A to Z	**fall**	*0.00*	0.07	*0.01*	0.07
	rise	0.00	*0.10*	0.01	*0.10*
EN to Z	**fall**	*0.00*	0.07	*0.01*	0.07
	rise	0.00	*0.10*	0.00	*0.10*

Capacitance [fF]	
A	3.1099
EN	2.2493
Z	3.1882

Leakage [nW]
15.10

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		3.5482	210.0972	3.5482	210.0972
A to Z	**fall**	*6.67*	8.39	*6.43*	8.89
	rise	4.11	*5.12*	4.05	*5.03*
EN to Z	**fall**	*6.39*	7.57	*5.71*	7.72
	rise	6.82	*6.74*	6.47	*6.88*

TBUF_X8

Databook Build Date: Thursday Feb 17 15:07 2011

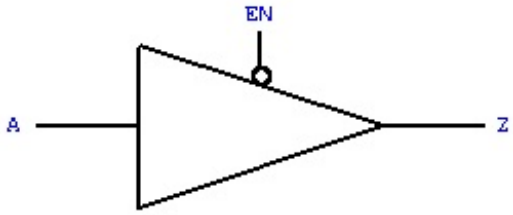
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	4.788 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		7.0285	419.2735	7.0285	419.2735
A to Z	**fall**	*0.02*	0.11	*0.09*	0.18
	rise	0.02	*0.14*	0.06	*0.19*
EN to Z	**fall**	*0.03*	0.11	*0.10*	0.19
	rise	0.04	*0.15*	0.11	*0.23*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		7.0285	419.2735	7.0285	419.2735
A to Z	**fall**	*0.00*	0.07	*0.01*	0.07
	rise	0.00	*0.10*	0.01	*0.10*
EN to Z	**fall**	*0.00*	0.07	*0.01*	0.07
	rise	0.00	*0.10*	0.00	*0.10*

Capacitance [fF]	
A	6.1819
EN	4.6416
Z	6.6755

Leakage [nW]
30.24

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		7.0285	419.2735	7.0285	419.2735
A to Z	**fall**	*13.53*	17.12	*13.29*	18.16
	rise	8.20	*9.45*	8.21	*8.80*
EN to Z	**fall**	*13.03*	15.38	*11.66*	15.70
	rise	14.02	*15.19*	13.33	*14.47*

TINV_X1

Databook Build Date: Thursday Feb 17 15:07 2011

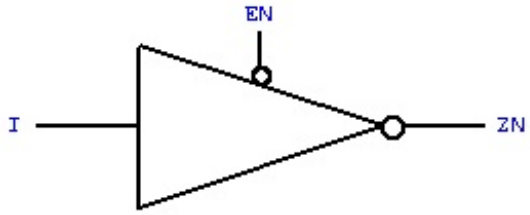
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!I", when !(EN)
Type	Tristate
Input	EN, I
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
I	**EN**	**ZN**
L	L	H
H	L	L
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		1.1578	23.4139	1.1578	23.4139
EN to ZN	**fall**	0.01	*0.07*	0.07	*0.13*
	rise	*0.01*	0.11	*0.03*	0.21
I to ZN	**fall**	0.01	*0.07*	0.04	*0.15*
	rise	*0.02*	0.11	*0.08*	0.22

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		1.1578	23.4139	1.1578	23.4139
EN to ZN	**fall**	0.00	*0.05*	0.01	*0.05*
	rise	*0.00*	0.09	*0.04*	0.10
I to ZN	**fall**	0.00	*0.05*	0.02	*0.06*
	rise	*0.01*	0.09	*0.02*	0.10

Capacitance [fF]	
EN	1.6934
I	1.3640
ZN	0.7927

Leakage [nW]
4.31

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		1.1578	23.4139	1.1578	23.4139
EN to ZN	**fall**	1.87	*1.99*	1.53	*2.13*
	rise	*2.17*	2.19	*1.70*	2.44
I to ZN	**fall**	0.28	*0.40*	0.40	*0.34*
	rise	*1.77*	1.67	*1.53*	1.68

TLAT_X1

Databook Build Date: Thursday Feb 17 15:07 2011

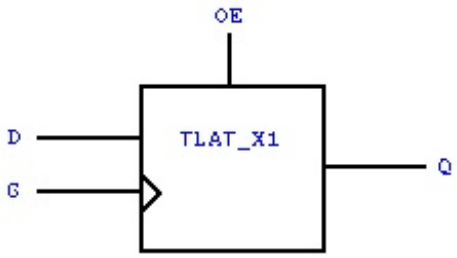
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	High enable Latch with drive strength X1
Strength	1
Cell Area	3.458 um^2
Equation	Q = "(D)", when !(!OE)
Enable	G
Type	Sequential
Input	D, OE
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
OE	**D**	**G**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	-	-	-	-	Z
H	L	H	-	-	L
H	H	H	-	-	H
H	-	L	L	H	L
H	-	L	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		1.1504	23.369	1.1504	23.369
D to Q	**fall**	*0.04*	0.10	*0.12*	0.18
	rise	0.04	*0.14*	0.09	*0.19*
G to Q	**fall**	*0.05*	0.10	*0.09*	0.15
	rise	0.05	*0.15*	0.10	*0.20*
OE to Q	**fall**	*0.00*	0.06	*0.00*	0.14
	rise	0.02	*0.11*	0.05	*0.15*

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		1.1504	23.369	1.1504	23.369
D to Q	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.01	*0.09*	0.01	*0.09*
G to Q	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.01	*0.09*	0.01	*0.09*
OE to Q	**fall**	*0.00*	0.05	*0.03*	0.08
	rise	0.01	*0.09*	0.01	*0.09*

Constraints Time [ns]		
Setup G to D	**fall**	0.11
	rise	0.07
Hold G to D	**fall**	0.19
	rise	0.10

Capacitance [fF]	
D	1.0722
G	0.9339
OE	1.4085
Q	0.7860

Leakage [nW]
11.02

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0012	0.1792
D	fall	1025.96	1018.33
	rise	357.42	344.07
G	fall	2078.40	2252.54
	rise	2055.82	2178.57
OE	fall	1251.49	1460.09
	rise	611.67	759.04

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		1.1504	23.369	1.1504	23.369
D to Q	fall	*3.15*	3.53	*2.81*	3.48
	rise	2.94	*3.09*	2.91	*2.99*
G to Q	fall	*3.83*	4.22	*3.94*	4.30
	rise	4.59	*4.77*	4.76	*4.95*
OE to Q	fall	*0.29*	0.57	*0.41*	0.73
	rise	1.02	*1.08*	1.07	*1.15*

XNOR2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

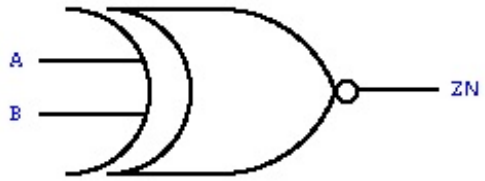
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "!(A ^ B)"
Type	Combinational
Input	A, B
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**ZN**
L	L	H
L	H	L
H	L	L
H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	26.0162	0.3656	26.0162
A to ZN	**fall**	0.03	0.08	0.10	0.16
	rise	0.03	0.11	0.08	0.22
B to ZN	**fall**	0.03	0.08	0.11	0.16
	rise	0.03	0.11	0.08	0.22

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	26.0162	0.3656	26.0162
A to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.04	0.10
B to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A	2.1728
B	2.4787

Leakage [nW]
8.18

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	26.0162	0.3656	26.0162
A to ZN	**fall**	1.79	1.68	1.69	1.77
	rise	2.41	2.36	2.37	2.33
B to ZN	**fall**	1.91	2.02	1.89	2.07
	rise	2.47	2.46	2.19	2.28

XNOR2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

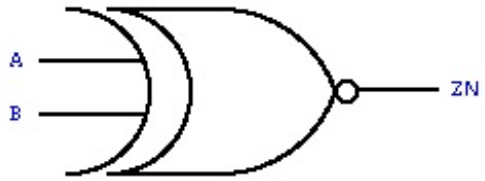
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.660 um^2
Equation	ZN = "!(A ^ B)"
Type	Combinational
Input	A, B
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**ZN**
L	L	H
L	H	L
H	L	L
H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	52.0325	0.3656	52.0325
A to ZN	**fall**	0.03	0.08	0.10	0.16
	rise	0.03	0.11	0.07	0.22
B to ZN	**fall**	0.03	0.08	0.10	0.16
	rise	0.03	0.11	0.08	0.22

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	52.0325	0.3656	52.0325
A to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.10
B to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A	3.8945
B	4.6552

Leakage [nW]
16.41

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	52.0325	0.3656	52.0325
A to ZN	**fall**	3.33	3.11	3.15	3.30
	rise	4.79	4.77	4.75	4.74
B to ZN	**fall**	3.59	3.80	3.56	3.91
	rise	5.00	4.95	4.41	4.62

XOR2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

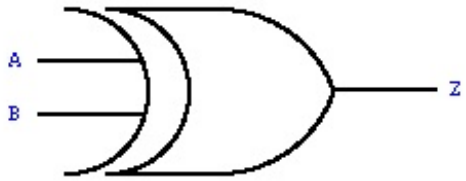
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	Z = "(A ^ B)"
Type	Combinational
Input	A, B
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**Z**
L	L	L
L	H	H
H	L	H
H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	25.3296	0.3656	25.3296
A to Z	**fall**	0.04	0.07	0.12	0.16
	rise	0.03	0.12	0.08	0.21
B to Z	**fall**	0.04	0.08	0.12	0.16
	rise	0.03	0.12	0.08	0.22

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	25.3296	0.3656	25.3296
A to Z	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.04	0.10
B to Z	**fall**	0.01	0.05	0.02	0.06
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A	2.1886
B	2.3398

Leakage [nW]
7.82

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	25.3296	0.3656	25.3296
A to Z	**fall**	1.79	1.81	1.76	1.86
	rise	2.43	2.38	2.40	2.33
B to Z	**fall**	1.87	1.89	1.58	1.85
	rise	2.60	2.58	2.68	2.69

XOR2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

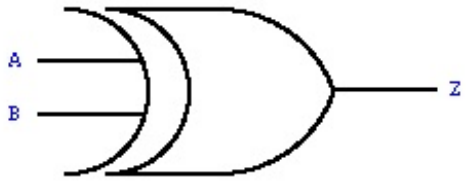
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_worst_low_worst_low**: Vdd= **.95**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_low_temp_low_temp].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	Z = "(A ^ B)"
Type	Combinational
Input	A, B
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**Z**
L	L	L
L	H	H
H	L	H
H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	50.5066	0.3656	50.5066
A to Z	**fall**	0.04	0.08	0.12	0.16
	rise	0.03	0.12	0.08	0.21
B to Z	**fall**	0.04	0.08	0.12	0.16
	rise	0.03	0.12	0.08	0.22

Output Transition [ns]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	50.5066	0.3656	50.5066
A to Z	**fall**	0.01	0.05	0.03	0.07
	rise	0.01	0.09	0.03	0.10
B to Z	**fall**	0.01	0.05	0.02	0.06
	rise	0.01	0.09	0.03	0.10

Capacitance [fF]	
A	4.2377
B	4.3419

Leakage [nW]
15.70

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0012		0.1792	
Load Capacitance [fF]		0.3656	50.5066	0.3656	50.5066
A to Z	**fall**	3.70	3.77	3.64	3.85
	rise	4.95	4.72	4.88	4.77
B to Z	**fall**	3.85	3.94	3.26	3.85
	rise	5.30	5.32	5.45	5.40

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Library

NangateOpenCellLibrary_low_temp

Corner

low_temp

AND2_X1

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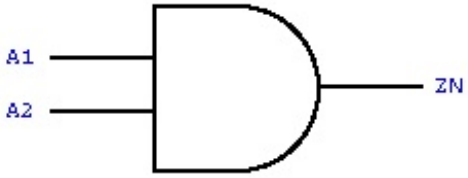
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	L
H	H	H
-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.01	0.07	0.03	0.10
A2 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.01	0.08	0.03	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A1	0.9339
A2	0.9797

Leakage [nW]
64.56

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	3.85	4.39	11.00	11.18
	rise	2.56	2.47	10.34	8.79
A2 to ZN	**fall**	4.28	4.88	12.10	11.73
	rise	2.55	2.64	8.85	8.26

AND2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

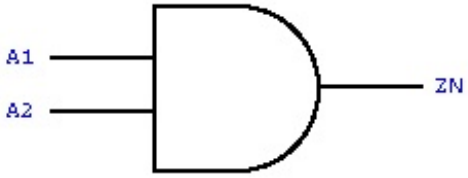
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	L
H	H	H
-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.01	0.07	0.03	0.09
A2 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.01	0.07	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A1	1.6896
A2	1.7600

Leakage [nW]
129.69

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	**fall**	7.35	8.36	21.80	22.10
	rise	4.97	5.26	20.95	18.77
A2 to ZN	**fall**	8.20	9.32	23.96	23.20
	rise	4.97	4.72	17.68	17.00

AND2_X4

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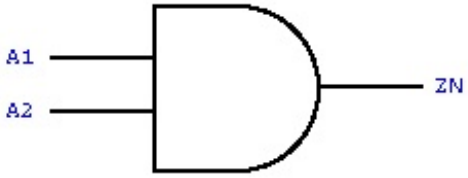
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	L
H	H	H
-	L	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.01	0.07	0.03	0.09
A2 to ZN	**fall**	0.01	0.07	0.03	0.10
	rise	0.01	0.07	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A1	3.2665
A2	3.5957

Leakage [nW]
259.38

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	**fall**	14.16	16.13	43.39	43.79
	rise	9.72	10.12	41.69	36.54
A2 to ZN	**fall**	15.91	18.10	47.74	45.96
	rise	9.72	9.08	35.08	33.62

AND3_X1

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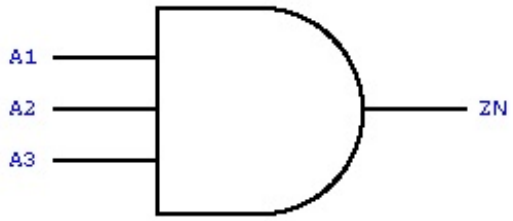
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	L
H	H	H	H
-	L	-	L
-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.4248**	**0.3656**	**60.4248**
A1 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.02	0.08	0.04	0.11
A2 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.02	0.08	0.04	0.11
A3 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.02	0.08	0.03	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.4248**	**0.3656**	**60.4248**
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A1	0.9295
A2	0.9348
A3	0.9777

Leakage [nW]
69.82

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.4248**	**0.3656**	**60.4248**
A1 to ZN	**fall**	4.05	4.67	10.57	10.52
	rise	3.19	2.51	10.29	8.58
A2 to ZN	**fall**	4.54	5.20	11.48	10.98
	rise	3.17	2.92	9.00	7.69
A3 to ZN	**fall**	4.97	5.65	12.52	12.01
	rise	3.21	3.09	8.83	8.19

AND3_X2

Databook Build Date: Thursday Feb 17 15:07 2011

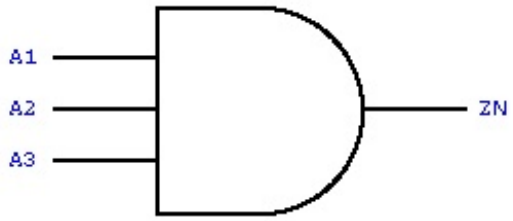
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.596 um^2
Equation	ZN = "((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	L
H	H	H	H
-	L	-	L
-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
A1 to ZN	**fall**	0.01	0.07	0.02	0.09
	rise	0.02	0.08	0.04	0.11
A2 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.02	0.08	0.04	0.11
A3 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.02	0.08	0.03	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.06

Capacitance [fF]	
A1	1.6962
A2	1.6767
A3	1.7395

Leakage [nW]
140.22

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
A1 to ZN	**fall**	7.72	8.84	20.86	20.64
	rise	6.23	5.26	20.70	16.81
A2 to ZN	**fall**	8.71	9.90	22.66	21.55
	rise	6.29	5.19	18.11	14.59
A3 to ZN	**fall**	9.58	10.81	24.85	23.67
	rise	6.28	4.81	17.65	15.49

AND3_X4

Databook Build Date: Thursday Feb 17 15:07 2011

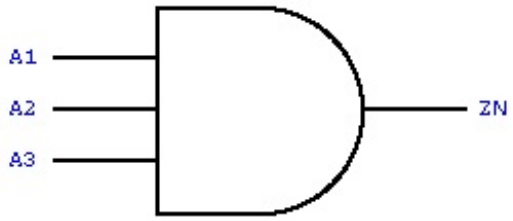
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	ZN = "((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	L
H	H	H	H
-	L	-	L
-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.01	0.07	0.02	0.09
	rise	0.02	0.08	0.04	0.11
A2 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.02	0.08	0.04	0.10
A3 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.02	0.08	0.03	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.06

Capacitance [fF]	
A1	3.2702
A2	3.3458
A3	3.6025

Leakage [nW]
280.44

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	14.93	17.15	41.41	40.87
	rise	12.11	10.23	41.12	34.33
A2 to ZN	**fall**	17.01	19.27	44.81	42.62
	rise	12.05	9.53	36.02	30.75
A3 to ZN	**fall**	18.67	21.11	49.33	46.91
	rise	12.09	10.94	35.45	31.10

AND4_X1

Databook Build Date: Thursday Feb 17 15:07 2011

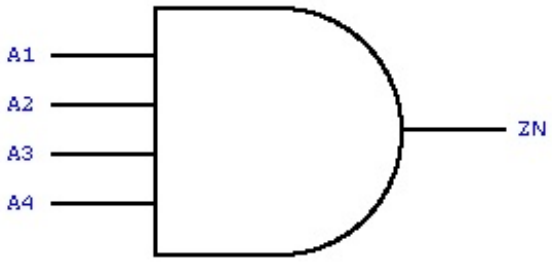
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	L
H	H	H	H	H
-	L	-	-	L
-	-	L	-	L
-	-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A1 to ZN	**fall**	0.01	0.07	0.02	0.09
	rise	0.02	0.09	0.06	0.12
A2 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.03	0.09	0.05	0.12
A3 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.03	0.09	0.05	0.11
A4 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.03	0.09	0.04	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.06
A4 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.06

Capacitance [fF]	
A1	0.9195
A2	0.9386
A3	0.9258
A4	0.9575

Leakage [nW]
72.07

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.2722	0.3656	60.2722
A1 to ZN	fall	4.30	4.96	10.28	10.13
	rise	3.87	3.24	10.33	7.94
A2 to ZN	fall	4.79	5.50	11.11	10.54
	rise	3.89	3.17	9.17	6.80
A3 to ZN	fall	5.27	6.00	11.94	11.41
	rise	3.90	3.26	8.99	7.58
A4 to ZN	fall	5.60	6.44	12.98	12.38
	rise	3.87	2.77	8.97	7.39

AND4_X2

Databook Build Date: Thursday Feb 17 15:07 2011

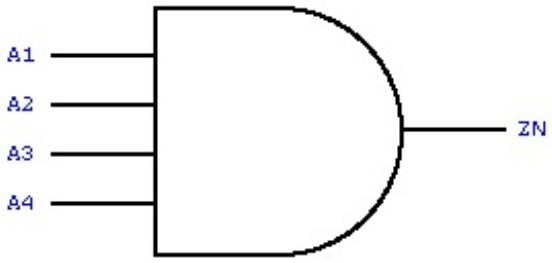
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	L
H	H	H	H	H
-	L	-	-	L
-	-	L	-	L
-	-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**120.392**	**0.3656**	**120.392**
A1 to ZN	**fall**	0.01	0.07	0.02	0.09
	rise	0.02	0.09	0.05	0.12
A2 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.02	0.09	0.05	0.12
A3 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.03	0.09	0.04	0.11
A4 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.03	0.09	0.03	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**120.392**	**0.3656**	**120.392**
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.06
A4 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.06

Capacitance [fF]	
A1	1.6781
A2	1.6776
A3	1.6548
A4	1.7289

Leakage [nW]
144.73

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.392	0.3656	120.392
A1 to ZN	fall	8.00	9.20	20.19	19.62
	rise	7.45	6.62	20.49	15.44
A2 to ZN	fall	9.04	10.27	21.64	20.41
	rise	7.40	6.39	18.23	14.27
A3 to ZN	fall	9.97	11.28	23.48	22.18
	rise	7.43	6.57	18.00	14.81
A4 to ZN	fall	10.65	12.16	25.68	24.22
	rise	7.43	5.72	17.84	15.70

AND4_X4

Databook Build Date: Thursday Feb 17 15:07 2011

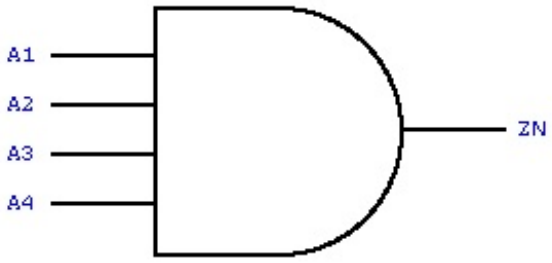
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	L
H	H	H	H	H
-	L	-	-	L
-	-	L	-	L
-	-	-	L	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.01	0.07	0.02	0.08
	rise	0.02	0.09	0.05	0.12
A2 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.02	0.09	0.05	0.12
A3 to ZN	**fall**	0.01	0.07	0.03	0.09
	rise	0.02	0.09	0.04	0.11
A4 to ZN	**fall**	0.01	0.07	0.04	0.10
	rise	0.03	0.09	0.03	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**241.089**	**0.3656**	**241.089**
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.06
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.06
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.06
A4 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.06

Capacitance [fF]	
A1	3.2737
A2	3.3969
A3	3.4919
A4	3.7904

Leakage [nW]
289.46

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	241.089	0.3656	241.089
A1 to ZN	fall	15.26	17.54	39.84	38.48
	rise	14.37	10.17	40.98	31.94
A2 to ZN	fall	17.34	19.70	42.99	40.03
	rise	14.52	13.05	36.19	29.17
A3 to ZN	fall	19.22	21.72	46.59	43.58
	rise	14.46	11.58	35.79	29.91
A4 to ZN	fall	20.50	23.50	50.83	47.70
	rise	14.29	12.00	35.44	31.04

NANGATE ANTENNA_X1

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.266 um^2
Type	Physical
Input	A
PG Pins	VDD (primary_power), VSS (primary_ground)



Capacitance [fF]	
A	0.0236

Leakage [nW]
0.00

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1251**
A	**fall**	0.00	0.00
	rise	0.00	0.00

AOI211_X1

Databook Build Date: Thursday Feb 17 15:07 2011

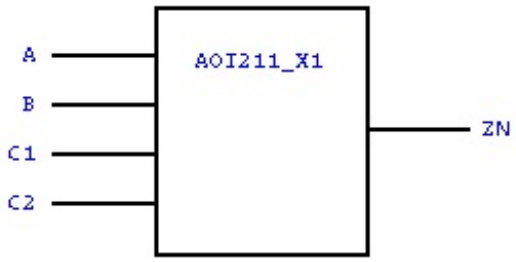
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((C1 & C2) | B) | A)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	L	L	-	H
L	L	-	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	14.4959	0.3656	14.4959
A to ZN	**fall**	0.01	0.02	0.01	0.04
	rise	0.02	0.05	0.04	0.08
B to ZN	**fall**	0.01	0.02	0.00	0.03
	rise	0.02	0.05	0.05	0.09
C1 to ZN	**fall**	0.01	0.03	0.00	0.05
	rise	0.01	0.04	0.04	0.09
C2 to ZN	**fall**	0.01	0.03	-0.00	0.04
	rise	0.01	0.04	0.05	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	14.4959	0.3656	14.4959
A to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.01	0.04	0.02	0.05
B to ZN	**fall**	0.00	0.02	0.03	0.04
	rise	0.01	0.04	0.02	0.05
C1 to ZN	**fall**	0.00	0.02	0.02	0.04
	rise	0.01	0.04	0.02	0.05
C2 to ZN	**fall**	0.00	0.02	0.02	0.04
	rise	0.01	0.04	0.02	0.05

Capacitance [fF]	
A	1.6511
B	1.6960
C1	1.6743
C2	1.7252

Leakage [nW]
87.43

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	14.4959	0.3656	14.4959
A to ZN	fall	1.57	1.89	10.31	9.21
	rise	6.61	6.69	13.96	11.82
B to ZN	fall	1.29	1.44	9.00	8.27
	rise	5.62	5.69	12.85	10.91
C1 to ZN	fall	0.29	0.37	7.42	6.09
	rise	3.67	3.73	12.33	10.37
C2 to ZN	fall	0.28	0.37	5.98	5.19
	rise	4.59	4.55	13.52	11.83

AOI211_X2

Databook Build Date: Thursday Feb 17 15:07 2011

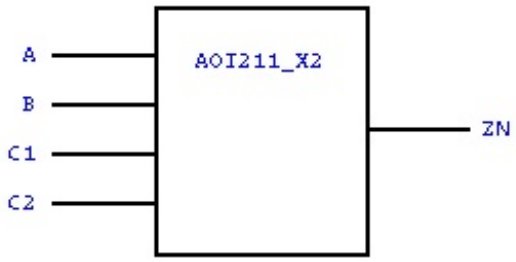
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((C1 & C2) | B) | A)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	L	L	-	H
L	L	-	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	28.9917	0.3656	28.9917
A to ZN	**fall**	0.01	0.02	0.01	0.04
	rise	0.02	0.05	0.04	0.08
B to ZN	**fall**	0.01	0.02	0.00	0.03
	rise	0.01	0.05	0.05	0.09
C1 to ZN	**fall**	0.01	0.03	0.00	0.05
	rise	0.01	0.04	0.04	0.09
C2 to ZN	**fall**	0.01	0.03	-0.00	0.04
	rise	0.01	0.04	0.05	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	28.9917	0.3656	28.9917
A to ZN	**fall**	0.00	0.02	0.03	0.04
	rise	0.01	0.04	0.02	0.05
B to ZN	**fall**	0.00	0.02	0.03	0.04
	rise	0.01	0.04	0.02	0.05
C1 to ZN	**fall**	0.00	0.02	0.02	0.04
	rise	0.01	0.04	0.02	0.05
C2 to ZN	**fall**	0.00	0.02	0.02	0.04
	rise	0.01	0.04	0.02	0.05

Capacitance [fF]	
A	3.1661
B	3.5095
C1	3.1956
C2	3.5375

Leakage [nW]
174.85

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	28.9917	0.3656	28.9917
A to ZN	fall	3.04	3.66	20.50	18.24
	rise	13.38	13.46	28.13	23.80
B to ZN	fall	2.43	2.73	17.81	16.38
	rise	11.39	11.61	25.98	22.07
C1 to ZN	fall	0.56	0.74	14.72	12.15
	rise	7.37	7.67	24.67	20.65
C2 to ZN	fall	0.56	0.74	11.83	10.35
	rise	9.26	9.20	27.07	23.57

AOI211_X4

Databook Build Date: Thursday Feb 17 15:07 2011

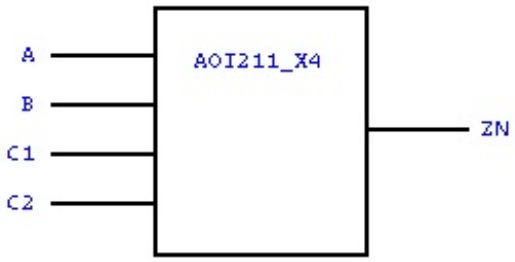
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	ZN = "!(!(!(((C1 & C2) | B) | A)))"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	L	L	-	H
L	L	-	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	**fall**	0.02	0.08	0.03	0.09
	rise	0.04	0.10	0.07	0.13
B to ZN	**fall**	0.02	0.08	0.03	0.08
	rise	0.03	0.10	0.07	0.13
C1 to ZN	**fall**	0.02	0.08	0.03	0.09
	rise	0.03	0.09	0.07	0.13
C2 to ZN	**fall**	0.02	0.08	0.03	0.08
	rise	0.03	0.09	0.07	0.13

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
B to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
C1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
C2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05

Capacitance [fF]	
A	1.6961
B	1.7374
C1	1.6466
C2	1.7916

Leakage [nW]
317.60

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	fall	15.71	17.74	30.85	33.07
	rise	22.08	23.08	31.80	33.77
B to ZN	fall	15.13	17.03	29.27	31.62
	rise	21.21	22.14	31.43	31.54
C1 to ZN	fall	14.14	15.93	28.09	30.64
	rise	19.33	20.98	32.86	32.63
C2 to ZN	fall	14.17	15.94	24.68	27.91
	rise	20.39	21.45	34.01	34.46

AOI21_X1

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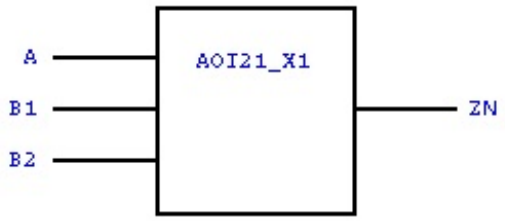
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!(A | (B1 & B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	L	-	H
L	-	L	H
H	-	-	L
-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to ZN	**fall**	0.01	0.03	0.01	0.05
	rise	0.01	0.05	0.04	0.09
B1 to ZN	**fall**	0.01	0.04	0.01	0.07
	rise	0.01	0.05	0.03	0.09
B2 to ZN	**fall**	0.01	0.04	0.00	0.06
	rise	0.01	0.05	0.04	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.00	0.04	0.02	0.05
B1 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.00	0.04	0.02	0.06
B2 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.00	0.04	0.02	0.06

Capacitance [fF]	
A	1.6532
B1	1.6515
B2	1.7132

Leakage [nW]
69.10

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to ZN	**fall**	1.16	1.35	10.97	8.49
	rise	5.17	5.26	14.02	10.99
B1 to ZN	**fall**	0.24	0.37	8.79	5.61
	rise	3.25	3.34	12.92	9.98
B2 to ZN	**fall**	0.26	0.37	7.23	4.72
	rise	4.16	4.19	14.39	11.51

AOI21_X2

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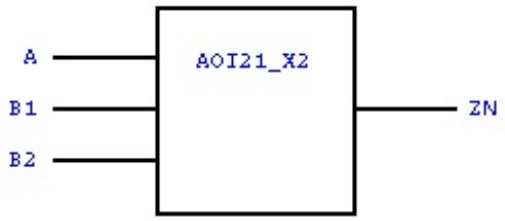
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!(A | (B1 & B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	L	-	H
L	-	L	H
H	-	-	L
-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**50.6592**	**0.3656**	**50.6592**
A to ZN	**fall**	0.01	0.03	0.01	0.05
	rise	0.01	0.05	0.04	0.09
B1 to ZN	**fall**	0.00	0.04	0.00	0.07
	rise	0.01	0.05	0.03	0.09
B2 to ZN	**fall**	0.01	0.04	0.00	0.06
	rise	0.01	0.05	0.04	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**50.6592**	**0.3656**	**50.6592**
A to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.00	0.04	0.02	0.05
B1 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.00	0.04	0.02	0.06
B2 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.00	0.04	0.02	0.06

Capacitance [fF]	
A	3.1593
B1	3.1341
B2	3.5585

Leakage [nW]
138.20

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**50.6592**	**0.3656**	**50.6592**
A to ZN	**fall**	2.20	2.55	21.84	16.86
	rise	10.13	10.24	27.96	21.92
B1 to ZN	**fall**	0.50	0.74	17.59	11.19
	rise	6.41	6.68	25.80	20.06
B2 to ZN	**fall**	0.53	0.74	14.52	9.43
	rise	8.24	8.58	28.60	23.14

AOI21_X4

Databook Build Date: Thursday Feb 17 15:07 2011

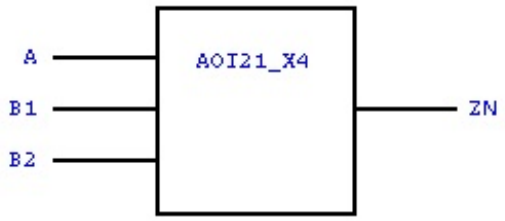
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(A | (B1 & B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	L	-	H
L	-	L	H
H	-	-	L
-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**101.013**	**0.3656**	**101.013**
A to ZN	**fall**	0.01	0.03	0.01	0.05
	rise	0.01	0.05	0.04	0.09
B1 to ZN	**fall**	0.00	0.04	0.00	0.07
	rise	0.01	0.05	0.03	0.09
B2 to ZN	**fall**	0.01	0.04	0.00	0.06
	rise	0.01	0.05	0.04	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**101.013**	**0.3656**	**101.013**
A to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.00	0.04	0.02	0.05
B1 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.00	0.04	0.02	0.06
B2 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.00	0.04	0.02	0.06

Capacitance [fF]	
A	6.1677
B1	6.4216
B2	6.8418

Leakage [nW]
276.41

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**101.013**	**0.3656**	**101.013**
A to ZN	**fall**	4.35	5.09	43.42	33.50
	rise	20.70	20.82	56.49	43.92
B1 to ZN	**fall**	0.98	1.48	34.81	22.22
	rise	12.89	13.58	51.47	39.42
B2 to ZN	**fall**	1.02	1.47	28.90	18.74
	rise	16.55	16.97	56.97	45.63

AOI221_X1

Databook Build Date: Thursday Feb 17 15:07 2011

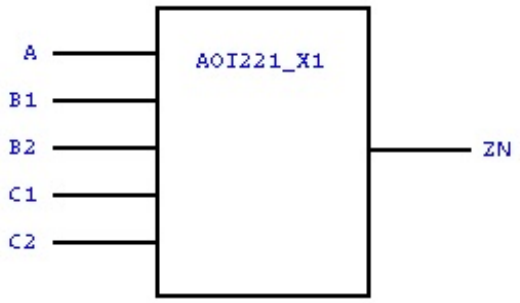
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "!(((C1 & C2) | A) | (B1 & B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	L	-	H
L	L	-	-	L	H
L	-	L	L	-	H
L	-	L	-	L	H
H	-	-	-	-	L
-	H	H	-	-	L
-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**13.8092**	**0.3656**	**13.8092**
A to ZN	**fall**	0.01	0.02	0.00	0.03
	rise	0.02	0.05	0.05	0.09
B1 to ZN	**fall**	0.01	0.03	0.02	0.06
	rise	0.02	0.05	0.04	0.08
B2 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.02	0.05	0.05	0.08
C1 to ZN	**fall**	0.01	0.03	0.00	0.05
	rise	0.01	0.04	0.04	0.09
C2 to ZN	**fall**	0.01	0.03	0.00	0.04
	rise	0.01	0.04	0.05	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**13.8092**	**0.3656**	**13.8092**
A to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.01	0.04	0.02	0.05
B1 to ZN	**fall**	0.01	0.02	0.03	0.05
	rise	0.01	0.04	0.02	0.05
B2 to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.01	0.04	0.02	0.05
C1 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.04	0.02	0.05
C2 to ZN	**fall**	0.00	0.02	0.02	0.04
	rise	0.01	0.04	0.02	0.05

Capacitance [fF]	
A	1.7079
B1	1.6560
B2	1.6522
C1	1.6486
C2	1.7514

Leakage [nW]
105.50

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	13.8092	0.3656	13.8092
A to ZN	fall	1.14	1.26	9.27	8.46
	rise	5.98	6.06	13.40	11.50
B1 to ZN	fall	1.92	2.19	9.71	8.03
	rise	7.11	7.19	12.85	11.05
B2 to ZN	fall	1.92	2.19	8.26	7.02
	rise	8.02	8.06	13.96	12.47
C1 to ZN	fall	0.28	0.37	7.70	6.31
	rise	4.00	4.16	12.75	10.86
C2 to ZN	fall	0.29	0.37	6.23	5.39
	rise	4.92	4.99	13.99	12.31

AOI221_X2

Databook Build Date: Thursday Feb 17 15:07 2011

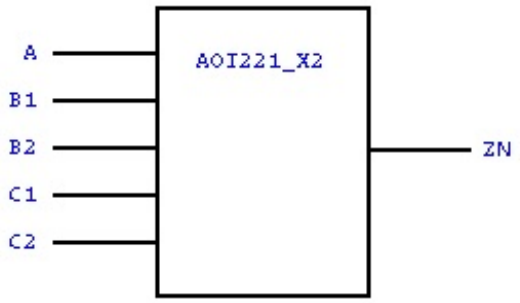
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.926 um^2
Equation	ZN = "!(((C1 & C2) | A) | (B1 & B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	L	-	H
L	L	-	-	L	H
L	-	L	L	-	H
L	-	L	-	L	H
H	-	-	-	-	L
-	H	H	-	-	L
-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**27.6184**	**0.3656**	**27.6184**
A to ZN	**fall**	0.01	0.02	0.00	0.03
	rise	0.02	0.05	0.05	0.09
B1 to ZN	**fall**	0.01	0.03	0.02	0.06
	rise	0.02	0.05	0.04	0.08
B2 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.02	0.05	0.05	0.08
C1 to ZN	**fall**	0.01	0.03	0.00	0.05
	rise	0.01	0.04	0.04	0.09
C2 to ZN	**fall**	0.01	0.03	-0.00	0.04
	rise	0.01	0.04	0.05	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**27.6184**	**0.3656**	**27.6184**
A to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.01	0.04	0.02	0.05
B1 to ZN	**fall**	0.01	0.02	0.03	0.05
	rise	0.01	0.04	0.02	0.05
B2 to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.01	0.04	0.02	0.05
C1 to ZN	**fall**	0.00	0.02	0.03	0.05
	rise	0.01	0.04	0.02	0.05
C2 to ZN	**fall**	0.00	0.02	0.02	0.04
	rise	0.01	0.04	0.02	0.05

Capacitance [fF]	
A	3.5749
B1	3.3790
B2	3.1777
C1	3.1731
C2	3.5944

Leakage [nW]
211.00

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	27.6184	0.3656	27.6184
A to ZN	fall	2.39	2.67	18.61	17.01
	rise	11.97	12.09	26.88	22.99
B1 to ZN	fall	3.85	4.38	19.47	16.05
	rise	14.15	14.35	25.69	22.03
B2 to ZN	fall	3.84	4.38	16.58	14.04
	rise	15.93	15.95	27.94	24.84
C1 to ZN	fall	0.55	0.74	15.41	12.59
	rise	8.09	8.24	25.57	21.64
C2 to ZN	fall	0.57	0.74	12.47	10.75
	rise	9.93	10.03	28.05	24.69

AOI221_X4

Databook Build Date: Thursday Feb 17 15:07 2011

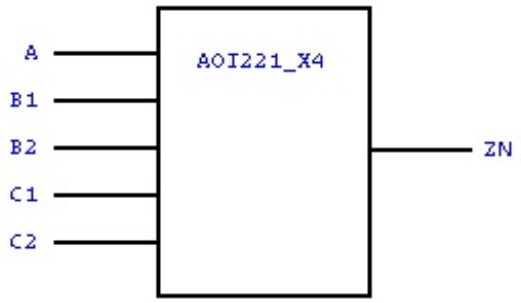
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(!(!(((C1 & C2) | A) | (B1 & B2))))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	L	-	H
L	L	-	-	L	H
L	-	L	L	-	H
L	-	L	-	L	H
H	-	-	-	-	L
-	H	H	-	-	L
-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	**fall**	0.02	0.08	0.03	0.09
	rise	0.04	0.10	0.08	0.14
B1 to ZN	**fall**	0.03	0.09	0.05	0.11
	rise	0.04	0.10	0.06	0.12
B2 to ZN	**fall**	0.03	0.09	0.04	0.10
	rise	0.04	0.10	0.07	0.13
C1 to ZN	**fall**	0.02	0.08	0.04	0.09
	rise	0.03	0.09	0.07	0.13
C2 to ZN	**fall**	0.02	0.08	0.03	0.09
	rise	0.03	0.09	0.08	0.14

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
B1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
B2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
C1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
C2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05

Capacitance [fF]	
A	1.6827
B1	1.6297
B2	1.6776
C1	1.6079
C2	1.7209

Leakage [nW]
329.68

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	fall	15.21	17.06	29.87	32.13
	rise	21.53	22.29	31.93	33.15
B1 to ZN	fall	17.06	19.06	31.06	32.57
	rise	22.81	24.15	31.28	32.57
B2 to ZN	fall	17.08	19.07	27.80	29.75
	rise	23.99	24.44	32.68	33.50
C1 to ZN	fall	14.30	16.03	28.64	31.11
	rise	19.67	20.57	33.45	34.01
C2 to ZN	fall	14.30	16.04	25.17	28.32
	rise	20.89	21.32	34.62	35.38

AOI222_X1

Databook Build Date: Thursday Feb 17 15:07 2011

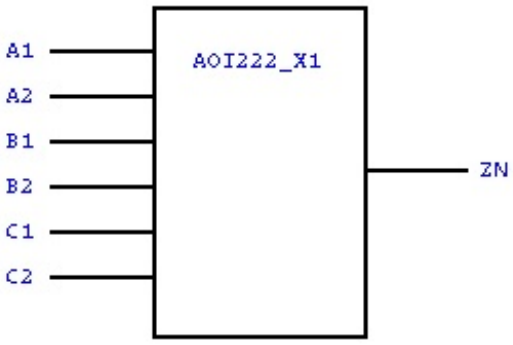
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.128 um^2
Equation	ZN = "!(((A1 & A2) | (B1 & B2)) | (C1 & C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	L	-	L	-	H
L	-	L	-	-	L	H
L	-	-	L	L	-	H
L	-	-	L	-	L	H
H	H	-	-	-	-	L
-	L	L	-	L	-	H
-	L	L	-	-	L	H
-	L	-	L	L	-	H
-	L	-	L	-	L	H
-	-	H	H	-	-	L
-	-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**13.0081**	**0.3656**	**13.0081**
A1 to ZN	**fall**	0.01	0.02	0.01	0.05
	rise	0.01	0.04	0.05	0.09
A2 to ZN	**fall**	0.01	0.03	0.00	0.04
	rise	0.01	0.04	0.06	0.09
B1 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.02	0.05	0.05	0.09
B2 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.02	0.05	0.06	0.09
C1 to ZN	**fall**	0.01	0.03	0.03	0.06
	rise	0.02	0.05	0.04	0.08
C2 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.02	0.05	0.05	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**13.0081**	**0.3656**	**13.0081**
A1 to ZN	**fall**	0.01	0.02	0.03	0.05
	rise	0.01	0.04	0.03	0.05
A2 to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.01	0.04	0.02	0.05
B1 to ZN	**fall**	0.01	0.02	0.03	0.05
	rise	0.01	0.04	0.03	0.05
B2 to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.01	0.04	0.02	0.05
C1 to ZN	**fall**	0.01	0.03	0.03	0.05
	rise	0.01	0.04	0.02	0.05
C2 to ZN	**fall**	0.01	0.03	0.03	0.04
	rise	0.01	0.04	0.02	0.05

Capacitance [fF]	
A1	1.6479
A2	1.7350
B1	1.5893
B2	1.6666
C1	1.6129
C2	1.6123

Leakage [nW]
119.62

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	13.0081	0.3656	13.0081
A1 to ZN	fall	0.27	0.34	7.97	6.55
	rise	4.26	4.45	13.20	11.39
A2 to ZN	fall	0.27	0.34	6.49	5.59
	rise	5.20	5.30	14.49	12.93
B1 to ZN	fall	1.55	1.67	8.98	7.55
	rise	6.45	6.54	12.49	10.84
B2 to ZN	fall	1.55	1.67	7.52	6.54
	rise	7.37	7.43	13.64	12.22
C1 to ZN	fall	2.47	2.80	10.19	8.62
	rise	8.23	8.34	13.46	11.88
C2 to ZN	fall	2.47	2.80	8.77	7.59
	rise	9.13	9.21	14.62	13.27

AOI222_X2

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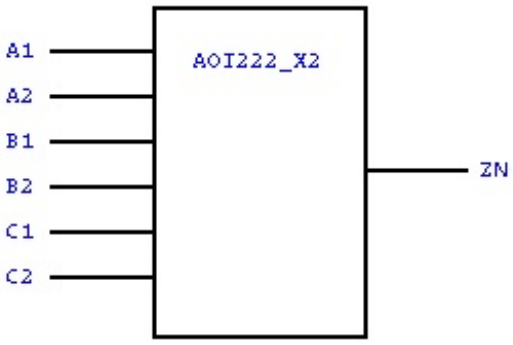
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	3.724 um^2
Equation	ZN = "!(((A1 & A2) | (B1 & B2)) | (C1 & C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	L	-	L	-	H
L	-	L	-	-	L	H
L	-	-	L	L	-	H
L	-	-	L	-	L	H
H	H	-	-	-	-	L
-	L	L	-	L	-	H
-	L	L	-	-	L	H
-	L	-	L	L	-	H
-	L	-	L	-	L	H
-	-	H	H	-	-	L
-	-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**25.6348**	**0.3656**	**25.6348**
A1 to ZN	**fall**	0.01	0.02	0.01	0.05
	rise	0.01	0.04	0.05	0.09
A2 to ZN	**fall**	0.01	0.03	0.00	0.04
	rise	0.01	0.04	0.06	0.09
B1 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.02	0.05	0.05	0.09
B2 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.02	0.05	0.06	0.09
C1 to ZN	**fall**	0.01	0.03	0.03	0.06
	rise	0.02	0.05	0.04	0.08
C2 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.02	0.05	0.05	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**25.6348**	**0.3656**	**25.6348**
A1 to ZN	**fall**	0.01	0.02	0.03	0.05
	rise	0.01	0.04	0.03	0.05
A2 to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.01	0.04	0.02	0.05
B1 to ZN	**fall**	0.01	0.02	0.03	0.05
	rise	0.01	0.04	0.03	0.05
B2 to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.01	0.04	0.02	0.05
C1 to ZN	**fall**	0.01	0.03	0.04	0.05
	rise	0.01	0.04	0.02	0.05
C2 to ZN	**fall**	0.01	0.03	0.03	0.04
	rise	0.01	0.04	0.02	0.05

Capacitance [fF]	
A1	3.1874
A2	3.2837
B1	3.0419
B2	3.4868
C1	3.0938
C2	3.3537

Leakage [nW]
239.24

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	25.6348	0.3656	25.6348
A1 to ZN	fall	0.53	0.69	15.88	13.07
	rise	8.77	9.08	26.65	22.94
A2 to ZN	fall	0.55	0.69	12.86	11.07
	rise	10.88	11.10	29.37	26.21
B1 to ZN	fall	3.09	3.35	17.97	15.09
	rise	13.15	13.42	25.30	21.96
B2 to ZN	fall	3.10	3.35	15.05	13.09
	rise	14.97	15.18	27.56	24.73
C1 to ZN	fall	4.93	5.61	20.38	17.21
	rise	16.86	17.03	27.32	24.11
C2 to ZN	fall	4.94	5.61	17.51	15.18
	rise	18.66	18.84	29.58	26.79

AOI222_X4

Databook Build Date: Thursday Feb 17 15:07 2011

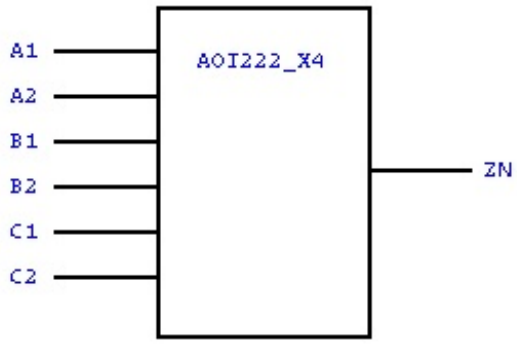
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.724 um^2
Equation	ZN = "!(!(!(((A1 & A2) | (B1 & B2)) | (C1 & C2))))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	L	-	L	-	H
L	-	L	-	-	L	H
L	-	-	L	L	-	H
L	-	-	L	-	L	H
H	H	-	-	-	-	L
-	L	L	-	L	-	H
-	L	L	-	-	L	H
-	L	-	L	L	-	H
-	L	-	L	-	L	H
-	-	H	H	-	-	L
-	-	-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.02	0.08	0.04	0.10
	rise	0.03	0.09	0.07	0.13
A2 to ZN	**fall**	0.02	0.08	0.03	0.09
	rise	0.03	0.09	0.08	0.14
B1 to ZN	**fall**	0.03	0.09	0.05	0.11
	rise	0.04	0.10	0.08	0.14
B2 to ZN	**fall**	0.03	0.09	0.04	0.10
	rise	0.04	0.10	0.08	0.14
C1 to ZN	**fall**	0.03	0.09	0.06	0.11
	rise	0.04	0.10	0.07	0.13
C2 to ZN	**fall**	0.03	0.09	0.05	0.11
	rise	0.04	0.10	0.07	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
A2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
B1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
B2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
C1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
C2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05

Capacitance [fF]	
A1	1.6175
A2	1.7523
B1	1.6114
B2	1.7213
C1	1.6444
C2	1.6755

Leakage [nW]
334.82

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	fall	14.39	16.09	29.11	31.48
	rise	19.95	21.45	33.87	35.22
A2 to ZN	fall	14.39	16.09	25.59	28.67
	rise	21.10	22.23	35.04	35.65
B1 to ZN	fall	16.28	18.10	30.29	31.72
	rise	21.94	22.60	31.64	32.72
B2 to ZN	fall	16.29	18.10	26.94	28.85
	rise	23.08	24.64	32.93	34.48
C1 to ZN	fall	17.81	19.90	31.84	33.31
	rise	23.65	24.56	31.76	34.33
C2 to ZN	fall	17.82	19.90	28.52	30.44
	rise	24.82	25.90	33.15	35.51

AOI22_X1

Databook Build Date: Thursday Feb 17 15:07 2011

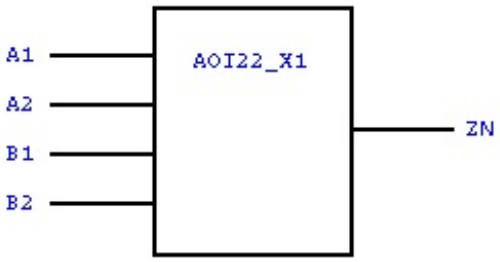
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!((A1 & A2) | (B1 & B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**24.6048**	**0.3656**	**24.6048**
A1 to ZN	**fall**	0.01	0.04	0.01	0.07
	rise	0.01	0.05	0.03	0.09
A2 to ZN	**fall**	0.01	0.04	0.00	0.06
	rise	0.01	0.05	0.04	0.10
B1 to ZN	**fall**	0.01	0.04	0.02	0.08
	rise	0.01	0.05	0.03	0.09
B2 to ZN	**fall**	0.01	0.05	0.02	0.07
	rise	0.01	0.05	0.04	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**24.6048**	**0.3656**	**24.6048**
A1 to ZN	**fall**	0.00	0.03	0.03	0.06
	rise	0.00	0.04	0.02	0.06
A2 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.01	0.04	0.02	0.06
B1 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.00	0.04	0.02	0.05
B2 to ZN	**fall**	0.01	0.04	0.02	0.05
	rise	0.01	0.04	0.02	0.05

Capacitance [fF]	
A1	1.6856
A2	1.7215
B1	1.6317
B2	1.6427

Leakage [nW]
81.53

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	24.6048	0.3656	24.6048
A1 to ZN	fall	0.23	0.35	9.22	5.84
	rise	3.56	3.59	13.41	10.59
A2 to ZN	fall	0.25	0.35	7.67	4.92
	rise	4.49	4.57	14.93	12.23
B1 to ZN	fall	1.46	1.63	10.23	7.15
	rise	5.44	5.46	12.41	9.94
B2 to ZN	fall	1.46	1.64	8.67	6.17
	rise	6.30	6.30	13.73	11.35

AOI22_X2

Databook Build Date: Thursday Feb 17 15:07 2011

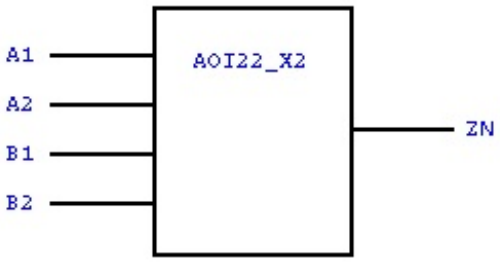
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!((A1 & A2) | (B1 & B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**49.1333**	**0.3656**	**49.1333**
A1 to ZN	**fall**	0.01	0.04	0.01	0.07
	rise	0.01	0.05	0.03	0.09
A2 to ZN	**fall**	0.01	0.04	0.00	0.06
	rise	0.01	0.05	0.04	0.10
B1 to ZN	**fall**	0.01	0.05	0.02	0.08
	rise	0.01	0.05	0.03	0.09
B2 to ZN	**fall**	0.01	0.05	0.02	0.07
	rise	0.01	0.05	0.04	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**49.1333**	**0.3656**	**49.1333**
A1 to ZN	**fall**	0.00	0.03	0.03	0.06
	rise	0.00	0.04	0.02	0.06
A2 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.01	0.04	0.02	0.06
B1 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.00	0.04	0.02	0.05
B2 to ZN	**fall**	0.01	0.04	0.02	0.05
	rise	0.01	0.04	0.02	0.05

Capacitance [fF]	
A1	3.1449
A2	3.5388
B1	3.0914
B2	3.4985

Leakage [nW]
163.05

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	49.1333	0.3656	49.1333
A1 to ZN	fall	0.46	0.70	18.35	11.64
	rise	7.16	7.43	26.86	21.21
A2 to ZN	fall	0.48	0.70	15.30	9.82
	rise	9.01	8.96	29.87	24.24
B1 to ZN	fall	2.89	3.26	20.36	14.21
	rise	11.27	11.45	25.20	20.27
B2 to ZN	fall	2.91	3.27	17.26	12.27
	rise	13.04	13.05	27.81	23.00

AOI22_X4

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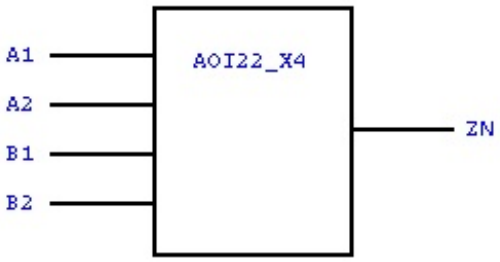
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Equation	ZN = "!((A1 & A2) | (B1 & B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**97.9614**	**0.3656**	**97.9614**
A1 to ZN	**fall**	0.00	0.04	0.01	0.07
	rise	0.01	0.05	0.03	0.10
A2 to ZN	**fall**	0.01	0.04	0.00	0.06
	rise	0.01	0.05	0.04	0.10
B1 to ZN	**fall**	0.01	0.04	0.02	0.08
	rise	0.01	0.05	0.03	0.09
B2 to ZN	**fall**	0.01	0.05	0.02	0.07
	rise	0.01	0.05	0.04	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**97.9614**	**0.3656**	**97.9614**
A1 to ZN	**fall**	0.00	0.03	0.03	0.06
	rise	0.00	0.04	0.02	0.06
A2 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.01	0.04	0.02	0.06
B1 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.00	0.04	0.02	0.05
B2 to ZN	**fall**	0.01	0.04	0.02	0.05
	rise	0.01	0.04	0.02	0.05

Capacitance [fF]	
A1	6.4154
A2	6.9005
B1	6.2921
B2	6.6978

Leakage [nW]
326.11

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	97.9614	0.3656	97.9614
A1 to ZN	fall	0.92	1.41	36.45	23.08
	rise	14.34	14.70	53.54	42.17
A2 to ZN	fall	0.96	1.41	30.28	19.49
	rise	18.08	18.26	59.45	48.16
B1 to ZN	fall	5.74	6.52	40.69	28.30
	rise	22.25	22.46	50.17	39.61
B2 to ZN	fall	5.78	6.53	34.46	24.46
	rise	25.83	26.02	55.36	46.01

BUF_X16

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	6.650 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	965.576	0.3656	965.576
A to Z	**fall**	0.01	0.07	0.03	0.10
	rise	0.01	0.07	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	965.576	0.3656	965.576
A to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.06

Capacitance [fF]	
A	12.0923

Leakage [nW]
840.71

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	965.576	0.3656	965.576
A to Z	**fall**	51.48	59.30	187.87	192.63
	rise	31.98	36.38	168.24	168.34

BUF_X1

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
A to Z	**fall**	0.01	0.07	0.03	0.10
	rise	0.01	0.07	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
A to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A	0.9702

Leakage [nW]
52.31

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
A to Z	**fall**	3.62	4.12	11.85	12.25
	rise	2.27	2.39	10.56	10.67

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BUF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.064 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A to Z	**fall**	0.01	0.07	0.03	0.09
	rise	0.01	0.07	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A	1.7787

Leakage [nW]
105.09

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A to Z	**fall**	6.86	7.86	23.63	24.41
	rise	4.38	4.80	21.33	21.27

BUF_X32

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	32
Cell Area	13.034 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	1904.30	0.3656	1904.30
A to Z	**fall**	0.01	0.07	0.03	0.10
	rise	0.01	0.07	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	1904.30	0.3656	1904.30
A to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.06	0.01	0.06

Capacitance [fF]	
A	26.4525

Leakage [nW]
1681.43

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	1904.30	0.3656	1904.30
A to Z	**fall**	101.95	115.86	369.15	375.17
	rise	62.53	64.76	325.45	311.55

BUF_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	1.862 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to Z	**fall**	0.01	0.07	0.03	0.09
	rise	0.01	0.07	0.01	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A	3.3790

Leakage [nW]
210.17

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to Z	**fall**	13.10	14.92	46.89	48.29
	rise	8.24	9.05	42.35	43.12

BUF_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	3.458 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	484.009	0.3656	484.009
A to Z	**fall**	0.01	0.07	0.03	0.10
	rise	0.01	0.07	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	484.009	0.3656	484.009
A to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A	6.4945

Leakage [nW]
420.36

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	484.009	0.3656	484.009
A to Z	**fall**	25.85	29.54	94.23	96.48
	rise	16.02	15.23	84.83	85.96

CLKBUF_X1

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
A to Z	**fall**	0.01	0.13	0.02	0.14
	rise	0.01	0.07	0.04	0.11

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
A to Z	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A	0.7746

Leakage [nW]
28.43

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
A to Z	**fall**	3.02	3.33	8.01	8.48
	rise	1.86	1.76	7.44	6.79

CLKBUF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.064 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to Z	**fall**	0.01	0.13	0.02	0.14
	rise	0.01	0.07	0.04	0.10

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to Z	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A	1.3857

Leakage [nW]
58.03

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
A to Z	**fall**	5.70	6.34	16.16	17.20
	rise	3.54	3.37	15.19	13.59

CLKBUF_X3

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	3
Cell Area	1.330 um^2
Equation	Z = "A"
Type	Combinational
Input	A
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**Z**
L	L
H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	181.885	0.3656	181.885
A to Z	**fall**	0.01	0.13	0.02	0.15
	rise	0.01	0.08	0.04	0.11

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	181.885	0.3656	181.885
A to Z	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A	1.4035

Leakage [nW]
77.37

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	181.885	0.3656	181.885
A to Z	**fall**	7.16	8.09	20.23	18.71
	rise	5.74	4.76	19.75	16.19

NANGATE CLKGATETST_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	3.990 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	0.01	0.13	0.03	0.16
	rise	0.01	0.07	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.05	0.01	0.05

Constraints Time [ns]		
Setup CK to E	**fall**	0.07
	rise	0.05
Hold CK to E	**fall**	0.09
	rise	0.07
Setup CK to SE	**fall**	0.07
	rise	0.05
Hold CK to SE	**fall**	0.09
	rise	0.06

Capacitance [fF]	
CK	1.8038
E	0.8789
SE	0.7785

Leakage [nW]
147.80

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	8158.66	17734.55
	rise	2570.87	11438.00
E	fall	2712.46	5314.65
	rise	1244.83	4117.92
SE	fall	2209.66	5443.89
	rise	1207.37	4484.20

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to GCK	fall	7.00	8.68	19.42	24.24
	rise	4.19	4.24	18.43	20.62

NANGATE CLKGATETST_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	4.256 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
CK to GCK	**fall**	0.01	0.13	0.03	0.15
	rise	0.01	0.07	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.05	0.01	0.05

Constraints Time [ns]		
Setup CK to E	**fall**	0.06
	rise	0.05
Hold CK to E	**fall**	0.09
	rise	0.07
Setup CK to SE	**fall**	0.07
	rise	0.05
Hold CK to SE	**fall**	0.09
	rise	0.06

Capacitance [fF]	
CK	2.7905
E	0.8742
SE	0.8104

Leakage [nW]
196.49

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	9413.97	19492.45
	rise	1558.36	10412.28
E	fall	2712.47	5346.21
	rise	1185.98	4077.65
SE	fall	2209.00	5473.48
	rise	1147.03	4456.63

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.46	0.3656	121.46
CK to GCK	fall	10.12	12.50	28.46	34.73
	rise	5.40	5.86	26.31	28.43

NANGATE CLKGATETST_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	5.320 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.92	0.3656	242.92
CK to GCK	**fall**	0.01	0.13	0.04	0.16
	rise	0.01	0.07	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.92	0.3656	242.92
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.05	0.01	0.06

Constraints Time [ns]		
Setup CK to E	**fall**	0.07
	rise	0.07
Hold CK to E	**fall**	0.07
	rise	0.06
Setup CK to SE	**fall**	0.08
	rise	0.06
Hold CK to SE	**fall**	0.08
	rise	0.06

Capacitance [fF]	
CK	4.4601
E	0.9336
SE	0.8111

Leakage [nW]
321.20

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	13505.88	23178.99
	rise	154.11	8957.25
E	fall	3590.08	6139.56
	rise	1745.84	4574.00
SE	fall	3087.64	6263.09
	rise	1709.84	4928.08

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.92	0.3656	242.92
CK to GCK	fall	17.41	20.84	52.11	56.51
	rise	10.42	12.32	37.71	45.92

NANGATE CLKGATETST_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	7.714 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
CK	**E**	**SE**	**$IQ_{(int)}$**	**GCK**
L	L	L	-	L
L	L	H	-	L
L	H	L	-	L
L	H	H	-	L
H	L	L	L	L
H	L	L	H	H
H	L	H	L	L
H	L	H	H	H
H	H	L	L	L
H	H	L	H	H
H	H	H	L	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	484.619	0.3656	484.619
CK to GCK	**fall**	0.01	0.13	0.04	0.16
	rise	0.01	0.07	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	484.619	0.3656	484.619
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.05	0.01	0.06

Constraints Time [ns]		
Setup CK to E	**fall**	0.08
	rise	0.09
Hold CK to E	**fall**	0.05
	rise	0.05
Setup CK to SE	**fall**	0.09
	rise	0.09
Hold CK to SE	**fall**	0.06
	rise	0.05

Capacitance [fF]	
CK	8.0139
E	0.8992
SE	0.8027

Leakage [nW]
561.49

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	23597.71	33165.74
	rise	-2529.28	6232.06
E	fall	5122.43	7680.42
	rise	3180.80	5996.72
SE	fall	4619.88	7795.86
	rise	3143.44	6359.84

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	484.619	0.3656	484.619
CK to GCK	fall	31.14	37.70	95.11	100.55
	rise	18.59	22.19	66.63	78.17

NANGATE CLKGATE_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	3.458 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	0.01	0.13	0.03	0.16
	rise	0.01	0.07	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.05	0.01	0.05

Constraints Time [ns]		
Setup CK to E	**fall**	0.05
	rise	0.05
Hold CK to E	**fall**	0.09
	rise	0.09

Capacitance [fF]	
CK	1.8284
E	0.9240

Leakage [nW]
121.88

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1251**
CK	**fall**	6392.40	16321.02
	rise	1683.15	10765.85
E	**fall**	2035.98	3708.92
	rise	863.02	2511.09

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
CK to GCK	**fall**	7.00	9.02	19.38	24.59
	rise	4.00	4.12	17.99	20.04

NANGATE CLKGATE_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	3.724 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**121.307**	**0.3656**	**121.307**
CK to GCK	**fall**	0.01	0.13	0.03	0.15
	rise	0.01	0.07	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**121.307**	**0.3656**	**121.307**
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.05	0.01	0.05

Constraints Time [ns]		
Setup CK to E	**fall**	0.05
	rise	0.05
Hold CK to E	**fall**	0.08
	rise	0.09

Capacitance [fF]	
CK	2.5404
E	0.8996

Leakage [nW]
168.51

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1251**
CK	**fall**	7976.11	18159.13
	rise	1377.55	10401.44
E	**fall**	2276.82	3891.80
	rise	1346.70	2956.87

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**121.307**	**0.3656**	**121.307**
CK to GCK	**fall**	10.07	12.98	28.59	35.07
	rise	5.93	5.93	26.51	27.76

NANGATE CLKGATE_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
CK to GCK	**fall**	0.01	0.13	0.03	0.15
	rise	0.01	0.07	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.05	0.01	0.05

Constraints Time [ns]		
Setup CK to E	**fall**	0.06
	rise	0.06
Hold CK to E	**fall**	0.08
	rise	0.08

Capacitance [fF]	
CK	4.2542
E	0.8937

Leakage [nW]
308.44

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1251**
CK	**fall**	10765.31	21666.74
	rise	308.42	9342.77
E	**fall**	3253.36	5111.41
	rise	2164.18	4059.43

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
CK to GCK	**fall**	15.77	20.90	47.08	56.21
	rise	9.63	7.93	43.33	44.18

NANGATE CLKGATE_X8

Databook Build Date: Thursday Feb 17 15:07 2011

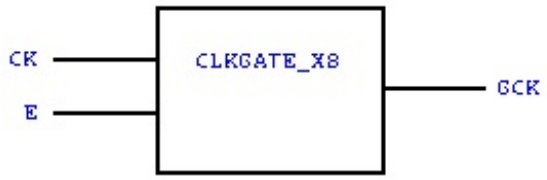
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	6.916 um^2
Enable	E
Type	Sequential
Output	GCK
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
CK	**E**	**$IQ_{(int)}$**	**GCK**
L	L	-	L
L	H	-	L
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	0.01	0.13	0.03	0.16
	rise	0.01	0.07	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	0.00	0.11	0.01	0.11
	rise	0.00	0.05	0.01	0.06

Constraints Time [ns]		
Setup CK to E	**fall**	0.06
	rise	0.06
Hold CK to E	**fall**	0.08
	rise	0.08

Capacitance [fF]	
CK	7.7098
E	1.1806

Leakage [nW]
504.77

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1251**
CK	**fall**	16983.50	28851.57
	rise	-1380.57	7388.92
E	**fall**	5608.04	8597.38
	rise	3597.55	6652.95

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**484.619**	**0.3656**	**484.619**
CK to GCK	**fall**	28.02	37.94	84.83	99.60
	rise	16.68	13.14	76.68	76.22

DFFRS_X1

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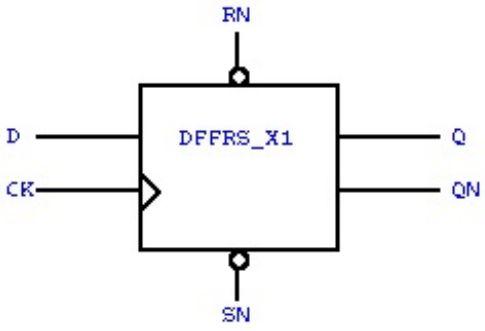
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, and drive strength X1
Strength	1
Cell Area	6.384 um^2
Equation	Q = "(!(!RN)*((D)+(!SN)))" QN = "(!(!SN)*(!(D)+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	RN	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	H	-	-	L	H
R	H	H	H	-	-	H	L
F	-	H	H	L	H	L	H
F	-	H	H	H	L	H	L
-	-	L	L	-	-	L	L
-	-	L	H	-	-	L	H
-	-	H	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.04	0.10	0.05	0.11
	rise	0.04	0.10	0.05	0.11
CK to QN	fall	0.05	*0.10*	0.06	*0.11*
	rise	0.05	*0.11*	0.07	*0.12*
RN to Q	fall	0.01	0.07	0.04	0.10
	rise	0.07	0.14	0.09	0.15
RN to QN	rise	0.03	*0.09*	0.06	*0.11*
SN to Q	rise	0.08	0.14	0.10	0.16
SN to QN	fall	0.01	*0.07*	0.04	*0.10*
	rise	0.01	*0.08*	0.03	*0.09*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
CK to QN	fall	0.00	*0.05*	0.00	*0.05*
	rise	0.00	*0.05*	0.00	*0.05*
RN to Q	fall	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
RN to QN	rise	0.00	*0.05*	0.00	*0.05*
SN to Q	rise	0.00	0.05	0.00	0.05
SN to QN	fall	0.00	*0.05*	0.01	*0.05*
	rise	0.00	*0.05*	0.01	*0.05*

Constraints Time [ns]		
Setup CK to D	fall	0.04
	rise	0.04
Hold CK to D	fall	0.09
	rise	0.09
Recovery CK to RN	rise	0.07
Removal CK to RN	rise	0.11
Recovery CK to SN	rise	-0.01
Removal CK to SN	rise	0.26

Capacitance [fF]	
CK	0.9653
D	1.1656
RN	1.4578
SN	2.1885

Leakage [nW]
252.69

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	7299.10	16025.20
	rise	9341.44	18049.59
D	fall	3195.00	6115.47
	rise	1645.16	4573.49
RN	fall	899.17	902.04
	rise	-607.00	-624.20
SN	fall	2081.57	3747.41
	rise	-957.10	648.68

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	7.98	7.84	12.13	11.97
	rise	8.02	7.78	12.23	12.39
CK to QN	fall	8.02	*7.81*	12.23	*12.34*
	rise	7.98	*7.83*	12.13	*12.09*
RN to Q	fall	4.99	7.63	11.27	13.30
	rise	6.04	5.92	10.72	9.32
RN to QN	rise	5.11	*7.04*	10.35	*11.08*
SN to Q	rise	7.66	7.77	15.14	13.93
SN to QN	fall	7.19	*7.92*	16.97	*17.28*
	rise	3.40	*3.73*	13.76	*15.19*

DFFRS_X2

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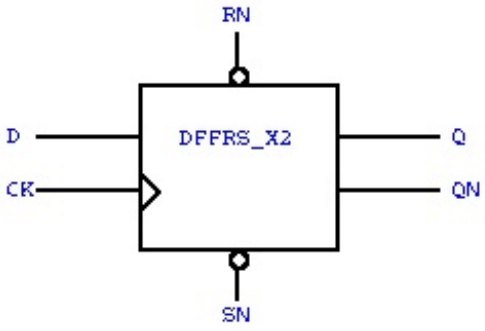
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, and drive strength X2
Strength	2
Cell Area	6.916 um^2
Equation	Q = "(!(!RN)*((D)+(!SN)))" QN = "(!(!SN)*(!(D)+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	RN	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	H	-	-	L	H
R	H	H	H	-	-	H	L
F	-	H	H	L	H	L	H
F	-	H	H	H	L	H	L
-	-	L	L	-	-	L	L
-	-	L	H	-	-	L	H
-	-	H	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.04	0.10	0.05	0.11
	rise	0.04	0.10	0.05	0.11
CK to QN	fall	0.05	0.11	0.06	0.11
	rise	0.06	0.12	0.07	0.13
RN to Q	fall	0.01	0.07	0.04	0.10
	rise	0.08	0.14	0.09	0.15
RN to QN	rise	0.03	0.09	0.06	0.12
SN to Q	rise	0.08	0.14	0.11	0.17
SN to QN	fall	0.01	0.07	0.04	0.10
	rise	0.02	0.08	0.03	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
RN to Q	fall	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
RN to QN	rise	0.00	0.05	0.00	0.05
SN to Q	rise	0.00	0.05	0.00	0.05
SN to QN	fall	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Constraints Time [ns]		
Setup CK to D	fall	0.04
	rise	0.04
Hold CK to D	fall	0.09
	rise	0.09
Recovery CK to RN	rise	0.03
Removal CK to RN	rise	0.11
Recovery CK to SN	rise	-0.01
Removal CK to SN	rise	0.26

Capacitance [fF]	
CK	0.9419
D	1.1750
RN	1.8455
SN	2.6064

Leakage [nW]
359.74

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	7185.51	15933.79
	rise	10420.63	19139.90
D	fall	3232.12	6147.76
	rise	1603.13	4529.80
RN	fall	1303.69	1223.26
	rise	-1126.89	-1145.74
SN	fall	2445.06	4092.94
	rise	-1310.80	308.20

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	11.14	10.45	15.34	14.93
	rise	11.74	11.10	16.11	15.66
CK to QN	fall	11.74	11.10	16.11	15.66
	rise	11.14	10.45	15.34	14.93
RN to Q	fall	8.49	11.24	20.01	19.72
	rise	9.74	9.22	15.78	14.10
RN to QN	rise	8.80	10.55	18.00	16.63
SN to Q	rise	11.67	11.19	22.83	19.22
SN to QN	fall	10.50	11.05	25.07	23.16
	rise	6.16	6.06	21.53	20.85

DFFR_X1

Databook Build Date: Thursday Feb 17 15:07 2011

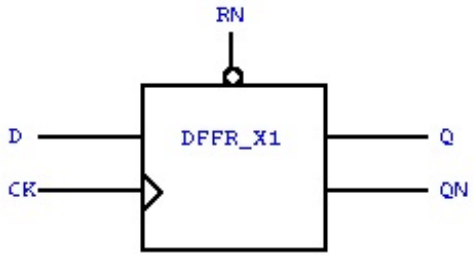
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, and drive strength X1
Strength	1
Cell Area	5.320 um^2
Equation	Q = "(!(!RN)*(D))" QN = "((!RN)+!(D))"
Clock	CK
Reset	RN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	RN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.04	0.10	0.05	0.11
	rise	0.04	0.10	0.05	0.11
CK to QN	fall	0.03	*0.09*	0.04	*0.10*
	rise	0.03	*0.09*	0.04	*0.11*
RN to Q	fall	0.01	0.07	0.04	0.10
RN to QN	rise	0.07	*0.13*	0.12	*0.18*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
CK to QN	fall	0.00	*0.05*	0.00	*0.05*
	rise	0.00	*0.05*	0.00	*0.05*
RN to Q	fall	0.00	0.05	0.01	0.05
RN to QN	rise	0.01	*0.05*	0.01	*0.05*

Constraints Time [ns]		
Setup CK to D	fall	0.04
	rise	0.04
Hold CK to D	fall	0.09
	rise	0.09
Recovery CK to RN	rise	0.01
Removal CK to RN	rise	0.24

Capacitance [fF]	
CK	0.9760
D	1.1446
RN	1.7976

Leakage [nW]
214.31

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	7022.26	15799.56
	rise	8814.61	17650.35
D	fall	3325.34	6549.91
	rise	1761.96	4969.64
RN	fall	903.61	850.48
	rise	-757.65	-774.62

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	6.46	6.41	10.48	10.60
	rise	6.91	6.40	11.46	11.01
CK to QN	fall	6.91	*6.49*	11.46	*11.11*
	rise	6.46	*6.23*	10.48	*10.69*
RN to Q	fall	6.88	6.55	12.02	10.37
RN to QN	rise	6.88	*6.44*	12.02	*10.32*

DFFR_X2

Databook Build Date: Thursday Feb 17 15:07 2011

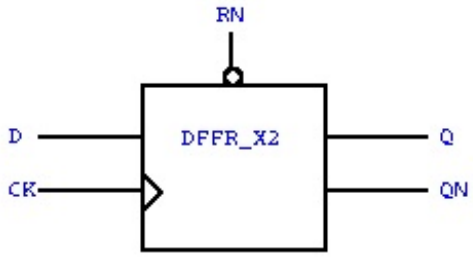
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, and drive strength X2
Strength	2
Cell Area	5.852 um^2
Equation	Q = "(!(!RN)*(D))" QN = "((!RN)+!(D))"
Clock	CK
Reset	RN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	RN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.05	0.10	0.06	0.12
	rise	0.05	0.11	0.06	0.12
CK to QN	fall	0.03	*0.10*	0.05	*0.11*
	rise	0.04	*0.10*	0.05	*0.11*
RN to Q	fall	0.01	0.07	0.04	0.10
RN to QN	rise	0.08	*0.14*	0.12	*0.19*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
CK to QN	fall	0.01	*0.05*	0.01	*0.05*
	rise	0.00	*0.05*	0.00	*0.05*
RN to Q	fall	0.00	0.05	0.01	0.05
RN to QN	rise	0.01	*0.06*	0.01	*0.06*

Constraints Time [ns]		
Setup CK to D	fall	0.04
	rise	0.04
Hold CK to D	fall	0.09
	rise	0.09
Recovery CK to RN	rise	-0.00
Removal CK to RN	rise	0.24

Capacitance [fF]	
CK	0.9650
D	1.1465
RN	2.4947

Leakage [nW]
313.04

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	6948.62	15748.47
	rise	10558.40	20361.31
D	fall	3302.98	6535.44
	rise	1715.12	4932.89
RN	fall	1657.04	1544.80
	rise	-1411.39	-1430.86

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	11.10	9.74	15.19	13.81
	rise	12.41	9.91	17.09	14.13
CK to QN	fall	12.41	*10.00*	17.09	*14.16*
	rise	11.10	*9.48*	15.19	*13.86*
RN to Q	fall	12.31	10.32	22.38	17.42
RN to QN	rise	12.31	*10.31*	22.38	*17.43*

DFFS_X1

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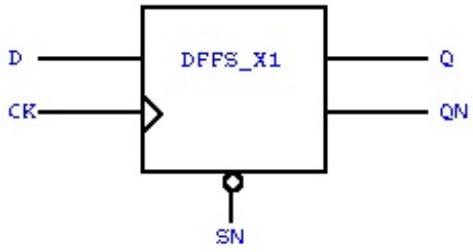
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, and drive strength X1
Strength	1
Cell Area	5.320 um^2
Equation	Q = "((!SN)+(D))" QN = "(!((!SN)+(D)))"
Clock	CK
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table

CK	D	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	H	L

Propagation Delay [ns]

Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.04	0.10	0.05	0.11
	rise	0.04	0.10	0.05	0.11
CK to QN	fall	0.03	*0.09*	0.04	*0.10*
	rise	0.03	*0.09*	0.05	*0.11*
SN to Q	rise	0.07	0.13	0.12	0.17
SN to QN	fall	0.06	*0.13*	0.10	*0.17*

Output Transition [ns]

Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
CK to QN	fall	0.00	*0.05*	0.00	*0.05*
	rise	0.00	*0.05*	0.00	*0.05*
SN to Q	rise	0.00	0.05	0.00	0.05
SN to QN	fall	0.01	*0.05*	0.01	*0.05*

Constraints Time [ns]

Setup CK to D	fall	0.04
	rise	0.03
Hold CK to D	fall	0.10
	rise	0.10
Recovery CK to SN	rise	-0.02
Removal CK to SN	rise	0.23

Capacitance [fF]

CK	0.9684
D	1.1834
SN	1.3598

Leakage [nW]

209.34

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	7965.76	16585.24
	rise	9006.89	17369.36
D	fall	3110.03	6155.47
	rise	1326.11	4303.07
SN	fall	1255.93	2958.44
	rise	-429.24	1102.88

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.6537	0.3656	60.6537
CK to Q	fall	5.93	5.73	9.85	9.95
	rise	6.20	5.81	10.68	10.28
CK to QN	fall	6.20	*5.93*	10.67	*10.37*
	rise	5.93	*5.85*	9.85	*10.06*
SN to Q	rise	8.10	6.59	11.22	9.40
SN to QN	fall	8.10	*6.59*	11.22	*9.37*

DFFS_X2

Databook Build Date: Thursday Feb 17 15:07 2011

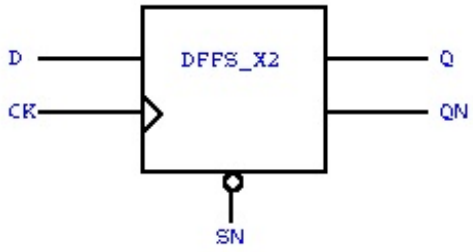
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, and drive strength X2
Strength	2
Cell Area	5.586 um^2
Equation	Q = "((!SN)+(D))" QN = "(!((!SN)+(D)))"
Clock	CK
Set	SN
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
CK	D	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	-	-	L	H
R	H	H	-	-	H	L
F	-	H	L	H	L	H
F	-	H	H	L	H	L
-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.05	0.11	0.06	0.12
	rise	0.05	0.10	0.06	0.12
CK to QN	fall	0.04	*0.10*	0.05	*0.11*
	rise	0.04	*0.10*	0.05	*0.12*
SN to Q	rise	0.11	0.16	0.16	0.21
SN to QN	fall	0.09	*0.17*	0.14	*0.21*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
CK to QN	fall	0.01	*0.05*	0.01	*0.05*
	rise	0.00	*0.05*	0.00	*0.05*
SN to Q	rise	0.01	0.05	0.01	0.05
SN to QN	fall	0.02	*0.06*	0.02	*0.06*

Constraints Time [ns]		
Setup CK to D	fall	0.04
	rise	0.03
Hold CK to D	fall	0.10
	rise	0.10
Recovery CK to SN	rise	-0.02
Removal CK to SN	rise	0.22

Capacitance [fF]	
CK	0.9687
D	1.1829
SN	1.3404

Leakage [nW]
299.95

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	8118.73	16710.72
	rise	9523.80	17845.92
D	fall	3170.87	6209.22
	rise	1313.61	4282.82
SN	fall	1244.61	2949.60
	rise	-426.98	1108.51

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	10.67	9.41	14.70	13.49
	rise	11.76	8.89	16.42	13.46
CK to QN	fall	11.76	*9.17*	16.42	*13.65*
	rise	10.67	*9.41*	14.70	*13.48*
SN to Q	rise	17.96	12.20	21.35	15.00
SN to QN	fall	17.96	*12.03*	21.35	*15.12*

DFF_X1

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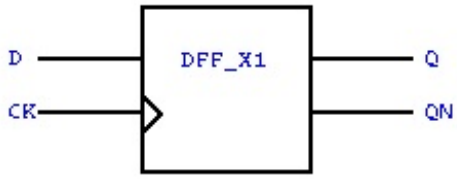
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with drive strength X1
Strength	1
Cell Area	4.522 um^2
Equation	Q = "(D)" QN = "(!(D))"
Clock	CK
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
CK	D	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	-	-	L	H
R	H	-	-	H	L
F	-	L	H	L	H
F	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to Q	fall	0.04	0.10	0.05	0.11
	rise	0.04	0.09	0.05	0.11
CK to QN	fall	0.03	*0.09*	0.04	*0.10*
	rise	0.03	*0.09*	0.04	*0.11*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
CK to QN	fall	0.00	*0.05*	0.00	*0.05*
	rise	0.00	*0.05*	0.00	*0.05*

Constraints Time [ns]		
Setup CK to D	fall	0.04
	rise	0.03
Hold CK to D	fall	0.10
	rise	0.09

Capacitance [fF]	
CK	0.9510
D	1.1588

Leakage [nW]
194.10

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	7738.05	16470.45
	rise	3325.61	11793.59
D	fall	3210.66	6468.61
	rise	1724.61	4930.24

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.73	0.3656	60.73
CK to Q	fall	5.84	5.68	9.78	10.07
	rise	6.06	5.80	10.60	10.35
CK to QN	fall	6.06	*5.47*	10.60	*10.25*
	rise	5.84	*5.69*	9.78	*9.83*

DFF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

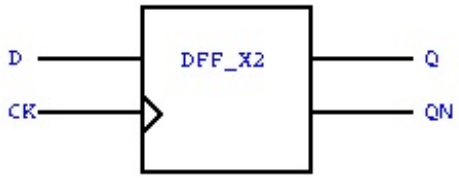
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with drive strength X2
Strength	2
Cell Area	5.054 um^2
Equation	Q = "(D)" QN = "(!(D))"
Clock	CK
Type	Sequential
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
CK	**D**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**	**QN**
R	L	-	-	L	H
R	H	-	-	H	L
F	-	L	H	L	H
F	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
CK to Q	**fall**	0.05	0.10	0.06	0.12
	rise	0.05	0.10	0.06	0.12
CK to QN	**fall**	0.03	*0.10*	0.05	*0.11*
	rise	0.04	*0.10*	0.05	*0.12*

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**121.155**	**0.3656**	**121.155**
CK to Q	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
CK to QN	**fall**	0.01	*0.05*	0.01	*0.05*
	rise	0.00	*0.05*	0.00	*0.05*

Constraints Time [ns]		
Setup CK to D	**fall**	0.04
	rise	0.03
Hold CK to D	**fall**	0.10
	rise	0.09

Capacitance [fF]	
CK	0.9294
D	1.1415

Leakage [nW]
282.00

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	7765.99	16488.21
	rise	3346.21	11808.22
D	fall	3256.42	6500.02
	rise	1782.33	4973.91

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	10.63	8.90	14.69	13.09
	rise	11.63	9.20	16.35	13.99
CK to QN	fall	11.63	*9.39*	16.35	*13.69*
	rise	10.63	*9.09*	14.69	*13.19*

DLH_X1

Databook Build Date: Thursday Feb 17 15:07 2011

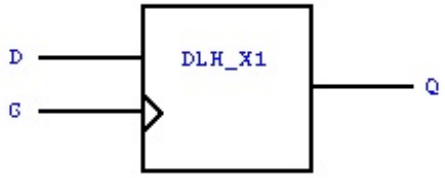
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	High enable Latch with drive strength X1
Strength	1
Cell Area	2.660 um^2
Equation	Q = "(D)"
Enable	G
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**G**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	H	-	-	L
H	H	-	-	H
-	L	L	H	L
-	L	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
D to Q	**fall**	0.02	0.08	0.05	0.11
	rise	0.02	0.08	0.03	0.09
G to Q	**fall**	0.02	0.08	0.03	0.09
	rise	0.03	0.09	0.03	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
D to Q	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
G to Q	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05

Constraints Time [ns]		
Setup G to D	**fall**	0.06
	rise	0.03
Hold G to D	**fall**	0.13
	rise	0.08

Capacitance [fF]	
D	0.9341
G	0.9845

Leakage [nW]
100.12

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1251**
D	**fall**	666.28	650.44
	rise	-520.51	-597.31
G	**fall**	3903.88	12826.47
	rise	2909.82	11840.38

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
D to Q	**fall**	5.35	6.01	10.78	10.59
	rise	3.60	3.99	9.58	8.47
G to Q	**fall**	7.05	7.73	15.45	17.15
	rise	6.59	6.57	14.15	16.01

DLH_X2

Databook Build Date: Thursday Feb 17 15:07 2011

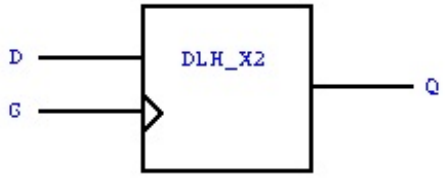
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	High enable Latch with drive strength X2
Strength	2
Cell Area	2.926 um^2
Equation	Q = "(D)"
Enable	G
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**G**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	H	-	-	L
H	H	-	-	H
-	L	L	H	L
-	L	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**120.697**	**0.3656**	**120.697**
D to Q	**fall**	0.02	0.08	0.05	0.12
	rise	0.02	0.08	0.03	0.10
G to Q	**fall**	0.02	0.08	0.03	0.09
	rise	0.03	0.09	0.04	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**120.697**	**0.3656**	**120.697**
D to Q	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
G to Q	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05

Constraints Time [ns]		
Setup G to D	**fall**	0.06
	rise	0.03
Hold G to D	**fall**	0.12
	rise	0.08

Capacitance [fF]	
D	1.1886
G	0.9864

Leakage [nW]
140.70

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1251**
D	**fall**	897.32	874.35
	rise	-701.15	-802.43
G	**fall**	3802.96	12755.94
	rise	2918.39	11845.75

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**120.697**	**0.3656**	**120.697**
D to Q	**fall**	9.12	9.21	17.73	15.34
	rise	6.81	5.34	16.70	11.73
G to Q	**fall**	10.31	10.51	19.65	19.99
	rise	9.84	9.05	17.44	18.82

DLL_X1

Databook Build Date: Thursday Feb 17 15:07 2011

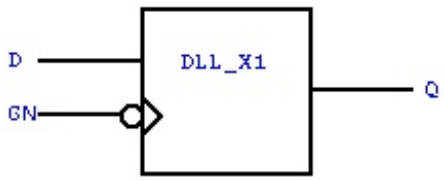
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Low enable Latch with drive strength X1
Strength	1
Cell Area	2.660 um^2
Equation	Q = "(D)"
Enable	GN
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**GN**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	L	-	-	L
H	L	-	-	H
-	H	L	H	L
-	H	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
D to Q	**fall**	0.02	0.08	0.05	0.11
	rise	0.02	0.08	0.03	0.09
GN to Q	**fall**	0.03	0.09	0.05	0.11
	rise	0.02	0.08	0.05	0.11

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
D to Q	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
GN to Q	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05

Constraints Time [ns]		
Setup GN to D	**fall**	0.05
	rise	0.06
Hold GN to D	**fall**	0.09
	rise	0.09

Capacitance [fF]	
D	0.9055
GN	0.9852

Leakage [nW]
100.12

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1251**
D	**fall**	665.90	650.40
	rise	-521.02	-597.33
GN	**fall**	4200.46	13250.41
	rise	2328.51	11315.17

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
D to Q	**fall**	5.34	5.94	10.86	10.55
	rise	3.73	4.41	9.66	9.20
GN to Q	**fall**	8.46	9.05	16.30	18.08
	rise	7.95	8.36	16.54	17.59

DLL_X2

Databook Build Date: Thursday Feb 17 15:07 2011

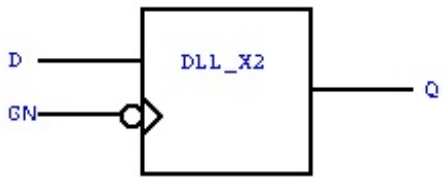
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Low enable Latch with drive strength X2
Strength	2
Cell Area	2.926 um^2
Equation	Q = "(D)"
Enable	GN
Type	Sequential
Input	D
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
D	**GN**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	L	-	-	L
H	L	-	-	H
-	H	L	H	L
-	H	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
D to Q	**fall**	0.02	0.08	0.05	0.11
	rise	0.02	0.08	0.03	0.09
GN to Q	**fall**	0.03	0.09	0.05	0.11
	rise	0.02	0.08	0.05	0.11

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
D to Q	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
GN to Q	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05

Constraints Time [ns]		
Setup GN to D	**fall**	0.05
	rise	0.06
Hold GN to D	**fall**	0.09
	rise	0.08

Capacitance [fF]	
D	1.1575
GN	0.9796

Leakage [nW]
140.69

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		**0.0009**	**0.1251**
D	**fall**	897.19	874.31
	rise	-701.82	-802.73
GN	**fall**	4367.81	13426.65
	rise	2212.94	11184.88

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**120.544**	**0.3656**	**120.544**
D to Q	**fall**	8.82	8.86	17.53	15.04
	rise	6.66	6.02	16.70	12.39
GN to Q	**fall**	11.66	11.70	19.53	20.74
	rise	11.09	10.09	20.69	20.21

FA_X1

Databook Build Date: Thursday Feb 17 15:07 2011

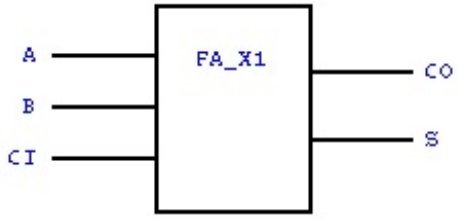
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	4.256 um^2
Equation	CO = "((A & B) | (CI & (A | B)))" S = "(CI ^ (A ^ B))"
Type	Combinational
Input	A, B, CI
Output	CO, S
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**CI**	**CO**	**S**
L	L	L	L	L
L	L	H	L	H
L	H	L	L	H
L	H	H	H	L
H	L	L	L	H
H	L	H	H	L
H	H	L	H	L
H	H	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A to CO	**fall**	0.02	*0.08*	0.06	*0.13*
	rise	0.02	*0.09*	0.05	*0.11*
A to S	**fall**	0.04	0.10	0.06	0.12
	rise	0.04	0.10	0.08	0.14
B to CO	**fall**	0.02	*0.08*	0.06	*0.12*
	rise	0.02	*0.09*	0.04	*0.10*
B to S	**fall**	0.04	0.10	0.07	0.13
	rise	0.04	0.10	0.08	0.14
CI to CO	**fall**	0.02	*0.08*	0.06	*0.12*
	rise	0.02	*0.08*	0.04	*0.10*
CI to S	**fall**	0.04	0.10	0.07	0.13
	rise	0.04	0.10	0.08	0.14

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.2722**	**0.3656**	**60.2722**
A to CO	**fall**	0.00	*0.05*	0.01	*0.05*
	rise	0.00	*0.05*	0.01	*0.05*
A to S	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
B to CO	**fall**	0.00	*0.05*	0.01	*0.05*
	rise	0.00	*0.05*	0.01	*0.05*
B to S	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
CI to CO	**fall**	0.00	*0.05*	0.01	*0.05*
	rise	0.00	*0.05*	0.01	*0.05*
CI to S	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A	3.7755
B	3.4770
CI	2.7844

Leakage [nW]
188.50

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.2722	0.3656	60.2722
A to CO	fall	7.12	*6.54*	10.44	*9.07*
	rise	3.70	*3.51*	7.71	*6.19*
A to S	fall	5.90	6.01	9.76	9.12
	rise	4.77	4.26	9.12	7.43
B to CO	fall	7.37	*6.77*	10.18	*8.99*
	rise	3.90	*3.75*	7.42	*6.03*
B to S	fall	5.75	5.88	9.37	8.71
	rise	4.75	4.42	8.84	7.23
CI to CO	fall	6.72	*6.19*	10.76	*9.11*
	rise	4.22	*3.96*	8.71	*7.06*
CI to S	fall	5.37	5.52	9.89	9.02
	rise	4.85	4.41	9.66	7.77

FILLCELL_X16

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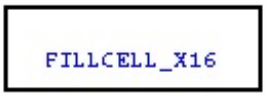
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	4.256 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X1

Databook Build Date: Thursday Feb 17 15:07 2011

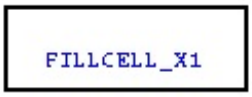
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.266 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X2

Databook Build Date: Thursday Feb 17 15:07 2011

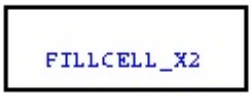
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	0.266 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X32

Databook Build Date: Thursday Feb 17 15:07 2011

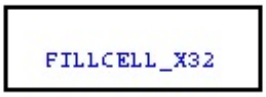
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	32
Cell Area	8.512 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X4

Databook Build Date: Thursday Feb 17 15:07 2011

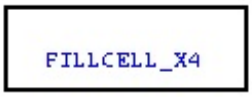
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	1.064 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



FILLCELL_X8

Databook Build Date: Thursday Feb 17 15:07 2011

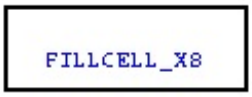
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	2.128 um^2
Type	Physical
PG Pins	VDD (primary_power), VSS (primary_ground)



HA_X1

Databook Build Date: Thursday Feb 17 15:07 2011

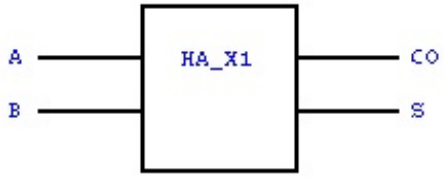
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.660 um^2
Equation	CO = "(A & B)" S = "(A ^ B)"
Type	Combinational
Input	A, B
Output	CO, S
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B**	**CO**	**S**
L	L	L	L
L	H	L	H
H	L	L	H
H	H	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A to CO	**fall**	0.01	0.07	0.03	0.09
	rise	0.01	0.07	0.03	0.10
A to S	**fall**	0.02	*0.09*	0.05	*0.15*
	rise	0.02	*0.12*	0.03	*0.17*
B to CO	**fall**	0.01	0.07	0.04	0.10
	rise	0.01	0.07	0.03	0.09
B to S	**fall**	0.02	*0.09*	0.05	*0.13*
	rise	0.02	*0.12*	0.04	*0.17*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A to CO	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A to S	**fall**	0.00	*0.08*	0.02	*0.09*
	rise	0.01	*0.09*	0.02	*0.10*
B to CO	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
B to S	**fall**	0.00	*0.08*	0.02	*0.08*
	rise	0.00	*0.09*	0.02	*0.11*

Capacitance [fF]	
A	3.1570
B	3.4530

Leakage [nW]
151.74

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A to CO	fall	3.71	3.92	13.02	9.88
	rise	1.36	1.50	10.07	6.48
A to S	fall	4.46	*4.20*	11.58	*9.31*
	rise	4.08	*3.69*	11.42	*9.61*
B to CO	fall	4.66	4.77	14.24	10.87
	rise	1.18	1.33	8.26	5.56
B to S	fall	4.57	*4.44*	9.97	*8.64*
	rise	4.12	*4.07*	12.10	*9.37*

INV_X16

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	4.522 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	969.238	0.3656	969.238
A to ZN	**fall**	0.00	0.06	-0.00	0.10
	rise	0.00	0.06	0.02	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	969.238	0.3656	969.238
A to ZN	**fall**	0.00	0.05	0.02	0.07
	rise	0.00	0.05	0.02	0.07

Capacitance [fF]	
A	25.0612

Leakage [nW]
560.42

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	969.238	0.3656	969.238
A to ZN	**fall**	-0.00	-0.02	182.87	85.06
	rise	35.37	35.93	263.26	164.08

INV_X1

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.532 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
A to ZN	**fall**	0.00	0.06	-0.00	0.10
	rise	0.00	0.06	0.02	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
A to ZN	**fall**	0.00	0.05	0.02	0.07
	rise	0.00	0.05	0.02	0.07

Capacitance [fF]	
A	1.7040

Leakage [nW]
35.03

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.73**	**0.3656**	**60.73**
A to ZN	**fall**	-0.00	-0.00	11.36	5.18
	rise	2.35	2.12	16.48	10.03

INV_X2

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	0.798 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
A to ZN	**fall**	0.00	0.06	-0.00	0.10
	rise	0.00	0.06	0.02	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
A to ZN	**fall**	0.00	0.05	0.02	0.07
	rise	0.00	0.05	0.02	0.07

Capacitance [fF]	
A	3.2329

Leakage [nW]
70.05

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**121.46**	**0.3656**	**121.46**
A to ZN	**fall**	-0.00	-0.00	23.18	10.57
	rise	4.61	3.66	32.81	20.22

INV_X32

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	32
Cell Area	8.778 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	1923.83	0.3656	1923.83
A to ZN	**fall**	0.00	0.06	-0.00	0.10
	rise	0.00	0.07	0.02	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	1923.83	0.3656	1923.83
A to ZN	**fall**	0.00	0.05	0.02	0.07
	rise	0.00	0.06	0.02	0.07

Capacitance [fF]	
A	48.5810

Leakage [nW]
1120.85

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	1923.83	0.3656	1923.83
A to ZN	**fall**	-0.00	-0.05	359.62	172.76
	rise	69.27	58.25	517.56	322.38

INV_X4

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	1.330 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**242.92**	**0.3656**	**242.92**
A to ZN	**fall**	0.00	0.06	-0.00	0.10
	rise	0.00	0.06	0.02	0.12

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**242.92**	**0.3656**	**242.92**
A to ZN	**fall**	0.00	0.05	0.02	0.07
	rise	0.00	0.05	0.02	0.07

Capacitance [fF]	
A	6.2077

Leakage [nW]
140.11

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**242.92**	**0.3656**	**242.92**
A to ZN	**fall**	-0.00	-0.01	46.38	21.17
	rise	8.86	7.06	65.64	42.10

INV_X8

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	2.394 um^2
Equation	ZN = "!A"
Type	Combinational
Input	A
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table	
A	**ZN**
L	H
H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	485.229	0.3656	485.229
A to ZN	**fall**	0.00	0.06	-0.00	0.10
	rise	0.00	0.06	0.02	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	485.229	0.3656	485.229
A to ZN	**fall**	0.00	0.05	0.02	0.07
	rise	0.00	0.05	0.02	0.07

Capacitance [fF]	
A	11.5276

Leakage [nW]
280.22

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	485.229	0.3656	485.229
A to ZN	**fall**	-0.00	-0.01	93.45	42.47
	rise	17.09	19.53	132.16	84.49

LOGIC0_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.532 um^2
Equation	Z = "0"
Type	Constant
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)

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LOGIC1_X1

Databook Build Date: Thursday Feb 17 15:07 2011

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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.532 um^2
Equation	Z = "1"
Type	Constant
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)

NANGATE MUX2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

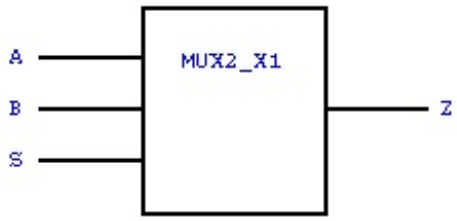
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.862 um^2
Equation	Z = "((S & B) | (A & !S))"
Type	Combinational
Input	A, B, S
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B**	**S**	**Z**
L	-	L	L
H	-	L	H
-	L	H	L
-	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.5011**	**0.3656**	**60.5011**
A to Z	**fall**	0.02	0.08	0.05	0.11
	rise	0.02	0.08	0.03	0.09
B to Z	**fall**	0.02	0.08	0.05	0.11
	rise	0.02	0.08	0.03	0.09
S to Z	**fall**	0.02	0.08	0.05	0.11
	rise	0.02	0.08	0.05	0.11

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.5011**	**0.3656**	**60.5011**
A to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
B to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
S to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A	0.9747
B	0.9662
S	1.8963

Leakage [nW]
88.19

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.5011**	**0.3656**	**60.5011**
A to Z	**fall**	4.82	5.23	9.94	9.55
	rise	3.41	3.55	9.04	8.07
B to Z	**fall**	5.17	5.69	10.25	9.99
	rise	3.07	3.13	8.77	7.00
S to Z	**fall**	5.35	5.85	15.83	17.08
	rise	4.86	4.55	15.59	16.06

MUX2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

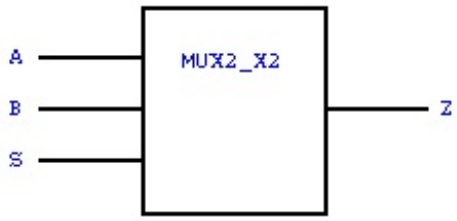
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	Z = "((S & B) | (A & !S))"
Type	Combinational
Input	A, B, S
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B**	**S**	**Z**
L	-	L	L
H	-	L	H
-	L	H	L
-	H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A to Z	**fall**	0.02	0.08	0.04	0.11
	rise	0.02	0.08	0.04	0.10
B to Z	**fall**	0.02	0.08	0.05	0.12
	rise	0.02	0.08	0.02	0.09
S to Z	**fall**	0.03	0.09	0.04	0.10
	rise	0.03	0.09	0.05	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
B to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
S to Z	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A	1.6368
B	1.7725
S	2.5933

Leakage [nW]
169.58

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A to Z	**fall**	9.58	11.06	18.99	19.47
	rise	6.95	6.45	21.20	16.75
B to Z	**fall**	9.52	10.78	24.08	22.57
	rise	5.21	4.38	17.01	15.77
S to Z	**fall**	10.93	12.22	24.76	26.09
	rise	9.39	8.50	23.44	22.75

NANGATE NAND2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

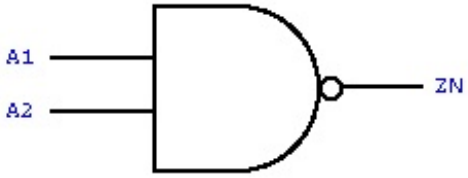
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	ZN = "!(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	59.3567	0.3656	59.3567
A1 to ZN	**fall**	0.00	0.09	0.01	0.13
	rise	0.00	0.06	0.01	0.11
A2 to ZN	**fall**	0.01	0.09	0.01	0.12
	rise	0.00	0.06	0.02	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	59.3567	0.3656	59.3567
A1 to ZN	**fall**	0.00	0.07	0.02	0.09
	rise	0.00	0.05	0.02	0.07
A2 to ZN	**fall**	0.00	0.07	0.02	0.08
	rise	0.00	0.05	0.02	0.07

Capacitance [fF]	
A1	1.6192
A2	1.6795

Leakage [nW]
43.63

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	59.3567	0.3656	59.3567
A1 to ZN	**fall**	0.15	0.28	10.82	4.49
	rise	2.80	2.55	13.57	8.66
A2 to ZN	**fall**	0.16	0.29	9.39	3.57
	rise	3.69	3.29	15.43	10.11

NANGATE NAND2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

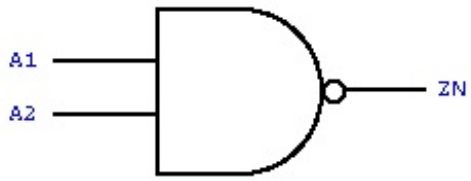
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "!(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	118.713	0.3656	118.713
A1 to ZN	**fall**	0.00	0.09	0.01	0.13
	rise	0.00	0.06	0.01	0.11
A2 to ZN	**fall**	0.01	0.09	0.01	0.12
	rise	0.00	0.06	0.02	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	118.713	0.3656	118.713
A1 to ZN	**fall**	0.00	0.08	0.02	0.09
	rise	0.00	0.05	0.02	0.07
A2 to ZN	**fall**	0.00	0.08	0.02	0.08
	rise	0.00	0.05	0.02	0.07

Capacitance [fF]	
A1	3.0995
A2	3.4867

Leakage [nW]
87.27

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	118.713	0.3656	118.713
A1 to ZN	**fall**	0.28	0.57	21.82	8.98
	rise	5.69	5.69	27.36	18.17
A2 to ZN	**fall**	0.30	0.57	18.91	7.14
	rise	7.45	7.43	30.99	20.33

NANGATE NAND2_X4

Databook Build Date: Thursday Feb 17 15:07 2011

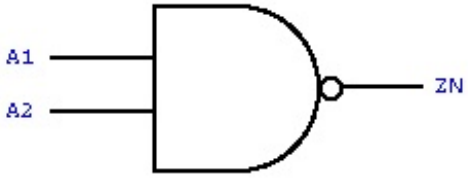
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "!(A1 & A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	237.427	0.3656	237.427
A1 to ZN	**fall**	0.00	0.09	0.01	0.13
	rise	0.00	0.06	0.01	0.11
A2 to ZN	**fall**	0.01	0.09	0.01	0.12
	rise	0.00	0.06	0.02	0.12

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	237.427	0.3656	237.427
A1 to ZN	**fall**	0.00	0.08	0.02	0.09
	rise	0.00	0.05	0.02	0.07
A2 to ZN	**fall**	0.00	0.08	0.02	0.08
	rise	0.00	0.05	0.02	0.07

Capacitance [fF]	
A1	6.0070
A2	6.2126

Leakage [nW]
174.53

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	237.427	0.3656	237.427
A1 to ZN	**fall**	0.51	1.13	43.74	17.84
	rise	10.45	11.02	53.67	35.33
A2 to ZN	**fall**	0.56	1.11	37.83	14.07
	rise	14.50	14.35	61.09	40.07

NANGATE NAND3_X1

Databook Build Date: Thursday Feb 17 15:07 2011

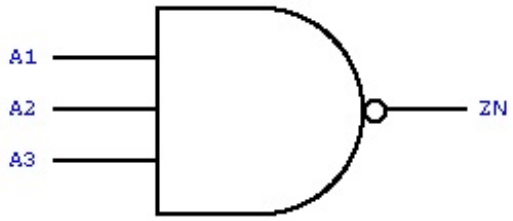
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	H
H	H	H	L
-	L	-	H
-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**58.3649**	**0.3656**	**58.3649**
A1 to ZN	**fall**	0.01	0.12	0.02	0.16
	rise	0.00	0.06	0.01	0.11
A2 to ZN	**fall**	0.01	0.12	0.02	0.15
	rise	0.00	0.06	0.02	0.11
A3 to ZN	**fall**	0.01	0.12	0.01	0.13
	rise	0.01	0.06	0.02	0.11

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**58.3649**	**0.3656**	**58.3649**
A1 to ZN	**fall**	0.00	0.10	0.02	0.10
	rise	0.00	0.05	0.02	0.07
A2 to ZN	**fall**	0.00	0.10	0.02	0.10
	rise	0.00	0.05	0.02	0.07
A3 to ZN	**fall**	0.00	0.10	0.02	0.10
	rise	0.00	0.05	0.02	0.07

Capacitance [fF]	
A1	1.6621
A2	1.6168
A3	1.6721

Leakage [nW]
46.17

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**58.3649**	**0.3656**	**58.3649**
A1 to ZN	**fall**	0.55	0.65	10.20	4.13
	rise	3.17	3.01	12.00	8.24
A2 to ZN	**fall**	0.56	0.65	8.80	3.20
	rise	4.12	3.83	13.48	8.90
A3 to ZN	**fall**	0.56	0.65	9.63	3.46
	rise	4.82	5.09	15.81	10.69

NANGATE NAND3_X2

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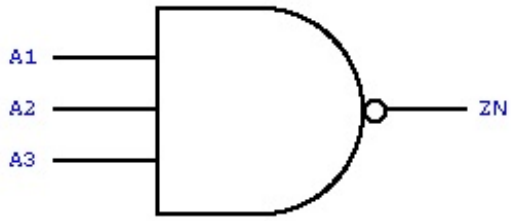
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	H
H	H	H	L
-	L	-	H
-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**116.272**	**0.3656**	**116.272**
A1 to ZN	**fall**	0.01	0.12	0.02	0.16
	rise	0.00	0.06	0.01	0.11
A2 to ZN	**fall**	0.01	0.12	0.02	0.15
	rise	0.00	0.06	0.02	0.11
A3 to ZN	**fall**	0.01	0.12	0.01	0.13
	rise	0.01	0.06	0.02	0.11

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**116.272**	**0.3656**	**116.272**
A1 to ZN	**fall**	0.00	0.10	0.02	0.10
	rise	0.00	0.05	0.02	0.07
A2 to ZN	**fall**	0.00	0.10	0.02	0.10
	rise	0.00	0.05	0.02	0.07
A3 to ZN	**fall**	0.00	0.10	0.02	0.10
	rise	0.00	0.05	0.02	0.07

Capacitance [fF]	
A1	3.1189
A2	3.2701
A3	3.5845

Leakage [nW]
92.34

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**116.272**	**0.3656**	**116.272**
A1 to ZN	**fall**	1.30	1.51	20.62	8.45
	rise	6.56	6.05	24.24	16.00
A2 to ZN	**fall**	1.32	1.52	17.84	6.61
	rise	8.50	8.47	27.28	18.85
A3 to ZN	**fall**	1.33	1.52	19.44	7.15
	rise	9.94	10.01	31.87	21.50

NANGATE NAND3_X4

Databook Build Date: Thursday Feb 17 15:07 2011

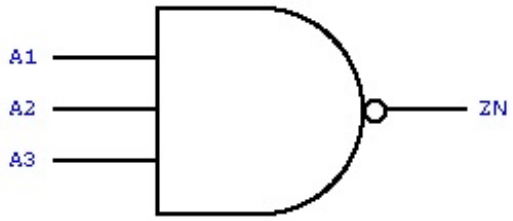
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!((A1 & A2) & A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	-	-	H
H	H	H	L
-	L	-	H
-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**233.154**	**0.3656**	**233.154**
A1 to ZN	**fall**	0.01	0.12	0.02	0.16
	rise	0.00	0.06	0.01	0.11
A2 to ZN	**fall**	0.01	0.12	0.02	0.15
	rise	0.00	0.06	0.02	0.11
A3 to ZN	**fall**	0.01	0.12	0.01	0.13
	rise	0.00	0.06	0.02	0.11

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**233.154**	**0.3656**	**233.154**
A1 to ZN	**fall**	0.00	0.10	0.02	0.10
	rise	0.00	0.05	0.02	0.07
A2 to ZN	**fall**	0.00	0.10	0.02	0.10
	rise	0.00	0.05	0.02	0.07
A3 to ZN	**fall**	0.00	0.10	0.02	0.10
	rise	0.00	0.05	0.02	0.07

Capacitance [fF]	
A1	6.5288
A2	6.8905
A3	7.2076

Leakage [nW]
184.68

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**233.154**	**0.3656**	**233.154**
A1 to ZN	**fall**	2.60	3.05	41.37	16.98
	rise	12.81	11.70	48.31	32.95
A2 to ZN	**fall**	2.63	3.05	36.06	13.27
	rise	16.73	16.13	53.88	36.02
A3 to ZN	**fall**	2.66	3.06	39.23	14.36
	rise	19.41	19.99	63.18	42.37

NANGATE NAND4_X1

Databook Build Date: Thursday Feb 17 15:07 2011

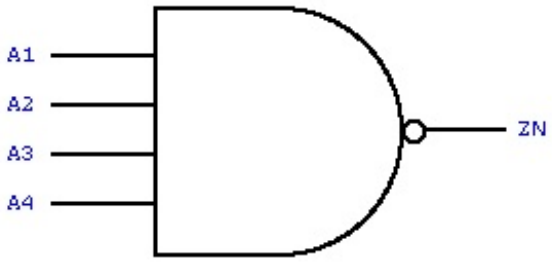
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	H
H	H	H	H	L
-	L	-	-	H
-	-	L	-	H
-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**55.9998**	**0.3656**	**55.9998**
A1 to ZN	**fall**	0.01	0.14	0.03	0.18
	rise	0.00	0.06	0.01	0.11
A2 to ZN	**fall**	0.01	0.14	0.03	0.18
	rise	0.01	0.06	0.01	0.11
A3 to ZN	**fall**	0.01	0.15	0.03	0.17
	rise	0.01	0.06	0.02	0.11
A4 to ZN	**fall**	0.01	0.15	0.02	0.15
	rise	0.01	0.06	0.03	0.11

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**55.9998**	**0.3656**	**55.9998**
A1 to ZN	**fall**	0.01	0.12	0.02	0.12
	rise	0.00	0.05	0.03	0.07
A2 to ZN	**fall**	0.01	0.12	0.02	0.12
	rise	0.00	0.05	0.02	0.07
A3 to ZN	**fall**	0.01	0.12	0.02	0.12
	rise	0.00	0.05	0.02	0.07
A4 to ZN	**fall**	0.01	0.12	0.02	0.12
	rise	0.00	0.05	0.03	0.07

Capacitance [fF]	
A1	1.6258
A2	1.6240
A3	1.6352
A4	1.6868

Leakage [nW]
46.68

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	55.9998	0.3656	55.9998
A1 to ZN	fall	1.16	1.23	9.73	4.27
	rise	3.55	3.43	11.10	7.53
A2 to ZN	fall	1.16	1.24	8.40	3.37
	rise	4.49	4.25	12.37	8.95
A3 to ZN	fall	1.17	1.24	8.86	3.43
	rise	5.30	5.57	14.11	10.17
A4 to ZN	fall	1.17	1.24	9.72	3.69
	rise	5.90	6.13	16.15	11.46

NANGATE NAND4_X2

Databook Build Date: Thursday Feb 17 15:07 2011

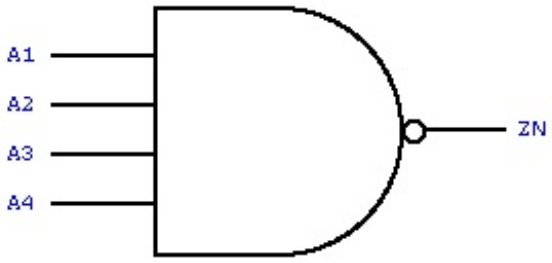
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	H
H	H	H	H	L
-	L	-	-	H
-	-	L	-	H
-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**111.542**	**0.3656**	**111.542**
A1 to ZN	**fall**	0.01	0.14	0.03	0.18
	rise	0.00	0.06	0.01	0.11
A2 to ZN	**fall**	0.01	0.14	0.03	0.18
	rise	0.01	0.06	0.01	0.11
A3 to ZN	**fall**	0.01	0.15	0.03	0.17
	rise	0.01	0.06	0.02	0.11
A4 to ZN	**fall**	0.01	0.15	0.02	0.15
	rise	0.01	0.06	0.03	0.11

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**111.542**	**0.3656**	**111.542**
A1 to ZN	**fall**	0.01	0.12	0.03	0.12
	rise	0.00	0.05	0.03	0.07
A2 to ZN	**fall**	0.01	0.12	0.02	0.12
	rise	0.00	0.05	0.02	0.07
A3 to ZN	**fall**	0.01	0.12	0.02	0.12
	rise	0.00	0.05	0.02	0.07
A4 to ZN	**fall**	0.01	0.12	0.02	0.12
	rise	0.00	0.05	0.03	0.07

Capacitance [fF]	
A1	3.1284
A2	3.3412
A3	3.4721
A4	3.8613

Leakage [nW]
93.37

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	111.542	0.3656	111.542
A1 to ZN	fall	2.30	2.47	19.46	8.52
	rise	7.67	7.68	22.72	15.22
A2 to ZN	fall	2.35	2.47	16.75	6.73
	rise	9.64	9.54	25.34	18.73
A3 to ZN	fall	2.33	2.47	17.67	6.84
	rise	11.29	11.63	28.77	20.80
A4 to ZN	fall	2.34	2.47	19.38	7.37
	rise	12.42	13.41	32.85	23.30

NANGATE NAND4_X4

Databook Build Date: Thursday Feb 17 15:07 2011

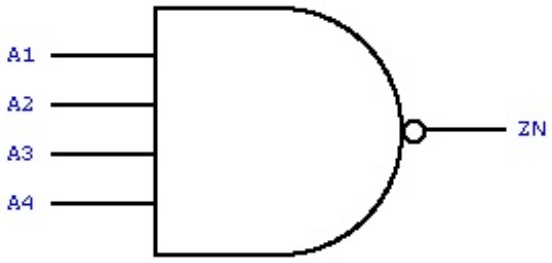
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.788 um^2
Equation	ZN = "!(((A1 & A2) & A3) & A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	-	-	-	H
H	H	H	H	L
-	L	-	-	H
-	-	L	-	H
-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**222.778**	**0.3656**	**222.778**
A1 to ZN	**fall**	0.01	0.14	0.03	0.18
	rise	0.00	0.06	0.00	0.11
A2 to ZN	**fall**	0.01	0.15	0.03	0.18
	rise	0.01	0.06	0.01	0.11
A3 to ZN	**fall**	0.01	0.15	0.02	0.17
	rise	0.01	0.06	0.02	0.11
A4 to ZN	**fall**	0.01	0.15	0.02	0.15
	rise	0.01	0.06	0.03	0.11

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**222.778**	**0.3656**	**222.778**
A1 to ZN	**fall**	0.01	0.12	0.02	0.13
	rise	0.00	0.05	0.02	0.07
A2 to ZN	**fall**	0.01	0.12	0.02	0.12
	rise	0.00	0.05	0.02	0.07
A3 to ZN	**fall**	0.01	0.12	0.02	0.12
	rise	0.00	0.05	0.02	0.07
A4 to ZN	**fall**	0.01	0.12	0.02	0.12
	rise	0.00	0.05	0.03	0.07

Capacitance [fF]	
A1	6.0353
A2	5.8990
A3	5.8611
A4	6.1611

Leakage [nW]
186.73

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	222.778	0.3656	222.778
A1 to ZN	fall	4.49	4.94	39.21	17.07
	rise	13.37	13.14	43.25	28.41
A2 to ZN	fall	4.57	4.96	33.54	13.28
	rise	17.86	18.12	48.71	36.07
A3 to ZN	fall	4.60	4.97	35.07	13.36
	rise	21.32	21.98	56.11	40.35
A4 to ZN	fall	4.58	4.94	38.46	14.29
	rise	23.58	24.96	64.48	45.97

NANGATE NOR2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

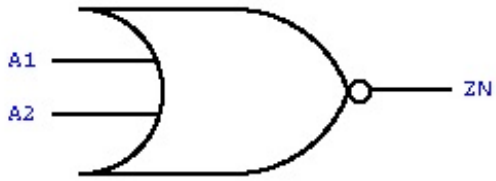
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	0.798 um^2
Equation	ZN = "!(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	26.7029	0.3656	26.7029
A1 to ZN	**fall**	0.00	0.03	-0.01	0.05
	rise	0.00	0.05	0.03	0.10
A2 to ZN	**fall**	0.00	0.03	-0.00	0.05
	rise	0.01	0.05	0.03	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	26.7029	0.3656	26.7029
A1 to ZN	**fall**	0.00	0.02	0.02	0.05
	rise	0.00	0.04	0.02	0.06
A2 to ZN	**fall**	0.00	0.02	0.02	0.05
	rise	0.00	0.04	0.02	0.05

Capacitance [fF]	
A1	1.7384
A2	1.6781

Leakage [nW]
52.30

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	26.7029	0.3656	26.7029
A1 to ZN	**fall**	-0.00	-0.00	8.64	6.51
	rise	2.86	2.80	14.73	11.30
A2 to ZN	**fall**	0.16	0.32	9.81	7.41
	rise	4.00	4.13	13.17	10.00

NOR2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

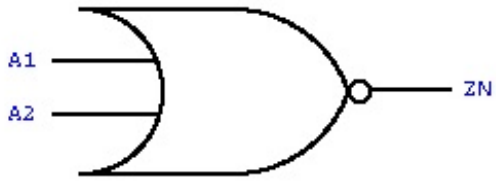
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "!(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	53.4058	0.3656	53.4058
A1 to ZN	**fall**	0.00	0.03	-0.01	0.05
	rise	0.00	0.05	0.03	0.10
A2 to ZN	**fall**	0.00	0.03	-0.00	0.05
	rise	0.01	0.05	0.03	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	53.4058	0.3656	53.4058
A1 to ZN	**fall**	0.00	0.02	0.02	0.05
	rise	0.00	0.04	0.02	0.06
A2 to ZN	**fall**	0.00	0.02	0.02	0.05
	rise	0.00	0.04	0.02	0.05

Capacitance [fF]	
A1	3.3342
A2	3.3923

Leakage [nW]
104.61

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	53.4058	0.3656	53.4058
A1 to ZN	**fall**	-0.00	-0.00	17.54	13.23
	rise	5.46	5.89	29.34	21.98
A2 to ZN	**fall**	0.54	0.85	19.69	15.02
	rise	7.69	8.03	26.36	19.79

NANGATE NOR2_X4

Databook Build Date: Thursday Feb 17 15:07 2011

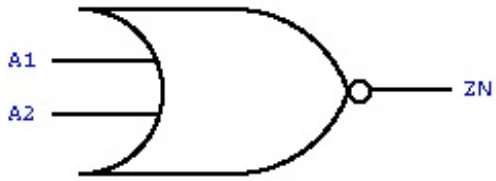
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "!(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	106.811	0.3656	106.811
A1 to ZN	**fall**	0.00	0.03	-0.01	0.05
	rise	0.00	0.05	0.03	0.10
A2 to ZN	**fall**	0.00	0.03	-0.00	0.05
	rise	0.01	0.05	0.03	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	106.811	0.3656	106.811
A1 to ZN	**fall**	0.00	0.02	0.02	0.05
	rise	0.00	0.04	0.02	0.06
A2 to ZN	**fall**	0.00	0.02	0.02	0.05
	rise	0.00	0.04	0.02	0.05

Capacitance [fF]	
A1	6.8476
A2	6.7435

Leakage [nW]
209.21

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	106.811	0.3656	106.811
A1 to ZN	**fall**	-0.00	-0.00	34.83	26.54
	rise	10.75	10.62	58.62	44.32
A2 to ZN	**fall**	1.06	1.71	39.26	30.11
	rise	15.14	15.61	52.47	39.31

NANGATE NOR3_X1

Databook Build Date: Thursday Feb 17 15:07 2011

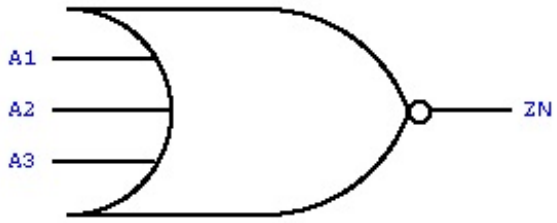
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	H
H	-	-	L
-	H	-	L
-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	16.0217	0.3656	16.0217
A1 to ZN	**fall**	0.00	0.02	-0.01	0.03
	rise	0.01	0.04	0.04	0.09
A2 to ZN	**fall**	0.00	0.02	-0.01	0.03
	rise	0.01	0.05	0.04	0.09
A3 to ZN	**fall**	0.00	0.02	0.00	0.04
	rise	0.01	0.05	0.03	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	16.0217	0.3656	16.0217
A1 to ZN	**fall**	0.00	0.01	0.02	0.04
	rise	0.00	0.04	0.02	0.05
A2 to ZN	**fall**	0.00	0.02	0.02	0.04
	rise	0.00	0.04	0.02	0.05
A3 to ZN	**fall**	0.00	0.02	0.02	0.04
	rise	0.00	0.04	0.02	0.05

Capacitance [fF]	
A1	1.7958
A2	1.7073
A3	1.6483

Leakage [nW]
67.93

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	16.0217	0.3656	16.0217
A1 to ZN	**fall**	-0.00	-0.00	6.93	6.55
	rise	3.19	3.39	13.30	10.94
A2 to ZN	**fall**	0.29	0.41	7.84	7.23
	rise	4.29	4.27	12.02	9.82
A3 to ZN	**fall**	0.57	0.91	9.21	8.14
	rise	5.28	5.42	13.20	10.60

NANGATE NOR3_X2

Databook Build Date: Thursday Feb 17 15:07 2011

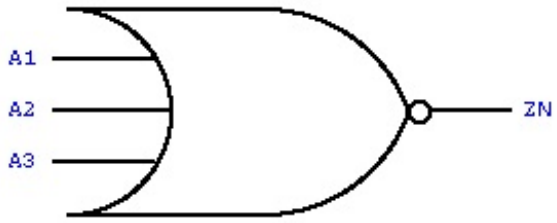
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	H
H	-	-	L
-	H	-	L
-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	31.7383	0.3656	31.7383
A1 to ZN	**fall**	0.00	0.02	-0.01	0.03
	rise	0.01	0.04	0.04	0.09
A2 to ZN	**fall**	0.00	0.02	-0.01	0.03
	rise	0.01	0.05	0.04	0.09
A3 to ZN	**fall**	0.00	0.02	0.00	0.04
	rise	0.01	0.05	0.03	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	31.7383	0.3656	31.7383
A1 to ZN	**fall**	0.00	0.01	0.02	0.04
	rise	0.00	0.04	0.02	0.05
A2 to ZN	**fall**	0.00	0.02	0.02	0.04
	rise	0.00	0.04	0.02	0.05
A3 to ZN	**fall**	0.00	0.02	0.02	0.04
	rise	0.00	0.04	0.02	0.05

Capacitance [fF]	
A1	3.4218
A2	3.5107
A3	3.5164

Leakage [nW]
135.85

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	31.7383	0.3656	31.7383
A1 to ZN	**fall**	-0.00	-0.00	14.19	13.30
	rise	6.56	6.86	26.85	22.00
A2 to ZN	**fall**	0.80	1.04	15.90	14.68
	rise	8.69	8.71	24.25	19.59
A3 to ZN	**fall**	1.36	2.04	18.59	16.50
	rise	10.67	10.98	26.62	21.65

NANGATE NOR3_X4

Databook Build Date: Thursday Feb 17 15:07 2011

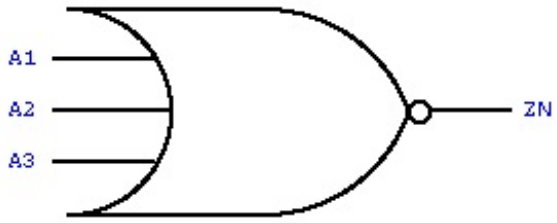
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.724 um^2
Equation	ZN = "!((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	H
H	-	-	L
-	H	-	L
-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	63.324	0.3656	63.324
A1 to ZN	**fall**	0.00	0.02	-0.01	0.03
	rise	0.01	0.04	0.04	0.09
A2 to ZN	**fall**	0.00	0.02	-0.01	0.03
	rise	0.01	0.05	0.04	0.09
A3 to ZN	**fall**	0.00	0.02	0.00	0.04
	rise	0.01	0.05	0.03	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	63.324	0.3656	63.324
A1 to ZN	**fall**	0.00	0.01	0.02	0.04
	rise	0.00	0.04	0.02	0.05
A2 to ZN	**fall**	0.00	0.02	0.02	0.04
	rise	0.00	0.04	0.02	0.05
A3 to ZN	**fall**	0.00	0.02	0.02	0.04
	rise	0.00	0.04	0.02	0.05

Capacitance [fF]	
A1	6.6211
A2	6.2956
A3	6.1649

Leakage [nW]
271.70

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	63.324	0.3656	63.324
A1 to ZN	**fall**	-0.00	-0.00	27.76	26.27
	rise	12.96	13.68	53.53	44.07
A2 to ZN	**fall**	1.58	2.09	31.35	28.98
	rise	18.23	18.78	48.59	39.51
A3 to ZN	**fall**	2.60	4.06	37.00	32.59
	rise	22.50	23.17	53.99	43.30

NOR4_X1

Databook Build Date: Thursday Feb 17 15:07 2011

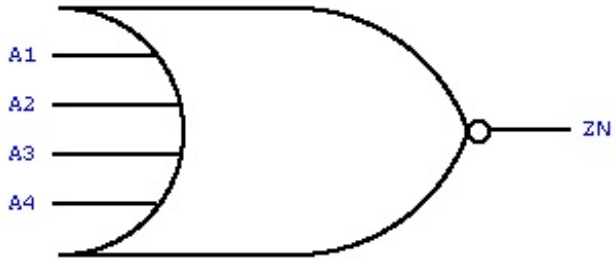
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	-	L
-	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**10.4713**	**0.3656**	**10.4713**
A1 to ZN	**fall**	0.00	0.01	-0.01	0.02
	rise	0.01	0.04	0.05	0.09
A2 to ZN	**fall**	0.00	0.01	-0.01	0.02
	rise	0.01	0.05	0.05	0.09
A3 to ZN	**fall**	0.01	0.02	-0.00	0.02
	rise	0.02	0.05	0.05	0.08
A4 to ZN	**fall**	0.01	0.02	0.00	0.03
	rise	0.02	0.05	0.04	0.08

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**10.4713**	**0.3656**	**10.4713**
A1 to ZN	**fall**	0.00	0.01	0.02	0.04
	rise	0.01	0.04	0.02	0.05
A2 to ZN	**fall**	0.00	0.01	0.02	0.04
	rise	0.01	0.04	0.02	0.05
A3 to ZN	**fall**	0.00	0.01	0.02	0.04
	rise	0.01	0.04	0.02	0.04
A4 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.01	0.04	0.02	0.04

Capacitance [fF]	
A1	1.7832
A2	1.7272
A3	1.6892
A4	1.6356

Leakage [nW]
84.14

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	10.4713	0.3656	10.4713
A1 to ZN	fall	-0.00	-0.00	5.79	5.95
	rise	3.87	3.95	12.61	11.09
A2 to ZN	fall	0.33	0.41	6.57	6.59
	rise	4.95	5.06	11.42	10.03
A3 to ZN	fall	0.72	0.98	7.63	7.42
	rise	5.93	5.91	12.27	10.62
A4 to ZN	fall	0.83	1.35	8.69	8.23
	rise	6.86	6.99	13.36	11.54

NOR4_X2

Databook Build Date: Thursday Feb 17 15:07 2011

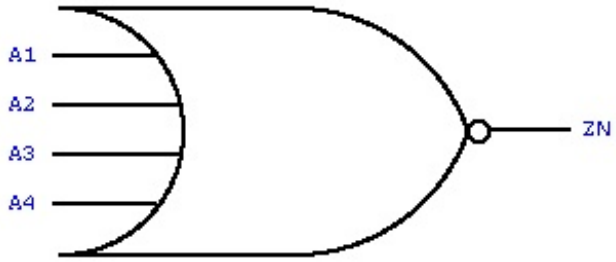
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	-	L
-	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**20.9045**	**0.3656**	**20.9045**
A1 to ZN	**fall**	0.00	0.01	-0.02	0.02
	rise	0.01	0.04	0.05	0.09
A2 to ZN	**fall**	0.00	0.01	-0.01	0.02
	rise	0.01	0.05	0.05	0.09
A3 to ZN	**fall**	0.01	0.02	-0.00	0.02
	rise	0.02	0.05	0.05	0.08
A4 to ZN	**fall**	0.00	0.02	0.00	0.03
	rise	0.02	0.05	0.04	0.08

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**20.9045**	**0.3656**	**20.9045**
A1 to ZN	**fall**	0.00	0.01	0.02	0.04
	rise	0.01	0.04	0.02	0.05
A2 to ZN	**fall**	0.00	0.01	0.02	0.04
	rise	0.01	0.04	0.02	0.05
A3 to ZN	**fall**	0.00	0.01	0.02	0.04
	rise	0.01	0.04	0.02	0.04
A4 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.01	0.04	0.02	0.04

Capacitance [fF]	
A1	3.4813
A2	3.4954
A3	3.6139
A4	3.6694

Leakage [nW]
168.29

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	20.9045	0.3656	20.9045
A1 to ZN	fall	-0.00	-0.00	11.78	12.14
	rise	7.56	7.74	25.14	21.97
A2 to ZN	fall	0.88	1.05	13.39	13.40
	rise	9.72	9.80	22.76	19.85
A3 to ZN	fall	1.63	2.19	15.43	15.03
	rise	11.67	11.77	24.44	21.14
A4 to ZN	fall	1.89	2.93	17.50	16.66
	rise	13.55	13.77	26.64	22.93

NOR4_X4

Databook Build Date: Thursday Feb 17 15:07 2011

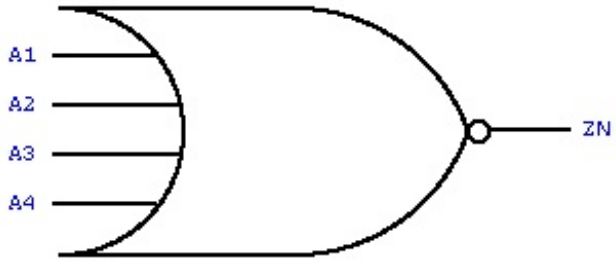
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.788 um^2
Equation	ZN = "!(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	H
H	-	-	-	L
-	H	-	-	L
-	-	H	-	L
-	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**41.5039**	**0.3656**	**41.5039**
A1 to ZN	**fall**	0.00	0.01	-0.02	0.02
	rise	0.01	0.04	0.05	0.09
A2 to ZN	**fall**	0.00	0.01	-0.01	0.02
	rise	0.01	0.05	0.05	0.09
A3 to ZN	**fall**	0.01	0.02	-0.00	0.02
	rise	0.02	0.05	0.05	0.09
A4 to ZN	**fall**	0.01	0.02	0.00	0.03
	rise	0.02	0.05	0.04	0.08

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**41.5039**	**0.3656**	**41.5039**
A1 to ZN	**fall**	0.00	0.01	0.02	0.04
	rise	0.01	0.04	0.02	0.05
A2 to ZN	**fall**	0.00	0.01	0.02	0.04
	rise	0.01	0.04	0.02	0.05
A3 to ZN	**fall**	0.00	0.01	0.02	0.04
	rise	0.01	0.04	0.02	0.04
A4 to ZN	**fall**	0.00	0.01	0.03	0.04
	rise	0.01	0.04	0.02	0.04

Capacitance [fF]	
A1	6.6961
A2	6.3694
A3	6.2058
A4	6.1033

Leakage [nW]
336.57

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	41.5039	0.3656	41.5039
A1 to ZN	fall	-0.00	-0.00	23.10	23.87
	rise	14.94	15.47	50.10	43.77
A2 to ZN	fall	1.75	2.09	26.27	26.41
	rise	20.07	20.28	45.77	39.90
A3 to ZN	fall	3.25	4.35	30.37	29.63
	rise	24.63	24.75	49.51	42.98
A4 to ZN	fall	3.64	5.80	34.44	32.75
	rise	29.13	29.56	54.40	46.66

OAI211_X1

Databook Build Date: Thursday Feb 17 15:07 2011

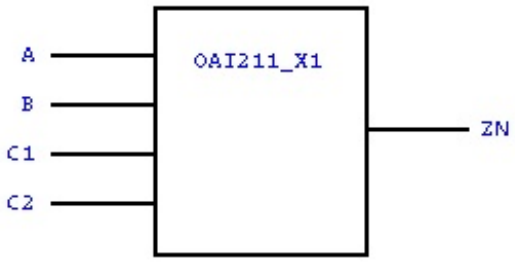
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!(((C1 | C2) & A) & B)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	-	-	-	H
H	H	H	-	L
H	H	-	H	L
-	L	-	-	H
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**25.5585**	**0.3656**	**25.5585**
A to ZN	**fall**	0.01	0.06	0.03	0.09
	rise	0.01	0.03	0.02	0.07
B to ZN	**fall**	0.01	0.06	0.02	0.08
	rise	0.01	0.03	0.03	0.08
C1 to ZN	**fall**	0.01	0.06	0.02	0.09
	rise	0.01	0.05	0.02	0.09
C2 to ZN	**fall**	0.01	0.06	0.03	0.10
	rise	0.01	0.05	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**25.5585**	**0.3656**	**25.5585**
A to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.00	0.03	0.03	0.05
B to ZN	**fall**	0.01	0.05	0.02	0.05
	rise	0.01	0.03	0.03	0.05
C1 to ZN	**fall**	0.00	0.05	0.03	0.06
	rise	0.00	0.04	0.02	0.06
C2 to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.00	0.04	0.02	0.05

Capacitance [fF]	
A	1.6271
B	1.6799
C1	1.5922
C2	1.6380

Leakage [nW]
56.58

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	25.5585	0.3656	25.5585
A to ZN	fall	0.80	0.90	8.75	4.73
	rise	5.61	5.60	15.15	12.88
B to ZN	fall	0.81	0.90	9.40	5.20
	rise	6.31	6.48	17.38	14.55
C1 to ZN	fall	0.66	0.76	8.69	5.01
	rise	3.59	3.67	11.84	9.18
C2 to ZN	fall	1.23	1.30	9.26	5.80
	rise	4.65	4.77	10.69	8.45

OAI211_X2

Databook Build Date: Thursday Feb 17 15:07 2011

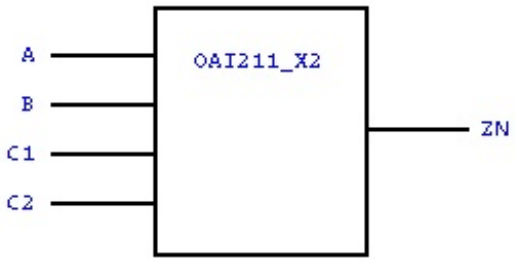
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!(((C1 | C2) & A) & B)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	-	-	-	H
H	H	H	-	L
H	H	-	H	L
-	L	-	-	H
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	50.8118	0.3656	50.8118
A to ZN	**fall**	0.01	0.06	0.03	0.09
	rise	0.01	0.03	0.02	0.07
B to ZN	**fall**	0.01	0.06	0.02	0.08
	rise	0.01	0.03	0.03	0.08
C1 to ZN	**fall**	0.01	0.06	0.02	0.09
	rise	0.01	0.05	0.02	0.09
C2 to ZN	**fall**	0.01	0.06	0.03	0.10
	rise	0.01	0.05	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	50.8118	0.3656	50.8118
A to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.00	0.03	0.03	0.05
B to ZN	**fall**	0.01	0.05	0.02	0.05
	rise	0.00	0.03	0.03	0.05
C1 to ZN	**fall**	0.00	0.05	0.02	0.06
	rise	0.00	0.04	0.02	0.06
C2 to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.00	0.04	0.02	0.05

Capacitance [fF]	
A	3.3560
B	3.2407
C1	2.9860
C2	3.3956

Leakage [nW]
113.17

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	50.8118	0.3656	50.8118
A to ZN	fall	1.83	2.02	17.76	9.69
	rise	11.32	11.32	30.30	25.87
B to ZN	fall	1.82	2.02	19.09	10.64
	rise	12.69	12.98	34.88	29.09
C1 to ZN	fall	1.55	1.74	17.59	10.19
	rise	7.26	7.20	23.66	18.51
C2 to ZN	fall	2.66	2.82	18.74	11.79
	rise	9.49	9.64	21.58	16.96

OAI211_X4

Databook Build Date: Thursday Feb 17 15:07 2011

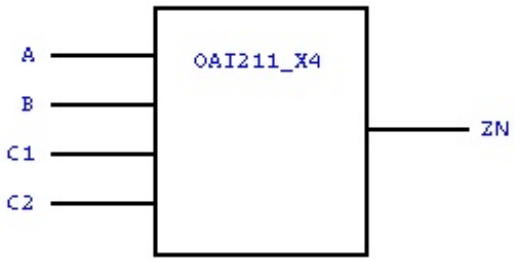
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Equation	ZN = "!(((C1 | C2) & A) & B)"
Type	Combinational
Input	A, B, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A	**B**	**C1**	**C2**	**ZN**
L	-	-	-	H
H	H	H	-	L
H	H	-	H	L
-	L	-	-	H
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	101.624	0.3656	101.624
A to ZN	**fall**	0.01	0.06	0.03	0.09
	rise	0.01	0.03	0.02	0.07
B to ZN	**fall**	0.01	0.06	0.02	0.08
	rise	0.01	0.03	0.03	0.08
C1 to ZN	**fall**	0.01	0.06	0.02	0.09
	rise	0.01	0.05	0.02	0.09
C2 to ZN	**fall**	0.01	0.06	0.03	0.10
	rise	0.01	0.05	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	101.624	0.3656	101.624
A to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.00	0.03	0.03	0.05
B to ZN	**fall**	0.01	0.05	0.02	0.05
	rise	0.01	0.03	0.03	0.05
C1 to ZN	**fall**	0.00	0.05	0.02	0.06
	rise	0.00	0.04	0.02	0.06
C2 to ZN	**fall**	0.01	0.05	0.02	0.06
	rise	0.00	0.04	0.02	0.05

Capacitance [fF]	
A	6.4142
B	6.6190
C1	6.1835
C2	6.7559

Leakage [nW]
226.33

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	101.624	0.3656	101.624
A to ZN	fall	3.65	3.98	35.44	19.14
	rise	22.93	22.99	60.80	52.01
B to ZN	fall	3.62	3.97	38.12	21.01
	rise	25.71	26.74	69.70	58.31
C1 to ZN	fall	3.04	3.46	35.03	20.20
	rise	14.91	14.52	47.44	37.26
C2 to ZN	fall	5.33	5.65	37.34	23.40
	rise	19.06	19.54	43.12	34.64

OAI21_X1

Databook Build Date: Thursday Feb 17 15:07 2011

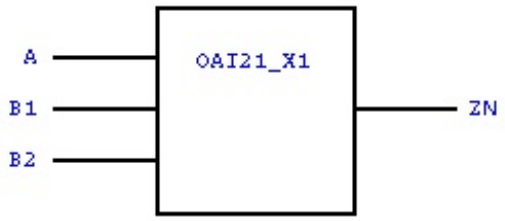
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!(A & (B1 | B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	-	-	H
H	H	-	L
H	-	H	L
-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**26.0544**	**0.3656**	**26.0544**
A to ZN	**fall**	0.01	0.05	0.01	0.07
	rise	0.01	0.03	0.03	0.08
B1 to ZN	**fall**	0.01	0.04	0.01	0.07
	rise	0.01	0.05	0.03	0.10
B2 to ZN	**fall**	0.01	0.04	0.02	0.08
	rise	0.01	0.05	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**26.0544**	**0.3656**	**26.0544**
A to ZN	**fall**	0.00	0.04	0.02	0.05
	rise	0.00	0.03	0.03	0.05
B1 to ZN	**fall**	0.00	0.03	0.02	0.06
	rise	0.00	0.04	0.02	0.06
B2 to ZN	**fall**	0.00	0.04	0.02	0.06
	rise	0.00	0.04	0.02	0.05

Capacitance [fF]	
A	1.6898
B1	1.6703
B2	1.6390

Leakage [nW]
57.02

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**26.0544**	**0.3656**	**26.0544**
A to ZN	**fall**	0.41	0.53	9.42	5.63
	rise	5.22	5.25	17.14	14.37
B1 to ZN	**fall**	0.28	0.39	8.87	5.54
	rise	3.23	3.44	12.98	9.94
B2 to ZN	**fall**	0.81	0.92	9.67	6.38
	rise	4.32	4.47	11.60	9.06

OAI21_X2

Databook Build Date: Thursday Feb 17 15:07 2011

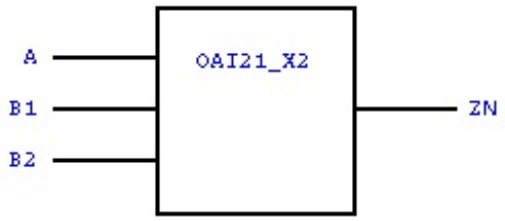
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "!(A & (B1 | B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	-	-	H
H	H	-	L
H	-	H	L
-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**52.1088**	**0.3656**	**52.1088**
A to ZN	**fall**	0.01	0.05	0.01	0.07
	rise	0.01	0.03	0.03	0.08
B1 to ZN	**fall**	0.00	0.04	0.01	0.07
	rise	0.01	0.05	0.03	0.10
B2 to ZN	**fall**	0.01	0.04	0.01	0.08
	rise	0.01	0.05	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**52.1088**	**0.3656**	**52.1088**
A to ZN	**fall**	0.00	0.04	0.02	0.05
	rise	0.00	0.03	0.03	0.05
B1 to ZN	**fall**	0.00	0.03	0.02	0.06
	rise	0.00	0.04	0.02	0.06
B2 to ZN	**fall**	0.00	0.04	0.02	0.06
	rise	0.00	0.04	0.02	0.05

Capacitance [fF]	
A	3.1951
B1	3.1077
B2	3.4700

Leakage [nW]
114.04

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**52.1088**	**0.3656**	**52.1088**
A to ZN	**fall**	0.80	1.05	18.88	11.27
	rise	10.29	10.58	34.20	28.69
B1 to ZN	**fall**	0.54	0.78	17.57	11.03
	rise	6.53	6.86	25.86	19.59
B2 to ZN	**fall**	1.63	1.85	19.33	12.74
	rise	8.71	8.87	23.21	17.93

OAI21_X4

Databook Build Date: Thursday Feb 17 15:07 2011

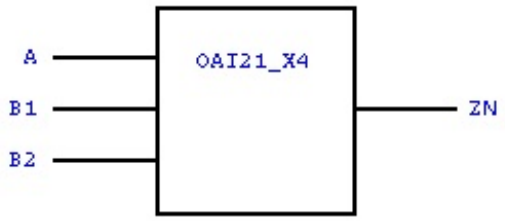
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(A & (B1 | B2))"
Type	Combinational
Input	A, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A	**B1**	**B2**	**ZN**
L	-	-	H
H	H	-	L
H	-	H	L
-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**104.065**	**0.3656**	**104.065**
A to ZN	**fall**	0.01	0.05	0.01	0.07
	rise	0.01	0.03	0.03	0.08
B1 to ZN	**fall**	0.00	0.04	0.00	0.07
	rise	0.01	0.05	0.02	0.10
B2 to ZN	**fall**	0.01	0.04	0.01	0.08
	rise	0.01	0.05	0.02	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**104.065**	**0.3656**	**104.065**
A to ZN	**fall**	0.00	0.04	0.02	0.05
	rise	0.00	0.03	0.03	0.05
B1 to ZN	**fall**	0.00	0.04	0.02	0.06
	rise	0.00	0.04	0.02	0.06
B2 to ZN	**fall**	0.00	0.04	0.02	0.06
	rise	0.00	0.04	0.02	0.05

Capacitance [fF]	
A	6.2013
B1	6.3695
B2	6.7718

Leakage [nW]
228.07

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**104.065**	**0.3656**	**104.065**
A to ZN	**fall**	1.55	2.04	37.73	22.33
	rise	20.58	21.38	68.30	56.51
B1 to ZN	**fall**	1.06	1.55	35.20	21.86
	rise	12.94	12.87	51.33	39.76
B2 to ZN	**fall**	3.28	3.69	38.36	25.23
	rise	17.44	17.77	46.44	35.70

OAI221_X1

Databook Build Date: Thursday Feb 17 15:07 2011

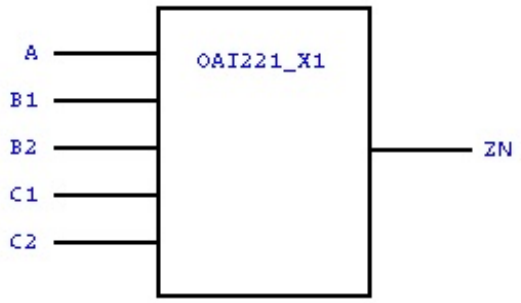
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "!(((C1 | C2) & A) & (B1 | B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	-	-	-	H
H	H	-	H	-	L
H	H	-	-	H	L
H	-	H	H	-	L
H	-	H	-	H	L
-	L	L	-	-	H
-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	22.1634	0.3656	22.1634
A to ZN	**fall**	0.02	0.06	0.03	0.09
	rise	0.01	0.03	0.02	0.07
B1 to ZN	**fall**	0.01	0.06	0.01	0.07
	rise	0.01	0.05	0.05	0.10
B2 to ZN	**fall**	0.02	0.06	0.02	0.07
	rise	0.01	0.05	0.04	0.09
C1 to ZN	**fall**	0.01	0.05	0.02	0.09
	rise	0.01	0.04	0.03	0.09
C2 to ZN	**fall**	0.01	0.05	0.03	0.09
	rise	0.01	0.05	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	22.1634	0.3656	22.1634
A to ZN	**fall**	0.01	0.04	0.02	0.05
	rise	0.01	0.03	0.03	0.05
B1 to ZN	**fall**	0.01	0.04	0.02	0.05
	rise	0.01	0.04	0.03	0.06
B2 to ZN	**fall**	0.01	0.04	0.02	0.05
	rise	0.01	0.04	0.03	0.05
C1 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.04	0.03	0.06
C2 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.04	0.03	0.05

Capacitance [fF]	
A	1.6346
B1	1.7067
B2	1.6459
C1	1.5654
C2	1.6708

Leakage [nW]
86.82

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	22.1634	0.3656	22.1634
A to ZN	fall	1.04	1.15	9.04	5.45
	rise	5.80	5.76	15.87	13.70
B1 to ZN	fall	0.91	1.01	7.44	4.68
	rise	6.76	6.88	16.15	13.64
B2 to ZN	fall	1.47	1.55	8.02	5.44
	rise	7.77	7.95	14.83	12.77
C1 to ZN	fall	0.90	1.01	8.88	5.64
	rise	3.85	3.94	12.45	9.85
C2 to ZN	fall	1.45	1.54	9.53	6.45
	rise	4.94	5.02	11.26	9.14

NANGATE OAI221_X2

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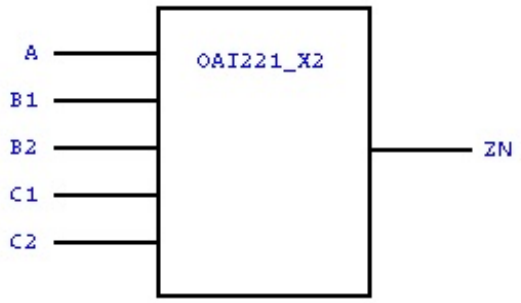
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.926 um^2
Equation	ZN = "!(((C1 | C2) & A) & (B1 | B2))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	-	-	-	H
H	H	-	H	-	L
H	H	-	-	H	L
H	-	H	H	-	L
H	-	H	-	H	L
-	L	L	-	-	H
-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	43.869	0.3656	43.869
A to ZN	**fall**	0.02	0.06	0.03	0.09
	rise	0.01	0.03	0.02	0.07
B1 to ZN	**fall**	0.01	0.06	0.01	0.07
	rise	0.01	0.05	0.05	0.10
B2 to ZN	**fall**	0.02	0.06	0.02	0.07
	rise	0.01	0.05	0.04	0.09
C1 to ZN	**fall**	0.01	0.05	0.02	0.09
	rise	0.01	0.04	0.03	0.09
C2 to ZN	**fall**	0.01	0.05	0.03	0.09
	rise	0.01	0.05	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	43.869	0.3656	43.869
A to ZN	**fall**	0.01	0.04	0.02	0.05
	rise	0.01	0.03	0.03	0.05
B1 to ZN	**fall**	0.01	0.04	0.02	0.05
	rise	0.01	0.04	0.03	0.06
B2 to ZN	**fall**	0.01	0.04	0.02	0.05
	rise	0.01	0.04	0.03	0.05
C1 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.04	0.03	0.06
C2 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.04	0.03	0.05

Capacitance [fF]	
A	3.5497
B1	3.5624
B2	3.1749
C1	2.9971
C2	3.4171

Leakage [nW]
173.65

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	43.869	0.3656	43.869
A to ZN	fall	2.29	2.51	18.19	11.03
	rise	12.08	12.27	32.15	27.80
B1 to ZN	fall	2.04	2.24	15.02	9.54
	rise	13.86	14.29	32.61	27.68
B2 to ZN	fall	3.16	3.31	16.21	11.05
	rise	15.92	16.34	30.02	25.81
C1 to ZN	fall	2.02	2.24	17.96	11.47
	rise	8.01	8.27	25.17	20.00
C2 to ZN	fall	3.14	3.31	19.24	13.09
	rise	10.17	10.10	22.84	18.72

OAI221_X4

Databook Build Date: Thursday Feb 17 15:07 2011

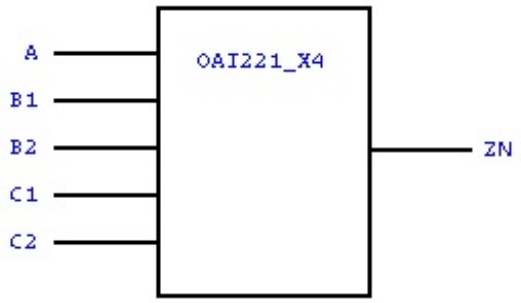
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "!(!(!(((C1 | C2) & A) & (B1 | B2))))"
Type	Combinational
Input	A, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
A	**B1**	**B2**	**C1**	**C2**	**ZN**
L	-	-	-	-	H
H	H	-	H	-	L
H	H	-	-	H	L
H	-	H	H	-	L
H	-	H	-	H	L
-	L	L	-	-	H
-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	**fall**	0.03	0.09	0.06	0.12
	rise	0.02	0.08	0.05	0.11
B1 to ZN	**fall**	0.03	0.09	0.04	0.10
	rise	0.03	0.09	0.07	0.13
B2 to ZN	**fall**	0.04	0.09	0.05	0.10
	rise	0.03	0.09	0.06	0.13
C1 to ZN	**fall**	0.03	0.09	0.05	0.11
	rise	0.02	0.08	0.05	0.11
C2 to ZN	**fall**	0.03	0.09	0.06	0.12
	rise	0.03	0.09	0.05	0.11

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
B1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
B2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
C1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
C2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05

Capacitance [fF]	
A	1.6512
B1	1.7195
B2	1.6735
C1	1.5541
C2	1.6683

Leakage [nW]
283.03

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A to ZN	fall	15.83	17.67	27.93	30.10
	rise	20.69	21.02	36.73	38.21
B1 to ZN	fall	15.66	17.50	26.08	28.89
	rise	22.58	23.84	37.32	37.84
B2 to ZN	fall	16.67	18.50	26.92	28.80
	rise	23.60	24.71	34.27	36.21
C1 to ZN	fall	15.64	17.47	29.72	31.60
	rise	18.95	20.09	33.69	34.43
C2 to ZN	fall	16.62	18.48	30.31	31.62
	rise	20.04	20.16	30.37	31.81

OAI222_X1

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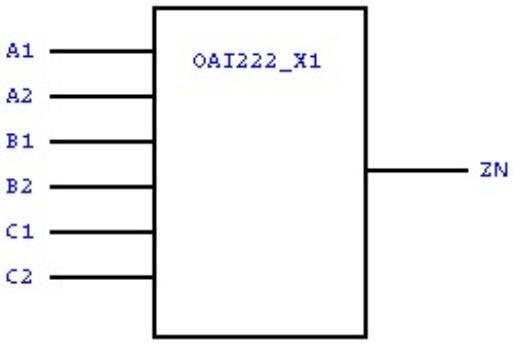
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.128 um^2
Equation	ZN = "!(((A1 | A2) & (B1 | B2)) & (C1 | C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	-	-	-	H
H	-	H	-	H	-	L
H	-	H	-	-	H	L
H	-	-	H	H	-	L
H	-	-	H	-	H	L
-	H	H	-	H	-	L
-	H	H	-	-	H	L
-	H	-	H	H	-	L
-	H	-	H	-	H	L
-	-	L	L	-	-	H
-	-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**20.0653**	**0.3656**	**20.0653**
A1 to ZN	**fall**	0.01	0.05	0.03	0.09
	rise	0.01	0.04	0.03	0.08
A2 to ZN	**fall**	0.02	0.05	0.04	0.09
	rise	0.01	0.04	0.03	0.08
B1 to ZN	**fall**	0.02	0.05	0.03	0.08
	rise	0.01	0.04	0.04	0.09
B2 to ZN	**fall**	0.02	0.06	0.04	0.09
	rise	0.01	0.05	0.04	0.08
C1 to ZN	**fall**	0.02	0.06	0.02	0.07
	rise	0.01	0.05	0.05	0.10
C2 to ZN	**fall**	0.02	0.06	0.03	0.07
	rise	0.02	0.05	0.05	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**20.0653**	**0.3656**	**20.0653**
A1 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.04	0.04	0.06
A2 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.04	0.03	0.05
B1 to ZN	**fall**	0.01	0.04	0.03	0.05
	rise	0.01	0.04	0.04	0.06
B2 to ZN	**fall**	0.01	0.04	0.03	0.05
	rise	0.01	0.04	0.03	0.05
C1 to ZN	**fall**	0.01	0.04	0.02	0.05
	rise	0.01	0.04	0.03	0.06
C2 to ZN	**fall**	0.01	0.04	0.02	0.05
	rise	0.01	0.04	0.03	0.05

Capacitance [fF]	
A1	1.5827
A2	1.6782
B1	1.6384
B2	1.6697
C1	1.6479
C2	1.6077

Leakage [nW]
109.47

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	20.0653	0.3656	20.0653
A1 to ZN	fall	1.14	1.26	9.12	6.22
	rise	4.14	4.20	13.02	10.49
A2 to ZN	fall	1.68	1.79	9.81	7.04
	rise	5.22	5.29	11.84	9.75
B1 to ZN	fall	1.15	1.26	7.53	4.84
	rise	6.21	6.22	14.97	12.83
B2 to ZN	fall	1.70	1.79	8.11	5.59
	rise	7.24	7.29	13.72	11.93
C1 to ZN	fall	1.15	1.26	7.39	4.92
	rise	7.86	8.06	17.22	15.08
C2 to ZN	fall	1.70	1.79	8.00	5.67
	rise	8.94	9.15	15.92	14.07

OAI222_X2

Databook Build Date: Thursday Feb 17 15:07 2011

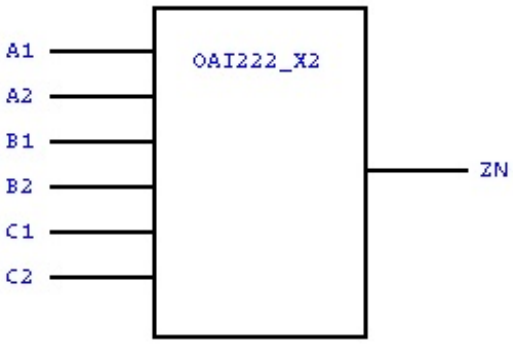
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	3.724 um^2
Equation	ZN = "!(((A1 | A2) & (B1 | B2)) & (C1 | C2))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	-	-	-	H
H	-	H	-	H	-	L
H	-	H	-	-	H	L
H	-	-	H	H	-	L
H	-	-	H	-	H	L
-	H	H	-	H	-	L
-	H	H	-	-	H	L
-	H	-	H	H	-	L
-	H	-	H	-	H	L
-	-	L	L	-	-	H
-	-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**39.5966**	**0.3656**	**39.5966**
A1 to ZN	**fall**	0.01	0.05	0.03	0.09
	rise	0.01	0.04	0.03	0.08
A2 to ZN	**fall**	0.02	0.05	0.04	0.09
	rise	0.01	0.04	0.03	0.08
B1 to ZN	**fall**	0.02	0.05	0.03	0.08
	rise	0.01	0.04	0.04	0.09
B2 to ZN	**fall**	0.02	0.06	0.04	0.09
	rise	0.01	0.05	0.04	0.08
C1 to ZN	**fall**	0.02	0.06	0.02	0.07
	rise	0.01	0.05	0.05	0.10
C2 to ZN	**fall**	0.02	0.06	0.03	0.07
	rise	0.02	0.05	0.05	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**39.5966**	**0.3656**	**39.5966**
A1 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.04	0.04	0.06
A2 to ZN	**fall**	0.01	0.04	0.03	0.06
	rise	0.01	0.04	0.03	0.05
B1 to ZN	**fall**	0.01	0.04	0.03	0.05
	rise	0.01	0.04	0.04	0.06
B2 to ZN	**fall**	0.01	0.04	0.03	0.05
	rise	0.01	0.04	0.03	0.05
C1 to ZN	**fall**	0.01	0.04	0.02	0.05
	rise	0.01	0.04	0.03	0.06
C2 to ZN	**fall**	0.01	0.04	0.02	0.05
	rise	0.01	0.04	0.03	0.05

Capacitance [fF]	
A1	3.2376
A2	3.1559
B1	3.3457
B2	3.0954
C1	3.4613
C2	3.1325

Leakage [nW]
218.94

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	39.5966	0.3656	39.5966
A1 to ZN	fall	2.51	2.74	18.43	12.62
	rise	8.70	8.91	26.41	21.45
A2 to ZN	fall	3.60	3.81	19.80	14.22
	rise	10.87	10.99	23.98	19.80
B1 to ZN	fall	2.53	2.73	15.26	9.88
	rise	12.80	12.74	30.33	26.01
B2 to ZN	fall	3.62	3.80	16.39	11.36
	rise	14.91	14.92	27.76	24.14
C1 to ZN	fall	2.52	2.72	15.01	10.04
	rise	16.28	16.63	34.87	30.48
C2 to ZN	fall	3.62	3.80	16.17	11.52
	rise	18.32	18.71	32.25	28.45

OAI222_X4

Databook Build Date: Thursday Feb 17 15:07 2011

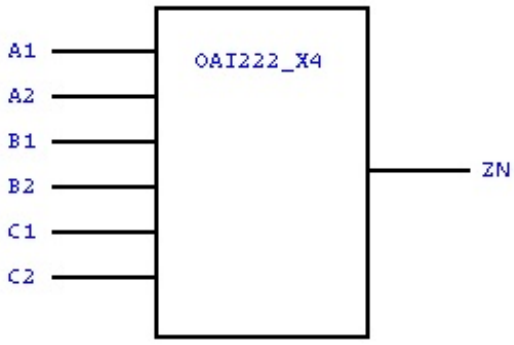
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.724 um^2
Equation	ZN = "!(!(!(((A1 | A2) & (B1 | B2)) & (C1 | C2))))"
Type	Combinational
Input	A1, A2, B1, B2, C1, C2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**B1**	**B2**	**C1**	**C2**	**ZN**
L	L	-	-	-	-	H
H	-	H	-	H	-	L
H	-	H	-	-	H	L
H	-	-	H	H	-	L
H	-	-	H	-	H	L
-	H	H	-	H	-	L
-	H	H	-	-	H	L
-	H	-	H	H	-	L
-	H	-	H	-	H	L
-	-	L	L	-	-	H
-	-	-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.03	0.09	0.06	0.12
	rise	0.03	0.09	0.06	0.12
A2 to ZN	**fall**	0.04	0.09	0.07	0.12
	rise	0.03	0.09	0.05	0.11
B1 to ZN	**fall**	0.04	0.09	0.06	0.12
	rise	0.03	0.09	0.07	0.13
B2 to ZN	**fall**	0.04	0.10	0.06	0.12
	rise	0.03	0.09	0.06	0.12
C1 to ZN	**fall**	0.04	0.10	0.05	0.10
	rise	0.03	0.09	0.07	0.13
C2 to ZN	**fall**	0.04	0.10	0.05	0.11
	rise	0.03	0.09	0.07	0.13

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**242.31**	**0.3656**	**242.31**
A1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
A2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
B1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
B2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
C1 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
C2 to ZN	**fall**	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05

Capacitance [fF]	
A1	1.5771
A2	1.6601
B1	1.6488
B2	1.6622
C1	1.7008
C2	1.6717

Leakage [nW]
314.67

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	fall	15.98	17.80	29.94	31.91
	rise	19.54	19.85	34.74	35.90
A2 to ZN	fall	16.90	18.77	30.63	31.99
	rise	20.57	21.18	31.35	33.04
B1 to ZN	fall	16.00	17.83	27.13	29.23
	rise	22.10	23.03	36.86	37.22
B2 to ZN	fall	16.94	18.79	27.83	29.31
	rise	23.10	23.48	33.65	34.48
C1 to ZN	fall	15.98	17.81	26.09	28.78
	rise	24.17	25.14	39.11	39.90
C2 to ZN	fall	16.91	18.78	26.91	28.75
	rise	25.22	26.87	35.96	37.58

OAI22_X1

Databook Build Date: Thursday Feb 17 15:07 2011

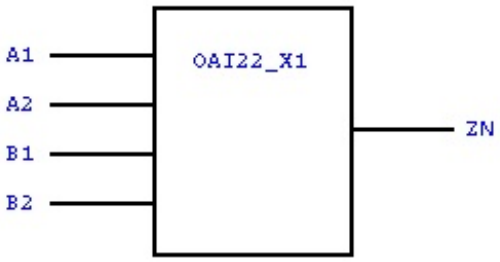
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "!((A1 | A2) & (B1 | B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	L	-	-	H
H	-	H	-	L
H	-	-	H	L
-	H	H	-	L
-	H	-	H	L
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**23.2315**	**0.3656**	**23.2315**
A1 to ZN	**fall**	0.01	0.04	0.01	0.07
	rise	0.01	0.04	0.03	0.09
A2 to ZN	**fall**	0.01	0.04	0.02	0.07
	rise	0.01	0.05	0.03	0.09
B1 to ZN	**fall**	0.01	0.04	0.01	0.06
	rise	0.01	0.05	0.04	0.10
B2 to ZN	**fall**	0.01	0.04	0.02	0.07
	rise	0.01	0.05	0.04	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**23.2315**	**0.3656**	**23.2315**
A1 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.01	0.04	0.03	0.06
A2 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.01	0.04	0.02	0.05
B1 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.01	0.04	0.03	0.06
B2 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.01	0.04	0.02	0.05

Capacitance [fF]	
A1	1.6805
A2	1.6422
B1	1.7056
B2	1.6543

Leakage [nW]
85.39

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	23.2315	0.3656	23.2315
A1 to ZN	fall	0.51	0.64	9.06	6.28
	rise	3.47	3.54	13.58	10.72
A2 to ZN	fall	1.04	1.16	9.96	7.12
	rise	4.57	4.68	12.24	9.55
B1 to ZN	fall	0.51	0.64	7.42	4.98
	rise	5.40	5.43	15.55	13.13
B2 to ZN	fall	1.06	1.16	8.21	5.74
	rise	6.44	6.45	14.09	12.00

OAI22_X2

Databook Build Date: Thursday Feb 17 15:07 2011

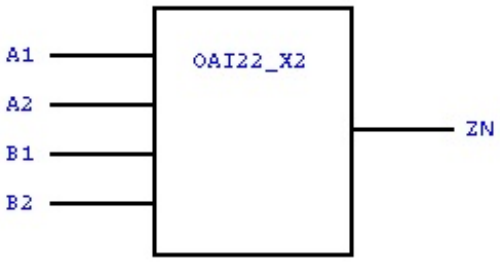
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	ZN = "!((A1 | A2) & (B1 | B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	L	-	-	H
H	-	H	-	L
H	-	-	H	L
-	H	H	-	L
-	H	-	H	L
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**46.3104**	**0.3656**	**46.3104**
A1 to ZN	**fall**	0.01	0.04	0.01	0.07
	rise	0.01	0.04	0.03	0.09
A2 to ZN	**fall**	0.01	0.04	0.02	0.07
	rise	0.01	0.05	0.03	0.09
B1 to ZN	**fall**	0.01	0.04	0.01	0.06
	rise	0.01	0.05	0.04	0.10
B2 to ZN	**fall**	0.01	0.04	0.02	0.07
	rise	0.01	0.05	0.04	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**46.3104**	**0.3656**	**46.3104**
A1 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.01	0.04	0.03	0.06
A2 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.01	0.04	0.02	0.05
B1 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.01	0.04	0.03	0.06
B2 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.01	0.04	0.02	0.05

Capacitance [fF]	
A1	3.1431
A2	3.5171
B1	3.2216
B2	3.3991

Leakage [nW]
170.78

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	46.3104	0.3656	46.3104
A1 to ZN	fall	1.01	1.27	18.04	12.49
	rise	7.13	7.32	27.31	21.41
A2 to ZN	fall	2.09	2.33	19.90	14.20
	rise	9.27	9.19	24.64	19.22
B1 to ZN	fall	1.01	1.27	14.75	9.92
	rise	11.04	11.05	31.28	26.36
B2 to ZN	fall	2.10	2.33	16.39	11.44
	rise	13.17	13.23	28.40	23.97

NANGATE OAI22_X4

Databook Build Date: Thursday Feb 17 15:07 2011

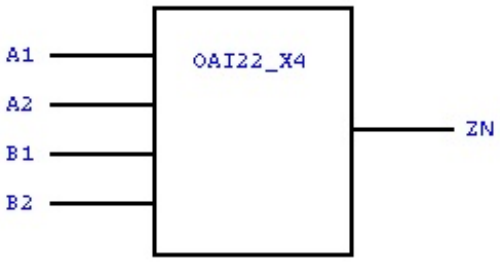
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	4.522 um^2
Equation	ZN = "!((A1 | A2) & (B1 | B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**B1**	**B2**	**ZN**
L	L	-	-	H
H	-	H	-	L
H	-	-	H	L
-	H	H	-	L
-	H	-	H	L
-	-	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**92.4683**	**0.3656**	**92.4683**
A1 to ZN	**fall**	0.01	0.04	0.01	0.07
	rise	0.01	0.04	0.03	0.09
A2 to ZN	**fall**	0.01	0.04	0.02	0.08
	rise	0.01	0.05	0.03	0.09
B1 to ZN	**fall**	0.01	0.04	0.01	0.06
	rise	0.01	0.05	0.04	0.10
B2 to ZN	**fall**	0.01	0.04	0.02	0.07
	rise	0.01	0.05	0.04	0.09

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**92.4683**	**0.3656**	**92.4683**
A1 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.01	0.04	0.03	0.06
A2 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.01	0.04	0.02	0.05
B1 to ZN	**fall**	0.00	0.03	0.02	0.05
	rise	0.01	0.04	0.03	0.06
B2 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.01	0.04	0.02	0.05

Capacitance [fF]	
A1	6.4682
A2	6.7665
B1	6.6259
B2	6.6176

Leakage [nW]
341.56

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	92.4683	0.3656	92.4683
A1 to ZN	fall	1.99	2.54	36.05	24.68
	rise	14.43	14.67	54.52	42.53
A2 to ZN	fall	4.16	4.67	39.64	28.07
	rise	18.73	19.28	49.15	38.44
B1 to ZN	fall	2.01	2.48	29.57	19.67
	rise	22.21	22.72	62.55	52.43
B2 to ZN	fall	4.20	4.65	32.70	22.68
	rise	26.48	26.96	56.85	48.37

OAI33_X1

Databook Build Date: Thursday Feb 17 15:07 2011

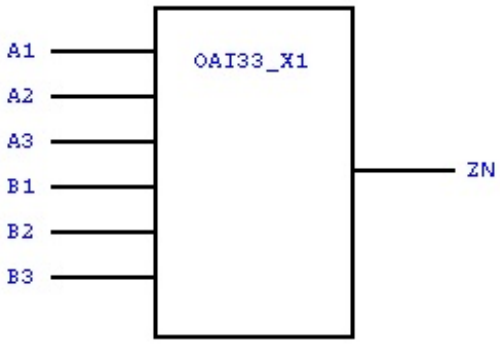
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.862 um^2
Equation	ZN = "!(((A1 | A2) | A3) & ((B1 | B2) | B3))"
Type	Combinational
Input	A1, A2, A3, B1, B2, B3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table						
A1	**A2**	**A3**	**B1**	**B2**	**B3**	**ZN**
L	L	L	-	-	-	H
H	-	-	H	-	-	L
H	-	-	-	H	-	L
H	-	-	-	-	H	L
-	H	-	H	-	-	L
-	H	-	-	H	-	L
-	H	-	-	-	H	L
-	-	H	H	-	-	L
-	-	H	-	H	-	L
-	-	H	-	-	H	L
-	-	-	L	L	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**11.4822**	**0.3656**	**11.4822**
A1 to ZN	**fall**	0.01	0.03	0.01	0.05
	rise	0.01	0.04	0.04	0.08
A2 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.01	0.04	0.04	0.08
A3 to ZN	**fall**	0.01	0.03	0.03	0.06
	rise	0.01	0.04	0.04	0.07
B1 to ZN	**fall**	0.01	0.03	0.01	0.04
	rise	0.02	0.04	0.06	0.09
B2 to ZN	**fall**	0.01	0.03	0.02	0.05
	rise	0.02	0.05	0.06	0.09
B3 to ZN	**fall**	0.02	0.03	0.02	0.05
	rise	0.02	0.05	0.05	0.08

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**11.4822**	**0.3656**	**11.4822**
A1 to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.01	0.04	0.04	0.05
A2 to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.01	0.04	0.03	0.05
A3 to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.01	0.04	0.03	0.05
B1 to ZN	**fall**	0.01	0.02	0.02	0.04
	rise	0.02	0.04	0.03	0.06
B2 to ZN	**fall**	0.01	0.02	0.02	0.04
	rise	0.02	0.04	0.03	0.05
B3 to ZN	**fall**	0.01	0.02	0.03	0.04
	rise	0.02	0.04	0.03	0.05

Capacitance [fF]	
A1	1.7061
A2	1.6372
A3	1.6468
B1	1.7066
B2	1.6598
B3	1.6120

Leakage [nW]
123.43

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	11.4822	0.3656	11.4822
A1 to ZN	fall	0.68	0.80	7.80	7.10
	rise	4.15	4.23	13.22	11.38
A2 to ZN	fall	1.30	1.41	8.60	7.75
	rise	5.21	5.32	12.04	10.36
A3 to ZN	fall	1.71	1.92	9.73	8.56
	rise	6.17	6.26	13.00	11.19
B1 to ZN	fall	0.68	0.80	6.23	5.54
	rise	6.93	7.05	15.57	14.39
B2 to ZN	fall	1.32	1.41	6.91	6.12
	rise	8.00	8.09	14.29	13.24
B3 to ZN	fall	1.74	1.92	7.94	6.87
	rise	8.98	9.08	15.25	14.14

OR2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

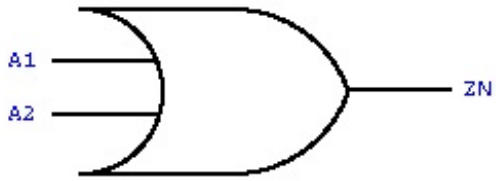
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	L
H	-	H
-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.01	0.07	0.05	0.11
	rise	0.01	0.07	0.01	0.07
A2 to ZN	**fall**	0.02	0.07	0.05	0.11
	rise	0.01	0.07	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A1	0.9567
A2	0.9606

Leakage [nW]
52.86

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	**fall**	3.84	4.38	11.21	10.72
	rise	2.36	2.34	9.56	9.03
A2 to ZN	**fall**	4.36	4.90	9.74	9.92
	rise	2.66	2.86	10.28	8.87

OR2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

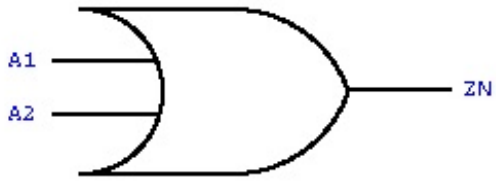
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.330 um^2
Equation	ZN = "(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	L
H	-	H
-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	0.01	0.07	0.05	0.11
	rise	0.01	0.07	0.01	0.07
A2 to ZN	**fall**	0.01	0.07	0.05	0.11
	rise	0.01	0.07	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A1	1.7682
A2	1.7090

Leakage [nW]
106.37

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	7.33	8.35	22.22	21.27
	rise	4.50	4.68	19.28	18.61
A2 to ZN	**fall**	8.41	9.42	19.31	19.71
	rise	5.21	5.35	20.47	18.17

NANGATE OR2_X4

Databook Build Date: Thursday Feb 17 15:07 2011

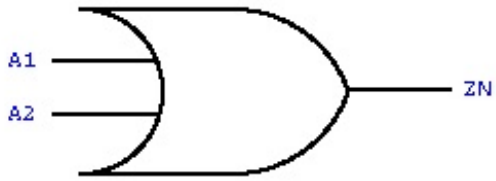
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.394 um^2
Equation	ZN = "(A1 | A2)"
Type	Combinational
Input	A1, A2
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A1	**A2**	**ZN**
L	L	L
H	-	H
-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.01	0.07	0.05	0.11
	rise	0.01	0.07	0.01	0.07
A2 to ZN	**fall**	0.01	0.07	0.04	0.11
	rise	0.01	0.07	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A1	3.4137
A2	3.4994

Leakage [nW]
212.75

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	14.36	16.31	44.38	42.30
	rise	8.38	8.62	37.99	37.22
A2 to ZN	**fall**	16.57	18.45	38.42	39.23
	rise	9.79	9.47	40.88	36.24

NANGATE OR3_X1

Databook Build Date: Thursday Feb 17 15:07 2011

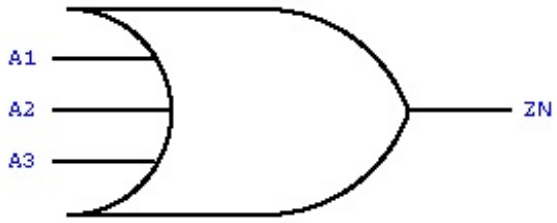
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.330 um^2
Equation	ZN = "((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	L
H	-	-	H
-	H	-	H
-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.02	0.08	0.06	0.13
	rise	0.01	0.07	0.01	0.07
A2 to ZN	**fall**	0.02	0.08	0.06	0.12
	rise	0.01	0.07	0.01	0.08
A3 to ZN	**fall**	0.02	0.09	0.05	0.12
	rise	0.01	0.07	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A1	0.9715
A2	0.9695
A3	0.9394

Leakage [nW]
56.58

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	4.45	4.89	10.93	10.06
	rise	2.37	2.10	8.97	8.01
A2 to ZN	**fall**	4.98	5.41	9.69	9.32
	rise	2.75	2.62	9.51	8.08
A3 to ZN	**fall**	5.47	5.89	9.93	9.91
	rise	3.03	2.78	10.33	8.87

NANGATE OR3_X2

Databook Build Date: Thursday Feb 17 15:07 2011

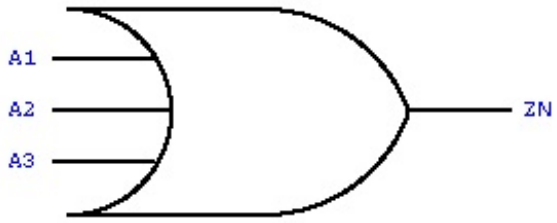
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.596 um^2
Equation	ZN = "((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	L
H	-	-	H
-	H	-	H
-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	0.02	0.08	0.06	0.12
	rise	0.01	0.07	0.01	0.07
A2 to ZN	**fall**	0.02	0.08	0.06	0.12
	rise	0.01	0.07	0.01	0.07
A3 to ZN	**fall**	0.02	0.08	0.05	0.12
	rise	0.01	0.07	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A1	1.7999
A2	1.7328
A3	1.6937

Leakage [nW]
114.00

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
A1 to ZN	**fall**	8.43	9.21	21.60	19.74
	rise	4.56	5.15	17.88	16.77
A2 to ZN	**fall**	9.57	10.27	19.16	18.28
	rise	5.36	4.83	18.98	16.58
A3 to ZN	**fall**	10.47	11.24	19.63	19.52
	rise	5.88	5.62	20.64	17.84

NANGATE OR3_X4

Databook Build Date: Thursday Feb 17 15:07 2011

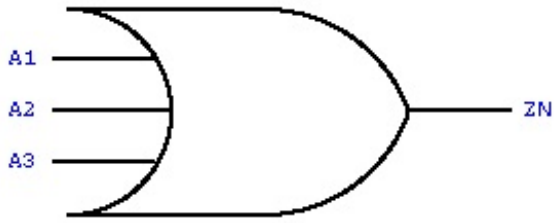
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	ZN = "((A1 | A2) | A3)"
Type	Combinational
Input	A1, A2, A3
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table			
A1	**A2**	**A3**	**ZN**
L	L	L	L
H	-	-	H
-	H	-	H
-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.02	0.08	0.06	0.12
	rise	0.01	0.07	0.01	0.07
A2 to ZN	**fall**	0.02	0.08	0.06	0.12
	rise	0.01	0.07	0.01	0.07
A3 to ZN	**fall**	0.02	0.08	0.05	0.12
	rise	0.01	0.07	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A1	3.4342
A2	3.4610
A3	3.4759

Leakage [nW]
228.00

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	242.31	0.3656	242.31
A1 to ZN	**fall**	16.13	17.60	42.61	38.89
	rise	8.56	8.27	35.32	33.76
A2 to ZN	**fall**	18.30	19.74	37.92	35.99
	rise	10.05	9.14	37.31	31.64
A3 to ZN	**fall**	20.27	21.70	38.75	38.49
	rise	11.25	12.09	40.95	33.59

NANGATE OR4_X1

Databook Build Date: Thursday Feb 17 15:07 2011

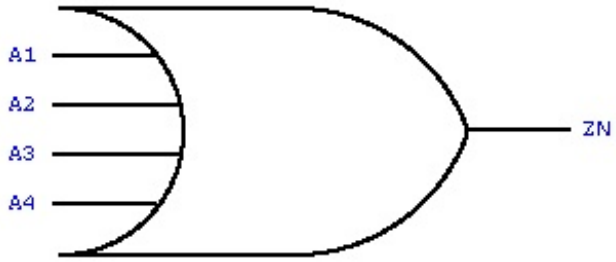
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	L
H	-	-	-	H
-	H	-	-	H
-	-	H	-	H
-	-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.02	0.09	0.07	0.14
	rise	0.01	0.07	0.01	0.07
A2 to ZN	**fall**	0.03	0.09	0.07	0.14
	rise	0.01	0.07	0.01	0.07
A3 to ZN	**fall**	0.03	0.10	0.07	0.13
	rise	0.01	0.07	0.02	0.08
A4 to ZN	**fall**	0.03	0.10	0.06	0.13
	rise	0.01	0.07	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**60.5774**	**0.3656**	**60.5774**
A1 to ZN	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A3 to ZN	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A4 to ZN	**fall**	0.01	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A1	0.9593
A2	0.9693
A3	0.9490
A4	0.9317

Leakage [nW]
62.59

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
A1 to ZN	fall	5.00	5.23	10.71	9.66
	rise	2.41	2.10	8.52	7.67
A2 to ZN	fall	5.53	5.75	9.71	9.00
	rise	2.77	2.44	9.00	7.56
A3 to ZN	fall	6.03	6.23	9.86	9.37
	rise	3.09	3.34	9.65	8.00
A4 to ZN	fall	6.50	6.70	10.21	9.91
	rise	3.24	3.40	10.34	8.48

OR4_X2

Databook Build Date: Thursday Feb 17 15:07 2011

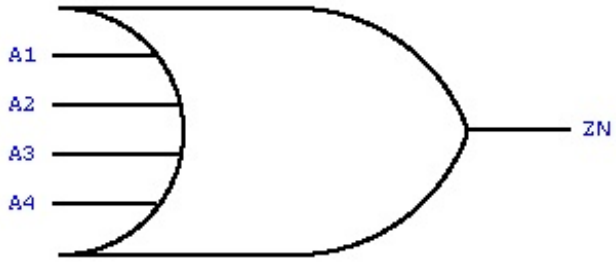
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	1.862 um^2
Equation	ZN = "(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	L
H	-	-	-	H
-	H	-	-	H
-	-	H	-	H
-	-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**120.85**	**0.3656**	**120.85**
A1 to ZN	**fall**	0.02	0.09	0.07	0.13
	rise	0.01	0.07	0.00	0.07
A2 to ZN	**fall**	0.03	0.09	0.07	0.13
	rise	0.01	0.07	0.01	0.07
A3 to ZN	**fall**	0.03	0.09	0.06	0.13
	rise	0.01	0.07	0.01	0.08
A4 to ZN	**fall**	0.03	0.10	0.06	0.13
	rise	0.01	0.07	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**120.85**	**0.3656**	**120.85**
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A4 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A1	1.7774
A2	1.7363
A3	1.6939
A4	1.6637

Leakage [nW]
126.22

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
A1 to ZN	fall	9.46	9.78	21.14	18.79
	rise	4.63	5.15	16.80	15.35
A2 to ZN	fall	10.53	10.84	19.01	17.48
	rise	5.39	5.27	17.82	14.86
A3 to ZN	fall	11.51	11.81	19.28	18.29
	rise	6.07	6.41	19.16	16.05
A4 to ZN	fall	12.43	12.78	20.03	19.40
	rise	6.30	7.07	20.66	17.28

OR4_X4

Databook Build Date: Thursday Feb 17 15:07 2011

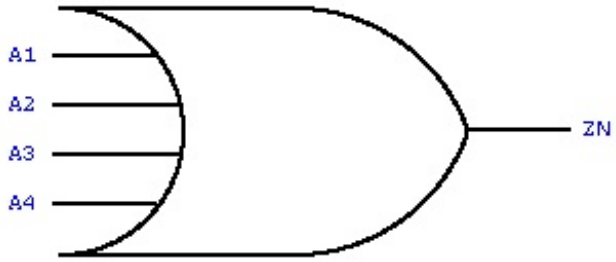
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	3.458 um^2
Equation	ZN = "(((A1 | A2) | A3) | A4)"
Type	Combinational
Input	A1, A2, A3, A4
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table				
A1	**A2**	**A3**	**A4**	**ZN**
L	L	L	L	L
H	-	-	-	H
-	H	-	-	H
-	-	H	-	H
-	-	-	H	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**241.699**	**0.3656**	**241.699**
A1 to ZN	**fall**	0.02	0.09	0.07	0.13
	rise	0.01	0.07	0.00	0.07
A2 to ZN	**fall**	0.03	0.09	0.07	0.13
	rise	0.01	0.07	0.01	0.07
A3 to ZN	**fall**	0.03	0.09	0.06	0.13
	rise	0.01	0.07	0.01	0.08
A4 to ZN	**fall**	0.03	0.09	0.06	0.12
	rise	0.01	0.07	0.02	0.08

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**241.699**	**0.3656**	**241.699**
A1 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A2 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A3 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
A4 to ZN	**fall**	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Capacitance [fF]	
A1	3.4373
A2	3.4770
A3	3.5957
A4	3.6723

Leakage [nW]
252.44

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	241.699	0.3656	241.699
A1 to ZN	fall	18.47	19.15	42.13	37.29
	rise	8.91	7.62	33.42	30.45
A2 to ZN	fall	20.65	21.27	37.76	34.65
	rise	10.44	10.11	35.60	29.12
A3 to ZN	fall	22.58	23.22	38.46	36.29
	rise	11.70	11.26	38.20	30.63
A4 to ZN	fall	24.54	25.15	39.94	38.54
	rise	12.30	13.40	41.15	34.24

SDFFRS_X1

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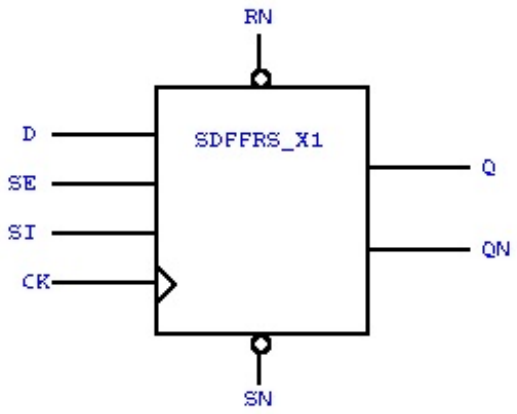
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, scan, and drive strength X1
Strength	1
Cell Area	7.714 um^2
Equation	Q = "(!(!RN)*((((SE * SI) + (D * !SE)))+(!SN)))" QN = "(!(!SN)*(!((((SE * SI) + (D * !SE)))+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table									
CK	D	RN	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	H	-	-	L	H
R	H	H	L	-	H	-	-	H	L
R	-	H	H	L	H	-	-	L	H
R	-	H	H	H	H	-	-	H	L
F	-	H	-	-	H	L	H	L	H
F	-	H	-	-	H	H	L	H	L
-	-	L	-	-	L	-	-	L	L
-	-	L	-	-	H	-	-	L	H
-	-	H	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.05	*0.10*	0.06	*0.11*
	rise	0.05	*0.11*	0.06	*0.12*
CK to QN	fall	0.04	0.10	0.05	0.11
	rise	0.04	0.10	0.05	0.11
RN to Q	fall	0.01	*0.07*	0.04	*0.10*
	rise	0.02	*0.08*	0.03	*0.09*
RN to QN	rise	0.08	0.14	0.10	0.17
SN to Q	rise	0.03	*0.09*	0.06	*0.12*
SN to QN	fall	0.01	0.07	0.04	0.11
	rise	0.07	0.14	0.09	0.15

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.00	*0.05*	0.00	*0.05*
	rise	0.00	*0.05*	0.00	*0.05*
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
RN to Q	fall	0.00	*0.05*	0.01	*0.05*
	rise	0.00	*0.05*	0.01	*0.05*
RN to QN	rise	0.00	0.05	0.00	0.05
SN to Q	rise	0.00	*0.05*	0.00	*0.05*
SN to QN	fall	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Constraints Time [ns]		
Setup CK to D	fall	0.07
	rise	0.04
Hold CK to D	fall	0.10
	rise	0.06
Recovery CK to RN	rise	-0.01
Removal CK to RN	rise	0.26
Setup CK to SE	fall	0.06
	rise	0.05
Hold CK to SE	fall	0.08
	rise	0.08
Setup CK to SI	fall	0.07
	rise	0.04
Hold CK to SI	fall	0.10
	rise	0.06
Recovery CK to SN	rise	0.03
Removal CK to SN	rise	0.11

Capacitance [fF]	
CK	0.9546
D	1.1572
RN	2.2156
SE	2.1307
SI	0.8686
SN	1.5189

Leakage [nW]
308.40

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	7307.30	15933.49
	rise	9320.30	17974.85
D	fall	3959.41	7307.26
	rise	2188.57	5386.60
RN	fall	2061.11	3732.27
	rise	-970.33	643.56
SE	fall	5223.78	15963.54
	rise	2332.30	12695.09
SI	fall	3546.99	5598.77
	rise	1751.19	3719.15
SN	fall	905.29	850.87
	rise	-756.28	-774.28

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	8.12	*8.03*	12.38	*12.36*
	rise	7.89	*7.66*	12.02	*11.99*
CK to QN	fall	7.89	7.76	12.02	11.90
	rise	8.13	7.93	12.38	12.33
RN to Q	fall	7.26	*7.92*	17.04	*17.21*
	rise	3.47	*3.75*	13.84	*15.29*
RN to QN	rise	7.79	7.79	15.27	14.12
SN to Q	rise	5.49	*7.50*	10.77	*11.57*
SN to QN	fall	5.43	8.16	12.08	13.79
	rise	6.09	6.04	9.90	9.90

NANGATE SDFFRS_X2

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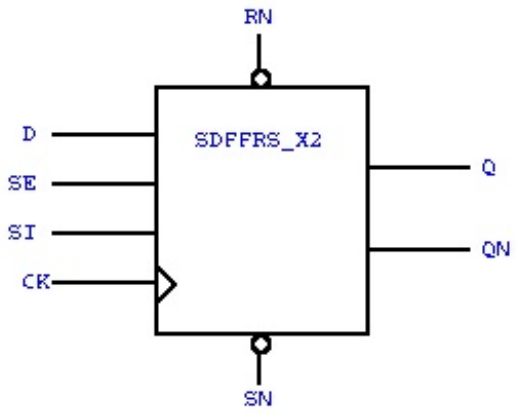
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, active low set, scan, and drive strength X2
Strength	2
Cell Area	8.246 um^2
Equation	Q = "(!(!RN)*((((SE * SI) + (D * !SE)))+(!SN)))" QN = "(!(!SN)*(!((((SE * SI) + (D * !SE)))+(!RN)))"
Clock	CK
Reset	RN
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table									
CK	D	RN	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	H	-	-	L	H
R	H	H	L	-	H	-	-	H	L
R	-	H	H	L	H	-	-	L	H
R	-	H	H	H	H	-	-	H	L
F	-	H	-	-	H	L	H	L	H
F	-	H	-	-	H	H	L	H	L
-	-	L	-	-	L	-	-	L	L
-	-	L	-	-	H	-	-	L	H
-	-	H	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.05	0.11	0.06	0.12
	rise	0.06	0.12	0.07	0.13
CK to QN	fall	0.04	0.10	0.05	0.11
	rise	0.04	0.10	0.05	0.11
RN to Q	fall	0.01	0.07	0.03	0.09
	rise	0.01	0.08	0.04	0.10
RN to QN	rise	0.08	0.14	0.11	0.17
SN to Q	rise	0.03	0.09	0.06	0.11
SN to QN	fall	0.01	0.07	0.04	0.10
	rise	0.07	0.14	0.09	0.15

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
RN to Q	fall	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05
RN to QN	rise	0.00	0.05	0.00	0.05
SN to Q	rise	0.00	0.05	0.00	0.05
SN to QN	fall	0.00	0.05	0.01	0.05
	rise	0.00	0.05	0.01	0.05

Constraints Time [ns]		
Setup CK to D	fall	0.07
	rise	0.04
Hold CK to D	fall	0.10
	rise	0.06
Recovery CK to RN	rise	0.02
Removal CK to RN	rise	0.26
Setup CK to SE	fall	0.06
	rise	0.05
Hold CK to SE	fall	0.08
	rise	0.08
Setup CK to SI	fall	0.07
	rise	0.04
Hold CK to SI	fall	0.10
	rise	0.06
Recovery CK to SN	rise	0.06
Removal CK to SN	rise	0.11

Capacitance [fF]	
CK	0.9349
D	1.1332
RN	2.6427
SE	2.0211
SI	0.8714
SN	1.9068

Leakage [nW]
408.91

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	7481.20	16098.15
	rise	10455.61	19089.29
D	fall	4030.87	7363.15
	rise	2132.56	5321.97
RN	fall	2450.93	4179.77
	rise	-1102.91	514.22
SE	fall	5191.71	15945.99
	rise	2330.30	12687.96
SI	fall	3605.49	5649.10
	rise	1695.31	3655.68
SN	fall	1295.39	1300.73
	rise	-878.42	-893.03

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	12.14	10.86	16.43	15.92
	rise	11.84	10.99	16.00	15.06
CK to QN	fall	11.84	10.99	16.00	15.06
	rise	12.14	10.86	16.43	15.92
RN to Q	fall	10.18	10.98	24.14	22.95
	rise	5.81	5.56	24.48	20.91
RN to QN	rise	11.25	11.01	22.54	19.02
SN to Q	rise	8.82	10.17	17.43	16.40
SN to QN	fall	8.60	11.15	19.33	19.55
	rise	9.29	8.22	17.34	14.09

SDFFR_X1

Databook Build Date: Thursday Feb 17 15:07 2011

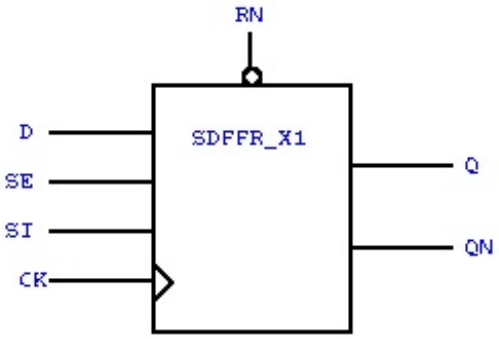
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, scan, and drive strength X1
Strength	1
Cell Area	6.650 um^2
Equation	Q = "(!(!RN)*(((SE * SI) + (D * !SE))))" QN = "((!RN)+!(((SE * SI) + (D * !SE))))"
Clock	CK
Reset	RN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	RN	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	-	-	L	H
R	H	H	L	-	-	-	H	L
R	-	H	H	L	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	H	-	-	L	H	L	H
F	-	H	-	-	H	L	H	L
-	-	L	-	-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
CK to Q	fall	0.03	0.09	0.04	0.10
	rise	0.04	0.10	0.05	0.11
CK to QN	fall	0.05	0.10	0.06	0.12
	rise	0.04	0.10	0.05	0.11
RN to Q	fall	0.07	0.13	0.11	0.17
RN to QN	rise	0.08	0.14	0.12	0.18

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
RN to Q	fall	0.00	0.05	0.01	0.05
RN to QN	rise	0.00	0.05	0.00	0.05

Constraints Time [ns]		
Setup CK to D	fall	0.07
	rise	0.04
Hold CK to D	fall	0.10
	rise	0.06
Recovery CK to RN	rise	-0.01
Removal CK to RN	rise	0.22
Setup CK to SE	fall	0.06
	rise	0.05
Hold CK to SE	fall	0.08
	rise	0.08
Setup CK to SI	fall	0.07
	rise	0.04
Hold CK to SI	fall	0.09
	rise	0.06

Capacitance [fF]	
CK	1.0285
D	1.1627
RN	1.4860
SE	1.9876
SI	0.8861

Leakage [nW]
260.52

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	7432.28	16035.99
	rise	7584.45	16146.29
D	fall	3485.78	6425.93
	rise	1560.25	4364.24
RN	fall	1395.70	1856.33
	rise	-645.78	-268.68
SE	fall	4972.05	15219.15
	rise	1976.82	11835.88
SI	fall	3354.20	5505.19
	rise	1354.08	3412.32

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.4248	0.3656	60.4248
CK to Q	fall	7.14	6.67	11.68	11.21
	rise	7.00	6.80	11.03	11.14
CK to QN	fall	7.00	6.80	11.03	11.14
	rise	7.14	6.67	11.68	11.21
RN to Q	fall	7.25	6.65	9.41	8.53
RN to QN	rise	7.25	6.65	9.41	8.53

SDFFR_X2

Databook Build Date: Thursday Feb 17 15:07 2011

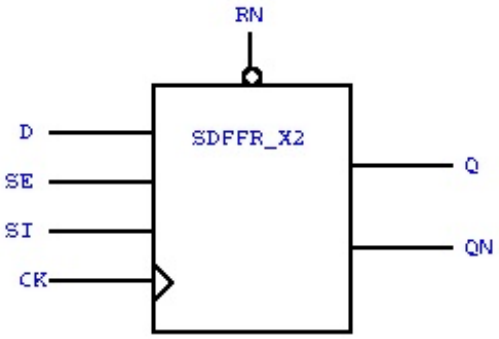
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low reset, scan, and drive strength X2
Strength	2
Cell Area	6.916 um^2
Equation	Q = "(!(!RN)*(((SE * SI) + (D * !SE))))" QN = "((!RN)+!(((SE * SI) + (D * !SE))))"
Clock	CK
Reset	RN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	RN	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	H	L	-	-	-	L	H
R	H	H	L	-	-	-	H	L
R	-	H	H	L	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	H	-	-	L	H	L	H
F	-	H	-	-	H	L	H	L
-	-	L	-	-	-	-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.04	*0.10*	0.05	*0.11*
	rise	0.04	*0.11*	0.05	*0.12*
CK to QN	fall	0.05	0.11	0.07	0.12
	rise	0.05	0.11	0.06	0.12
RN to Q	fall	0.07	*0.14*	0.11	*0.18*
RN to QN	rise	0.09	0.14	0.13	0.18

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.01	*0.05*	0.01	*0.05*
	rise	0.00	*0.05*	0.00	*0.05*
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
RN to Q	fall	0.01	*0.05*	0.01	*0.05*
RN to QN	rise	0.00	0.05	0.00	0.05

Constraints Time [ns]		
Setup CK to D	fall	0.07
	rise	0.04
Hold CK to D	fall	0.10
	rise	0.06
Recovery CK to RN	rise	-0.01
Removal CK to RN	rise	0.04
Setup CK to SE	fall	0.06
	rise	0.05
Hold CK to SE	fall	0.08
	rise	0.08
Setup CK to SI	fall	0.07
	rise	0.04
Hold CK to SI	fall	0.10
	rise	0.06

Capacitance [fF]	
CK	0.9631
D	1.1486
RN	1.5241
SE	1.9011
SI	0.8897

Leakage [nW]
351.08

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	7255.45	15866.01
	rise	7770.88	16347.20
D	fall	3439.26	6382.45
	rise	1525.50	4335.16
RN	fall	1346.54	1813.70
	rise	-644.62	-256.33
SE	fall	4940.99	15194.31
	rise	1937.08	11794.27
SI	fall	3307.43	5461.91
	rise	1319.48	3381.30

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	12.80	*10.46*	17.44	*15.00*
	rise	11.78	*10.12*	15.90	*14.17*
CK to QN	fall	11.78	10.42	15.90	14.63
	rise	12.80	10.46	17.44	14.91
RN to Q	fall	12.80	*10.12*	15.87	*12.15*
RN to QN	rise	12.80	10.04	15.87	12.33

SDFFS_X1

Databook Build Date: Thursday Feb 17 15:07 2011

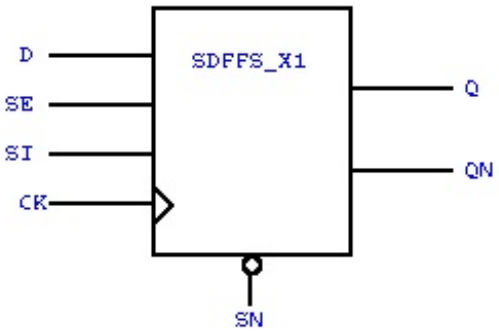
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, scan, and drive strength X1
Strength	1
Cell Area	6.650 um^2
Equation	Q = "((!SN)+(((SE * SI) + (D * !SE))))" QN = "(!((!SN)+(((SE * SI) + (D * !SE)))))"
Clock	CK
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	H	-	-	L	H
R	H	L	-	H	-	-	H	L
R	-	H	L	H	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	-	-	H	L	H	L	H
F	-	-	-	H	H	L	H	L
-	-	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.03	*0.09*	0.04	*0.10*
	rise	0.03	*0.10*	0.04	*0.11*
CK to QN	fall	0.04	0.10	0.06	0.11
	rise	0.04	0.10	0.05	0.11
SN to Q	rise	0.07	*0.13*	0.12	*0.18*
SN to QN	fall	0.01	0.07	0.04	0.10

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.00	*0.05*	0.00	*0.05*
	rise	0.00	*0.05*	0.00	*0.05*
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
SN to Q	rise	0.01	*0.05*	0.01	*0.05*
SN to QN	fall	0.00	0.05	0.01	0.05

Constraints Time [ns]		
Setup CK to D	fall	0.07
	rise	0.04
Hold CK to D	fall	0.10
	rise	0.06
Setup CK to SE	fall	0.06
	rise	0.05
Hold CK to SE	fall	0.08
	rise	0.07
Setup CK to SI	fall	0.07
	rise	0.04
Hold CK to SI	fall	0.09
	rise	0.06
Recovery CK to SN	rise	0.01
Removal CK to SN	rise	0.24

Capacitance [fF]	
CK	0.9845
D	1.1671
SE	1.9927
SI	0.9094
SN	1.3548

Leakage [nW]
267.36

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	7316.51	15996.39
	rise	8935.40	17711.46
D	fall	3846.30	6793.88
	rise	1775.09	4583.10
SE	fall	5487.43	15619.45
	rise	2399.91	12127.90
SI	fall	3717.78	5876.09
	rise	1566.33	3630.01
SN	fall	906.09	850.61
	rise	-757.64	-774.62

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	7.08	*6.70*	11.60	*11.14*
	rise	6.67	*6.55*	10.69	*10.92*
CK to QN	fall	6.68	6.63	10.70	10.91
	rise	7.09	6.69	11.60	11.08
SN to Q	rise	6.97	*6.62*	12.13	*10.28*
SN to QN	fall	6.97	6.57	12.13	10.40

SDFFS_X2

Databook Build Date: Thursday Feb 17 15:07 2011

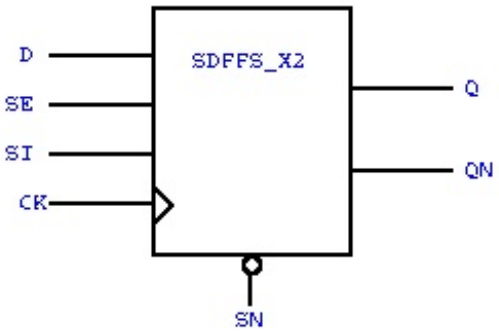
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with active low set, scan, and drive strength X2
Strength	2
Cell Area	7.182 um^2
Equation	Q = "((!SN)+(((SE * SI) + (D * !SE))))" QN = "(!((!SN)+(((SE * SI) + (D * !SE)))))"
Clock	CK
Set	SN
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table								
CK	D	SE	SI	SN	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	H	-	-	L	H
R	H	L	-	H	-	-	H	L
R	-	H	L	H	-	-	L	H
R	-	H	H	H	-	-	H	L
F	-	-	-	H	L	H	L	H
F	-	-	-	H	H	L	H	L
-	-	-	-	L	-	-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.03	*0.10*	0.05	*0.11*
	rise	0.04	*0.10*	0.05	*0.12*
CK to QN	fall	0.05	0.11	0.06	0.12
	rise	0.05	0.11	0.06	0.12
SN to Q	rise	0.08	*0.15*	0.13	*0.19*
SN to QN	fall	0.01	0.07	0.03	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	0.01	*0.05*	0.01	*0.05*
	rise	0.00	*0.05*	0.00	*0.05*
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
SN to Q	rise	0.01	*0.06*	0.01	*0.06*
SN to QN	fall	0.00	0.05	0.01	0.05

Constraints Time [ns]		
Setup CK to D	fall	0.07
	rise	0.04
Hold CK to D	fall	0.10
	rise	0.06
Setup CK to SE	fall	0.06
	rise	0.06
Hold CK to SE	fall	0.08
	rise	0.07
Setup CK to SI	fall	0.07
	rise	0.04
Hold CK to SI	fall	0.09
	rise	0.06
Recovery CK to SN	rise	0.01
Removal CK to SN	rise	0.24

Capacitance [fF]	
CK	0.9641
D	1.1377
SE	1.8868
SI	0.9239
SN	2.2897

Leakage [nW]
348.46

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	7459.70	16165.33
	rise	11023.37	20575.64
D	fall	4068.05	7011.30
	rise	1838.85	4640.86
SE	fall	5577.27	15727.99
	rise	2540.74	12277.82
SI	fall	3938.73	6092.40
	rise	1629.77	3687.84
SN	fall	1644.10	1648.56
	rise	-1103.35	-1118.06

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	120.85	0.3656	120.85
CK to Q	fall	12.50	*10.10*	17.15	*14.04*
	rise	12.17	*10.70*	16.29	*14.91*
CK to QN	fall	12.17	10.60	16.29	14.75
	rise	12.50	10.11	17.15	14.60
SN to Q	rise	12.66	*10.25*	23.03	*17.92*
SN to QN	fall	12.66	10.52	23.03	17.83

SDFF_X1

Databook Build Date: Thursday Feb 17 15:07 2011

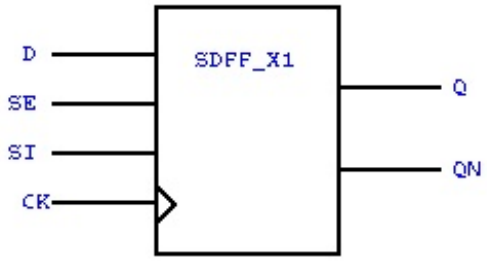
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with scan, and drive strength X1
Strength	1
Cell Area	6.118 um^2
Equation	Q = "(((SE * SI) + (D * !SE)))" QN = "(!(((SE * SI) + (D * !SE))))"
Clock	CK
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	-	-	L	H
R	H	L	-	-	-	H	L
R	-	H	L	-	-	L	H
R	-	H	H	-	-	H	L
F	-	-	-	L	H	L	H
F	-	-	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.03	0.09	0.04	0.10
	rise	0.03	0.09	0.04	0.10
CK to QN	fall	0.04	*0.10*	0.05	*0.11*
	rise	0.04	*0.09*	0.05	*0.11*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05
CK to QN	fall	0.00	*0.05*	0.00	*0.05*
	rise	0.00	*0.05*	0.00	*0.05*

Constraints Time [ns]		
Setup CK to D	fall	0.07
	rise	0.04
Hold CK to D	fall	0.10
	rise	0.06
Setup CK to SE	fall	0.06
	rise	0.05
Hold CK to SE	fall	0.08
	rise	0.07
Setup CK to SI	fall	0.07
	rise	0.04
Hold CK to SI	fall	0.09
	rise	0.06

Capacitance [fF]	
CK	0.9594
D	1.1265
SE	1.8622
SI	0.9249

Leakage [nW]
247.25

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	6886.22	15721.85
	rise	3492.46	12135.17
D	fall	3813.53	6758.73
	rise	1809.20	4619.71
SE	fall	5191.50	15477.27
	rise	2270.91	12167.35
SI	fall	3682.87	5841.62
	rise	1601.32	3666.55

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	60.5774	0.3656	60.5774
CK to Q	fall	6.36	5.96	10.92	10.52
	rise	6.22	5.98	10.25	10.24
CK to QN	fall	6.23	*6.02*	10.25	*10.29*
	rise	6.36	*6.01*	10.93	*10.51*

SDFF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

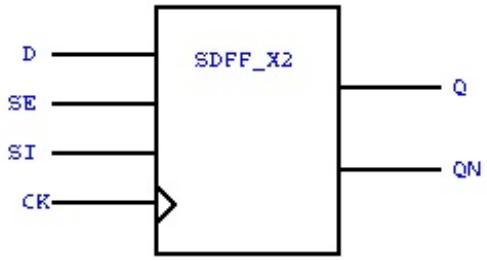
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	Pos. edge Flip-Flop with scan, and drive strength X2
Strength	2
Cell Area	6.384 um^2
Equation	Q = "(((SE * SI) + (D * !SE)))" QN = "(!(((SE * SI) + (D * !SE))))"
Clock	CK
Type	Sequential
Scan Enable	SE
Scan Data	SI
Input	D
Output	Q, QN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table							
CK	D	SE	SI	$IQ_{(int)}$	$IQN_{(int)}$	Q	QN
R	L	L	-	-	-	L	H
R	H	L	-	-	-	H	L
R	-	H	L	-	-	L	H
R	-	H	H	-	-	H	L
F	-	-	-	L	H	L	H
F	-	-	-	H	L	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.03	*0.10*	0.04	*0.11*
	rise	0.04	*0.10*	0.05	*0.11*
CK to QN	fall	0.05	0.10	0.06	0.11
	rise	0.05	0.10	0.06	0.11

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	0.01	*0.05*	0.01	*0.05*
	rise	0.00	*0.05*	0.00	*0.05*
CK to QN	fall	0.00	0.05	0.00	0.05
	rise	0.00	0.05	0.00	0.05

Constraints Time [ns]		
Setup CK to D	fall	0.07
	rise	0.04
Hold CK to D	fall	0.10
	rise	0.06
Setup CK to SE	fall	0.06
	rise	0.05
Hold CK to SE	fall	0.08
	rise	0.07
Setup CK to SI	fall	0.07
	rise	0.04
Hold CK to SI	fall	0.09
	rise	0.06

Capacitance [fF]	
CK	0.9825
D	1.1445
SE	1.8402
SI	0.9063

Leakage [nW]
335.15

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
CK	fall	6976.17	15767.64
	rise	3594.15	12195.96
D	fall	3823.25	6762.47
	rise	1814.79	4619.53
SE	fall	5179.18	15466.50
	rise	2277.83	12175.23
SI	fall	3692.96	5846.39
	rise	1606.81	3667.80

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	121.155	0.3656	121.155
CK to Q	fall	11.85	*9.65*	16.68	*14.26*
	rise	10.92	*9.24*	15.02	*13.20*
CK to QN	fall	10.92	9.48	15.02	13.85
	rise	11.85	9.71	16.68	14.26

TBUF_X16

Databook Build Date: Thursday Feb 17 15:07 2011

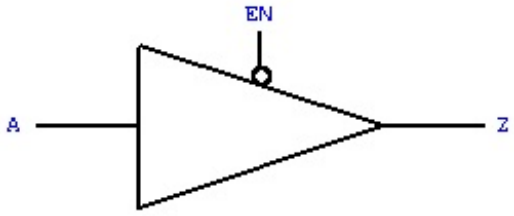
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	16
Cell Area	6.916 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		12.9359	832.913	12.9359	832.913
A to Z	**fall**	*0.01*	0.07	*0.05*	0.11
	rise	0.01	*0.08*	0.03	*0.11*
EN to Z	**fall**	*0.01*	0.08	*0.04*	0.11
	rise	0.02	*0.09*	0.05	*0.12*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		12.9359	832.913	12.9359	832.913
A to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.06*	0.01	*0.06*
EN to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.06*	0.00	*0.06*

Capacitance [fF]	
A	6.5096
EN	4.9659
Z	12.6010

Leakage [nW]
471.99

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		12.9359	832.913	12.9359	832.913
A to Z	**fall**	*46.72*	47.86	*97.82*	95.43
	rise	23.52	*32.71*	63.99	*78.32*
EN to Z	**fall**	*30.07*	39.25	*54.58*	74.75
	rise	31.26	*40.13*	44.39	*56.83*

TBUF_X1

Databook Build Date: Thursday Feb 17 15:07 2011

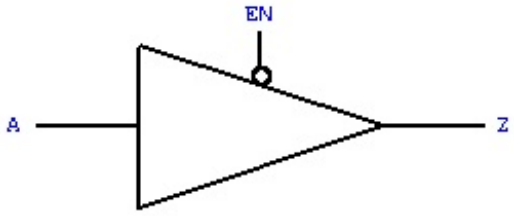
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	2.128 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		1.3837	52.5936	1.3837	52.5936
A to Z	**fall**	*0.01*	0.07	*0.03*	0.10
	rise	0.01	*0.07*	0.02	*0.09*
EN to Z	**fall**	*0.01*	0.07	*0.03*	0.10
	rise	0.02	*0.08*	0.04	*0.10*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		1.3837	52.5936	1.3837	52.5936
A to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.05*	0.01	*0.05*
EN to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.05*	0.00	*0.05*

Capacitance [fF]	
A	1.8786
EN	1.7320
Z	1.0189

Leakage [nW]
79.97

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		1.3837	52.5936	1.3837	52.5936
A to Z	**fall**	*5.77*	6.57	*13.82*	18.56
	rise	2.34	*2.55*	12.02	*14.91*
EN to Z	**fall**	*5.37*	5.97	*15.94*	20.83
	rise	5.19	*5.17*	11.62	*13.98*

TBUF_X2

Databook Build Date: Thursday Feb 17 15:07 2011

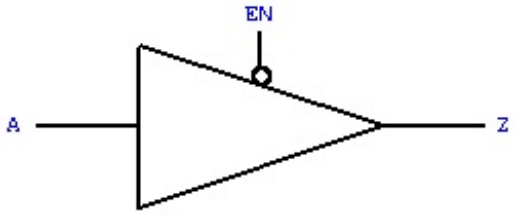
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		1.9423	105.1854	1.9423	105.1854
A to Z	**fall**	*0.01*	0.07	*0.03*	0.10
	rise	0.01	*0.07*	0.02	*0.09*
EN to Z	**fall**	*0.01*	0.07	*0.03*	0.10
	rise	0.02	*0.08*	0.04	*0.10*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		1.9423	105.1854	1.9423	105.1854
A to Z	**fall**	*0.00*	0.05	*0.00*	0.05
	rise	0.00	*0.05*	0.01	*0.05*
EN to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.05*	0.00	*0.05*

Capacitance [fF]	
A	3.2963
EN	2.7298
Z	1.5784

Leakage [nW]
143.87

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		1.9423	105.1854	1.9423	105.1854
A to Z	**fall**	*10.14*	11.81	*26.19*	36.25
	rise	4.21	*4.78*	23.33	*30.12*
EN to Z	**fall**	*7.53*	8.94	*21.92*	30.32
	rise	7.81	*8.08*	14.37	*17.40*

TBUF_X4

Databook Build Date: Thursday Feb 17 15:07 2011

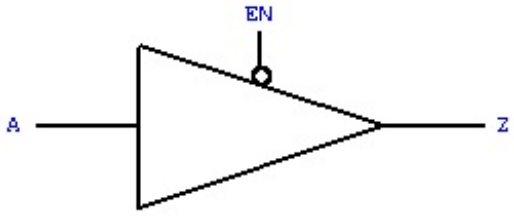
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	4
Cell Area	2.926 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		3.4692	210.0172	3.4692	210.0172
A to Z	**fall**	*0.01*	0.07	*0.04*	0.11
	rise	0.01	*0.07*	0.02	*0.09*
EN to Z	**fall**	*0.01*	0.07	*0.03*	0.10
	rise	0.02	*0.08*	0.04	*0.11*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		3.4692	210.0172	3.4692	210.0172
A to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.05*	0.01	*0.05*
EN to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.05*	0.00	*0.05*

Capacitance [fF]	
A	3.3931
EN	2.4456
Z	3.1082

Leakage [nW]
174.52

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		3.4692	210.0172	3.4692	210.0172
A to Z	**fall**	*12.06*	14.65	*32.18*	40.34
	rise	6.29	*8.62*	24.75	*32.90*
EN to Z	**fall**	*10.85*	13.17	*22.22*	31.64
	rise	11.03	*12.13*	17.31	*20.13*

TBUF_X8

Databook Build Date: Thursday Feb 17 15:07 2011

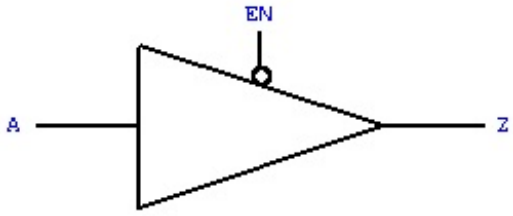
Copyright © 2004-2010 Nangate Inc.

Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	8
Cell Area	4.788 um^2
Equation	Z = "A", when !(EN)
Type	Tristate
Input	A, EN
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**EN**	**Z**
L	L	L
H	L	H
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		6.8681	419.1113	6.8681	419.1113
A to Z	**fall**	*0.01*	0.07	*0.04*	0.11
	rise	0.01	*0.07*	0.02	*0.09*
EN to Z	**fall**	*0.01*	0.07	*0.03*	0.10
	rise	0.02	*0.08*	0.04	*0.11*

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		6.8681	419.1113	6.8681	419.1113
A to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.05*	0.01	*0.05*
EN to Z	**fall**	*0.00*	0.05	*0.01*	0.05
	rise	0.00	*0.05*	0.00	*0.05*

Capacitance [fF]	
A	6.7518
EN	5.0373
Z	6.5133

Leakage [nW]
349.49

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		6.8681	419.1113	6.8681	419.1113
A to Z	**fall**	*24.83*	29.83	*65.41*	81.06
	rise	12.41	*16.11*	49.61	*66.91*
EN to Z	**fall**	*22.01*	26.80	*44.83*	63.82
	rise	22.65	*26.48*	35.45	*44.17*

TINV_X1

Databook Build Date: Thursday Feb 17 15:07 2011

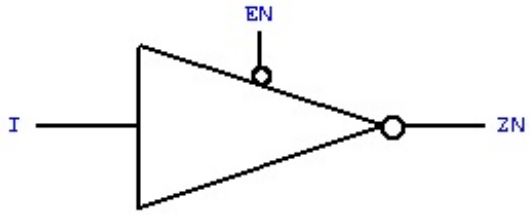
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.064 um^2
Equation	ZN = "!I", when !(EN)
Type	Tristate
Input	EN, I
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
I	**EN**	**ZN**
L	L	H
H	L	L
-	H	Z

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		1.1403	23.3962	1.1403	23.3962
EN to ZN	**fall**	0.01	*0.04*	0.02	*0.07*
	rise	*0.00*	0.05	*-0.04*	0.07
I to ZN	**fall**	0.01	*0.04*	0.01	*0.07*
	rise	*0.01*	0.05	*0.03*	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		1.1403	23.3962	1.1403	23.3962
EN to ZN	**fall**	0.00	*0.04*	0.01	*0.04*
	rise	*0.00*	0.04	*0.03*	0.07
I to ZN	**fall**	0.00	*0.04*	0.02	*0.05*
	rise	*0.00*	0.04	*0.02*	0.05

Capacitance [fF]	
EN	1.7476
I	1.4692
ZN	0.7750

Leakage [nW]
44.10

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		1.1403	23.3962	1.1403	23.3962
EN to ZN	**fall**	3.39	*3.54*	8.17	*11.78*
	rise	*3.86*	3.88	*3.93*	12.30
I to ZN	**fall**	0.65	*0.81*	6.81	*4.72*
	rise	*3.23*	3.27	*9.83*	7.61

TLAT_X1

Databook Build Date: Thursday Feb 17 15:07 2011

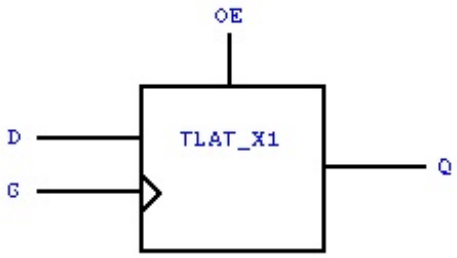
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Description	High enable Latch with drive strength X1
Strength	1
Cell Area	3.458 um^2
Equation	Q = "(D)", when !(!OE)
Enable	G
Type	Sequential
Input	D, OE
Output	Q
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table					
OE	**D**	**G**	**$IQ_{(int)}$**	**$IQN_{(int)}$**	**Q**
L	-	-	-	-	Z
H	L	H	-	-	L
H	H	H	-	-	H
H	-	L	L	H	L
H	-	L	H	L	H

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**1.1335**	**23.352**	**1.1335**	**23.352**
D to Q	**fall**	*0.02*	0.06	*0.05*	0.09
	rise	0.02	*0.06*	0.03	*0.07*
G to Q	**fall**	*0.02*	0.06	*0.03*	0.07
	rise	0.03	*0.07*	0.04	*0.08*
OE to Q	**fall**	*0.00*	0.04	*-0.05*	0.06
	rise	0.01	*0.05*	0.00	*0.06*

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**1.1335**	**23.352**	**1.1335**	**23.352**
D to Q	**fall**	*0.00*	0.04	*0.01*	0.04
	rise	0.00	*0.04*	0.01	*0.04*
G to Q	**fall**	*0.00*	0.04	*0.00*	0.04
	rise	0.00	*0.04*	0.00	*0.04*
OE to Q	**fall**	*0.00*	0.04	*0.02*	0.06
	rise	0.00	*0.04*	0.01	*0.04*

Constraints Time [ns]		
Setup G to D	**fall**	0.05
	rise	0.02
Hold G to D	**fall**	0.13
	rise	0.08

Capacitance [fF]	
D	1.1543
G	1.0132
OE	1.4836
Q	0.7690

Leakage [nW]
118.55

Input Pin Power Consumption [nW/GHz]			
Input Transition [ns]		0.0009	0.1251
D	fall	1909.17	3363.78
	rise	240.92	1635.97
G	fall	3659.55	12673.55
	rise	3683.98	12685.67
OE	fall	2154.00	10450.28
	rise	1086.60	9239.44

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		1.1335	23.352	1.1335	23.352
D to Q	fall	*5.92*	6.43	*11.12*	12.34
	rise	5.24	*5.37*	10.74	*11.60*
G to Q	fall	*7.13*	7.65	*15.21*	17.26
	rise	8.14	*8.52*	16.06	*17.62*
OE to Q	fall	*0.64*	1.07	*2.21*	9.27
	rise	1.79	*1.90*	6.42	*9.79*

XNOR2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

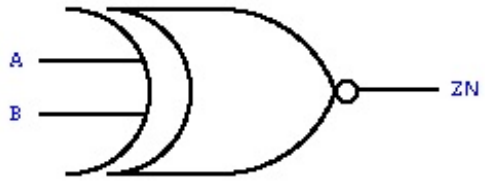
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	ZN = "!(A ^ B)"
Type	Combinational
Input	A, B
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**ZN**
L	L	H
L	H	L
H	L	L
H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	26.0162	0.3656	26.0162
A to ZN	**fall**	0.02	0.05	0.04	0.07
	rise	0.02	0.05	0.04	0.10
B to ZN	**fall**	0.01	0.05	0.04	0.08
	rise	0.02	0.05	0.03	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	26.0162	0.3656	26.0162
A to ZN	**fall**	0.00	0.04	0.02	0.06
	rise	0.00	0.04	0.02	0.06
B to ZN	**fall**	0.00	0.04	0.02	0.06
	rise	0.00	0.04	0.02	0.05

Capacitance [fF]	
A	2.2174
B	2.5768

Leakage [nW]
90.91

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	26.0162	0.3656	26.0162
A to ZN	**fall**	3.25	3.04	10.08	8.95
	rise	4.41	4.18	12.58	11.12
B to ZN	**fall**	3.40	3.57	10.97	9.59
	rise	4.53	4.48	10.87	9.79

XNOR2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

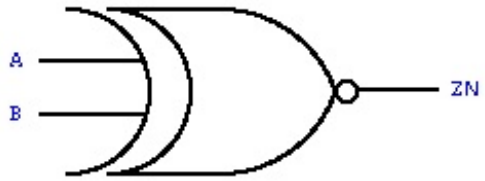
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.660 um^2
Equation	ZN = "!(A ^ B)"
Type	Combinational
Input	A, B
Output	ZN
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**ZN**
L	L	H
L	H	L
H	L	L
H	H	H

Propagation Delay [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	52.0325	0.3656	52.0325
A to ZN	**fall**	0.01	0.05	0.03	0.07
	rise	0.02	0.05	0.03	0.10
B to ZN	**fall**	0.01	0.05	0.04	0.08
	rise	0.02	0.05	0.03	0.09

Output Transition [ns]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	52.0325	0.3656	52.0325
A to ZN	**fall**	0.00	0.04	0.02	0.06
	rise	0.00	0.04	0.02	0.06
B to ZN	**fall**	0.00	0.04	0.02	0.06
	rise	0.00	0.04	0.02	0.05

Capacitance [fF]	
A	3.9542
B	4.8245

Leakage [nW]
182.39

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		0.0009		0.1251	
Load Capacitance [fF]		0.3656	52.0325	0.3656	52.0325
A to ZN	**fall**	6.08	5.64	19.74	17.53
	rise	8.78	8.62	25.11	22.21
B to ZN	**fall**	6.42	6.74	21.49	18.78
	rise	9.17	8.77	21.80	19.72

XOR2_X1

Databook Build Date: Thursday Feb 17 15:07 2011

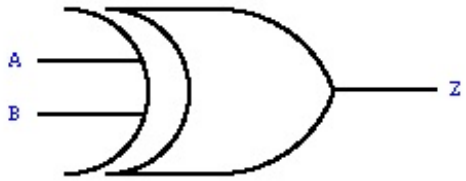
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	1.596 um^2
Equation	Z = "(A ^ B)"
Type	Combinational
Input	A, B
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**Z**
L	L	L
L	H	H
H	L	H
H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to Z	**fall**	0.02	0.04	0.05	0.08
	rise	0.02	0.06	0.03	0.09
B to Z	**fall**	0.02	0.04	0.05	0.08
	rise	0.02	0.06	0.04	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to Z	**fall**	0.00	0.03	0.02	0.05
	rise	0.01	0.04	0.02	0.06
B to Z	**fall**	0.00	0.03	0.02	0.05
	rise	0.00	0.04	0.02	0.06

Capacitance [fF]	
A	2.2342
B	2.4726

Leakage [nW]
87.17

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**25.3296**	**0.3656**	**25.3296**
A to Z	**fall**	3.36	3.29	10.59	9.14
	rise	4.33	4.21	11.70	10.80
B to Z	**fall**	3.51	3.45	8.96	8.00
	rise	4.62	4.69	12.70	11.24

XOR2_X2

Databook Build Date: Thursday Feb 17 15:07 2011

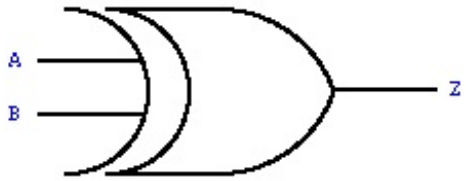
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Conditions for characterization library **NangateOpenCellLibrary**, corner **NangateOpenCellLibrary_low_temp_low_temp**: Vdd= **1.25**V, Tj= **-40.0** deg. C . Additional corners: NangateOpenCellLibrary [NangateOpenCellLibrary_typical_typical], NangateOpenCellLibrary [NangateOpenCellLibrary_slow_slow], NangateOpenCellLibrary [NangateOpenCellLibrary_fast_fast], NangateOpenCellLibrary [NangateOpenCellLibrary_worst_low_worst_low].

Output transition is defined from **30%** to **70%** (rising) and from **70%** to **30%** (falling) output voltage.

Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	2
Cell Area	2.394 um^2
Equation	Z = "(A ^ B)"
Type	Combinational
Input	A, B
Output	Z
PG Pins	VDD (primary_power), VSS (primary_ground)



State Table		
A	**B**	**Z**
L	L	L
L	H	H
H	L	H
H	H	L

Propagation Delay [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**50.5066**	**0.3656**	**50.5066**
A to Z	**fall**	0.02	0.04	0.05	0.08
	rise	0.02	0.06	0.03	0.09
B to Z	**fall**	0.02	0.04	0.05	0.08
	rise	0.02	0.06	0.04	0.10

Output Transition [ns]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**50.5066**	**0.3656**	**50.5066**
A to Z	**fall**	0.00	0.03	0.02	0.05
	rise	0.00	0.04	0.02	0.06
B to Z	**fall**	0.00	0.03	0.02	0.05
	rise	0.00	0.04	0.02	0.06

Capacitance [fF]	
A	4.3341
B	4.6074

Leakage [nW]
175.01

Dynamic Power Consumption [uW/GHz]					
Input Transition [ns]		**0.0009**		**0.1251**	
Load Capacitance [fF]		**0.3656**	**50.5066**	**0.3656**	**50.5066**
A to Z	**fall**	6.91	6.85	21.33	18.53
	rise	8.83	8.45	23.68	21.68
B to Z	**fall**	7.28	7.17	18.15	16.27
	rise	9.34	9.42	25.54	22.92